



Intelligent Security API (General)

Developer Guide

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Contents

Chapter 1 Overview	1
1.1 Introduction	1
1.2 Update History	2
1.2.1 Version 20.04-Version 20.06	2
1.2.2 Version 20.01-Version 20.03	5
Chapter 2 ISAPI Description	9
2.1 Operation Method	9
2.2 URL Format	12
2.3 Message Format	13
2.4 Others	15
Chapter 3 Security	16
3.1 Authentication	16
3.2 User Permission	17
3.3 Encryption	17
3.3.1 Encrypt Stream	19
3.3.2 Encrypt Sensitive Information	21
3.4 Security Service	31
Chapter 4 Login	34
Chapter 5 Device Management	35
5.1 Device Activation	35
5.1.1 Directly Activate Device	35
5.1.2 Activate Network Camera via NVR	37
5.2 Reset Password by Setting Recovery Email	37
5.3 Online Upgrade Device	39
5.4 IoT Devices Access	41
5.4.1 Add IoT Devices to NVR/DVR	42

5.4.2 Configure Alarm of IoT Device	44
Chapter 6 Device Maintenance	47
6.1 Basic	47
6.1.1 Securely Export and Import Configuration Files	48
6.2 Log	50
6.3 Status	51
6.4 Calibration	52
6.5 Auto Power On or Off	53
6.6 Advanced	53
Chapter 7 User Management	55
7.1 Configure Double Verification	56
Chapter 8 Video and Audio	60
8.1 Search for Recording Start and End Time by Channel	60
8.2 Live View and Playback	61
8.3 Start Two-Way Audio	68
8.4 Stream Encoding	69
Chapter 9 Capture and Recording	71
Chapter 10 PTZ Control	73
10.1 Configure Master-Slave Tracking	77
10.2 Configure Panorama Tracking	78
Chapter 11 Storage Management	79
11.1 Configure Log Storage Mode	79
11.2 HDD Management	80
11.3 Others	82
Chapter 12 File Management	83
12.1 Video Files	83
12.2 Audio Files	84
Chapter 13 Alarm/Event	86

13.1 Configure Exception Alarm	86
13.2 Configure Certificate Expiry Alarm	87
13.3 Configure Network Traffic Pre-alarm	90
13.4 Configure Motion Detection Alarm	91
13.5 Configure Vibration Detection Alarm	93
13.6 Receive Alarm/Event in Arming Mode	96
13.7 Receive Alarm/Event in Listening Mode	97
13.8 Subscribe Alarm/Event in Arming Mode	99
Chapter 14 Parameter Configuration	102
14.1 I/O Settings	102
14.2 Video/Image Settings	104
14.3 Audio Settings	109
14.4 Channel Settings	110
14.5 Peripherals Settings	112
14.6 Picture Storage Server Settings	113
Chapter 15 System Configuration	114
15.1 Network Settings	114
15.1.1 Email	115
15.1.2 Network Access	115
15.1.3 Network Interface	118
15.1.4 Wireless Network	119
15.1.5 Network Service	121
15.2 Time Settings	123
15.3 System Diagnose	124
15.4 Data Replenishment	125
Chapter 16 Request URIs	127
16.1 /ISAPI/AUXInfo	127
16.1.1 /ISAPI/AUXInfo/attributes/Channels	127

16.1.2 /ISAPI/AUXInfo/attributes/Channels/<ID>	127
16.2 /ISAPI/ContentMgmt	128
16.2.1 /ISAPI/ContentMgmt/capabilities	128
16.2.2 /ISAPI/ContentMgmt/download	128
16.2.3 /ISAPI/ContentMgmt/download/capabilities	128
16.2.4 /ISAPI/ContentMgmt/download/toUSB/<taskId>/progress?format=json	129
16.2.5 /ISAPI/ContentMgmt/download/toUSB/capabilities?format=json	129
16.2.6 /ISAPI/ContentMgmt/download/toUSB?format=json	130
16.2.7 /ISAPI/ContentMgmt/InputProxy/channels	130
16.2.8 /ISAPI/ContentMgmt/InputProxy/channels/<ID>	131
16.2.9 /ISAPI/ContentMgmt/InputProxy/channels/<ID>/capabilities	132
16.2.10 /ISAPI/ContentMgmt/InputProxy/channels/<ID>/chanCtrl	132
16.2.11 /ISAPI/ContentMgmt/InputProxy/channels/<ID>/deviceInfo	133
16.2.12 /ISAPI/ContentMgmt/InputProxy/channels/<ID>/portMapParam	134
16.2.13 /ISAPI/ContentMgmt/InputProxy/channels/<ID>/reboot	134
16.2.14 /ISAPI/ContentMgmt/InputProxy/channels/<ID>/status	135
16.2.15 /ISAPI/ContentMgmt/InputProxy/channels/<ID>/video/overlays	135
16.2.16 /ISAPI/ContentMgmt/InputProxy/channels/<ID>/video/overlays/capabilities	136
16.2.17 /ISAPI/ContentMgmt/InputProxy/channels/activate	136
16.2.18 /ISAPI/ContentMgmt/InputProxy/channels/activate/capabilities	137
16.2.19 /ISAPI/ContentMgmt/InputProxy/channels/capabilities	137
16.2.20 /ISAPI/ContentMgmt/InputProxy/channels/status	138
16.2.21 /ISAPI/ContentMgmt/InputProxy/deviceInfo	138
16.2.22 /ISAPI/ContentMgmt/InputProxy/search	138
16.2.23 /ISAPI/ContentMgmt/InputProxy/sourceCapability	139
16.2.24 /ISAPI/ContentMgmt/IOProxy/inputs	139
16.2.25 /ISAPI/ContentMgmt/IOProxy/inputs/<ID>	140
16.2.26 /ISAPI/ContentMgmt/IOProxy/inputs/<ID>/status	141

16.2.27 /ISAPI/ContentMgmt/IOProxy/outputs	141
16.2.28 /ISAPI/ContentMgmt/IOProxy/outputs/<ID>	142
16.2.29 /ISAPI/ContentMgmt/IOProxy/outputs/<ID>/status	143
16.2.30 /ISAPI/ContentMgmt/IOProxy/outputs/<ID>/trigger	143
16.2.31 /ISAPI/ContentMgmt/logConfig	144
16.2.32 /ISAPI/ContentMgmt/logConfig/capabilities	144
16.2.33 /ISAPI/ContentMgmt/logSearch	145
16.2.34 /ISAPI/ContentMgmt/logSearch?format=json	145
16.2.35 /ISAPI/ContentMgmt/logSearch/dataPackage	145
16.2.36 /ISAPI/ContentMgmt/record/control/cleanUp/capabilities?format=json	146
16.2.37 /ISAPI/ContentMgmt/record/control/cleanUp?format=json	146
16.2.38 /ISAPI/ContentMgmt/record/control/locks/capabilities	147
16.2.39 /ISAPI/ContentMgmt/record/control/locks/name	147
16.2.40 /ISAPI/ContentMgmt/record/control/manual/start/tracks/<ID>	148
16.2.41 /ISAPI/ContentMgmt/record/control/manual/stop/tracks/<ID>	148
16.2.42 /ISAPI/ContentMgmt/record/control/manualRefresh/channels/<ID>	149
16.2.43 /ISAPI/ContentMgmt/record/control/remark/capabilities?format=json	149
16.2.44 /ISAPI/ContentMgmt/record/control/remark?format=json	149
16.2.45 /ISAPI/ContentMgmt/record/profile	150
16.2.46 /ISAPI/ContentMgmt/record/storageMounts	150
16.2.47 /ISAPI/ContentMgmt/record/tracks	151
16.2.48 /ISAPI/ContentMgmt/record/tracks/<ID>	151
16.2.49 /ISAPI/ContentMgmt/record/tracks/<ID>/capabilities	152
16.2.50 /ISAPI/ContentMgmt/record/tracks/<ID>/dailyDistribution	153
16.2.51 /ISAPI/ContentMgmt/search	153
16.2.52 /ISAPI/ContentMgmt/search/profile	155
16.2.53 /ISAPI/ContentMgmt/security/logSearch	156
16.2.54 /ISAPI/ContentMgmt/SmartSearch	156

16.2.55 /ISAPI/ContentMgmt/SmartSearch/capabilities	156
16.2.56 /ISAPI/ContentMgmt/Storage/extension	157
16.2.57 /ISAPI/ContentMgmt/Storage/extension/capabilities	157
16.2.58 /ISAPI/ContentMgmt/Storage/ExtraInfo	158
16.2.59 /ISAPI/ContentMgmt/Storage/ExtraInfo/capabilities	158
16.2.60 /ISAPI/ContentMgmt/Storage/hdd	159
16.2.61 /ISAPI/ContentMgmt/Storage/hdd/<ID>	159
16.2.62 /ISAPI/ContentMgmt/Storage/hdd/<ID>/BadSectorsTest/pause	160
16.2.63 /ISAPI/ContentMgmt/Storage/hdd/<ID>/BadSectorsTest/resume	160
16.2.64 /ISAPI/ContentMgmt/Storage/hdd/<ID>/BadSectorsTest/start	161
16.2.65 /ISAPI/ContentMgmt/Storage/hdd/<ID>/BadSectorsTest/status	161
16.2.66 /ISAPI/ContentMgmt/Storage/hdd/<ID>/BadSectorsTest/stop	162
16.2.67 /ISAPI/ContentMgmt/Storage/hdd/<ID>/encryptFormat?format=json	162
16.2.68 /ISAPI/ContentMgmt/Storage/hdd/<ID>/encryptVerify?format=json	163
16.2.69 /ISAPI/ContentMgmt/Storage/hdd/<ID>/formatStatus	163
16.2.70 /ISAPI/ContentMgmt/Storage/hdd/<ID>/format?formatType=	164
16.2.71 /ISAPI/ContentMgmt/Storage/hdd/<ID>/logProTest?format=json	164
16.2.72 /ISAPI/ContentMgmt/Storage/hdd/<ID>/SMARTTest/start	165
16.2.73 /ISAPI/ContentMgmt/Storage/hdd/<ID>/SMARTTest/status	165
16.2.74 /ISAPI/ContentMgmt/Storage/hdd/capabilities	166
16.2.75 /ISAPI/ContentMgmt/storage/hdd/clearingSpace/capabilities?format=json	166
16.2.76 /ISAPI/ContentMgmt/storage/hdd/clearingSpace?format=json	167
16.2.77 /ISAPI/ContentMgmt/Storage/hdd/format	167
16.2.78 /ISAPI/ContentMgmt/Storage/hdd/SMARTTest/config	168
16.2.79 /ISAPI/ContentMgmt/Storage/hdd/specifyHddFormat?format=json	168
16.2.80 /ISAPI/ContentMgmt/Storage/quota	169
16.2.81 /ISAPI/ContentMgmt/StreamingProxy/channels/<ID>/Picture	169
16.2.82 /ISAPI/ContentMgmt/time/search/capabilities?format=json	169

16.2.83 /ISAPI/ContentMgmt/time/search?format=json	170
16.3 /ISAPI/Event	170
16.3.1 /ISAPI/Event/capabilities	170
16.3.2 /ISAPI/Event/channels/<ID>/capabilities	171
16.3.3 /ISAPI/Event/channels/capabilities	171
16.3.4 /ISAPI/Event/IOT/channels/<ID>/capabilities?format=json	172
16.3.5 /ISAPI/Event/notification/alertStream	172
16.3.6 /ISAPI/Event/notification/httpHosts	173
16.3.7 /ISAPI/Event/notification/httpHosts/<ID>	174
16.3.8 /ISAPI/Event/notification/httpHosts/<ID>/test	175
16.3.9 /ISAPI/Event/notification/httpHosts/capabilities	176
16.3.10 /ISAPI/Event/notification/subscribeEvent	176
16.3.11 /ISAPI/Event/notification/subscribeEvent/<ID>	177
16.3.12 /ISAPI/Event/notification/subscribeEventCap	178
16.3.13 /ISAPI/Event/notification/unSubscribeEvent	178
16.3.14 /ISAPI/Event/schedules/motionDetections	178
16.3.15 /ISAPI/Event/schedules/motionDetections/<ID>	179
16.3.16 /ISAPI/Event/schedules/vibrationDetection/<ID>	180
16.3.17 /ISAPI/Event/schedules/<EventType>/<ID>	180
16.3.18 /ISAPI/Event/triggers/hdBadBlock	181
16.3.19 /ISAPI/Event/triggers/hdImpact	182
16.3.20 /ISAPI/Event/triggers/highHDTemperature	182
16.3.21 /ISAPI/Event/triggers/lowHDTemperature	183
16.3.22 /ISAPI/Event/triggers/severeHDFailure	183
16.3.23 /ISAPI/Event/triggers/<ID>	184
16.3.24 /ISAPI/Event/triggers/<ID>/notifications	185
16.3.25 /ISAPI/Event/triggersCap/IOT	186
16.3.26 /ISAPI/Event/triggers/notifications/AudioAlarm/<audioID>/test?format=json ...	186

16.3.27 /ISAPI/Event/triggers/notifications/AudioAlarm/capabilities?format=json	187
16.3.28 /ISAPI/Event/triggers/notifications/AudioAlarm/customAudio/<customAudioID>?format=json	187
16.3.29 /ISAPI/Event/triggers/notifications/AudioAlarm/customAudio?format=json	188
16.3.30 /ISAPI/Event/triggers/notifications/AudioAlarm/customAudioInfo/<customAudioID>?format=json	188
16.3.31 /ISAPI/Event/triggers/notifications/AudioAlarm/customAudioInfo?format=json	189
16.3.32 /ISAPI/Event/triggers/notifications/AudioAlarm?format=json	189
16.3.33 /ISAPI/Event/triggers/notifications/whiteLightAlarm?format=json	190
16.3.34 /ISAPI/Event/triggers/notifications/whiteLightAlarm/capabilities?format=json ..	191
16.3.35 /ISAPI/Event/triggersCap	191
16.3.36 http://ipAddress:portNo/url	192
16.4 /ISAPI/Image	192
16.4.1 /ISAPI/Image/channels	192
16.4.2 /ISAPI/Image/channels/<ID>	193
16.4.3 /ISAPI/Image/channels/<ID>/BLC	193
16.4.4 /ISAPI/Image/channels/<ID>/capabilities	194
16.4.5 /ISAPI/Image/channels/<ID>/capturemode	194
16.4.6 /ISAPI/Image/channels/<ID>/color	195
16.4.7 /ISAPI/Image/channels/<ID>/color/capabilities	196
16.4.8 /ISAPI/Image/channels/<ID>/corridor	196
16.4.9 /ISAPI/Image/channels/<ID>/dehaze	197
16.4.10 /ISAPI/Image/channels/<ID>/EPTZ	198
16.4.11 /ISAPI/Image/channels/<ID>/EPTZ/mode/capabilities?format=json	198
16.4.12 /ISAPI/Image/channels/<ID>/EPTZ/mode?format=json	199
16.4.13 /ISAPI/Image/channels/<ID>/exposure	199
16.4.14 /ISAPI/Image/channels/<ID>/focusConfiguration	200
16.4.15 /ISAPI/Image/channels/<ID>/focusConfiguration/capabilities	201

16.4.16 /ISAPI/Image/channels/<ID>/gain	201
16.4.17 /ISAPI/Image/channels/<ID>/imageCap	202
16.4.18 /ISAPI/Image/channels/<ID>/ImageFlip	202
16.4.19 /ISAPI/Image/channels/<ID>/imageMode/<ID>	203
16.4.20 /ISAPI/Image/channels/<ID>/imageModes	203
16.4.21 /ISAPI/Image/channels/<ID>/IrcutFilter	204
16.4.22 /ISAPI/Image/channels/<ID>/ISPMode	205
16.4.23 /ISAPI/Image/channels/<ID>/lensDistortionCorrection	205
16.4.24 /ISAPI/Image/channels/<ID>/lensDistortionCorrection/capabilities	206
16.4.25 /ISAPI/Image/channels/<ID>/mountingScenario	206
16.4.26 /ISAPI/Image/channels/<ID>/mountingScenario/capabilities	207
16.4.27 /ISAPI/Image/channels/<ID>/noiseReduce	208
16.4.28 /ISAPI/Image/channels/<ID>/Palettes	208
16.4.29 /ISAPI/Image/channels/<ID>/powerLineFrequency	209
16.4.30 /ISAPI/Image/channels/<ID>/regionalFocus	210
16.4.31 /ISAPI/Image/channels/<ID>/reset	210
16.4.32 /ISAPI/Image/channels/<ID>/restore	210
16.4.33 /ISAPI/Image/channels/<ID>/sharpness	211
16.4.34 /ISAPI/Image/channels/<ID>/shutter	212
16.4.35 /ISAPI/Image/channels/<ID>/supplementLight	212
16.4.36 /ISAPI/Image/channels/<ID>/supplementLight/capabilities	213
16.4.37 /ISAPI/Image/channels/<ID>/targetEnhancement	213
16.4.38 /ISAPI/Image/channels/<ID>/targetEnhancement/capabilities	214
16.4.39 /ISAPI/Image/channels/<ID>/tempRange	215
16.4.40 /ISAPI/Image/channels/<ID>/tempRange/capabilities	215
16.4.41 /ISAPI/Image/channels/<ID>/WDR	216
16.4.42 /ISAPI/Image/channels/<ID>/whiteBalance	217
16.5 /ISAPI/MasterSlaveTracking	217

16.5.1 /ISAPI/MasterSlaveTracking/capabilities	217
16.5.2 /ISAPI/MasterSlaveTracking/channels/<ID>/slaveCamera/<ID>	218
16.5.3 /ISAPI/MasterSlaveTracking/channels/<ID>/slaveCamera/<ID>/tracking	218
16.5.4 /ISAPI/MasterSlaveTracking/channels/<ID>/slaveCamera/<ID>/trackingRatio	219
16.5.5 /ISAPI/MasterSlaveTracking/channels/<ID>/slaveCamera/capabilities	220
16.5.6 /ISAPI/MasterSlaveTracking/channels/<ID>/slaveCameraStatus	220
16.5.7 /ISAPI/MasterSlaveTracking/channels/<ID>/tracking/capabilities	220
16.5.8 /ISAPI/MasterSlaveTracking/linkedTracking/capabilities?format=json	221
16.5.9 /ISAPI/MasterSlaveTracking/linkedTracking?format=json	221
16.6 /ISAPI/SDT	222
16.6.1 /ISAPI/SDT/Management/capabilities?format=json	222
16.6.2 /ISAPI/SDT/Management/EventSearch?format=json	222
16.6.3 /ISAPI/SDT/Management/EventSearch/capabilities?format=json	223
16.6.4 /ISAPI/SDT/Management/IntelligentSearch/capabilities?format=json	223
16.6.5 /ISAPI/SDT/Management/IntelligentSearch?format=json	223
16.7 /ISAPI/PTZCtrl	224
16.7.1 /ISAPI/PTZCtrl/channels/<ID>	224
16.7.2 /ISAPI/PTZCtrl/channels/<ID>/auxcontrols	225
16.7.3 /ISAPI/PTZCtrl/channels/<ID>/auxcontrols/<ID>	225
16.7.4 /ISAPI/PTZCtrl/channels/<ID>/capabilities	226
16.7.5 /ISAPI/PTZCtrl/channels/<ID>/EagleFocusing/auto/capabilities?format=json	226
16.7.6 /ISAPI/PTZCtrl/channels/<ID>/EagleFocusing/auto?format=json	227
16.7.7 /ISAPI/PTZCtrl/channels/<ID>/EagleFocusing/capabilities	227
16.7.8 /ISAPI/PTZCtrl/channels/<ID>/GPS?format=json	228
16.7.9 /ISAPI/PTZCtrl/channels/<ID>/gyroVerify/capabilities?format=json	228
16.7.10 /ISAPI/PTZCtrl/channels/<ID>/gyroVerify?format=json	229
16.7.11 /ISAPI/PTZCtrl/channels/<ID>/lensCorrection/capabilities?format=json	229
16.7.12 /ISAPI/PTZCtrl/channels/<ID>/lensCorrection?format=json	230

16.7.13 /ISAPI/PTZCtrl/channels/<ID>/lockPTZ	230
16.7.14 /ISAPI/PTZCtrl/channels/<ID>/maxelevation	231
16.7.15 /ISAPI/PTZCtrl/channels/<ID>/maxelevation/capabilities	232
16.7.16 /ISAPI/PTZCtrl/channels/<ID>/moveAutoTracking?format=json	232
16.7.17 /ISAPI/PTZCtrl/channels/<ID>/onpushfoucs/reset	233
16.7.18 /ISAPI/PTZCtrl/channels/<ID>/parkaction	233
16.7.19 /ISAPI/PTZCtrl/channels/<ID>/parkaction/capabilities	234
16.7.20 /ISAPI/PTZCtrl/channels/<ID>/presets	234
16.7.21 /ISAPI/PTZCtrl/channels/<ID>/presets/<ID>	235
16.7.22 /ISAPI/PTZCtrl/channels/<ID>/PTZOSDDisplay	236
16.7.23 /ISAPI/PTZCtrl/channels/<ID>/save	237
16.7.24 /ISAPI/PTZCtrl/channels/<ID>/save?format=json	237
16.7.25 /ISAPI/PTZCtrl/channels/<ID>/status	238
16.7.26 /ISAPI/PTZCtrl/channels/<ID>/trackingRatio/capabilities?format=json	238
16.7.27 /ISAPI/PTZCtrl/channels/<ID>/trackingRatio?format=json	238
16.7.28 /ISAPI/PTZCtrl/channels/<ID>/zoomFocus	239
16.8 /ISAPI/Security	240
16.8.1 /ISAPI/Security/adminAccesses	240
16.8.2 /ISAPI/Security/adminAccesses/<ID>	240
16.8.3 /ISAPI/Security/adminAccesses/capabilities	241
16.8.4 /ISAPI/Security/advanced?format=json	242
16.8.5 /ISAPI/Security/capabilities	242
16.8.6 /ISAPI/Security/certificate/select/<functinName>?format=json	243
16.8.7 /ISAPI/Security/certificate/select/capabilities?format=json	243
16.8.8 /ISAPI/Security/challenge	244
16.8.9 /ISAPI/Security/CommuMode/capabilities?format=json	244
16.8.10 /ISAPI/Security/CommuMode?format=json	244
16.8.11 /ISAPI/Security/deviceCertificate	245

16.8.12 /ISAPI/Security/deviceCertificate/capabilities?format=json	246
16.8.13 /ISAPI/Security/deviceCertificate/certificateRevocation/capabilities?format=json	246
16.8.14 /ISAPI/Security/deviceCertificate/certificateRevocation?format=json	247
16.8.15 /ISAPI/Security/deviceCertificate/certificates/<customID>?format=json	247
16.8.16 /ISAPI/Security/deviceCertificate/certificates/<ID>	248
16.8.17 /ISAPI/Security/deviceCertificate/certificates/<ID>/recreate?format=json	248
16.8.18 /ISAPI/Security/deviceCertificate/certificates/<ID>/status?format=json	249
16.8.19 /ISAPI/Security/deviceCertificate/certificates/capabilities?format=json	249
16.8.20 /ISAPI/Security/deviceCertificate/certificates/recreate?format=json	250
16.8.21 /ISAPI/Security/deviceCertificate/certificates/status?format=json	250
16.8.22 /ISAPI/Security/deviceCertificate/certificates?format=json	250
16.8.23 /ISAPI/Security/deviceCertificate?customID=	251
16.8.24 /ISAPI/Security/doubleVerification/users/<ID>?format=json	252
16.8.25 /ISAPI/Security/doubleVerification/users/capabilities?format=json	253
16.8.26 /ISAPI/Security/doubleVerification/users?format=json	253
16.8.27 /ISAPI/Security/doubleVerification/UsersPermission/<ID>?format=json	254
16.8.28 /ISAPI/Security/doubleVerification/UsersPermission/capabilities?format=json ..	255
16.8.29 /ISAPI/Security/doubleVerification?format=json	255
16.8.30 /ISAPI/Security/email/parameter/capabilities?format=json	256
16.8.31 /ISAPI/Security/email/parameter?format=json	257
16.8.32 /ISAPI/Security/email/qrCode?format=json	257
16.8.33 /ISAPI/Security/emailCertification?format=json	258
16.8.34 /ISAPI/Security/extern/capabilities	258
16.8.35 /ISAPI/Security/illegalLoginLock	259
16.8.36 /ISAPI/Security/loginLinkNum?format=json	259
16.8.37 /ISAPI/Security/questionConfiguration	260
16.8.38 /ISAPI/Security/RTSPCertificate	261

16.8.39 /ISAPI/Security/RTSPCertificate/capabilities	261
16.8.40 /ISAPI/Security/serverCertificate/capabilities?format=json	262
16.8.41 /ISAPI/Security/serverCertificate/certificate	262
16.8.42 /ISAPI/Security/serverCertificate/certificate?customID=	263
16.8.43 /ISAPI/Security/serverCertificate/certificates?format=json	264
16.8.44 /ISAPI/Security/serverCertificate/certificates/<customID>?format=json	265
16.8.45 /ISAPI/Security/serverCertificate/certSignReq	265
16.8.46 /ISAPI/Security/serverCertificate/certSignReq?customID=	266
16.8.47 /ISAPI/Security/serverCertificate/downloadCertSignReq	267
16.8.48 /ISAPI/Security/serverCertificate/downloadCertSignReq?customID=	267
16.8.49 /ISAPI/Security/serverCertificate/selfSignCert?customID=	268
16.8.50 /ISAPI/Security/userCheck	268
16.8.51 /ISAPI/Security/UserPermission	269
16.8.52 /ISAPI/Security/UserPermission/<ID>	269
16.8.53 /ISAPI/Security/UserPermission/<ID>/localPermission	270
16.8.54 /ISAPI/Security/UserPermission/<ID>/remotePermission	271
16.8.55 /ISAPI/Security/UserPermission/adminCap	271
16.8.56 /ISAPI/Security/UserPermission/anonymouslogin	272
16.8.57 /ISAPI/Security/UserPermission/operatorCap	272
16.8.58 /ISAPI/Security/UserPermission/viewerCap	273
16.8.59 /ISAPI/Security/users	273
16.8.60 /ISAPI/Security/users/<ID>	277
16.8.61 /ISAPI/Security/webCertificate	278
16.8.62 /ISAPI/SecurityCP/ReportCenterCfg/capabilities?format=json	279
16.8.63 /ISAPI/SecurityCP/ReportCenterCfg/<ID>?format=json	279
16.9 /ISAPI/Smart	280
16.9.1 /ISAPI/Smart/capabilities	280
16.10 /ISAPI/Streaming	281

16.10.1 /ISAPI/Streaming/channels	281
16.10.2 /ISAPI/Streaming/channels/<ID>	282
16.10.3 /ISAPI/Streaming/channels/<ID>/capabilities	284
16.10.4 /ISAPI/Streaming/channels/<ID>/dynamicCap	285
16.10.5 /ISAPI/Streaming/channels/<ID>/picture	286
16.10.6 /ISAPI/Streaming/channels/<ID>/regionClip	287
16.10.7 /ISAPI/Streaming/channels/<ID>/regionClip/capabilities	288
16.10.8 /ISAPI/Streaming/channels/<ID>/smartOverlap/capabilities?format=json	288
16.10.9 /ISAPI/Streaming/channels/<ID>/smartOverlap?format=json	289
16.10.10 /ISAPI/Streaming/channels/<ID>/status	289
16.10.11 /ISAPI/Streaming/encryption/capabilities?format=json	290
16.10.12 /ISAPI/Streaming/encryption/secretKey?format=json	290
16.10.13 /ISAPI/Streaming/encryption?format=json	291
16.10.14 /ISAPI/Streaming/status	292
16.10.15 rtsp://<host>[:port]/ISAPI/Streaming/channels/<ID>	292
16.10.16 rtsp://<host>[:port]/ISAPI/Streaming/channels/<ID>?npq=	293
16.10.17 rtsp://<host>[:port]/ISAPI/Streaming/tracks/<ID>?starttime=&endtime=	295
16.11 /ISAPI/System	297
16.11.1 /ISAPI/System/accessDevice/associatedChannel?format=json	297
16.11.2 /ISAPI/System/AcsUpdate/capabilities	297
16.11.3 /ISAPI/System/activate	297
16.11.4 /ISAPI/System/algorithmsInfo/capabilities?format=json	298
16.11.5 /ISAPI/System/algorithmsInfo?format=json	298
16.11.6 /ISAPI/System/algorithmsVersion	299
16.11.7 /ISAPI/System/Audio/AudioIn/channels/<ID>	299
16.11.8 /ISAPI/System/Audio/AudioIn/channels/<ID>/capabilities	300
16.11.9 /ISAPI/System/Audio/AudioOut/channels/<ID>	300
16.11.10 /ISAPI/System/Audio/AudioOut/channels/<ID>/capabilities	301

16.11.11 /ISAPI/System/Audio/capabilities	301
16.11.12 /ISAPI/System/Audio/channels	302
16.11.13 /ISAPI/System/Audio/channels/<ID>	302
16.11.14 /ISAPI/System/Audio/channels/<ID>/dynamicCap	302
16.11.15 /ISAPI/System/autoMaintenance/capabilities?format=json	303
16.11.16 /ISAPI/System/autoMaintenance?format=json	303
16.11.17 /ISAPI/System/autoSwitch/capabilities?format=json	304
16.11.18 /ISAPI/System/autoSwitch?format=json	304
16.11.19 /ISAPI/System/Bluetooth/capabilities	305
16.11.20 /ISAPI/System/Bluetooth/ibeaconParam	305
16.11.21 /ISAPI/System/Bluetooth/ibeaconParam/capabilities	306
16.11.22 /ISAPI/System/capabilities	306
16.11.23 /ISAPI/System/channels/<ID>/vibrationDetection/capabilities?format=json	306
16.11.24 /ISAPI/System/channels/<ID>/vibrationDetection?format=json	307
16.11.25 /ISAPI/System/configurationData?secretkey=	308
16.11.26 /ISAPI/System/configurationData?type=	309
16.11.27 /ISAPI/System/deviceInfo	310
16.11.28 /ISAPI/System/deviceInfo/capabilities	311
16.11.29 /ISAPI/System/DeviceLanguage	311
16.11.30 /ISAPI/System/DeviceLanguage/capabilities	312
16.11.31 /ISAPI/System/diagnosedData	312
16.11.32 /ISAPI/System/diagnosedData/exportStatus	313
16.11.33 /ISAPI/System/diagnosedData/parameter	313
16.11.34 /ISAPI/System/diagnosedData/parameter/capabilities	314
16.11.35 /ISAPI/System/diagnosis/capabilities?format=json	314
16.11.36 /ISAPI/System/diagnosis?format=json	315
16.11.37 /ISAPI/System/doubleLensParking/capabilities	315
16.11.38 /ISAPI/System/dumpData	315

16.11.39 /ISAPI/System/exportLogToUSB/capabilities?format=json	316
16.11.40 /ISAPI/System/exportLogToUSB/mode?format=json	316
16.11.41 /ISAPI/System/exportLogToUSB/status?format=json	317
16.11.42 /ISAPI/System/externalDevice/capabilities	317
16.11.43 /ISAPI/System/factoryReset?mode=	317
16.11.44 /ISAPI/System/fileExport/capabilities?format=json	318
16.11.45 /ISAPI/System/fileExport?format=json	319
16.11.46 /ISAPI/System/firmwareCodeV2	319
16.11.47 /ISAPI/System/GPSCalibration/channels/<ID>/capabilities?format=json	319
16.11.48 /ISAPI/System/GPSCalibration/channels/<ID>/points/<PID>?format=json	320
16.11.49 /ISAPI/System/GPSCalibration/channels/<ID>/points/status?format=json	321
16.11.50 /ISAPI/System/GPSCalibration/channels/<ID>/points?format=json	321
16.11.51 /ISAPI/System/GPSPTZCalibration/channels/<ID>/calibration?format=json	322
16.11.52 /ISAPI/System/GPSVerification/channels/<ID>/display?format=json	323
16.11.53 /ISAPI/System/GPSVerification/channels/<ID>/points/<ID>?format=json	323
16.11.54 /ISAPI/System/GPSVerification/channels/<ID>/points/capabilities?format=json	324
16.11.55 /ISAPI/System/GPSVerification/channels/<ID>/points/resultInfo?format=json .	325
16.11.56 /ISAPI/System/GPSVerification/channels/<ID>/points?format=json	325
16.11.57 /ISAPI/System/guardAgainstTheft	326
16.11.58 /ISAPI/System/guardAgainstTheft/capabilities	327
16.11.59 /ISAPI/System/guideConfig/<guideEvent>/capabilities?format=json	327
16.11.60 /ISAPI/System/guideConfig/<guideEvent>?format=json	328
16.11.61 /ISAPI/System/Hardware	328
16.11.62 /ISAPI/System/Hardware/defog	329
16.11.63 /ISAPI/System/installationAngleCalibration/channels/<ID>/capabilities? format=json	330
16.11.64 /ISAPI/System/installationAngleCalibration/channels/<ID>?format=json	330

16.11.65 /ISAPI/System/IO/capabilities	331
16.11.66 /ISAPI/System/IO/inputs	331
16.11.67 /ISAPI/System/IO/inputs/<ID>	331
16.11.68 /ISAPI/System/IO/inputs/<ID>/status	332
16.11.69 /ISAPI/System/IO/inputs/capabilities	332
16.11.70 /ISAPI/System/IO/outputs	333
16.11.71 /ISAPI/System/IO/outputs/capabilities	333
16.11.72 /ISAPI/System/IO/outputs/<ID>	334
16.11.73 /ISAPI/System/IO/outputs/<ID>/status	334
16.11.74 /ISAPI/System/IO/outputs/<ID>/trigger	335
16.11.75 /ISAPI/System/IO/status	335
16.11.76 /ISAPI/System/IOT/channelConfig?format=json	335
16.11.77 /ISAPI/System/IOT/channels/<ID>/<EventType>/capabilities?format=json	337
16.11.78 /ISAPI/System/IOT/channels/<ID>/<EventType>?format=json	337
16.11.79 /ISAPI/System/IOT/channels/<ID>/alarmStatistics?format=json	338
16.11.80 /ISAPI/System/IOT/channels/<ID>/all?format=json	338
16.11.81 /ISAPI/System/IOT/channels/<ID>/basicParam?format=json	338
16.11.82 /ISAPI/System/IOT/channels/<ID>/OSD/capabilities?format=json	339
16.11.83 /ISAPI/System/IOT/channels/<ID>/OSD?format=json	339
16.11.84 /ISAPI/System/IOT/channels/<ID>/status?format=json	340
16.11.85 /ISAPI/System/IOT/channels/<ID>?format=json	340
16.11.86 /ISAPI/System/IOT/channels/status?format=json	342
16.11.87 /ISAPI/System/IOT/channels/status?format=json&deviceInductiveType=	342
16.11.88 /ISAPI/System/IOT/channels?format=json	343
16.11.89 /ISAPI/System/IOT/channels?format=json&deviceInductiveType=	344
16.11.90 /ISAPI/System/IOT/linkageChannels?format=json	344
16.11.91 /ISAPI/System/IOT/search?format=json	345
16.11.92 /ISAPI/System/IOT/sourceCapabilities?format=json	345

16.11.93 /ISAPI/System/IOT/sourceSupport/capabilities?format=json	346
16.11.94 /ISAPI/System/laserOpticalAxis/capabilities?format=json	346
16.11.95 /ISAPI/System/laserOpticalAxis/goto?format=json	347
16.11.96 /ISAPI/System/laserOpticalAxis?format=json	347
16.11.97 /ISAPI/System/logServer	348
16.11.98 /ISAPI/System/logServer/capabilities	348
16.11.99 /ISAPI/System/logServer/test	349
16.11.100 /ISAPI/System/Network/adaption/capabilities?format=json	349
16.11.101 /ISAPI/System/Network/adaption?format=json&streamType=	350
16.11.102 /ISAPI/System/Network/Bond	350
16.11.103 /ISAPI/System/Network/Bond/<ID>	351
16.11.104 /ISAPI/System/Network/capabilities	351
16.11.105 /ISAPI/System/Network/channels/<ID>/buletooth/status	352
16.11.106 /ISAPI/System/Network/DDNS	353
16.11.107 /ISAPI/System/Network/DDNS/capabilities	353
16.11.108 /ISAPI/System/Network/DDNS/<ID>	354
16.11.109 /ISAPI/System/Network/DDNS/<ID>/capabilities	354
16.11.110 /ISAPI/System/Network/Ehome	355
16.11.111 /ISAPI/System/Network/Ehome/capabilities	355
16.11.112 /ISAPI/System/Network/Ehome?centerID=	356
16.11.113 /ISAPI/System/Network/EZVIZ	356
16.11.114 /ISAPI/System/Network/EZVIZ/secretKey?format=json	357
16.11.115 /ISAPI/System/Network/ftp	358
16.11.116 /ISAPI/System/Network/ftp/<ID>	358
16.11.117 /ISAPI/System/Network/ftp/capabilities	359
16.11.118 /ISAPI/System/Network/ftp/test	359
16.11.119 /ISAPI/System/Network/interfaces	360
16.11.120 /ISAPI/System/Network/interfaces/<ID>	361

16.11.121 /ISAPI/System/Network/interfaces/<ID>/capabilities	361
16.11.122 /ISAPI/System/Network/interfaces/<ID>/discovery	362
16.11.123 /ISAPI/System/Network/interfaces/<ID>/discovery/capabilities	362
16.11.124 /ISAPI/System/Network/interfaces/<ID>/dynamicHostName?format=json	363
16.11.125 /ISAPI/System/Network/interfaces/<ID>/dynamicHostName/capabilities? format=json	364
16.11.126 /ISAPI/System/Network/interfaces/<ID>/ieee802.1x	364
16.11.127 /ISAPI/System/Network/interfaces/<ID>/ieee802.1x/capabilities	365
16.11.128 /ISAPI/System/Network/interfaces/<ID>/ipAddress	365
16.11.129 /ISAPI/System/Network/interfaces/<ID>/ipAddress/capabilities	366
16.11.130 /ISAPI/System/Network/interfaces/<ID>/link	367
16.11.131 /ISAPI/System/Network/interfaces/<ID>/link/capabilities	367
16.11.132 /ISAPI/System/Network/interfaces/<ID>/NetworkMode	368
16.11.133 /ISAPI/System/Network/interfaces/<ID>/wireless	368
16.11.134 /ISAPI/System/Network/interfaces/<ID>/wireless/accessDeviceList/<ID>	369
16.11.135 /ISAPI/System/Network/interfaces/<ID>/wireless/accessDeviceList/capabilities	370
16.11.136 /ISAPI/System/Network/interfaces/<ID>/wireless/accessPointList	370
16.11.137 /ISAPI/System/Network/interfaces/<ID>/wireless/accessPointList/<ID>	371
16.11.138 /ISAPI/System/Network/interfaces/<ID>/wireless/capabilities	371
16.11.139 /ISAPI/System/Network/interfaces/<ID>/wirelessServer	372
16.11.140 /ISAPI/System/Network/interfaces/<ID>/wirelessServer/accessDeviceList	372
16.11.141 /ISAPI/System/Network/interfaces/<ID>/wirelessServer/capabilities	373
16.11.142 /ISAPI/System/Network/interfaces/<ID>/wirelessServerStatus	373
16.11.143 /ISAPI/System/Network/interfaces/<ID>/wirelessServerStatus/capabilities ...	374
16.11.144 /ISAPI/System/Network/interfaces/<ID>/wirelessStatus	374
16.11.145 /ISAPI/System/Network/interfaces/<ID>/wirelessStatus/capabilities	375
16.11.146 /ISAPI/System/Network/interfaces/<ID>/WPS/ApPinCode	376
16.11.147 /ISAPI/System/Network/interfaces/<ID>/WPS/AutoConnect	376

16.11.148 /ISAPI/System/Network/interfaces/<ID>/WPS/devicePinCode	377
16.11.149 /ISAPI/System/Network/ipFilter	378
16.11.150 /ISAPI/System/Network/ipFilter/capabilities	378
16.11.151 /ISAPI/System/Network/MACFilter	379
16.11.152 /ISAPI/System/Network/MACFilter/capabilities	379
16.11.153 /ISAPI/System/Network/mailling	379
16.11.154 /ISAPI/System/Network/mailling/<ID>	380
16.11.155 /ISAPI/System/Network/mailling/capabilities	381
16.11.156 /ISAPI/System/Network/mailling/test	381
16.11.157 /ISAPI/System/Network/pingtest	381
16.11.158 /ISAPI/System/Network/POE/capabilities?format=json	382
16.11.159 /ISAPI/System/Network/POE?format=json	382
16.11.160 /ISAPI/System/Network/resourceStatistics?format=json	383
16.11.161 /ISAPI/System/Network/SIP	383
16.11.162 /ISAPI/System/Network/SIP/<ID>	384
16.11.163 /ISAPI/System/Network/SIP/<ID>/capabilities	384
16.11.164 /ISAPI/System/Network/SIP/<ID>/SIPInfo	385
16.11.165 /ISAPI/System/Network/SIP/<ID>/SIPInfo/capabilities	386
16.11.166 /ISAPI/System/Network/SIP/<ID>/SIPInfo/multiInfo	386
16.11.167 /ISAPI/System/Network/SNMP	387
16.11.168 /ISAPI/System/Network/SNMP/advanced	388
16.11.169 /ISAPI/System/Network/SNMP/advanced/trapReceivers	388
16.11.170 /ISAPI/System/Network/SNMP/advanced/trapReceivers/<ID>	389
16.11.171 /ISAPI/System/Network/socketIP	390
16.11.172 /ISAPI/System/Network/ssh	390
16.11.173 /ISAPI/System/Network/StaticRoute	391
16.11.174 /ISAPI/System/Network/telnetd	392
16.11.175 /ISAPI/System/Network/telnetd/capabilities	392

16.11.176 /ISAPI/System/Network/UPnP	393
16.11.177 /ISAPI/System/Network/UPnP/ports	393
16.11.178 /ISAPI/System/Network/UPnP/ports/status	394
16.11.179 /ISAPI/System/Network/UPnP/ports/<ID>	394
16.11.180 /ISAPI/System/Network/UPnP/ports/<ID>/status	395
16.11.181 /ISAPI/System/Network/verificationCodeCheck	396
16.11.182 /ISAPI/System/Network/WirelessDial/Interfaces	396
16.11.183 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>	397
16.11.184 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/dial	397
16.11.185 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/dial/capabilities	398
16.11.186 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/dialstatus	399
16.11.187 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/keepAlive/capabilities? format=json	399
16.11.188 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/keepAlive/test? format=json	400
16.11.189 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/keepAlive?format=json	400
16.11.190 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/messageConfig	401
16.11.191 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/messageConfig/ messageConfigCap	402
16.11.192 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/messageConfig/WhiteList	402
16.11.193 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/messageConfig/WhiteList/ <ID>	403
16.11.194 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/schedule	404
16.11.195 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/trafficMonitor/ capabilities?format=json	405
16.11.196 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/trafficMonitor/status? format=json	405
16.11.197 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/trafficMonitor? format=json	405

16.11.198 /ISAPI/System/Network/WPS/capabilities	406
16.11.199 /ISAPI/System/onlineUpgrade/capabilities	407
16.11.200 /ISAPI/System/onlineUpgrade/deviceParameter?format=json	407
16.11.201 /ISAPI/System/onlineUpgrade/downloadPackage/pause?format=json	407
16.11.202 /ISAPI/System/onlineUpgrade/downloadPackage/resume?format=json	408
16.11.203 /ISAPI/System/onlineUpgrade/downloadPackage/status?format=json	408
16.11.204 /ISAPI/System/onlineUpgrade/downloadPackage?format=json	409
16.11.205 /ISAPI/System/onlineUpgrade/ignoreCurrentVersion?format=json	409
16.11.206 /ISAPI/System/onlineUpgrade/server	410
16.11.207 /ISAPI/System/onlineUpgrade/status	410
16.11.208 /ISAPI/System/onlineUpgrade/upgradeWithoutDownload?format=json	410
16.11.209 /ISAPI/System/onlineUpgrade/version	411
16.11.210 /ISAPI/System/PictureServer/capabilities?format=json	411
16.11.211 /ISAPI/System/PictureServer?format=json	411
16.11.212 /ISAPI/System/reboot	412
16.11.213 /ISAPI/System/Serial/authentication/capabilities?format=json	413
16.11.214 /ISAPI/System/Serial/authentication?format=json	413
16.11.215 /ISAPI/System/Serial/capabilities	414
16.11.216 /ISAPI/System/Serial/ports	414
16.11.217 /ISAPI/System/Serial/ports/<ID>	415
16.11.218 /ISAPI/System/Serial/ports/<ID>/capabilities	415
16.11.219 /ISAPI/System/Serial/ports/command	416
16.11.220 /ISAPI/System/Serial/ports/command/capabilities	417
16.11.221 /ISAPI/System/serialLogCfg/capabilities?format=json	417
16.11.222 /ISAPI/System/serialLogCfg?format=json	417
16.11.223 /ISAPI/System/shutdown?format=json	418
16.11.224 /ISAPI/System/Software/channels/<ID>	418
16.11.225 /ISAPI/System/Software/channels/<ID>/capabilities	419

16.11.226 /ISAPI/System/status	420
16.11.227 /ISAPI/System/syncSignalOutput/<ID>	420
16.11.228 /ISAPI/System/syncSignalOutput/<ID>/capabilities	421
16.11.229 /ISAPI/System/time	421
16.11.230 /ISAPI/System/time/capabilities	422
16.11.231 /ISAPI/System/time/localTime	422
16.11.232 /ISAPI/System/time/ntpServers	423
16.11.233 /ISAPI/System/time/ntpServers/<ID>	424
16.11.234 /ISAPI/System/time/ntpServers/<ID>/capabilities	424
16.11.235 /ISAPI/System/time/ntpServers/capabilities	425
16.11.236 /ISAPI/System/time/ntpServers/test	425
16.11.237 /ISAPI/System/time/timeType?format=json	426
16.11.238 /ISAPI/System/time/timeZone	426
16.11.239 /ISAPI/System/TwoWayAudio/channels	427
16.11.240 /ISAPI/System/TwoWayAudio/channels/<ID>	428
16.11.241 /ISAPI/System/TwoWayAudio/channels/<ID>/audioData	428
16.11.242 /ISAPI/System/TwoWayAudio/channels/<ID>/capabilities	430
16.11.243 /ISAPI/System/TwoWayAudio/channels/<ID>/close	430
16.11.244 /ISAPI/System/TwoWayAudio/channels/<ID>/open	431
16.11.245 /ISAPI/System/unitConfig/capabilities?format=json	432
16.11.246 /ISAPI/System/unitConfig?format=json	433
16.11.247 /ISAPI/System/updateFirmware	433
16.11.248 /ISAPI/System/updateFirmware?type=&id=	434
16.11.249 /ISAPI/System/updateFirmware?type=&moduleAddress=	435
16.11.250 /ISAPI/System/upgradeStatus	436
16.11.251 /ISAPI/System/usb/capabilities?format=json	436
16.11.252 /ISAPI/System/usb?format=json	436
16.11.253 /ISAPI/System/userLock/config/capabilities?format=json	437

16.11.254 /ISAPI/System/userLock/config?format=json	437
16.11.255 /ISAPI/System/userLock/lockedUsers?format=json	438
16.11.256 /ISAPI/System/userLock/unlockUser?format=json	438
16.11.257 /ISAPI/System/Video/capabilities	439
16.11.258 /ISAPI/System/Video/inputs	439
16.11.259 /ISAPI/System/Video/inputs/channels	439
16.11.260 /ISAPI/System/Video/inputs/channels/<ID>	440
16.11.261 /ISAPI/System/Video/inputs/channels?format=json	441
16.11.262 /ISAPI/System/Video/inputs/channels/<ID>/focus	441
16.11.263 /ISAPI/System/Video/inputs/channels/<ID>?format=json	441
16.11.264 /ISAPI/System/Video/inputs/channels/<ID>/capabilities	442
16.11.265 /ISAPI/System/Video/inputs/channels/<ID>/iris	442
16.11.266 /ISAPI/System/Video/inputs/channels/<ID>/motionDetection	443
16.11.267 /ISAPI/System/Video/inputs/channels/<ID>/motionDetection/capabilities	444
16.11.268 /ISAPI/System/Video/inputs/channels/<ID>/motionDetection/layout	444
16.11.269 /ISAPI/System/Video/inputs/channels/<ID>/motionDetection/layout/capabilities	445
16.11.270 /ISAPI/System/Video/inputs/channels/<ID>/motionDetection/layout/gridLayout	445
16.11.271 /ISAPI/System/Video/inputs/channels/<ID>/overlays	446
16.11.272 /ISAPI/System/Video/inputs/channels/<ID>/overlays/BatteryPowerOverlay ..	448
16.11.273 /ISAPI/System/Video/inputs/channels/<ID>/overlays/BatteryPowerOverlay/ capabilities	449
16.11.274 /ISAPI/System/Video/inputs/channels/<ID>/overlays/capabilities	449
16.11.275 /ISAPI/System/Video/inputs/channels/<ID>/overlays/channelNameOverlay ..	449
16.11.276 /ISAPI/System/Video/inputs/channels/<ID>/overlays/dateTimeOverlay	450
16.11.277 /ISAPI/System/Video/inputs/channels/<ID>/overlays/dateTimeOverlay/ capabilities	451
16.11.278 /ISAPI/System/Video/inputs/channels/<ID>/overlays/text	451

16.11.279 /ISAPI/System/Video/inputs/channels/<ID>/overlays/text/<ID>	453
16.11.280 /ISAPI/System/Video/inputs/channels/<ID>/privacyMask	453
16.11.281 /ISAPI/System/Video/inputs/channels/<ID>/privacyMask/privacyMaskCap ...	454
16.11.282 /ISAPI/System/Video/inputs/channels/<ID>/privacyMask/regions	454
16.11.283 /ISAPI/System/Video/inputs/channels/<ID>/privacyMask/regions/<ID>	456
16.11.284 /ISAPI/System/Video/inputs/channels/<ID>/VCAResource	456
16.11.285 /ISAPI/System/Video/inputs/channels/<ID>/VCAResource/capabilities	457
16.11.286 /ISAPI/System/Video/inputs/channels/counting/collection/capabilities? format=json	458
16.11.287 /ISAPI/System/Video/inputs/channels/counting/collection?format=json	458
16.11.288 /ISAPI/System/Video/inputs/channels/heatMap/collection/capabilities? format=json	459
16.11.289 /ISAPI/System/Video/inputs/channels/heatMap/collection?format=json	459
16.11.290 /ISAPI/System/Video/inputs/OSDLanguage	459
16.11.291 /ISAPI/System/Video/outputs	460
16.11.292 /ISAPI/System/Video/outputs/channels	460
16.11.293 /ISAPI/System/Video/outputs/channels/<ID>	461
16.11.294 /ISAPI/System/Video/outputs/channels/<ID>/capabilities	462
16.11.295 /ISAPI/System/Video/outputs/PreviewSwitch/capabilities	462
16.11.296 /ISAPI/System/Video/outputs/PreviewSwitch/capabilities?videoOutType=	462
16.11.297 /ISAPI/System/Video/outputs/PreviewSwitch?groupID=&videoOutType=	463
16.11.298 /ISAPI/System/Video/outputs/PreviewSwitch? groupID=&videoOutType=&previewFrameNo=	464
16.11.299 /ISAPI/System/wiresharkToUSB/capabilities?format=json	465
16.11.300 /ISAPI/System/wiresharkToUSB/mode?format=json	466
16.11.301 /ISAPI/System/wiresharkToUSB/status?format=json	466
16.11.302 /ISAPI/System/workingstatus/hdStatus?format=json	466
16.11.303 /ISAPI/System/workingstatus/chanStatus?format=json	467
16.11.304 /ISAPI/System/workingstatus/capabilities?format=json	468

16.11.305 /ISAPI/System/workingstatus?format=json	468
16.11.306 /ISAPI/System/workingstatus/IOStatus?format=json	468
16.11.307 /ISAPI/System/zeroBiasCalibration/channels/<ID>?format=json	469
16.12 /ISAPI/Thermal	469
16.12.1 /ISAPI/Thermal/capabilities	469
16.12.2 /ISAPI/Thermal/temperature/collection/capabilities?format=json	470
16.12.3 /ISAPI/Thermal/temperature/collection?format=json	470
Appendix A. Appendixes	471
A.1 Request and Response Message	471
A.1.1 JSON Messages	471
A.1.2 XML Messages	599
A.2 Log Types for ISAPI	861
A.3 Supported Alarm/Event Types	880
A.4 ISAPI Response Codes	884

Chapter 1 Overview

1.1 Introduction

The OPEN Intelligent Security Application Programming Interface (ISAPI) is a text protocol in RESTful style based on HTTP for communicating between security devices/servers (e.g., cameras, DVR, NVR, etc.) and client software/system. It defines the communication standard between device/server and client software/system via the Internet Protocol (IP), see the network topology and communication protocol layer framework in the figures below.

 Note

REST (REpresentational State Transfer) is a protocol design method which abstracts all information as the resources. The abstracted resources are marked by the uniform identifies, i.e., URI (Uniform Resource Identifiers) for simple and extendable management.

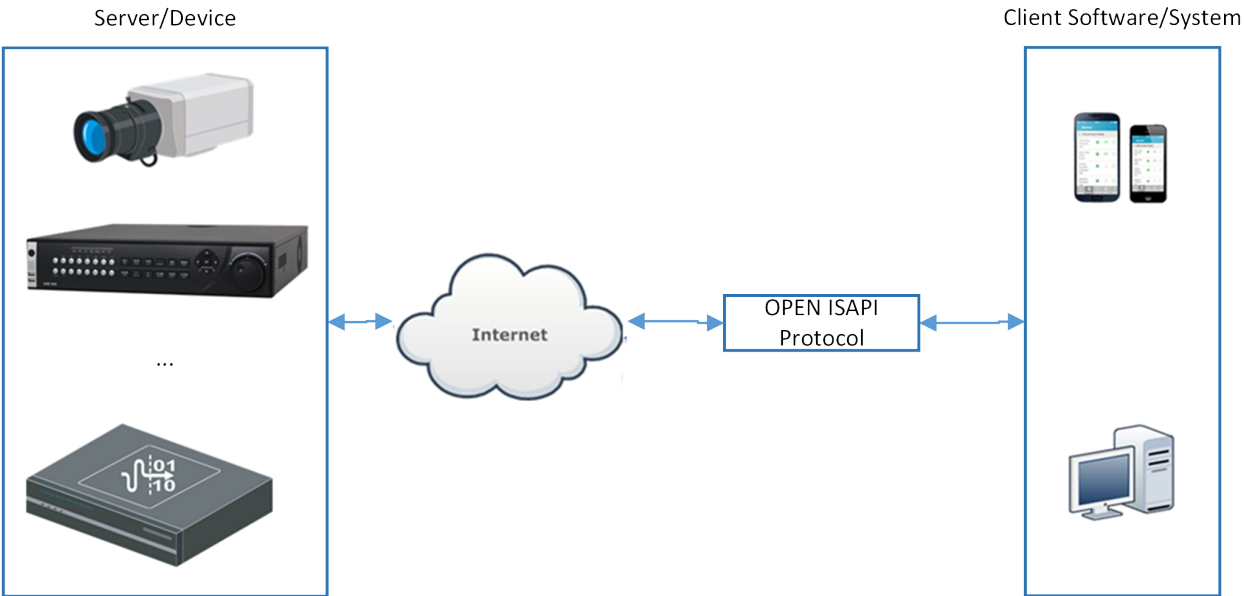


Figure 1-1 Network Topology

Search Online Device	Interaction Commands	Streaming
SADP	HTTP	RTSP/RTP/RTCP
MCAST	TCP	TCP/UDP
IP		

Figure 1-2 Communication Protocol Layer Framework

1.2 Update History

1.2.1 Version 20.04-Version 20.06

Summary of Changes in Version 2.6_June, 2020

Related Product: Radar PTZ Camera with Model of iDS-2SR8141XS-AB(40X) in Software Version 5.5.21

Extended the tracking parameters **JSON_TrackingRatio** and its configuration capability **JSON_TrackingRatioCap** (related URIs: **/ISAPI/PTZCtrl/channels/<ID>/trackingRatio?format=json** and **/ISAPI/PTZCtrl/channels/<ID>/trackingRatio/capabilities?format=json**):
added a node "trackingTakeoverEnabled" (whether to enable tracking takeover).

Summary of Changes in Version 2.6_June, 2020

Related Product: iDS-8104AHQXLI-F4 Series ATM DVR

1. Extended the device storage capability **XML_RacmCap** (related URI: GET **/ISAPI/ContentMgmt/capabilities**):
added four search conditions to node **<pictureSearchType>** and **<recordSearchType>**:
"ATMPanel" (ATM panel mode), "ATMSurround" (ATM environment mode), ATMFace (ATM human face mode), and "ATMSafetyCabin" (ATM safety cabin mode).
2. Extended the search conditions of video and picture **XML_VideoPic_CMSearchDescription** (related URI: **/ISAPI/ContentMgmt/search**):
added four search conditions to the sub node **<metadataDescriptor>** of **<metadataList>**:
"ATMPanel" (ATM panel mode), "ATMSurround" (ATM environment mode), ATMFace (ATM human face mode), and "ATMSafetyCabin" (ATM safety cabin mode).
3. Extended the log types, see details in **Log Types for ISAPI** :
added eight minor types to alarm logs: "ATMPanelStart" (ATM panel alarm started), "ATMPanelStop" (ATM panel alarm ended), "ATMSurroundStart" (ATM surround alarm started), "ATMSurroundStop" (ATM surround alarm ended), "ATMFaceStart" (ATM face alarm started), "ATMFaceStop" (ATM face alarm ended), "ATMSafetyCabinStart" (ATM safety cabin alarm started), and "ATMSafetyCabinStop" (ATM safety cabin alarm ended).

Summary of Changes in Version 2.6_June, 2020

Related Products: NVR with Models of iDS-9632NXI-I8/BA and iDS-6704NXI-I/BA in Software Version 4.1.73

1. Extended the device storage capability **XML_RacmCap** (related URI: **/ISAPI/ContentMgmt/capabilities**):
added a value "loitering" (loitering detection) to the nodes **<perimeterEventPicSearch>** (perimeter protection events that support picture search) and **<pictureSearchType>** (picture search conditions).
2. Added two error codes, i.e., 0x400012C8-"ruleEventTypeDuplicate" and 0x400015C2-"polygonNotAllowCrossing", to **ISAPI Response Codes** .

Summary of Changes in Version 2.6_June, 2020

Related Product: DS-2TD1217B series and DS-2TD2636B series Thermal & Optical Bi-spectrum Network Camera

1. Extended the channel attributes ***XML_ChannelInfo*** (related URI: GET ***/ISAPI/AUXInfo/attributes/Channels/<ID>***):
added one sub node ***<FaceSnapThermometry>*** (upload the captured face picture with temperature information) to node ***<Thermal>***.
2. Extended the audible warning configuration capability ***JSON_AudioAlarmCap*** (related URI: GET ***/ISAPI/Event/triggers/notifications/AudioAlarm/capabilities?format=json***) and audible warning configurations ***JSON_AudioAlarm*** (related URI: ***/ISAPI/Event/triggers/notifications/AudioAlarm?format=json***):
added one alarm type "noMaskDetection" (no wearing mask detection) to node ***alarmType***.
3. Extended the device event capability ***XML_EventCap*** (related URI: GET ***/ISAPI/Event/capabilities***):
added two nodes ***<isSupportNoMaskDetection>*** (whether device supports no wearing mask detection) and ***<isSupportTMPA>*** (whether device supports temperature measurement pre-alarm).

Summary of Changes in Version 2.6_May, 2020

Related Product: iDS-2SK8144IXS-D/J(40X/2812) Smart Linkage PTZ Camera with software version 5.5.40

1. Extended device capability ***XML_DeviceCap*** (related URI: GET ***/ISAPI/System/capabilities***):
added five nodes: ***<isSupportPersonArmingTrack>*** (whether device supports auto person arming), ***<isSupportManualPersonArmingTrack>*** (whether device supports manual person arming), ***<isSupportGPSCalibrationMode>*** (whether device supports GPS calibration), ***<isSupportGPSVerification>*** (whether device supports GPS verification), and ***<isSupportHBDLib>*** (whether device supports human body picture library).
2. Extended message of event capability supported by a single channel ***XML_ChannelEventCap*** (related URI: GET ***/ISAPI/Event/channels/<ID>/capabilities***):
added two event types "personArmingTrack" (auto person arming) and "manualPersonArmingTrack" (manual person arming) to the node ***<eventType>***.
3. Extended the linkage capabilities of different alarm categories ***XML_EventTriggersCap*** (related URI: GET ***/ISAPI/Event/triggersCap***):
added one node ***<HBDLibTriggerCap>*** (alarm linkage configuration capability of human body picture library).
4. Added two event types "personArmingTrack" (auto person arming) and "manualPersonArmingTrack" (manual person arming) to ***Supported Alarm/Event Types***.
5. Extended the master-slave tracking capability ***XML_MasterSlaveTrackingCap*** (related URI: GET ***/ISAPI/MasterSlaveTracking/capabilities***):
added one node ***<isSupportLinkedTracking>*** (whether device supports advanced parameters configuration of linked tracking capture).
6. Added a function of configuring advanced parameters of linked tracking capture.

Get capability: GET */ISAPI/MasterSlaveTracking/linkedTracking/capabilities?format=json* ;

Get or set advanced parameters of linked tracking capture: GET or PUT */ISAPI/MasterSlaveTracking/linkedTracking?format=json* .

7. Extended **ISAPI Response Codes** :

added three sub status codes to status code 4 (Invalid Operation):

0x40002092-"HBDLibNumOverLimit" (the number of human body picture libraries reaches the upper limit), 0x4000209C-pointsOnStraightLine (calibrating failed. The calibration points cannot be one the same line), and 0x4000209D-TValueLessThanOrEqualZero (calibration failed. The T value of the calibration points should be larger than 0);

added nine sub status codes to status code 6 (Invalid Message Content):

0x6000008b-"customHBPIDError" (incorrect ID of custom human body picture library), 0x6000008c-"customHBPIDRepeat" (duplicated ID of custom human body picture library), 0x6000008d-"dataVersionsInHBDLibMismatches" (database versions mismatches of human body picture library), 0x6000008e-"invalidHBPID" (invalid human body picture PID), 0x6000008f-"invalidHBDID" (invalid ID of human body picture library), 0x60000090-"humanLibraryError" (error of human body picture library), 0x60000091-"humanLibraryNumError" (no more human body picture library can be added), 0x60000092-"humanImagesNumError" (no more human body picture can be added), and 0x60000093-"noHumanInThePicture" (modeling failed, no human body in the picture).

8. Extended the capability of exporting device diagnose information

XML_Cap_DiagnosedDataParameter (related URI: */ISAPI/System/diagnosedData/parameter/capabilities*) and device diagnose information exporting parameters

XML_DiagnosedDataParameter (related URI: */ISAPI/System/diagnosedData/parameter*):

added three nodes <HardwareInfo> (whether to enable exporting hardware information), <SoftwareInfo> (whether to enable exporting software information), and <logInfo> (whether to enable exporting startup log).

9. Added the URIs for verifying GPS calibration, refer to **GPS Calibration** for details.

10. Extended the device protocol capability **XML_Cap_AdminAccessProtocolList** (related URI:

GET */ISAPI/Security/adminAccesses/capabilities*) and device supported protocol parameters

XML_AdminAccessProtocol (related URI: */ISAPI/Security/adminAccesses*):

added one protocol "Bonjour" to node <protocol>.

11. Extended the device security capability **XML_SecurityCap** (related URI: GET */ISAPI/Security/capabilities*) and **XML_IllegalLoginLock** (related URI: */ISAPI/Security/illegalLoginLock*):

added one node <maxIllegalLoginLockTime> (lock duration when maximum illegal login attempts reached the upper limit).

12. Extended the PTZ channel capability **XML_PTZChanelCap** (related URI: GET */ISAPI/PTZCtrl/channels/<ID>/capabilities*):

added one nodes <isSupportGyroVerify> (whether device supports gyroscope attitude calibration).

13. Added URIs for calibrating gyroscope attitude:

Get capability: GET */ISAPI/PTZCtrl/channels/<ID>/gyroVerify/capabilities?format=json*

Set gyroscope attitude calibration parameters: PUT */ISAPI/PTZCtrl/channels/<ID>/gyroVerify?format=json* .

14. Extended the intelligent resource switch capability *XML_Cap_VCResource* (related URI: GET */ISAPI/System/Video/inputs/channels/<ID>/VCResource/capabilities*) and intelligent resource switch parameters *XML_VCResource* (related URI: */ISAPI/System/Video/inputs/channels/<ID>/VCResource*):
added one intelligent resource types "personArming" (person arming) to node *<type>*.

Summary of Changes in Version 2.6_May, 2020

Related Product: TB-4117-3-S Temperature Screening Thermal Camera Module with Software Version 5.5.2

Added URIs for configuring device USB parameters:

Get USB parameters configuration capability: GET */ISAPI/System/usb/capabilities?format=json* ;

Get or set USB parameters: GET or PUT */ISAPI/System/usb?format=json* .

Summary of Changes in Version 2.6_April, 2020

Related Product: DS-2DE Series IR Network Speed Dome with software version 5.5.23

Extended the supplement light configuration capability *XML_Cap_SupplementLight* and supplement light configuration parameters *XML_SupplementLight* (related URIs: */ISAPI/Image/channels/<ID>/supplementLight/capabilities* and */ISAPI/Image/channels/<ID>/supplementLight*):

added two nodes: *<irLightbrightnessLimit>* (IR light brightness limit) and *<whiteLightbrightnessLimit>* (white light brightness limit).

1.2.2 Version 20.01-Version 20.03

Summary of Changes in Version 2.6_Mar., 2020

Related Product: iDS-SR8141MXS-A/T2(40×) Radar PTZ Camera with Software Version 5.5.20

1. Extended device capability message *XML_DeviceCap* (related URI: */ISAPI/System/capabilities*):
added two nodes: *<radarIPDForm>* (radar form) and *<mixedTargetDetectionWebNoDisplay>* (whether to enable not displaying multi-target-type recognition).
2. Extended PTZ control capability message *XML_PTZChanelCap* (related URI: */ISAPI/PTZCtrl/channels/<ID>/capabilities*):
added a node *<isSupportTrackingRatioCfg>* (whether it supports configuring tracking parameters).
3. Added two URIs of configuring tracking parameters:
Get configuration capability: GET */ISAPI/PTZCtrl/channels/<ID>/trackingRatio/capabilities?format=json* ;
Get or set parameters: GET or PUT */ISAPI/PTZCtrl/channels/<ID>/trackingRatio?format=json* .
4. Extended sub status codes in *ISAPI Response Codes* :
added a sub status code to status code 3 (Device Error): 0x30002051-"calibrationTimeout" (Calibration timed out);

added four sub status codes to status code 4 (Invalid Operation):

0x4000206A-"completeTheLevelCalibrationFirst" (The level calibration is not set),

0x4000206B-"completeTheRadarCameraCalibrationFirst" (The radar-camera calibration is not

set), 0x40002093-"theShieldRegionError" (Saving failed. The shielded area should be the ground

area where the shielded object is located), and 0x40002094-"theDetectionAreaError" (Saving

failed. The detection area should only cover the ground area);

added two sub status codes to status code 6 (Invalid Content):

0x60008016-"uploadImageSizeError" (The size of the uploaded picture is larger than 5 MB) and

0x6000202C-"width/heightRatioOfPictureError" (The width/height ratio of the uploaded picture

should be in the range from 1:2 to 2:1).

Summary of Changes in Version 2.6_Mar., 2020

Related Product: iDS-2CD70 series box camera, iDS-2CD70 series shield, iDS-2CD71 and iDS-2CD75 series dome camera, iDS-2CD7A series bullet camera with software version 5.5.120

1. Added URIs of configuring vibration detection alarm, see **Configure Vibration Detection Alarm** for details.
2. Extended device capability message **XML_DeviceCap** (related URI: **/ISAPI/System/capabilities**): added one node **<isSupportVibrationDetection>** (whether it supports vibration detection).
3. Extended message about event capability of all channels **XML_ChannelEventCapList** and message about event capability of a single channel **XML_ChannelEventCap** (related URIs: **/ISAPI/Event/channels/capabilities** and **/ISAPI/Event/channels/<ID>/capabilities**): added one event type "vibrationDetection" (vibration detection) to the node **<eventType>**.
4. Extended message about linkage capabilities of different alarm categories **XML_EventTriggersCap** (related URI: **/ISAPI/Event/triggersCap**): added one node **<VibrationDetectionTriggerCap>** (alarm linkage capability of vibration detection).
5. Extended linkage parameter message **XML_EventTrigger** (related URI: **/ISAPI/Event/triggers/<ID>**): added one event type "vibrationDetection" (vibration detection) to the node **<eventType>**.
6. Extended device storage capability message **XML_RacmCap** (related URI: **/ISAPI/ContentMgmt/capabilities**): added a type "vibrationDetection" (vibration detection) to the node **<pictureSearchType>** (picture search condition) and **<recordSearchType>** (video search condition).
7. Extended capability message **XML_Cap_VCResource** and configuration message **XML_VCResource** about intelligent resources switch (related URIs: **/ISAPI/System/Video/inputs/channels/<ID>/VCResource/capabilities** and **/ISAPI/System/Video/inputs/channels/<ID>/VCResource**): added one intelligent resource type "faceCounting" (face counting) to the node **<type>**.
8. Extended the log types, see details in **Log Types for ISAPI**: added two minor types "vibrationDetectionStart" (Vibration Detection Alarm Started) and "vibrationDetectionStop" (Vibration Detection Alarm Stopped) to Alarm Logs.
9. Extended error codes, see details in **ISAPI Response Codes**:

added one error code 0x40002096-"invalidLaneLine" (Saving failed. Invalid lane line.) to status code 4 (Invalid Operation).

Summary of Changes in Version 2.6_Feb., 2020

Related Product: Thermal & Optical Bi-spectrum Network Speed Dome; Thermal & Optical Bi-spectrum Positioning System; Thermal & Optical Bi-spectrum Network Stable PTZ Camera in Version 5.5.25

Added URIs of configuring laser optical axis:

Get capability: GET */ISAPI/System/laserOpticalAxis/capabilities?format=json* ;

Get or set laser optical axis parameters: GET or PUT */ISAPI/System/laserOpticalAxis?format=json* ;

Adjust laser optical axis: PUT */ISAPI/System/laserOpticalAxis/goto?format=json* .

Summary of Changes in Version 2.6_Feb., 2020

Related Product: DS-K1T804A Series and DS-K1T8003 Series Fingerprint Access Control Terminal with Software Version 1.3.0; DS-K1A802A Series and DS-K1A8503 Series Fingerprint Time Attendance Terminal with Software Version 1.3.0

1. Extended configuration capability message *XML_Cap_DeviceLanguage* and parameter message *XML_DeviceLanguage* of device language (related URIs: */ISAPI/System/DeviceLanguage/capabilities* and */ISAPI/System/DeviceLanguage*):
added a language supported by the device "Ukrainian" to the node *<language>*.
2. Extended Wi-Fi configuration capability message *XML_Cap_Wireless* and Wi-Fi parameter message *XML_Wireless* of a specific network interface (related URIs: */ISAPI/System/Network/interfaces/<ID>/wireless/capabilities* and */ISAPI/System/Network/interfaces/<ID>/wireless*):
added a security mode "WPA/WPA2" to the sub node *<securityMode>* of the node *<WirelessSecurity>*.

Summary of Changes in Version 2.6_Jan., 2020

Related Product: iDS-9632NXI-I8/X(B) DeepinMind NVR and iDS-6708NXI-I/8F(B) DeepinMind NVS with Software Version V4.1.72

1. Extended the log types in *Log Types for ISAPI* :
added two alarm minor log types: "sleepOnDutyStart" (Sleeping on Duty Alarm Started) and "sleepOnDutyStop" (Sleeping on Duty Alarm Stopped).
2. Extended message about device storage capability *XML_RacmCap* (related URI: */ISAPI/ContentMgmt/capabilities*):
added two nodes *<behaviorEventPicSearch>* (behavior analysis events supported by picture search) and *<perimeterEventPicSearch>* (perimeter events supported by picture search).
added one search condition "playCellphone" (playing mobile phone) to the node *<pictureSearchType>*.
added 6 video types "running", "failDown", "violentMotion", "retention", "allPerimeterEvent", "allBehaviorEvent" to the node *<recordSearchType>* (video file type to be searched for).
3. Extended message about version information of algorithm library *XML_AlgorithmsVersion* (related URI: */ISAPI/System/algorithmsVersion*):

added two algorithm library names "streetBehavior" (street behavior) and "workBehavior" (work behavior)

to the node <name> of <AlgorithmsVersionInfo>.

4. Extended error codes in **ISAPI Response Codes** :

added sub error codes to status code 4 (Invalid Operation):

0x400012a6-"vehicleEnginesNoResource" (Insufficient vehicle engine resources),

0x400050df-"noDetectionArea" (No detection area),

0x4000128A-"noMorePeopleNumChangeRulesAdded" (Maximum number of people number changing rules reached), 0x4000128D-"noMoreViolentMotionRulesAdded" (Maximum number of violent motion rules reached), 0x4000128E-"noMoreLeavePositionRulesAdded" (Maximum number of leaving position rules reached), 0x400012a9-"noMoreRunningRulesAdded" (Maximum number of running rules reached), 0x400012aa-"noMoreGroupRulesAdded" (Maximum number of people gathering rules reached),

0x400012ab-"noMoreFailDownRulesAdded" (Maximum number of people falling down rules reached), 0x400012ac-"noMorePlayCellphoneRulesAdded" (Maximum number of playing cellphone rules reached), 0x400015ad-"noMoreRetentionRulesAdded" (Maximum number of people retention rules reached), 0x400015ae-"noMoreSleepOnDutyRulesAdded" (Maximum number of sleeping on duty rules reached).

added sub errors to status code 6 (Invalid Message Content):

0x60002023-"noSafetyHelmetRegion" (The hard hat detection area is not configured),

0x60001053-"behaviorAnalysisRuleInfoError" (Behavior analysis rule parameters error),

0x60001054-"safetyHelmetParamError" (Hard hat parameters error),

0x60001077-"OneChannelOnlyCanBindOneEngine" (No more engines can be bound).

Summary of Changes in Version 2.6_Jan., 2020

Related Product: DS-K1T341AM, DS-K1T341AMF, DS-K1T642M, DS-K1T642MF, DS-K1T642E, DS-K1T642EF, DS-K1T642MW, DS-K1T642MFW, DS-K1T642EW, and DS-K1T642EFW Face Recognition Terminal with Software Version 1.0

1. Added two URIs of getting the algorithm library information:

Get capability: GET **/ISAPI/System/algorithmsInfo/capabilities?format=json** ;

Get algorithm library information: GET **/ISAPI/System/algorithmsInfo?format=json** .

2. Extended device capability message **XML_DeviceCap** (related URI: **/ISAPI/System/capabilities**) :

added two nodes: <isSupportCertificateCaptureEvent>> (whether it supports certificate capture and comparison events) and <isSupportAlgorithmsInfo> (whether it supports getting the algorithm library version information).

Chapter 2 ISAPI Description

The design of Intelligent Security API (hereafter referred as to ISAPI) adopts RESTful style, so this part introduces the predefined resource operation methods, API (URL) format, interaction message format, time format, namespace, and error processing method.

2.1 Operation Method

The methods to operate resources via ISAPI are same as those of HTTP (Hyper Text Transport Protocol) and RTSP (Real Time Streaming Protocol).



Note

The RTSP operation methods are mainly used to get the real-time stream for live view, two-way audio, and playback in this manual. For details about HTTP and RTSP, please refer to <https://tools.ietf.org/html/rfc2612> and <https://tools.ietf.org/html/rfc2326>.

Table 2-1 HTTP Operation Method

Method	Description
POST	Create resources. This method is only available for adding resource that does not exist before.
GET	Retrieve resources. This method cannot change the system status, only return data as the response to the requester.
PUT	Update resources. This method is usually for update the resource that already exists, but it can also be used to create the resource if the specific resource does not exist.
DELETE	Delete resources.

Table 2-2 RTSP Operation Method

Method	Description
OPTIONS	Get the supported RTSP operation methods. See the request and response message format below when interacting between client software and server. <pre>OPTIONS %s RTSP/1.0\r\n //Request URL CSeq:%u\r\n //Command No. User-Agent:%s\r\n //Client software name \r\n /*Succeeded*/ RTSP/1.0 200 OK\r\n CSeq: %u\r\n //Command No.</pre>

Method	Description
	Public: %s\r\n //Supported operation methods Date: %s\r\n //Date and time \r\n /*Failed*/ RTSP/1.0 4XX/5XX %s\r\n CSeq: %u\r\n //Command No. Date: %s\r\n //Date and time \r\n
DESCRIBE	Transfer basic information by SDP (Session Description Protocol, see https://tools.ietf.org/html/rfc2327) files, such as URL with SETUP command and so on. See the request and response message format below when interacting between client software and server. DESCRIBE %s RTSP/1.0\r\n //URL CSeq: %u\r\n //Command No. Accept: application/sdp\r\n //The SDP description is accepted Authorization: %s\r\n //Authentication information User-Agent: %s\r\n //Client software name \r\n /*Succeeded*/ RTSP/1.0 200 OK\r\n CSeq: %u\r\n //Command No. Content-Type: application/sdp\r\n //The SDP description exists behind the command Content-Base: %s\r\n //URL Content-Length: %d\r\n //The length of contents behind the command \r\n [content] //SDP description /*Failed*/ RTSP/1.0 4XX/5XX %s\r\n CSeq: %u\r\n //Command No. \r\n
SETUP	Interact the session information, such as transmission mode, port number, and so on. See the request and response message format below when interacting between client software and server. SETUP %s RTSP/1.0\r\n //URL CSeq: %u\r\n //Command No. Authorization: %s\r\n //Authentication information Session: %s\r\n //Session ID is only returned at the even number of times Transport: %s\r\n //Transmission protocol User-Agent: %s\r\n //Client software name \r\n /*Succeeded*/ RTSP/1.0 200 OK \r\n

Method	Description
	<pre> CSeq: %u\r\n Session:%s\r\n //Session ID Transport: s% //Transmission method Date: s% //Date and time /*Failed*/ RTSP/1.0 4XX/5XX %s\r\n CSeq: %u\r\n //Command No. \r\n </pre>
PLAY	Start the stream transmission. See the request and response message format below when interacting between client software and server. <pre> PLAY %s RTSP/1.0\r\n //URL CSeq:%u\r\n //Command No. Authorization:%s\r\n //Authentication information Session:%s\r\n //Session ID Range: npt=%f-%f\r\n //Determine the play range User-Agent:%s\r\n //Client software name \r\n /*Succeeded*/ RTSP/1.0 200 OK \r\n CSeq: %u\r\n Session:%s\r\n RTP-Info:%s Date: %s /*Failed*/ RTSP/1.0 4XX/5XX %s\r\n CSeq: %u\r\n //Command No. Session:%s\r\n \r\n </pre>
PAUSE	Pause the stream transmission.
TEARDOWN	Stop the stream transmission. See the request and response message format below when interacting between client software and server. <pre> TEARDOWN %s RTSP/1.0\r\n //URL CSeq: %u\r\n //Command No. Authorization:%s\r\n //Authentication information Session:%s\r\n //Session ID User-Agent:%s\r\n //Client software name \r\n /*Succeeded*/ RTSP/1.0 200 OK \r\n CSeq: %u\r\n Session:%s\r\n </pre>

Method	Description
	Date:%s\r\n\r\n /*Failed*/ RTSP/1.0 4XX/5XX %s\r\n CSeq: %u\r\n //Command No. Session:%s\r\n\r\n

2.2 URL Format

URL (Uniform Resource Locator) is a further class of URIs, it can identify a resource and locate the resource by describing its primary access mechanism.

The format of URL is defined as the follows: **<protocol>://<host>[:port][abs_path [?query]]**.

protocol

Protocol types, i.e., HTTP (version 1.1) and RTSP (version 1.0).

host

Host name, IP address, or the FQDN (Fully Qualified Domain Name) of network devices.

port

Port number of host service for listening the connection status of TCP (Transmission Control Protocol, see <https://tools.ietf.org/html/rfc793>) or UDP (User Datagram Protocol, see <https://tools.ietf.org/html/rfc768>). If this field is not configured, the default port number will be adopted. For HTTP, the default port number is 80, and for RTSP, the default port number is 554.

abs_path

Resource URI: /ServiceName/ResourceType/resource. Here, the **ServiceName** is ISAPI; the **ResourceType** is predefined with upper camel case according to different functions, see details in the following table; the resource is defined with lower camel case and can be extended in actual applications. E.g., /ISAPI/System/Network/interfaces.

Predefined URI Model	Description
/ISAPI/System/...	System related resources
/ISAPI/Security/...	Security related resources
/ISAPI/Event...	Event/alarm related resources
/ISAPI/Image/...	Video encoding and image related resources
/ISAPI/ContentMgmt/...	Storage management related resources

query

Strings for describing resources information, including related parameters. The parameter names and values must be listed as the following format in this field: ?p1=v1&p2=v2&...&pn=vn.



Note

- To locate the connected device, when operating lower-level device via the URL, the **query** field should be filled as ?devIndex=uuid&p1=v1&p2=v2&...&pn=vn. The uuid (or guid) is a 32-byte (128 bits) random number, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
- For message in JSON format, the **query** field should be filled as ?format=json&p1=v1&p2=v2&...&pn=vn. For details about message format, refer to the next section below. E.g., <http://10.17.132.22/ISAPI/System/time?format=json&devIndex=550e8400e29b41d4a716446655440000>.

2.3 Message Format

During the ISAPI integration, the request and response messages generated among the interaction between devices and platform are data in XML format or JSON format.



Note

The message format here is only available for URLs based on HTTP.

XML Format

- For the previous integration, XML is a common format which may only cause a little changes in the later integration.
- Generally, for configuration information, the **Content-Type** in the XML format message is "application/xml; charset='UTF-8'", see details below.

```
//Request Message
GET /ISAPI/System/status HTTP/1.1
...

//Response Message
HTTP/1.1 200 OK
...
Content-Type: application/xml; charset="UTF-8"
...
<?xml version="1.0" encoding="UTF-8"?>
<DeviceStatus version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
...
</DeviceStatus>
```

For data (e.g., firmware, configuration files), the **Content-Type** in the XML format message is "application/octet-stream", see details below.

```
//Request Message
PUT /ISAPI/System/configurationData HTTP/1.1
```

```
...
Content-Type: application/octet-stream
...
[proprietary configuration file data content]

//Response Message
HTTP/1.1 200 OK
...

Content-Type: application/xml; charset="UTF-8"
...
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
...
</ResponseStatus>
```

JSON Format Message

- The leaf node (without any sub node) in the message is named by lower camel case, while the non-leaf node in the message is named by upper camel case.
- To communicate by the messages in JSON format, the devices must support the public specifications in <http://www.ecma-international.org/publications/files/ECMA-ST/ECMA-404.pdf> and HTTP with version 1.1.



Note

JSON is a lightweight data format which is a subset of JavaScript language and is small, fast, and easy to be parsed.

- Generally, for configuration information, the **Content-Type** of message is "application/json", see the example below:

```
//Request message
GET /ISAPI/System/status HTTP/1.1
...

//Response message
HTTP/1.1 200 OK
...
Content-Type: application/json
...
"DeviceStatus":""
...
```

For data (e.g., firmware, configuration files), the **Content-Type** of message is "application/octet-stream", see the example below:

```
//Request message
PUT /ISAPI/System/configurationData HTTP/1.1
...
Content-Type: application/octet-stream
...
[proprietary configuration file data content]
```

```
//Response message
HTTP/1.1 200 OK
...
Content-Type: application/json
...
"ResponseStatus":""
...
```

2.4 Others

Time Format

The time format during ISAPI integration adopts ISO8601 standard (see details in <http://www.w3.org/TR/NOTE-datetime-970915>), that is, YYYY-MM-DDThh:mm:ss.sTZD (e.g., 2017-08-16T20:17:06+08:00).

Namespace

For message in XML format, namespace is required. The following namespaces are available:

- xmlns=http://www.isapi.org/ver20/XMLSchema
- xmlns:xs="http://www.w3.org/2001/XMLSchema"
- xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
- xmlns:xlink="http://www.w3.org/1999/xlink"

Error Processing

During the integration applications of ISAPI protocol, when the error of URL based on HTTP occurred, the ResponseStatus message (in XML or JSON format) which contains error code will be returned. If the error of URL based on RTSP occurs, the corresponding status code will directly be returned, for details, refer to <https://tools.ietf.org/html/rfc2326> .

Chapter 3 Security

This part mainly introduces the authentication, user permission, and encryption in the integration applications of ISAPI.

3.1 Authentication

When communicating via ISAPI protocol, the digest of the session must be authenticated.



- The authentication must be based on *HTTP Authentication: Basic and Digest Access Authentication*, see <https://tools.ietf.org/html/rfc2617> for details.
 - The request session must contain authentication information, otherwise, the device will return 401 error code.
-

The message digest, which contains user name, password, specific nonce value, HTTP or RTSP operation methods, and request URL, is generated by the MD5 algorithm, see the calculation rules below.

qop=Undefined

Digest=MD5(MD5(A1):<nonce>:MD5(A2))

qop="auth:"

Digest=MD5(MD5(A1):<nonce>:<nc>:<cnonce>:<qop>:MD5(A2))

qop="auth-int:"

Digest=MD5(MD5(A1):<nonce>:<nc>:<cnonce>:<qop>:MD5(A2))



- The **qop** is a value for determining whether the authentication is required.
- A1 and A2 are two data blocks required for digest calculation.
 - A1: Data block about security, which contains user name, password, security domain, random number, and so on. If the digest calculation algorithm is MD5, A1=<user>:<realm>:<password>; if the algorithm is MD5-sess, A1=MD5(<user>:<realm>:<password>):<nonce>:<cnonce>.
 - A2: Data block about message, such as URL, repeated requests, message body, and so on, it helps to prevent repeated, and realize the resource/message tamper-proof. If the **qop** is not defined or it is "auth:", A2=<request-method>:<uri-directive-value>; if the **qop** is "auth-int:", A2=<request-method>:<uri-directive-value>:MD5(<request-entity-body>).
- The **nonce** is the random number generated by service, the following generation formula is suggested: nonce = BASE64(time-stamp MD5(time-stamp ":" ETag ":" private-key)). The **time-stamp** in the formula is the time stamp generated by service or the unique serial No.; the **ETag** is

the value of HTTP ETag header in the request message; the **private-key** is the data that only known by service.

If authentication failed, the device will return the ***XML_ResponseStatus_AuthenticationFailed*** message, and the remaining authentication attempts will also be returned. If the remaining attempts is 0, the user will be locked at the next authentication attempt.

3.2 User Permission

Three user types with different permissions are adopted in the security specifications to control the access.

Administrator

The account of this user type is called "admin", which has the permissions to access all supported resources. And it must be activated at any time.

Operator

The account of this user type has the permissions to access common resources and less advanced resources, see details in the description of each resource.

Viewer

The account of this user type has the permissions to access common resources only, see details in the description of each resource.

3.3 Encryption

Two encryption modes are available, including all information encryption (based on HTTPS) and sensitive information (e.g., user name, password, card No. and so on) encryption.



Note

- The sensitive information varies with different applications, see details in the description of protocol messages.
 - Encryption will not affect the escape behavior of contents in XML format, which means that the contents should be escaped before encryption.
-

Encryption Capability

Before encrypting the information, the encryption capability of devices or client software should be obtained via the URI: GET ***/ISAPI/Security/capabilities*** to know the available encryption algorithm. The encryption algorithm information is returned by the **securityVersion** node in response message ***XML_SecurityCap***.

Note

Different values of **securityVersion** node correspond to different encrypted contents and encryption modes. If node value is 1, it indicates that the encryption mode is AES128 CBC, and only the password will be encrypted; if the node value is 2, the AES256 CBC mode is adopted, and both the user name and password will be encrypted.

Encryption Flow

Flow of encrypting source data: **Source Data** → **Encode by Base64 Method** → **Encrypt in AES128/AEC256 CBC Mode** .

Note

Encoding data by Base64 method is to be compatible with binary data and get the data size before encrypting by AES128/AEC256 CBC mode (the data size is not real after encrypting as 0 will be supplied).

Encryption Key

The key used for encryption is generated by iteratively encrypting the (Password + "AaBbCcDd1234!@#\$") according to SHA256 algorithm. E.g., if the device password is abcd1234, then the key is SHA(SHA(SHA("abcd1234AaBbCcDd1234!@#\$"))...).

Note

- The number of iterations is determined by device capability, see the node **keyIterateNum** in **XML_SecurityCap** message.
 - If the password is changed when generating the key, the previous password will be used.
 - The device gets the key when it is authenticated, and the client software/system calculates the key after logging in.
-

Encryption URI Format

The encryption modes are displayed by adding **query** parameters, i.e., **security** and **iv**, to the URI (e.g., /ISAPI/xxx?security=1&iv=xxx).

Note

- The parameter **security** is the version No. of encryption scheme. When **security** does not exist, it indicates that the data is not encrypted; when **security**=1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when **security**=2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode.
- The parameter **iv** is the initialization vector, and it is required when **security**=1 or 2. The 16-byte hexadecimal iv of AES128/AES256 CBC mode is generated by client software or system, and it will be transformed to 32-byte characters when transmitting it to devices via URI. When encrypting in AES128/AES256 CBC mode, loop encryption by data block size is not required,

and after encrypting for once, the **iv** will be changed, so if there are two or more nodes to be encrypted, you should copy the iv during encoding to make sure the iv is same for encrypting each node.

- When you want to determine the value of **security** in the URI, you must compare the encryption scheme version No. of client software and device. If the supported versions are different, you should adopt the lower version. E.g., if the client software supports **security=2**, the device supports **security=1**, then **security=1** will be adopted.
-

Suggestion About Extension

If there are other existing nodes in the message are regarded as sensitive information, you can add values to the parameter **security** in the URI, e.g., **security=3** (encrypt added sensitive information in AES128 CBC mode) and **security=4** (encrypt added sensitive information in AES256 CBC mode).

3.3.1 Encrypt Stream

To improve the security of real-time or history stream from devices and comply with the laws and regulations of some parts of oversea markets, we have provided the stream encryption function based on AES algorithm to encrypt the stream.

Steps

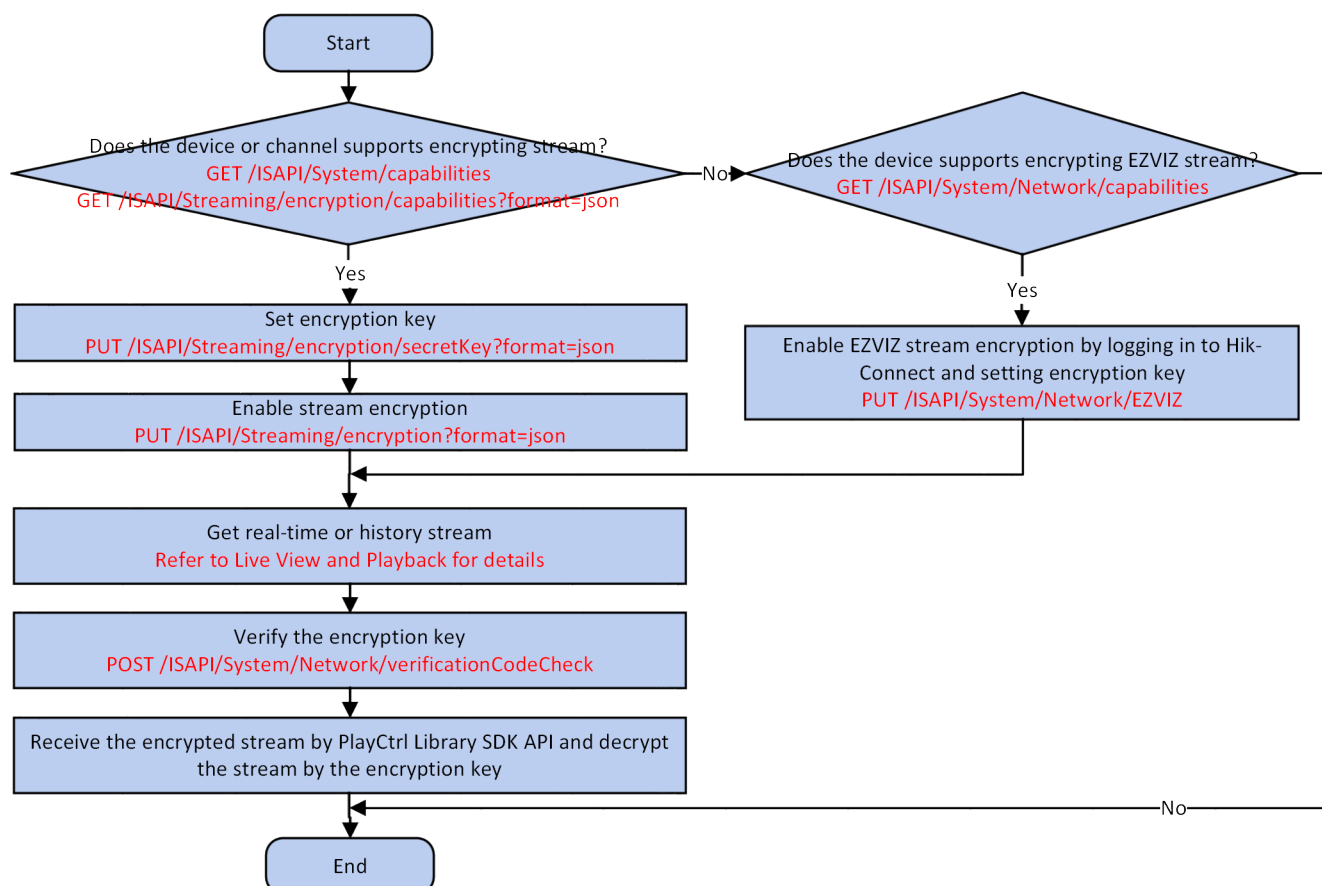


Figure 3-1 Programming Flow of Encrypting Stream



Note

- For devices with Hisilicon chips, up to 16 channels' stream can be encrypted. If the number of channels is larger than 16, the error "insufficient resources" will be returned when starting live view or playback.
- For Sky series devices, all channels' stream can be encrypted.
- The PlayCtrl library will not check if the stream is encrypted. If the stream is encrypted, but the decryption key is not configured or incorrect decryption key is configured, the display image will be black.
- If the encryption algorithm of the external stream is known, you can decrypt and play the stream without the PlayCtrl library.

1. Call **`/ISAPI/System/capabilities`** or **`/ISAPI/Streaming/encryption/capabilities?format=json`** by GET method to get device capability or encryption capability to check if the device or channel supports stream encryption.

- The node **<isSupportEncryption>** is not returned in the capability message ***XML_DeviceCap*** or the value of node **<enable>** in the capability message ***JSON_EncryptionCap*** is "false": perform the steps below.
 - a. Call ***/ISAPI/System/Network/capabilities*** by GET method to get network capability to check if the device supports encrypting EZVIZ stream.

Note

If the node **<isSupportEZVIZ>** is returned in the capability message ***XML_NetworkCap*** and its value is "true", it indicates that encrypting EZVIZ stream is supported. Otherwise, please end this task.

- b. Call ***/ISAPI/System/Network/EZVIZ*** by PUT method to log in to Hik-Connect and set encryption key to enable EZVIZ stream encryption.
 - The node **<isSupportEncryption>** is returned in the capability message ***XML_DeviceCap*** and its value is "true" or the value of node **<enable>** in the capability message ***JSON_EncryptionCap*** is "true": the device or channel supports stream encryption, and continue to perform the steps below.
 - a. Set encryption key: PUT ***/ISAPI/Streaming/encryption/secretKey?format=json*** .
 - b. Enable stream encryption: PUT ***/ISAPI/Streaming/encryption?format=json*** .
2. Get real-time or history stream, refer to ***Live View and Playback*** for details.
 3. Call ***/ISAPI/System/Network/verificationCodeCheck*** by POST method to verify the encryption key.

Note

To check if verifying verification code is supported, you can call ***/ISAPI/System/Network/capabilities*** by GET method to get the device network capability (***XML_NetworkCap***). If supports, the node **<isSupportVerificationCodeCheck>** is returned in the capability and its value is "true".

4. Receive the encrypted stream by calling encrypted stream callback function (PlayM4_SetEncryptTypeCallBack) of PlayCtrl Library SDK and decrypt the stream by the encryption key.

Note

Refer to *Cross-Platform PlayCtrl Library SDK User Manual* for details of the encrypted stream callback function and playing of the decrypted stream.

3.3.2 Encrypt Sensitive Information

Some sensitive information (such as user name, password, and so on) in the messages should be encrypted to improve security during interaction with devices via ISAPI.

Steps

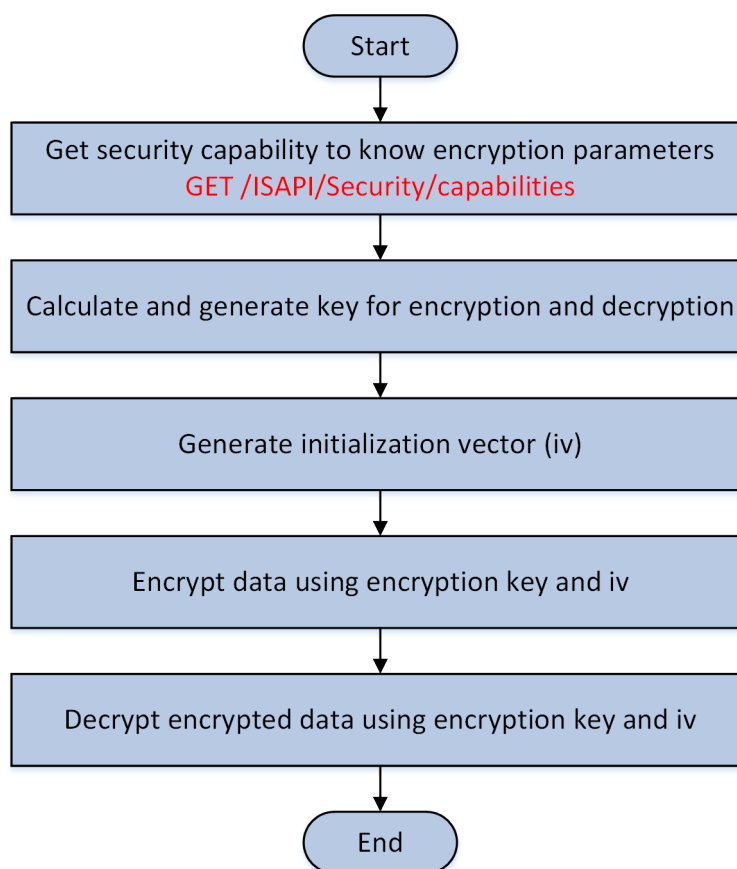


Figure 3-2 Programming Flow of Encrypting Sensitive Information

Note

Values of the following parameters in query parameters, XML messages, and JSON messages are sensitive information that needs to be encrypted.

Table 3-1 Sensitive Information to be Encrypted

Format	Parameters
Query	identityKey (identity key, which is the key of the private face picture library) and secretkey (secret key)
XML	<userName> (user name), <password> (password), <loginPassword> (login password), <snmpAuthenticationPassword> (SNMP authentication password), <snmpPrivacyPassword> (SNMP encryption password), <accountName> (account name), <encryptionKey> (encryption key), <sharedKey> (shared key), <Username> (user name), <passwd> (password), <name> (name), <authPasswd> (authentication

Format	Parameters
	password), <passWord> (password), <key> (key), <tokenKey> (token key), <oldPwd> (old password), <newPwd> (new password), and <terminalPwd> (terminal password)
JSON	"userName" (user name), "password" (password), "loginPassword" (login password), "snmpAuthenticationPassword" (SNMP authentication password), "snmpPrivacyPassword" (SNMP encryption password), "accountName" (account name), "encryptionKey" (encryption key), "sharedKey" (shared key), "Username" (user name), "passwd" (password), "authPasswd" (authentication password), "passWord" (password), "key" (key), "tokenKey" (token key), "randomCode" (random code), "AK" (access key), "SK" (encryption key), "Card" (card No.), "Password" (password), "FingerPrint" (fingerprint), "TemporaryPassword" (temporary password), "RemoteControl" (remote control), and "WirelessInfo" (wireless information)

1. Call **/ISAPI/Security/capabilities** by GET method to get the device security capability to know the encryption parameters.

The security capability is returned in the message **XML_SecurityCap**. Nodes of encryption parameters include <securityVersion> (encryption scheme version), <keyIterateNum> (iteration times), <isIrreversible> (whether the password is irreversible), and <salt> (salt value).

Note

The salt value will be returned only when the request URL contains the query parameter **username**.

2. Calculate and generate the key for encryption and decryption according to the encryption parameters.
 - If <isIrreversible> is "true" and <salt> is also configured, the key is calculated and generated by iteratively encrypting the value of (user name+salt+password) with SHA256 algorithm.

Note

- The value of <salt> should be different for different users.
 - The password encrypted by this method is irreversible.
- If <isIrreversible> is "false" or does not exist, and <salt> is null, the key is calculated and generated by iteratively encrypting the password with SHA256 algorithm.

Note

The password encrypted by this method is reversible.

3. Generate a byte array containing 16 elements randomly as the initialization vector (iv) and covert the byte array to character array.
4. Encrypt the data.
 - 1) Convert iv from character array to byte array.

- 2) Encode the data to be encrypted by using base64.
- 3) Set the AES key based on the encryption key and the encryption version.



Note

Currently, only AES-128 encryption is supported, which indicates that the **securityVersion** is 1.

- 4) Encrypt the data by using iv and AES key in AES-CBC mode.
 - 5) Convert the encrypted data to character array.
5. Decrypt the encrypted data.
- 1) Convert iv from character array to byte array.
 - 2) Convert the encrypted data from character array to byte array.
 - 3) Set the AES key based on the encryption key and the encryption version.



Note

Currently, only AES-128 encryption is supported, which indicates that the **securityVersion** is 1.

- 4) Decrypt the encrypted data by using iv and AES key in AES-CBC mode.
- 5) Decode the decrypted data by using base64.

Example

Sample Code of Encrypting Sensitive Information

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Security.Cryptography;
using System.Text;
using System.Threading.Tasks;

namespace SystemManagement.InformationEncryption
{
    class InfoEncryption
    {
        /// <summary>
        /// Generate initialization vector (iv)
        /// </summary>
        /// <param name="strInitVector"></param>
        /// <returns></returns>
        public static void GetInitVector(out string strInitVector)
        {
            byte[] szInitVector = new byte[16];
            Random ra = new Random();
            ra.NextBytes(szInitVector);
            byte[] byHexAes = AESEncryption.converByteArrayToCharArray(szInitVector, szInitVector.Length);
            strInitVector = Encoding.UTF8.GetString(byHexAes).ToLower();//This must be in lowercase.
        }

        /// <summary>
        /// Encrypt data with SHA256 algorithm. The data encrypted by this algorithm is reversible.
    }
}
```

```
/// </summary>
/// <param name="strData"></param>
/// <returns></returns>
public static string sha256(string strData)
{
    byte[] szData = Encoding.UTF8.GetBytes(strData);
    byte[] szHash = SHA256Managed.Create().ComputeHash(szData);
    byte[] szSha256 = AESEncryption.converByteArrayToCharArray(szHash, 32); //The standard algorithm is Hash. The
length 32 is set according to the sample document.
    return Encoding.UTF8.GetString(szSha256);
}

/// <summary>
/// Encrypt (user name+salt+password) with SHA256 algorithm. The password encrypted by this algorithm is
irreversible.
/// </summary>
/// <param name="strUser"></param>
/// <param name="szSalt"></param>
/// <param name="strPassword"></param>
/// <returns></returns>
public static string calcSha256(string strUser, byte[] szSalt, string strPassword)
{
    string strSrcData = strUser;
    byte[] szRealSalt = new byte[64]; //The size of the salt value is 64 bits.
    if (szSalt.Length > 64)
    {
        return null;
    }
    Array.Copy(szSalt, szRealSalt, szSalt.Length);
    strSrcData = strSrcData + Encoding.UTF8.GetString(szRealSalt) + strPassword;
    return sha256(strSrcData);
}

/// <summary>
/// Generate encryption key
/// </summary>
/// <param name="strUserName"></param>
/// <param name="strSalt"></param>
/// <param name="strPassword"></param>
/// <param name="szOut"></param>
/// <param name="iKeyIterateNum"></param>
/// <param name="blIrreversible"></param>
/// <returns></returns>
public static void getEncryptKey(string strUserName, string strSalt, string strPassword, out byte[] szOut, int
iKeyIterateNum, bool blIrreversible)
{
    byte[] szSalt = null;
    if (strSalt != null)
    {
        szSalt = Encoding.UTF8.GetBytes(strSalt);
    }
    string strSrcData = string.Empty;
```

```
if (blrreversible && szSalt.Length > 0)
{
    string strIrrPsw = calcSha256(strUserName, szSalt, strPassword);
    if (strIrrPsw.Length > 64)
    {
        strSrcData = strIrrPsw.Substring(0,64);
    }
    else
    {
        strSrcData = strIrrPsw;
    }
}
else
{
    if (strPassword.Length > 64)
    {
        strSrcData = strPassword.Substring(0,64);
    }
    else
    {
        strSrcData = strPassword;
    }
}
strSrcData += "AaBbCcDd1234!@#$$";
if (iKeyIterateNum <= 0)//Iterations
{
    iKeyIterateNum = 100;
}

//For the SHA256 iteration, iKeyIterateNum represents the number of iterations returned by the capability set.
for (int i = 0; i < iKeyIterateNum; i++)
{
    strSrcData = sha256(strSrcData);
}
byte[] szSHA256 = Encoding.UTF8.GetBytes(strSrcData);

//The result calculated with SHA256 algorithm is converted to a byte array for the last time.
byte[] szByteArray = AESEncryption.convertCharArrayToByteArray(szSHA256, szSHA256.Length);

if (szByteArray.Length > 32)
{
    szOut = new byte[32];
    Array.Copy(szByteArray, szOut, 32);
}
else
{
    szOut = new byte[szByteArray.Length];
    Array.Copy(szByteArray, szOut, szByteArray.Length);
}
}

/// <summary>
```

```
/// Encrypt content (sensitive information)
/// </summary>
/// <param name="strInitVexor"></param>
/// <param name="szAESKey"></param>
/// <param name="strSrcContent"></param>
/// <param name="strOut"></param>
/// <param name="iSecurityVersion"></param>
/// <returns></returns>
public static void getEncryptContent(string strInitVexor, byte[] szAESKey, string strSrcContent, out string strOut, int
iSecurityVersion)
{
    if (iSecurityVersion != 1)//1 refers to the AES-128 algorithm. Currently only 1 is supported.
    {
        strOut = strSrcContent;
        return;
    }

    //Convert to UTF-8
    byte[] szInitVexor = Encoding.UTF8.GetBytes(strInitVexor);
    byte[] szInitVexorByteArray = AESEncryption.convertCharArrayToByteArray(szInitVexor, szInitVexor.Length);
    byte[] szSrcBytes = Encoding.UTF8.GetBytes(strSrcContent);
    string strSrcBase64 = Convert.ToBase64String(szSrcBytes);

    //AES encryption
    byte[] szAesData = AESEncryption.AesEncrypt(strSrcBase64, szAESKey, szInitVexorByteArray);
    byte[] szOut = AESEncryption.converByteArrayToCharArray(szAesData, szAesData.Length);
    strOut = Encoding.UTF8.GetString(szOut);
}

/// <summary>
/// Decrypt content (sensitive information)
/// </summary>
/// <param name="strInitVexor"></param>
/// <param name="szAESKey"></param>
/// <param name="strSrcContent"></param>
/// <param name="strOut"></param>
/// <param name="iSecurityVersion"></param>
/// <returns></returns>
public static void getDecryptContent(string strInitVexor, byte[] szAESKey, string strSrcContent, out string strOut, int
iSecurityVersion)
{
    if (iSecurityVersion != 1)//1 refers to the AES-128 algorithm. Currently only 1 is supported.
    {
        strOut = strSrcContent;
        return;
    }

    //Convert to UTF-8
    byte[] szInitVexor = Encoding.UTF8.GetBytes(strInitVexor);
    byte[] szInitVexorByteArray = AESEncryption.convertCharArrayToByteArray(szInitVexor, szInitVexor.Length);
    byte[] szSrcBytes = Encoding.UTF8.GetBytes(strSrcContent);
    byte[] szSrcByteArray = AESEncryption.convertCharArrayToByteArray(szSrcBytes, szSrcBytes.Length);
```

```
string strAesData = AesEncryption.AesDecrypt(szSrcByteArray, szAESKey, szInitVextorByteArray);
byte[] szOut = Convert.FromBase64String(strAesData);
strOut = Encoding.UTF8.GetString(szOut);
}
}
}
```

Note

The sample code of the class AesEncryption is as shown below.

```
using System;
using System.Collections.Generic;
using System.IO;
using System.Linq;
using System.Security.Cryptography;
using System.Text;
using System.Threading.Tasks;

namespace SystemManagement.InformationEncryption
{
    class AesEncryption
    {
        /// <summary>
        /// Convert hexadecimal to binary
        /// </summary>
        /// <param name="chstr"></param>
        /// <returns></returns>
        public static byte hexToBinary(byte chstr)
        {
            char crtn = '\0';
            if (('0' <= chstr) && ('9' >= chstr))
            {
                crtn = (char)(chstr & 0x0F);
            }
            else if (('A' <= chstr) && ('F' >= chstr))
            {
                crtn = (char)(chstr - 'A' + 10);
            }
            else if (('a' <= chstr) && ('f' >= chstr))
            {
                crtn = (char)(chstr - 'a' + 10);
            }
            return (byte)crtn;
        }

        /// <summary>
        /// Convert a character array to a byte array
        /// </summary>
        /// <param name="pSrc"></param>
        /// <param name="nSrcLen"></param>
```



```
/// <returns></returns>
public static byte[] convertCharArrayToByteArray(byte[] pSrc, int nSrcLen)
{
    byte[] byChallengeDst2 = new byte[nSrcLen / 2];
    for (int i = 0; i < nSrcLen; i = i + 2)
    {
        byChallengeDst2[i / 2] = (byte)(hexToBinary(pSrc[i]) << 4);
        byChallengeDst2[i / 2] += (byte)hexToBinary(pSrc[i + 1]);
    }
    return byChallengeDst2;
}

/// <summary>
/// Convert a byte array to a character array
/// </summary>
/// <param name="pSrc"></param>
/// <param name="nSrcLen"></param>
/// <returns></returns>
public static byte[] converByteArrayToCharArray(byte[] pSrc, int nSrcLen)
{
    StringBuilder strB = new StringBuilder();
    for (int i = 0; i < nSrcLen; i++)
    {
        strB.Append(pSrc[i].ToString("x2")); //Here x must be in lowercase, which means converting to a lowercase
        hexadecimal character.
    }
    return Encoding.UTF8.GetBytes(strB.ToString());
}

/// <summary>
/// Encrypt in AES-CBC mode
/// </summary>
/// <param name="strSrcContent"></param>
/// <param name="szAESKey"></param>
/// <param name="iv"></param>
/// <returns></returns>
public static byte[] AesEncrypt(string strSrcContent, byte[] szAESKey, byte[] iv)
{
    RijndaelManaged rijndaelCipher = new RijndaelManaged();
    rijndaelCipher.Mode = CipherMode.CBC;
    rijndaelCipher.Padding = PaddingMode.Zeros; //The end is all-zero padding
    rijndaelCipher.KeySize = 128;
    rijndaelCipher.BlockSize = 128;
    byte[] keyBytes = new byte[16];
    int len = szAESKey.Length;
    if (len > keyBytes.Length)
    {
        len = keyBytes.Length;
    }
    System.Array.Copy(szAESKey, keyBytes, len);
}
```

```
rijndaelCipher.Key = keyBytes;
rijndaelCipher.IV = iv;
ICryptoTransform transform = rijndaelCipher.CreateEncryptor();
byte[] szSrcContent = Encoding.UTF8.GetBytes(strSrcContent);
byte[] szDstContent = transform.TransformFinalBlock(szSrcContent, 0, szSrcContent.Length);
return szDstContent;
}

/// <summary>
/// Decrypt in AES-CBC mode
/// </summary>
/// <param name="szSrcContent"></param>
/// <param name="szAESKey"></param>
/// <param name="iv"></param>
/// <returns></returns>
public static string AesDecrypt(byte[] szSrcContent, byte[] szAESKey, byte[] iv)
{
    RijndaelManaged rijndaelCipher = new RijndaelManaged();
    rijndaelCipher.Mode = CipherMode.CBC;
    rijndaelCipher.Padding = PaddingMode.Zeros; //The end is all-zero padding
    rijndaelCipher.KeySize = 128;
    rijndaelCipher.BlockSize = 128;
    byte[] keyBytes = new byte[16];
    int len = szAESKey.Length;
    if (len > keyBytes.Length)
    {
        len = keyBytes.Length;
    }
    System.Array.Copy(szAESKey, keyBytes, len);
    rijndaelCipher.Key = keyBytes;
    rijndaelCipher.IV = iv;
    ICryptoTransform transform = rijndaelCipher.CreateDecryptor();
    byte[] szDstContent = transform.TransformFinalBlock(szSrcContent, 0, szSrcContent.Length);

    //After decrypting the last 16 bytes of the ciphertext, the decrypted data of the last 16 bytes will be checked.
    //If the value of the last byte is 16, the 16-byte string data will be discarded.
    //If the value of the last byte is smaller than 16, it indicates that the original text was padded during encryption.
    The padding will be discarded.
    for (int i = 0; i < szDstContent.Length; i++)
    {
        if (szDstContent[i] <= 16)
        {
            szDstContent[i] = 0;
        }
    }
    //The \0 at the end needs to be removed.
    return Encoding.UTF8.GetString(szDstContent).Replace("\0", "");
}
}
```

3.4 Security Service

Management and Configuration

Function	Request URI
Get or set advanced parameters of security management	GET or PUT <i>/ISAPI/Security/advanced?format=json</i>
Get or set security questions in batch	GET or PUT <i>/ISAPI/Security/questionConfiguration</i>
Search for security logs	POST <i>/ISAPI/ContentMgmt/security/logSearch</i>
Security Mode Level of Private Protocol	• Get configuration capability of security mode level of private protocol: GET <i>/ISAPI/Security/CommuMode/capabilities?format=json</i> • Get parameters of security mode level of private protocol: GET <i>/ISAPI/Security/CommuMode?format=json</i> • Set parameters of security mode level of private protocol: PUT <i>/ISAPI/Security/CommuMode?format=json</i>
Get capability of selecting certificate	GET <i>/ISAPI/Security/certificate/select/capabilities?format=json</i>
Get or set parameters of selecting certificate	GET or PUT <i>/ISAPI/Security/certificate/select/<functionName>?format=json</i>
Get or set certificate type parameters of web service	GET or PUT <i>/ISAPI/Security/webCertificate</i>

Network Certificate

Function	Request URI
Get CA (Certificate Authority) certificate capability	GET <i>/ISAPI/Security/deviceCertificate/capabilities?format=json</i>
Import network certificate to device	PUT <i>/ISAPI/Security/deviceCertificate</i>
Import CA (Certificate Authority) certificate to device	PUT <i>/ISAPI/Security/deviceCertificate?customID=</i>

Function	Request URI
Network Certificate Search	- Get capability of certificate search: GET <i>/ISAPI/Security/deviceCertificate/certificates/capabilities?format=json</i> - Search for certificate information in a batch: GET <i>/ISAPI/Security/deviceCertificate/certificates?format=json</i> - Search for information of a specific device certificate: GET <i>/ISAPI/Security/deviceCertificate/certificates/<customID>?format=json</i>
Delete network certificate	DELETE <i>/ISAPI/Security/deviceCertificate/certificates/<ID></i> or <i>/ISAPI/Security/deviceCertificate/certificates/<customID>?format=json</i>

Authentication Certificate

Function	Request URI
Get, generate, or delete certificate signature request	GET, PUT , or DELETE <i>/ISAPI/Security/serverCertificate/certSignReq</i>
Download authentication certificate	GET <i>/ISAPI/Security/serverCertificate/downloadCertSignReq</i>
Get, install, or delete authentication certificate	GET, PUT , or DELETE <i>/ISAPI/Security/serverCertificate/certificate</i>
Authentication Certificate Status Search	- Get status of one authentication certificate: GET <i>/ISAPI/Security/deviceCertificate/certificates/<ID>/status?format=json</i> - Get status of all authentication certificates: GET <i>/ISAPI/Security/deviceCertificate/certificates/status?format=json</i>
Generate one authentication certificate	PUT <i>/ISAPI/Security/deviceCertificate/certificates/<ID>/recreate?format=json</i>
Generate all authentication certificates	PUT <i>/ISAPI/Security/deviceCertificate/certificates/recreate?format=json</i>

Client/Server Certificate

Function	Request URI
Get client/server certificate capability	GET <i>/ISAPI/Security/serverCertificate/capabilities?format=json</i>
Generate PKCS#10 signature request of client/server certificate	POST <i>/ISAPI/Security/serverCertificate/certSignReq?customID=</i>

Function	Request URI
Get or generate PKCS#10 signature request of client/server self-signed certificate	GET or PUT <i>/ISAPI/Security/serverCertificate/selfSignCert?customID=</i>
Get information of multiple client/server certificates in a batch	GET <i>/ISAPI/Security/serverCertificate/certificates?format=json</i>
Get or delete information of a specific client/server certificate	GET or DELETE <i>/ISAPI/Security/serverCertificate/certificates/<customID>?format=json</i>
Import client/server certificate to device	POST <i>/ISAPI/Security/serverCertificate/certificate?customID=</i>
Export client/server certificate	GET <i>/ISAPI/Security/serverCertificate/downloadCertSignReq?customID=</i>

Chapter 4 Login

Function	Request URI
Login by digest	GET <i>/ISAPI/Security/userCheck</i>
Lock for illegal login	GET or PUT <i>/ISAPI/Security/illegalLoginLock</i>
Get or set maximum failed login attempts	GET or PUT <i>/ISAPI/Security/loginLinkNum?format=json</i>

Chapter 5 Device Management

This chapter lists the request URIs for getting device information, managing network cameras, getting the capabilities of network cameras, and so on.

Function	Request URI
Get device capability	GET <i>/ISAPI/System/deviceInfo/capabilities</i>
Get device information	GET <i>/ISAPI/System/deviceInfo</i>

5.1 Device Activation

5.1.1 Directly Activate Device

For the connected devices, you must activate them to make sure the device password is reconfigured and the password string conforms to the security rule.

Steps

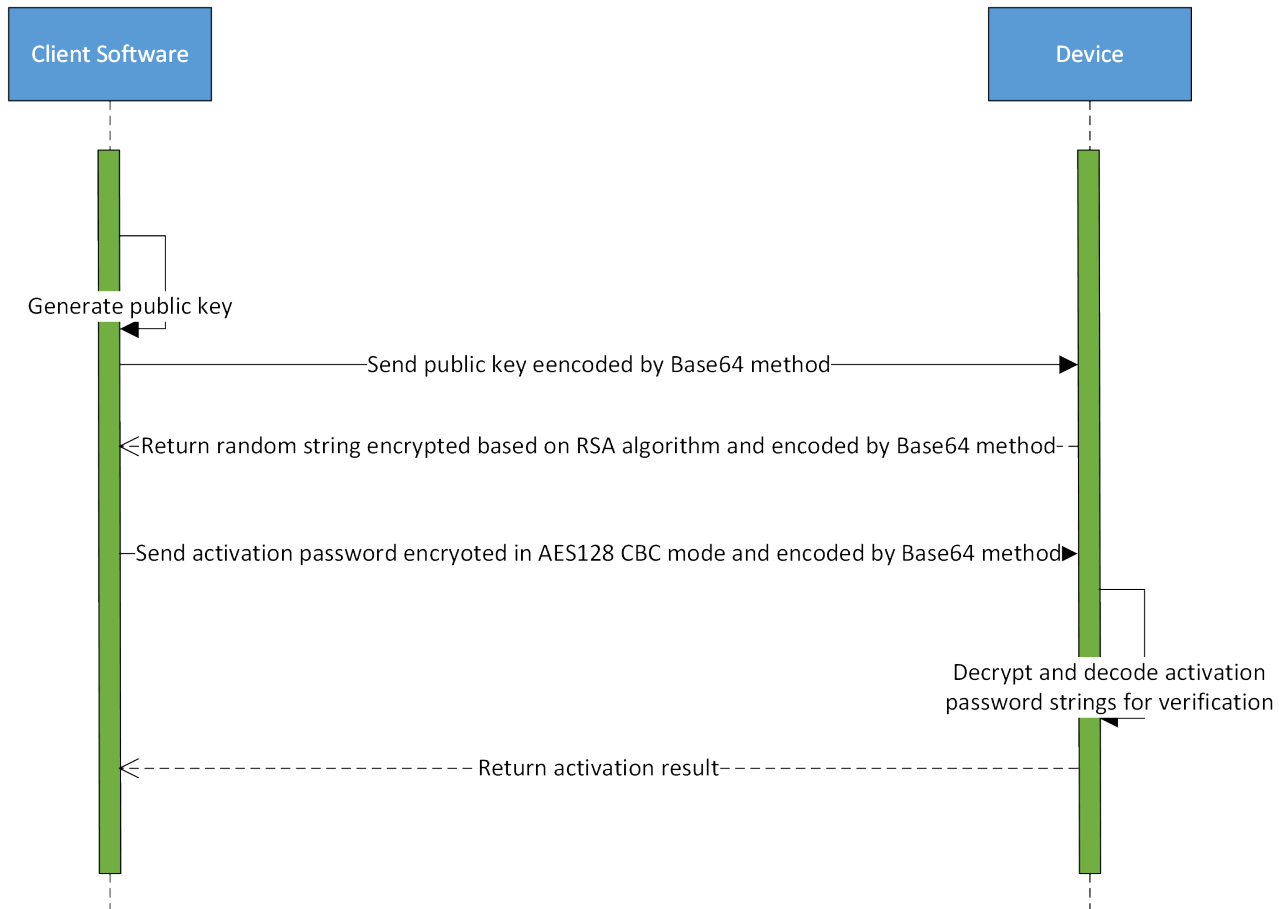


Figure 5-1 Direct Activation Sequence Diagram

1. Client software generates public and private key.

Note

The size of public key is 1024 bits (128 bytes). When transforming to the hexadecimal string, each byte corresponds to two characters (e.g., **127** → **FF**), so the public key is transformed to a 256-byte string finally.

2. Client software encodes the public key (a 256-byte string) by Base64 method and sends it to device.
3. Device decodes the encoded public key by Base64 method and generates random strings.
4. Device encrypts the random strings based on RSA algorithm and encodes the encrypted strings by Base64 method.
5. Call **/ISAPI/Security/challenge** by POST method to get the encrypted and encoded random strings.
6. Client software decodes the obtained random strings by Base64 method and decrypts the random strings by using private key.

7. Client software encrypts the activation password with the random strings in AES128 CBC mode and encodes the password by Base64 method.
8. Call ***/ISAPI/System/activate*** by PUT method to send the activation password strings to device.



Note

The sent activation password strings consist of first 16 characters of random strings and actual password.

Example

If the first 16 characters of random strings is "aaaabbbbccccdddd", and the actual password is "Abc12345", the encrypted activation password strings is "aaaabbbbccccddddAbc12345".

9. Device decodes the activation password by Base64 method and decrypts the password with the random string in AES128 CBC mode for verification.
10. Device starts activating and sends the activation result to client software.

5.1.2 Activate Network Camera via NVR

For network devices, except directly activating them, you can also activate them via NVR if they can be searched on the same network domain of NVR.

Steps

1. Call ***/ISAPI/ContentMgmt/InputProxy/search*** by GET method to search for the online network devices in the same network domain with the NVR.
2. **Optional:** Call ***/ISAPI/ContentMgmt/InputProxy/channels/activate/capabilities*** by GET method to get the activation capability of network devices for reference.
3. Call ***/ISAPI/ContentMgmt/InputProxy/channels/activate*** by PUT method to activate the searched online devices via NVR.

5.2 Reset Password by Setting Recovery Email

A recovery email is added or configured for resetting the password as required. The admin user can set a recovery email after activating the device, and then receive the verification code from the manufacturer via the recovery email to reset the device password.

Before You Start

Make sure you have activated the device.

Steps

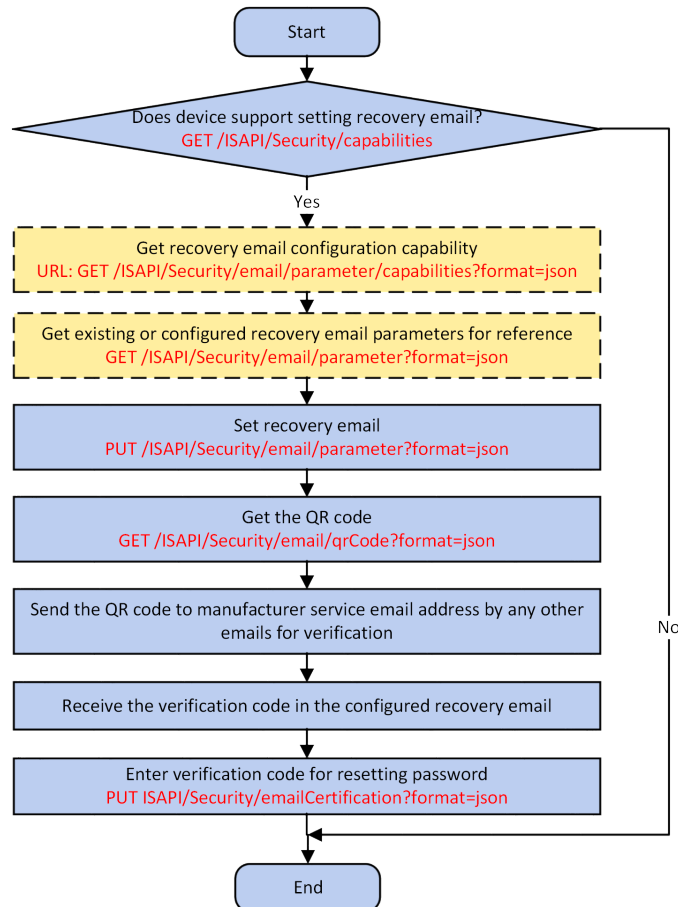


Figure 5-2 Programming Flow of Resetting Password by Setting Recovery Email

Note

You can reset password by other methods, such as answering security question, importing GUID files, and so on. But all methods should be supported by device, so you should call ***/ISAPI/Security/extern/capabilities*** by GET method to get the other security capability (***XML_externSecurityCap***) before choosing method to reset password.

1. Call ***/ISAPI/Security/capabilities*** by GET method to check if the device supports setting recovery email.
The security capability is returned in the message ***XML_SecurityCap*** .
If the node ***<isSupportSecurityEmail>*** exists in the returned message and its value is "true", it indicates that setting recovery email is supported by the device, and you can continue to perform the following steps; otherwise, end this task.
2. **Optional:** Call ***/ISAPI/Security/email/parameter/capabilities?format=json*** by GET method to get recovery email configuration capability.

The recovery email configuration capability is returned in the message *JSON_SecurityEmailCap* by the output parameter pointer (**IpOutputParam**).

3. **Optional:** Call */ISAPI/Security/email/parameter?format=json* by GET method to get the existing or configured recovery email parameters for reference.
4. Call */ISAPI/Security/email/parameter?format=json* by PUT method to set the recovery email for the device.
5. Call */ISAPI/Security/email/qrCode?format=json* by GET method to get the QR code of the recovery email.
6. Send the QR code to manufacturer service email address by any other emails for verification.
7. Receive the verification code in the configured recovery email.
8. Call */ISAPI/Security/emailCertification?format=json* by PUT method to Enter the received verification code for resetting the device password.

5.3 Online Upgrade Device

For the devices that can connect to Guarding Vision Server, you can online upgrade their firmware via Guarding Vision Client, and get the upgrade progress. Besides, in the condition of bad network, you can also enable automatic download of upgrade package in the background to improve the upgrade speed.

Before You Start

Make sure you have activated the device.

Steps

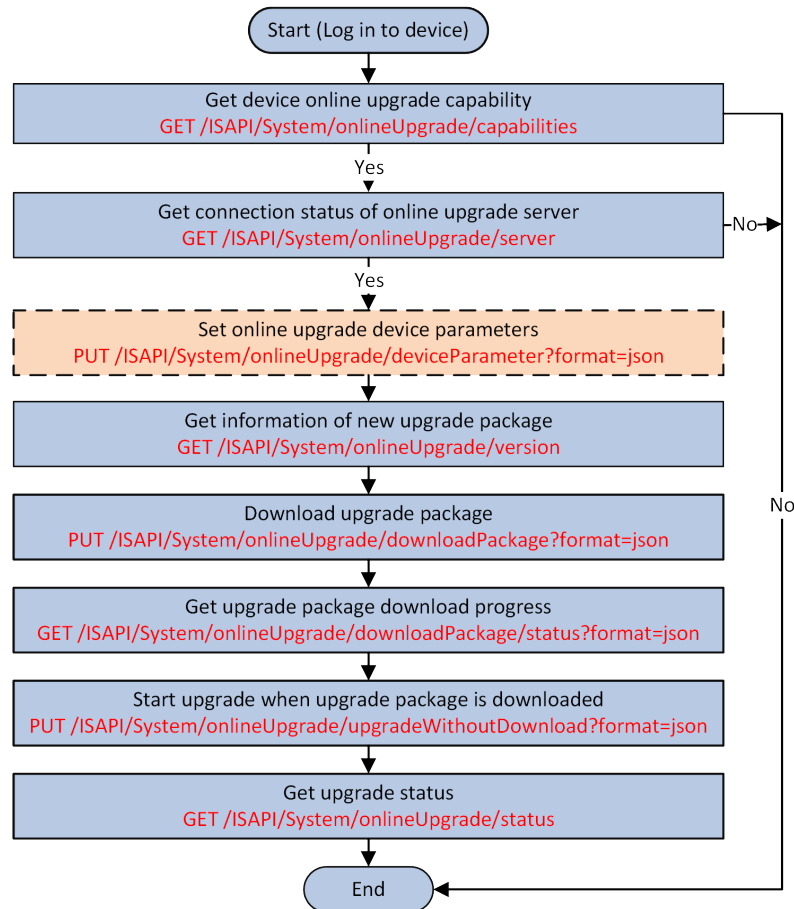


Figure 5-3 Programming Flow of Online Upgrade Device

1. Call `/ISAPI/System/onlineUpgrade/capabilities` by GET method to check if the device support online upgrade.
The online upgrade capability set is returned in *XML_OnlineUpgradeCap*.
2. Call `/ISAPI/System/onlineUpgrade/server` by GET method to get the connection status of online upgrade server.



Note

Only when the node `<connectStatus>` is returned and values "true", you can go on for next step. Otherwise, you should end this task.

The upgrade server connection status is returned in *XML_OnlineUpgradeServer*.

3. **Optional:** Call `/ISAPI/System/onlineUpgrade/deviceParameter?format=json` by PUT method to set the device online upgrade parameters.



Note

Before setting the device online upgrade parameters, you can get these parameters by calling ***/ISAPI/System/onlineUpgrade/deviceParameter?format=json*** via GET method.

4. Call ***/ISAPI/System/onlineUpgrade/version*** by GET method to get the new upgrade package information.
5. Call ***/ISAPI/System/onlineUpgrade/downloadPackage?format=json*** by PUT method to download the upgrade package.
6. **Optional:** During downloading the upgrade package, you can perform the following operations.

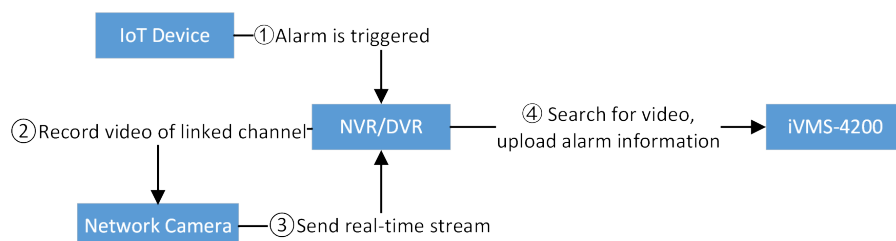
Pause Download	PUT <i>/ISAPI/System/onlineUpgrade/downloadPackage/pause?format=json</i>
Resume Download	PUT <i>/ISAPI/System/onlineUpgrade/downloadPackage/resume?format=json</i>
Cancel Download	DELETE <i>/ISAPI/System/onlineUpgrade/downloadPackage?format=json</i>
7. Call ***/ISAPI/System/onlineUpgrade/downloadPackage/status?format=json*** by GET method to get the upgrade package download progress.
8. Call ***/ISAPI/System/onlineUpgrade/upgradeWithoutDownload?format=json*** by PUT method to start upgrade when the upgrade package is downloaded.
9. Call ***/ISAPI/System/onlineUpgrade/status*** by GET method to get the upgrade status.

5.4 IoT Devices Access

The IoT devices here include access control terminal, security control panel, video intercom, analog camera RF, etc. The NVR/DVR supports accessing IoT devices for device management, status search, event/alarm linkage, and alarm receiving. When the IoT devices are added to NVR/DVR, the NVR/DVR can receive the alarm of IoT device, link the channel for recording, and receive the real-time stream when alarm is triggered.

This chapter mainly introduces the methods of adding IoT devices, and configuring event/alarm of IoT devices.

Application Scenario



5.4.1 Add IoT Devices to NVR/DVR

For convenient management of IoT device, such as status searching, event/alarm linkage, alarm receiving, and so on, you should add the IoT device to NVR/DVR first.

Before You Start

Make sure you have logged in to the device.

Steps

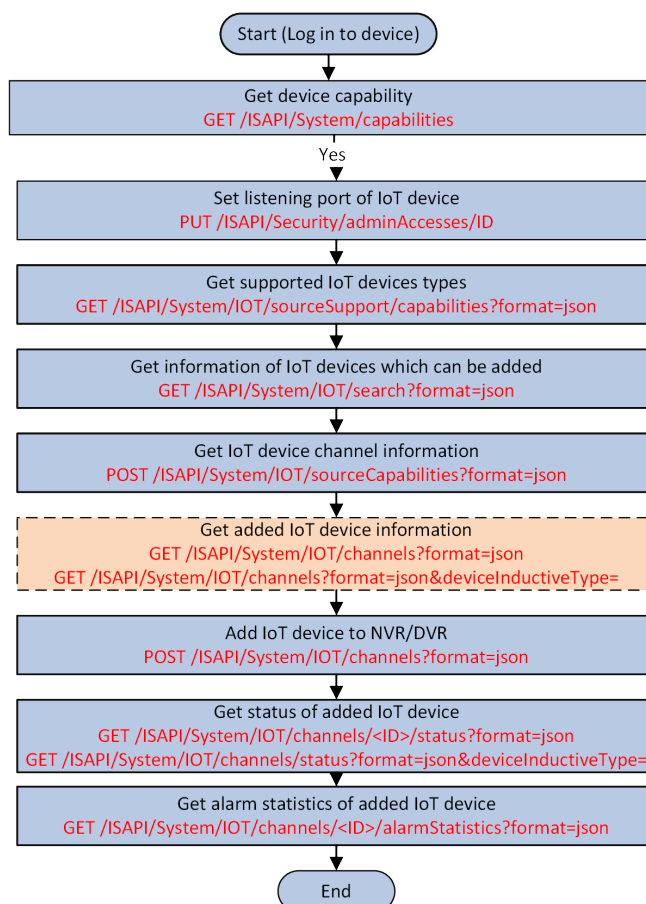


Figure 5-4 Programming Flow of Adding IoT Device to NVR/DVR

1. Call **/ISAPI/System/capabilities** by GET method to get the device capability for checking if the IoT device access function is supported via the node **<IOTCap>**.
The device capability **XML_DeviceCap** is returned.
2. Call **/ISAPI/Security/adminAccesses/<ID>** by PUT method to configure listening port before adding the OPTeX and Luminite IoT devices and set the value of **<protocol>** in **XML_AdminAccessProtocol** to "IOT".
3. Call **/ISAPI/System/IOT/sourceSupport/capabilities?format=json** by GET method to get the supported IoT devices types of device.
The supported IoT devices types are returned in **JSON_IOTSourceSupport**.

4. Call **`/ISAPI/System/IOT/search?format=json`** by GET method to get information of IoT devices which can be added.
The IoT devices which can be added is returned in **`JSON_IOTSourceList`**.
5. Call **`/ISAPI/System/IOT/sourceCapabilities?format=json`** by POST method to get the number of IoT device channels which can be added.
The supported source capability is returned in **`JSON_SourceCapabilities`**.
6. **Optional:** Get the information of added IoT devices.
 - Call **`/ISAPI/System/IOT/channels?format=json`** by GET method to get the information of added IoT devices.
 - Call **`/ISAPI/System/IOT/channels?format=json&deviceInductiveType=`** by GET method to get the information of added IoT devices according to inductive type
7. Call **`/ISAPI/System/IOT/channels?format=json`** by POST method to Add the IoT device to NVR/DVR.
8. Call **`/ISAPI/System/IOT/channels/<ID>/basicParam?format=json`** by PUT method to set the basic parameters of added IoT device according to channel ID.
9. **Optional:** Call **`/ISAPI/System/IOT/channels/<ID>?format=json`** by PUT method to Set the specified added IoT device according to channel ID.



Note

Before setting the specified IoT device, you can get the IoT device by calling **`/ISAPI/System/IOT/channels/<ID>?format=json`** via GET method.

10. **Optional:** Delete the added IoT device.
 - Call **`/ISAPI/System/IOT/channels/<ID>?format=json`** by DELETE method to delete the specified IoT channel.
 - Call **`/ISAPI/System/IOT/channels/<ID>/all?format=json`** by DELETE method to delete all channels (video channel and IoT channel) of added IoT device.
11. Get the status of added IoT device.
 - Call **`/ISAPI/System/IOT/channels/status?format=json`** by GET method to get the status of added IoT device.
 - Call **`/ISAPI/System/IOT/channels/status?format=json&deviceInductiveType=`** by GET method to get the status of added IoT device according to inductive type.
 - Call **`/ISAPI/System/IOT/channels/<ID>/status?format=json`** by GET method to get the status of specified channel of added IoT device.
12. Call **`/ISAPI/System/IOT/channels/<ID>/alarmStatistics?format=json`** by GET method to get the alarm statistics of added IoT device according to channel ID.
13. **Optional:** Call **`/ISAPI/System/accessDevice/associatedChannel?format=json`** by POST method to get the linked channel information of added IoT device.
14. **Optional:** Import or export the list of added IoT devices.

Import Call **`/ISAPI/System/IOT/channelConfig?format=json`** by PUT method to import the list of added IoT devices

Export Call ***/ISAPI/System/IOT/channelConfig?format=json*** by GET method to export the list of added IoT devices

Example

```
PUT /ISAPI/System/IOT/channelConfig?format=json
Accept: text/html, application/xhtml+xml,
Accept-Language: us-EN
Content-Type: File/ Opaque Data
User-Agent: Mozilla/5.0 (compatible; MSIE 9.0; Windows NT 6.1; WOW64; Trident/5.0)
Accept-Encoding: gzip, deflate
Host: 10.10.36.29:8080
Content-Length: 9907
Connection: Keep-Alive
Cache-Control: no-cache
Content-Disposition: form-data; name="XX";
Content-Length: 9907

.....JFIF.....` `.....C.....
..
..... $' "#..(7),01444.'9=82<.342...C. ....
```

5.4.2 Configure Alarm of IoT Device

When the IoT device is added to NVR/DVR, you can configure the alarm parameters such as arming schedule, linkage method, OSD, and so on for receiving the alarm of IoT device.

Before You Start

- Make sure you have logged in to the device.
- Make sure the IoT device is added to NVR/DVR.

Steps

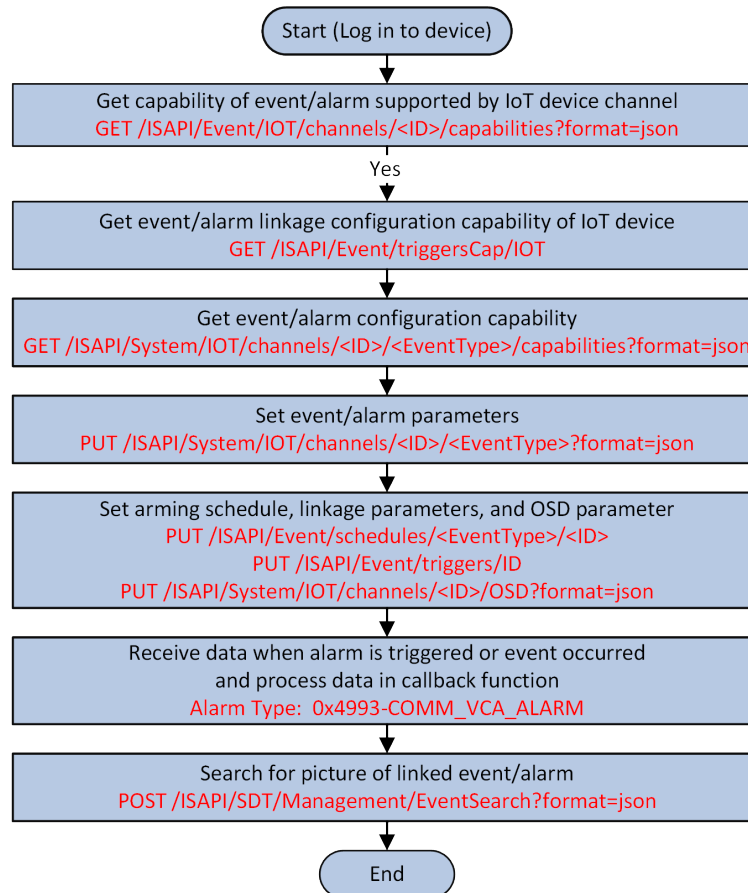


Figure 5-5 Programming Flow of Configuring Alarm of IoT Device

1. Call **`/ISAPI/Event/IOT/channels/<ID>/capabilities?format=json`** by GET method to get the IoT device channel capability.
The event capability supported by IoT device channel is returned in **`JSON_IOTChannelEventCap`**.
2. Call **`/ISAPI/Event/triggersCap/IOT`** by GET method to get the event/alarm linkage configuration capability of IoT device.
The capability is returned in **`XML_IOTTriggersCap`**.
3. Call **`/ISAPI/System/IOT/channels/<ID>/<EventType>/capabilities?format=json`** by GET method to get the event configuration capability.
The event configuration capability is returned in **`JSON_XXCap`**.
4. Call **`/ISAPI/System/IOT/channels/<ID>/<EventType>?format=json`** by PUT method to set the event/alarm parameters.
5. Call **`/ISAPI/Event/schedules/<EventType>/<ID>`** by PUT method to set the arming schedule
6. Call **`/ISAPI/Event/triggers/<ID>`** by PUT method to set the event/alarm linkage parameters.
7. Configure OSD.

- 1) Call **`/ISAPI/System/IOT/channels/<ID>/OSD/capabilities?format=json`** by GET method to get the OSD configuration capability.
 - 2) Call **`/ISAPI/System/IOT/channels/<ID>/OSD?format=json`** by GET method to Get the OSD parameters.
 - 3) Call **`/ISAPI/System/IOT/channels/<ID>/OSD?format=json`** by PUT method to set the OSD parameters.
8. Receive event/alarm information.
- Receive event/alarm in arming mode, see details in ***Receive Alarm/Event in Arming Mode*** .
 - Receive event/alarm in listening mode, see details in ***Receive Alarm/Event in Listening Mode*** .
9. Call **`/ISAPI/SDT/Management/EventSearch?format=json`** by POST method to search for the picture of linked event/alarm.



Note

Before searching for event, you can get the event search capability by calling **`/ISAPI/SDT/Management/EventSearch/capabilities?format=json`** via GET method.

10. **Optional:** Call **`/ISAPI/System/IOT/linkageChannels?format=json`** by POST method to Get linked channel of searched event.

Chapter 6 Device Maintenance

6.1 Basic

Shut Down, Reboot and Restore to Default Settings

Function	Request URI
Shut down device	PUT <i>/ISAPI/System/shutdown?format=json</i>
Reboot device	PUT <i>/ISAPI/System/reboot</i>
Restore device to default settings	PUT <i>/ISAPI/System/factoryReset?mode=</i>

Upgrade

Function	Request URI
Get upgrade capability of peripherals' firmwares	GET <i>/ISAPI/System/AcsUpdate/capabilities</i>
Upgrade device's firmware	PUT or POST <i>/ISAPI/System/updateFirmware</i>
Upgrade slave device's firmware	PUT or POST <i>/ISAPI/System/updateFirmware?type=&id=</i>
Upgrade peripheral module's firmware	PUT or POST <i>/ISAPI/System/updateFirmware?type=&moduleAddress=</i>
Get device upgrading status	GET <i>/ISAPI/System/upgradeStatus</i>

Configuration File

Function	Request URI
Get capability of exporting files from device	GET <i>/ISAPI/System/fileExport/capabilities?format=json</i>
Securely export files from device	POST <i>/ISAPI/System/fileExport?format=json</i>  Note To check if exporting files from device is supported by the device, you can call GET <i>/ISAPI/System/capabilities</i> to get the device capability (<i>XML_DeviceCap</i>). If supports, the node <i><isSptFileExport></i> will be returned.

Algorithm

Function	Request URI
Get algorithm version information of device	GET <i>/ISAPI/System/algorithmsVersion</i>
Get capability of getting algorithm library information	GET <i>/ISAPI/System/algorithmsInfo/capabilities?format=json</i>
Get algorithm library information	GET <i>/ISAPI/System/algorithmsInfo?format=json</i>

6.1.1 Securely Export and Import Configuration Files

You can export configuration files of devices to back up basic device parameters (such as network parameters, stream parameters, etc.) timely, and the configuration files will be encrypted with the encryption key when being exported. If several devices adopt the same configuration, you can import configuration files to configure multiple devices in a batch. The devices will decrypt the encrypted configuration file using the encryption key which ensures that the configuration file cannot be used by unauthorized devices.

Steps

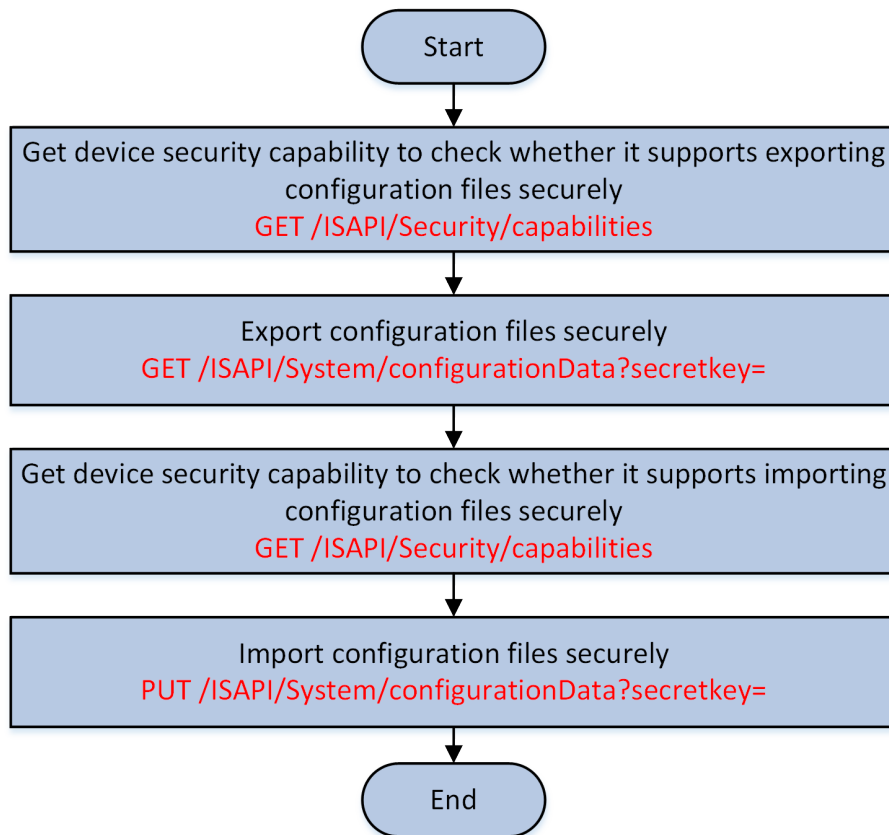


Figure 6-1 Programming Flow of Securely Exporting and Importing Configuration Files

1. Call **/ISAPI/Security/capabilities** by GET method to get the device security capability for checking whether the device supports securely exporting configuration files.
The device security capability is returned in the message **XML_SecurityCap** .
If this function is supported by the device, the node **<isSupportConfigFileExport>** will be returned and its value is "true", then you can perform the following steps.
Otherwise, please end this task.
2. Call **/ISAPI/System/configurationData?secretkey=** by GET method to export the configuration file from the device securely.

Note

The query parameter **secretKey** is the sensitive information that should be encrypted for transmission, and the parameter range to be encrypted is determined by the attributes "min" and "max" of the node **<cfgFileSecretKeyLenLimit>** in message **XML_SecurityCap** . For details about encrypting sensitive information, refer to **Encrypt Sensitive Information** .

3. Call **/ISAPI/Security/capabilities** by GET method to get the device security capability for checking whether the device supports securely importing configuration files.
The device security capability is returned in the message **XML_SecurityCap** .

If this function is supported by the device, the node **<isSupportConfigFileImport>** will be returned and its value is "true", then you can perform the following steps.

Otherwise, please end this task.

4. Call ***/ISAPI/System/configurationData?secretkey=*** by PUT method to import the configuration file to the device securely.



Note

- The query parameter **secretKey** is the sensitive information that should be encrypted for transmission, and the parameter range to be encrypted is determined by the attributes "min" and "max" of the field **<cfgFileSecretKeyLenLimit>** in the message ***XML_SecurityCap*** . For details about encrypting sensitive information, refer to ***Encrypt Sensitive Information*** .
 - The value of **secretKey** for importing the configuration file should be the same as that for exporting the file. Otherwise, importing will fail.
-

6.2 Log

Search and Export Log Files

Function	Request URI
Search for log files	Option1: POST <i>/ISAPI/ContentMgmt/logSearch</i> Option 2: POST <i>/ISAPI/ContentMgmt/logSearch?format=json</i>
Export log files	POST <i>/ISAPI/ContentMgmt/logSearch/dataPackage</i> Note To check whether the device supports exporting log files, you can call <i>/ISAPI/ContentMgmt/capabilities</i> by GET method to get the storage capability (<i>XML_RacmCap</i>). If it supports, the node <isSupportLogDataPackage> is returned in the capability message and its value is "true".

Serial Port Log Redirection

Function	Request URI
Get configuration capability of serial port log redirection	GET <i>/ISAPI/System/serialLogCfg/capabilities?format=json</i>
Get or set parameters of serial port log redirection	GET or PUT <i>/ISAPI/System/serialLogCfg?format=json</i>

Function	Request URI
	 Note To check whether the device supports configuring serial port log redirection, you can call <i>/ISAPI/System/capabilities</i> by GET method to get the device capability (<i>XML_DeviceCap</i>). If it supports, the node <i><isSptSerialLogCfg></i> will be returned and its value is "true".

Log Settings

Function	Request URI
Get log configuration capability	GET <i>/ISAPI/ContentMgmt/logConfig/capabilities</i>
Get or set log parameters	GET or PUT <i>/ISAPI/ContentMgmt/logConfig</i>

6.3 Status

Device Status

Function	Request URI
Get device status capability	GET <i>/ISAPI/System/workingstatus/capabilities?format=json</i>
Get all statuses of device	GET <i>/ISAPI/System/workingstatus?format=json</i>

Channel Status

Function	Request URI
Get status(es) of all channels or one channel	GET or POST <i>/ISAPI/System/workingstatus/chanStatus?format=json</i>
Get statuses of all channels by condition	POST <i>/ISAPI/System/Video/inputs/channels?format=json</i>
Get status of one channel by condition	POST <i>/ISAPI/System/Video/inputs/channels/<ID>?format=json</i>

Note

To check if getting channel status by condition is supported by device, you can call GET ***/ISAPI/System/capabilities*** to get the device capability (***XML_DeviceCap***). If supports, the node ***<isSupportFlexible>*** will be returned.

Others

Function	Request URI
Get status(es) of all HDDs or one HDD	GET or POST <i>/ISAPI/System/workingstatus/hdStatus?format=json</i>
Get triggered I/O status	GET <i>/ISAPI/System/workingstatus/IOStatus?format=json</i>
Get CPU and memory status	GET <i>/ISAPI/System/status</i>
Upload status changes	Refer to Subscribe Alarm/Event in Arming Mode for integration process of uploading device status changes, and the node <type> in the request message <i>XML_SubscribeEvent</i> for the URIs (POST <i>/ISAPI/Event/notification/subscribeEvent</i> and PUT <i>/ISAPI/Event/notification/subscribeEvent/<ID></i>) should be set to "statusChanged".

The changes details can be obtained by calling */ISAPI/Event/notification/alertStream* via GET method.



Note

To check if getting device status and uploading status changes are supported by device, you can call */ISAPI/System/capabilities* by GET method to get the device capability (*XML_DeviceCap*). If supports, the nodes **<isSupportSimpleDevStatus>**, **<isSupportSimpleDevStatus>**, and **<isSupportChangedUpload>** will be returned and their values are "true".

6.4 Calibration

Accelerometer Zero Bias Calibration

Function	Request URI
Get device capability to check if device supports accelerometer zero bias calibration	GET <i>/ISAPI/System/capabilities</i> ; the device capability is returned in the message <i>XML_DeviceCap</i> . If supports, the node <isSupportZeroBiasCalibration> is returned in the message <i>XML_DeviceCap</i> and its value is "true".
Calibrate accelerometer zero bias	PUT <i>/ISAPI/System/zeroBiasCalibration/channels/<ID>?format=json</i>

Installation Angle Calibration

Function	Request URI
Get the device capability to check if device supports installation angle calibration	GET <i>/ISAPI/System/capabilities</i> ; the device capability is returned in the message <i>XML_DeviceCap</i> . If supports, the node <i><isSupportInstallationAngleCalibration></i> is returned in the message <i>XML_DeviceCap</i> and its value is "true".
Get capability of installation angle calibration	GET <i>/ISAPI/System/installationAngleCalibration/channels/<ID>/capabilities?format=json</i>
Get installation angle calibration status or calibrate installation angle	GET or PUT <i>/ISAPI/System/installationAngleCalibration/channels/<ID>?format=json</i>

Laser Optical Axis

Function	Request URI
Get configuration capability of laser optical axis	GET <i>/ISAPI/System/laserOpticalAxis/capabilities?format=json</i>
Get or set laser optical axis parameters	GET or PUT <i>/ISAPI/System/laserOpticalAxis?format=json</i>
Adjust laser optical axis	PUT <i>/ISAPI/System/laserOpticalAxis/goto?format=json</i>

6.5 Auto Power On or Off

Function	Request URI
Get capability of auto power on or off	GET <i>/ISAPI/System/autoSwitch/capabilities?format=json</i>
Get or set parameters of auto power on or off	GET or PUT <i>/ISAPI/System/autoSwitch?format=json</i>

6.6 Advanced

Automatic Maintenance

Function	Request URI
Get configuration capability of automatic maintenance	GET <i>/ISAPI/System/autoMaintenance/capabilities?format=json</i>
Get or set parameters of automatic maintenance	GET or PUT <i>/ISAPI/System/autoMaintenance?format=json</i>

Note

To check if the device supports automatic maintenance, you can call the GET */ISAPI/System/capabilities* to get the device capability (*XML_DeviceCap*). If supports, the node **<isSupportAutoMaintenance>** will be returned.

Device Diagnosis

Function	Request URI
Get device diagnosis capability	GET <i>/ISAPI/System/diagnosis/capabilities?format=json</i>
Diagnose device	POST <i>/ISAPI/System/diagnosis?format=json</i>

Note

To check if configuring device diagnosis is supported by the device, you can call GET */ISAPI/System/capabilities* to get the device capability (*XML_DeviceCap*). If supports, the node **<isSptDiagnosis>** will be returned.

Device Anti-Theft

Function	Request URI
Get device anti-theft configuration capability	GET <i>/ISAPI/System/guardAgainstTheft/capabilities</i>
Get or set device anti-theft parameters	GET or PUT <i>/ISAPI/System/guardAgainstTheft</i>

Note

To check if the anti-theft configuration is supported by the device, you can call GET */ISAPI/System/capabilities* to get the device capability (*XML_DeviceCap*). If this function is supported, the node **<isSupportGuardAgainstTheft>** will be returned and its value is "true".

Chapter 7 User Management

You can add, edit, and delete users to manage for logging in to the device via ISAPI protocol. And you can also assign permission to users according to actual needs.

Manage Users

Function	Request URI
Add one user	GET <i>/ISAPI/Security/users</i>
Get, edit, or delete all users	GET, PUT, or DELETE <i>/ISAPI/Security/users</i>
Get, edit, or delete one user	GET, PUT, or DELETE <i>/ISAPI/Security/users/<ID></i>



Note

The function of managing users cannot be integrated via EZVIZ Open Platform.

User Permission

Function	Request URI
Get guest's permission capability	GET <i>/ISAPI/Security/UserPermission/viewerCap</i>
Get operator's permission capability	GET <i>/ISAPI/Security/UserPermission/operatorCap</i>
Get administrator's permission capability	GET <i>/ISAPI/Security/UserPermission/adminCap</i>
Get User Permission	• Get permissions of all users: GET <i>/ISAPI/Security/UserPermission</i> • Get permission of one user: GET <i>/ISAPI/Security/UserPermission/<ID></i> • Get local permission of one user: GET <i>/ISAPI/Security/UserPermission/<ID>/localPermission</i> • Get remote permission of one user: GET <i>/ISAPI/Security/UserPermission/<ID>/remotePermission</i> • Get permission of anonymous user: GET <i>/ISAPI/Security/UserPermission/anonymouslogin</i>
Assign Permission	• Assign permission to all users: PUT <i>/ISAPI/Security/UserPermission</i> • Assign permission to one user: PUT <i>/ISAPI/Security/UserPermission/<ID></i>

Function	Request URI
	- Assign local permission to one user: PUT <i>/ISAPI/Security/UserPermission/<ID>/localPermission</i> - Assign remote permission to one user: PUT <i>/ISAPI/Security/UserPermission/<ID>/remotePermission</i> - Assign permission to anonymous user: PUT <i>/ISAPI/Security/UserPermission/anonymouslogin</i>



Note

The user permission function cannot be integrated via EZVIZ Open Platform.

Lock and Unlock User

Function	Request URI
Get configuration capability of user locking and unlocking	GET <i>/ISAPI/System/userLock/config/capabilities?format=json</i> .
Get information of all locked users	GET <i>/ISAPI/System/userLock/lockedUsers?format=json</i>
Get user locking parameter	GET <i>/ISAPI/System/userLock/config?format=json</i>
Lock user	PUT <i>/ISAPI/System/userLock/config?format=json</i>
Unlock user	PUT <i>/ISAPI/System/userLock/unlockUser?format=json</i>

7.1 Configure Double Verification

Double verification helps to protect the critical video files of NVR/DVR by limiting playback and download. The basic concept is that two users should always be required to start playback and download. For example, when a normal user A (operator or guest) wants to play back the video of a channel which requires double verification, he/she should ask a double verification user to enter the correct user name and password for double verification.

Before You Start

Make sure you have logged in to the device.

Steps

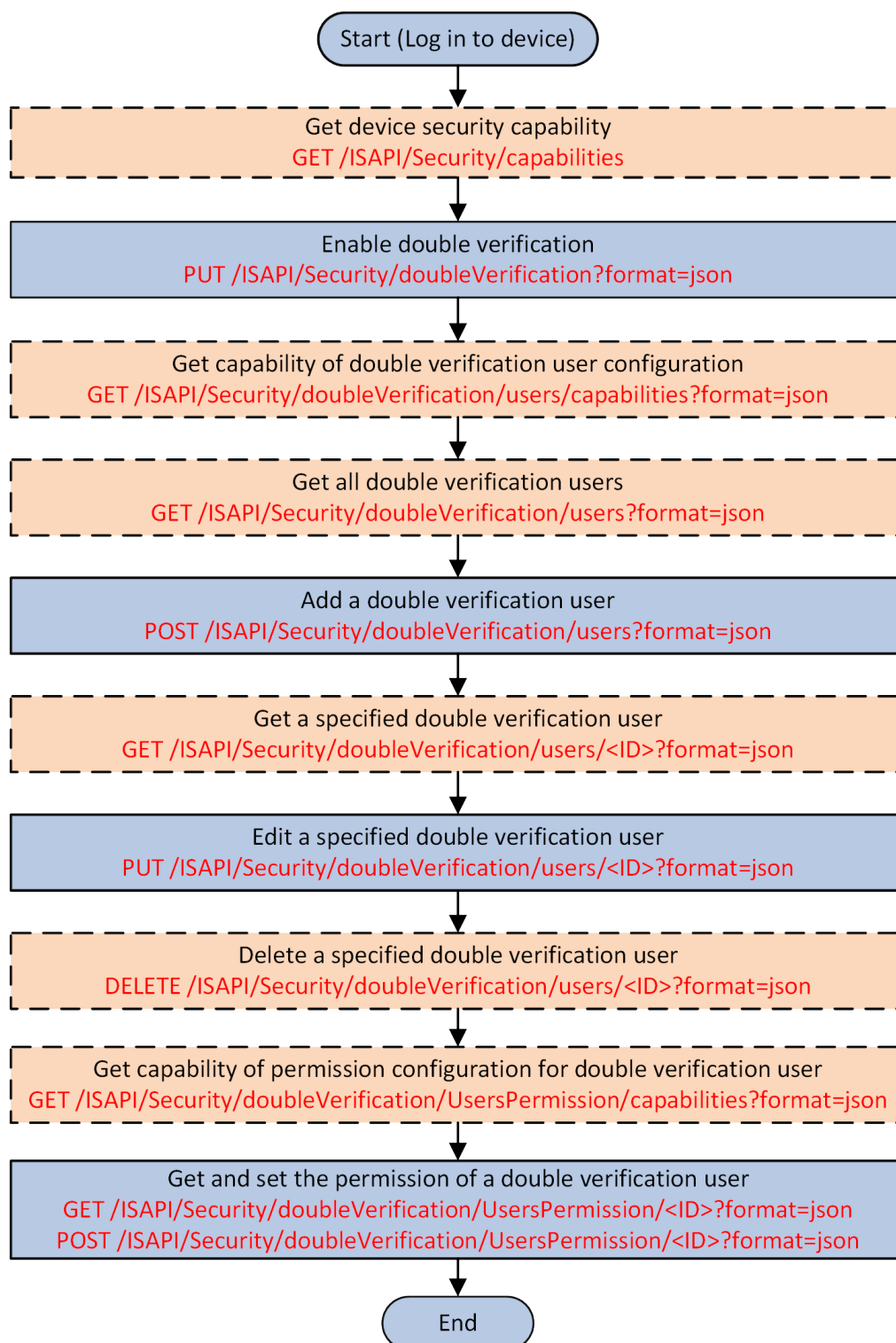


Figure 7-1 Programming Flow of Configuring Double Verification

Note

- Only the admin can configure double verification.
- The admin is not required for double verification.
- The double verification user name and password is only for double verification, and cannot be used for login.

-
1. **Optional:** Call ***/ISAPI/Security/capabilities*** by GET method to get the device security capability to check whether the double verification function is supported.

The security capability ***XML_SecurityCap*** is returned.

2. Call ***/ISAPI/Security/doubleVerification?format=json*** by PUT method to enable double verification



Note

Before enabling double verification, you can check whether double verification is enabled by calling ***/ISAPI/Security/doubleVerification?format=json*** via GET method.

-
3. **Optional:** Call ***/ISAPI/Security/doubleVerification/users/capabilities?format=json*** by GET method to get the capability of double verification user configuration.
 4. Call ***/ISAPI/Security/doubleVerification/users?format=json*** by POST method to add a double verification user.



Note

Before adding a double verification user, you can get all verification users by calling ***/ISAPI/Security/doubleVerification/users?format=json*** via GET method.

The ID of added double verification user is returned in ***JSON_id***.

-
5. Call ***/ISAPI/Security/doubleVerification/users/<ID>?format=json*** by PUT method to edit a specified double verification user.



Note

Before editing the double verification user, you can get the user information by calling ***/ISAPI/Security/doubleVerification/users/<ID>?format=json*** via GET method.

-
6. **Optional:** Call ***/ISAPI/Security/doubleVerification/users/<ID>?format=json*** by DELETE method to delete a specified double verification user.
 7. **Optional:** Call ***/ISAPI/Security/doubleVerification/UsersPermission/capabilities?format=json*** by GET method to get the capability of permission configuration for double verification users.
 8. Call ***/ISAPI/Security/doubleVerification/UsersPermission/<ID>?format=json*** by PUT method to set the permission of a specified double verification user.



Note

Before setting the permission of a double verification user, you can get the permission parameters by calling ***/ISAPI/Security/doubleVerification/UsersPermission/<ID>?format=json*** via GET method.

What to do next

Configure the user permission, determine whether double verification are required for them during playback and download.

Chapter 8 Video and Audio

This part provides the request URIs to implement the video and audio related functions, such as video search, live view, playback, two-way audio, control video in live view (e.g., focus and iris adjustment), and so on.

8.1 Search for Recording Start and End Time by Channel

You can check whether the channel has recorded videos and get the recording start and end time.

Steps

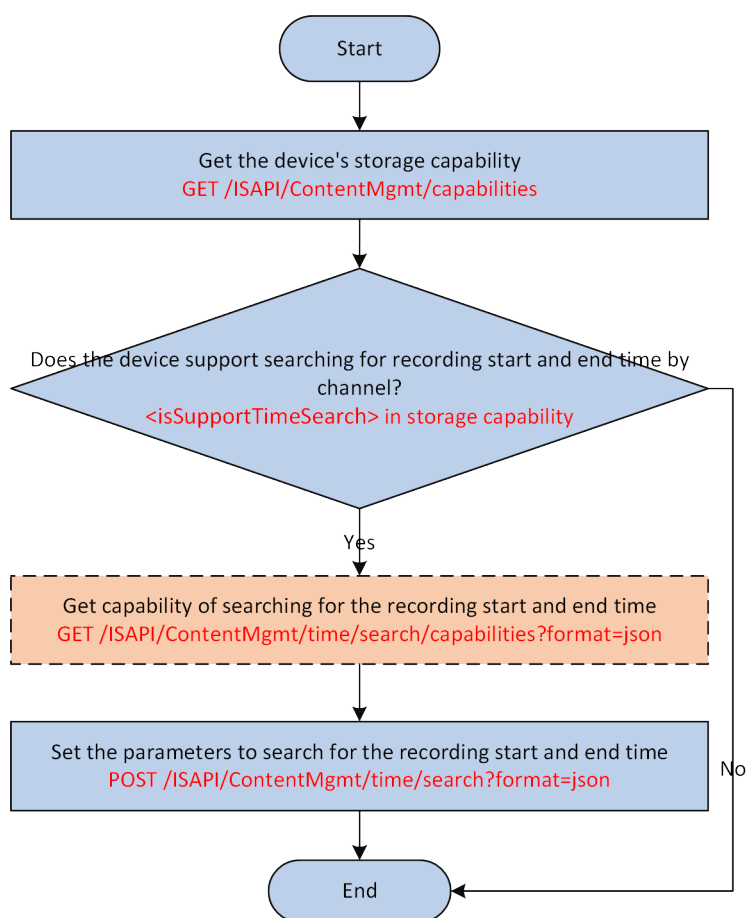


Figure 8-1 API Calling Flow of Searching for Recording Start and End Time by Channel

1. Call `/ISAPI/ContentMgmt/capabilities` by GET method to get the device's storage capability to check whether it supports searching for recording start and end time.

The device's storage capability is returned in the message *XML_RacmCap*.

If the node **<isSupportTimeSearch>** is returned in the message and its value is "true", it indicates that searching for recording start and end time is supported and you can continue to perform the following steps.

Otherwise, searching for recording start and end time is not supported, please end this task.

2. **Optional:** Call ***/ISAPI/ContentMgmt/time/search/capabilities?format=json*** by GET method to get the capability of searching for the recording start and end time for reference.

The capability is returned in the message ***JSON_Cap_TimeSearchCond***.

3. Call ***/ISAPI/ContentMgmt/time/search?format=json*** by POST method to set the parameters to search for the recording start and end time.

8.2 Live View and Playback

Live view helps to remotely get the real-time video or audio of monitoring area and know the status of human, vehicle, object, and so on. During live view, you can record and store the remarkable videos for remotely playback.

Live View

Table 8-1 Normal Operation

Function	Request URI
Start or stop live view	PLAY or TEARDOW

Table 8-2 Auto-Switch in Live View

Function	Request URI
Get configuration capability of live view auto-switch	GET <i>/ISAPI/System/Video/outputs/PreviewSwitch/capabilities</i>
Get configuration capability of live view auto-switch by video output	GET <i>/ISAPI/System/Video/outputs/PreviewSwitch/capabilities?videoOutType=</i>
Get or set parameters of live view auto-switch	GET or PUT <i>/ISAPI/System/Video/outputs/PreviewSwitch?groupID=&videoOutType=</i> GET <i>/ISAPI/System/Video/outputs/PreviewSwitch?groupID=&videoOutType=&previewFrameNo=</i>

Table 8-3 Control in Live View

Function	Request URI
Perform manual focus	PUT <i>/ISAPI/System/Video/inputs/channels/<ID>/focus</i>
Manually adjust iris	PUT <i>/ISAPI/System/Video/inputs/channels/<ID>/iris</i>

Playback

Function	Request URI
Start or stop playback	PLAY or TEARDOW <i>rtsp://<host>[:port]/ISAPI/Streaming/tracks/<ID>?starttime=&endtime=</i>
Refresh video index before playback for instant playback	PUT <i>/ISAPI/ContentMgmt/record/control/manualRefresh/channels/<ID></i>

Example

Sample Code for Starting and Stopping Live View

Start Live View:

DESCRIBE rtsp://10.17.133.46:554/ISAPI/streaming/channels/101 RTSP/1.0

CSeq: 0

Accept:application/sdp

User-Agent:NKPlayer-1.00.00.081112

RTSP/1.0 401 Unauthorized

CSeq: 0

WWW-Authenticate: Digest realm="4419b66d2485", nonce="47f172cffa4ce6e6942bbdcfa5931d0f", stale="FALSE"

Date: Tue, Mar 14 2017 10:42:35 GMT

DESCRIBE rtsp://10.17.133.46:554/ISAPI/streaming/channels/101 RTSP/1.0

CSeq:1

Accept:application/sdp

Authorization: Digest username="admin",

realm="4419b66d2485",

nonce="47f172cffa4ce6e6942bbdcfa5931d0f",

uri="rtsp://10.17.133.46:554/ISAPI/streaming/channels/101",

response="da5deb4033515f74ed0483d2da3f7fe0"

User-Agent:NKPlayer-1.00.00.081112

RTSP/1.0 200 OK

CSeq: 1

Content-Type: application/sdp

Content-Length: 604

v=0

o=- 1489488155469547 1489488155469547 IN IP4 10.17.133.46

s=Media Presentation

e=NONE

b=AS:5100

t=0 0

a=control:*

m=video 0 RTP/AVP 96

c=IN IP4 0.0.0.0

b=AS:5000

a=recvonly

a=x-dimensions:1280,720

a=control:trackID=1

[illegible]

```
Session:846509115;timeout=60
Rate-Control:yes
Scale:1.000
User-Agent:NKPlayer-1.00.00.081112

RTSP/1.0 200 OK
CSeq: 4
Session: 846509115
Scale: 1.0
RTP-Info: url=trackID=1;seq=55143,url=trackID=2;seq=38217
Date: Tue, Mar 14 2017 10:42:35 GMT

Stop Live View
TEARDOWN rtsp://10.17.133.46:554/ISAPI/streaming/channels/101 RTSP/1.0
CSeq: 5
Authorization: Digest username="admin",
realm="4419b66d2485",
nonce="47f172cffa4ce6e6942bbdcfa5931d0f",
uri="rtsp://10.17.133.46:554/ISAPI/streaming/channels/101",
response="31d5e172b87db528aa5b1dc07af3f530"
Session:846509115;timeout=60
User-Agent: NKPlayer-1.00.00.081112

RTSP/1.0 200 OK
CSeq: 5
Session: 846509115
Date: Tue, Mar 14 2017 10:42:36 GMT
```

Example

Sample Code for Starting and Stopping Playback

```
Enable:
DESCRIBE rtsp://10.17.133.46:554/ISAPI/streaming/tracks/101?
starttime=20170313T230652Z&endtime=20170314T025706Z RTSP/1.0
CSeq:0
Accept:application/sdp
User-Agent:NKPlayer-1.00.00.081112

RTSP/1.0 401 Unauthorized
CSeq: 0
WWW-Authenticate: Digest realm="4419b66d2485",
nonce="a0ecd9b1586ff9461f02f910035d0486",
stale="FALSE"
Date: Tue, Mar 14 2017 10:57:23 GMT

DESCRIBE rtsp://10.17.133.46:554/ISAPI/streaming/tracks/101?
starttime=20170313T230652Z&endtime=20170314T025706Z RTSP/1.0
CSeq:1
Accept:application/sdp
Authorization: Digest username="admin",
realm="4419b66d2485",
nonce="a0ecd9b1586ff9461f02f910035d0486",
```

```
uri="rtsp://10.17.133.46:554/ISAPI/streaming/tracks/101?
starttime=20170313T230652Z&endtime=20170314T025706Z",
response="1ebae505abbafd431fb433cec6e85189"
User-Agent:NKPlayer-1.00.00.081112

RTSP/1.0 200 OK
CSeq: 1
Content-Type: application/sdp
Content-Length: 604

v=0
o=- 1489489043122939 1489489043122939 IN IP4 10.17.133.46
s=Media Presentation
e=NONE
b=AS:5100
t=0 0
a=control:*
m=video 0 RTP/AVP 96
c=IN IP4 0.0.0.0
b=AS:5000
a=recvonly
a=x-dimensions:1280,720
a=control:trackID=1
a=rtpmap:96 H264/90000
a=fmtp:96 profile-level-id=420029; packetization-mode=1; sprop-parameter-
sets=Z2QAH6wXKgFAFuhAAAHCAABX5CE=,aP44sA==
m=audio 0 RTP/AVP 8
c=IN IP4 0.0.0.0
b=AS:50
a=recvonly
a=control:trackID=2
a=rtpmap:8 PCMA/8000
a=Media_header:MEDIAINFO=494D4B48010200000400000111710110401F000000FA00000000000000000000000000
0000000000;
a=appversion:1.0

SETUP rtsp://10.17.133.46:554/ISAPI/streaming/tracks/101?
starttime=20170313T230652Z&endtime=20170314T025706Z/trackID=1 RTSP/1.0
CSeq:2
Authorization: Digest username="admin",
realm="4419b66d2485",
nonce="a0ecd9b1586ff9461f02f910035d0486",
uri="rtsp://10.17.133.46:554/ISAPI/streaming/tracks/101?
starttime=20170313T230652Z&endtime=20170314T025706Z",
response="6a7a733b41908bd9549a4ac07c498509"
Transport:RTP/AVP/UDP;unicast;client_port=20056-20057;ssrc=0
User-Agent:NKPlayer-1.00.00.081112

RTSP/1.0 200 OK
CSeq: 2
Session: 2049381566;timeout=60
Transport: RTP/AVP/UDP;unicast;client_port=20056-20057;ssrc=7d3c07bc;server_port=8220-8221;mode="play"
```

Date: Tue, Mar 14 2017 10:57:23 GMT

SETUP rtsp://10.17.133.46:554/ISAPI/streaming/tracks/101?
starttime=20170313T230652Z&endtime=20170314T025706Z/trackID=2 RTSP/1.0
CSeq:3
Authorization: Digest username="admin",
realm="4419b66d2485",
nonce="a0ecd9b1586ff9461f02f910035d0486",
uri="rtsp://10.17.133.46:554/ISAPI/streaming/tracks/101?
starttime=20170313T230652Z&endtime=20170314T025706Z",
response="6a7a733b41908bd9549a4ac07c498509"
Session:2049381566;timeout=60
Transport:RTP/AVP/UDP;unicast;client_port=20058-20059;ssrc=0
User-Agent:NKPlayer-1.00.00.081112

RTSP/1.0 200 OK
CSeq: 3
Session: 2049381566;timeout=60
Transport: RTP/AVP/UDP;unicast;client_port=20058-20059;ssrc=06557745;server_port=8226-8227;mode="play"
Date: Tue, Mar 14 2017 10:57:23 GMT

PLAY rtsp://10.17.133.46:554/ISAPI/streaming/tracks/101?
starttime=20170313T230652Z&endtime=20170314T025706Z RTSP/1.0
CSeq:4
Authorization: Digest username="admin",
realm="4419b66d2485",
nonce="a0ecd9b1586ff9461f02f910035d0486",
uri="rtsp://10.17.133.46:554/ISAPI/streaming/tracks/101?
starttime=20170313T230652Z&endtime=20170314T025706Z",
response="fb986d385a7d839052ec4f0b2b70c631"
Session:2049381566;timeout=60
Range:clock=20170313T230652Z-20170314T025706Z
Rate-Control:yes
Scale:1.000
User-Agent:NKPlayer-1.00.00.081112

RTSP/1.0 200 OK
CSeq: 4
Session: 2049381566
Scale: 1.000
RTP-Info: url=trackID=1;seq=1,url=trackID=2;seq=1
Date: Tue, Mar 14 2017 10:57:23 GMT

Pause:
PAUSE rtsp://10.17.133.46:554/ISAPI/streaming/tracks/101?
starttime=20170313T230652Z&endtime=20170314T025706Z RTSP/1.0
CSeq: 5
Authorization: Digest username="admin",
realm="4419b66d2485",
nonce="a0ecd9b1586ff9461f02f910035d0486",
uri="rtsp://10.17.133.46:554/ISAPI/streaming/tracks/101?
starttime=20170313T230652Z&endtime=20170314T025706Z",

response="d229c4d2419df553212bc7d74bbfd432"

Session:2049381566;timeout=60

User-Agent: NKPlayer-1.00.00.081112

RTSP/1.0 200 OK

CSeq: 5

Session: 2049381566

Date: Tue, Mar 14 2017 10:57:24 GMT

Play in 2× Speed:

PLAY rtsp://10.17.133.46:554/ISAPI/streaming/tracks/101?

starttime=20170313T230652Z&endtime=20170314T025706Z RTSP/1.0

CSeq:6

Authorization: Digest username="admin",

realm="4419b66d2485",

nonce="a0ecd9b1586ff9461f02f910035d0486",

uri="rtsp://10.17.133.46:554/ISAPI/streaming/tracks/101?

starttime=20170313T230652Z&endtime=20170314T025706Z",

response="fb986d385a7d839052ec4f0b2b70c631"

Session:2049381566;timeout=60

Scale:2.000

User-Agent:NKPlayer-1.00.00.081112

RTSP/1.0 200 OK

CSeq: 6

Session: 2049381566

Scale: 2.000

RTP-Info: url=trackID=1;seq=1,url=trackID=2;seq=1

Date: Tue, Mar 14 2017 10:57:24 GMT

Pause:

PAUSE rtsp://10.17.133.46:554/ISAPI/streaming/tracks/101?

starttime=20170313T230652Z&endtime=20170314T025706Z RTSP/1.0

CSeq: 7

Authorization: Digest username="admin",

realm="4419b66d2485",

nonce="a0ecd9b1586ff9461f02f910035d0486",

uri="rtsp://10.17.133.46:554/ISAPI/streaming/tracks/101?

starttime=20170313T230652Z&endtime=20170314T025706Z",

response="d229c4d2419df553212bc7d74bbfd432"

Session:2049381566;timeout=60

User-Agent: NKPlayer-1.00.00.081112

RTSP/1.0 200 OK

CSeq: 7

Session: 2049381566

Date: Tue, Mar 14 2017 10:57:25 GMT

Play in 4× Speed:

PLAY rtsp://10.17.133.46:554/ISAPI/streaming/tracks/101?

starttime=20170313T230652Z&endtime=20170314T025706Z RTSP/1.0

CSeq:8

```
Authorization: Digest username="admin",
realm="4419b66d2485",
nonce="a0ecd9b1586ff9461f02f910035d0486",
uri="rtsp://10.17.133.46:554/ISAPI/streaming/tracks/101?
starttime=20170313T230652Z&endtime=20170314T025706Z",
response="fb986d385a7d839052ec4f0b2b70c631"
Session:2049381566;timeout=60
Scale:4.000
User-Agent:NKPlayer-1.00.00.081112

RTSP/1.0 200 OK
CSeq: 8
Session: 2049381566
Scale: 4.000
RTP-Info: url=trackID=1;seq=1,url=trackID=2;seq=1
Date: Tue, Mar 14 2017 10:57:25 GMT

Disable:
TEARDOWN rtsp://10.17.133.46:554/ISAPI/streaming/tracks/101?
starttime=20170313T230652Z&endtime=20170314T025706Z RTSP/1.0
CSeq: 9
Authorization: Digest username="admin",
realm="4419b66d2485",
nonce="a0ecd9b1586ff9461f02f910035d0486",
uri="rtsp://10.17.133.46:554/ISAPI/streaming/tracks/101?
starttime=20170313T230652Z&endtime=20170314T025706Z",
response="50c86f44780a497ee07622b70c050bcd"
Session:2049381566;timeout=60
User-Agent: NKPlayer-1.00.00.081112

RTSP/1.0 200 OK
CSeq: 9
Session: 2049381566
Date: Tue, Mar 14 2017 10:57:28 GMT
```

8.3 Start Two-Way Audio

Two-way audio function enables the voice talk of the camera. You can get not only the live video but also the real-time audio from the camera. If the device has multiple two-way audio channels, you can select the channel to start two-way audio.

Steps

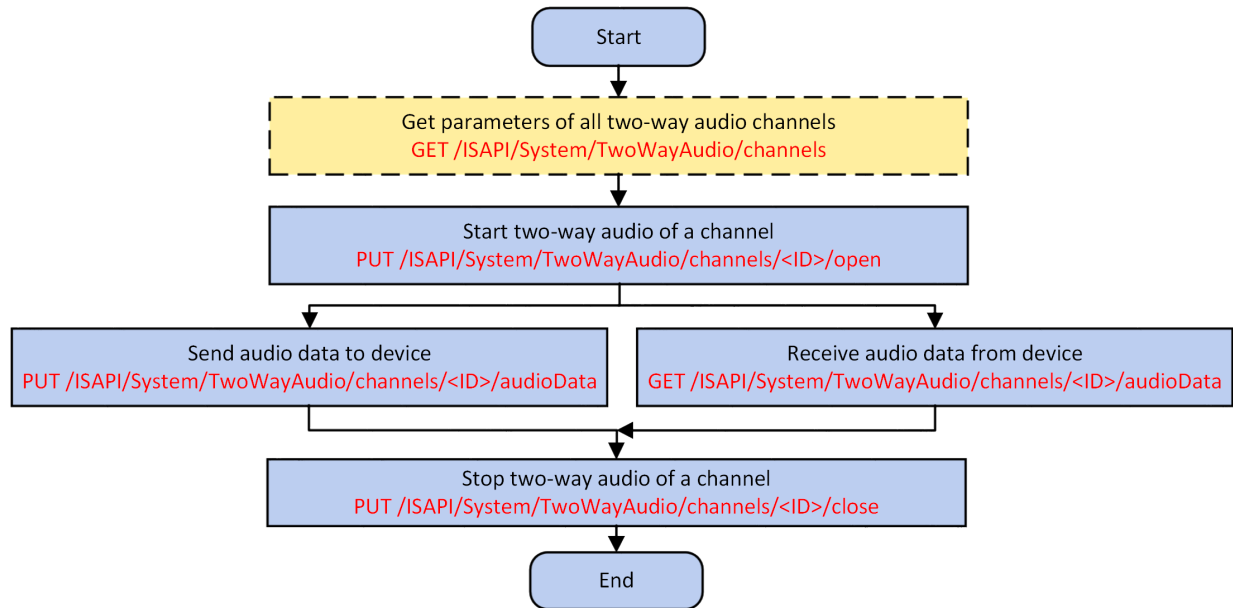


Figure 8-2 Programming Flow of Starting Two-Way Audio

1. **Optional:** Call `/ISAPI/System/TwoWayAudio/channels` by GET method to get parameters, including channel No., encoding mode, and so on, of all two-way audio channels.
2. Call `/ISAPI/System/TwoWayAudio/channels/<ID>/open` by PUT method to specify a channel of device to start the two-way audio.
3. Perform the following operations to transmit the audio data between platform and device.

Send Audio Data to Device	PUT <code>/ISAPI/System/TwoWayAudio/channels/<ID>/audioData</code>
---------------------------	--

Receive Audio Data from Device	GET <code>/ISAPI/System/TwoWayAudio/channels/<ID>/audioData</code>
--------------------------------	--

4. Call `/ISAPI/System/TwoWayAudio/channels/<ID>/close` by PUT method to stop two-way audio of the specific channel.

8.4 Stream Encoding

The following request URIs are used for configuring encoding parameters of video and audio stream by one channel or multiple channels.

Encode Stream of Multiple Channels

Function	Request URI
Get or set encoding parameters of multiple channels	GET or PUT <i>/ISAPI/Streaming/channels</i>
Get encoding status of multiple channels	GET <i>/ISAPI/Streaming/status</i>

Encode Stream of One Channel

Function	Request URI
Get encoding capability of one channel	GET <i>/ISAPI/Streaming/channels/<ID>/capabilities</i>
Get dynamic encoding capability of one channel	GET <i>/ISAPI/Streaming/channels/<ID>/dynamicCap</i> .
Get or set encoding parameters of one channel	GET or PUT <i>/ISAPI/Streaming/channels/<ID></i>
Get encoding status of one channel	GET <i>/ISAPI/Streaming/channels/<ID>/status</i>

Chapter 9 Capture and Recording

For some remarkable views in the live view, you can manually capture the views or record the video segments and save the captured pictures or videos to the configured storage. For recording, you can also configure recording schedule to record videos continuously or based on command or event.

Capture

Function	Request URI
Capture picture (available for front-end devices)	GET <i>/ISAPI/Streaming/channels/<ID>/picture</i>
Capture picture (available for rear-end devices)	GET <i>/ISAPI/ContentMgmt/StreamingProxy/channels/<ID>/Picture</i>

Recording

Table 9-1 Normal Operation

Function	Request URI
Get video track type	GET <i>/ISAPI/ContentMgmt/record/profile</i>
Set storage for recorded files	PUT <i>/ISAPI/ContentMgmt/record/storageMounts</i>  Note To check if setting recording storage is supported, you can call the GET <i>/ISAPI/ContentMgmt/capabilities</i> to get the device storage management capability (<i>XML_RacmCap</i>).

Table 9-2 Recording Schedule

Function	Request URI
Get configuration capability of a recording schedule	GET <i>/ISAPI/ContentMgmt/record/tracks/<ID>/capabilities</i>
Get all recording schedules	GET <i>/ISAPI/ContentMgmt/record/tracks</i>
Add a recording schedule	POST <i>/ISAPI/ContentMgmt/record/tracks</i>
Set a recording schedule	PUT <i>/ISAPI/ContentMgmt/record/tracks/<ID></i>

Table 9-3 Manual Recording

Function	Request URI
Start manual recording	POST <i>/ISAPI/ContentMgmt/record/control/manual/start/tracks/<ID></i>
Stop manual recording	POST <i>/ISAPI/ContentMgmt/record/control/manual/stop/tracks/<ID></i>

Chapter 10 PTZ Control

PTZ control is to implement the panning, tilting, zoom functions of cameras, set presets for specific monitoring areas to fast switch the cameras as needed, and group the user-defined presets as a scanning track for patrol.

Basic PTZ Parameters

Function	Request URI
Get PTZ control capability	GET <i>/ISAPI/PTZCtrl/channels/<ID>/capabilities</i>
Get or set zoom/focus coordinate parameters	GET or PUT <i>/ISAPI/PTZCtrl/channels/<ID>/zoomFocus</i>
Maximum Tilt Angle	- Get configuration capability of maximum tilt angle: GET <i>/ISAPI/PTZCtrl/channels/<ID>/maxelevation/capabilities</i> - Get or set parameters of maximum tilt angle: GET or PUT <i>/ISAPI/PTZCtrl/channels/<ID>/maxelevation</i>
Get PTZ position information of current channel	GET <i>/ISAPI/PTZCtrl/channels/<ID>/save</i>
Set PTZ position information of current channel	PUT <i>/ISAPI/PTZCtrl/channels/<ID>/save?format=json</i>
Get PTZ status of a specific channel	GET <i>/ISAPI/PTZCtrl/channels/<ID>/status</i>
Preset Configuration	- Get all presets' parameters: GET <i>/ISAPI/PTZCtrl/channels/<ID>/presets</i> - Set all presets' parameters: PUT <i>/ISAPI/PTZCtrl/channels/<ID>/presets</i> - Added a preset: POST <i>/ISAPI/PTZCtrl/channels/<ID>/presets</i> - Delete all presets: DELETE <i>/ISAPI/PTZCtrl/channels/<ID>/presets</i> - Get a preset's parameters: GET <i>/ISAPI/PTZCtrl/channels/<ID>/presets/<ID></i> - Set a preset's parameters: PUT <i>/ISAPI/PTZCtrl/channels/<ID>/presets/<ID></i> - Delete a preset: DELETE <i>/ISAPI/PTZCtrl/channels/<ID>/presets/<ID></i>
Get parking configuration capability	GET <i>/ISAPI/PTZCtrl/channels/<ID>/parkaction/capabilities</i>
Get or set parking parameters	GET or PUT <i>/ISAPI/PTZCtrl/channels/<ID>/parkaction</i>

Other PTZ Parameters

Function	Request URI
Lens	- Initialize lens: PUT <i>/ISAPI/PTZCtrl/channels/<ID>/onepushfoucs/reset</i> - Get lens correction capability: GET <i>/ISAPI/PTZCtrl/channels/<ID>/lensCorrection/capabilities?format=json</i> - Get or set lens correction parameters: GET or PUT <i>/ISAPI/PTZCtrl/channels/<ID>/lensCorrection?format=json</i>  Note To check if the lens correction is supported by device, you can call the GET <i>/ISAPI/PTZCtrl/channels/<ID>/capabilities</i> to get the PTZ control capability (<i>XML_PTZChanelCap</i>). If supports, the node <i><isSupportLensCorrection></i> will be returned.
Get or set PTZ locking time duration	GET or PUT <i>/ISAPI/PTZCtrl/channels/<ID>/lockPTZ</i>  Note You can call the GET <i>/ISAPI/PTZCtrl/channels/<ID>/capabilities</i> to get the supported locking time duration (<i><lockTime></i>) from the PTZ control capability (<i>XML_PTZChanelCap</i>).
Get or set PTZ OSD parameters	GET or PUT <i>/ISAPI/PTZCtrl/channels/<ID>/PTZOSDDisplay</i>
Get or set moving object tracking status	GET or PUT <i>/ISAPI/PTZCtrl/channels/<ID>/moveAutoTracking?format=json</i>
Get configuration capability of tracking parameters	GET <i>/ISAPI/PTZCtrl/channels/<ID>/trackingRatio/capabilities?format=json</i>
Get or set tracking parameters	GET or PUT <i>/ISAPI/PTZCtrl/channels/<ID>/trackingRatio?format=json</i>

PTZ Auxiliaries Control

Function	Request URI
Get or set all PTZ auxiliaries status	GET or PUT <i>/ISAPI/PTZCtrl/channels/<ID>/auxcontrols</i>
Get or set the status of single PTZ auxiliary	GET or PUT <i>/ISAPI/PTZCtrl/channels/<ID>/auxcontrols/<ID></i>

E-PTZ Parameters

Function	Request URI
Get or set e-PTZ parameters	GET or PUT <i>/ISAPI/Image/channels/<ID>/EPTZ</i>
Get capability of switching e-PTZ mode	GET <i>/ISAPI/Image/channels/<ID>/EPTZ/mode/capabilities?format=json</i>
Get or set parameters of switching e-PTZ mode	GET or PUT <i>/ISAPI/Image/channels/<ID>/EPTZ/mode?format=json</i>

Gyroscope Calibration

Function	Request URI
Get gyroscope attitude calibration	GET <i>/ISAPI/PTZCtrl/channels/<ID>/gyroVerify/capabilities?format=json</i>
Set gyroscope attitude calibration parameters	PUT <i>/ISAPI/PTZCtrl/channels/<ID>/gyroVerify?format=json</i>

GPS Calibration

Function	Request URI
Get capability of verifying GPS calibration	GET <i>/ISAPI/System/GPSVerification/channels/<ID>/points/capabilities?format=json</i>
Get GPS calibration configuration capability	GET <i>/ISAPI/System/GPSCalibration/channels/<ID>/capabilities?format=json</i>
Configure a single GPS calibration point.	- Get parameters of a single GPS calibration point: GET <i>/ISAPI/System/GPSCalibration/channels/<ID>/points/<PID>?format=json</i> - Set a single GPS calibration point: PUT <i>/ISAPI/System/GPSCalibration/channels/<ID>/points/<PID>?format=json</i> - Delete a single GPS calibration point: DELETE <i>/ISAPI/System/GPSCalibration/channels/<ID>/points/<PID>?format=json</i>
Configure multiple GPS calibration points	- Get parameters of multiple GPS calibration points in a batch: GET <i>/ISAPI/System/GPSCalibration/channels/<ID>/points?format=json</i> - Set multiple GPS calibration points in a batch:

Function	Request URI
	PUT <i>/ISAPI/System/GPSCalibration/channels/<ID>/points?format=json</i> • Delete multiple GPS calibration points in a batch DELETE <i>/ISAPI/System/GPSCalibration/channels/<ID>/points?format=json</i>
Enable GPS calibration	PUT <i>/ISAPI/System/GPSPTZCalibration/channels/<ID>/calibration?format=json</i>
Get capability of verifying GPS calibration	GET <i>/ISAPI/System/GPSVerification/channels/<ID>/points/capabilities?format=json</i>
Configure multiple verification points	• Add a verification points: POST <i>/ISAPI/System/GPSVerification/channels/<ID>/points?format=json</i> • Get information of multiple verification points: GET <i>/ISAPI/System/GPSVerification/channels/<ID>/points?format=json</i> • Set multiple verification points: PUT <i>/ISAPI/System/GPSVerification/channels/<ID>/points?format=json</i> • Delete multiple verification points: DELETE <i>/ISAPI/System/GPSVerification/channels/<ID>/points?format=json</i>
Configure one specified verification point	• Get information of one specified verification point: GET <i>/ISAPI/System/GPSVerification/channels/<ID>/points/<ID>?format=json</i> • Set information of one specified verification point: PUT <i>/ISAPI/System/GPSVerification/channels/<ID>/points/<ID>?format=json</i> • Delete one specified verification point: DELETE <i>/ISAPI/System/GPSVerification/channels/<ID>/points/<ID>?format=json</i>
Get GPS calibration verification results in a batch	GET <i>/ISAPI/System/GPSVerification/channels/<ID>/points/resultInfo?format=json</i>
Get or set parameters of displaying GPS calibration verification result	GET or PUT <i>/ISAPI/System/GPSVerification/channels/<ID>/display?format=json</i>

10.1 Configure Master-Slave Tracking

The master-slave tracking system is formed by one fixed camera and one PTZ camera. The PTZ camera smartly tracks multiple moving targets, and the fixed camera offers the panoramic view. The master-slave tracking system provides detection and tracking of moving targets, and multiple targets can be tracked simultaneously. The system mainly applies to major industrial infrastructures requiring perimeter prevention, such as airport, prison, railway, museum, enterprise, high-end villas, etc.

Steps

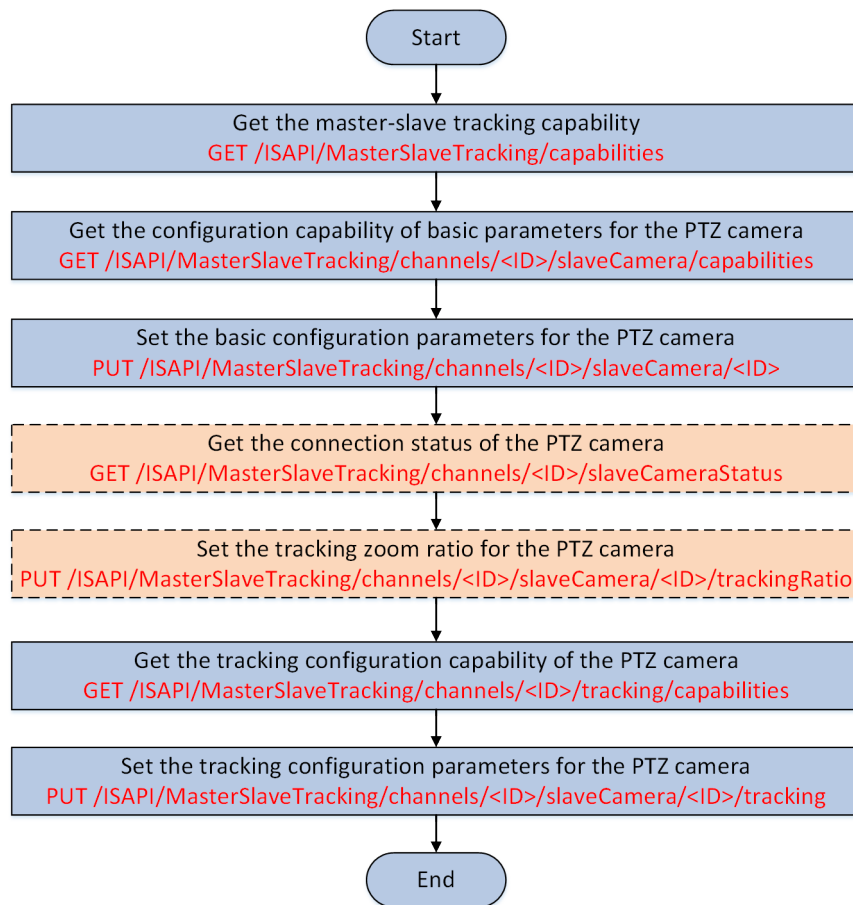


Figure 10-1 API Calling Flow of Configuring Master-Slave Tracking

1. Call **`/ISAPI/MasterSlaveTracking/capabilities`** by GET method to get the master-slave tracking capability to know the supported configurations.
2. Call **`/ISAPI/MasterSlaveTracking/channels/<ID>/slaveCamera/capabilities`** by GET method to get the configuration capability of basic parameters for the PTZ camera to know the basic parameters that can be configured.
3. Call **`/ISAPI/MasterSlaveTracking/channels/<ID>/slaveCamera/<ID>`** by PUT method to set the basic configuration parameters for the PTZ camera.

4. **Optional:** Call ***/ISAPI/MasterSlaveTracking/channels/<ID>/slaveCameraStatus*** by GET method to get the connection status of the PTZ camera.
5. **Optional:** Call ***/ISAPI/MasterSlaveTracking/channels/<ID>/slaveCamera/<ID>/trackingRatio*** by PUT method to set the tracking zoom ratio for the PTZ camera.
6. Call ***/ISAPI/MasterSlaveTracking/channels/<ID>/tracking/capabilities*** by GET method to get the tracking configuration capability of the PTZ camera to know the tracking parameters that can be configured.
7. Call ***/ISAPI/MasterSlaveTracking/channels/<ID>/slaveCamera/<ID>/tracking*** by PUT method to set the tracking configuration parameters for the PTZ camera.

10.2 Configure Panorama Tracking

Tracking Takeover and Linked Tracking Capture are two mutually exclusive modes for panorama tracking.

Linked Tracking Capture Parameters

Function	Request URI
Get capability of linked tracking capture configuration	GET <i>/ISAPI/MasterSlaveTracking/linkedTracking/capabilities?format=json</i>
Get or set linked tracking capture parameters	GET or PUT <i>/ISAPI/MasterSlaveTracking/linkedTracking?format=json</i>

Chapter 11 Storage Management

11.1 Configure Log Storage Mode

You can save the log files in default mode or custom mode. In the default mode, each HDD can save up to 64 MB log files, and the storage space will automatically be overwritten when the space usage reaches 64 MB. In the custom mode, it is available to specify HDDs to save log files.

Steps

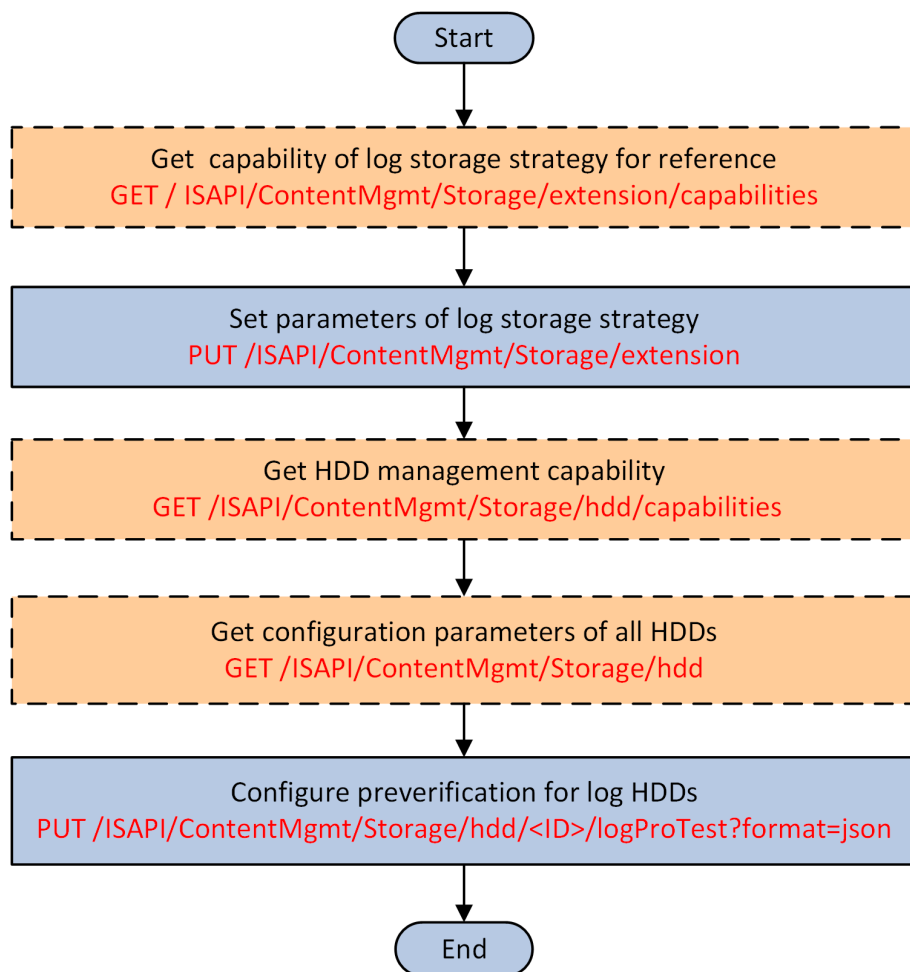


Figure 11-1 Programming Flow of Configuring Log Storage Mode

- Optional:** Call `/ISAPI/ContentMgmt/Storage/extension/capabilities` by GET method to get the capability of log storage strategy for reference.
The capability is returned in the message `XML_Cap_storageExtension`.
- Call `/ISAPI/ContentMgmt/Storage/extension` by PUT method, and set the node `<logStorageMode>` as "system" (system default mode) or "custom" (custom mode) in the message `XML_storageExtension`.

3. **Optional:** Call ***/ISAPI/ContentMgmt/Storage/hdd/capabilities*** by GET method to get the HDD management capability.
The capability is returned in the message ***XML_Cap_hddList*** .
4. **Optional:** Call ***/ISAPI/ContentMgmt/Storage/hdd*** by GET method to get configuration parameters of all HDDs.
The parameters are returned in the message ***XML_hddList*** .
5. Call ***/ISAPI/ContentMgmt/Storage/hdd/<ID>/logProTest?format=json*** by PUT method to configure preverification for log HDDs

11.2 HDD Management

The HDD (Hard Disk Drive) is used for storage. Before storing information in HDD, you must format it, and you can also reallocate its space for better storage management.

HDD Configuration

Function	Request URI
Get HDD management capability	GET <i>/ISAPI/ContentMgmt/Storage/hdd/capabilities</i>
Get parameters of all HDDs	GET <i>/ISAPI/ContentMgmt/Storage/hdd</i>
Get or set parameters of one HDD	GET or PUT <i>/ISAPI/ContentMgmt/Storage/hdd/<ID></i>
Get parameters of all HDD quotas	GET <i>/ISAPI/ContentMgmt/Storage/quota</i>
Verify HDD encryption password	PUT <i>/ISAPI/ContentMgmt/Storage/hdd/<ID>/encryptVerify?format=json</i>

HDD Formatting

Function	Request URI
Format all HDDs	PUT <i>/ISAPI/ContentMgmt/Storage/hdd/format</i>
Specify formatting type to format one HDD	PUT <i>/ISAPI/ContentMgmt/Storage/hdd/<ID>/format?formatType=</i>
Format multiple specified HDDs	PUT <i>/ISAPI/ContentMgmt/Storage/hdd/specifyHddFormat?format=json</i>

Function	Request URI
Format an encrypted HDD	PUT <i>/ISAPI/ContentMgmt/Storage/hdd/<ID>/encryptFormat?format=json</i>
Get formatting status of one HDD	GET <i>/ISAPI/ContentMgmt/Storage/hdd/<ID>/formatStatus</i>

HDD Clearing

Function	Request URI
Get HDD clearing capability	GET <i>/ISAPI/ContentMgmt/storage/hdd/clearingSpace/capabilities?format=json</i>
Get or set HDD clearing parameters	GET or PUT <i>/ISAPI/ContentMgmt/storage/hdd/clearingSpace?format=json</i>

HDD Checking

Function	Request URI
Get or set parameters for checking HDD status	GET or PUT <i>/ISAPI/ContentMgmt/Storage/hdd/SMARTTest/config</i>
Start checking HDD status	PUT <i>/ISAPI/ContentMgmt/Storage/hdd/<ID>/SMARTTest/start</i>
Get HDD checking status	GET <i>/ISAPI/ContentMgmt/Storage/hdd/<ID>/SMARTTest/status</i>
Start checking bad sectors of a HDD	PUT <i>/ISAPI/ContentMgmt/Storage/hdd/<ID>/BadSectorsTest/start</i>
Get bad sector checking status of a HDD	GET <i>/ISAPI/ContentMgmt/Storage/hdd/<ID>/BadSectorsTest/status</i>
Pause checking bad sectors of a HDD	PUT <i>/ISAPI/ContentMgmt/Storage/hdd/<ID>/BadSectorsTest/pause</i>
Resume checking bad sectors of a HDD	PUT <i>/ISAPI/ContentMgmt/Storage/hdd/<ID>/BadSectorsTest/resume</i>
Stop checking bad sectors of a HDD	PUT <i>/ISAPI/ContentMgmt/Storage/hdd/<ID>/BadSectorsTest/stop</i>

SHM (SkyHawk Health Management) Alarm Linkage

Function	Request URI
Get or set linkage parameters of HDD high temperature detection	GET or PUT <i>/ISAPI/Event/triggers/highHDTemperature</i>
Get or set linkage parameters of HDD low temperature detection	GET or PUT <i>/ISAPI/Event/triggers/lowHDTemperature</i>
Get or set linkage parameters of HDD impact detection	GET or PUT <i>/ISAPI/Event/triggers/hdImpact</i>
Get or set linkage parameters of HDD bad sector detection	GET or PUT <i>/ISAPI/Event/triggers/hdBadBlock</i>
Get or set linkage parameters of HDD severe fault detection	GET or PUT <i>/ISAPI/Event/triggers/severeHDFailure</i>

11.3 Others**Table 11-1 APIs about Additional Information**

Function	API
Get storage capability of additional information	GET <i>/ISAPI/ContentMgmt/Storage/ExtraInfo/capabilities</i>
Get or set storage parameters of additional information	GET or PUT <i>/ISAPI/ContentMgmt/Storage/ExtraInfo</i>

Chapter 12 File Management

Table 12-1 Download Files

Function	Request URI
Get downloading capability	GET <i>/ISAPI/ContentMgmt/download/capabilities</i>
Download files	POST <i>/ISAPI/ContentMgmt/download</i>
Get capability of exporting video or picture files to devices via USB	GET <i>/ISAPI/ContentMgmt/download/toUSB/capabilities?format=json</i>  Note To check whether this function is supported by the device, you can call <i>/ISAPI/ContentMgmt/download/capabilities</i> by GET method to get the downloading capability (<i>XML_DownloadAbility</i>). If it supports, the node <i><isSupportDownloadToUSB></i> will be returned and its value is "true".
Export video or picture files to devices via USB	POST <i>/ISAPI/ContentMgmt/download/toUSB?format=json</i>
Get progress of exporting video or picture files to devices via USB	GET <i>/ISAPI/ContentMgmt/download/toUSB/<taskId>/progress?format=json</i>

12.1 Video Files

Search Video Files

Function	Request URI
Get search conditions	GET <i>/ISAPI/ContentMgmt/search/profile</i>
Search video files	POST <i>/ISAPI/ContentMgmt/search</i>  Note This URI is not only used to search video files, it can also be used to search any resources stored in the device.
Search video files by time	POST <i>/ISAPI/ContentMgmt/record/tracks/<ID>/dailyDistribution</i>

Function	Request URI
Get capability of searching video files by VCA event	GET <i>/ISAPI/ContentMgmt/SmartSearch/capabilities</i>
Search video files by VCA event	POST <i>/ISAPI/ContentMgmt/SmartSearch</i>

Lock or Unlock Video Files

Function	Request URI
Get video file locking and unlocking capability	GET <i>/ISAPI/ContentMgmt/record/control/locks/capabilities</i>
Lock or unlock video files by name	PUT <i>/ISAPI/ContentMgmt/record/control/locks/name</i>

Add Remarks to Video File

Function	Request URI
Get capability of adding remarks to video files	GET <i>/ISAPI/ContentMgmt/record/control/remark/capabilities?format=json</i>
Add remarks to video files	PUT <i>/ISAPI/ContentMgmt/record/control/remark?format=json</i>



Note

To check whether the device supports adding remarks to the video files, you can call */ISAPI/ContentMgmt/capabilities* by GET method to get the device storage capability (*XML_RacmCap*). If it supports, the node **<isSupportRemark>** will be returned and its value is "true".

Clear Stored Video Files

Function	Request URI
Get the capability of deleting files in a batch	GET <i>/ISAPI/ContentMgmt/record/control/cleanUp/capabilities?format=json</i>
Delete files in a batch	POST <i>/ISAPI/ContentMgmt/record/control/cleanUp?format=json</i>

12.2 Audio Files

Custom Audio File Operations

Function	Request URI
Check whether custom audio operation is supported	Call <i>/ISAPI/Event/triggers/notifications/AudioAlarm/capabilities?format=json</i> by GET method. The capability is returned in <i>JSON_AudioAlarmCap</i> and if "true" is returned in the node <i>isSupportCustomAudio</i> , it indicates that custom audio operation is supported.
Download or delete custom audio file	Call <i>/ISAPI/Event/triggers/notifications/AudioAlarm/customAudio/<customAudioID>?format=json</i> by GET or DELETE method.
Get or edit custom audio file information	Call <i>/ISAPI/Event/triggers/notifications/AudioAlarm/customAudioInfo/<customAudioID>?format=json</i> by GET or PUT method.
Import custom audio file	Call <i>/ISAPI/Event/triggers/notifications/AudioAlarm/customAudio?format=json</i> by POST method.

Audio Test

Function	Request URI
Check whether audio test is supported	Call <i>/ISAPI/Event/triggers/notifications/AudioAlarm/capabilities?format=json</i> by GET method. The capability is returned in <i>JSON_AudioAlarmCap</i> and if "true" is returned in the node <i>isSupportAudioTest</i> , it indicates that audio test is supported.
Audio test	Call <i>/ISAPI/Event/triggers/notifications/AudioAlarm/<audioID>/test?format=json</i> by PUT method.

Chapter 13 Alarm/Event

When the alarm is triggered or the event occurred, if you have configured alarm/event uploading parameters, you can receive and process the alarm/event information in the third-party platform or system by calling the request URLs of ISAPI. Two modes are available for receiving alarms, including arming mode and listening mode.

Arming Mode

When the alarm is triggered or event occurred, the third-party platform or system can send the request URL to the device for getting the alarm/event stream, and then the device uploads the response message with alarm/event information.

Listening Mode

When alarm is triggered or event occurred, the device uploads the alarm information automatically, and then the third-party platform or system can receives the alarm/event by configuring listening port of HTTP host server.

13.1 Configure Exception Alarm

To monitor the device status, you can configure the exception alarm, such as supply voltage exception, PoE power exception, and so on. When the exception occurs, the configured linkage action will be triggered and the alarm information will be uploaded automatically.

Supply Voltage Exception

Function	Request URI
Get configuration capability of alarm linkage action	GET <i>/ISAPI/Event/triggersCap</i>
Get or set the linkage action of supply voltage exception alarm	GET or PUT <i>/ISAPI/Event/triggers/<ID></i> , and the <ID> in the URI is "voltageinstable"
Receive supply voltage exception alarm in listening mode	Refer to <i>Receive Alarm/Event in Listening Mode</i> for details, the event type is "voltageinstable", and the alarm information is returned in message <i>JSON_EventNotificationAlert_voltageinstable</i>



Note

To check if the device supports supply voltage exception alarm, you can call */ISAPI/Event/capabilities* by GET method to get the device event capability (*XML_EventCap*). If supports, the node **<isSupportVoltageInstable>** will be returned.

13.2 Configure Certificate Expiry Alarm

Generally, the device certificate is only valid in a specific period of time. You can configure the certificate expiry alarm to remind the user a few days in advance. When the certificate is expired, the alarm will be triggered and uploaded automatically.

Steps

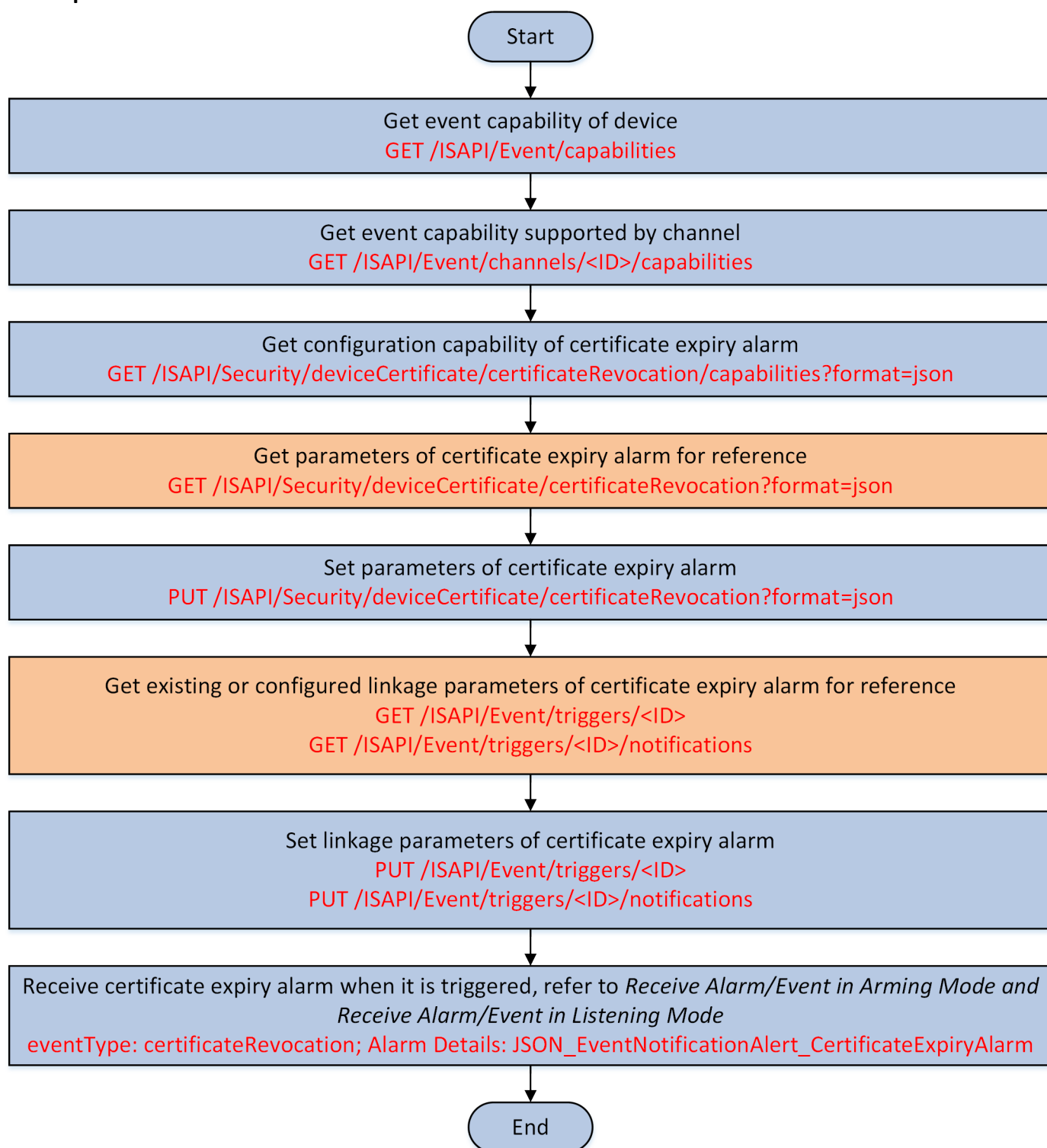


Figure 13-1 Programming Flow of Configuring Certificate Expiry Alarm

1. Call ***/ISAPI/Event/capabilities*** by GET method to get the event capability of the device for checking whether the device supports certificate expiry alarm.
The event capability of the device is returned in the message ***XML_EventCap*** . If the certificate expiry alarm is supported, the node ***<isSupportCertificateRevocation>*** will be returned and its value is "true", then you can perform the following steps. Otherwise, please end this task.
2. Call ***/ISAPI/Event/channels/<ID>/capabilities*** by GET method to get the event capability of the channel for checking whether the certificate expiry alarm is supported by the channel.
The event capability supported by the channel is returned in the message ***XML_ChannelEventCap*** . If the certificate expiry alarm is supported, the node ***<eventType>*** will be returned and it contains "certificateRevocation".
3. Call ***/ISAPI/Security/deviceCertificate/certificateRevocation/capabilities?format=json*** by GET method to get the configuration capability of certificate expiry alarm to know the supported parameters of certificate expiry alarm that can be configured.
4. **Optional:** Call ***/ISAPI/Security/deviceCertificate/certificateRevocation?format=json*** by GET method to get the parameters of certificate expiry alarm to get the existing or configured parameters for reference.
5. Call ***/ISAPI/Security/deviceCertificate/certificateRevocation?format=json*** by PUT method to set the parameters of certificate expiry alarm .
6. **Optional:** Call ***/ISAPI/Event/triggers/<ID>*** or ***/ISAPI/Event/triggers/<ID>/notifications*** by GET method to get the linkage parameters of certificate expiry alarm to get the existing or configured parameters for reference.



Note

The ***<ID>*** in the request URI refers to the channel No., and it should be set to the format "certificateRevocation-***<channelID>***".

7. Call ***/ISAPI/Event/triggers/<ID>*** or ***/ISAPI/Event/triggers/<ID>/notifications*** by PUT method to Set the linkage parameters of certificate expiry alarm.



Note

The ***<ID>*** in the request URI refers to the channel No., and it should be set to the format "certificateRevocation-***<channelID>***".

8. **Optional:** Receive the uploaded certificate expiry alarm in arming mode (see details in ***Receive Alarm/Event in Arming Mode***) or in listening mode (see details in ***Receive Alarm/Event in Listening Mode***) when the certificate is expired.

The certificate expiry alarm details are returned in the message ***JSON_EventNotificationAlert_CertificateExpiryAlarmMsg*** , and the corresponding alarm type (***eventType***) is "certificateRevocation".

13.3 Configure Network Traffic Pre-alarm

You can set or search for data plans, monitor data usage, and set a traffic limit. When the limit is exceeded, the alarm will be triggered and uploaded automatically.

Steps

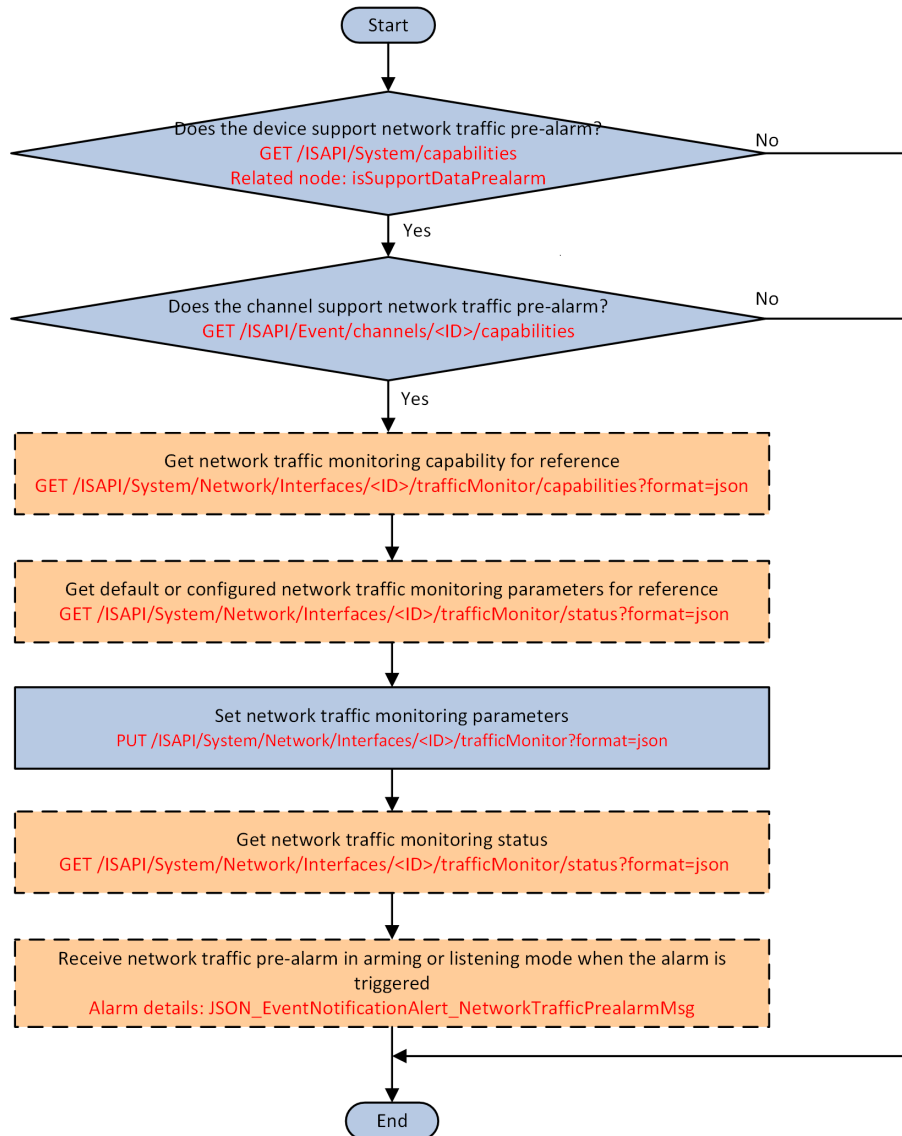


Figure 13-2 Programming Flow of Configuring Network Traffic Pre-alarm

1. Call `/ISAPI/System/capabilities` by GET to check whether the device supports network traffic pre-alarm.
2. Call `/ISAPI/Event/channels/<ID>/capabilities` by GET to check whether the channel supports network traffic pre-alarm.
3. **Optional:** Call `/ISAPI/System/Network/WirelessDial/Interfaces/<ID>/trafficMonitor/capabilities?format=json` by GET to get network traffic monitoring capability for reference.

The capability is returned in the message *JSON_TrafficMonitorCap* .

4. **Optional:** Call */ISAPI/System/Network/WirelessDial/Interfaces/<ID>/trafficMonitor?format=json* by GET to get default or configured network traffic monitoring parameters for reference.

The parameters are returned in the message *JSON_TrafficMonitor* .

5. Call */ISAPI/System/Network/WirelessDial/Interfaces/<ID>/trafficMonitor?format=json* by PUT to set parameters of network traffic monitoring.
6. **Optional:** Call */ISAPI/System/Network/WirelessDial/Interfaces/<ID>/trafficMonitor/status?format=json* by GET to get network traffic monitoring status.

The status is returned in the message *JSON_TrafficMonitorStatus* .

7. **Optional:** Receive the uploaded network traffic alarm in arming mode (see details in *Receive Alarm/Event in Arming Mode*) or in listening mode (see details in *Receive Alarm/Event in Listening Mode*).



Note

The traffic pre-alarm only supports the default linkage action (upload to center) and arming schedule (all-day schedule).

The network traffic alarm details are returned in the message *JSON_EventNotificationAlert_NetworkTrafficPrealarmMsg* , and the corresponding alarm type (*eventType*) is "dataPrealarm".

13.4 Configure Motion Detection Alarm

Motion detection detects the moving objects in the configured surveillance area, and a series of actions can be taken when the alarm is triggered. In order to detect the moving objects accurately and reduce the false alarm rate, detection target types (including human and vehicle), and detection region types (including region, grid, and ROI) are selectable for different motion detection environment.

Steps

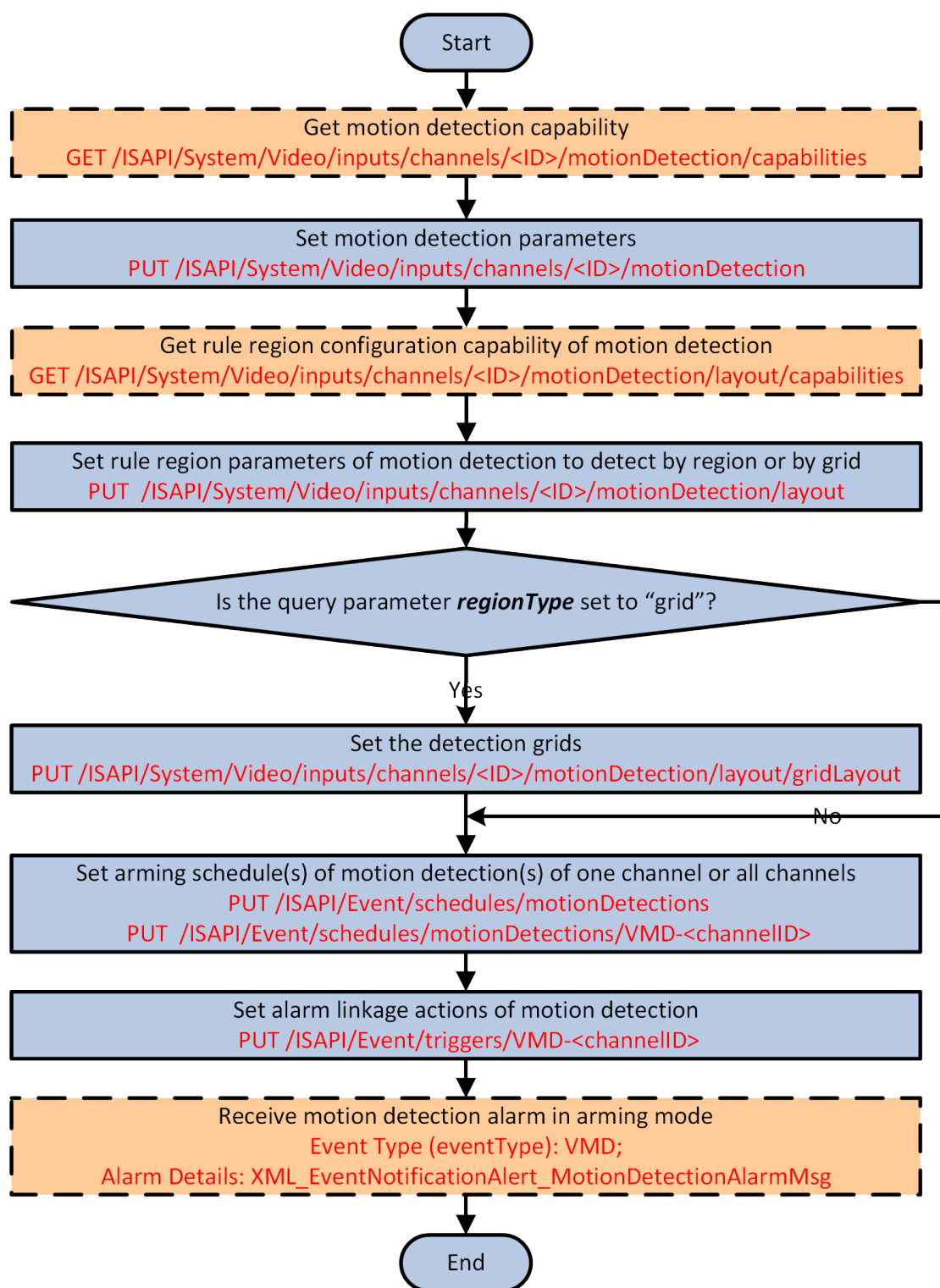


Figure 13-3 API Calling Flow of Configuring Motion Detection Alarm

1. Configure motion detection.

- 1) **Optional:** Call `/ISAPI/System/Video/inputs/channels/<ID>/motionDetection/capabilities` by GET method to get the motion detection capability for reference.
- 2) **Optional:** Call `/ISAPI/System/Video/inputs/channels/<ID>/motionDetection` by GET method to get default or configured motion detection parameters for reference.
- 3) Call `/ISAPI/System/Video/inputs/channels/<ID>/motionDetection` by PUT method to set motion detection parameters.
2. Configure rule region of motion detection to detect by region or by grid.
 - 1) **Optional:** Call `/ISAPI/System/Video/inputs/channels/<ID>/motionDetection/layout/capabilities` by GET method to get rule region configuration capability of motion detection for reference.
 - 2) **Optional:** Call `/ISAPI/System/Video/inputs/channels/<ID>/motionDetection/layout` by GET method to get default or configured rule region parameters of motion detection for reference.
 - 3) Call `/ISAPI/System/Video/inputs/channels/<ID>/motionDetection/layout` by PUT method to set rule region of motion detection to detect by region or by grid.
3. Check the value of query parameter **regionType** in the request URI of `/ISAPI/System/Video/inputs/channels/<ID>/motionDetection/layout/gridLayout`.
 - If the query parameter **regionType** in the request URI is set to "grid", call `/ISAPI/System/Video/inputs/channels/<ID>/motionDetection/layout/gridLayout` by PUT method to set the detection grids.
 - If the query parameter **regionType** in the request URI is set to "region", perform the following steps.
4. Configure arming schedule(s) of motion detection(s).
 - Call `/ISAPI/Event/schedules/motionDetections` by PUT method to set arming schedules of all channels' motion detection.
 - Call `/ISAPI/Event/schedules/motionDetections/<ID>` by PUT method and set the **<ID>** in the request URI to "VMD-**<channelID>**" (e.g., `/ISAPI/Event/schedules/motionDetections/VMD-101`) to set arming schedule of a channel's motion detection.
5. Call `/ISAPI/Event/triggers/<ID>` by PUT method and set the **<ID>** in the request URI to "VMD-**<channelID>**" (e.g., `/ISAPI/Event/triggers/VMD-101`) to set alarm linkage action of a channel's motion detection.
6. Call the flow of **Receive Alarm/Event in Arming Mode** to receive the motion detection alarm information if it is triggered and uploaded.

The motion detection alarm information with the event type (**eventType**) of "VMD" will be returned in the message **XML_EventNotificationAlert_MotionDetectionMsg**.

13.5 Configure Vibration Detection Alarm

Vibration detection function can be applied to the scenes in which the real-time vibration status of devices should be detected in case people damage the devices on purpose. You can configure the arming schedule and linkage actions of the vibration detection alarm. For those triggered alarms, you can also search for the related videos or pictures.

Steps

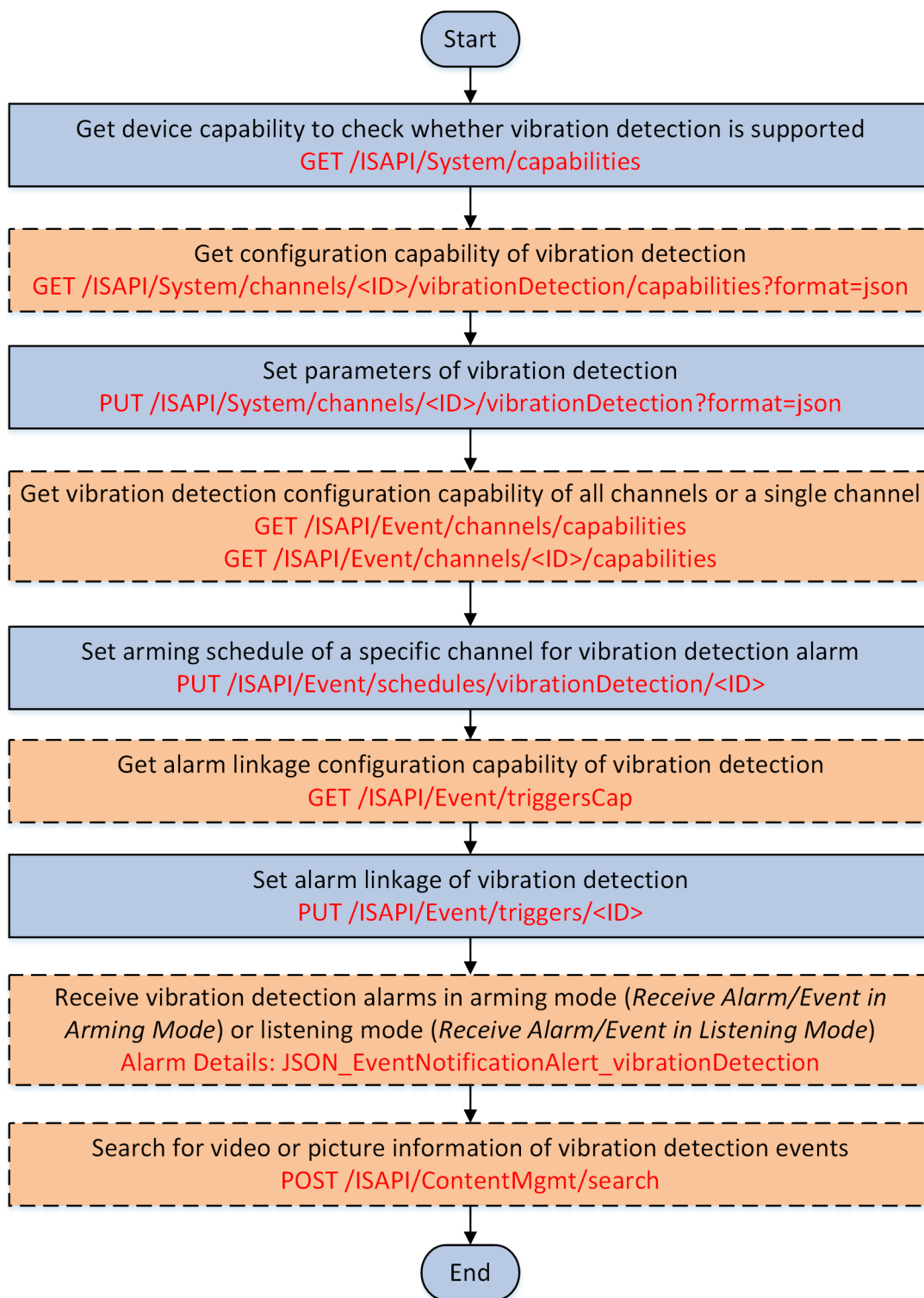


Figure 13-4 API Calling Flow of Configuring Vibration Detection Alarm

1. Call ***/ISAPI/System/capabilities*** by GET method to get the device capability and check whether vibration detection is supported by the device.

The device capability is returned in the message ***XML_DeviceCap***.

If the node ***<isSupportVibrationDetection>*** is returned in the message and its value is "true", it indicates that the vibration detection is supported, and then you can continue to perform the following steps.

Otherwise, the vibration detection is not supported, please end this task.

2. **Optional:** Call ***/ISAPI/System/channels/<ID>/vibrationDetection/capabilities?format=json*** by GET method to get configuration capability of vibration detection.
3. Call ***/ISAPI/System/channels/<ID>/vibrationDetection?format=json*** by PUT method to set parameters of vibration detection.
4. **Optional:** Call ***/ISAPI/Event/channels/capabilities*** or ***/ISAPI/Event/channels/<ID>/capabilities*** by GET method to get event configuration capability of all channels or a single channel for knowing the configuration details or notices.
5. Call ***/ISAPI/Event/schedules/vibrationDetection/<ID>*** by PUT method and set the ***<ID>*** in the format of "vibrationDetection-***<channelID>***", e.g., ***/ISAPI/Event/schedules/vibrationDetection/vibrationDetection-1***, to set arming schedule of a specific channel for vibration detection alarm.
6. **Optional:** Call ***/ISAPI/Event/triggersCap*** by GET method to get alarm linkage configuration capability of vibration detection.

The alarm linkage capability is returned in the message ***XML_EventTriggersCap***.

7. Call ***/ISAPI/Event/triggers/<ID>*** by PUT method and set the ***<ID>*** in the format of "vibrationDetection-***<channelID>***", e.g., ***/ISAPI/Event/triggers/vibrationDetection-1***, to set the alarm linkages.
8. **Optional:** Receive vibration detection alarm in arming mode (refer to ***Receive Alarm/Event in Arming Mode*** for details) or in listening mode (refer to ***Receive Alarm/Event in Listening Mode*** for details) when alarm is triggered.



Note

The vibration detection alarm details are uploaded in the message ***JSON_EventNotificationAlert_vibrationDetection*** and the corresponding alarm type (***eventType***) is "vibrationDetection".

9. **Optional:** Call ***/ISAPI/ContentMgmt/search*** by POST method to search for video or picture information of the vibration detection alarms.



Note

To check whether searching for video or picture information of vibration detection events is supported, you can call ***/ISAPI/ContentMgmt/capabilities*** by GET method to get the storage capability message ***XML_RacmCap*** and check whether the event type "vibrationDetection" exists in the nodes ***recordSearchType*** and ***pictureSearchType***.

13.6 Receive Alarm/Event in Arming Mode

When alarm is triggered or event occurred, and the alarm/event linkage is configured, you can send request message to device for getting the alarm/event stream, and then the device uploads the corresponding response message, which contains alarm/event information.

Before You Start

Make sure you have configured alarm/event and triggered the alarm/event. For configuring alarm/event parameters, refer to the some typical applications of alarm/event configuration.

Steps

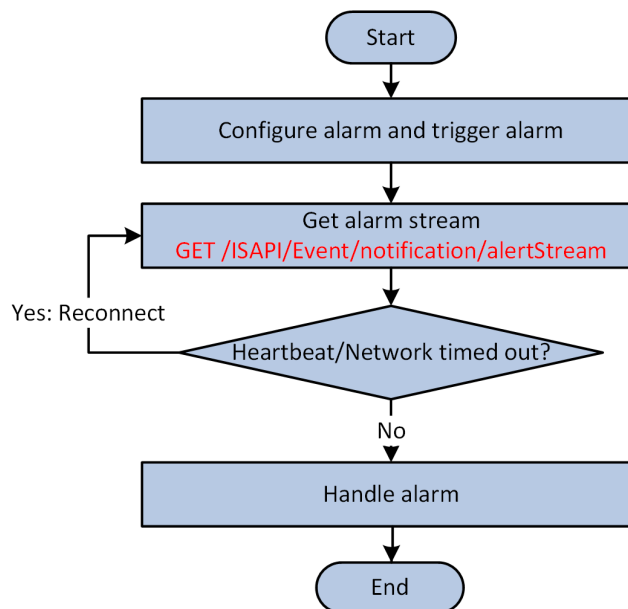


Figure 13-5 Programming Flow of Receiving Alarm/Event in Arming Mode

1. Call `/ISAPI/Event/notification/alertStream` by GET to get the alarm/event stream.
2. Check if the heartbeat receiving timed out or network disconnected.
 - If the heartbeat keeps alive and the network still connected, perform the following step to continue.
 - If the heartbeat receiving timed out or network disconnected, perform the above step repeatedly until reconnected.
3. Receive and process the alarm/event information.

Example

Sample Code of Receiving Alarm/Event in Arming Mode (without Binary Picture Data)

```
GET /ISAPI/Event/notification/alertStream HTTP/1.1
Host: data_gateway_ip
Connection: Keep-Alive
```

```
HTTP/1.1 401 Unauthorized
Date: Sun, 01 Apr 2018 18:58:53 GMT
```

```
Server:
Content-Length: 178
Content-Type: text/html
Connection: keep-alive
Keep-Alive: timeout=10, max=99
WWW-Authenticate: Digest qop="auth",
realm="IP Camera(C2183)",
nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d",
stale="FALSE"

GET /ISAPI/Event/notification/alertStream HTTP/1.1
Authorization: Digest username="admin",
realm="IP Camera(C2183)",
nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d",
uri="/ISAPI/Event/notification/alertStream",
cnonce="3d183a245b8729121ae4ca3d41b90f18",
nc=00000001,
qop="auth",
response="f2e0728991bb031f83df557a8f185178"
Host: 10.6.165.192

HTTP/1.1 200 OK
MIME-Version: 1.0
Connection: close
Content-Type: multipart/mixed; boundary=<frontier>

--<frontier>
Content-Type: application/xml; charset="UTF-8"
Content-Length: text_length

<EventNotificationAlert/>
--<frontier>
```

Note

Some alarm data is in JSON format, so the **Content-Type** may be "application/xml" or "application/json".

13.7 Receive Alarm/Event in Listening Mode

When alarm is triggered or event occurred, and the alarm/event linkage is configured, the device uploads the alarm/event information automatically, you can receive the alarm/event by configuring the listening port of HTTP host server.

Before You Start

Make sure you have configured alarm/event and triggered the alarm/event. For configuring alarm/event parameters, refer to the some typical applications of alarm/event configuration.

Steps

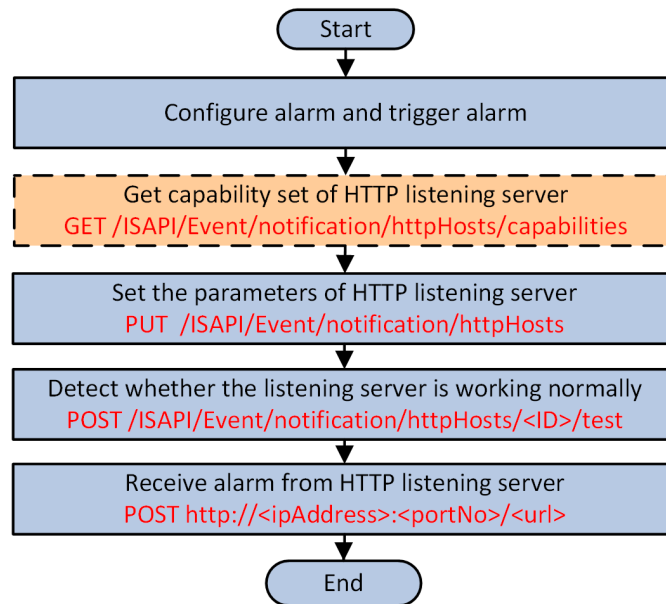


Figure 13-6 Programming Manual of Receiving Alarm/Event in Listening Mode

1. **Optional:** Call **/ISAPI/Event/notification/httpHosts/capabilities** by GET method to get the capability of HTTP listening server.
2. Call **/ISAPI/Event/notification/httpHosts** by PUT method to set the parameters (including listening address and listening port) of HTTP listening server.



Note

Before setting the listening server, you'd better call the URI by GET method to get the default or configured parameters for reference.

3. Call **/ISAPI/Event/notification/httpHosts/<ID>/test** by POST method to check if the listening server is working normally.
4. Call **http://ipAddress:portNo/url** by POST method to receive the alarm/event information from the listening server.

Example

Sample Code of Receiving Alarm/Event in Listening Mode

- with Binary Picture Data

```
//Request
POST requestUrl HTTP/1.1
Host: data_gateway_ip:port
Accept-Language: en-US
Date: YourDate
Content-Type: multipart/form-data;boundary=<frontier>
Content-Length: body_length
Connection: keep-alive

--<frontier>
```

```
Content-Disposition: form-data; name="Event_Type"
Content-Type: text/xml
Content-Length: xml_length

<EventNotificationAlert/>
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Length: image_length
Content-Type: image/jpeg

[binary picture data]
--<frontier>--

//Response
HTTP/1.1 HTTP statusCode
Date: YourDate
Connection: close
```

- without Binary Picture Data

```
//Request
POST requestUrl HTTP/1.1
Host: data_gateway_ip:port
Accept-Language: en-US
Date: YourDate
Content-Type: text/xml;
Content-Length: text_length
Connection: Close

<EventNotificationAlert/>

//Response
HTTP/1.1 HTTP statusCode
Date: YourDate
Connection: close
```

Note

- The **Host** is the HTTP server domain name or IP address and port No.
 - Some alarm data is in JSON format, so the **Content-Type** may be "text/xml" or "text/json".
-

13.8 Subscribe Alarm/Event in Arming Mode

For arming mode, the platform will connect to the devices automatically and send commands to the devices for uploading alarm/event information when the alarm is triggered or event occurred. To reduce the CPU and bandwidth usage of platform, and improve the device processing performance, the platform can subscribe alarm/event types to receive alarm/event information as required.

Before You Start

Configure alarm/event and trigger the alarm/event. For configuring alarm/event parameters, refer to the some typical applications of alarm/event configuration.

Steps

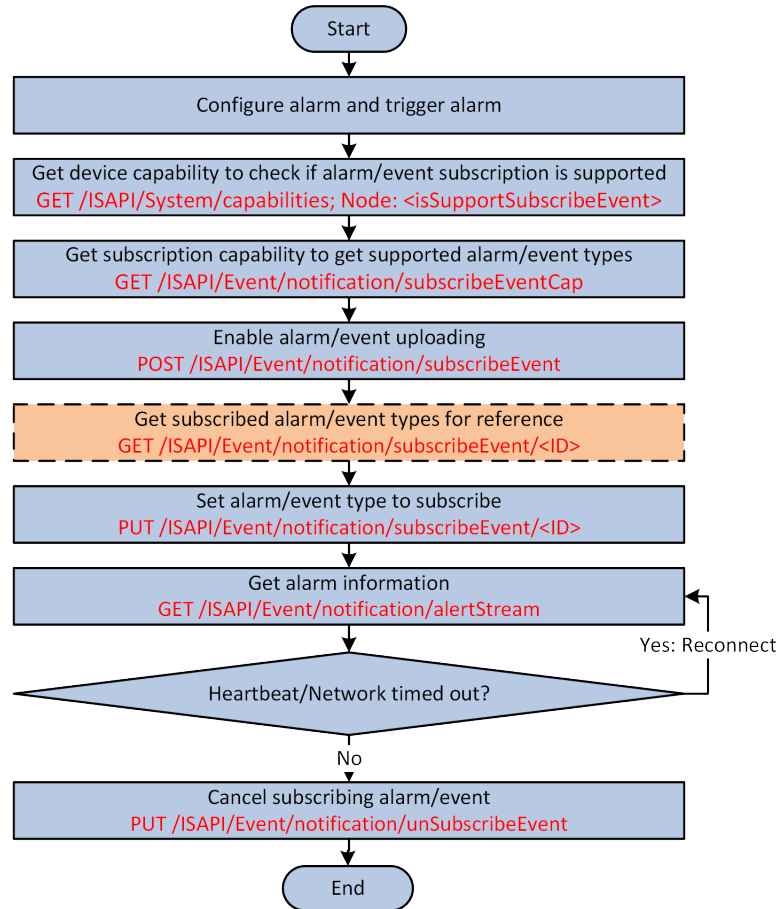


Figure 13-7 Programming Flow of Subscribing Alarm/Event in Arming Mode

1. Call **/ISAPI/System/capabilities** by GET method to get device capability to check whether the alarm/event subscription is supported.

The device capability is returned in the message **XML_DeviceCap**.

If the node **<isSupportSubscribeEvent>** is also returned in the message and its value is "true", it indicates that alarm/event subscription is supported by device, and you can continue to perform the following steps;

Otherwise, alarm/event subscription is not supported, please end the task.

2. Call **/ISAPI/Event/notification/subscribeEventCap** by GET method to get subscription capability to get supported alarm/event types.

The alarm/event subscription capability is returned in the message **XML_SubscribeEventCap**.

3. Call **/ISAPI/Event/notification/subscribeEvent** by POST method to enable alarm/event uploading.

Note

The sub node **<type>** of event type list (**EventList**) in the request message **XML_SubscribeEvent** can be set to different values (refer to *Supported Alarm/Event Types* for details) to subscribe different types of alarm/event information.

4. **Optional:** Call **/ISAPI/Event/notification/subscribeEvent/<ID>** by GET method to get subscribed alarm/event types for reference.
 5. Call **/ISAPI/Event/notification/subscribeEvent/<ID>** by PUT to set alarm/event type to subscribe.
-

Note

The sub node **<type>** of event type list (**EventList**) in the request message **XML_SubscribeEvent** can be set to different values (refer to *Supported Alarm/Event Types* for details) to subscribe different types of alarm/event information.

6. Call **/ISAPI/Event/notification/alertStream** by GET method to receive the alarm/event information.
7. Check if the heartbeat receiving timed out or network disconnected.
 - If the heartbeat keeps alive and the network still connected, perform the following step to continue.
 - If the heartbeat receiving timed out or network disconnected, perform the above step repeatedly until reconnected.
8. Call **/ISAPI/Event/notification/unSubscribeEvent** by PUT method to cancel subscribing alarm/event.

Chapter 14 Parameter Configuration

14.1 I/O Settings

I/O (Alarm Input and Output)

Function	Request URI
Get I/O configuration capability	GET <i>/ISAPI/System/IO/capabilities</i> .
Get I/O status	GET <i>/ISAPI/System/IO/status</i>

Alarm Input

Function	Request URI
Get configuration capability of alarm inputs	GET <i>/ISAPI/System/IO/inputs/capabilities</i> .
Get parameters of all alarm inputs	GET <i>/ISAPI/System/IO/inputs</i>
Get or set parameters of one alarm input	GET or PUT <i>/ISAPI/System/IO/inputs/<ID></i>
Get status of one alarm input	GET <i>/ISAPI/System/IO/inputs/<ID>/status</i>
Alarm Input Configuration of Digital Channel	 Note - To check if the alarm input configuration of digital channel is supported, you can call the GET <i>/ISAPI/ContentMgmt/capabilities</i> to get the device storage management capability (<i>XML_RacmCap</i>). - If this function is supported, the node <code><isSupportIOInputProxy></code> is returned and its value is "true". - Get or set alarm input parameters of all digital channels: GET or PUT <i>/ISAPI/ContentMgmt/IOProxy/inputs</i> - Added alarm input configuration of a specific digital channel: POST <i>/ISAPI/ContentMgmt/IOProxy/inputs</i> - Get or set alarm input parameters of a specific digital channel: GET or PUT <i>/ISAPI/ContentMgmt/IOProxy/inputs/<ID></i> - Delete alarm input configuration of a specific digital channel: DELETE <i>/ISAPI/ContentMgmt/IOProxy/inputs/<ID></i> - Get alarm input status of a specific digital channel: GET <i>/ISAPI/ContentMgmt/IOProxy/inputs/<ID>/status</i>

Alarm Output

Function	Request URI
Basic Alarm Output Parameters	- Get configuration capability of alarm outputs: GET <i>/ISAPI/System/IO/outputs/capabilities</i> - Get parameters of all alarm outputs: GET <i>/ISAPI/System/IO/outputs</i> - Get or set parameters of one alarm output: GET or PUT <i>/ISAPI/System/IO/outputs/<ID></i> - Get status of one alarm output: GET <i>/ISAPI/System/IO/outputs/<ID>/status</i>
Supplement Light Alarm Output	- Get configuration capability of supplement light alarm outputs: GET <i>/ISAPI/Event/triggers/notifications/whiteLightAlarm/capabilities?format=json</i> - Get or set parameters of supplement light alarm outputs: GET or PUT <i>/ISAPI/Event/triggers/notifications/whiteLightAlarm?format=json</i>
Audible Alarm Output	- Get configuration capability of audible alarm outputs: GET <i>/ISAPI/Event/triggers/notifications/AudioAlarm/capabilities?format=json</i> - Get or set parameters of audible alarm outputs: GET or PUT <i>/ISAPI/Event/triggers/notifications/AudioAlarm?format=json</i>
Manually trigger an alarm output	PUT <i>/ISAPI/System/IO/outputs/<ID>/trigger</i>
Alarm Output Configuration of Digital Channel	 Note To check if the alarm output configuration of digital channel is supported, you can call the GET <i>/ISAPI/ContentMgmt/capabilities</i> to get the device storage management capability (<i>XML_RacmCap</i>). If this function is supported, the node <isSupportIOOutputProxy> is returned and its value is "true". - Get or set alarm output parameters of all digital channels: GET or PUT <i>/ISAPI/ContentMgmt/IOProxy/outputs</i> - Added alarm output configuration of a specific digital channel: POST <i>/ISAPI/ContentMgmt/IOProxy/outputs</i> - Get or set alarm output parameters of a specific digital channel: GET or PUT <i>/ISAPI/ContentMgmt/IOProxy/outputs/<ID></i>

Function	Request URI
	- Delete alarm output configuration of a specific digital channel: DELETE <i>/ISAPI/ContentMgmt/IOProxy/outputs/<ID></i> - Get alarm output status of a specific digital channel: GET <i>/ISAPI/ContentMgmt/IOProxy/outputs/<ID>/status</i> - Trigger alarm output of a specific digital channel: PUT <i>/ISAPI/ContentMgmt/IOProxy/outputs/<ID>/trigger</i>  Note The alarm details are uploaded by the message <i>XML_EventNotificationAlert_AlarmInputAlarmMsg</i> .

14.2 Video/Image Settings

Basic Image Parameters

Function	Request URI
Get image configuration capability	GET <i>/ISAPI/Image/channels/<ID>/capabilities</i>
Get or set image parameters of all channels	GET or PUT <i>/ISAPI/Image/channels</i>
Get or set image parameters of one channel	GET or PUT <i>/ISAPI/Image/channels/<ID></i>
Get basic image parameters of all channels	GET <i>/ISAPI/Image/channels/<ID>/imageModes</i>
Get basic image parameters of one channel	GET <i>/ISAPI/Image/channels/<ID>/imageMode/<ID></i>
Reset image parameters	PUT <i>/ISAPI/Image/channels/<ID>/reset</i>
Restore to default image settings	PUT <i>/ISAPI/Image/channels/<ID>/restore</i>

Camera Video/Image Parameters

Function	Request URI
Get or set day/night mode parameters	GET or PUT <i>/ISAPI/Image/channels/<ID>/ISPMode</i>
Get or set image auto flip parameters	GET or PUT <i>/ISAPI/Image/channels/<ID>/ImageFlip</i>
Get or set wide dynamic range (WDR) parameters	GET or PUT <i>/ISAPI/Image/channels/<ID>/WDR</i>
Get or set backlight compensation (BLC) parameters	GET or PUT <i>/ISAPI/Image/channels/<ID>/BLC</i>
Get or set day/night auto switch parameters	GET or PUT <i>/ISAPI/Image/channels/<ID>/IrcutFilter</i>
Get or set 3-dimension digital noise reduction (3D DNR) parameters	GET or PUT <i>/ISAPI/Image/channels/<ID>/noiseReduce</i>
Get or set white balance (WB) parameters	GET or PUT <i>/ISAPI/Image/channels/<ID>/whiteBalance</i>
Get or set sharpness parameters	GET or PUT <i>/ISAPI/Image/channels/<ID>/sharpness</i>
Get or set defog parameters	GET or PUT <i>/ISAPI/Image/channels/<ID>/dehaze</i>
Get or set image standard	GET or PUT <i>/ISAPI/Image/channels/<ID>/powerLineFrequency</i>
Get or set image rotate mode parameters	GET or PUT <i>/ISAPI/Image/channels/<ID>/corridor</i>
Get or set video input mode parameters	GET or PUT <i>/ISAPI/Image/channels/<ID>/capturemode</i>

Table 14-1 Exposure

Function	Request URI
Get or set exposure parameters	GET or PUT <i>/ISAPI/Image/channels/<ID>/exposure</i>
Get or set shutter parameters in exposure mode	GET or PUT <i>/ISAPI/Image/channels/<ID>/shutter</i>
Get or set gain parameters in exposure mode	GET or PUT <i>/ISAPI/Image/channels/<ID>/gain</i>

Table 14-2 Image Adjustment

Function	Request URI
Get image adjustment capability	GET <i>/ISAPI/Image/channels/<ID>/color/capabilities</i>
Get or set image adjustment parameters	GET or PUT <i>/ISAPI/Image/channels/<ID>/color</i>

Other Video/Image Parameters

Table 14-3 Focus

Function	Request URI
Get focus configuration capability	GET <i>/ISAPI/Image/channels/<ID>/focusConfiguration/capabilities</i>
Get or set focus parameters	GET or PUT <i>/ISAPI/Image/channels/<ID>/focusConfiguration</i>
Perform regional focus	PUT <i>/ISAPI/Image/channels/<ID>/regionalFocus</i>  Note To check if the regional focus is supported by device, you can call the GET to get the image capability (<i>XML_ImageCap</i>). If supports, the node <isSupportRegionalFocus> will be returned in the capability.

Table 14-4 Target Coloration

Function	Request URI
Get image configuration capability of a specific channel	GET <i>/ISAPI/Image/channels/<ID>/capabilities</i>
Get or set the palettes parameters	GET or PUT <i>/ISAPI/Image/channels/<ID>/Palettes</i>

Table 14-5 On Screen Display (OSD)

Function	Request URI
Get or set OSD parameters	GET or PUT <i>/ISAPI/System/Video/inputs/channels/<ID>/overlays</i>  Note To check if the OSD language configuration is supported by device, you can call the GET / to get the video capability (<i>XML_VideoCap</i>). If supports, the node <i><OSDLanguage></i> will be returned.
Get text overlay configuration capability of a specific video digital input channel	GET <i>/ISAPI/ContentMgmt/InputProxy/channels/<ID>/video/overlays/capabilities</i>
Get or set text overlay parameters of a specific video digital input channel	GET or PUT <i>/ISAPI/ContentMgmt/InputProxy/channels/<ID>/video/overlays</i>
Get or set text language of OSD	GET <i>/ISAPI/System/Video/inputs/OSDLanguage</i>

Table 14-6 Stream Configuration for Displaying VCA Rules of Smart Events

Function	Request URI
Get capability of configuring stream for displaying VCA rules of smart events	GET <i>/ISAPI/Streaming/channels/<ID>/smartOverlap/capabilities?format=json</i>  Note To check if the stream configuration for displaying VCA rules of smart events is supported by the device, you can call the GET <i>/ISAPI/System/capabilities</i> to get the device capability (<i>XML_DeviceCap</i>). If this function is supported, the node

Function	Request URI
	<supportSmartOverlapChannels> will be returned and its value is "true".
Get or set stream parameters for displaying VCA rules of smart events	GET or PUT <i>/ISAPI/Streaming/channels/<ID>/smartOverlap?format=json</i>

Table 14-7 Target Enhancement

Function	Request URI
Get target enhancement capability	GET <i>/ISAPI/Image/channels/<ID>/targetEnhancement/capabilities</i>
Get or set target enhancement parameters	GET or PUT <i>/ISAPI/Image/channels/<ID>/targetEnhancement</i>

Table 14-8 Privacy Mask

Function	Request URI
Get configuration capability of privacy mask	GET <i>/ISAPI/System/Video/inputs/channels/<ID>/privacyMask/privacyMaskCap</i>
Get or set privacy mask parameters	GET or PUT <i>/ISAPI/System/Video/inputs/channels/<ID>/privacyMask</i>
Get or set multiple privacy mask regions	GET or PUT <i>/ISAPI/System/Video/inputs/channels/<ID>/privacyMask/regions</i>
Get or set one privacy mask region	GET or PUT <i>/ISAPI/System/Video/inputs/channels/<ID>/privacyMask/regions/<ID></i>

Table 14-9 Distortion Correction

Function	Request URI
Get configuration capability of distortion correction	GET <i>/ISAPI/Image/channels/<ID>/lensDistortionCorrection/capabilities</i>
Get or set distortion correction parameters	GET or PUT <i>/ISAPI/Image/channels/<ID>/lensDistortionCorrection</i>

Table 14-10 Target Cropping

Function	Request URI
Get configuration capability of target cropping	GET <i>/ISAPI/Streaming/channels/<ID>/regionClip/capabilities</i>
Get or set target cropping parameters	GET or PUT <i>/ISAPI/Streaming/channels/<ID>/regionClip</i>

Table 14-11 Temperature Range

Function	Request URI
Get temperature range configuration capability	GET <i>/ISAPI/Image/channels/<ID>/tempRange/capabilities</i>
Get or set temperature range	GET or PUT <i>/ISAPI/Image/channels/<ID>/tempRange</i>

Table 14-12 Mounting Scenario

Function	Request URI
Get configuration capability of mounting scenario mode	GET <i>/ISAPI/Image/channels/<ID>/mountingScenario/capabilities</i>
Get or set mounting scenario mode by channel	GET or PUT <i>/ISAPI/Image/channels/<ID>/mountingScenario</i>

14.3 Audio Settings

Basic Audio Configuration

Function	Request URI
Get audio capability	GET <i>/ISAPI/System/Audio/capabilities</i>
Get dynamic audio capability by channel	GET <i>/ISAPI/System/Audio/channels/<ID>/dynamicCap</i>
Get parameters of all audio channels	GET <i>/ISAPI/System/Audio/channels</i>
Get parameters of a specific audio channel	GET <i>/ISAPI/System/Audio/channels/<ID></i>

Audio Input Configuration

Function	Request URI
Get audio input capability of a specific channel	GET <i>/ISAPI/System/Audio/AudioIn/channels/<ID>/capabilities</i>
Get or set audio input parameters of a specific channel	GET or PUT <i>/ISAPI/System/Audio/AudioIn/channels/<ID></i>

Audio Output Configuration

Function	Request URI
Get audio output capability of a specific channel	GET <i>/ISAPI/System/Audio/AudioOut/channels/<ID>/capabilities</i>
Get or set audio output parameters of a specific channel	GET or PUT <i>/ISAPI/System/Audio/AudioOut/channels/<ID></i>

14.4 Channel Settings

Basic Channel Parameters

Function	Request URI
Get all channels' attributes	GET <i>/ISAPI/AUXInfo/attributes/Channels</i>
Get one channel's attributes	GET <i>/ISAPI/AUXInfo/attributes/Channels/<ID></i>

Video Input Channel

Function	Request URI
Get configuration capability by video input channel	GET <i>/ISAPI/System/Video/inputs/channels/<ID>/capabilities</i>
Get parameters of all video input channels	GET <i>/ISAPI/System/Video/inputs/channels?format=json</i>
Get parameters of one video input channel	<i>/ISAPI/System/Video/inputs/channels/<ID>?format=json</i>

Video Output Channel

Function	Request URI
Get parameters of all video outputs	GET <i>/ISAPI/System/Video/outputs</i>
Get configuration capability by video output channel	GET <i>/ISAPI/System/Video/outputs/channels/<ID>/capabilities</i>
Get parameters of all video output channels	GET <i>/ISAPI/System/Video/outputs/channels</i>
Get parameters of one video output channel	GET <i>/ISAPI/System/Video/outputs/channels/<ID></i>

Digital Channel



Note

To check if the digital channel management or configuration is supported, you can call the GET */ISAPI/ContentMgmt/capabilities* to get the device storage management capability (*XML_RacmCap*).

Function	Request URI
Get management capability of all digital channels	GET <i>/ISAPI/ContentMgmt/InputProxy/channels/capabilities</i>
Get management capability of one digital channel	GET <i>/ISAPI/ContentMgmt/InputProxy/channels/<ID>/capabilities</i>
Get supported number of digital channels	POST <i>/ISAPI/ContentMgmt/InputProxy/sourceCapability</i>
Get parameters of all digital channels	GET <i>/ISAPI/ContentMgmt/InputProxy/channels</i>
Get status of all digital channels	GET <i>/ISAPI/ContentMgmt/InputProxy/channels/status</i>
Get, add, or delete parameters of one digital channel	GET, POST, or PUT <i>/ISAPI/ContentMgmt/InputProxy/channels</i>
Get status of one digital channel	GET <i>/ISAPI/ContentMgmt/InputProxy/channels/<ID>/status</i>
Get or set control parameters of a digital channel	GET or PUT <i>/ISAPI/ContentMgmt/InputProxy/channels/<ID>/chanCtrl</i>
Reboot a digital channel	GET <i>/ISAPI/ContentMgmt/InputProxy/channels/<ID>/reboot</i>

Function	Request URI
Check if port mapping is required for connecting to a digital channel	POST <i>/ISAPI/ContentMgmt/InputProxy/channels/<ID>/portMapParam</i>
Get or set the device parameters of a digital channel	GET or PUT <i>/ISAPI/ContentMgmt/InputProxy/channels/<ID>/deviceInfo</i>
Get device parameters of all digital channels	POST <i>/ISAPI/ContentMgmt/InputProxy/deviceInfo</i>
Get firmware code list of the digital channel	GET <i>/ISAPI/System/firmwareCodeV2</i>

14.5 Peripherals Settings

Function	Request URI
Get peripherals configuration capability	GET <i>/ISAPI/System/externalDevice/capabilities</i>
Get configuration capability of built-in supplement light	GET <i>/ISAPI/Image/channels/<ID>/supplementLight/capabilities</i>
Get or set parameters of built-in supplement light	GET or PUT <i>/ISAPI/Image/channels/<ID>/supplementLight</i>  Note To check if configuring built-in supplement light is supported, you can call the GET <i>/ISAPI/Image/channels/<ID>/capabilities</i> to get the image channel capability (<i>XML_Cap_ImageChannel</i>). If supports, the node <SupplementLight> will be returned in the message.

14.6 Picture Storage Server Settings

Function	Request URI
Get picture storage server capability	GET <i>/ISAPI/System/PictureServer/capabilities?format=json</i>
Get picture storage server parameters	GET <i>/ISAPI/System/PictureServer?format=json</i>
Set picture storage server parameters	PUT <i>/ISAPI/System/PictureServer?format=json</i>

Chapter 15 System Configuration

Function	Request URI
Get Configuration Capability of Device Language	GET <i>/ISAPI/System/DeviceLanguage/capabilities</i>
Get or Set Device Language	GET or PUT <i>/ISAPI/System/DeviceLanguage</i>
Get USB Mode Configuration Capability	GET <i>/ISAPI/System/usb/capabilities?format=json</i>
Get or Set USB Mode	GET or PUT <i>/ISAPI/System/usb?format=json</i>

15.1 Network Settings

This chapter lists the request URIs for communication and network configurations of devices, including network interface settings, wireless network settings, network access settings, email settings, and so on.

Function	Request URI
Get network capability	GET <i>/ISAPI/System/Network/capabilities</i>
Import network certificate	GET <i>/ISAPI/Security/deviceCertificate</i>
Remotely get connection socket IP	GET <i>/ISAPI/System/Network/socketIP</i>  Note To check if getting socket IP is supported by device, you can call the GET <i>/ISAPI/System/Network/capabilities</i> to get the device capability (<i>XML_DeviceCap</i>). If supports, the node <i><isSupportGetLinkSocketIP></i> will be returned.
Network Self-Adaptive	- Get self-adaptive configuration capability: GET <i>/ISAPI/System/Network/adaption/capabilities?format=json</i> - Get self-adaptive parameters: GET <i>/ISAPI/System/Network/adaption?format=json&streamType=</i> - Set self-adaptive parameters: PUT <i>/ISAPI/System/Network/adaption?format=json&streamType=</i>

Function	Request URI
	 Note To check if network self-adaptive configuration is supported, you can get the network capability <i>XML_NetworkCap</i> by the <i>/ISAPI/System/Network/capabilities</i> . If supports, the node <i><Adaption></i> will be returned in the capability message.

15.1.1 Email

Function	Request URI
Get email configuration capability	GET <i>/ISAPI/System/Network/mailing/capabilities</i>
Get or set parameters of multiple emails	GET or PUT <i>/ISAPI/System/Network/mailing</i>
Get or set parameters of one email	GET or PUT <i>/ISAPI/System/Network/mailing/<ID></i>
Execute email test	POST <i>/ISAPI/System/Network/mailing/test</i>

15.1.2 Network Access

Network Access Test by Ping

Function	Request URI
Set ping test parameters	POST <i>/ISAPI/System/Network/pingtest</i>

Protocol Access

Function	Request URI
Get configuration capability of protocol access	GET <i>/ISAPI/Security/adminAccesses/capabilities</i>
Get or set access parameters of multiple protocols	GET or PUT <i>/ISAPI/Security/adminAccesses</i>
Get or set access parameters of one protocol	GET or PUT <i>/ISAPI/Security/adminAccesses/<ID></i>

Function	Request URI
Get 802.1 protocol access capability	GET <i>/ISAPI/System/Network/interfaces/<ID>/ieee802.1x/capabilities</i>
Get or set access parameters of 802.1 protocol	GET or PUT <i>/ISAPI/System/Network/interfaces/<ID>/ieee802.1x</i>

Hik-Connect Access

Function	Request URI
Get or set access parameters of Hik-Connect	GET or PUT <i>/ISAPI/System/Network/EZVIZ</i>  Note To check if the device supports accessing to Hik-Connect, you can call the request URI: GET <i>/ISAPI/System/Network/capabilities</i> to get the network capability (<i>XML_NetworkCap</i>). If supports, the node <i><isSupportEhome></i> will be returned in the capability and its value is "true".
Edit verification code for Hik-Connect	PUT <i>/ISAPI/System/Network/EZVIZ/secretKey?format=json</i>  Note To check if the device supports editing verification code for Hik-Connect, you can call the request URI: GET <i>/ISAPI/System/Network/capabilities</i> to get the network capability sets <i>XML_NetworkCap</i> . If supports, the node <i><secretKey></i> will be returned.

EHome Server Access

Function	Request URI
Get configuration capability of EHome server access	GET <i>/ISAPI/System/Network/Ehome/capabilities</i>
Get or set access parameters of EHome server	GET or PUT <i>/ISAPI/System/Network/Ehome</i>
Get or set access parameters of EHome server according to the center ID	GET or PUT <i>/ISAPI/System/Network/Ehome?centerID=</i>

Function	Request URI
Get configuration capability of report uploading method	GET <i>/ISAPI/SecurityCP/ReportCenterCfg/capabilities?format=json</i>
Get or set parameters of report uploading method	GET or PUT <i>/ISAPI/SecurityCP/ReportCenterCfg/<ID>?format=json</i>

Note

To check if the device supports accessing to EHome server, you can call the request URI: GET */ISAPI/System/Network/capabilities* to get the network capability (*XML_NetworkCap*). If supports, the node *<isSupportEZVIZ>* will be returned in the capability and its value is "true".

SSH Server Access

Function	Request URI
Get or set access parameters of SSH server	GET or PUT <i>/ISAPI/System/Network/ssh</i>

Note

To check if the device supports accessing to SSH server, you can call the request URI: GET */ISAPI/System/Network/capabilities* to get the network capability (*XML_NetworkCap*). If supports, the node *<isSupportSSH>* will be returned in the capability and its value is "true".

SNMP Access

Function	Request URI
Get batch configuration parameters of SNMP	GET <i>/ISAPI/System/Network/SNMP</i>
Get advanced configuration parameters of SNMP	GET <i>/ISAPI/System/Network/SNMP/advanced</i>
Get all trap address parameters of SNMP in version 3	GET <i>/ISAPI/System/Network/SNMP/advanced/trapReceivers</i>
Set all trap address parameters of SNMP in version 3	PUT <i>/ISAPI/System/Network/SNMP/advanced/trapReceivers</i>
Add a new trap address of SNMP in version 3	POST <i>/ISAPI/System/Network/SNMP/advanced/trapReceivers</i>
Delete all trap addresses of SNMP in version 3	DELETE <i>/ISAPI/System/Network/SNMP/advanced/trapReceivers</i>

Function	Request URI
Get a single trap address's parameters of SNMP in version 3	GET <i>/ISAPI/System/Network/SNMP/advanced/trapReceivers/<ID></i>
Set parameters for a single trap address of SNMP in version 3	PUT <i>/ISAPI/System/Network/SNMP/advanced/trapReceivers/<ID></i>
Delete a single address of SNMP in version 3	DELETE <i>/ISAPI/System/Network/SNMP/advanced/trapReceivers/<ID></i>

15.1.3 Network Interface

Network Interface

Function	Request URI
Get configuration capability of one network interface	GET <i>/ISAPI/System/Network/interfaces/<ID>/capabilities</i>
Get information of multiple network interfaces	GET <i>/ISAPI/System/Network/interfaces</i>
Get or set parameters of one network interface	GET or PUT <i>/ISAPI/System/Network/interfaces/<ID></i>
Get IP address of one network interface	GET or PUT <i>/ISAPI/System/Network/interfaces/<ID>/ipAddress</i>
Get connection configuration capability of a specific network interface	GET <i>/ISAPI/System/Network/interfaces/<ID>/link/capabilities</i>
Get or set connection parameters of a specific network interface	GET or PUT <i>/ISAPI/System/Network/interfaces/<ID>/link</i>

PoE

Function	Request URI
Get configuration capability of PoE port	GET <i>/ISAPI/System/Network/POE/capabilities?format=json</i>
Get or set PoE port parameters	GET or PUT <i>/ISAPI/System/Network/POE?format=json</i>

UPnP

Function	Request URI
Get or set UPnP parameters	GET or PUT <i>/ISAPI/System/Network/UPnP</i>
Get or set parameters of all UPnP ports	GET or PUT <i>/ISAPI/System/Network/UPnP/ports</i>
Get or set parameters of one UPnP port	GET or PUT <i>/ISAPI/System/Network/UPnP/ports/<ID></i>
Get mapping statuses of all UPnP ports	GET <i>/ISAPI/System/Network/UPnP/ports/status</i>
Get mapping status of one UPnP port	GET <i>/ISAPI/System/Network/UPnP/ports/<ID>/status</i>

15.1.4 Wireless Network

Wi-Fi

Function	Request URI
Get Wi-Fi configuration capability	GET <i>/ISAPI/System/Network/interfaces/<ID>/wireless/capabilities</i>
Get or set Wi-Fi parameters of one network interface	GET or PUT <i>/ISAPI/System/Network/interfaces/<ID>/wireless</i>
WPS (Wi-Fi Protected Setup)	• Get WPS configuration capability: GET <i>/ISAPI/System/Network/WPS/capabilities</i> • Set parameters of PBC (Push Button Configuration) automatic connection of WPS: PUT <i>/ISAPI/System/Network/interfaces/<ID>/WPS/AutoConnect</i> • Get or set parameters of PIN (Personal Identification Number) code of wireless access point: GET or PUT <i>/ISAPI/System/Network/interfaces/<ID>/WPS/AppPinCode</i> • Get or set PIN (Personal Identification Number) code of WPS of device: GET or PUT <i>/ISAPI/System/Network/interfaces/<ID>/WPS/devicePinCode</i>

Wireless Hotspot

Function	Request URI
Get configuration capability of wireless hotspot	GET <i>/ISAPI/System/Network/interfaces/<ID>/wirelessServer/capabilities</i>
Get or set wireless hotspot parameters	GET or PUT <i>/ISAPI/System/Network/interfaces/<ID>/wirelessServer</i>
Get list of devices connected to hotspot	GET <i>/ISAPI/System/Network/interfaces/<ID>/wirelessServer/accessDeviceList</i>
Get information of a device connected to hotspot	GET <i>/ISAPI/System/Network/interfaces/<ID>/wireless/accessDeviceList/<ID></i>
Get capability of devices connected to hotspot	GET <i>/ISAPI/System/Network/interfaces/<ID>/wireless/accessDeviceList/capabilities</i>
Get hotspot list	GET <i>/ISAPI/System/Network/interfaces/<ID>/wireless/accessPointList</i>
Get one hotspot information	GET <i>/ISAPI/System/Network/interfaces/<ID>/wireless/accessPointList/<ID></i>

Wireless Dial

Function	Request URI
Get configuration capability of wireless dial	GET <i>/ISAPI/System/Network/WirelessDial/Interfaces/<ID>/dial/capabilities</i>
Get or set wireless dial parameters	GET or PUT <i>/ISAPI/System/Network/WirelessDial/Interfaces/<ID>/dial</i>
Get wireless dial schedule	GET <i>/ISAPI/System/Network/WirelessDial/Interfaces/<ID>/schedule</i>

Network Keepalive

Function	Request URI
Get network keepalive capability	GET <i>/ISAPI/System/Network/WirelessDial/Interfaces/<ID>/keepAlive/capabilities?format=json</i>
Get or set network keepalive parameters	GET or PUT <i>/ISAPI/System/Network/WirelessDial/Interfaces/<ID>/keepAlive?format=json</i> .
Test network keepalive configuration	POST <i>/ISAPI/System/Network/WirelessDial/Interfaces/<ID>/keepAlive/test?format=json</i>

Network Traffic Monitoring

Function	Request URI
Get network traffic monitoring capability	GET <i>/ISAPI/System/Network/WirelessDial/Interfaces/<ID>/trafficMonitor/capabilities?format=json ;</i>
Get or set network traffic monitoring parameters	GET or PUT <i>/ISAPI/System/Network/WirelessDial/Interfaces/<ID>/trafficMonitor?format=json .</i>
Get network traffic monitoring status	GET <i>/ISAPI/System/Network/WirelessDial/Interfaces/<ID>/trafficMonitor/status?format=json .</i>
Network traffic pre-alarm to be uploaded	Alarm Details: <i>JSON_EventNotificationAlert_NetworkTrafficPrealarmMsg</i>

SMS Configuration

Function	Request URI
Get dial information configuration capability	GET <i>/ISAPI/System/Network/WirelessDial/Interfaces/<ID>/messageConfig/messageConfigCap):</i>
Get or set parameters of dial information configuration	GET or PUT <i>/ISAPI/System/Network/WirelessDial/Interfaces/<ID>/messageConfig</i>
Get all dial whitelist configuration	GET <i>/ISAPI/System/Network/WirelessDial/Interfaces/<ID>/messageConfig/WhiteList/<ID>)</i>
Get or set specific dial whitelist configuration	GET or PUT <i>/ISAPI/System/Network/WirelessDial/Interfaces/<ID>/messageConfig/WhiteList</i>

15.1.5 Network Service

Software Service

Function	Request URI
Get configuration capability of software service	GET <i>/ISAPI/System/Software/channels/<ID>/capabilities</i>
Get or set parameters of software service	GET or PUT <i>/ISAPI/System/Software/channels/<ID></i>

Session Initialization Protocol (SIP) Server

Function	Request URI
Get SIP server capability	GET <i>/ISAPI/System/Network/SIP/<ID>/capabilities</i>
Get or set parameters of all SIP servers	GET or PUT <i>/ISAPI/System/Network/SIP</i>
Get or set parameters of one SIP server	GET or PUT <i>/ISAPI/System/Network/SIP/<ID></i>
Get configuration capability of SIP server	GET <i>/ISAPI/System/Network/SIP/<ID>/SIPInfo/capabilities</i>
Get or set information of one SIP server	GET or PUT <i>/ISAPI/System/Network/SIP/<ID>/SIPInfo</i>
Get or set information of all SIP servers	GET or PUT <i>/ISAPI/System/Network/SIP/<ID>/SIPInfo/multiInfo</i>

File Transfer Protocol (FTP) Server

Function	Request URI
Get configuration capability of FTP server	GET <i>/ISAPI/System/Network/ftp/capabilities</i>
Get or set parameters of one FTP server	GET or PUT <i>/ISAPI/System/Network/ftp/<ID></i>
Get or set parameters of all FTP servers	GET or PUT <i>/ISAPI/System/Network/ftp</i>
Perform FTP server test	POST <i>/ISAPI/System/Network/ftp/test</i>

Dynamic Domain Name Server (DDNS)

Function	Request URI
Get DDNS configuration capability	GET <i>/ISAPI/System/Network/DDNS/capabilities</i>
Get or set parameters of all DDNSs	GET or PUT <i>/ISAPI/System/Network/DDNS</i>
Get or set parameters of one DDNS	GET or PUT <i>/ISAPI/System/Network/DDNS/<ID></i>

Function	Request URI
Get configuration capability of dynamic domain name	GET <i>/ISAPI/System/Network/interfaces/<ID>/dynamicHostName/capabilities?format=json</i>
Get or set dynamic domain name	GET or PUT <i>/ISAPI/System/Network/interfaces/<ID>/dynamicHostName?format=json</i>

Note

To check if setting dynamic domain name is supported, you can call the GET */ISAPI/System/Network/capabilities* to get the network capability (*XML_NetworkCap*). If supports, the node *<isSupportDynamicHostName>* is returned and its value is "true".

Log Server

Function	Request URI
Get log server configuration capability	GET <i>/ISAPI/System/logServer/capabilities</i>
Get or set log server parameters	GET or PUT <i>/ISAPI/System/logServer</i>
Perform log server test	POST <i>/ISAPI/System/logServer/test</i>

Static Routing

Function	Request URI
Get, set, or delete static routing parameters	GET, PUT, or DELETE <i>/ISAPI/System/Network/StaticRoute</i>

15.2 Time Settings

This chapter lists the request URIs for device time configuration, including setting time, setting time zone, and so on.

Function	Request URI
Get time configuration capability	GET <i>/ISAPI/System/time/capabilities</i>  Note To check if time configuration is supported, you can call the GET <i>/ISAPI/System/capabilities</i> to get the device capability

Function	Request URI
	(<i>XML_DeviceCap</i>). If supports, the node <i><isSupportTimeCap></i> will be returned.
Get or set time type	GET or PUT <i>/ISAPI/System/time/timeType?format=json</i>
Get or set time parameters	GET or PUT <i>/ISAPI/System/time</i>
Get or set time zone	GET or PUT <i>/ISAPI/System/time/timeZone</i>
Get or set local time parameters	GET or PUT <i>/ISAPI/System/time/localTime</i>
NTP Server	• Get NTP server capability: GET <i>/ISAPI/System/time/ntpServers/capabilities</i> • Get, set, or delete multiple NTP servers: GET, PUT, or DELETE <i>/ISAPI/System/time/ntpServers</i> • Get set, or delete one NTP server: GET, PUT, or DELETE <i>/ISAPI/System/time/ntpServers/<ID></i> • Add one NTP server: POST <i>/ISAPI/System/time/ntpServers</i> • Execute NTP server test: POST <i>/ISAPI/System/time/ntpServers/test</i>

15.3 System Diagnose

This chapter lists the request URIs for getting system diagnose information and the exporting status.

Function	Request URI
Get diagnose information	GET <i>/ISAPI/System/diagnosedData</i>
Get process of exporting diagnose information	GET <i>/ISAPI/System/diagnosedData/exportStatus</i>  Note To check if getting diagnose information is supported, you can call <i>/ISAPI/System/capabilities</i> by GET method to get the device capability (<i>XML_DeviceCap</i>). If supports, the node <i><isSupportDiagnosedData></i> will be returned.

15.4 Data Replenishment

In the condition of disconnection between platform and NVR, the NVR saves the data of people counting, heat map, temperature, and vehicle. When the connection is resumed, the platform can get the specified data from NVR during disconnection period.

People Counting Data Replenishment

Function	Request URI
Get data replenishment capability of people counting	GET <i>/ISAPI/System/Video/inputs/channels/counting/collection/capabilities?format=json</i>  Note Before getting people counting data replenishment capability, you should call <i>/ISAPI/System/Video/capabilities</i> by GET method to check whether this function is supported by the device via the node <code><isSupportCountingCollection></code> in <i>XML_VideoCap</i> .
Perform data replenishment of people counting	POST <i>/ISAPI/System/Video/inputs/channels/counting/collection?format=json</i>

Heat Map Data Replenishment

Function	Request URI
Get data replenishment capability of heat map	GET <i>/ISAPI/System/Video/inputs/channels/heatMap/collection/capabilities?format=json</i>  Note Before getting people counting data replenishment capability, you should call <i>/ISAPI/System/Video/capabilities</i> by GET method to check whether this function is supported by the device via the node <code><isSupportHeatmapCollection></code> in <i>XML_VideoCap</i> .
Perform data replenishment of heat map	POST <i>/ISAPI/System/Video/inputs/channels/heatMap/collection?format=json</i>

Temperature Data Replenishment

Function	Request URI
Get data replenishment capability of temperature	GET <i>/ISAPI/Thermal/temperature/collection/capabilities?format=json</i>  Note Before getting temperature data replenishment capability, you should call <i>/ISAPI/Thermal/capabilities</i> by GET method to check whether this function is supported by the device via the node <code><isSupportTemperatureCollection></code> in <i>XML_ThermalCap</i> .
Perform temperature data replenishment	POST <i>/ISAPI/Thermal/temperature/collection?format=json</i>

Vehicle Detection Data Replenishment

Function	Request URI
Get intelligent management capability	GET <i>/ISAPI/SDT/Management/IntelligentSearch/capabilities?format=json</i>
Perform vehicle detection data replenishment	POST <i>/ISAPI/SDT/Management/IntelligentSearch?format=json</i>  Note Before searching for vehicle detection data, you should check whether the device supports this function. If it supports, the node <code><isSupportIntelligentSearch></code> will be returned in the intelligent management capability <i>JSON_Cap_IntelliManagement</i> (call <i>/ISAPI/SDT/Management/capabilities?format=json</i> by GET method to get) and its value is "true".

Chapter 16 Request URIs

16.1 /ISAPI/AUXInfo

16.1.1 /ISAPI/AUXInfo/attributes/Channels

Get the attributes of channels.

Request URI Definition

Table 16-1 GET /ISAPI/AUXInfo/attributes/Channels

Method	GET
Description	Get the attributes of channels.
Query	None.
Request	None.
Response	<i>XML_ChannelInfoList</i>

16.1.2 /ISAPI/AUXInfo/attributes/Channels/<ID>

Get the attributes of a specific channel according to channel ID.

Request URI Definition

Table 16-2 GET /ISAPI/AUXInfo/attributes/Channels/<ID>

Method	GET
Description	Get the attributes of a specific channel according to channel ID.
Query	None.
Request	None.
Response	<i>XML_ChannelInfo</i>

Remarks

The <ID> in the request URI is the channel ID.

16.2 /ISAPI/ContentMgmt

16.2.1 /ISAPI/ContentMgmt/capabilities

Get storage capability.

Request URI Definition

Table 16-3 GET /ISAPI/ContentMgmt/capabilities

Method	GET
Description	Get storage capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_RacmCap</i> Failed: <i>XML_ResponseStatus</i>

16.2.2 /ISAPI/ContentMgmt/download

Download the file via plug-in.

Request URI Definition

Table 16-4 POST /ISAPI/ContentMgmt/download

Method	POST
Description	Download the file.
Query	None.
Request	<i>XML_downloadRequest</i>
Response	File

16.2.3 /ISAPI/ContentMgmt/download/capabilities

Get the downloading capability sets.

Request URI Definition

Table 16-5 GET /ISAPI/ContentMgmt/download/capabilities

Method	GET
Description	Get the downloading capability sets.
Query	None.
Request	None.
Response	Succeeded: <i>XML_DownloadAbility</i> Failed: <i>XML_ResponseStatus</i>

16.2.4 /ISAPI/ContentMgmt/download/toUSB/<taskId>/progress?format=json

Get the progress of exporting files to devices via USB.

Request URI Definition

Table 16-6 GET /ISAPI/ContentMgmt/download/toUSB/<taskId>/progress?format=json

Method	GET
Description	Get the progress of exporting files to devices via USB.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_ProgressInfo</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <taskId> in the request URI refers to the task ID, and it is returned in the message *JSON_ExporttoUSB_TaskInfo* after calling the request URI: POST */ISAPI/ContentMgmt/download/toUSB?format=json* .

16.2.5 /ISAPI/ContentMgmt/download/toUSB/capabilities?format=json

Get the capability of exporting files to devices via USB.

Request URI Definition

Table 16-7 GET /ISAPI/ContentMgmt/download/toUSB/capabilities?format=json

Method	GET
Description	Get the capability of exporting files to devices via USB.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_Cap_ExportInfo</i> Failed: <i>JSON_ResponseStatus</i>

16.2.6 /ISAPI/ContentMgmt/download/toUSB?format=json

Export files to devices via USB.

Request URI Definition

Table 16-8 POST /ISAPI/ContentMgmt/download/toUSB?format=json

Method	POST
Description	Export files to devices via USB.
Query	format : determine the format of request or response message.
Request	<i>JSON_ExportInfo</i>
Response	Succeeded: <i>JSON_ExporttoUSB_TaskInfo</i> Failed: <i>JSON_ResponseStatus</i>

16.2.7 /ISAPI/ContentMgmt/InputProxy/channels

Operations about management of all digital channels.

Request URI Definition

Table 16-9 GET /ISAPI/ContentMgmt/InputProxy/channels

Method	GET
Description	Get parameters of all digital channels.
Query	None.

Request	None.
Response	<i>XML_InputProxyChannelList</i>

Table 16-10 PUT /ISAPI/ContentMgmt/InputProxy/channels

Method	PUT
Description	Set parameters of all digital channels.
Query	None.
Request	<i>XML_InputProxyChannelList</i>
Response	<i>XML_ResponseStatus</i>

Table 16-11 POST /ISAPI/ContentMgmt/InputProxy/channels

Method	POST
Description	Add a digital channel.
Query	None.
Request	<i>XML_InputProxyChannel</i>
Response	<i>XML_ResponseStatus</i>

16.2.8 /ISAPI/ContentMgmt/InputProxy/channels/<ID>

Operations about management of a specific digital channel.

Request URI Definition

Table 16-12 GET /ISAPI/ContentMgmt/InputProxy/channels/<ID>

Method	GET
Description	Get parameter of a specific digital channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_InputProxyChannel</i> Failed: <i>XML_ResponseStatus</i>

Table 16-13 PUT /ISAPI/ContentMgmt/InputProxy/channels/<ID>

Method	PUT
Description	Set parameters of a specific digital channel.
Query	None.

Request	<i>XML_InputProxyChannel</i>
Response	<i>XML_ResponseStatus</i>

Table 16-14 DELETE /ISAPI/ContentMgmt/InputProxy/channels/<ID>

Method	DELETE
Description	Delete a specific digital channel.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the digital channel ID.

16.2.9 /ISAPI/ContentMgmt/InputProxy/channels/<ID>/capabilities

Get management capability of a specific digital channel.

Request URI Definition

Table 16-15 GET /ISAPI/ContentMgmt/InputProxy/channels/<ID>/capabilities

Method	GET
Description	Get management capability of a specific digital channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_InputProxyChannel</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the digital channel ID.

16.2.10 /ISAPI/ContentMgmt/InputProxy/channels/<ID>/chanCtrl

Get or set control parameters of a specific channel.

Request URI Definition

Table 16-16 GET /ISAPI/ContentMgmt/InputProxy/channels/<ID>/chanCtrl

Method	GET
Description	Get control parameters of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_ChanCtrl</i> Failed: <i>XML_ResponseStatus</i>

Table 16-17 PUT /ISAPI/ContentMgmt/InputProxy/channels/<ID>/chanCtrl

Method	PUT
Description	Set control parameters of a specific channel.
Query	None.
Request	<i>XML_ChanCtrl</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel ID.

16.2.11 /ISAPI/ContentMgmt/InputProxy/channels/<ID>/deviceInfo

Get or set the device parameters of a digital channel.

Request URI Definition

Table 16-18 GET /ISAPI/ContentMgmt/InputProxy/channels/<ID>/deviceInfo

Method	GET
Description	Get the device parameters of a digital channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_DeviceInfo</i> Failed: <i>XML_ResponseStatus</i>

Table 16-19 PUT /ISAPI/ContentMgmt/InputProxy/channels/<ID>/deviceInfo

Method	PUT
Description	Set the device parameters of a digital channel.
Query	None.
Request	<i>XML_DeviceInfo</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to channel ID.

16.2.12 /ISAPI/ContentMgmt/InputProxy/channels/<ID>/portMapParam

Check if port mapping is required for connecting to camera.

Request URI Definition
Table 16-20 POST /ISAPI/ContentMgmt/InputProxy/channels/<ID>/portMapParam

Method	POST
Description	Check if port mapping is required for connecting to camera according to the IP address of terminal.
Query	None.
Request	<i>XML_PortMapParam</i>
Response	Succeeded: <i>XML_PortMapParamRet</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel ID.

16.2.13 /ISAPI/ContentMgmt/InputProxy/channels/<ID>/reboot

Reboot a specific channel.

Request URI Definition
Table 16-21 GET /ISAPI/ContentMgmt/InputProxy/channels/<ID>/reboot

Method	GET
Description	Reboot a specific channel.

Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel ID.

16.2.14 /ISAPI/ContentMgmt/InputProxy/channels/<ID>/status

Get status of a specific digital channel.

Request URI Definition

Table 16-22 GET /ISAPI/ContentMgmt/InputProxy/channels/<ID>/status

Method	GET
Description	Get status of a specific digital channel.
Query	None.
Request	None.
Response	<i>XML_InputProxyChannelStatus</i>

Remarks

The <ID> in the request URI refers to the digital channel ID.

16.2.15 /ISAPI/ContentMgmt/InputProxy/channels/<ID>/video/overlays

Get or set text overlay parameters of a specific video digital input channel.

Request URI Definition

Table 16-23 GET /ISAPI/ContentMgmt/InputProxy/channels/<ID>/video/overlays

Method	GET
Description	Get text overlay parameters of a specific video digital input channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Channel_VideoOverlay</i> Failed: <i>XML_ResponseStatus</i>

Table 16-24 PUT /ISAPI/ContentMgmt/InputProxy/channels/<ID>/video/overlays

Method	PUT
Description	Set text overlay parameters of a specific video digital input channel.
Query	None.
Request	<i>XML_Channel_VideoOverlay</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

16.2.16 /ISAPI/ContentMgmt/InputProxy/channels/<ID>/video/overlays/capabilities

Get the text overlay configuration capability of a specific video digital input channel.

Request URI Definition
Table 16-25 GET /ISAPI/ContentMgmt/InputProxy/channels/<ID>/video/overlays/capabilities

Method	GET
Description	Get the text overlay configuration capability of a specific video digital input channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_ChannelCap_VideoOverlay</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

16.2.17 /ISAPI/ContentMgmt/InputProxy/channels/activate

Activate the network devices.

Request URI Definition

Table 16-26 PUT /ISAPI/ContentMgmt/InputProxy/channels/activate

Method	PUT
Description	Activate the network devices.
Query	None
Request	<i>XML_VideoSourceActivation</i>
Response	<i>XML_ResponseStatus</i>

16.2.18 /ISAPI/ContentMgmt/InputProxy/channels/activate/capabilities

Get capability of activating network devices.

Request URI Definition

Table 16-27 GET /ISAPI/ContentMgmt/InputProxy/channels/activate/capabilities

Method	GET
Description	Get capability of activating network devices.
Query	None
Request	None.
Response	Succeeded: <i>XML_VideoSourceActivationCapability</i> Failed: <i>XML_ResponseStatus</i>

16.2.19 /ISAPI/ContentMgmt/InputProxy/channels/capabilities

Get management capability of all digital channels.

Request URI Definition

Table 16-28 GET /ISAPI/ContentMgmt/InputProxy/channels/capabilities

Method	GET
Description	Get management capability of all digital channels.
Query	None.
Request	None.
Response	<i>XML_Cap_InputProxyChannelListCap</i>

16.2.20 /ISAPI/ContentMgmt/InputProxy/channels/status

Get status of all digital channels.

Request URI Definition

Table 16-29 GET /ISAPI/ContentMgmt/InputProxy/channels/status

Method	GET
Description	Get status of all digital channels.
Query	None.
Request	None.
Response	<i>XML_InputProxyChannelStatusList</i>

16.2.21 /ISAPI/ContentMgmt/InputProxy/deviceInfo

Get device parameters of all digital channels.

Request URI Definition

Table 16-30 POST /ISAPI/ContentMgmt/InputProxy/deviceInfo

Method	POST
Description	Get device parameters of all digital channels.
Query	None.
Request	<i>XML_ProxyDevicesInfoCondition</i>
Response	Succeeded: <i>XML_ProxyDevicesInfoResult</i> Failed: <i>XML_ResponseStatus</i>

16.2.22 /ISAPI/ContentMgmt/InputProxy/search

Search for network cameras in the LAN (Local Area Network) that can be connected.

Request URI Definition

Table 16-31 GET /ISAPI/ContentMgmt/InputProxy/search

Method	GET
Description	Search for network cameras in the LAN (Local Area Network) that can be connected.

Query	None.
Request	None.
Response	Succeeded: <i>XML_VideoSourceList</i> Failed: <i>XML_ResponseStatus</i>

16.2.23 /ISAPI/ContentMgmt/InputProxy/sourceCapability

Get supported number of digital channels.

Request URI Definition

Table 16-32 POST /ISAPI/ContentMgmt/InputProxy/sourceCapability

Method	POST
Description	Get supported number of digital channels.
Query	None.
Request	<i>XML_sourceDescriptor</i>
Response	<i>XML_sourceCapability</i>

16.2.24 /ISAPI/ContentMgmt/IOProxy/inputs

Operations about the alarm input configuration of all digital channels.

Request URI Definition

Table 16-33 GET /ISAPI/ContentMgmt/IOProxy/inputs

Method	GET
Description	Get the alarm input parameters of all digital channels.
Query	None.
Request	None.
Response	Succeeded: <i>XML_IOProxyInputPortList</i> Failed: <i>XML_ResponseStatus</i>

Table 16-34 PUT /ISAPI/ContentMgmt/IOProxy/inputs

Method	PUT
Description	Set the alarm input parameters of all digital channels.
Query	None.

Request	<i>XML_IOProxyInputPortList</i>
Response	<i>XML_ResponseStatus</i>

Table 16-35 POST /ISAPI/ContentMgmt/IOProxy/inputs

Method	POST
Description	Add the alarm input configuration of a specific digital channel.
Query	None.
Request	<i>XML_IOProxyInputPort</i>
Response	<i>XML_ResponseStatus</i>

16.2.25 /ISAPI/ContentMgmt/IOProxy/inputs/<ID>

Operations about the alarm input configuration of a specific digital channel.

Request URI Definition

Table 16-36 GET /ISAPI/ContentMgmt/IOProxy/inputs/<ID>

Method	GET
Description	Get the alarm input parameters of a specific digital channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_IOProxyInputPort</i> Failed: <i>XML_ResponseStatus</i>

Table 16-37 PUT /ISAPI/ContentMgmt/IOProxy/inputs/<ID>

Method	PUT
Description	Set the alarm input parameters of a specific digital channel.
Query	None.
Request	<i>XML_IOProxyInputPort</i>
Response	<i>XML_ResponseStatus</i>

Table 16-38 DELETE /ISAPI/ContentMgmt/IOProxy/inputs/<ID>

Method	DELETE
Description	Delete the alarm input configuration of a specific digital channel.
Query	None.

Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the digital channel No.

16.2.26 /ISAPI/ContentMgmt/IOProxy/inputs/<ID>/status

Get the alarm input status of a specific digital channel.

Request URI Definition

Table 16-39 GET /ISAPI/ContentMgmt/IOProxy/inputs/<ID>/status

Method	GET
Description	Get the alarm input status of a specific digital channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_IOPortStatus</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the digital channel No.

16.2.27 /ISAPI/ContentMgmt/IOProxy/outputs

Operations about the alarm output configuration of all digital channels.

Request URI Definition

Table 16-40 GET /ISAPI/ContentMgmt/IOProxy/outputs

Method	GET
Description	Get the alarm output parameters of all digital channels.
Query	None.
Request	None.
Response	Succeeded: <i>XML_IOProxyOutputPortList</i> Failed: <i>XML_ResponseStatus</i>

Table 16-41 PUT /ISAPI/ContentMgmt/IOProxy/outputs

Method	PUT
Description	Set the alarm output parameters of all digital channels.
Query	None.
Request	<i>XML_IOProxyOutputPortList</i>
Response	<i>XML_ResponseStatus</i>

Table 16-42 POST /ISAPI/ContentMgmt/IOProxy/outputs

Method	POST
Description	Add the alarm output configuration of a specific digital channel.
Query	None.
Request	<i>XML_IOProxyOutputPort</i>
Response	<i>XML_ResponseStatus</i>

16.2.28 /ISAPI/ContentMgmt/IOProxy/outputs/<ID>

Operations about the alarm output configuration of a specific digital channel.

Request URI Definition

Table 16-43 GET /ISAPI/ContentMgmt/IOProxy/outputs/<ID>

Method	GET
Description	Get the alarm output parameters of a specific digital channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_IOProxyOutputPort</i> Failed: <i>XML_ResponseStatus</i>

Table 16-44 PUT /ISAPI/ContentMgmt/IOProxy/outputs/<ID>

Method	PUT
Description	Set the alarm output parameters of a specific digital channel.
Query	None.
Request	<i>XML_IOProxyOutputPort</i>
Response	<i>XML_ResponseStatus</i>

Table 16-45 DELETE /ISAPI/ContentMgmt/IOProxy/outputs/<ID>

Method	DELETE
Description	Delete the alarm output configuration of a specific digital channel.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the digital channel No.

16.2.29 /ISAPI/ContentMgmt/IOProxy/outputs/<ID>/status

Get the alarm output status of a specific digital channel.

Request URI Definition
Table 16-46 GET /ISAPI/ContentMgmt/IOProxy/outputs/<ID>/status

Method	GET
Description	Get the alarm output status of a specific digital channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_IOPortStatus</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the digital channel No.

16.2.30 /ISAPI/ContentMgmt/IOProxy/outputs/<ID>/trigger

Trigger alarm output of a specific digital channel.

Request URI Definition
Table 16-47 PUT /ISAPI/ContentMgmt/IOProxy/outputs/<ID>/trigger

Method	PUT
Description	Trigger alarm output of a specific digital channel.
Query	None.

Request	<i>XML_IOPortData</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the digital channel No.

16.2.31 /ISAPI/ContentMgmt/logConfig

Get or set log parameters.

Request URI Definition

Table 16-48 GET /ISAPI/ContentMgmt/logConfig

Method	GET
Description	Get log parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_LogConfig</i> Failed: <i>XML_ResponseStatus</i>

Table 16-49 PUT /ISAPI/ContentMgmt/logConfig

Method	PUT
Description	Set log parameters.
Query	None.
Request	<i>XML_LogConfig</i>
Response	<i>XML_ResponseStatus</i>

16.2.32 /ISAPI/ContentMgmt/logConfig/capabilities

Get log configuration capability.

Request URI Definition

Table 16-50 GET /ISAPI/ContentMgmt/logConfig/capabilities

Method	GET
Description	Get log configuration capability.
Query	None.

Request	None.
Response	Succeeded: <i>XML_Cap_LogConfig</i> Failed: <i>XML_ResponseStatus</i>

16.2.33 /ISAPI/ContentMgmt/logSearch

Search for log files.

Request URI Definition

Table 16-51 POST /ISAPI/ContentMgmt/logSearch

Method	POST
Description	Search for log files.
Query	None.
Request	<i>XML_Log_CMSearchDescription</i>
Response	Succeeded: <i>XML_Log_CMSearchResult</i> Failed: <i>XML_ResponseStatus</i>

16.2.34 /ISAPI/ContentMgmt/logSearch?format=json

Search log information.

Request URI Definition

Table 16-52 POST /ISAPI/ContentMgmt/logSearch?format=json

Method	POST
Description	Search log information.
Query	format: determine the format of request or response message.
Request	<i>JSON_SearchCondition</i>
Response	Succeeded: <i>JSON_MatchList</i> Failed: <i>JSON_ResponseStatus</i>

16.2.35 /ISAPI/ContentMgmt/logSearch/dataPackage

Export device log files.

Request URI Definition

Table 16-53 POST /ISAPI/ContentMgmt/logSearch/dataPackage

Method	POST
Description	Export device log files.
Query	None.
Request	<i>XML_CMSearchDataPackage</i>
Response	Succeeded: <i>XML_CMSearchDataPackageResult</i> Failed: <i>XML_ResponseStatus</i>

16.2.36 /ISAPI/ContentMgmt/record/control/cleanUp/capabilities?format=json

Get the capability of deleting saved files in a batch.

Request URI Definition

Table 16-54 GET /ISAPI/ContentMgmt/record/control/cleanUp/capabilities?format=json

Method	GET
Description	Get the capability of deleting saved files in a batch.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_Cap_CleanUpFileList</i> Failed: <i>JSON_ResponseStatus</i>

16.2.37 /ISAPI/ContentMgmt/record/control/cleanUp?format=json

Delete saved files in a batch.

Request URI Definition

Table 16-55 POST /ISAPI/ContentMgmt/record/control/cleanUp?format=json

Method	POST
Description	Delete saved files in a batch.
Query	format: determine the format of request or response message.

Request	<i>JSON_CleanUpFileList</i>
Response	<i>JSON_ResponseStatus</i>

16.2.38 /ISAPI/ContentMgmt/record/control/locks/capabilities

Get video locking or unlocking capability.

Request URI Definition

Table 16-56 GET /ISAPI/ContentMgmt/record/control/locks/capabilities

Method	GET
Description	Get video locking or unlocking capability.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	None
Response	Succeeded: <i>XML_LockCap</i> Failed: <i>XML_ResponseStatus</i>

16.2.39 /ISAPI/ContentMgmt/record/control/locks/name

Lock or unlock videos by file name.

Request URI Definition

Table 16-57 PUT /ISAPI/ContentMgmt/record/control/locks/name

Method	PUT
Description	Lock or unlock videos by file name.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode.

	iv : the initialization vector, and it is required when security is 1 or 2.
Request	<i>XML_LockByName</i>
Response	<i>XML_ResponseStatus</i>

16.2.40 /ISAPI/ContentMgmt/record/control/manual/start/tracks/<ID>

Start manual recording.

Request URI Definition

Table 16-58 POST /ISAPI/ContentMgmt/record/control/manual/start/tracks/<ID>

Method	POST
Description	Start manual recording.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the recording channel ID.

16.2.41 /ISAPI/ContentMgmt/record/control/manual/stop/tracks/<ID>

Stop manual recording of a specific channel.

Request URI Definition

Table 16-59 POST /ISAPI/ContentMgmt/record/control/manual/stop/tracks/<ID>

Method	POST
Description	Stop manual recording of a specific channel.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the recording channel ID.

16.2.42 /ISAPI/ContentMgmt/record/control/manualRefresh/channels/<ID>

Refresh the video mode manually before playback.

Request URI Definition

Table 16-60 PUT /ISAPI/ContentMgmt/record/control/manualRefresh/channels/<ID>

Method	PUT
Description	Refresh the video mode manually before playback.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

16.2.43 /ISAPI/ContentMgmt/record/control/remark/capabilities?format=json

Get capability of adding remarks for videos by video ID.

Request URI Definition

Table 16-61 GET /ISAPI/ContentMgmt/record/control/remark/capabilities?format=json

Method	GET
Description	Get capability of adding remarks for videos by video ID.
Query	format: determine the format of request or response message.
Request	None
Response	Succeeded: <i>JSON_Cap_RemarkList</i> Failed: <i>XML_ResponseStatus</i>

16.2.44 /ISAPI/ContentMgmt/record/control/remark?format=json

Add remarks for videos by video ID.

Request URI Definition

Table 16-62 PUT /ISAPI/ContentMgmt/record/control/remark?format=json

Method	PUT
Description	Add remarks for videos by video ID.
Query	format : determine the format of request or response message.
Request	<i>JSON_RemarkList</i>
Response	<i>XML_ResponseStatus</i>

16.2.45 /ISAPI/ContentMgmt/record/profile

Get video track type.

Request URI Definition

Table 16-63 GET /ISAPI/ContentMgmt/record/profile

Method	GET
Description	Get video track type.
Query	None.
Request	None.
Response	Succeeded: <i>XML_CMSRecordProfile</i> Failed: <i>XML_ResponseStatus</i>

16.2.46 /ISAPI/ContentMgmt/record/storageMounts

Set recording storage parameters.

Request URI Definition

Table 16-64 PUT /ISAPI/ContentMgmt/record/storageMounts

Method	PUT
Description	Set recording storage parameters, including root directory, size, and so on.
Query	None.
Request	<i>XML_MountList</i>
Response	<i>XML_ResponseStatus</i>

16.2.47 /ISAPI/ContentMgmt/record/tracks

Operations about recording schedule configuration.

Request URI Definition

Table 16-65 GET /ISAPI/ContentMgmt/record/tracks

Method	GET
Description	Get all recording schedules.
Query	None.
Request	None.
Response	Succeeded: <i>XML_TrackList</i> Failed: <i>XML_ResponseStatus</i>

Table 16-66 PUT /ISAPI/ContentMgmt/record/tracks

Method	PUT
Description	Set all recording schedules.
Query	None.
Request	<i>XML_TrackList</i>
Response	<i>XML_ResponseStatus</i>

Table 16-67 POST /ISAPI/ContentMgmt/record/tracks

Method	POST
Description	Add a recording schedule.
Query	None.
Request	<i>XML_Track</i>
Response	<i>XML_ResponseStatus</i>

16.2.48 /ISAPI/ContentMgmt/record/tracks/<ID>

Operations about a recording schedule.

Request URI Definition

Table 16-68 GET /ISAPI/ContentMgmt/record/tracks/<ID>

Method	GET
Description	Get a recording schedule.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Track</i> Failed: <i>XML_ResponseStatus</i>

Table 16-69 PUT /ISAPI/ContentMgmt/record/tracks/<ID>

Method	PUT
Description	Set a recording schedule.
Query	None.
Request	<i>XML_Track</i>
Response	<i>XML_ResponseStatus</i>

Table 16-70 DELETE /ISAPI/ContentMgmt/record/tracks/<ID>

Method	DELETE
Description	Delete a recording schedule.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the recording schedule ID.

16.2.49 /ISAPI/ContentMgmt/record/tracks/<ID>/capabilities

Get the configuration capability of the recording schedule.

Request URI Definition

Table 16-71 GET /ISAPI/ContentMgmt/record/tracks/<ID>/capabilities

Method	GET
Description	Get the configuration capability of the recording schedule.

Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_Track</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the recording schedule ID.

16.2.50 /ISAPI/ContentMgmt/record/tracks/<ID>/dailyDistribution

Search recorded video files by date.

Request URI Definition

Table 16-72 POST /ISAPI/ContentMgmt/record/tracks/<ID>/dailyDistribution

Method	POST
Description	Search recorded video files by date.
Query	None.
Request	<i>XML_trackDailyParam</i>
Response	<i>XML_trackDailyDistribution</i>

Remarks

The <ID> in the request URI refers to the recording channel ID.

16.2.51 /ISAPI/ContentMgmt/search

Search for specified resources.

Request URI Definition

Table 16-73 GET or POST /ISAPI/ContentMgmt/search

Method	GET or POST
Description	Search for specific resources.
Query	None.
Request	<i>XML_VideoPic_CMSearchDescription</i>
Response	Succeeded: <i>XML_VideoPic_CMSearchResult</i> Failed: <i>XML_ResponseStatus</i>

Example

Sample Code for Searching Video Files Stored in Device

```
POST /ISAPI/ContentMgmt/search HTTP/1.1
Host: 10.17.132.49
Content-Length: 493
Connection: Keep-Alive
Authorization: Digest username="admin",
realm="DS-2CD2F12FWD-IWS",
nonce="4e7a46474e305a454d5445365a4445314e6a51345a54413d",
uri="/ISAPI/ContentMgmt/search",
cnonce="ce22590094d2f2bb352fc3c4cd2a1ca3",
nc=00000019,
response="ad6f2c23636f25c6db5911a113375ea9",
qop="auth"

<?xml version="1.0" encoding="utf-8"?>
<CMSearchDescription>
  <searchID>C77384AD-66A0-0001-E7C2-1151F04F90B0</searchID>
  <trackIDList>
    <trackID>101</trackID>
  </trackIDList>
  <timeSpanList>
    <timeSpan>
      <startTime>2017-03-13T16:00:00Z</startTime>
      <endTime>2017-03-16T15:59:59Z</endTime>
    </timeSpan>
  </timeSpanList>
  <maxResults>40</maxResults>
  <searchResultPostion>0</searchResultPostion>
  <metadataList>
    <metadataDescriptor>//recordType.meta.std-cgi.com</metadataDescriptor>
  </metadataList>
</CMSearchDescription>

HTTP/1.1 200 OK
Date: Wed, 15 Mar 2017 09:40:02 GMT
Connection: close
Content-Length: 1649
Content-Type: application/xml

<?xml version="1.0" encoding="UTF-8"?>
<CMSearchResult version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <searchID>{C77384AD-66A0-0001-E7C2-1151F04F90B0}</searchID>
  <responseStatus>true</responseStatus>
  <responseStatusStrg>OK</responseStatusStrg>
  <numOfMatches>29</numOfMatches>
  <matchList>
    <searchMatchItem>
      <sourceID>{0000000000-0000-0000-0000-000000000000}</sourceID>
      <trackID>101</trackID>
      <timeSpan>
```

```

    <startTime>2017-03-14T10:32:01Z</startTime>
    <endTime>2017-03-14T10:40:42Z</endTime>
  </timeSpan>
  <mediaSegmentDescriptor>
    <contentType>video</contentType>
    <codecType>H.264-BP</codecType>
    <playbackURI>rtsp://10.17.132.49/Streaming/tracks/101/?
starttime=20170314T103201Z&endtime=20170314T104042Z&name=ch01_08000000016000000&size
=260358144</playbackURI>
  </mediaSegmentDescriptor>
  <metadataMatches>
    <metadataDescriptor>recordType.meta.isapi.com/timing</metadataDescriptor>
  </metadataMatches>
</searchMatchItem>
<searchMatchItem>
  <sourceID>{0000000000-0000-0000-0000-000000000000}</sourceID>
  <trackID>101</trackID>
  <timeSpan>
    <startTime>2017-03-14T10:40:42Z</startTime>
    <endTime>2017-03-14T10:53:14Z</endTime>
  </timeSpan>
  <mediaSegmentDescriptor>
    <contentType>video</contentType>
    <codecType>H.264-BP</codecType>
    <playbackURI>rtsp://10.17.132.49/Streaming/tracks/101/?
starttime=20170314T104042Z&endtime=20170314T105314Z&name=ch01_08000000017000000&size
=260603904</playbackURI>
  </mediaSegmentDescriptor>
  <metadataMatches>
    <metadataDescriptor>recordType.meta.isapi.com/timing</metadataDescriptor>
  </metadataMatches>
</searchMatchItem>
</matchList>
</CMSearchResult>

```

16.2.52 /ISAPI/ContentMgmt/search/profile

Get video search conditions.

Request URI Definition

Table 16-74 GET /ISAPI/ContentMgmt/search/profile

Method	GET
Description	Get video search conditions.
Query	None.

Request	None.
Response	Succeeded: <i>XML_CMSearchProfile</i> Failed: <i>XML_ResponseStatus</i>

16.2.53 /ISAPI/ContentMgmt/security/logSearch

Search for security logs.

Request URI Definition

Table 16-75 POST /ISAPI/ContentMgmt/security/logSearch

Method	POST
Description	Search for security logs.
Query	None.
Request	<i>XML_VideoPic_CMSearchDescription</i>
Response	Succeeded: <i>XML_VideoPic_CMSearchResult</i> Failed: <i>XML_ResponseStatus</i>

16.2.54 /ISAPI/ContentMgmt/SmartSearch

Search for video files by VCA event.

Request URI Definition

Table 16-76 POST /ISAPI/ContentMgmt/SmartSearch

Method	POST
Description	Search for video files by VCA event.
Query	None.
Request	<i>XML_SmartSearchDescription</i>
Response	<i>XML_SmartSearchResult</i>

16.2.55 /ISAPI/ContentMgmt/SmartSearch/capabilities

Get capability of searching videos by VCA event.

Request URI Definition

Table 16-77 GET /ISAPI/ContentMgmt/SmartSearch/capabilities

Method	GET
Description	Get capability of searching videos by VCA event.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_SmartSearchDescription</i> Failed: <i>XML_ResponseStatus</i>

16.2.56 /ISAPI/ContentMgmt/Storage/extension

Get or set parameters of storage strategy.

Request URI Definition

Table 16-78 GET /ISAPI/ContentMgmt/Storage/extension

Method	GET
Description	Get parameters of storage strategy.
Query	None.
Request	None.
Response	Succeeded: <i>XML_storageExtension</i> Failed: <i>XML_ResponseStatus</i>

Table 16-79 PUT /ISAPI/ContentMgmt/Storage/extension

Method	PUT
Description	Set parameters of storage strategy.
Query	None.
Request	<i>XML_storageExtension</i>
Response	<i>XML_ResponseStatus</i>

16.2.57 /ISAPI/ContentMgmt/Storage/extension/capabilities

Get configuration capability of log storage.

Request URI Definition

Table 16-80 GET /ISAPI/ContentMgmt/Storage/extension/capabilities

Method	GET
Description	Get configuration capability of log storage.
Query	None.
Request	None
Response	Succeeded: <i>XML_Cap_storageExtension</i> Failed: <i>JSON_ResponseStatus</i>

16.2.58 /ISAPI/ContentMgmt/Storage/ExtraInfo

Operations about the configuration for storing additional information.

Request URI Definition

Table 16-81 GET /ISAPI/ContentMgmt/Storage/ExtraInfo

Method	GET
Description	Get the parameters for storing additional information.
Query	None.
Request	None.
Response	Succeeded: <i>XML_ExtraInfo</i> Failed: <i>XML_ResponseStatus</i>

Table 16-82 PUT /ISAPI/ContentMgmt/Storage/ExtraInfo

Method	PUT
Description	Set the parameters for storing additional information.
Query	None.
Request	<i>XML_ExtraInfo</i>
Response	<i>XML_ResponseStatus</i>

16.2.59 /ISAPI/ContentMgmt/Storage/ExtraInfo/capabilities

Get the capability of storing additional information.

Request URI Definition

Table 16-83 GET /ISAPI/ContentMgmt/Storage/ExtraInfo/capabilities

Method	GET
Description	Get the capability of storing additional information.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_ExtraInfo</i> Failed: <i>XML_ResponseStatus</i>

16.2.60 /ISAPI/ContentMgmt/Storage/hdd

Get configuration parameters of all HDDs.

Request URI Definition

Table 16-84 GET /ISAPI/ContentMgmt/Storage/hdd

Method	GET
Description	Get configuration parameters of all HDDs.
Query	None.
Request	None.
Response	Succeeded: <i>XML_hddList</i> Failed: <i>XML_ResponseStatus</i>

16.2.61 /ISAPI/ContentMgmt/Storage/hdd/<ID>

Get and set parameters for a specific HDD.

Request URI Definition

Table 16-85 GET /ISAPI/ContentMgmt/Storage/hdd/<ID>

Method	GET
Description	Get parameters for a specific HDD.
Query	None.

Request	None.
Response	Succeeded: <i>XML_hdd</i> Failed: <i>XML_ResponseStatus</i>

Table 16-86 PUT /ISAPI/ContentMgmt/Storage/hdd/<ID>

Method	PUT
Description	Set parameters for a specific HDD.
Query	None.
Request	<i>XML_hdd</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the HDD ID.

16.2.62 /ISAPI/ContentMgmt/Storage/hdd/<ID>/BadSectorsTest/pause

Pause checking bad sectors of a HDD.

Request URI Definition

Table 16-87 PUT /ISAPI/ContentMgmt/Storage/hdd/<ID>/BadSectorsTest/pause

Method	PUT
Description	Pause checking bad sectors of a HDD.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the HDD No.

16.2.63 /ISAPI/ContentMgmt/Storage/hdd/<ID>/BadSectorsTest/resume

Resume checking bad sectors of a HDD.

Request URI Definition**Table 16-88 PUT /ISAPI/ContentMgmt/Storage/hdd/<ID>/BadSectorsTest/resume**

Method	PUT
Description	Resume checking bad sectors of a HDD.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the HDD No.

16.2.64 /ISAPI/ContentMgmt/Storage/hdd/<ID>/BadSectorsTest/start

Start checking bad sectors of a HDD.

Request URI Definition**Table 16-89 PUT /ISAPI/ContentMgmt/Storage/hdd/<ID>/BadSectorsTest/start**

Method	PUT
Description	Start checking bad sectors of a HDD.
Query	None.
Request	<i>XML_BadSectorsTest</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the HDD No.

16.2.65 /ISAPI/ContentMgmt/Storage/hdd/<ID>/BadSectorsTest/status

Get bad sector checking status of a HDD.

Request URI Definition**Table 16-90 GET /ISAPI/ContentMgmt/Storage/hdd/<ID>/BadSectorsTest/status**

Method	GET
Description	Get bad sector checking status of a HDD.

Query	None.
Request	None.
Response	Succeeded: <i>XML_BadSectorsTestStatus</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the HDD No.

16.2.66 /ISAPI/ContentMgmt/Storage/hdd/<ID>/BadSectorsTest/stop

Stop checking bad sectors of a HDD.

Request URI Definition

Table 16-91 PUT /ISAPI/ContentMgmt/Storage/hdd/<ID>/BadSectorsTest/stop

Method	PUT
Description	Stop checking bad sectors of a HDD.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the HDD No.

16.2.67 /ISAPI/ContentMgmt/Storage/hdd/<ID>/encryptFormat?format=json

Format an encrypted HDD.

Request URI Definition

Table 16-92 PUT /ISAPI/ContentMgmt/Storage/hdd/<ID>/encryptFormat?format=json

Method	PUT
Description	Format an encrypted HDD.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates

	that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2. format : determine the format of request or response message.
Request	<i>JSON_EncryptFormat</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the HDD ID.

16.2.68 /ISAPI/ContentMgmt/Storage/hdd/<ID>/encryptVerify?format=json

Verify the HDD encryption password.

Request URI Definition

Table 16-93 PUT /ISAPI/ContentMgmt/Storage/hdd/<ID>/encryptVerify?format=json

Method	PUT
Description	Verify the HDD encryption password.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2. format : determine the format of request or response message.
Request	<i>JSON_EncryptVerify</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the HDD ID.

16.2.69 /ISAPI/ContentMgmt/Storage/hdd/<ID>/formatStatus

Get the formatting status of a specific HDD.

Request URI Definition

Table 16-94 GET /ISAPI/ContentMgmt/Storage/hdd/<ID>/formatStatus

Method	GET
Description	Get the formatting status of a specific HDD.
Query	None.
Request	None.
Response	Succeeded: <i>XML_formatStatus</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the HDD ID.

16.2.70 /ISAPI/ContentMgmt/Storage/hdd/<ID>/format?formatType=

Format a specific HDD according to the specified formatting type. This URI is only available for SD card.

Request URI Definition

Table 16-95 PUT /ISAPI/ContentMgmt/Storage/hdd/<ID>/format?formatType=

Method	PUT
Description	Format a specific HDD according to the specified formatting type.
Query	formatType : formatting type, the available values are "EXT4" and "FAT32". But if you adopt the default formatting type "FAT32", this query parameter is not required in the URI.
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the HDD ID.

16.2.71 /ISAPI/ContentMgmt/Storage/hdd/<ID>/logProTest?format=json

Configure preverification for log HDDs.

Request URI Definition**Table 16-96 PUT /ISAPI/ContentMgmt/Storage/hdd/<ID>/logProTest?format=json**

Method	PUT
Description	Configure preverification for log HDD.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_logProTestResult</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the HDD No.

16.2.72 /ISAPI/ContentMgmt/Storage/hdd/<ID>/SMARTTest/start

Start checking HDD status.

Request URI Definition**Table 16-97 PUT /ISAPI/ContentMgmt/Storage/hdd/<ID>/SMARTTest/start**

Method	PUT
Description	Start checking HDD status.
Query	None.
Request	<i>XML_HDDSMARTTest</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the HDD No.

16.2.73 /ISAPI/ContentMgmt/Storage/hdd/<ID>/SMARTTest/status

Get HDD checking status.

Request URI Definition

Table 16-98 GET /ISAPI/ContentMgmt/Storage/hdd/<ID>/SMARTTest/status

Method	GET
Description	Get HDD checking status.
Query	None.
Request	None.
Response	Succeeded: <i>XML_SMARTTestStatus</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the HDD No.

16.2.74 /ISAPI/ContentMgmt/Storage/hdd/capabilities

Get HDD management capability.

Request URI Definition

Table 16-99 GET /ISAPI/ContentMgmt/Storage/hdd/capabilities

Method	GET
Description	Get HDD management capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_hddList</i> Failed: <i>XML_ResponseStatus</i>

16.2.75 /ISAPI/ContentMgmt/storage/hdd/clearingSpace/capabilities?format=json

Get configuration capability of clearing HDD.

Request URI Definition

Table 16-100 GET /ISAPI/ContentMgmt/storage/hdd/clearingSpace/capabilities?format=json

Method	GET
Description	Get configuration capability of clearing HDD.

Query	format: determine the format of request or response message.
Request	None
Response	Succeeded: <i>JSON_Cap_ClearingSpaceConfig</i> Failed: <i>JSON_ResponseStatus</i>

16.2.76 /ISAPI/ContentMgmt/storage/hdd/clearingSpace?format=json

Get or set parameters of clearing HDD.

Request URI Definition

Table 16-101 GET /ISAPI/ContentMgmt/storage/hdd/clearingSpace?format=json

Method	GET
Description	Get parameters of clearing HDD.
Query	format: determine the format of request or response message.
Request	None
Response	Succeeded: <i>JSON_ClearingSpaceConfig</i> Failed: <i>JSON_ResponseStatus</i>

Table 16-102 PUT /ISAPI/ContentMgmt/storage/hdd/clearingSpace?format=json

Method	PUT
Description	Set parameters of clearing HDD.
Query	format: determine the format of request or response message.
Request	<i>JSON_ClearingSpaceConfig</i>
Response	<i>JSON_ResponseStatus</i>

16.2.77 /ISAPI/ContentMgmt/Storage/hdd/format

Format all HDDs.

Request URI Definition

Table 16-103 PUT /ISAPI/ContentMgmt/Storage/hdd/format

Method	PUT
Description	Format all HDDs.
Query	None.

Request	None.
Response	<i>XML_ResponseStatus</i>

16.2.78 /ISAPI/ContentMgmt/Storage/hdd/SMARTTest/config

Operations about the configuration of HDD SMART status detection.

Request URI Definition

Table 16-104 GET /ISAPI/ContentMgmt/Storage/hdd/SMARTTest/config

Method	GET
Description	Get the parameters of HDD SMART status detection.
Query	None.
Request	None.
Response	Succeeded: <i>XML_SMARTTestConfig</i> Failed: <i>XML_ResponseStatus</i>

Table 16-105 PUT /ISAPI/ContentMgmt/Storage/hdd/SMARTTest/config

Method	PUT
Description	Set the parameters of HDD SMART status detection.
Query	None.
Request	<i>XML_SMARTTestConfig</i>
Response	<i>XML_ResponseStatus</i>

16.2.79 /ISAPI/ContentMgmt/Storage/hdd/specifyHddFormat?format=json

Specify multiple HDDs to be formatted.

Request URI Definition

Table 16-106 PUT /ISAPI/ContentMgmt/Storage/hdd/specifyHddFormat?format=json

Method	PUT
Description	Specify multiple HDDs to format.
Query	format: determine the format of request or response message.
Request	<i>JSON_HddFormatList</i>
Response	<i>JSON_ResponseStatus</i>

16.2.80 /ISAPI/ContentMgmt/Storage/quota

Get parameters of all HDD quotas.

Request URI Definition

Table 16-107 GET /ISAPI/ContentMgmt/Storage/quota

Method	GET
Description	Get parameters of all HDD quotas.
Query	None.
Request	None.
Response	Succeeded: <i>XML_diskQuota</i> Failed: <i>XML_ResponseStatus</i>

16.2.81 /ISAPI/ContentMgmt/StreamingProxy/channels/<ID>/Picture

Capture pictures via digital channel.

Request URI Definition

Table 16-108 GET /ISAPI/ContentMgmt/StreamingProxy/channels/<ID>/Picture

Method	GET
Description	Capture pictures via digital channel.
Query	None
Request	None
Response	Succeeded: Captured picture data, which is transmitted over HTTP. Failed: <i>XML_ResponseStatus</i>

Remarks

- The <ID> in the request URI refers to the digital channel ID.
- The supported picture resolutions include QCIF (176 × 144), CIF (352 × 288) , and D1 (704 × 576).

16.2.82 /ISAPI/ContentMgmt/time/search/capabilities?format=json

Get the capability of searching for recording start and end time.

Request URI Definition

Table 16-109 GET /ISAPI/ContentMgmt/time/search/capabilities?format=json

Method	GET
Description	Get the capability of searching for recording start and end time.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_Cap_TimeSearchCond</i> Failed: <i>JSON_ResponseStatus</i>

16.2.83 /ISAPI/ContentMgmt/time/search?format=json

Search for the recording start and end time.

Request URI Definition

Table 16-110 POST /ISAPI/ContentMgmt/time/search?format=json

Method	POST
Description	Search for the recording start and end time.
Query	format : determine the format of request or response message.
Request	<i>JSON_TimeSearchCond</i>
Response	Succeeded: <i>JSON_TimeSearchResult</i> Failed: <i>JSON_ResponseStatus</i>

16.3 /ISAPI/Event

16.3.1 /ISAPI/Event/capabilities

Get the device event capability.

Request URI Definition

Table 16-111 GET /ISAPI/Event/capabilities

Method	GET
Description	Get the device event capability set.

Query	None.
Request	None.
Response	Succeeded: <i>XML_EventCap</i> Failed: <i>XML_ResponseStatus</i>

16.3.2 /ISAPI/Event/channels/<ID>/capabilities

Get event capabilities supported by the channel.

Request URI Definition

Table 16-112 GET /ISAPI/Event/channels/<ID>/capabilities

Method	GET
Description	Get event capabilities supported by the channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_ChannelEventCap</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI refers to the channel ID.

16.3.3 /ISAPI/Event/channels/capabilities

Get the event capability of all channels.

Request URL Definition

Table 16-113 GET /ISAPI/Event/channels/capabilities

Method	GET
Description	Get the event capability of all channels.
Query	None.
Request	None.
Response	Succeeded: <i>XML_ChannelEventCapList</i> Failed: <i>XML_ResponseStatus</i>

16.3.4 /ISAPI/Event/IOT/channels/<ID>/capabilities?format=json

Get the event capabilities supported by IoT device channel.

Request URI Definition

Table 16-114 GET /ISAPI/Event/IOT/channels/<ID>/capabilities?format=json

Method	GET
Description	Get the event capabilities supported by IoT device channel.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_IOTChannelEventCap</i> Filed: <i>JSON_ResponseStatus</i>

16.3.5 /ISAPI/Event/notification/alertStream

Get the uploaded heartbeat or alarm/event information.

Request URI Definition

Table 16-115 GET /ISAPI/Event/notification/alertStream

Method	GET
Description	Get the heartbeat or uploaded alarm/event information.
Query	None.
Request	None.
Response	Option 1: <i>XML_EventNotificationAlert_AlarmEventInfo</i> or <i>XML_EventNotificationAlert_HeartbeatInfo</i> Option 2: <i>JSON_EventNotificationAlert_Alarm/EventInfo</i>  Note The messages here only show the format of alarm/event information to be uploaded. For details, refer to the corresponding alarm/event configuration chapters.

Remarks

- After calling this URI, a persistent connection is set up between the device and the platform, and the alarm or event information will be uploaded from device continuously once the alarm is triggered or event occurred.
- You can check if the XML response message is the heartbeat information according to the nodes **<eventType>** and **<eventState>**. If the values of these two node are "videoloss" and "inactive", respectively, the returned message is the heartbeat information.

16.3.6 /ISAPI/Event/notification/httpHosts

Get or set parameters of all HTTP listening servers, add a HTTP listening server, and delete all HTTP listening servers.

Request URI Definition

Table 16-116 GET /ISAPI/Event/notification/httpHosts

Method	GET
Description	Get parameters of all HTTP listening servers.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode.
Request	None
Response	Succeeded: <i>XML_HttpHostNotificationList</i> Failed: <i>XML_ResponseStatus</i>

Table 16-117 PUT /ISAPI/Event/notification/httpHosts

Method	PUT
Description	Set parameters of all HTTP listening servers.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode.

Request	<i>XML_HttpHostNotificationList</i>
Response	<i>XML_ResponseStatus</i>

Table 16-118 POST /ISAPI/Event/notification/httpHosts

Method	POST
Description	Add a HTTP listening server.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode.
Request	<i>XML_HttpHostNotification</i>
Response	<i>XML_ResponseStatus</i>

Table 16-119 DELETE /ISAPI/Event/notification/httpHosts

Method	DELETE
Description	Delete all HTTP listening servers.
Query	None
Request	None
Response	<i>XML_ResponseStatus</i>

16.3.7 /ISAPI/Event/notification/httpHosts/<ID>

Get or set the parameters of a HTTP listening server, and delete a HTTP listening server.

Request URI Definition

Table 16-120 GET /ISAPI/Event/notification/httpHosts/<ID>

Method	GET
Description	Get the parameters of a HTTP listening server.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode.

Request	None
Response	Succeeded: <i>XML_HttpHostNotification</i> Failed: <i>XML_ResponseStatus</i>

Table 16-121 PUT /ISAPI/Event/notification/httpHosts/<ID>

Method	PUT
Description	Set the parameters of a HTTP listening server.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode.
Request	<i>XML_HttpHostNotification</i>
Response	<i>XML_ResponseStatus</i>

Table 16-122 DELETE /ISAPI/Event/notification/httpHosts/<ID>

Method	DELETE
Description	Delete a HTTP listening server.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode.
Request	None
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the HTTP listening server No.

16.3.8 /ISAPI/Event/notification/httpHosts/<ID>/test

Check whether the HTTP listening server is working normally.

Request URI Definition

Table 16-123 POST /ISAPI/Event/notification/httpHosts/<ID>/test

Method	POST
Description	Check whether the HTTP listening server is working normally.
Query	None
Request	<i>XML_HttpHostNotification</i>
Response	Succeeded: <i>XML_HttpHostTestResult</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the HTTP listening server ID.

16.3.9 /ISAPI/Event/notification/httpHosts/capabilities

Get the configuration capabilities of all HTTP listening servers.

Request URI Definition

Table 16-124 GET /ISAPI/Event/notification/httpHosts/capabilities

Method	GET
Description	Get the configuration capabilities of all HTTP listening servers.
Query	None
Request	None
Response	Succeeded: <i>XML_HttpHostNotificationCap</i> Failed: <i>XML_ResponseStatus</i>

16.3.10 /ISAPI/Event/notification/subscribeEvent

Subscribe events/alarms in arming mode.

Request URI Definition

Table 16-125 POST /ISAPI/Event/notification/subscribeEvent

Method	POST
Description	Subscribe events/alarms in arming mode.

Query	None.
Request	<i>XML_SubscribeEvent</i>
Response	Succeeded: <i>XML_SubscribeEventResponse</i> or <i>XML_EventNotificationAlert_SubscriptionHeartbeat</i> and alarm/event details message Failed: <i>XML_ResponseStatus</i>

Remarks

The ***XML_EventNotificationAlert_SubscriptionHeartbeat*** and alarm/event details message is uploaded repeatedly. The default time interval of uploading heartbeat information is 30s.

16.3.11 /ISAPI/Event/notification/subscribeEvent/<ID>

Get or set alarm/event subscription parameters.

Request URI Definition**Table 16-126 GET /ISAPI/Event/notification/subscribeEvent/<ID>**

Method	GET
Description	Get alarm/event subscription parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_SubscribeEvent</i> Failed: <i>XML_ResponseStatus</i>

Table 16-127 PUT /ISAPI/Event/notification/subscribeEvent/<ID>

Method	PUT
Description	Set alarm/event subscription parameters.
Query	None.
Request	<i>XML_SubscribeEvent</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the subscription No. which is returned by the device. After the persistent connection for receiving events or alarms in arming mode is closed, the device will release the resource used by the <ID>.

16.3.12 /ISAPI/Event/notification/subscribeEventCap

Get event/alarm subscription capability.

Request URI Definition

Table 16-128 GET /ISAPI/Event/notification/subscribeEventCap

Method	GET
Description	Get event/alarm subscription capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_SubscribeEventCap</i> Failed: <i>XML_ResponseStatus</i>

16.3.13 /ISAPI/Event/notification/unSubscribeEvent

Cancel subscribing alarm/event.

Request URI Definition

Table 16-129 PUT /ISAPI/Event/notification/unSubscribeEvent

Method	PUT
Description	Cancel subscribing alarm/event.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

16.3.14 /ISAPI/Event/schedules/motionDetections

Get or set arming schedule of motion detection of all channels.

Request URI Definition

Table 16-130 GET /ISAPI/Event/schedules/motionDetections

Method	GET
Description	Get arming schedule of motion detection of all channels.
Query	None

Request	None
Response	Succeeded: <i>XML_MotionDetectionScheduleList</i> Failed: <i>XML_ResponseStatus</i>

Table 16-131 PUT /ISAPI/Event/schedules/motionDetections

Method	PUT
Description	Set arming schedule of motion detection of all channels.
Query	None
Request	<i>XML_MotionDetectionScheduleList</i>
Response	<i>XML_ResponseStatus</i>

16.3.15 /ISAPI/Event/schedules/motionDetections/<ID>

Get or set the arming schedule of motion detection by channel.

Request URI Definition

Table 16-132 GET /ISAPI/Event/schedules/motionDetections/<ID>

Method	GET
Description	Get the arming schedule of motion detection by channel.
Query	None
Request	None
Response	Succeeded: <i>XML_Schedule</i> Failed: <i>XML_ResponseStatus</i>

Table 16-133 PUT /ISAPI/Event/schedules/motionDetections/<ID>

Method	PUT
Description	Set the arming schedule of motion detection by channel.
Query	None
Request	<i>XML_Schedule</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the ID of channel that triggered motion detection alarm. It consists of alarm type name ("VMD") and channel ID, its format is "VMD-<channelID>", e.g., "VMD-101".

16.3.16 /ISAPI/Event/schedules/vibrationDetection/<ID>

Get or set arming schedule of vibration detection by channel.

Request URI Definition

Table 16-134 GET /ISAPI/Event/schedules/vibrationDetection/<ID>

Method	GET
Description	Get arming schedule of vibration detection by channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Schedule</i> Failed: <i>XML_ResponseStatus</i>

Table 16-135 PUT /ISAPI/Event/schedules/vibrationDetection/<ID>

Method	PUT
Description	Set arming schedule of vibration detection by channel.
Query	None.
Request	<i>XML_Schedule</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI should be set in the format of "vibrationDetection-<channelID>", e.g., /ISAPI/Event/schedules/vibrationDetection/vibrationDetection-101.

16.3.17 /ISAPI/Event/schedules/<EventType>/<ID>

Operations about arming schedule of specified event type.

Request URI Definition

Table 16-136 GET /ISAPI/Event/schedules/<EventType>/<ID>

Method	GET
Description	Get the arming schedule of specified event type.
Query	None.

Request	None.
Response	Succeeded: <i>XML_Schedule</i> Failed: <i>XML_ResponseStatus</i>

Table 16-137 PUT /ISAPI/Event/schedules/<EventType>/<ID>

Method	PUT
Description	Set the arming schedule of specified event type.
Query	None.
Request	<i>XML_Schedule</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <**EventType**> in the URI refers to the specified event type.

The <**ID**> in the URI refers to the ID of channel, which is triggered by alarm.

16.3.18 /ISAPI/Event/triggers/hdBadBlock

Operations about the linkage configuration of the HDD bad sector detection.

Request URI Definition**Table 16-138 GET /ISAPI/Event/triggers/hdBadBlock**

Method	GET
Description	Get the linkage parameters of the HDD bad sector detection.
Query	None.
Request	None.
Response	Succeeded: <i>XML_EventTrigger</i> Failed: <i>XML_ResponseStatus</i>

Table 16-139 PUT /ISAPI/Event/triggers/hdBadBlock

Method	PUT
Description	Set the linkage parameters of the HDD bad sector detection.
Query	None.
Request	<i>XML_EventTrigger</i>
Response	<i>XML_ResponseStatus</i>

16.3.19 /ISAPI/Event/triggers/hdImpact

Operations about the linkage configuration of the HDD impact detection.

Request URI Definition

Table 16-140 GET /ISAPI/Event/triggers/hdImpact

Method	GET
Description	Get the linkage parameters of the HDD impact detection.
Query	None.
Request	None.
Response	Succeeded: <i>XML_EventTrigger</i> Failed: <i>XML_ResponseStatus</i>

Table 16-141 PUT /ISAPI/Event/triggers/hdImpact

Method	PUT
Description	Set the linkage parameters of the HDD impact detection.
Query	None.
Request	<i>XML_EventTrigger</i>
Response	<i>XML_ResponseStatus</i>

16.3.20 /ISAPI/Event/triggers/highHDTemperature

Operations about the linkage configuration of the HDD high temperature detection.

Request URI Definition

Table 16-142 GET /ISAPI/Event/triggers/highHDTemperature

Method	GET
Description	Get the linkage parameters of the HDD high temperature detection.
Query	None.
Request	None.
Response	Succeeded: <i>XML_EventTrigger</i> Failed: <i>XML_ResponseStatus</i>

Table 16-143 PUT /ISAPI/Event/triggers/highHDTemperature

Method	PUT
Description	Set the linkage parameters of the HDD high temperature detection.
Query	None.
Request	<i>XML_EventTrigger</i>
Response	<i>XML_ResponseStatus</i>

16.3.21 /ISAPI/Event/triggers/lowHDTemperature

Operations about the linkage configuration of the HDD low temperature detection.

Request URI Definition

Table 16-144 GET /ISAPI/Event/triggers/lowHDTemperature

Method	GET
Description	Get the linkage parameters of the HDD low temperature detection.
Query	None.
Request	None.
Response	Succeeded: <i>XML_EventTrigger</i> Failed: <i>XML_ResponseStatus</i>

Table 16-145 PUT /ISAPI/Event/triggers/lowHDTemperature

Method	PUT
Description	Set the linkage parameters of the HDD low temperature detection.
Query	None.
Request	<i>XML_EventTrigger</i>
Response	<i>XML_ResponseStatus</i>

16.3.22 /ISAPI/Event/triggers/severeHDFailure

Operations about the linkage configuration of the HDD severe fault detection.

Request URI Definition**Table 16-146 GET /ISAPI/Event/triggers/severeHDFailure**

Method	GET
Description	Get the linkage parameters of the HDD severe fault detection.
Query	None.
Request	None.
Response	Succeeded: <i>XML_EventTrigger</i> Failed: <i>XML_ResponseStatus</i>

Table 16-147 PUT /ISAPI/Event/triggers/severeHDFailure

Method	PUT
Description	Set the linkage parameters of the HDD severe fault detection.
Query	None.
Request	<i>XML_EventTrigger</i>
Response	<i>XML_ResponseStatus</i>

16.3.23 /ISAPI/Event/triggers/<ID>

Get, set, or delete the alarm linkage action by channel.

Request URI Definition**Table 16-148 GET /ISAPI/Event/triggers/<ID>**

Method	GET
Description	Get the alarm linkage action by channel.
Query	None
Request	None
Response	Succeeded: <i>XML_EventTrigger</i> Failed: <i>XML_ResponseStatus</i>

Table 16-149 PUT /ISAPI/Event/triggers/<ID>

Method	PUT
Description	Set the alarm linkage action by channel.
Query	None

Request	<i>XML_EventTrigger</i>
Response	<i>XML_ResponseStatus</i>

Table 16-150 DELETE /ISAPI/Event/triggers/<ID>

Method	DELETE
Description	Delete the alarm linkage action by channel.
Query	None
Request	None
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI is defined as the ID of the alarm triggered channel. For example, if the No. of the video input channel triggered by face capture alarm is 1, the <ID> is "faceSnap-1".

16.3.24 /ISAPI/Event/triggers/<ID>/notifications

Operations about configurations of alarm/event linkage actions.

Request URL Definition**Table 16-151 GET /ISAPI/Event/triggers/<ID>/notifications**

Method	GET
Description	Get the configuration parameters of alarm/event linkage actions.
Query	None.
Request	None.
Response	Succeeded: <i>XML_EventTriggerNotificationList</i> Failed: <i>XML_ResponseStatus</i>

Table 16-152 PUT /ISAPI/Event/triggers/<ID>/notifications

Method	PUT
Description	Set the configuration parameters for alarm/event linkage actions.
Query	None.
Request	<i>XML_EventTriggerNotificationList</i>
Response	<i>XML_ResponseStatus</i>

Table 16-153 DELETE /ISAPI/Event/triggers/<ID>/notifications

Method	DELETE
Description	Delete the configuration parameters of alarm/event linkage actions.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URL refers to No. of alarm triggered channel.

16.3.25 /ISAPI/Event/triggersCap/IOT

Get the alarm linkage configuration capability of IoT device.

Request URI Definition**Table 16-154 GET /ISAPI/Event/triggersCap/IOT**

Method	GET
Description	Get the alarm linkage configuration capability of IoT device.
Query	None.
Request	None.
Response	<i>XML_IOTTriggersCap</i>

16.3.26 /ISAPI/Event/triggers/notifications/AudioAlarm/<audioID>/test?format=json

Audio test.

Request URI Definition**Table 16-155 PUT /ISAPI/Event/triggers/notifications/AudioAlarm/<audioID>/test?format=json**

Method	PUT
Description	Audio test.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

Remarks

The <audioID> in the URI refers to audio ID.

16.3.27 /ISAPI/Event/triggers/notifications/AudioAlarm/capabilities?format=json

Get audible warning configuration capability.

Request URI Definition

Table 16-156 GET /ISAPI/Event/triggers/notifications/AudioAlarm/capabilities?format=json

Method	GET
Description	Get audible warning configuration capability.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_AudioAlarmCap</i> Failed: <i>JSON_ResponseStatus</i>

16.3.28 /ISAPI/Event/triggers/notifications/AudioAlarm/customAudio/<customAudioID>?format=json

Download or delete custom alarm sound file.

Request URI Definition

Table 16-157 GET /ISAPI/Event/triggers/notifications/AudioAlarm/customAudio/<customAudioID>?format=json

Method	GET
Description	Download custom alarm sound file.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: The custom alarm sound file information is returned. Failed: <i>JSON_ResponseStatus</i>

Table 16-158 DELETE /ISAPI/Event/triggers/notifications/AudioAlarm/customAudio/<customAudioID>?format=json

Method	DELETE
Description	Delete custom audio file.

Query	format : determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

Remarks

<customAudioID> in the URI refers to custom alarm sound ID and the value can be 1 to 3.

16.3.29 /ISAPI/Event/triggers/notifications/AudioAlarm/customAudio?format=json

Import custom alarm sound file.

Request URI Definition

Table 16-159 POST /ISAPI/Event/triggers/notifications/AudioAlarm/customAudio?format=json

Method	POST
Description	Import custom alarm sound file.
Query	format : determine the format of request or response message.
Request	<i>JSON_ImportCustomAudio</i>
Response	Succeeded: <i>JSON_CustomAudioID</i> Failed: <i>JSON_ResponseStatus</i>

16.3.30 /ISAPI/Event/triggers/notifications/AudioAlarm/customAudioInfo/<customAudioID>?format=json

Get or edit custom alarm sound file information.

Request URI Definition

Table 16-160 GET /ISAPI/Event/triggers/notifications/AudioAlarm/customAudioInfo/<customAudioID>?format=json

Method	GET
Description	Get custom alarm sound file information.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_CustomAudioInfo</i> Failed: <i>JSON_ResponseStatus</i>

Table 16-161 PUT /ISAPI/Event/triggers/notifications/AudioAlarm/customAudioInfo/<customAudioID>?format=json

Method	PUT
Description	Edit custom audio file information.
Query	format: determine the format of request or response message.
Request	<i>JSON_CustomAudioInfo</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The <customAudioID> in the URI refers to custom alarm sound ID and the value can be 1 to 3.

16.3.31 /ISAPI/Event/triggers/notifications/AudioAlarm/customAudioInfo?format=json

Get the custom alarm sound file.

Request URI Definition**Table 16-162 GET /ISAPI/Event/triggers/notifications/AudioAlarm/customAudioInfo?format=json**

Method	GET
Description	Get the custom alarm sound file.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_GetCustomAudio</i> Failed: <i>JSON_ResponseStatus</i>

16.3.32 /ISAPI/Event/triggers/notifications/AudioAlarm?format=json

Get or set audible warning parameters.

Request URI Definition**Table 16-163 GET /ISAPI/Event/triggers/notifications/AudioAlarm?format=json**

Method	GET
Description	Get parameters of audible warning.

Query	format: determine the format of request or response message. alarmType: alarm type, including "behavior"-behavior analysis, "thermometry"-temperature measurement, "dynamicFire"-fire detection, "smokingMode"-smoke detection, "noMaskDetection"-no wearing mask detection.
Request	None.
Response	Succeeded: <i>JSON_AudioAlarm</i> Failed: <i>JSON_ResponseStatus</i>

Note

For example, if you want to get the audible warning parameters of thermometry, the request URI is: GET /ISAPI/Event/triggers/notifications/AudioAlarm?format=json&alarmType=thermometry

Table 16-164 PUT /ISAPI/Event/triggers/notifications/AudioAlarm?format=json

Method	PUT
Description	Set parameters of audible warning.
Query	format: determine the format of request or response message.
Request	<i>JSON_AudioAlarm</i>
Response	<i>JSON_ResponseStatus</i>

16.3.33 /ISAPI/Event/triggers/notifications/whiteLightAlarm?format=json

Operations about configuration of supplement light alarm linkage.

Request URI Definition

Table 16-165 GET /ISAPI/Event/triggers/notifications/whiteLightAlarm?format=json

Method	GET
Description	Get parameters of supplement light alarm linkage.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_WhiteLightAlarm</i> Failed: <i>JSON_ResponseStatus</i>

Table 16-166 PUT /ISAPI/Event/triggers/notifications/whiteLightAlarm?format=json

Method	PUT
Description	Set parameters of supplement light alarm linkage.
Query	format: determine the format of request or response message.
Request	<i>JSON_WhiteLightAlarm</i>
Response	<i>JSON_WhiteLightAlarm</i>

16.3.34 /ISAPI/Event/triggers/notifications/whiteLightAlarm/capabilities?format=json

Get the configuration capability of supplement light alarm linkage.

Request URI Definition

Table 16-167 GET /ISAPI/Event/triggers/notifications/whiteLightAlarm/capabilities?format=json

Method	GET
Description	Get the configuration capability of supplement light alarm linkage.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_WhiteLightAlarmCap</i>

16.3.35 /ISAPI/Event/triggersCap

Get alarm linkage capability.

Request URI Definition

Table 16-168 GET /ISAPI/Event/triggersCap

Method	GET
Description	Get alarm linkage capability.
Query	None
Request	None
Response	Succeeded: <i>XML_EventTriggersCap</i> Failed: <i>XML_ResponseStatus</i>

16.3.36 http://ipAddress:portNo/url

Listening sever sends alarm information to alarm center.

Request URL Definition

Table 16-169 POST http://ipAddress:portNo/url

Method	POST
Description	Listening sever sends alarm information to alarm center.
Query	None.
Request	None.
Response	Succeeded: <i>XML_EventNotificationAlert_AlarmEventInfo</i> or <i>JSON_EventNotificationAlert_Alarm/EventInfo</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The default port number in the URL is 80, so the URL without port No. is also valid.

16.4 /ISAPI/Image

16.4.1 /ISAPI/Image/channels

Operations about image configuration of all channels.

Request URI Definition

Table 16-170 GET /ISAPI/Image/channels

Method	GET
Description	Get image parameters of all channels.
Query	None.
Request	None.
Response	Succeeded: <i>XML_ImageChannellist</i> Failed: <i>XML_ResponseStatus</i>

Table 16-171 PUT /ISAPI/Image/channels

Method	GET
Description	Set image parameters of all channels.

Query	None.
Request	<i>XML_ImageChannellist</i>
Response	<i>XML_ResponseStatus</i>

16.4.2 /ISAPI/Image/channels/<ID>

Operations about image configuration of a specific channel.

Request URI Definition

Table 16-172 GET /ISAPI/Image/channels/<ID>

Method	GET
Description	Get image parameters of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_ImageChannel</i> Failed: <i>XML_ResponseStatus</i>

Table 16-173 PUT /ISAPI/Image/channels/<ID>

Method	PUT
Description	Set image parameters of a specific channel.
Query	None.
Request	<i>XML_ImageChannel</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel ID.

16.4.3 /ISAPI/Image/channels/<ID>/BLC

Operations about BLC (Blacklist Compensation) configuration of a specific channel.

Request URI Definition

Table 16-174 GET /ISAPI/Image/channels/<ID>/BLC

Method	GET
Description	Get BLC (Blacklist Compensation) parameters of a specific channel.

Query	None.
Request	None.
Response	Succeeded: <i>XML_BLC</i> Failed: <i>XML_ResponseStatus</i>

Table 16-175 PUT /ISAPI/Image/channels/<ID>/BLC

Method	PUT
Description	Set BLC (Blacklist Compensation) parameters of a specific channel.
Query	None.
Request	<i>XML_BLC</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel ID.

16.4.4 /ISAPI/Image/channels/<ID>/capabilities

Get image configuration capability of a specific channel.

Request URI Definition**Table 16-176 GET /ISAPI/Image/channels/<ID>/capabilities**

Method	GET
Description	Get image configuration capability of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_ImageChannel</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

16.4.5 /ISAPI/Image/channels/<ID>/capturemode

Operations about video input mode configuration of a specific channel.

Request URI Definition**Table 16-177 GET /ISAPI/Image/channels/<ID>/capturemode**

Method	GET
Description	Get video input modes of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_CaptureMode</i> Failed: <i>XML_ResponseStatus</i>

Table 16-178 PUT /ISAPI/Image/channels/<ID>/capturemode

Method	PUT
Description	Set video input mode of a specific channel.
Query	None.
Request	<i>XML_CaptureMode</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel ID.

16.4.6 /ISAPI/Image/channels/<ID>/color

Operations about the image adjustment parameters of a specific channel.

Request URI Definition**Table 16-179 GET /ISAPI/Image/channels/<ID>/color**

Method	GET
Description	Get the image adjustment parameters of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Color</i> Failed: <i>XML_ResponseStatus</i>

Table 16-180 PUT /ISAPI/Image/channels/<ID>/color

Method	PUT
Description	Set the image adjustment parameters of a specific channel.
Query	None.
Request	<i>XML_Color</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

16.4.7 /ISAPI/Image/channels/<ID>/color/capabilities

Get the image adjustment capability of a specified channel.

Request URI Definition

Table 16-181 GET /ISAPI/Image/channels/<ID>/color/capabilities

Method	GET
Description	Get the image adjustment capability of a specified channel.
Query	None.
Request	None.
Response	<i>XML_Cap_Color</i>

Remarks

The <ID> in the request URI refers to the channel No.

16.4.8 /ISAPI/Image/channels/<ID>/corridor

Operations about image rotate mode configuration of a specific channel.

Request URI Definition

Table 16-182 GET /ISAPI/Image/channels/<ID>/corridor

Method	GET
Description	Get image rotate mode of a specific channel.
Query	None.

Request	None.
Response	Succeeded: <i>XML_corridor</i> Failed: <i>XML_ResponseStatus</i>

Table 16-183 PUT /ISAPI/Image/channels/<ID>/corridor

Method	PUT
Description	Get image rotate mode of a specific channel.
Query	None.
Request	<i>XML_corridor</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel ID.

16.4.9 /ISAPI/Image/channels/<ID>/dehaze

Operations about defog mode configuration of a specific channel.

Request URI Definition

Table 16-184 GET /ISAPI/Image/channels/<ID>/dehaze

Method	GET
Description	Get defog mode parameters of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_De haze</i> Failed: <i>XML_ResponseStatus</i>

Table 16-185 PUT /ISAPI/Image/channels/<ID>/dehaze

Method	PUT
Description	Set defog mode parameters of a specific channel.
Query	None.
Request	<i>XML_De haze</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel ID.

16.4.10 /ISAPI/Image/channels/<ID>/EPTZ

Get or set the e-PTZ configuration.

Request URI Definition

Table 16-186 GET /ISAPI/Image/channels/<ID>/EPTZ

Method	GET
Description	Get the e-PTZ parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_EPTZ</i> Failed: <i>XML_ResponseStatus</i>

Table 16-187 PUT /ISAPI/Image/channels/<ID>/EPTZ

Method	PUT
Description	Set the e-PTZ parameters.
Query	None.
Request	<i>XML_EPTZ</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

16.4.11 /ISAPI/Image/channels/<ID>/EPTZ/mode/capabilities?format=json

Get the capability of switching e-PTZ mode.

Request URI Definition

Table 16-188 GET /ISAPI/Image/channels/<ID>/EPTZ/mode/capabilities?format=json

Method	GET
Description	Get the capability of switching e-PTZ mode.
Query	format: determine the format of request or response message.

Request	None.
Response	Succeeded: <i>JSON_EPTZModeCap</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

16.4.12 /ISAPI/Image/channels/<ID>/EPTZ/mode?format=json

Get or set the configuration for switching e-PTZ mode.

Request URI Definition

Table 16-189 GET /ISAPI/Image/channels/<ID>/EPTZ/mode?format=json

Method	GET
Description	Get the parameters of switching e-PTZ mode.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_EPTZMode</i> Failed: <i>JSON_ResponseStatus</i>

Table 16-190 PUT /ISAPI/Image/channels/<ID>/EPTZ/mode?format=json

Method	PUT
Description	Set the parameters for switching e-PTZ mode.
Query	format: determine the format of request or response message.
Request	<i>JSON_EPTZMode</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

16.4.13 /ISAPI/Image/channels/<ID>/exposure

Operations about exposure mode configuration of a specific channel.

Request URI Definition

Table 16-191 GET /ISAPI/Image/channels/<ID>/exposure

Method	GET
Description	Get the exposure mode of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Exposure</i> Failed: <i>XML_ResponseStatus</i>

Table 16-192 PUT /ISAPI/Image/channels/<ID>/exposure

Method	PUT
Description	Set the exposure mode of a specific channel.
Query	None.
Request	<i>XML_Exposure</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel ID.

16.4.14 /ISAPI/Image/channels/<ID>/focusConfiguration

Get or set the focus parameters of a specified channel.

Request URI Definition

Table 16-193 GET /ISAPI/Image/channels/<ID>/focusConfiguration

Method	GET
Description	Get the focus parameters of a specified channel.
Query	None.
Request	None.
Response	<i>XML_FocusConfiguration</i>

Table 16-194 PUT /ISAPI/Image/channels/<ID>/focusConfiguration

Method	PUT
Description	Set the focus parameters of a specified channel.

Query	None.
Request	<i>XML_FocusConfiguration</i>
Response	<i>XML_ResponseStatus</i>

16.4.15 /ISAPI/Image/channels/<ID>/focusConfiguration/capabilities

Get the focus configuration capability.

Request URI Definition

Table 16-195 GET /ISAPI/Image/channels/<ID>/focusConfiguration/capabilities

Method	GET
Description	Get the focus configuration capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_FocusConfiguration</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

16.4.16 /ISAPI/Image/channels/<ID>/gain

Operations about gain configuration in exposure of a specific channel.

Request URI Definition

Table 16-196 GET /ISAPI/Image/channels/<ID>/gain

Method	GET
Description	Get gain parameters in exposure of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Gain</i> Failed: <i>XML_ResponseStatus</i>

Table 16-197 PUT /ISAPI/Image/channels/<ID>/gain

Method	PUT
Description	Set gain parameters in exposure of a specific channel.
Query	None.
Request	<i>XML_Gain</i>
Response	<i>XML_ResponseStatus</i>

Remarks

- The <ID> in the request URI refers to the channel ID.
- PUT method is only valid when the node <ExposureType> in the message *XML_Exposure* is set to "GainFirst".

16.4.17 /ISAPI/Image/channels/<ID>/imageCap

Get the image capability of a specific channel.

Request URI Definition

Table 16-198 GET /ISAPI/Image/channels/<ID>/imageCap

Method	GET
Description	Get the image capability of a specific channel.
Query	None.
Request	NULL.
Response	<i>XML_ImageCap</i>

16.4.18 /ISAPI/Image/channels/<ID>/ImageFlip

Operations about the image flipping status of a specific channel.

Request URI Definition

Table 16-199 GET /ISAPI/Image/channels/<ID>/ImageFlip

Method	GET
Description	Get the image flipping status of a specific channel.
Query	None.

Request	None.
Response	Succeeded: <i>XML_ImageFlip</i> Failed: <i>XML_ResponseStatus</i>

Table 16-200 PUT /ISAPI/Image/channels/<ID>/ImageFlip

Method	PUT
Description	Update the image flipping status of a specific channel.
Query	None.
Request	<i>XML_ImageFlip</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

16.4.19 /ISAPI/Image/channels/<ID>/imageMode/<ID>

Get the default configuration parameters of a specific image mode of a specific channel.

Request URI Definition**Table 16-201 GET /ISAPI/Image/channels/<ID>/imageMode/<ID>**

Method	GET
Description	Get the default configuration parameters of a specific image mode of a specific channel.
Query	None.
Request	None.
Response	<i>XML_ImageMode</i>

Remarks

This function is not supported by the traffic camera.

16.4.20 /ISAPI/Image/channels/<ID>/imageModes

Get the default image mode parameters of a specific channel.

Request URI Definition

Table 16-202 GET /ISAPI/Image/channels/<ID>/imageModes

Method	GET
Description	Get the default image mode parameters of a specific channel.
Query	None.
Request	None.
Response	<i>XML_ImageModeList</i>

Remarks

- This API is used to get default image mode configuration parameters. For different scenes, you are recommended to set different image mode parameters for the device.
- The image mode is not supported by the traffic camera.

16.4.21 /ISAPI/Image/channels/<ID>/IrcutFilter

Operations about day/night auto switch configuration of a specific channel.

Request URI Definition

Table 16-203 GET /ISAPI/Image/channels/<ID>/IrcutFilter

Method	GET
Description	Get day/night auto switch parameters of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_IrcutFilter</i> Failed: <i>XML_ResponseStatus</i>

Table 16-204 PUT /ISAPI/Image/channels/<ID>/IrcutFilter

Method	PUT
Description	Set day/night auto switch parameters of a specific channel.
Query	None.
Request	<i>XML_IrcutFilter</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel ID.

16.4.22 /ISAPI/Image/channels/<ID>/ISPMode

Operations about the day/night mode configuration of a specific channel.

Request URI Definition**Table 16-205 GET /ISAPI/Image/channels/<ID>/ISPMode**

Method	GET
Description	Get the day/night mode parameters of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_ISPMode</i> Failed: <i>XML_ResponseStatus</i>

Table 16-206 PUT /ISAPI/Image/channels/<ID>/ISPMode

Method	PUT
Description	Set the day/night mode parameters of a specific channel.
Query	None.
Request	<i>XML_ISPMode</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

16.4.23 /ISAPI/Image/channels/<ID>/lensDistortionCorrection

Get or set parameters of image distortion correction.

Request URI Definition**Table 16-207 GET /ISAPI/Image/channels/<ID>/lensDistortionCorrection**

Method	GET
Description	Get parameters of image distortion correction.
Query	None.

Request	None.
Response	Succeeded: <i>XML_LensDistortionCorrection</i> Failed: <i>XML_ResponseStatus</i>

Table 16-208 PUT /ISAPI/Image/channels/<ID>/lensDistortionCorrection

Method	PUT
Description	Set parameters of image distortion correction.
Query	None.
Request	<i>XML_LensDistortionCorrection</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the video input channel ID.

16.4.24 /ISAPI/Image/channels/<ID>/lensDistortionCorrection/capabilities

Get capability of image distortion correction.

Request URI Definition

Table 16-209 GET /ISAPI/Image/channels/<ID>/lensDistortionCorrection/capabilities

Method	GET
Description	Get capability of image distortion correction.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_LensDistortionCorrection</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the video input channel ID.

16.4.25 /ISAPI/Image/channels/<ID>/mountingScenario

Operations about the mounting scenario mode of a camera.

Request URI Definition

Table 16-210 GET /ISAPI/Image/channels/<ID>/mountingScenario

Method	GET
Description	Get the mounting scenario mode of a camera.
Query	None.
Request	None.
Response	<i>XML_MountingScenario</i>

Table 16-211 PUT /ISAPI/Image/channels/<ID>/mountingScenario

Method	PUT
Description	Set the mounting scenario mode of a camera.
Query	None.
Request	<i>XML_MountingScenario</i>
Response	<i>XML_ResponseStatus</i>

Remarks

- The <ID> in the request URI refers to channel ID.
- The image parameters should be obtained again when mounting scenario mode changes.

16.4.26 /ISAPI/Image/channels/<ID>/mountingScenario/capabilities

Get the capability of configurations of mounting scenario mode.

Request URI Definition

Table 16-212 GET /ISAPI/Image/channels/<ID>/mountingScenario/capabilities

Method	GET
Description	Get the capability of configurations of mounting scenario mode.
Query	None.
Request	None.
Response	<i>XML_Cap_MountingScenario</i>

Remarks

The <ID> in the request URI refers to channel ID.

16.4.27 /ISAPI/Image/channels/<ID>/noiseReduce

Operations about 3D DNR (Digital Noise Reduction) configuration in auto mode of a specific channel.

Request URI Definition

Table 16-213 GET /ISAPI/Image/channels/<ID>/noiseReduce

Method	GET
Description	Get 3D DNR (Digital Noise Reduction) parameters in auto mode of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_NoiseReduce</i> Failed: <i>XML_ResponseStatus</i>

Table 16-214 PUT /ISAPI/Image/channels/<ID>/noiseReduce

Method	PUT
Description	Set 3D DNR (Digital Noise Reduction) parameters in auto mode of a specific channel.
Query	None.
Request	<i>XML_NoiseReduce</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel ID.

16.4.28 /ISAPI/Image/channels/<ID>/Palettes

Get or set the palettes parameters.

Request URI Definition

Table 16-215 GET /ISAPI/Image/channels/<ID>/Palettes

Method	GET
Description	Get the palettes configuration parameters.
Query	None.

Request	None.
Response	<i>XML_Palettes</i>

Table 16-216 PUT /ISAPI/Image/channels/<ID>/Palettes

Method	PUT
Description	Set the palettes parameters.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

16.4.29 /ISAPI/Image/channels/<ID>/powerLineFrequency

Operations about image standard configuration of a specific channel.

Request URI Definition

Table 16-217 GET /ISAPI/Image/channels/<ID>/powerLineFrequency

Method	GET
Description	Get image standard parameters of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_powerLineFrequency</i> Failed: <i>XML_ResponseStatus</i>

Table 16-218 PUT /ISAPI/Image/channels/<ID>/powerLineFrequency

Method	PUT
Description	Set image standard parameters of a specific channel.
Query	None.
Request	<i>XML_powerLineFrequency</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel ID.

16.4.30 /ISAPI/Image/channels/<ID>/regionalFocus

Focus on a specific region during live view.

Request URI Definition

Table 16-219 GET /ISAPI/Image/channels/<ID>/regionalFocus

Method	PUT
Description	Focus on a specific region during live view.
Query	None.
Request	<i>XML_RegionalFocus</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI is the channel ID.

16.4.31 /ISAPI/Image/channels/<ID>/reset

Reset the image parameters of a specific channel.

Request URI Definition

Table 16-220 PUT /ISAPI/Image/channels/<ID>/reset

Method	PUT
Description	Reset the image parameters of a specific channel.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel ID.

16.4.32 /ISAPI/Image/channels/<ID>/restore

Restore the image parameters of a specific channel to default settings.

Request URI Definition

Table 16-221 PUT /ISAPI/Image/channels/<ID>/restore

Method	PUT
Description	Restore the image parameters of a specific channel to default settings.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel ID.

16.4.33 /ISAPI/Image/channels/<ID>/sharpness

Operations about the sharpness parameters of s specific channel.

Request URI Definition

Table 16-222 GET /ISAPI/Image/channels/<ID>/sharpness

Method	GET
Description	Get the sharpness parameters of s specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Sharpness</i> Failed: <i>XML_ResponseStatus</i>

Table 16-223 PUT /ISAPI/Image/channels/<ID>/sharpness

Method	PUT
Description	Set the sharpness parameters of s specific channel.
Query	None.
Request	<i>XML_Sharpness</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel ID.

16.4.34 /ISAPI/Image/channels/<ID>/shutter

Operations about shutter configuration in exposure of a specific channel.

Request URI Definition

Table 16-224 GET /ISAPI/Image/channels/<ID>/shutter

Method	GET
Description	Get shutter parameters in exposure of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Shutter</i> Failed: <i>XML_ResponseStatus</i>

Table 16-225 PUT /ISAPI/Image/channels/<ID>/shutter

Method	PUT
Description	Set shutter parameters in exposure of a specific channel.
Query	None.
Request	<i>XML_Shutter</i>
Response	<i>XML_ResponseStatus</i>

Remarks

- The <ID> in the request URI refers to the channel ID.
- PUT method is only valid when the node <ExposureType> in the message *XML_Exposure* is set to "ShutterFirst".

16.4.35 /ISAPI/Image/channels/<ID>/supplementLight

Get or set supplement light parameters by channel.

Request URI Definition

Table 16-226 GET /ISAPI/Image/channels/<ID>/supplementLight

Method	GET
Description	Get supplement light parameters by channel.
Query	None.

Request	None.
Response	Succeeded: <i>XML_SupplementLight</i> Failed: <i>XML_ResponseStatus</i>

Table 16-227 PUT /ISAPI/Image/channels/<ID>/supplementLight

Method	PUT
Description	Set supplement light parameters by channel.
Query	None.
Request	<i>XML_SupplementLight</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel ID.

16.4.36 /ISAPI/Image/channels/<ID>/supplementLight/capabilities

Get supplement light configuration capability by channel.

Request URI Definition**Table 16-228 GET /ISAPI/Image/channels/<ID>/supplementLight/capabilities**

Method	GET
Description	Get supplement light configuration capability by channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_SupplementLight</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel ID.

16.4.37 /ISAPI/Image/channels/<ID>/targetEnhancement

Get or set the target enhancement parameters.

Request URI Definition

Table 16-229 GET /ISAPI/Image/channels/<ID>/targetEnhancement

Method	GET
Description	Get the target enhancement parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_TargetEnhancement</i> Failed: <i>XML_ResponseStatus</i>

Table 16-230 PUT /ISAPI/Image/channels/<ID>/targetEnhancement

Method	PUT
Description	Set the target enhancement parameters.
Query	None.
Request	<i>XML_TargetEnhancement</i>
Response	<i>XML_ResponseStatus</i>

Remarks

- This URI is used in combination with URI */ISAPI/Image/channels/<ID>* .
- The <ID> in the request URI refers to the channel No.

16.4.38 /ISAPI/Image/channels/<ID>/targetEnhancement/capabilities

Get the target enhancement capability.

Request URI Definition

Table 16-231 GET /ISAPI/Image/channels/<ID>/targetEnhancement/capabilities

Method	GET
Description	Get the target enhancement capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_TargetEnhancementCap</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

16.4.39 /ISAPI/Image/channels/<ID>/tempRange

Operation about temperature range parameters of the specified channel.

Request URI Definition**Table 16-232 GET /ISAPI/Image/channels/<ID>/tempRange**

Method	GET
Description	Get the temperature range parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_tempRange</i> Failed: <i>XML_ResponseStatus</i>

Table 16-233 PUT /ISAPI/Image/channels/<ID>/tempRange

Method	PUT
Description	Set the temperature range parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_tempRange</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

16.4.40 /ISAPI/Image/channels/<ID>/tempRange/capabilities

Get the temperature range configuration capability of a specified channel.

Request URI Definition

Table 16-234 GET

Method	GET
Description	Get the temperature range configuration capability of a specified channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_tempRange</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

16.4.41 /ISAPI/Image/channels/<ID>/WDR

Operations about the WRD (Wide Dynamic Range) configuration of a specific channel.

Request URI Definition

Table 16-235 GET /ISAPI/Image/channels/<ID>/WDR

Method	GET
Description	Get WRD (Wide Dynamic Range) parameters of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_WDR</i> Failed: <i>XML_ResponseStatus</i>

Table 16-236 PUT /ISAPI/Image/channels/<ID>/WDR

Method	PUT
Description	Set WRD (Wide Dynamic Range) parameters of a specific channel.
Query	None.
Request	<i>XML_WDR</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel ID.

16.4.42 /ISAPI/Image/channels/<ID>/whiteBalance

Operations about the white balance parameters of s specific channel.

Request URI Definition**Table 16-237 GET /ISAPI/Image/channels/<ID>/whiteBalance**

Method	GET
Description	Get the white balance parameters of s specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_WhiteBalance</i> Failed: <i>XML_ResponseStatus</i>

Table 16-238 PUT /ISAPI/Image/channels/<ID>/whiteBalance

Method	PUT
Description	Set the white balance parameters of s specific channel.
Query	None.
Request	<i>XML_WhiteBalance</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

16.5 /ISAPI/MasterSlaveTracking**16.5.1 /ISAPI/MasterSlaveTracking/capabilities**

Get the master-slave tracking capability.

Request URI Definition**Table 16-239 GET /ISAPI/MasterSlaveTracking/capabilities**

Method	GET
Description	Get the master-slave tracking capability.
Query	None.
Request	None.
Response	<i>XML_MasterSlaveTrackingCap</i>

16.5.2 /ISAPI/MasterSlaveTracking/channels/<ID>/slaveCamera/<ID>

Operations about the basic configuration of the PTZ camera.

Request URI Definition**Table 16-240 GET /ISAPI/MasterSlaveTracking/channels/<ID>/slaveCamera/<ID>**

Method	GET
Description	Get the basic configuration parameters of the PTZ camera.
Query	None.
Request	None.
Response	<i>XML_SlaveCamera</i>

Table 16-241 PUT /ISAPI/MasterSlaveTracking/channels/<ID>/slaveCamera/<ID>

Method	PUT
Description	Set the basic configuration parameters for the PTZ camera.
Query	None.
Request	<i>XML_SlaveCamera</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The first <ID> in the request URI refers to the channel No. The second <ID> in the request URI refers to the PTZ camera ID which ranges from 1 to 4.

16.5.3 /ISAPI/MasterSlaveTracking/channels/<ID>/slaveCamera/<ID>/tracking

Operations about the tracking configuration of the PTZ camera.

Request URI Definition**Table 16-242 GET /ISAPI/MasterSlaveTracking/channels/<ID>/slaveCamera/<ID>/tracking**

Method	GET
Description	Get the tracking configuration parameters of the PTZ camera.
Query	None.
Request	None.
Response	<i>XML_Tracking</i>

Table 16-243 PUT /ISAPI/MasterSlaveTracking/channels/<ID>/slaveCamera/<ID>/tracking

Method	PUT
Description	Set the tracking configuration parameters for the PTZ camera.
Query	None.
Request	<i>XML_Tracking</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The first <ID> in the request URI refers to the channel No. The second <ID> in the request URI refers to the PTZ camera ID.

16.5.4 /ISAPI/MasterSlaveTracking/channels/<ID>/slaveCamera/<ID>/trackingRatio

Set the tracking zoom ratio for the PTZ camera.

Request URI Definition**Table 16-244 PUT /ISAPI/MasterSlaveTracking/channels/<ID>/slaveCamera/<ID>/trackingRatio**

Method	PUT
Description	Set the tracking zoom ratio for the PTZ camera.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

16.5.5 /ISAPI/MasterSlaveTracking/channels/<ID>/slaveCamera/capabilities

Get the configuration capability of basic parameters for the PTZ camera.

Request URI Definition

Table 16-245 GET /ISAPI/MasterSlaveTracking/channels/<ID>/slaveCamera/capabilities

Method	GET
Description	Get the configuration capability of basic parameters for the PTZ camera.
Query	None.
Request	None.
Response	<i>XML_Cap_SlaveCamera</i>

Remarks

The <ID> in the request URI refers to the channel No.

16.5.6 /ISAPI/MasterSlaveTracking/channels/<ID>/slaveCameraStatus

Get the connection status of the PTZ camera.

Request URI Definition

Table 16-246 GET /ISAPI/MasterSlaveTracking/channels/<ID>/slaveCameraStatus

Method	GET
Description	Get the connection status of the PTZ camera.
Query	None.
Request	None.
Response	<i>XML_SlaveCameraStatus</i>

Remarks

The <ID> in the request URI refers to the PTZ camera ID.

16.5.7 /ISAPI/MasterSlaveTracking/channels/<ID>/tracking/capabilities

Get the tracking configuration capability of the PTZ camera.

Request URI Definition

Table 16-247 GET /ISAPI/MasterSlaveTracking/channels/<ID>/tracking/capabilities

Method	GET
Description	Get the tracking configuration capability of the PTZ camera.
Query	None.
Request	None.
Response	<i>XML_Cap_Tracking</i>

16.5.8 /ISAPI/MasterSlaveTracking/linkedTracking/capabilities?format=json

Get capability of configuring advanced parameters of linked tracking capture.

Request URI Definition

Table 16-248 GET /ISAPI/MasterSlaveTracking/linkedTracking/capabilities?format=json

Method	GET
Description	Get capability of configuring advanced parameters of linked tracking capture.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_LinkedTrackingCap</i> Failed: <i>JSON_ResponseStatus</i>

16.5.9 /ISAPI/MasterSlaveTracking/linkedTracking?format=json

Get or set the advanced parameters of linked tracking capture.

Request URI Definition

Table 16-249 GET /ISAPI/MasterSlaveTracking/linkedTracking?format=json

Method	GET
Description	Get the advanced parameters of linked tracking capture.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_LinkedTracking</i>

	Failed: <i>JSON_ResponseStatus</i>
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Table 16-250 PUT /ISAPI/MasterSlaveTracking/linkedTracking?format=json

Method	PUT
Description	Set the advanced parameters of linked tracking capture.
Query	format : determine the format of request or response message.
Request	<i>JSON_LinkedTracking</i>
Response	<i>JSON_ResponseStatus</i>

16.6 /ISAPI/SDT

16.6.1 /ISAPI/SDT/Management/capabilities?format=json

Get the intelligent management capability of the device.

Request URI Definition

Table 16-251 GET /ISAPI/SDT/Management/capabilities?format=json

Method	GET
Description	Get the intelligent management capability of the device to check the intelligent functions supported by the device.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_Cap_IntelliManagement</i> Failed: <i>JSON_ResponseStatus</i>

16.6.2 /ISAPI/SDT/Management/EventSearch?format=json

Search for event.

Request URI Definition

Table 16-252 POST /ISAPI/SDT/Management/EventSearch?format=json

Method	POST
Description	Search for event.
Query	format : determine the format of request or response message.

Request	<i>JSON_EventSearchCond</i>
Response	Succeeded: <i>JSON_EventSearchResult</i> Failed: <i>JSON_ResponseStatus</i>

16.6.3 /ISAPI/SDT/Management/EventSearch/capabilities?format=json

Get the event search capability.

Request URI Definition

Table 16-253 GET /ISAPI/SDT/Management/EventSearch/capabilities?format=json

Method	GET
Description	Get the event search capability.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_EventSearchCap</i>

16.6.4 /ISAPI/SDT/Management/IntelligentSearch/capabilities?format=json

Get intelligent search capability.

Request URI Definition

Table 16-254 GET /ISAPI/SDT/Management/IntelligentSearch/capabilities?format=json

Method	GET
Description	Get intelligent search capability.
Query	format: determine the format of request or response message.
Request	None
Response	Succeeded: <i>JSON_IntelligentSearchCap</i> Failed: <i>JSON_ResponseStatus</i>

16.6.5 /ISAPI/SDT/Management/IntelligentSearch?format=json

Start intelligent search.

Request URI Definition**Table 16-255 POST /ISAPI/SDT/Management/IntelligentSearch?format=json**

Method	POST
Description	Start intelligent search.
Query	format : determine the format of request or response message.
Request	<i>JSON_IntelligentSearchCondition</i>
Response	Succeeded: <i>JSON_IntelligentSearchResult</i> Failed: <i>JSON_ResponseStatus</i>

16.7 /ISAPI/PTZCtrl**16.7.1 /ISAPI/PTZCtrl/channels/<ID>**

Get or set the single PTZ control parameters.

Request URI Definition**Table 16-256 GET /ISAPI/PTZCtrl/channels/<ID>**

Method	GET
Description	Get the single PTZ control parameters.
Query	None
Request	None
Response	Succeeded: <i>XML_PTZChannel</i> Failed: <i>XML_ResponseStatus</i>

Table 16-257 PUT /ISAPI/PTZCtrl/channels/<ID>

Method	PUT
Description	Set the single PTZ control parameters.
Query	None
Request	<i>XML_PTZChannel</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI refers to the channel No.

16.7.2 /ISAPI/PTZCtrl/channels/<ID>/auxcontrols

Get or set all PTZ auxiliaries status.

Request URI Definition

Table 16-258 GET /ISAPI/PTZCtrl/channels/<ID>/auxcontrols

Method	GET
Description	Get all PTZ auxiliaries status.
Query	None.
Request	None.
Response	Succeeded: <i>XML_PTZAuxList</i> Failed: <i>XML_ResponseStatus</i>

Table 16-259 PUT /ISAPI/PTZCtrl/channels/<ID>/auxcontrols

Method	PUT
Description	Set all PTZ auxiliaries status.
Query	None.
Request	<i>XML_PTZAuxList</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI refers to the channel No.

16.7.3 /ISAPI/PTZCtrl/channels/<ID>/auxcontrols/<ID>

Get or set the specified PTZ auxiliary status.

Request URI Definition

Table 16-260 GET /ISAPI/PTZCtrl/channels/<ID>/auxcontrols/<ID>

Method	GET
Description	Get the specified PTZ auxiliary status.
Query	None.
Request	None.
Response	Succeeded: <i>XML_PTZAux</i>

	Failed: <i>XML_ResponseStatus</i>
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Table 16-261 PUT /ISAPI/PTZCtrl/channels/<ID>/auxcontrols/<ID>

Method	PUT
Description	Set the specified PTZ auxiliary status.
Query	None.
Request	<i>XML_PTZAux</i>
Response	<i>XML_ResponseStatus</i>

Remarks

- The first <ID> in the URI refers to the channel number.
- The second <ID> in refers to the auxiliary number, e.g., when there is only one wiper, the second <ID> is "1", when there are multiple wipers, the second <ID> varies according to the requirement.

16.7.4 /ISAPI/PTZCtrl/channels/<ID>/capabilities

Get the PTZ control capabilities.

Request URI Definition**Table 16-262 GET /ISAPI/PTZCtrl/channels/ID/capabilities**

Method	GET
Description	Get the PTZ control capabilities.
Query	None
Request	None.
Response	Succeeded: <i>XML_PTZChanelCap</i> Failed: <i>XML_ResponseStatus</i>

16.7.5 /ISAPI/PTZCtrl/channels/<ID>/EagleFocusing/auto/capabilities?format=json

Get the capability of rapid focus automatic calibration.

Request URI Definition

Table 16-263 GET /ISAPI/PTZCtrl/channels/<ID>/EagleFocusing/auto/capabilities?format=json

Method	GET
Description	Get the capability of rapid focus automatic calibration.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_AutoEagleFocusingCap</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URI refers to the channel ID.

16.7.6 /ISAPI/PTZCtrl/channels/<ID>/EagleFocusing/auto?format=json

Automatically calibrate for rapid focus.

Request URI Definition

Table 16-264 GET /ISAPI/PTZCtrl/channels/<ID>/EagleFocusing/auto?format=json

Method	GET
Description	Automatically calibrate for rapid focus.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_AutoEagleFocusing</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URI refers to the channel ID.

16.7.7 /ISAPI/PTZCtrl/channels/<ID>/EagleFocusing/capabilities

Get the rapid focus capability.

Request URI Definition

Table 16-265 GET /ISAPI/PTZCtrl/channels/<ID>/EagleFocusing/capabilities

Method	GET
Description	Get the rapid focus capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_EagleFocusing</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI refers to the channel ID.

16.7.8 /ISAPI/PTZCtrl/channels/<ID>/GPS?format=json

Control the device turn to corresponding place according to applied GPS information.

Request URI Definition

Table 16-266 PUT /ISAPI/PTZCtrl/channels/<ID>/GPS?format=json

Method	PUT
Description	Control the device turn to corresponding place according to applied GPS information.
Query	format: determine the format of request or response message.
Request	<i>JSON_GPSInfo</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URI refers to channel No.

16.7.9 /ISAPI/PTZCtrl/channels/<ID>/gyroVerify/capabilities?format=json

Get the gyroscope calibration capability.

Request URI Definition

Table 16-267 GET /ISAPI/PTZCtrl/channels/<ID>/gyroVerify/ capabilities?format=json

Method	GET
Description	Get the gyroscope calibration capability.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_GyroVerifyCap</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URI refers to channel No.

16.7.10 /ISAPI/PTZCtrl/channels/<ID>/gyroVerify?format=json

Set the gyroscope calibration parameters.

Request URI Definition

Table 16-268 PUT /ISAPI/PTZCtrl/channels/<ID>/gyroVerify?format=json

Method	PUT
Description	Set the gyroscope calibration parameters.
Query	format : determine the format of request or response message.
Request	<i>JSON_GyroVerify</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URI refers to channel No.

16.7.11 /ISAPI/PTZCtrl/channels/<ID>/lensCorrection/capabilities?format=json

Get the lens correction capability.

Request URI Definition**Table 16-269 GET /ISAPI/PTZCtrl/channels/<ID>/lensCorrection/capabilities?format=json**

Method	GET
Description	Get the lens correction capability.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_LensCorrectionCap</i>

16.7.12 /ISAPI/PTZCtrl/channels/<ID>/lensCorrection?format=json

Get or set the configuration parameters of lens correction.

Request URI Definition**Table 16-270 GET /ISAPI/PTZCtrl/channels/<ID>/lensCorrection?format=json**

Method	GET
Description	Get the configuration parameters of lens correction.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_LensCorrection</i> Failed: <i>XML_ResponseStatus</i>

Table 16-271 PUT /ISAPI/PTZCtrl/channels/<ID>/lensCorrection?format=json

Method	PUT
Description	Set the lens correction parameters.
Query	format: determine the format of request or response message.
Request	<i>JSON_LensCorrection</i>
Response	<i>XML_ResponseStatus</i>

16.7.13 /ISAPI/PTZCtrl/channels/<ID>/lockPTZ

Operations about locking PTZ of a specific channel.

Request URI Definition**Table 16-272 GET /ISAPI/PTZCtrl/channels/<ID>/lockPTZ**

Method	GET
Description	Get the remaining time before unlocking PTZ of a specific channel.
Query	None.
Request	None.
Response	<i>XML_LockPTZ</i>

Table 16-273 PUT /ISAPI/PTZCtrl/channels/<ID>/lockPTZ

Method	PUT
Description	Lock PTZ of a specific channel.
Query	None.
Request	<i>XML_LockPTZ</i>
Response	<i>XML_ResponseStatus</i>

16.7.14 /ISAPI/PTZCtrl/channels/<ID>/maxelevation

Get or set the max. tilt-angle parameters.

Request URI Definition**Table 16-274 GET /ISAPI/PTZCtrl/channels/<ID>/maxelevation**

Method	GET
Description	Get the max. tilt-angle configuration parameters
Query	None.
Request	None.
Response	<i>XML_MaxElevation</i>

Table 16-275 PUT /ISAPI/PTZCtrl/channels/<ID>/maxelevation

Method	PUT
Description	Set the max. tilt-angle
Query	None.
Request	<i>XML_MaxElevation</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI indicates the channel ID.

16.7.15 /ISAPI/PTZCtrl/channels/<ID>/maxelevation/capabilities

Get the max. tilt-angle capability.

Request URI Definition

Table 16-276 GET /ISAPI/PTZCtrl/channels/<ID>/maxelevation/capabilities

Method	GET
Description	Get the max. tilt-angle capability.
Query	None.
Request	None.
Response	<i>XML_Cap_MaxElevation</i>

Remarks

The <ID> in the URI indicates the channel ID.

16.7.16 /ISAPI/PTZCtrl/channels/<ID>/moveAutoTracking?format=json

Get or set moving object tracking status.

Request URI Definition

Table 16-277 GET /ISAPI/PTZCtrl/channels/<ID>/moveAutoTracking?format=json

Method	GET
Description	Get moving object tracking status.
Query	format: determine the format of request or response message.
Request	None
Response	Succeeded: <i>JSON_MoveAutoTracking</i> Failed: <i>JSON_ResponseStatus</i>

Table 16-278 PUT /ISAPI/PTZCtrl/channels/<ID>/moveAutoTracking?format=json

Method	PUT
Description	Set moving object tracking status.
Query	format: determine the format of request or response message.

Request	<i>JSON_MoveAutoTracking</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the PTZ channel ID.

16.7.17 /ISAPI/PTZCtrl/channels/<ID>/onpushfoucs/reset

Initialize the lens.

Request URI Definition

Table 16-279 PUT /ISAPI/PTZCtrl/channels/<ID>/onpushfoucs/reset

Method	PUT
Description	Initialize the lens.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

16.7.18 /ISAPI/PTZCtrl/channels/<ID>/parkaction

Get or set the parking parameters.

Request URI Definition

Table 16-280 GET /ISAPI/PTZCtrl/channels/<ID>/parkaction

Method	GET
Description	Get the parking parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_ParkAction</i> Failed: <i>XML_ResponseStatus</i>

Table 16-281 PUT /ISAPI/PTZCtrl/channels/<ID>/parkaction

Method	PUT
Description	Set the parking parameters.
Query	None.
Request	<i>XML_ParkAction</i>
Response	<i>XML_ResponseStatus</i>

16.7.19 /ISAPI/PTZCtrl/channels/<ID>/parkaction/capabilities

Get the parking configuration capability.

Request URI Definition

Table 16-282 GET /ISAPI/PTZCtrl/channels/<ID>/parkaction/capabilities

Method	GET
Description	Get the parking configuration capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_ParkAction</i> Failed: <i>XML_ResponseStatus</i>

16.7.20 /ISAPI/PTZCtrl/channels/<ID>/presets

Operations about all presets' configurations.

Request URI Definition

Table 16-283 GET /ISAPI/PTZCtrl/channels/<ID>/presets

Method	GET
Description	Get all presets' parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_PTZPresetList</i> Failed: <i>XML_ResponseStatus</i>

Table 16-284 PUT /ISAPI/PTZCtrl/channels/<ID>/presets

Method	PUT
Description	Set all presets' parameters.
Query	None.
Request	<i>XML_PTZPresetList</i>
Response	<i>XML_ResponseStatus</i>

Table 16-285 POST /ISAPI/PTZCtrl/channels/<ID>/presets

Method	POST
Description	Add a preset.
Query	None.
Request	<i>XML_PTZPreset</i>
Response	<i>XML_ResponseStatus</i>

Table 16-286 DELETE /ISAPI/PTZCtrl/channels/<ID>/presets

Method	DELETE
Description	Delete all presets.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI is the camera ID.

16.7.21 /ISAPI/PTZCtrl/channels/<ID>/presets/<ID>

Get or set the a preset's parameters, or delete a preset.

Request URI Definition**Table 16-287 GET /ISAPI/PTZCtrl/channels/<ID>/presets/<ID>**

Method	GET
Description	Get a preset's parameters.
Query	None.

Request	None.
Response	Succeeded: <i>XML_PTZPreset</i> Failed: <i>XML_ResponseStatus</i>

Table 16-288 PUT /ISAPI/PTZCtrl/channels/<ID>/presets/<ID>

Method	PUT
Description	Set a preset's parameters.
Query	None.
Request	<i>XML_PTZPreset</i>
Response	<i>XML_ResponseStatus</i>

Table 16-289 DELETE /ISAPI/PTZCtrl/channels/<ID>/presets/<ID>

Method	DELETE
Description	Delete a preset.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

16.7.22 /ISAPI/PTZCtrl/channels/<ID>/PTZOSDDisplay

Get or set the on-screen display parameters of the PTZ status.

Request URI Definition

Table 16-290 GET /ISAPI/PTZCtrl/channels/<ID>/PTZOSDDisplay

Method	GET
Description	Get the on-screen display parameters of the PTZ status.
Query	None.
Request	None.
Response	Succeeded: <i>XML_PTZOSDDisplay</i> Failed: <i>XML_ResponseStatus</i>

Table 16-291 PUT /ISAPI/PTZCtrl/channels/<ID>/PTZOSDDisplay

Method	PUT
Description	Set the on-screen display parameters of the PTZ status.

Query	None.
Request	<i>XML_PTZOSDDisplay</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the PTZ channel No.

16.7.23 /ISAPI/PTZCtrl/channels/<ID>/save

Get and save the PTZ position information of the current channel.

Request URI Definition

Table 16-292 GET /ISAPI/PTZCtrl/channels/<ID>/save

Method	GET
Description	Get and save the PTZ position information (including panning, tilting, and zooming parameters) of the current channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_PTZStatus</i> Failed: <i>XML_ResponseStatus</i>

16.7.24 /ISAPI/PTZCtrl/channels/<ID>/save?format=json

Save the PTZ position information of the current channel.

Request URI Definition

Table 16-293 PUT /ISAPI/PTZCtrl/channels/<ID>/save?format=json

Method	PUT
Description	Save the PTZ position information, including panning, tilting, and zooming parameters, of the current channel.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the PTZ channel No.

16.7.25 /ISAPI/PTZCtrl/channels/<ID>/status

Get the PTZ status of a specific channel of the device.

Request URI Definition**Table 16-294 GET /ISAPI/PTZCtrl/channels/<ID>/status**

Method	GET
Description	Get the PTZ status of a specific channel of the device.
Query	None.
Request	None.
Response	Succeeded: <i>XML_PTZStatus</i> Failed: <i>XML_ResponseStatus</i>

16.7.26 /ISAPI/PTZCtrl/channels/<ID>/trackingRatio/capabilities?format=json

Get the configuration capability of tracking parameters.

Request URI Definition**Table 16-295 GET /ISAPI/PTZCtrl/channels/<ID>/trackingRatio/capabilities?format=json**

Method	GET
Description	Get the configuration capability of tracking parameters.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_TrackingRatioCap</i> Failed: <i>JSON_ResponseStatus</i>

16.7.27 /ISAPI/PTZCtrl/channels/<ID>/trackingRatio?format=json

Get or set the tracking parameters.

Request URI Definition

Table 16-296 GET /ISAPI/PTZCtrl/channels/<ID>/trackingRatio?format=json

Method	GET
Description	Get the tracking parameters.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_TrackingRatio</i> Failed: <i>JSON_ResponseStatus</i>

Table 16-297 PUT /ISAPI/PTZCtrl/channels/<ID>/trackingRatio?format=json

Method	PUT
Description	Set the tracking parameters.
Query	format: determine the format of request or response message.
Request	<i>JSON_TrackingRatio</i>
Response	<i>JSON_ResponseStatus</i>

16.7.28 /ISAPI/PTZCtrl/channels/<ID>/zoomFocus

Operations about zoom and focus coordinates configuration for zoom camera module.

Request URI Definition

Table 16-298 GET /ISAPI/PTZCtrl/channels/<ID>/zoomFocus

Method	GET
Description	Get parameters of zoom and focus coordinates of zoom camera module.
Query	None.
Request	None.
Response	<i>XML_ZoomFocus</i>

Table 16-299 PUT /ISAPI/PTZCtrl/channels/<ID>/zoomFocus

Method	PUT
Description	Set parameters of zoom and focus coordinates of zoom camera module.

Query	None.
Request	<i>XML_ZoomFocus</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel ID.

16.8 /ISAPI/Security

16.8.1 /ISAPI/Security/adminAccesses

Get or set parameters of all protocols supported by device.

Request URI Definition

Table 16-300 GET /ISAPI/Security/adminAccesses

Method	GET
Description	Get parameters of all protocols supported by device.
Query	None.
Request	None.
Response	Succeeded: <i>XML_AdminAccessProtocolList</i> Failed: <i>XML_ResponseStatus</i>

Table 16-301 PUT /ISAPI/Security/adminAccesses

Method	PUT
Description	Set parameters of all protocols supported by device.
Query	None.
Request	<i>XML_AdminAccessProtocolList</i>
Response	<i>XML_ResponseStatus</i>

16.8.2 /ISAPI/Security/adminAccesses/<ID>

Get or set the parameters of a specific protocol that supported by device.

Request URI Definition

Table 16-302 GET /ISAPI/Security/adminAccesses/<ID>

Method	GET
Description	Get the parameters of a specific protocol that supported by device.
Query	None.
Request	None.
Response	Succeeded: <i>XML_AdminAccessProtocol</i> Failed: <i>XML_ResponseStatus</i>

Table 16-303 PUT /ISAPI/Security/adminAccesses/<ID>

Method	PUT
Description	Set the parameters of a specific protocol that supported by device.
Query	None.
Request	<i>XML_AdminAccessProtocol</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the protocol ID.

16.8.3 /ISAPI/Security/adminAccesses/capabilities

Get device protocol capability.

Request URI Definition

Table 16-304 GET /ISAPI/Security/adminAccesses/capabilities

Method	GET
Description	Get device protocol capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_AdminAccessProtocolList</i> Failed: <i>XML_ResponseStatus</i>

16.8.4 /ISAPI/Security/advanced?format=json

Get or set advanced parameters of security.

Request URI Definition

Table 16-305 GET /ISAPI/Security/advanced?format=json

Method	GET
Description	Get advanced configuration parameters of security.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_SecurityAdvanced</i> Failed: <i>JSON_ResponseStatus</i>

Table 16-306 PUT /ISAPI/Security/advanced?format=json

Method	PUT
Description	Set advanced parameters of security.
Query	format: determine the format of request or response message.
Request	<i>JSON_SecurityAdvanced</i>
Response	<i>JSON_ResponseStatus</i>

16.8.5 /ISAPI/Security/capabilities

Get the security capability of the device.

Request URI Definition

Table 16-307 GET /ISAPI/Security/capabilities

Method	GET
Description	Get the security capability of the device.
Query	username: user name, string, it should be encrypted.
Request	None.
Response	Succeeded: <i>XML_SecurityCap</i> Failed: <i>XML_ResponseStatus</i>

16.8.6 /ISAPI/Security/certificate/select/<functinName>?format=json

Get or set the parameters of selecting the certificate.

Request URI Definition

Table 16-308 GET /ISAPI/Security/certificate/select/<functinName>?format=json

Method	GET
Description	Get the parameters of selecting the certificate.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_CertificateSelect</i> Failed: <i>JSON_ResponseStatus</i>

Table 16-309 PUT /ISAPI/Security/certificate/select/<functinName>?format=json

Method	PUT
Description	Set the parameters of selecting the certificate.
Query	format: determine the format of request or response message.
Request	<i>JSON_CertificateSelect</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The <functinName> in the request URI refers to the function name.

16.8.7 /ISAPI/Security/certificate/select/capabilities?format=json

Get the capability of selecting the certificate.

Request URI Definition

Table 16-310 GET /ISAPI/Security/certificate/select/capabilities?format=json

Method	GET
Description	Get the capability of selecting the certificate.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_Cap_CertificateSelect</i>

	Failed: <i>JSON_ResponseStatus</i>
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16.8.8 /ISAPI/Security/challenge

Get the random challenge strings.

Request URI Definition

Table 16-311 POST /ISAPI/Security/challenge

Method	POST
Description	Get the random challenge strings.
Query	None.
Request	<i>XML_PublicKey</i>
Response	<i>XML_Challenge</i>

16.8.9 /ISAPI/Security/CommuMode/capabilities?format=json

Get the configuration capability of the security mode level of the private protocol.

Request URI Definition

Table 16-312 GET /ISAPI/Security/CommuMode/capabilities?format=json

Method	GET
Description	Get the configuration capability of the security mode level of the private protocol.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_Cap_CommMode</i> Failed: <i>JSON_ResponseStatus</i>

16.8.10 /ISAPI/Security/CommuMode?format=json

Operations about the configuration of the security mode level of the private protocol.

Request URI Definition

Table 16-313 GET /ISAPI/Security/CommuMode?format=json

Method	GET
Description	Get the parameters of the security mode level of the private protocol.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_CommuMode</i> Failed: <i>JSON_ResponseStatus</i>

Table 16-314 PUT /ISAPI/Security/CommuMode?format=json

Method	PUT
Description	Set the parameters of the security mode level of the private protocol.
Query	format: determine the format of request or response message.
Request	<i>JSON_CommuMode</i>
Response	<i>JSON_ResponseStatus</i>

16.8.11 /ISAPI/Security/deviceCertificate

Import network certificate to device.

Request URI Definition

Table 16-315 PUT /ISAPI/Security/deviceCertificate

Method	PUT
Description	Import network certificate to device.
Query	type: certificate type. It can be set to "wpa" (upload WPA certificates), "ieee802.1x" (upload ieee802.1x certificates), "securityLog" (upload CA certificates of security logs) or "LogServer" (log server certificate); if no value is assigned to this parameter, it indicates that the WPA certificate will be imported.
Request	Certificate data (in binary format)
Response	<i>XML_ResponseStatus</i>

Remarks

The imported certificate type is determined by "Content-type" in binary certificate data, which contains "x-509-ca-cert", "x-509-client-cert", and "x-509-client-key".

16.8.12 /ISAPI/Security/deviceCertificate/capabilities?format=json

Get the CA (Certificate Authority) certificate capability.

Request URI Definition

Table 16-316 GET /ISAPI/Security/deviceCertificate/capabilities?format=json

Method	GET
Description	Get the CA (Certificate Authority) certificate capability.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_DeviceCertificateCap</i> Failed: <i>JSON_ResponseStatus</i>

16.8.13 /ISAPI/Security/deviceCertificate/certificateRevocation/capabilities?format=json

Get the configuration capability of the certificate expiry alarm.

Request URI Definition

Table 16-317 GET /ISAPI/Security/deviceCertificate/certificateRevocation/capabilities?format=json

Method	GET
Description	Get the configuration capability of the certificate expiry alarm.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_CertificateRevocationCap</i> Failed: <i>JSON_ResponseStatus</i>

16.8.14 /ISAPI/Security/deviceCertificate/certificateRevocation?format=json

Get or set the configuration of certificate expiry alarm.

Request URI Definition

Table 16-318 GET /ISAPI/Security/deviceCertificate/certificateRevocation?format=json

Method	GET
Description	Get the parameters of certificate expiry alarm.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_CertificateRevocation</i> Failed: <i>JSON_ResponseStatus</i>

Table 16-319 PUT /ISAPI/Security/deviceCertificate/certificateRevocation?format=json

Method	PUT
Description	Set the parameters of certificate expiry alarm.
Query	format : determine the format of request or response message.
Request	<i>JSON_CertificateRevocation</i>
Response	<i>JSON_ResponseStatus</i>

16.8.15 /ISAPI/Security/deviceCertificate/certificates/<customID>?format=json

Get or delete the information of a specific device certificate.

Request URI Definition

Table 16-320 GET /ISAPI/Security/deviceCertificate/certificates/<customID>?format=json

Method	GET
Description	Get the information of a specific device certificate.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_DeviceCertificate</i> Failed: <i>JSON_ResponseStatus</i>

Table 16-321 DELETE /ISAPI/Security/deviceCertificate/certificates/<customID>?format=json

Method	DELETE
Description	Delete the information of a specific device certificate.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

Remarks

The <customID> in the request URI refers to the custom certificate ID.

16.8.16 /ISAPI/Security/deviceCertificate/certificates/<ID>

Delete the certificate of a specific device.

Request URI Definition**Table 16-322 DELETE /ISAPI/Security/deviceCertificate/certificates/<ID>**

Method	DELETE
Description	Delete the certificate of a specific device.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI is the device ID returned when searching certificate information.

16.8.17 /ISAPI/Security/deviceCertificate/certificates/<ID>/recreate?format=json

Regenerate a specific abnormal certificate.

Request URI Definition**Table 16-323 PUT /ISAPI/Security/deviceCertificate/certificates/<ID>/recreate?format=json**

Method	PUT
Description	Regenerate a specific certificate for authentication client.
Query	format: determine the format of request or response message.

Request	None.
Response	<i>JSON_ResponseStatus</i>

Remarks

- The <ID> in the request URI refers to the certificate ID.
- This URI is only supported by HTTPS certificate, and it is available when the certificate exception is detected.

16.8.18 /ISAPI/Security/deviceCertificate/certificates/<ID>/status?format=json

Get the status of a specific certificate.

Request URI Definition

Table 16-324 GET /ISAPI/Security/deviceCertificate/certificates/<ID>/status?format=json

Method	GET
Description	Get the status of a specific certificate.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_DeviceCertificateStatus</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI is the certificate ID, which is generated and returned by device.

16.8.19 /ISAPI/Security/deviceCertificate/certificates/capabilities?format=json

Get certificate search capability.

Request URI Definition

Table 16-325 GET /ISAPI/Security/deviceCertificate/certificates/capabilities?format=json

Method	GET
Description	Get certificate search capability.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_DeviceCertificatesCap</i>

	Failed: <i>JSON_ResponseStatus</i>
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16.8.20 /ISAPI/Security/deviceCertificate/certificates/recreate?format=json

Regenerate all abnormal certificates.

Request URI Definition

Table 16-326 PUT /ISAPI/Security/deviceCertificate/certificates/recreate?format=json

Method	PUT
Description	Regenerate all abnormal certificates.
Query	format : determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

Remarks

This URI is only supported by HTTPS certificate, and it is available when the certificate exception is detected.

16.8.21 /ISAPI/Security/deviceCertificate/certificates/status?format=json

Get status of all certificates.

Request URI Definition

Table 16-327 GET /ISAPI/Security/deviceCertificate/certificates/status?format=json

Method	GET
Description	Get status of all certificates.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_DeviceCertificateStatusList</i> Failed: <i>JSON_ResponseStatus</i>

16.8.22 /ISAPI/Security/deviceCertificate/certificates?format=json

Get device certificate information in a batch.

Request URI Definition**Table 16-328 GET /ISAPI/Security/deviceCertificate/certificates?format=json**

Method	GET
Description	Get device certificate information in a batch.
Query	format : determine the format of request or response message. type : certificate type, it can be set to "wpa" (get all WPA certificates), "ieee802.1x" (get all ieee802.1x certificates), and "securityLog" (get all CA certificates of security logs); if no value is assigned to type , all CA certificates will be searched and obtained.
Request	None.
Response	Succeeded: <i>JSON_DeviceCertificates</i> Failed: <i>JSON_ResponseStatus</i>

16.8.23 /ISAPI/Security/deviceCertificate?customID=

Import a CA (Certificate Authority) certificate to the device.

Request URI Definition**Table 16-329 PUT /ISAPI/Security/deviceCertificate?customID=**

Method	PUT
Description	Import a CA (Certificate Authority) certificate to the device. The CA certificate is used for 802.1x (radius) with various authentication mechanism.
Query	customID : custom certificate ID, string, it consists of digits and characters.
Request	Certificate data (its format depends on the device).
Response	<i>XML_ResponseStatus</i>

Remarks

- When importing CA certificate by calling this URI, the function of CA certificate will not be distinguished. The CA certificate and its function will be bound together afterward.
- The imported certificate type is determined by "Content-type" in the certificate data, which contains "x-x509-ca-cert" (CA certificate (root certificate)), "x-x509-client-cert" (client certificate), and "x-x509-client-key" (client password).

16.8.24 /ISAPI/Security/doubleVerification/users/<ID>?format=json

Operations about a specified double verification user.

Request URI Definition

Table 16-330 GET /ISAPI/Security/doubleVerification/users/<ID>?format=json

Method	GET
Description	Get a double verification user.
Query	format : determine the format of request or response message. security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	<i>JSON_User</i>

Table 16-331 PUT /ISAPI/Security/doubleVerification/users/<ID>?format=json

Method	PUT
Description	Set a double verification user.
Query	format : determine the format of request or response message. security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	<i>JSON_User</i>
Response	<i>JSON_ResponseStatus</i>

Table 16-332 DELETE /ISAPI/Security/doubleVerification/users/<ID>?format=json

Method	DELETE
Description	Delete a double verification user.

Query	format : determine the format of request or response message. security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URI indicates the user ID.

16.8.25 /ISAPI/Security/doubleVerification/users/capabilities?format=json

Get the double verification configuration capability.

Request URI Definition

Table 16-333 GET /ISAPI/Security/doubleVerification/users/capabilities?format=json

Method	GET
Description	Get the double verification configuration capability.
Query	format : determine the format of request or response message.
Request	None.
Response	<i>JSON_UserCap</i>

16.8.26 /ISAPI/Security/doubleVerification/users?format=json

Get all double verification users and add a double verification user.

Request URI Definition

Table 16-334 GET /ISAPI/Security/doubleVerification/users?format=json

Method	GET
Description	Get all double verification users.
Query	format : determine the format of request or response message.

	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	<i>JSON_UserList</i>

Table 16-335 POST /ISAPI/Security/doubleVerification/users?format=json

Method	POST
Description	Add a double verification user.
Query	format : determine the format of request or response message. security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	<i>JSON_User</i>
Response	Succeeded: <i>JSON_ResponseStatus</i> and <i>JSON_id</i> Failed: <i>JSON_ResponseStatus</i>

16.8.27 /ISAPI/Security/doubleVerification/UsersPermission/<ID>?format=json

Get and set the permission of double verification user.

Request URI Definition

Table 16-336 GET /ISAPI/Security/doubleVerification/UsersPermission/<ID>?format=json

Method	GET
Description	Get the configuration parameters of double verification user permission.
Query	format : determine the format of request or response message.

Request	None.
Response	Succeeded: <i>JSON_UserPermission</i> Failed: <i>JSON_ResponseStatus</i>

Table 16-337 PUT /ISAPI/Security/doubleVerification/UsersPermission/<ID>?format=json

Method	PUT
Description	Set the double verification user permission.
Query	format: determine the format of request or response message.
Request	<i>JSON_UserPermission</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URI indicates the user ID.

**16.8.28 /ISAPI/Security/doubleVerification/UsersPermission/capabilities?
format=json**

Get the capability of permission configuration for double verification user.

Request URI Definition**Table 16-338 GET /ISAPI/Security/doubleVerification/UsersPermission/capabilities?format=json**

Method	GET
Description	Get the capability of permission configuration for double verification user.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_UserPermissionCap</i> Failed: <i>JSON_ResponseStatus</i>

16.8.29 /ISAPI/Security/doubleVerification?format=json

Get the capability of double verification enabling configuration.

Request URI Definition

Table 16-339 GET /ISAPI/Security/doubleVerification?format=json

Method	GET
Description	Get the capability of double verification enabling configuration.
Query	format : determine the format of request or response message.
Request	None.
Response	<i>JSON_DoubleVerification</i>

Table 16-340 PUT /ISAPI/Security/doubleVerification?format=json

Method	PUT
Description	Enable or disable the double verification.
Query	format : determine the format of request or response message.
Request	<i>JSON_DoubleVerification</i>
Response	<i>JSON_ResponseStatus</i>

16.8.30 /ISAPI/Security/email/parameter/capabilities?format=json

Get recovery email configuration capability (only available in LAN and for admin user).

Request URI Definition

Table 16-341 GET /ISAPI/Security/email/parameter/capabilities?format=json

Method	GET
Description	Get recovery email configuration capability.
Query	format : determine the format of request or response message. security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	<i>JSON_SecurityEmailCap</i>

16.8.31 /ISAPI/Security/email/parameter?format=json

Operations about recovery email configuration (only available for LAN and for admin user).

Request URI Definition

Table 16-342 GET /ISAPI/Security/email/parameter?format=json

Method	GET
Description	Get recovery email parameters.
Query	format : determine the format of request or response message. security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	<i>JSON_SecurityEmail</i>

Table 16-343 PUT /ISAPI/Security/email/parameter?format=json

Method	PUT
Description	Set recovery email parameters.
Query	format : determine the format of request or response message. security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	<i>JSON_SecurityEmail</i>
Response	<i>JSON_ResponseStatus</i>

16.8.32 /ISAPI/Security/email/qrCode?format=json

Get the QR code of the configured recovery email (only available for LAN and for admin user).

Request URI Definition

Table 16-344 GET /ISAPI/Security/email/qrCode?format=json

Method	GET
Description	Get the QR code of the configured recovery email.
Query	format : determine the format of request or response message. security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	<i>JSON_SecurityEmailQrCode</i>

16.8.33 /ISAPI/Security/emailCertification?format=json

Reset password by the verification code via recovery email (only available for LAN and for admin user).

Request URI Definition

Table 16-345 PUT /ISAPI/Security/emailCertification?format=json

Method	PUT
Description	Reset password by the verification code via recovery email.
Query	format : determine the format of request or response message.
Request	<i>JSON_EmailCertification</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

For the encryption of passwords and security answers, first transform them by UTF8, and then transcode them by BASE64, finally, encrypt them by AES128CB.

16.8.34 /ISAPI/Security/extern/capabilities

Get capability of other security configuration.

Request URI Definition

Table 16-346 GET /ISAPI/Security/extern/capabilities

Method	GET
Description	Get capability of other security configuration.
Query	None.
Request	None.
Response	<i>XML_externSecurityCap</i>

16.8.35 /ISAPI/Security/illegalLoginLock

Get or set locking parameters of illegal login.

Request URI Definition

Table 16-347 GET /ISAPI/Security/illegalLoginLock

Method	GET
Description	Get locking parameters of illegal login.
Query	None.
Request	None.
Response	Succeeded: <i>XML_IllegalLoginLock</i> Failed: <i>XML_ResponseStatus</i>

Table 16-348 PUT /ISAPI/Security/illegalLoginLock

Method	PUT
Description	Set locking parameters of illegal login.
Query	None.
Request	<i>XML_IllegalLoginLock</i>
Response	<i>XML_ResponseStatus</i>

16.8.36 /ISAPI/Security/loginLinkNum?format=json

Get or set maximum number of logged in accounts.

Request URI Definition

Table 16-349 GET /ISAPI/Security/loginLinkNum?format=json

Method	GET
Description	Get maximum number of logged in accounts.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_LoginLinkNum</i> Failed: <i>JSON_ResponseStatus</i>

Table 16-350 PUT /ISAPI/Security/loginLinkNum?format=json

Method	PUT
Description	Set maximum number of logged in accounts.
Query	format : determine the format of request or response message.
Request	<i>JSON_LoginLinkNum</i>
Response	<i>JSON_ResponseStatus</i>

16.8.37 /ISAPI/Security/questionConfiguration

Get or set device security questions.

Request URI Definition

Table 16-351 GET /ISAPI/Security/questionConfiguration

Method	GET
Description	Get device security questions.
Query	None.
Request	None.
Response	Succeeded: <i>XML_SecurityQuestion</i> Failed: <i>XML_ResponseStatus</i>

Table 16-352 PUT /ISAPI/Security/questionConfiguration

Method	PUT
Description	Set device security questions.

Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	<i>XML_SecurityQuestion</i>
Response	<i>XML_ResponseStatus</i>

16.8.38 /ISAPI/Security/RTSPCertificate

Get or set parameters of devices' RTSP authentication mode.

Request URI Definition

Table 16-353 GET /ISAPI/Security/RTSPCertificate

Method	GET
Description	Get parameters of devices' RTSP authentication mode.
Query	None.
Request	None.
Response	Succeeded: <i>XML_RTSPCertificate</i> Failed: <i>XML_ResponseStatus</i>

Table 16-354 PUT /ISAPI/Security/RTSPCertificate

Method	PUT
Description	Set parameters of devices' RTSP authentication mode.
Query	None.
Request	<i>XML_RTSPCertificate</i>
Response	<i>XML_ResponseStatus</i>

16.8.39 /ISAPI/Security/RTSPCertificate/capabilities

Get the capability of devices' RTSP authentication mode.

Request URI Definition

Table 16-355 GET /ISAPI/Security/RTSPCertificate/capabilities

Method	GET
Description	Get the capability of devices' RTSP authentication mode.
Query	None.
Request	None.
Response	Succeeded: <i>XML_RTSPCertificateCap</i> Failed: <i>XML_ResponseStatus</i>

16.8.40 /ISAPI/Security/serverCertificate/capabilities?format=json

Get the client/server certificate capability.

Request URI Definition

Table 16-356 GET /ISAPI/Security/serverCertificate/capabilities?format=json

Method	GET
Description	Get the client/server certificate capability.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_ServerCertificateCap</i> Failed: <i>JSON_ResponseStatus</i>

16.8.41 /ISAPI/Security/serverCertificate/certificate

Get, upload, and delete device certificate information.

Request URI Definition

Table 16-357 GET /ISAPI/Security/serverCertificate/certificate

Method	GET
Description	Get device certificate information.
Query	type: certificate type, it can be set to "CCClientCertificate" (client certificate for CC authentication) or "LogServer" (log server

	certificate); if no value is assigned to type , it represents the HTTPS certificate.
Request	None.
Response	Succeeded: <i>XML_CertificateInfo</i> <i>XML_ResponseStatus</i>

Table 16-358 PUT /ISAPI/Security/serverCertificate/certificate

Method	PUT
Description	Upload certificate to device.
Query	type : certificate type, it can be set to "CCClientCertificate" (client certificate for CC authentication) or "LogServer" (log server certificate); if no value is assigned to type , it represents the HTTPS certificate.
Request	Certificate data (binary)
Response	<i>XML_ResponseStatus</i>

Table 16-359 DELETE /ISAPI/Security/serverCertificate/certificate

Method	DELETE
Description	Delete the installed certificate of device.
Query	type : certificate type, it can be set to "CCClientCertificate" (client certificate for CC authentication) or "LogServer" (log server certificate); if no value is assigned to type , it represents the HTTPS certificate.
Request	None.
Response	<i>XML_ResponseStatus</i>

16.8.42 /ISAPI/Security/serverCertificate/certificate?customID=

Import the client/server certificate to the device.

Request URI Definition

Table 16-360 POST /ISAPI/Security/serverCertificate/certificate?customID=

Method	POST
Description	Import the client/server certificate to the device.

Query	security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2. customID: custom certificate ID, string, it consists of digits and characters.
Request	<i>XML_CertificateReq_ImportCert</i> +certificate data.
Response	<i>XML_ResponseStatus</i>

Remarks

- When the node <certificateMode> in the request message *XML_CertificateReq_ImportCert* is set to "signingRequest", it means importing a self-signed certificate to the device.
- When the node <certificateMode> in the request message *XML_CertificateReq_ImportCert* is set to "privateKey", it means importing the certificate along with the private key by the following two steps.
 - Import the message *XML_CertificateReq_ImportCert* and the certificate file.

**Note**

The node <dataType> in the message *XML_CertificateReq_ImportCert* should be set to "certificate".

- Import the message *XML_CertificateReq_ImportCert* and the private key.

**Note**

- The node <dataType> in the message *XML_CertificateReq_ImportCert* should be set to "privateKey".
- This step depends on the first step. If exception occurs, the device will return error.

16.8.43 /ISAPI/Security/serverCertificate/certificates?format=json

Get the information of multiple client/server certificates in a batch.

Request URI Definition

Table 16-361 GET /ISAPI/Security/serverCertificate/certificates?format=json

Method	GET
Description	Get the information of multiple client/server certificates in a batch.

Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_ServerCertificates</i> Failed: <i>JSON_ResponseStatus</i>

16.8.44 /ISAPI/Security/serverCertificate/certificates/<customID>?format=json

Get or delete the information of a specific client/server certificate.

Request URI Definition

Table 16-362 GET /ISAPI/Security/serverCertificate/certificates/<customID>?format=json

Method	GET
Description	Get the information of a specific client/server certificate.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_DeviceCertificate</i> Failed: <i>JSON_ResponseStatus</i>

Table 16-363 DELETE /ISAPI/Security/serverCertificate/certificates/<customID>?format=json

Method	DELETE
Description	Delete the information of a specific client/server certificate.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

Remarks

The <customID> in the request URI refers to the custom certificate ID.

16.8.45 /ISAPI/Security/serverCertificate/certSignReq

Get signature request information, generate, and delete signature request of device certificate.

Request URI Definition

Table 16-364 GET /ISAPI/Security/serverCertificate/certSignReq

Method	GET
Description	Get signature request information of device certificate.
Query	type : certificate type, it can be set to "CCClientCertificate" (client certificate for CC authentication) or "LogServer" (log server certificate); if no value is assigned to type , it represents the HTTPS certificate.
Request	None.
Response	Succeeded: <i>XML_CertificateReqInfo</i> Failed: <i>XML_ResponseStatus</i>

Table 16-365 PUT /ISAPI/Security/serverCertificate/certSignReq

Method	PUT
Description	Generate signature request information of device certificate.
Query	type : certificate type, it can be set to "CCClientCertificate" (client certificate for CC authentication) or "LogServer" (log server certificate); if no value is assigned to type , it represents the HTTPS certificate.
Request	<i>XML_CertificateReq</i>
Response	<i>XML_ResponseStatus</i>

Table 16-366 DELETE /ISAPI/Security/serverCertificate/certSignReq

Method	DELETE
Description	Delete signature request of device certificate.
Query	type : certificate type, it can be set to "CCClientCertificate" (client certificate for CC authentication) or "LogServer" (log server certificate); if no value is assigned to type , it represents the HTTPS certificate.
Request	None.
Response	<i>XML_ResponseStatus</i>

16.8.46 /ISAPI/Security/serverCertificate/certSignReq?customID=

Create a PKCS#10 signature request of the client/server certificate.

Request URI Definition**Table 16-367 POST /ISAPI/Security/serverCertificate/certSignReq?customID=**

Method	POST
Description	Create a PKCS#10 signature request of the client/server certificate.
Query	customID : custom certificate ID, string, it consists of digits and characters.
Request	<i>XML_CertificateReq</i>
Response	Succeeded: <i>XML_CertificateResult</i> Failed: <i>XML_ResponseStatus</i>

16.8.47 /ISAPI/Security/serverCertificate/downloadCertSignReq

Download the certificate of authentication client.

Request URI Definition**Table 16-368 GET /ISAPI/Security/serverCertificate/downloadCertSignReq**

Method	GET
Description	Download the certificate of authenticated client.
Query	type : certificate type, it can be set to "CCClientCertificate" (client certificate for CC authentication) or "LogServer" (log server certificate); if no value is assigned to type , it represents the HTTPS certificate.
Request	None.
Response	Certificate data (binary)

16.8.48 /ISAPI/Security/serverCertificate/downloadCertSignReq?customID=

Export the client/server certificate.

Request URI Definition**Table 16-369 GET /ISAPI/Security/serverCertificate/downloadCertSignReq?customID=**

Method	GET
Description	Export the client/server certificate.

Query	customID: custom certificate ID, string, it consists of digits and characters.
Request	None.
Response	Certificate data.

Remarks

The returned certificate data should be either formatted exactly according to PKCS#10 standard or a PKCS#10 file encoded in PEM format.

16.8.49 /ISAPI/Security/serverCertificate/selfSignCert?customID=

Get or generate the PKCS#10 signature request of the client/server self-signed certificate.

Request URI Definition

Table 16-370 GET /ISAPI/Security/serverCertificate/selfSignCert?customID=

Method	GET
Description	Get the PKCS#10 signature request of the client/server self-signed certificate.
Query	customID: custom certificate ID, string, it consists of digits and characters.
Request	None.
Response	Succeeded: <i>XML_CertificateReq</i> Failed: <i>XML_ResponseStatus</i>

Table 16-371 PUT /ISAPI/Security/serverCertificate/selfSignCert?customID=

Method	PUT
Description	Generate the PKCS#10 signature request of the client/server self-signed certificate.
Query	customID: custom certificate ID, string, it consists of digits and characters.
Request	<i>XML_CertificateReq</i>
Response	<i>XML_ResponseStatus</i>

16.8.50 /ISAPI/Security/userCheck

Log in to the device by digest.

Request URI Definition

Table 16-372 GET /ISAPI/Security/userCheck

Method	GET
Description	Log in to the device by digest. This URI is used to check whether the user name matches with the password.
Query	None.
Request	None.
Response	Succeeded: <i>XML_userCheck</i> Failed: <i>XML_ResponseStatus</i>

16.8.51 /ISAPI/Security/UserPermission

Operations about the user permission of the device.

Request URI Definition

Table 16-373 GET /ISAPI/Security/UserPermission

Method	GET
Description	Get the user permission of the device.
Query	None.
Request	None.
Response	Succeeded: <i>XML_UserPermissionList</i> Failed: <i>XML_ResponseStatus</i>

Table 16-374 PUT /ISAPI/Security/UserPermission

Method	PUT
Description	Set the user permission of the device.
Query	None.
Request	<i>XML_UserPermissionList</i>
Response	<i>XML_ResponseStatus</i>

16.8.52 /ISAPI/Security/UserPermission/<ID>

Operations about a specific user's permission.

Request URI Definition

Table 16-375 GET /ISAPI/Security/UserPermission/<ID>

Method	GET
Description	Get a specific user's permission.
Query	None.
Request	None.
Response	Succeeded: <i>XML_UserPermission</i> Failed: <i>XML_ResponseStatus</i>

Table 16-376 PUT /ISAPI/Security/UserPermission/<ID>

Method	PUT
Description	Set a specific user's permission.
Query	None.
Request	<i>XML_UserPermission</i>
Response	<i>XML_ResponseStatus</i>

16.8.53 /ISAPI/Security/UserPermission/<ID>/localPermission

Get and set the local permission of a specified user.

Request URI Definition

Table 16-377 GET /ISAPI/Security/UserPermission/<ID>/localPermission

Method	GET
Description	Get the local permission of a specified user.
Query	None.
Request	None.
Response	<i>XML_localPermission</i>

Table 16-378 PUT /ISAPI/Security/UserPermission/<ID>/localPermission

Method	PUT
Description	Set the local permission of a specified user.
Query	None.

Request	<i>XML_localPermission</i>
Response	<i>XML_ResponseStatus</i>

16.8.54 /ISAPI/Security/UserPermission/<ID>/remotePermission

Get and set the remote permission of a specified user.

Request URI Definition

Table 16-379 GET /ISAPI/Security/UserPermission/<ID>/remotePermission

Method	GET
Description	Get the remote permission of a specified user.
Query	None.
Request	None.
Response	<i>XML_remotePermission</i>

Table 16-380 PUT /ISAPI/Security/UserPermission/<ID>/remotePermission

Method	PUT
Description	Set the remote permission of a specified user.
Query	None.
Request	<i>XML_remotePermission</i>
Response	<i>XML_ResponseStatus</i>

16.8.55 /ISAPI/Security/UserPermission/adminCap

Get the user permission capability of the administrator.

Request URI Definition

Table 16-381 GET /ISAPI/Security/UserPermission/adminCap

Method	GET
Description	Get the user permission capability of the administrator.
Query	None.
Request	None.
Response	Succeeded: <i>XML_UserPermissionCap</i> Failed: <i>XML_ResponseStatus</i>

Remarks

Only permissions that can be configured by the admin user should be returned.

16.8.56 /ISAPI/Security/UserPermission/anonymouslogin

Get or assign permission of the anonymous user.

Request URI Definition**Table 16-382 GET /ISAPI/Security/UserPermission/anonymouslogin**

Method	GET
Description	Get permission of the anonymous user.
Query	None.
Request	None.
Response	Succeeded: <i>XML_anonymouslogin</i> Failed: <i>XML_ResponseStatus</i>

Table 16-383 PUT /ISAPI/Security/UserPermission/anonymouslogin

Method	PUT
Description	Assign permission of the anonymous user.
Query	None.
Request	<i>XML_anonymouslogin</i>
Response	<i>XML_ResponseStatus</i>

16.8.57 /ISAPI/Security/UserPermission/operatorCap

Get the user permission capability of the operator.

Request URI Definition**Table 16-384 GET /ISAPI/Security/UserPermission/operatorCap**

Method	GET
Description	Get the user permission capability of the operator.
Query	None.
Request	None.
Response	Succeeded: <i>XML_UserPermissionCap</i>

	Failed: <i>XML_ResponseStatus</i>
--	-----------------------------------

Remarks

Only permissions that can be configured by the operator should be returned.

16.8.58 /ISAPI/Security/UserPermission/viewerCap

Get the user permission capability of the viewer.

Request URI Definition

Table 16-385 GET /ISAPI/Security/UserPermission/viewerCap

Method	GET
Description	Get the user permission capability of the viewer.
Query	None.
Request	None.
Response	<i>XML_UserPermissionCap</i>

16.8.59 /ISAPI/Security/users

Operations about the user list of the device.

Request URI Definition

Table 16-386 GET /ISAPI/Security/users

Method	GET
Description	Get the user list of the device.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	Succeeded: <i>XML_UserList</i> Failed: <i>XML_ResponseStatus</i>

Table 16-387 PUT /ISAPI/Security/users

Method	PUT
Description	Update the user list of the device.
Query	security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2.
Request	<i>XML_UserList</i>
Response	<i>XML_ResponseStatus</i>

Table 16-388 POST /ISAPI/Security/users

Method	POST
Description	Add a user to the user list of the device.
Query	security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2.
Request	<i>XML_User</i>
Response	<i>XML_ResponseStatus</i>

Table 16-389 DELETE /ISAPI/Security/users

Method	DELETE
Description	Delete the user list of the device.
Query	security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2. loginPassword: encrypted administrator password.

Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

- A default account "admin" must be provided with the administrator permission and it cannot be deleted.
- Passwords can only be uploaded and cannot be displayed when getting the user list of the device.
- The user ID should be returned when adding users to the user list.

Example

Sample Code for Adding User

```
POST /ISAPI/Security/users HTTP/1.1
Host: 10.17.132.49
Content-Length: 335
Connection: Keep-Alive
Authorization: Digest username="admin",
realm="DS-2CD2F12FWD-IWS",
nonce="4e6a4d774d6a63304f544936593255335a474d334f54673d",
uri="/ISAPI/Security/users",
cnonce="178e14d4977c835db891b5392b5d0a67",
nc=00000015,
response="d92518fb00cd2772cf212f5326b251be",
qop="auth"
```

```
<?xml version="1.0" encoding="UTF-8"?>
<User>
  <id>0</id>
  <userName>test</userName>
  <password>1qaz2wsx</password>
  <bondIpList>
    <bondIp>
      <id>1</id>
      <ipAddress>0.0.0.0</ipAddress>
      <ipv6Address>::</ipv6Address>
    </bondIp>
  </bondIpList>
  <macAddress></macAddress>
  <userLevel>Operator</userLevel>
  <attribute>
    <inherent>>false</inherent>
  </attribute>
</User>
```

```
HTTP/1.1 200 OK
Date: Tue, 14 Mar 2017 20:24:31 GMT
Server: App-webs/
Connection: close
Content-Length: 288
```

Content-Type: application/xml

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <requestURL>/ISAPI/Security/users</requestURL>
  <statusCode>1</statusCode>
  <statusString>OK</statusString>
  <id>2</id>
  <subStatusCode>ok</subStatusCode>
</ResponseStatus>
```

Example

Sample Code for Setting User Permission

```
PUT /ISAPI/Security/UserPermission/2 HTTP/1.1
Host: 10.17.132.49
Content-Length: 891
Connection: Keep-Alive
Authorization: Digest username="admin",
realm="DS-2CD2F12FWD-IWS",
nonce="4e6a4d774d6a63304f544936593255335a474d334f54673d",
uri="/ISAPI/Security/UserPermission/2",
cnonce="178e14d4977c835db891b5392b5d0a67",
nc=00000016,
response="f69a20db00b36e9bb46c26c06e9bae80",
qop="auth"
```

```
<?xml version="1.0" encoding="utf-8"?>
<UserPermission>
  <id>2</id>
  <userID>2</userID>
  <userType>operator</userType>
  <remotePermission>
    <parameterConfig>false</parameterConfig>
    <logOrStateCheck>true</logOrStateCheck>
    <upgrade>false</upgrade>
    <voiceTalk>true</voiceTalk>
    <restartOrShutdown>false</restartOrShutdown>
    <alarmOutOrUpload>false</alarmOutOrUpload>
    <contorlLocalOut>false</contorlLocalOut>
    <transParentChannel>false</transParentChannel>
    <preview>true</preview>
    <record>true</record>
    <ptzControl>true</ptzControl>
    <playBack>true</playBack>
    <videoChannelPermissionList>
      <videoChannelPermission>
        <id>1</id>
        <preview>true</preview>
        <record>true</record>
        <playBack>true</playBack>
      </videoChannelPermission>
    </videoChannelPermissionList>
  </remotePermission>
</UserPermission>
```



```

</videoChannelPermissionList>
<ptzChannelPermissionList>
  <ptzChannelPermission>
    <id>1</id>
    <ptzControl>true</ptzControl>
  </ptzChannelPermission>
</ptzChannelPermissionList>
</remotePermission>
</UserPermission>

HTTP/1.1 200 OK
Date: Tue, 14 Mar 2017 20:24:31 GMT
Server: App-webs/
Connection: close
Content-Length: 288
Content-Type: application/xml

<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <requestURL>/ISAPI/Security/UserPermission/2</requestURL>
  <statusCode>1</statusCode>
  <statusString>OK</statusString>
  <subStatusCode>ok</subStatusCode>
</ResponseStatus>

```

16.8.60 /ISAPI/Security/users/<ID>

Operations about the user's configuration of the device.

Request URI Definition

Table 16-390 GET /ISAPI/Security/users/<ID>

Method	GET
Description	Get a specific user's configuration of the device.
Query	security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	Succeeded: <i>XML_User</i> Failed: <i>XML_ResponseStatus</i>

Table 16-391 PUT /ISAPI/Security/users/<ID>

Method	PUT
Description	Set a specific user's configuration of the device.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	<i>XML_User</i>
Response	<i>XML_ResponseStatus</i>

Table 16-392 DELETE /ISAPI/Security/users/<ID>

Method	DELETE
Description	Delete a specific user's configuration of the device.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2. loginPassword : encrypted administrator password.
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The administrator account cannot be deleted.

16.8.61 /ISAPI/Security/webCertificate

Get or set the certificate type parameters of web service.

Request URI Definition

Table 16-393 GET /ISAPI/Security/webCertificate

Method	GET
Description	Get the certificate type parameters of web service.
Query	None.
Request	None.
Response	Succeeded: <i>XML_WebCertificate</i> Failed: <i>XML_ResponseStatus</i>

Table 16-394 PUT /ISAPI/Security/webCertificate

Method	PUT
Description	Set the certificate type parameters of web service.
Query	None.
Request	<i>XML_WebCertificate</i>
Response	<i>XML_ResponseStatus</i>

16.8.62 /ISAPI/SecurityCP/ReportCenterCfg/capabilities?format=json

Get the configuration capability of the report uploading method.

Request URI Definition

Table 16-395 GET /ISAPI/SecurityCP/ReportCenterCfg/capabilities?format=json

Method	GET
Description	Get the configuration capability of the report uploading method.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_Cap_ReportCenterCfg</i> Failed: <i>JSON_ResponseStatus</i>

16.8.63 /ISAPI/SecurityCP/ReportCenterCfg/<ID>?format=json

Operations about the configuration of the report uploading method.

Request URI Definition

Table 16-396 GET /ISAPI/SecurityCP/ReportCenterCfg/<ID>?format=json

Method	GET
Description	Get the parameters of the report uploading method.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_ReportCenterCfg</i> Failed: <i>JSON_ResponseStatus</i>

Table 16-397 PUT /ISAPI/SecurityCP/ReportCenterCfg/<ID>?format=json

Method	PUT
Description	Set the parameters of the report uploading method.
Query	format : determine the format of request or response message.
Request	<i>JSON_ReportCenterCfg</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the center group No.

16.9 /ISAPI/Smart

16.9.1 /ISAPI/Smart/capabilities

Get the capability of smart devices.

Request URI Definition

Table 16-398 GET /ISAPI/Smart/capabilities

Method	GET
Description	Get the capability of smart devices.
Query	None
Request	None
Response	Succeeded: <i>XML_SmartCap</i> Failed: <i>XML_ResponseStatus</i>

16.10 /ISAPI/Streaming

16.10.1 /ISAPI/Streaming/channels

Operations about the encoding configuration of multiple channels.

Request URI Definition

Table 16-399 GET /ISAPI/Streaming/channels

Method	GET
Description	Get the encoding parameters of multiple channels.
Query	None.
Request	None.
Response	<i>XML_StreamingChannelList</i>

Table 16-400 PUT /ISAPI/Streaming/channels

Method	PUT
Description	Set the encoding parameters of multiple channels.
Query	None.
Request	<i>XML_StreamingChannelList</i>
Response	<i>XML_ResponseStatus</i>

Table 16-401 POST /ISAPI/Streaming/channels

Method	POST
Description	Add an encoding channel.
Query	None.
Request	<i>XML_StreamingChannel</i>
Response	<i>XML_ResponseStatus</i>

Table 16-402 DELETE /ISAPI/Streaming/channels

Method	DELETE
Description	Delete all encoding channels.
Query	None.

Request	None.
Response	<i>XML_ResponseStatus</i>

16.10.2 /ISAPI/Streaming/channels/<ID>

Operations about the encoding configurations of a specific channel.

Request URI Definition

Table 16-403 GET /ISAPI/Streaming/channels/<ID>

Method	GET
Description	Get the encoding parameters of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_StreamingChannel</i> Failed: <i>XML_ResponseStatus</i>

Table 16-404 PUT /ISAPI/Streaming/channels/<ID>

Method	PUT
Description	Set the encoding parameters of a specific channel.
Query	None.
Request	<i>XML_StreamingChannel</i>
Response	<i>XML_ResponseStatus</i>

Table 16-405 DELETE /ISAPI/Streaming/channels/<ID>

Method	DELETE
Description	Delete the encoding parameters of a specific channel.
Query	None
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI refers to the channel ID.

Example

Interaction Example of Getting Encoding Parameters of A Specific Channel

```
GET /ISAPI/Streaming/channels/444 HTTP/1.1
...
HTTP/1.1 200 OK
Content-Type: application/xml; charset="UTF-8"
Content-Length: ISAPI

<?xml version="1.0" encoding="UTF-8"?>
<StreamingChannel version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id>444</id>
  <channelName>Input 1 MPEG-4 ASP</channelName>
  <enabled>true</enabled>
  <Transport>
    <rtspPortNo>554</rtspPortNo>
    <maxPacketSize>1446</maxPacketSize>
    <ControlProtocolList>
      <ControlProtocol>
        <streamingTransport>RTSP</streamingTransport>
      </ControlProtocol>
      <ControlProtocol>
        <streamingTransport>HTTP</streamingTransport>
      </ControlProtocol>
    </ControlProtocolList>
  </Transport>
  <Video>
    <enabled>true</enabled>
    <videoInputChannelID>2</videoInputChannelID>
    <videoCodecType>MPEG4</videoCodecType>
    <videoScanType>progressive</videoScanType>
    <videoResolutionWidth> 640</videoResolutionWidth>
    <videoResolutionHeight>480</videoResolutionHeight>
    <videoPositionX>0</videoPositionX>
    <videoPositionY>0</videoPositionY>
    <videoQualityControlType>CBR</videoQualityControlType>
    <constantBitRate>2000</constantBitRate>
    <maxFrameRate>2500</maxFrameRate>
    <keyFrameInterval>1000</keyFrameInterval>
    <rotationDegree>0</rotationDegree>
    <mirrorEnabled>false</mirrorEnabled>
    <snapshotImageType>JPEG</snapshotImageType>
  </Video>
  <Audio>
    <enabled>false</enabled>
    <audioInputChannelID>2</audioInputChannelID>
    <audioCompressionType> G.726</audioCompressionType>
    <audioBitRate>24</audioBitRate>
    <audioSamplingRate>8</audioSamplingRate>
  </Audio>
</StreamingChannel>
```

16.10.3 /ISAPI/Streaming/channels/<ID>/capabilities

Get encoding capability of a specific channel.

Request URI Definition

Table 16-406 GET /ISAPI/Streaming/channels/<ID>/capabilities

Method	GET
Description	Get the encoding capability of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_StreamingChannel</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI refers to the channel ID.

Example

Interaction Example of Getting Encoding Capability of A Specific Channel

```
GET /ISAPI/Streaming/channels/444/capabilities HTTP/1.1
...
HTTP/1.1 200 OK
Content-Type: application/xml; charset="UTF-8"
Content-Length: ISAPI

<?xml version="1.0" encoding="UTF-8"?>
<StreamingChannel version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id opt="111,222,333,444">444</id>
  <channelName min="0" max="64">Input 1 MPEG-4 ASP</channelName>
  <enabled opt="true,false" def="true">true</enabled>
  <Transport>
    <rtspPortNo min="0" max="65535" def="554">554</rtspPortNo>
    <maxPacketSize min="0" max="1500">1446</maxPacketSize>
    <audioPacketLength min="0" max="5000"/>
    <audioInboundPacketLength min="0" max="5000"/>
    <audioInboundPortNo min="0" max="65535"/>
    <videoSourcePortNo min="0" max="65535"/>
    <audioSourcePortNo min="0" max="65535"/>
  <ControlProtocolList>
    <ControlProtocol>
      <streamingTransport opt="RTSP/RTP,HTTP">RTSP</streamingTransport>
    </ControlProtocol>
    <ControlProtocol>
      <streamingTransport opt="RTSP/RTP,HTTP">HTTP</streamingTransport>
    </ControlProtocol>
  </ControlProtocolList>
</StreamingChannel>
```



```

</ControlProtocolList>
<Unicast>
  <enabled opt="true,false" def="false"/>
  <rtpTransportType opt="RTP/UDP,RTP/TCP"/>
</Unicast>
<Multicast>
  <enabled opt="true,false" def="false"/>
  <userTriggerThreshold/>
  <videoDestPortNo min="0" max="65535"/>
  <audioDestPortNo min="0" max="65535"/>
  <destIPAddress min="8" max="16"/>
  <destIPv6Address min="15" max="39"/>
  <ttl min="0" max="127" def="1"/>
</Multicast>
<Security>
  <enabled opt="true,false" def="false"/>
</Security>
</Transport>
<Video>
  <enabled opt="true,false">true</enabled>
  <videoInputChannelID opt="1,2,3,4">2</videoInputChannelID>
  <videoCodecType opt="MJPEG,MPEG4">MPEG4</videoCodecType>
  <videoScanType opt="interlaced,progressive">progressive</videoScanType>
  <videoResolutionWidth min="0" max="640">640</videoResolutionWidth>
  <videoResolutionHeight min="0" max="480">480</videoResolutionHeight>
  <videoPositionX min="0" max="640">0</videoPositionX>
  <videoPositionY min="0" max="480">0</videoPositionY>
  <videoQualityControlType opt="CBR,VBR">CBR</videoQualityControlType>
  <constantBitRate min="50" max="4000" dynamic="true">2000</constantBitRate>
  <maxFrameRate opt="2500,1250,625,312,156,78, 830" dynamic="true">2500</maxFrameRate>
  <keyFrameInterval min="0", max="10000">1000</keyFrameInterval>
  <rotationDegree opt="0,90,180,270" def="0">0</rotationDegree>
  <mirrorEnabled opt="true,false" def="false">false</mirrorEnabled>
  <snapShotImageType opt="JPEG" def="JPEG">JPEG</snapShotImageType>
</Video>
<Audio>
  <enabled opt="true,false" def="false">false</enabled>
  <audioInputChannelID opt="1,2,3,4">2</audioInputChannelID>
  <audioCompressionType opt="G.726,G.711ulaw" def="G.726">G.726</audioCompressionType>
  <audioBitRate opt="16,24,32,40" def="32" dynamic="true">24</audioBitRate>
  <audioSamplingRate opt="8" dynamic="true">8</audioSamplingRate>
  <audioResolution opt="3,4,5,6" dynamic="true"/>
</Audio>
</StreamingChannel>

```

16.10.4 /ISAPI/Streaming/channels/<ID>/dynamicCap

Get the dynamic encoding capability of a specific channel.

Request URI Definition

Table 16-407 GET /ISAPI/Streaming/channels/<ID>/dynamicCap

Method	GET
Description	Get the dynamic encoding capability of a specific channel.
Query	None.
Request	None.
Response	<i>XML_DynamicCap</i>

Remarks

The <ID> in the URI refers to the channel ID.

16.10.5 /ISAPI/Streaming/channels/<ID>/picture

Manually capture picture in the video stream of a specific channel.

Request URI Definition

Table 16-408 GET /ISAPI/Streaming/channels/<ID>/picture

Method	GET
Description	Manually capture picture in the video stream of a specific channel.
Query	(Optional) videoResolutionWidth : width of captured picture resolution. (Optional) videoResolutionHeight : height of captured picture resolution. (Optional) snapShotImageType : captured picture format, only the JPEG format is supported.
Request	None.
Response	Picture data, which is transmitted on HTTP.

Remarks

- The <ID> in the request URI refers to the streaming channel ID, which is equal to (channel No. × 100 + 1). For example, if a camera only has one channel, its ID is 101.
- This URI only supports capturing picture in main stream, and only JPEG format is supported.

Example

Sample Code for Manually Capturing Picture

```
GET /ISAPI/Streaming/channels/1/picture HTTP/1.1
Host: 10.17.132.49
```

```

Connection: Keep-Alive
Authorization: Digest username="admin",
realm="DS-2CD2F12FWD-IWS",
nonce="4d305134517a55344d6a55365a4445324f544130597a453d",
uri="/ISAPI/Streaming/channels/1/picture",
cnonce="481bb1afa48b24e512778b084bfe4977",
nc=00000001,
response="e5670d5d9d40c522afabb11b5fa35268",
qop="auth"

```

```

HTTP/1.1 200 OK
Content-Type: image/jpeg; charset="UTF-8"
Content-Length:19991

```

```

.....JFIF.....C.
.
...
.. .....!... .)10.)-,3:J>36F7,- (Binary picture data)

```

16.10.6 /ISAPI/Streaming/channels/<ID>/regionClip

Get or set target cropping parameters by streaming channel.

Request URI Definition

Table 16-409 GET /ISAPI/Streaming/channels/<ID>/regionClip

Method	GET
Description	Get target cropping parameters by streaming channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_RegionClip</i> Failed: <i>XML_ResponseStatus</i>

Table 16-410 PUT /ISAPI/Streaming/channels/<ID>/regionClip

Method	PUT
Description	Set target cropping parameters by streaming channel.
Query	None.
Request	<i>XML_RegionClip</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to streaming channel ID.

16.10.7 /ISAPI/Streaming/channels/<ID>/regionClip/capabilities

Get configuration capability of target cropping by streaming channel.

Request URI Definition

Table 16-411 GET /ISAPI/Streaming/channels/<ID>/regionClip/capabilities

Method	GET
Description	Get configuration capability of target cropping by streaming channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_RegionClip</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to streaming channel ID.

16.10.8 /ISAPI/Streaming/channels/<ID>/smartOverlap/capabilities?format=json

Get the capability of configuring the stream for displaying VCA rules of smart events.

Request URI Definition

Table 16-412 GET /ISAPI/Streaming/channels/<ID>/smartOverlap/capabilities?format=json

Method	GET
Description	Get the capability of configuring the stream for displaying VCA rules of smart events.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_SmartOverlapCap</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

16.10.9 /ISAPI/Streaming/channels/<ID>/smartOverlap?format=json

Get or set the stream parameters for displaying VCA rules of smart events.

Request URI Definition

Table 16-413 GET /ISAPI/Streaming/channels/<ID>/smartOverlap?format=json

Method	GET
Description	Get the stream parameters for displaying VCA rules of smart events.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_SmartOverlap</i> Failed: <i>JSON_ResponseStatus</i>

Table 16-414 PUT /ISAPI/Streaming/channels/<ID>/smartOverlap?format=json

Method	PUT
Description	Set the stream parameters for displaying VCA rules of smart events.
Query	format: determine the format of request or response message.
Request	<i>JSON_SmartOverlap</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

16.10.10 /ISAPI/Streaming/channels/<ID>/status

Get the streaming or encoding status of a specific channel.

Request URI Definition

Table 16-415 GET /ISAPI/Streaming/channels/<ID>/status

Method	GET
Description	Get the streaming or encoding status of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_StreamingSessionStatusList</i>

	Failed: <i>XML_ResponseStatus</i>
--	-----------------------------------

Remarks

The <ID> in the request URI refers to the streaming channel ID.

16.10.11 /ISAPI/Streaming/encryption/capabilities?format=json

Get stream encryption capability.

Request URI Definition

Table 16-416 GET /ISAPI/Streaming/encryption/capabilities?format=json

Method	GET
Description	Get stream encryption capability.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_EncryptionCap</i>

16.10.12 /ISAPI/Streaming/encryption/secretKey?format=json

Operations about the configuration of stream encryption key.

Request URI Definition

Table 16-417 GET /ISAPI/Streaming/encryption/secretKey?format=json

Method	GET
Description	Get the configuration parameters of stream encryption key.
Query	format: determine the format of request or response message. security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	<i>JSON_secretKey</i>

Table 16-418 PUT /ISAPI/Streaming/encryption/secretKey?format=json

Method	GET
Description	Set stream encryption key.
Query	format: determine the format of request or response message. security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2.
Request	<i>JSON_secretKey</i>
Response	<i>JSON_ResponseStatus</i>

16.10.13 /ISAPI/Streaming/encryption?format=json

Operations about the configuration of stream encryption.

Request URI Definition

Table 16-419 GET /ISAPI/Streaming/encryption?format=json

Method	GET
Description	Get the configuration parameters of stream encryption.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_EnableEncryption</i>

Table 16-420 PUT /ISAPI/Streaming/encryption?format=json

Method	GET
Description	Enable stream encryption.
Query	format: determine the format of request or response message.
Request	<i>JSON_EnableEncryption</i>
Response	<i>JSON_ResponseStatus</i>

16.10.14 /ISAPI/Streaming/status

Get the streaming or encoding status of all channels.

Request URI Definition

Table 16-421 GET /ISAPI/Streaming/status

Method	GET
Description	Get the streaming or encoding status of all channels.
Query	None.
Request	None.
Response	Succeeded: <i>XML_StreamingStatus</i> Failed: <i>XML_ResponseStatus</i>

16.10.15 rtsp://<host>[:port]/ISAPI/Streaming/channels/<ID>

Get basic parameter or session information for live view, or start and stop live view.

Request URL Definition

Table 16-422 DESCRIBE rtsp://<host>[:port]/ISAPI/Streaming/channels/<ID>

Method	DESCRIBE
Description	Get basic parameters for live view.
Query	None.
Request	None.
Response	None.

Table 16-423 SETUP rtsp://<host>[:port]/ISAPI/Streaming/channels/<ID>

Method	SETUP
Description	Get session information for live view.
Query	None.
Request	None.
Response	None.

Table 16-424 PLAY rtsp://<host>[:port]/ISAPI/Streaming/channels/<ID>

Method	PLAY
Description	Start live view.
Query	None.
Request	None.
Response	None.

Table 16-425 TEARDOW rtsp://<host>[:port]/ISAPI/Streaming/channels/<ID>

Method	TEARDOW
Description	Stop live view.
Query	None.
Request	None.
Response	None.

Remarks

The <ID> in the URL is defined as (channel No.)*100+(stream type value).

channel No.

For analog channel, the No. starts from 1, and for digital channel, the No. starts from the last analog channel No.

E.g., if the device has 16 analog channels and 16 digital channels, the analog channel No. is between 1 and 16, and the digital channel No. is between 17 and 32.

stream type value

The stream type values contain 1, 2, and 3. 1-main stream, 2-sub-stream, 3-third stream.

E.g., if ID is 101, it indicates the main stream of channel No.1; if ID is 102, it indicates the sub-stream of channel No. 2; if ID is 1601, it indicates the main stream of channel 16;...

16.10.16 rtsp://<host>[:port]/ISAPI/Streaming/channels/<ID>?npq=

Operations about multicast based on network protocol quality (NPQ).

Request URL Definition

Table 16-426 DESCRIBE rtsp://<host>[:port]/ISAPI/Streaming/channels/<ID>?npq=

Method	DESCRIBE
Description	Request for getting basic parameters for multicast based on NPQ.

Query	npq : Network Protocol Quality (NPQ), this query parameter is used to enable streaming according to NPQ and by FEC (Forward Error Correction) method if its value is set to "fec".
Request	None.
Response	None.

Table 16-427 SETUP `rtsp://<host>[:port]/ISAPI/Streaming/channels/<ID>?npq=`

Method	SETUP
Description	Request for getting session information for multicast based on NPQ.
Query	npq : Network Protocol Quality (NPQ), this query parameter is used to enable streaming according to NPQ and by FEC (Forward Error Correction) method if its value is set to "fec".
Request	None.
Response	None.

Table 16-428 PLAY `rtsp://<host>[:port]/ISAPI/Streaming/channels/<ID>?npq=`

Method	PLAY
Description	Start multicast based on NPQ.
Query	npq : Network Protocol Quality (NPQ), this query parameter is used to enable streaming according to NPQ and by FEC (Forward Error Correction) method if its value is set to "fec".
Request	None.
Response	None.

Table 16-429 TEARDOW `rtsp://<host>[:port]/ISAPI/Streaming/channels/<ID>?npq=`

Method	TEARDOW
Description	Stop multicast based on NPQ.
Query	npq : Network Protocol Quality (NPQ), this query parameter is used to enable streaming according to NPQ and by FEC (Forward Error Correction) method if its value is set to "fec".
Request	None.
Response	None.

Remarks

The <ID> in the request URL is the value of (channel No. × 100 + stream type value), e.g., if the channel No. is 1, and the stream type value is 1, the ID equals to 101.

- For channel No., if the device contains analog and digital channels, the analog channel No. starts from 1, and the digital channel No. starts from the next No. of the end analog channel. For example, if the device contains 16 analog channels and 16 digital channels, the analog channel No. is between 1 and 16, while the digital channel No. is between 17 and 32.
- For stream type, only two values are available, that is, 1 (main stream) and 2 (sub-stream).

16.10.17 rtsp://<host>[:port]/ISAPI/Streaming/tracks/<ID>?starttime=&endtime=

Operations about playback of a specific channel.

Request URL Definition

Table 16-430 DESCRIBE rtsp://<host>[:port]/ISAPI/Streaming/tracks/<ID>?starttime=&endtime=

Method	DESCRIBE
Description	Request for getting basic parameters for playback.
Query	starttime : start time of the target stream, whose format is based on the ISO 8601 standard. (Optional) endtime : end time of the target stream, whose format is based on the ISO 8601 standard. If this field is empty, the stream will be continuously obtained until the session is terminated or paused manually.
Request	None.
Response	None.

Table 16-431 SETUP rtsp://<host>[:port]/ISAPI/Streaming/tracks/<ID>?starttime=&endtime=

Method	SETUP
Description	Request for getting session information for playback.
Query	starttime : start time of the target stream, whose format is based on the ISO 8601 standard. (Optional) endtime : end time of the target stream, whose format is based on the ISO 8601 standard. If this field is empty, the stream will be continuously obtained until the session is terminated or paused manually.
Request	None.
Response	None.

Table 16-432 PLAY rtsp://<host>[:port]/ISAPI/Streaming/tracks/<ID>?starttime=&endtime=

Method	PLAY
Description	Start playback.
Query	starttime : start time of the target stream, whose format is based on the ISO 8601 standard. (Optional) endtime : end time of the target stream, whose format is based on the ISO 8601 standard. If this field is empty, the stream will be continuously obtained until the session is terminated or paused manually.
Request	None.
Response	None.

Table 16-433 TEARDOW rtsp://<host>[:port]/ISAPI/Streaming/tracks/<ID>?starttime=&endtime=

Method	TEARDOW
Description	Stop playback.
Query	starttime : start time of the target stream, whose format is based on the ISO 8601 standard. (Optional) endtime : end time of the target stream, whose format is based on the ISO 8601 standard. If this field is empty, the stream will be continuously obtained until the session is terminated or paused manually.
Request	None.
Response	None.

Remarks

The <ID> in the URL is defined as (channel No.)*100+(stream type value).

channel No.

For analog channel, the No. starts from 1, and for digital channel, the No. starts from the last analog channel No.

E.g., if the device has 16 analog channels and 16 digital channels, the analog channel No. is between 1 and 16, and the digital channel No. is between 17 and 32.

stream type value

The stream type values contain 1, 2, and 3. 1-main stream, 2-sub-stream, 3-third stream.

E.g., if **ID** is 101, it indicates the main stream of channel No.1; if **ID** is 102, it indicates the sub-stream of channel No. 2; if **ID** is 1601, it indicates the main stream of channel 16;...

16.11 /ISAPI/System

16.11.1 /ISAPI/System/accessDevice/associatedChannel?format=json

Get the linked channel information of added IoT device.

Request URI Definition

Table 16-434 GET /ISAPI/System/accessDevice/associatedChannel?format=json

Method	GET
Description	Get the linked channel information of added IoT device.
Query	format : determine the format of request or response message.
Request	<i>JSON_IOT_ChannelInfoList</i>
Response	<i>JSON_AssociatedChannelList</i>

16.11.2 /ISAPI/System/AcsUpdate/capabilities

Get the capability of upgrading device (slave access control device, peripheral module, etc.).

Request URI Definition

Table 16-435 GET /ISAPI/System/AcsUpdate/capabilities

Method	GET
Description	Get the capability of upgrading device (slave access control device, peripheral module, etc.).
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_AcsUpdate</i> Failed: <i>XML_ResponseStatus</i>

16.11.3 /ISAPI/System/activate

Device activation.

Request URI Definition**Table 16-436 PUT /ISAPI/System/activate**

Method	PUT
Description	Send activation password to activate the device.
Query	None.
Request	<i>XML_ActivateInfo</i>
Response	<i>XML_ResponseStatus</i>

16.11.4 /ISAPI/System/algorithmsInfo/capabilities?format=json

Get the capability of getting the algorithm library information.

Request URI Definition**Table 16-437 GET /ISAPI/System/algorithmsInfo/capabilities?format=json**

Method	GET
Description	Get the capability of getting the algorithm library information.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_AlgorithmsInfoCap</i> Failed: <i>JSON_ResponseStatus</i>

16.11.5 /ISAPI/System/algorithmsInfo?format=json

Get the algorithm library information.

Request URI Definition**Table 16-438 GET /ISAPI/System/algorithmsInfo?format=json**

Method	GET
Description	Get the algorithm library information.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_AlgorithmsInfo</i>

	Failed: <i>JSON_ResponseStatus</i>
--	------------------------------------

16.11.6 /ISAPI/System/algorithmsVersion

Get the version information of algorithm library.

Request URI Definition

Table 16-439 GET /ISAPI/System/algorithmsVersion

Method	GET
Description	Get the version information of algorithm library.
Query	None
Request	None
Response	Succeeded: <i>XML_AlgorithmsVersion</i> Failed: <i>XML_ResponseStatus</i>

16.11.7 /ISAPI/System/Audio/AudioIn/channels/<ID>

Get or set audio input parameters of a specific channel.

Request URI Definition

Table 16-440 GET /ISAPI/System/Audio/AudioIn/channels/<ID>

Method	GET
Description	Get audio input parameters of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_AudioIn</i> Failed: <i>XML_ResponseStatus</i>

Table 16-441 PUT /ISAPI/System/Audio/AudioIn/channels/<ID>

Method	PUT
Description	Set audio input parameters of a specific channel.
Query	None.
Request	<i>XML_AudioIn</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

16.11.8 /ISAPI/System/Audio/AudioIn/channels/<ID>/capabilities

Get the audio input capability of a specific channel.

Request URI Definition

Table 16-442 GET /ISAPI/System/Audio/AudioIn/channels/<ID>/capabilities

Method	GET
Description	Get the audio input capability of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_AudioInCap</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

16.11.9 /ISAPI/System/Audio/AudioOut/channels/<ID>

Get or set audio output parameters of a specific channel.

Request URI Definition

Table 16-443 GET /ISAPI/System/Audio/AudioOut/channels/<ID>

Method	GET
Description	Get audio output parameters of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_AudioOut</i> Failed: <i>XML_ResponseStatus</i>

Table 16-444 PUT /ISAPI/System/Audio/AudioOut/channels/<ID>

Method	PUT
Description	Set audio output parameters of a specific channel.

Query	None.
Request	<i>XML_AudioOut</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

16.11.10 /ISAPI/System/Audio/AudioOut/channels/<ID>/capabilities

Get the audio output capability of a specific channel.

Request URI Definition

Table 16-445 GET /ISAPI/System/Audio/AudioOut/channels/<ID>/capabilities

Method	GET
Description	Get the audio output capability of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_AudioOutCap</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

16.11.11 /ISAPI/System/Audio/capabilities

Get the audio capability.

Request URI Definition

Table 16-446 GET /ISAPI/System/Audio/capabilities

Method	GET
Description	Get the audio capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_AudioCap</i> Failed: <i>XML_ResponseStatus</i>

16.11.12 /ISAPI/System/Audio/channels

Get the configuration of all audio channels of the device.

Request URI Definition

Table 16-447 GET /ISAPI/System/Audio/channels

Method	GET
Description	Get the configuration of all audio channels of the device.
Query	None.
Request	None.
Response	Succeeded: <i>XML_AudioChannelList</i> Failed: <i>XML_ResponseStatus</i>

16.11.13 /ISAPI/System/Audio/channels/<ID>

Get the configuration of a specific audio channel of the device.

Request URI Definition

Table 16-448 GET /ISAPI/System/Audio/channels/<ID>

Method	GET
Description	Get the configuration of a specific audio channel of the device.
Query	None.
Request	None.
Response	Succeeded: <i>XML_AudioChannel</i> Failed: <i>XML_ResponseStatus</i>

16.11.14 /ISAPI/System/Audio/channels/<ID>/dynamicCap

Get dynamic audio capability by channel.

Request URI Definition

Table 16-449 GET /ISAPI/System/Audio/channels/<ID>/dynamicCap

Method	GET
Description	Get dynamic audio capability by channel.

Query	None.
Request	<i>XML_AudioDescriptor</i>
Response	Succeeded: <i>XML_DynamicCap</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the audio input channel ID.

16.11.15 /ISAPI/System/autoMaintenance/capabilities?format=json

Get the capability of automatic maintenance configuration.

Request URI Definition

Table 16-450 GET /ISAPI/System/autoMaintenance/capabilities?format=json

Method	GET
Description	Get the capability of automatic maintenance configuration.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_AutoMaintenanceCap</i>

16.11.16 /ISAPI/System/autoMaintenance?format=json

Get automatic maintenance configuration parameters.

Request URI Definition

Table 16-451 GET /ISAPI/System/autoMaintenance?format=json

Method	GET
Description	Get automatic maintenance configuration parameters.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_AutoMaintenance</i>

Table 16-452 PUT /ISAPI/System/autoMaintenance?format=json

Method	PUT
Description	Set automatic maintenance parameters.

Query	format: determine the format of request or response message.
Request	<i>JSON_AutoMaintenance</i>
Response	<i>JSON_ResponseStatus</i>

16.11.17 /ISAPI/System/autoSwitch/capabilities?format=json

Get configuration capability of auto power on or off.

Request URI Definition

Table 16-453 GET /ISAPI/System/autoSwitch/capabilities?format=json

Method	GET
Description	Get configuration capability of auto power on or off.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_AutoSwitchCap</i> Failed: <i>JSON_ResponseStatus</i>

16.11.18 /ISAPI/System/autoSwitch?format=json

Get or set parameters of auto power on or off.

Request URI Definition

Table 16-454 GET /ISAPI/System/autoSwitch?format=json

Method	GET
Description	Get parameters of auto power on or off.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_AutoSwitch</i> Failed: <i>JSON_ResponseStatus</i>

Table 16-455 PUT /ISAPI/System/autoSwitch?format=json

Method	PUT
Description	Set parameters of auto power on or off.
Query	None.

Request	<i>JSON_AutoSwitch</i>
Response	<i>JSON_ResponseStatus</i>

16.11.19 /ISAPI/System/Bluetooth/capabilities

Get the bluetooth configuration capability.

Request URI Definition

Table 16-456 GET /ISAPI/System/Bluetooth/capabilities

Method	GET
Description	Get the bluetooth configuration capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_BluetoothCap</i> Failed: <i>XML_ResponseStatus</i>

16.11.20 /ISAPI/System/Bluetooth/ibeaconParam

Operations about the ibeacon bluetooth configuration.

Request URI Definition

Table 16-457 GET /ISAPI/System/Bluetooth/ibeaconParam

Method	GET
Description	Get the ibeacon bluetooth parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_IbeaconParam</i> Failed: <i>XML_ResponseStatus</i>

Table 16-458 PUT /ISAPI/System/Bluetooth/ibeaconParam

Method	PUT
Description	Set the ibeacon bluetooth parameters.
Query	None.

Request	<i>XML_IbeaconParam</i>
Response	<i>XML_ResponseStatus</i>

16.11.21 /ISAPI/System/Bluetooth/ibeaconParam/capabilities

Get the ibeacon bluetooth configuration capability.

Request URI Definition

Table 16-459 GET /ISAPI/System/Bluetooth/ibeaconParam/capabilities

Method	GET
Description	Get the ibeacon bluetooth configuration capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_IbeaconParam</i> Failed: <i>XML_ResponseStatus</i>

16.11.22 /ISAPI/System/capabilities

Get device capability.

Request URI Definition

Table 16-460 GET /ISAPI/System/capabilities

Method	GET
Description	Get device capability.
Query	None
Request	None.
Response	Succeeded: <i>XML_DeviceCap</i> Failed: <i>XML_ResponseStatus</i>

16.11.23 /ISAPI/System/channels/<ID>/vibrationDetection/capabilities? format=json

Get the capability of vibration detection.

Request URI Definition

Table 16-461 GET /ISAPI/System/channels/<ID>/vibrationDetection/capabilities?format=json

Method	GET
Description	Get the capability of vibration detection.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_VibrationDetectionCap</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URI refers to the channel ID.

16.11.24 /ISAPI/System/channels/<ID>/vibrationDetection?format=json

Get or set the vibration detection parameters.

Request URI Definition

Table 16-462 GET /ISAPI/System/channels/<ID>/vibrationDetection?format=json

Method	GET
Description	Get vibration detection parameters.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_VibrationDetection</i> Failed: <i>JSON_ResponseStatus</i>

Table 16-463 PUT /ISAPI/System/channels/<ID>/vibrationDetection?format=json

Method	PUT
Description	Set vibration detection parameters.
Query	format: determine the format of request or response message.
Request	<i>JSON_VibrationDetection</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URI refers to the channel ID.

16.11.25 /ISAPI/System/configurationData?secretkey=

Import or export configuration files securely.

Request URI Definition

Table 16-464 GET /ISAPI/System/configurationData?secretkey=

Method	GET
Description	Export the configuration files securely.
Query	security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2. secretkey: the verification key, it is provided by the upper-layer. It should be encrypted for exporting and recorded for importing.
Request	None.
Response	Opaque data (binary data).

Table 16-465 PUT or POST /ISAPI/System/configurationData?secretkey=

Method	PUT, POST
Description	Import the configuration files securely.
Query	security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2. secretkey: the verification key, it must be the same as the exported one and should be encrypted for importing.
Request	Opaque data (binary data for PUT method, data in form format for POST method).
Response	<i>XML_ResponseStatus</i>

Remarks

The device may reboot after importing the configuration file.

16.11.26 /ISAPI/System/configurationData?type=

Import or export device configuration files safely.

Request URI Definition

Table 16-466 GET /ISAPI/System/configurationData?type=

Method	GET
Description	Export device configuration files safely.
Query	security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2. secretkey: the verification key, which is provided by the upper-layer and should be encrypted for importing. type: the type of configuration parameters to be exported or imported, which contains "OSD", "MODE", "IMAGE" and "IOOUT" types. For "OSD" type, the OSD (On Screen Display) parameters will be imported or exported; for "MODE" type, the application mode parameters will be imported or exported; for "IMAGE" type, the image parameters will be imported or exported; for "IOOUT" type, the IO output parameters will be imported or exported. Multiple types can be selected and they should be separated by commas, e.g., /ISAPI/System/configurationData?type=OSD,MODE,IMAGE,IOOUT. If no type is selected, it is recommended to return failure response message or set type to NULL.
Request	None.
Response	Succeeded: Opaque data. Failed: <i><u>XML_ResponseStatus</u></i>

Table 16-467 PUT or POST /ISAPI/System/configurationData?type=

Method	PUT, POST
Description	Import device configuration files safely.
Query	security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2. secretkey: the verification key, which is provided by the upper-layer and should be encrypted for importing. type: the type of configuration parameters to be exported or imported, which contains "OSD", "MODE", "IMAGE" and "IOOUT" types. For "OSD" type, the OSD (On Screen Display) parameters will be imported or exported; for "MODE" type, the application mode parameters will be imported or exported; for "IMAGE" type, the image parameters will be imported or exported; for "IOOUT" type, the IO output parameters will be imported or exported. Multiple types can be selected and they should be separated by commas, e.g., /ISAPI/System/configurationData?type=OSD,MODE,IMAGE,IOOUT. If no type is selected, it is recommended to return failure response message or set type to NULL.
Request	Opaque data.
Response	<i>XML_ResponseStatus</i>

Remarks

- This URI will import or export all configuration parameters by default.
- The device needs to reboot for the imported parameters to take effect.

16.11.27 /ISAPI/System/deviceInfo

Operations about the device information.

Request URI Definition

Table 16-468 GET /ISAPI/System/deviceInfo

Method	GET
Description	Get the device information.
Query	None
Request	None.
Response	Succeeded: <i>XML_DeviceInfo</i> Failed: <i>XML_ResponseStatus</i>

Table 16-469 PUT /ISAPI/System/deviceInfo

Method	PUT
Description	Set the device information.
Query	None
Request	<i>XML_DeviceInfo</i>
Response	<i>XML_ResponseStatus</i>

16.11.28 /ISAPI/System/deviceInfo/capabilities

Get the device information configuration capability.

Request URI Definition

Table 16-470 GET /ISAPI/System/deviceInfo/capabilities

Method	GET
Description	Get the device information configuration capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_DeviceInfo</i> Failed: <i>XML_ResponseStatus</i>

16.11.29 /ISAPI/System/DeviceLanguage

Operations about device language configuration.

Request URI Definition

Table 16-471 GET /ISAPI/System/DeviceLanguage

Method	GET
Description	Get device language configuration parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_DeviceLanguage</i> Failed: <i>XML_ResponseStatus</i>

Table 16-472 PUT /ISAPI/System/DeviceLanguage

Method	PUT
Description	Set device language parameters.
Query	None.
Request	<i>XML_DeviceLanguage</i>
Response	<i>XML_ResponseStatus</i>

16.11.30 /ISAPI/System/DeviceLanguage/capabilities

Get the capability of device language configuration.

Request URI Definition

Table 16-473 GET /ISAPI/System/DeviceLanguage/capabilities

Method	GET
Description	Get the capability of device language configuration.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_DeviceLanguage</i> Failed: <i>XML_ResponseStatus</i>

16.11.31 /ISAPI/System/diagnosedData

Get device diagnosis information.

Request URI Definition

Table 16-474 GET /ISAPI/System/diagnosedData

Method	GET
Description	Get device diagnosis information.
Query	None.
Request	None.
Response	Non-transparent data

Remarks

Before getting the device diagnosis information, you should get the information export status via URI GET */ISAPI/System/diagnosedData/exportStatus* , only when the node <status> in *XML_ExportStatus* is "unexport", you can call this URI to get the device diagnosis information.

16.11.32 /ISAPI/System/diagnosedData/exportStatus

Get the export status of device diagnosis information.

Request URI Definition

Table 16-475 GET /ISAPI/System/diagnosedData/exportStatus

Method	GET
Description	Get the export status of device diagnosis information.
Query	None.
Request	None.
Response	<i>XML_ExportStatus</i>

16.11.33 /ISAPI/System/diagnosedData/parameter

Get or set the parameters of exporting device diagnose information.

Request URI Definition

Table 16-476 GET /ISAPI/System/diagnosedData/parameter

Method	GET
Description	Get the parameters of exporting device diagnose information.
Query	None.

Request	None.
Response	Succeeded: <i>XML_DiagnosedDataParameter</i> Failed: <i>XML_ResponseStatus</i>

Table 16-477 PUT /ISAPI/System/diagnosedData/parameter

Method	PUT
Description	Set the parameters of exporting device diagnose information.
Query	None.
Request	<i>XML_DiagnosedDataParameter</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The speed dome does not support getting parameters of exporting diagnose information.

16.11.34 /ISAPI/System/diagnosedData/parameter/capabilities

Get the capability of exporting device diagnose information.

Request URI Definition

Table 16-478 GET /ISAPI/System/diagnosedData/parameter/capabilities

Method	GET
Description	Get the capability of exporting device diagnose information.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_DiagnosedDataParameter</i> Failed: <i>XML_ResponseStatus</i>

16.11.35 /ISAPI/System/diagnosis/capabilities?format=json

Get the device diagnosis capability.

Request URI Definition

Table 16-479 GET /ISAPI/System/diagnosis/capabilities?format=json

Method	GET
Description	Get the device diagnosis capability.

Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_DiagnosisCondCap</i> Failed: <i>JSON_ResponseStatus</i>

16.11.36 /ISAPI/System/diagnosis?format=json

Diagnose the device.

Request URI Definition

Table 16-480 POST /ISAPI/System/diagnosis?format=json

Method	POST
Description	Diagnose the device.
Query	format: determine the format of request or response message.
Request	<i>JSON_DiagnosisCond</i>
Response	<i>JSON_DiagnosisResult</i>

16.11.37 /ISAPI/System/doubleLensParking/capabilities

Get the capability of dual-lens parking space camera.

Request URI Definition

Table 16-481 GET /ISAPI/System/doubleLensParking/capabilities

Method	GET
Description	Get the capability of dual-lens parking space camera.
Query	None.
Request	None.
Response	Succeeded: <i>XML_DoubleLensParkingCap</i> Failed: <i>XML_ResponseStatus</i>

16.11.38 /ISAPI/System/dumpData

Get the dump file data of the device.

Request URI Definition

Table 16-482 GET /ISAPI/System/dumpData

Method	GET
Description	Get the dump file data of the device.
Query	None.
Request	None.
Response	Opaque data.

Remarks

The dump file data of the device may be in binary format or other format.

16.11.39 /ISAPI/System/exportLogToUSB/capabilities?format=json

Get the configuration capability of exporting log to USB flash drive.

Request URI Definition

Table 16-483 GET /ISAPI/System/exportLogToUSB/capabilities?format=json

Method	GET
Description	Get the configuration capability of exporting log to USB flash drive.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_ExportLogToUSBCap</i> Failed: <i>JSON_ResponseStatus</i>

16.11.40 /ISAPI/System/exportLogToUSB/mode?format=json

Set parameters of exporting logs to USB flash drive.

Request URI Definition

Table 16-484 PUT /ISAPI/System/exportLogToUSB/mode?format=json

Method	PUT
Description	Set parameters of exporting logs to USB flash drive.
Query	format: determine the format of request or response message.

Request	<i>JSON_ExportLogToUSB</i>
Response	<i>JSON_ResponseStatus</i>

16.11.41 /ISAPI/System/exportLogToUSB/status?format=json

Get the status of exporting logs to USB flash drive.

Table 16-485 GET /ISAPI/System/exportLogToUSB/status?format=json

Method	GET
Description	Get the status of exporting logs to USB flash drive.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_ExportLogToUSBStatus</i> Failed: <i>JSON_ResponseStatus</i>

16.11.42 /ISAPI/System/externalDevice/capabilities

Get peripherals configuration capability.

Request URI Definition

Table 16-486 GET /ISAPI/System/externalDevice/capabilities

Method	GET
Description	Get peripherals configuration capability.
Query	None.
Request	None.
Response	<i>XML_Cap_ExternalDevice</i>

16.11.43 /ISAPI/System/factoryReset?mode=

Restore the device to default settings.

Request URI Definition

Table 16-487 PUT /ISAPI/System/factoryReset?mode=

Method	PUT
Description	Restore the device to default settings.
Query	mode: The recovery mode, which contains "full", "basic", and "part" mode. The default mode is "full", and all the device parameters will be restored to default settings; for "basic" mode, the device parameters, except network and user parameters, will be restored to default settings; for "part" mode, only a part of default settings will be restored.
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

- For access control devices, this URI is only supported by facial recognition terminal, and the "full" and "basic" recovery modes are not supported by passthrough method.
- For facial recognition terminal, if the recovery mode is set to "part", i.e., /ISAPI/System/factoryReset?mode=part, the settings of card, face, event, fingerprint, schedule template, network, user name, and password will be reserved without being restored.

16.11.44 /ISAPI/System/fileExport/capabilities?format=json

Get the capability of exporting files from the device.

Request URI Definition

Table 16-488 GET /ISAPI/System/fileExport/capabilities?format=json

Method	GET
Description	Get the capability of exporting files from the device.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_FileExportCap</i> Failed: <i>JSON_ResponseStatus</i>

16.11.45 /ISAPI/System/fileExport?format=json

Export files from the device.

Request URI Definition

Table 16-489 POST /ISAPI/System/fileExport?format=json

Method	POST
Description	Export files from the device.
Query	format: determine the format of request or response message.
Request	<i>JSON_FileExport</i>
Response	<i>JSON_FileExportResult</i>

16.11.46 /ISAPI/System/firmwareCodeV2

Get firmware code list.

Request URI Definition

Table 16-490 GET /ISAPI/System/firmwareCodeV2

Method	GET
Description	Get firmware code list.
Query	None.
Request	None.
Response	Succeeded: <i>XML_FirmwareCodeList</i> Failed: <i>XML_ResponseStatus</i>

16.11.47 /ISAPI/System/GPSCalibration/channels/<ID>/capabilities?format=json

Get GPS calibration configuration capability.

Request URI Definition

Table 16-491 GET /ISAPI/System/GPSCalibration/channels/<ID>/capabilities?format=json

Method	GET
Description	Get GPS calibration configuration capability.
Query	format: determine the format of request or response message.

Request	None.
Response	Succeeded: <i>JSON_GPSCalibrationCap</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the video channel ID.

16.11.48 /ISAPI/System/GPSCalibration/channels/<ID>/points/<PID>?format=json

Get, set or delete configuration parameters of a single GPS calibration point.

Request URI Definition

Table 16-492 GET /ISAPI/System/GPSCalibration/channels/<ID>/points/<PID>?format=json

Method	GET
Description	Get the configuration parameters of a single GPS calibration point.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_GPSCalibration</i> Failed: <i>JSON_ResponseStatus</i>

Table 16-493 PUT /ISAPI/System/GPSCalibration/channels/<ID>/points/<PID>?format=json

Method	PUT
Description	Set the configuration parameters of a single GPS calibration point.
Query	format: determine the format of request or response message.
Request	<i>JSON_GPSCalibration</i>
Response	<i>JSON_ResponseStatus</i>

Table 16-494 DELETE /ISAPI/System/GPSCalibration/channels/<ID>/points/<PID>?format=json

Method	DELETE
Description	Delete the configuration parameters of a single GPS calibration point.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

Remarks

The first <ID> in the request URI refers to the video channel ID and the second <ID> refers to calibration point ID.

16.11.49 /ISAPI/System/GPSCalibration/channels/<ID>/points/status?format=json

Get GPS calibration status.

Request URI Definition

Table 16-495 GET /ISAPI/System/GPSCalibration/channels/<ID>/points/status?format=json

Method	GET
Description	Get GPS calibration status.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_GPSCalibrationStatus</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the video channel ID.

16.11.50 /ISAPI/System/GPSCalibration/channels/<ID>/points?format=json

Get, set or delete configuration parameters of multiple GPS calibration points in a batch.

Request URI Definition

Table 16-496 GET /ISAPI/System/GPSCalibration/channels/<ID>/points?format=json

Method	GET
Description	Get configuration parameters of multiple GPS calibration points in a batch.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_GPSCalibrationList</i> Failed: <i>JSON_ResponseStatus</i>

Table 16-497 PUT /ISAPI/System/GPSCalibration/channels/<ID>/points?format=json

Method	PUT
Description	Set configuration parameters of multiple GPS calibration points in a batch.
Query	format: determine the format of request or response message.
Request	<i>JSON_GPSCalibrationList</i>
Response	<i>JSON_ResponseStatus</i>

Table 16-498 DELETE /ISAPI/System/GPSCalibration/channels/<ID>/points?format=json

Method	DELETE
Description	Delete configuration parameters of multiple GPS calibration points in a batch.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the video channel ID.

16.11.51 /ISAPI/System/GPSPTZCalibration/channels/<ID>/calibration?format=json

Enable GPS calibration.

Request URI Definition**Table 16-499 PUT /ISAPI/System/GPSPTZCalibration/channels/<ID>/calibration?format=json**

Method	PUT
Description	Enable GPS calibration.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the video channel ID.

16.11.52 /ISAPI/System/GPSVerification/channels/<ID>/display?format=json

Get or set the parameters of displaying GPS calibration verification result.

Request URI Definition**Table 16-500 GET /ISAPI/System/GPSVerification/channels/<ID>/display?format=json**

Method	GET
Description	Get the parameters of displaying GPS calibration verification result.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_VerificationDisplay</i> Failed: <i>JSON_ResponseStatus</i>

Table 16-501 PUT /ISAPI/System/GPSVerification/channels/<ID>/display?format=json

Method	PUT
Description	Set the parameters of displaying GPS calibration verification result.
Query	format: determine the format of request or response message.
Request	<i>JSON_VerificationDisplay</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URI refers to channel No.

16.11.53 /ISAPI/System/GPSVerification/channels/<ID>/points/<ID>?format=json

Get, set, or delete a specified verification point.

Request URI Definition**Table 16-502 GET /ISAPI/System/GPSVerification/channels/<ID>/points/<ID>?format=json**

Method	GET
Description	Get the parameters of a specified verification point.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_VerificationPoint</i>

	Failed: <i>JSON_ResponseStatus</i>
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Table 16-503 PUT /ISAPI/System/GPSVerification/channels/<ID>/points/<ID>?format=json

Method	PUT
Description	Set the parameters of a specified verification point.
Query	format: determine the format of request or response message.
Request	<i>JSON_VerificationPoint</i>
Response	<i>JSON_ResponseStatus</i>

Table 16-504 DELETE /ISAPI/System/GPSVerification/channels/<ID>/points/<ID>?format=json

Method	DELETE
Description	Delete a specified verification point.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

Remarks

The first <ID> in the URI refers to channel No., and second <ID> refers to verification point No.

16.11.54 /ISAPI/System/GPSVerification/channels/<ID>/points/capabilities?format=json

Get the capability of verifying GPS calibration.

Request URI Definition**Table 16-505 GET /ISAPI/System/GPSVerification/channels/<ID>/points/capabilities?format=json**

Method	GET
Description	Get the capability of verifying GPS calibration.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_VerificationPointList</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URI refers to channel No.

16.11.55 /ISAPI/System/GPSVerification/channels/<ID>/points/resultInfo?format=json

Get the verification results of GPS calibration in a batch according to channel No.

Request URI Definition

Table 16-506 GET /ISAPI/System/GPSVerification/channels/<ID>/points/resultInfo?format=json

Method	GET
Description	Get the verification results of GPS calibration in a batch according to channel No.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_GPSVerification_ResultInfoList</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URI refers to channel No.

16.11.56 /ISAPI/System/GPSVerification/channels/<ID>/points?format=json

Get, set, or delete multiple verification points in a batch, or add a verification point.

Request URI Definition

Table 16-507 GET /ISAPI/System/GPSVerification/channels/<ID>/points?format=json

Method	GET
Description	Get the parameters of verification points in a batch according to channel No.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_VerificationPointList</i> Failed: <i>JSON_ResponseStatus</i>

Table 16-508 PUT /ISAPI/System/GPSVerification/channels/<ID>/points?format=json

Method	PUT
Description	Set the parameters of verification points in a batch according to channel No.
Query	format : determine the format of request or response message.
Request	<i>JSON_VerificationPointList</i>
Response	<i>JSON_ResponseStatus</i>

Table 16-509 DELETE /ISAPI/System/GPSVerification/channels/<ID>/points?format=json

Method	DELETE
Description	Delete all verification points in a batch according to channel No.
Query	format : determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

Table 16-510 POST /ISAPI/System/GPSVerification/channels/<ID>/points?format=json

Method	POST
Description	Add a verification point according to channel No.
Query	format : determine the format of request or response message.
Request	<i>JSON_VerificationPoint</i>
Response	Succeeded: <i>JSON_GPSVerification_ResultInfo</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URI refers to channel No.

16.11.57 /ISAPI/System/guardAgainstTheft

Get or set device anti-theft parameters.

Request URI Definition**Table 16-511 GET /ISAPI/System/guardAgainstTheft**

Method	GET
Description	Get the device anti-theft parameters.

Query	None.
Request	None.
Response	Succeeded: <i>XML_GuardAgainstTheft</i> Failed: <i>XML_ResponseStatus</i>

Table 16-512 PUT /ISAPI/System/guardAgainstTheft

Method	PUT
Description	Set the device anti-theft parameters.
Query	None.
Request	<i>XML_GuardAgainstTheft</i>
Response	<i>XML_ResponseStatus</i>

16.11.58 /ISAPI/System/guardAgainstTheft/capabilities

Get the device anti-theft configuration capability.

Request URI Definition

Table 16-513 GET /ISAPI/System/guardAgainstTheft/capabilities

Method	GET
Description	Get the device anti-theft configuration capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_GuardAgainstTheft</i> Failed: <i>XML_ResponseStatus</i>

16.11.59 /ISAPI/System/guideConfig/<guideEvent>/capabilities?format=json

Get the capability of quick setup instruction for specified event.

Request URI Definition

Table 16-514 GET /ISAPI/System/guideConfig/<guideEvent>/capabilities?format=json

Method	GET
Description	Get the capability of quick setup instruction for specified event.
Query	format: determine the format of request or response message.

Request	None.
Response	Succeeded: <i>JSON_GuideConfigCap</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <**guideEvent**> in the URI refers to the event, which supports quick setup, and now only "linkageCapture" (capture by linkage) is supported, i.e., the URI is: /ISAPI/System/guideConfig/linkageCapture/capabilities?format=json

16.11.60 /ISAPI/System/guideConfig/<guideEvent>?format=json

Get the parameters of quick setup instruction for specified event.

Request URI Definition

Table 16-515 GET /ISAPI/System/guideConfig/<guideEvent>?format=json

Method	GET
Description	Get the parameters of quick setup instruction for specified event.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_GuideConfig</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <**guideEvent**> in the URI refers to the event, which supports quick setup, and now only "linkageCapture" (capture by linkage) is supported, i.e., the URI is: /ISAPI/System/guideConfig/linkageCapture?format=json

16.11.61 /ISAPI/System/Hardware

Operations about the device hardware configuration.

Request URI Definition

Table 16-516 GET /ISAPI/System/Hardware

Method	GET
Description	Get the device hardware configuration parameters.
Query	None.

Request	None.
Response	Succeeded: <i>XML_HardwareService</i> Failed: <i>XML_ResponseStatus</i>

Table 16-517 PUT /ISAPI/System/Hardware

Method	PUT
Description	Set the device hardware parameters.
Query	None.
Request	<i>XML_HardwareService</i>
Response	<i>XML_ResponseStatus</i>

16.11.62 /ISAPI/System/Hardware/defog

Operations about the defogging control configuration.

Request URI Definition

Table 16-518 GET /ISAPI/System/Hardware/defog

Method	GET
Description	Get the defogging control configuration parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Defog</i> Failed: <i>XML_ResponseStatus</i>

Table 16-519 PUT /ISAPI/System/Hardware/defog

Method	PUT
Description	Set the defogging control parameters.
Query	None.
Request	<i>XML_Defog</i>
Response	<i>XML_ResponseStatus</i>

16.11.63 /ISAPI/System/installationAngleCalibration/channels/<ID>/capabilities?format=json

Get the capability of installation angle calibration.

Request URI Definition

Table 16-520 GET /ISAPI/System/installationAngleCalibration/channels/<ID>/capabilities?format=json

Method	GET
Description	Get the capability of installation angle calibration.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_Cap_CalibrationStatus</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

16.11.64 /ISAPI/System/installationAngleCalibration/channels/<ID>?format=json

Get the installation angle calibration status or calibrate the installation angle.

Request URI Definition

Table 16-521 GET /ISAPI/System/installationAngleCalibration/channels/<ID>?format=json

Method	GET
Description	Get the installation angle calibration status.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_CalibrationStatus</i> Failed: <i>JSON_ResponseStatus</i>

Table 16-522 PUT /ISAPI/System/installationAngleCalibration/channels/<ID>?format=json

Method	PUT
Description	Calibrate the installation angle.
Query	format: determine the format of request or response message.

Request	None.
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

16.11.65 /ISAPI/System/IO/capabilities

Get alarm input and output configuration capability.

Request URI Definition

Table 16-523 GET /ISAPI/System/IO/capabilities

Method	GET
Description	Get alarm input and output configuration capability.
Query	None.
Request	None.
Response	<i>XML_IOCap</i>

16.11.66 /ISAPI/System/IO/inputs

Get the information of all alarm input ports.

Request URI Definition

Table 16-524 Get the information of all alarm input ports.

Method	GET
Description	Get the information of all alarm input ports.
Query	None.
Request	None.
Response	<i>XML_IOInputPortList</i>

16.11.67 /ISAPI/System/IO/inputs/<ID>

Get or set an alarm input port.

Request URI Definition**Table 16-525 GET /ISAPI/System/IO/inputs/<ID>**

Method	GET
Description	Get an alarm input's information.
Query	None.
Request	None.
Response	<i>XML_IOInputPort</i>

Table 16-526 PUT /ISAPI/System/IO/inputs/<ID>

Method	PUT
Description	Set an alarm input's information.
Query	None.
Request	<i>XML_IOInputPort</i>
Response	<i>XML_ResponseStatus</i>

16.11.68 /ISAPI/System/IO/inputs/<ID>/status

Get the status of a specific alarm input.

Request URI Definition**Table 16-527 GET /ISAPI/System/IO/inputs/<ID>/status**

Method	GET
Description	Get the status of a specific alarm input.
Query	None.
Request	None.
Response	Succeeded: <i>XML_IOPortStatus</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the alarm input ID.

16.11.69 /ISAPI/System/IO/inputs/capabilities

Get alarm input configuration capability.

Request URI Definition

Table 16-528 GET /ISAPI/System/IO/inputs/capabilities

Method	GET
Description	Get the alarm input configuration capability.
Query	None.
Request	None.
Response	<i>XML_Cap_IOInputPortList</i>

16.11.70 /ISAPI/System/IO/outputs

Get the information of all I/O output ports.

Request URI Definition

Table 16-529 GET /ISAPI/System/IO/outputs

Method	GET
Description	Get the information of all I/O output ports.
Query	None.
Request	None.
Response	<i>XML_IOOutputPortList</i>

16.11.71 /ISAPI/System/IO/outputs/capabilities

Get the I/O output capability.

Request URI Definition

Table 16-530 GET /ISAPI/System/IO/outputs/capabilities

Method	GET
Description	Get the I/O output capability.
Query	None.
Request	None.
Response	<i>XML_Cap_IOOutputPortList</i>

16.11.72 /ISAPI/System/IO/outputs/<ID>

Get or set the information of specified I/O output port.

Table 16-531 GET /ISAPI/System/IO/outputs/<ID>

Method	GET
Description	Get the information of specified I/O output port.
Query	None.
Request	None.
Response	<i>XML_IOOutputPort</i>

Table 16-532 PUT /ISAPI/System/IO/outputs/<ID>

Method	PUT
Description	Set the information of specified alarm output port.
Query	<i>XML_IOOutputPort</i>
Request	None.
Response	<i>XML_ResponseStatus</i>

16.11.73 /ISAPI/System/IO/outputs/<ID>/status

Get status of a specific alarm output.

Request URI Definition

Table 16-533 GET /ISAPI/System/IO/outputs/<ID>/status

Method	GET
Description	Get status of a specific alarm output.
Query	None.
Request	None.
Response	Succeeded: <i>XML_IOPortStatus</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the alarm output ID.

16.11.74 /ISAPI/System/IO/outputs/<ID>/trigger

Manually trigger a specific alarm output.

Request URI Definition

Table 16-534 PUT /ISAPI/System/IO/outputs/<ID>/trigger

Method	PUT
Description	Manually trigger a specific alarm output.
Query	none.
Request	<i>XML_IOPortData</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the alarm output ID.

16.11.75 /ISAPI/System/IO/status

Get alarm input and output status.

Request URI Definition

Table 16-535 GET /ISAPI/System/IO/status

Method	GET
Description	Get alarm input and output status.
Query	None.
Request	None.
Response	Succeeded: <i>XML_IOPortStatusList</i> Failed: <i>XML_ResponseStatus</i>

16.11.76 /ISAPI/System/IOT/channelConfig?format=json

Import or export the list of added IoT devices.

Request URI Definition

Table 16-536 GET /ISAPI/System/IOT/channelConfig?format=json

Method	GET
Description	Export the list of added IoT devices.
Query	format: determine the format of request or response message. security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	Succeeded: Opaque Data Failed: <i>JSON_ResponseStatus</i>

Table 16-537 PUT /ISAPI/System/IOT/channelConfig?format=json

Method	PUT
Description	Import the list of added IoT devices.
Query	format: determine the format of request or response message. security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2.
Request	Opaque Data
Response	<i>JSON_ResponseStatus</i> <i>JSON_IOT_ErrorList</i>

Remarks

The Opaque Data should be encrypted.

16.11.77 /ISAPI/System/IOT/channels/<ID>/<EventType>/capabilities?format=json

Get the event configuration capability of IoT device.

Request URI Definition**Table 16-538 GET /ISAPI/System/IOT/channels/<ID>/<EventType>/capabilities?format=json**

Method	GET
Description	Get the event configuration capability of IoT device.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_XXCap</i> Failed: <i>JSON_ResponseStatus</i>

16.11.78 /ISAPI/System/IOT/channels/<ID>/<EventType>?format=json

Get or set the event/alarm parameters.

Request URI Definition**Table 16-539 GET /ISAPI/System/IOT/channels/<ID>/<EventType>?format=json**

Method	GET
Description	Get the event/alarm parameters.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_XX</i> Failed: <i>JSON_ResponseStatus</i>

Table 16-540 PUT /ISAPI/System/IOT/channels/<ID>/<EventType>?format=json

Method	PUT
Description	Set the event/alarm parameters.
Query	format: determine the format of request or response message.
Request	<i>JSON_XX</i>
Response	<i>JSON_ResponseStatus</i>

16.11.79 /ISAPI/System/IOT/channels/<ID>/alarmStatistics?format=json

Get the alarm statistics of added IoT device according to channel ID.

Request URI Definition

Table 16-541 GET /ISAPI/System/IOT/channels/<ID>/alarmStatistics?format=json

Method	GET
Description	Get the alarm statistics of added IoT device according to channel ID.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_AlarmStatistics</i> Failed: <i>JSON_ResponseStatus</i>

16.11.80 /ISAPI/System/IOT/channels/<ID>/all?format=json

Delete all channels (video channel and IoT channel) of added IoT device

Request URI Definition

Table 16-542 DELETE /ISAPI/System/IOT/channels/<ID>/all?format=json

Method	DELETE
Description	Delete all channels (video channel and IoT channel) of added IoT device
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

16.11.81 /ISAPI/System/IOT/channels/<ID>/basicParam?format=json

Set the basic parameters of added IoT device according to channel ID.

Request URI Definition

Table 16-543 PUT /ISAPI/System/IOT/channels/<ID>/basicParam?format=json

Method	PUT
Description	Set the basic parameters of added IoT device according to channel ID.
Query	format: determine the format of request or response message.
Request	<i>JSON_BasicParam</i>
Response	<i>JSON_ResponseStatus</i>

16.11.82 /ISAPI/System/IOT/channels/<ID>/OSD/capabilities?format=json

Get the OSD configuration capability of specified IoT device channel.

Request URI Definition

Table 16-544 GET /ISAPI/System/IOT/channels/<ID>/OSD/capabilities?format=json

Method	GET
Description	Get the OSD configuration capability of specified IoT device channel.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_OSDCap</i> Failed: <i>JSON_ResponseStatus</i>

16.11.83 /ISAPI/System/IOT/channels/<ID>/OSD?format=json

Get or set the OSD parameters.

Request URI Definition

Table 16-545 GET /ISAPI/System/IOT/channels/<ID>/OSD?format=json

Method	GET
Description	Get the OSD parameters.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_OSD</i>

Failed: *JSON_ResponseStatus*

Table 16-546 PUT /ISAPI/System/IOT/channels/<ID>/OSD?format=json

Method	PUT
Description	Set the OSD parameters.
Query	format : determine the format of request or response message.
Request	<i>JSON_OSD</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URI is defied as the channel ID.

16.11.84 /ISAPI/System/IOT/channels/<ID>/status?format=json

Get the status of specified channel of added IoT device.

Request URI Definition

Table 16-547 GET /ISAPI/System/IOT/channels/<ID>/status?format=json

Method	GET
Description	Get the status of specified channel of added IoT device.
Query	format : determine the format of request or response message. security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	<i>JSON_IOTChannelStatus</i>

Remarks

The <ID> in the URI is defined as channel ID.

16.11.85 /ISAPI/System/IOT/channels/<ID>?format=json

Operations about specified added IoT device.

Request URI Definition

Table 16-548 GET /ISAPI/System/IOT/channels/<ID>?format=json

Method	GET
Description	Get the information of added IoT device according to channel ID.
Query	format: determine the format of request or response message. security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	Succeeded: <i>JSON_IOTChannel</i> Failed: <i>JSON_ResponseStatus</i>

Table 16-549 PUT /ISAPI/System/IOT/channels/<ID>?format=json

Method	PUT
Description	Set the added IoT device according to channel ID.
Query	format: determine the format of request or response message. security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2.
Request	<i>JSON_IOTChannel</i>
Response	<i>JSON_ResponseStatus</i>

Table 16-550 DELETE /ISAPI/System/IOT/channels/<ID>?format=json

Method	DELETE
Description	Delete the specified IoT device according to channel ID.
Query	format: determine the format of request or response message.

Request	None.
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URI is defined as channel ID.

16.11.86 /ISAPI/System/IOT/channels/status?format=json

Get the status of added IoT device.

Request URI Definition

Table 16-551 GET /ISAPI/System/IOT/channels/status?format=json

Method	GET
Description	Get the status of added IoT device.
Query	format: determine the format of request or response message. security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	Succeeded: <i>JSON_IOTChannelStatusList</i> Failed: <i>JSON_ResponseStatus</i>

16.11.87 /ISAPI/System/IOT/channels/status?format=json&deviceInductiveType=

Get the status of added IoT device according to inductive type.

Request URI Definition

Table 16-552 GET /ISAPI/System/IOT/channels/status?format=json&deviceInductiveType=

Method	GET
Description	Get the status of added IoT device according to inductive type.
Query	format: determine the format of request or response message.

	security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2. deviceInductiveType: device inductive type, its value can be obtained by URI: GET <i>/ISAPI/System/IOT/sourceSupport/capabilities?format=json</i> , now there are two available values: "inductiveType1" and "inductiveType2"
Request	None.
Response	<i>JSON_IOTChannelStatusList</i>

16.11.88 /ISAPI/System/IOT/channels?format=json

Operations about IoT devices.

Request URI Definition

Table 16-553 GET /ISAPI/System/IOT/channels?format=json

Method	GET
Description	Get the information of added IoT devices.
Query	format: determine the format of request or response message. security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	<i>JSON_IOTChannelList</i>

Table 16-554 POST /ISAPI/System/IOT/channels?format=json

Method	POST
Description	Add the IoT devices to NVR/DVR.
Query	format : determine the format of request or response message.

	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	<i>JSON_IOTChannel</i>
Response	Succeeded: <i>JSON_IOT_ChannelInfo</i> Failed: <i>JSON_ResponseStatus</i>

16.11.89 /ISAPI/System/IOT/channels?format=json&deviceInductiveType=

Get the informaiton of added IoT devices according to device inductive type.

Request URI Definition

Table 16-555 GET /ISAPI/System/IOT/channels?format=json&deviceInductiveType=

Method	GET
Description	Get the added IoT devices according to device inductive type.
Query	format : determine the format of request or response message. deviceInductiveType : device inductive type, its value can be obtained by URI: GET <i>/ISAPI/System/IOT/sourceSupport/capabilities?format=json</i> , now there are two available values: "inductiveType1" and "inductiveType2" security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	<i>JSON_IOTChannelList</i>

16.11.90 /ISAPI/System/IOT/linkageChannels?format=json

Get the linked channel of searched event.

Request URI Definition

Table 16-556 POST /ISAPI/System/IOT/linkageChannels?format=json

Method	POST
Description	Get the linked channel of searched event.
Query	format: determine the format of request or response message.
Request	<i>JSON_LinkageChansCond</i>
Response	Succeeded: <i>JSON_Result</i> Failed: <i>JSON_ResponseStatus</i>

16.11.91 /ISAPI/System/IOT/search?format=json

Get the IoT devices which can be added.

Request URI Definition

Table 16-557 GET /ISAPI/System/IOT/search?format=json

Method	GET
Description	Get the IoT devices which can be added.
Query	format: determine the format of request or response message. security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	<i>JSON_IOTSourceList</i>

16.11.92 /ISAPI/System/IOT/sourceCapabilities?format=json

Get the number of IoT device channels.

Request URI Definition

Table 16-558 GET /ISAPI/System/IOT/sourceCapabilities?format=json

Method	GET
Description	Get the number of IoT device channels.
Query	format: determine the format of request or response message. security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2.
Request	<i>JSON_IOTSourceDescription</i>
Response	<i>JSON_SourceCapabilities</i>

16.11.93 /ISAPI/System/IOT/sourceSupport/capabilities?format=json

Get supported protocol type of IoT devices.

Request URI Definition

Table 16-559 GET /ISAPI/System/IOT/sourceSupport/capabilities?format=json

Method	GET
Description	Get supported protocol type of IoT devices.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_IOTSourceSupport</i>

16.11.94 /ISAPI/System/laserOpticalAxis/capabilities?format=json

Get the configuration capability of laser optical axis.

Request URI Definition

Table 16-560 GET /ISAPI/System/laserOpticalAxis/capabilities?format=json

Method	GET
Description	Get the configuration capability of laser optical axis.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_LaserOpticalAxisCap</i> Failed: <i>JSON_ResponseStatus</i>

16.11.95 /ISAPI/System/laserOpticalAxis/goto?format=json

Adjust the laser optical axis.

Request URI Definition

Table 16-561 PUT /ISAPI/System/laserOpticalAxis/goto?format=json

Method	PUT
Description	Adjust the laser optical axis.
Query	format: determine the format of request or response message.
Request	<i>JSON_LaserOpticalAxis_direction</i>
Response	<i>JSON_ResponseStatus</i>

16.11.96 /ISAPI/System/laserOpticalAxis?format=json

Get or set laser optical axis parameters.

Request URI Definition

Table 16-562 GET /ISAPI/System/laserOpticalAxis?format=json

Method	GET
Description	Get the laser optical axis parameters.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_LaserOpticalAxis</i>

	Failed: <i>JSON_ResponseStatus</i>
--	------------------------------------

Table 16-563 PUT /ISAPI/System/laserOpticalAxis?format=json

Method	PUT
Description	Configure the laser optical axis.
Query	format : determine the format of request or response message.
Request	<i>JSON_LaserOpticalAxis</i>
Response	<i>JSON_ResponseStatus</i>

16.11.97 /ISAPI/System/logServer

Get or set log server parameters.

Request URI Definition

Table 16-564 GET /ISAPI/System/logServer

Method	GET
Description	Get log server parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_LogServer</i> Failed: <i>XML_ResponseStatus</i>

Table 16-565 PUT /ISAPI/System/logServer

Method	PUT
Description	Set log server parameters.
Query	None.
Request	<i>XML_LogServer</i>
Response	<i>XML_ResponseStatus</i>

16.11.98 /ISAPI/System/logServer/capabilities

Get log server configuration capability.

Request URI Definition

Table 16-566 GET /ISAPI/System/logServer/capabilities

Method	GET
Description	Get log server configuration capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_LogServerCap</i> Failed: <i>XML_ResponseStatus</i>

16.11.99 /ISAPI/System/logServer/test

Perform security log server test.

Request URI Definition

Table 16-567 POST /ISAPI/System/logServer/test

Method	POST
Description	Perform security log server test.
Query	None.
Request	<i>XML_LogServerTestDescription</i>
Response	<i>XML_ResponseStatus</i>

16.11.100 /ISAPI/System/Network/adaption/capabilities?format=json

Get network self-adaptive configuration capability.

Request URI Definition

Table 16-568 GET /ISAPI/System/Network/adaption/capabilities?format=json

Method	GET
Description	Get network self-adaptive configuration capability.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_AdaptionCap</i>

	Failed: <i>JSON_ResponseStatus</i>
--	------------------------------------

16.11.101 /ISAPI/System/Network/adaption?format=json&streamType=

Operations about network self-adaptive configuration.

Request URI Definition

Table 16-569 GET /ISAPI/System/Network/adaption?format=json&streamType=

Method	GET
Description	Get network self-adaptive parameters.
Query	format : determine the format of request or response message. streamType : stream types, integer, the following values are available: 0-main stream, 1-sub-stream, 2-third stream, 3-virtual stream, 4-stream 5, 5-stream 6, 7-stream 7, 8-stream 8, ..., and so on.
Request	None.
Response	Succeeded: <i>JSON_Adaption</i> Failed: <i>JSON_ResponseStatus</i>

Table 16-570 PUT /ISAPI/System/Network/adaption?format=json&streamType=

Method	GET
Description	Set network self-adaptive parameters.
Query	format : determine the format of request or response message. streamType : stream types, integer, the following values are available: 0-main stream, 1-sub-stream, 2-third stream, 3-virtual stream, 4-stream 5, 5-stream 6, 7-stream 7, 8-stream 8, ..., and so on.
Request	<i>JSON_Adaption</i>
Response	<i>XML_ResponseStatus</i>

16.11.102 /ISAPI/System/Network/Bond

Get all network bonding configuration parameters.

Request URI Definition

Table 16-571 GET /ISAPI/System/Network/Bond

Method	GET
Description	Get all network bonding configuration parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_BondList</i> Failed: <i>XML_ResponseStatus</i>

16.11.103 /ISAPI/System/Network/Bond/<ID>

Operations about the bonding configuration of a specific network interface.

Request URI Definition

Table 16-572 GET /ISAPI/System/Network/Bond/<ID>

Method	GET
Description	Get the bonding parameters of a specific network interface.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Bond</i> Failed: <i>XML_ResponseStatus</i>

Table 16-573 PUT /ISAPI/System/Network/Bond/<ID>

Method	PUT
Description	Set the bonding parameters of a specific network interface.
Query	None.
Request	<i>XML_Bond</i>
Response	<i>XML_ResponseStatus</i>

16.11.104 /ISAPI/System/Network/capabilities

Get the network capability.

Request URI Definition

Table 16-574 GET /ISAPI/System/Network/capabilities

Method	GET
Description	Get the network capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_NetworkCap</i> Failed: <i>XML_ResponseStatus</i>

16.11.105 /ISAPI/System/Network/channels/<ID>/buletooth/status

Operations about the bluetooth status configuration of a specific channel.

Request URI Definition

Table 16-575 GET /ISAPI/System/Network/channels/<ID>/buletooth/status

Method	GET
Description	Get the bluetooth status parameters of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_BluetoothStatus</i> Failed: <i>XML_ResponseStatus</i>

Table 16-576 PUT /ISAPI/System/Network/channels/<ID>/buletooth/status

Method	PUT
Description	Set the bluetooth status parameters of a specific channel.
Query	None.
Request	<i>XML_BluetoothStatus</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

16.11.106 /ISAPI/System/Network/DDNS

Operations about all DDNS configuration parameters.

Request URI Definition

Table 16-577 GET /ISAPI/System/Network/DDNS

Method	GET
Description	Get all DDNS configuration parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_DDNSList</i> Failed: <i>XML_ResponseStatus</i>

Table 16-578 PUT /ISAPI/System/Network/DDNS

Method	PUT
Description	Set all DDNS configuration parameters.
Query	None.
Request	<i>XML_DDNSList</i>
Response	<i>XML_ResponseStatus</i>

16.11.107 /ISAPI/System/Network/DDNS/capabilities

Get DDNS (Dynamic Domain Name Server) configuration capability.

Request URI Definition

Table 16-579 GET /ISAPI/System/Network/DDNS/capabilities

Method	GET
Description	Get DDNS (Dynamic Domain Name Server) configuration capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_DDNSList</i> Failed: <i>XML_ResponseStatus</i>

16.11.108 /ISAPI/System/Network/DDNS/<ID>

Operations about the configuration of a specific DDNS.

Request URI Definition**Table 16-580 GET /ISAPI/System/Network/DDNS/<ID>**

Method	GET
Description	Get the configuration parameters of a specific DDNS.
Query	None.
Request	None.
Response	Succeeded: <i>XML_DDNS</i> Failed: <i>XML_ResponseStatus</i>

Table 16-581 PUT /ISAPI/System/Network/DDNS/<ID>

Method	PUT
Description	Set the parameters of a specific DDNS.
Query	None.
Request	<i>XML_DDNS</i>
Response	<i>XML_ResponseStatus</i>

16.11.109 /ISAPI/System/Network/DDNS/<ID>/capabilities

Get the configuration capability of a specific DDNS.

Request URI Definition**Table 16-582 GET /ISAPI/System/Network/DDNS/<ID>/capabilities**

Method	GET
Description	Get the configuration capability of a specific DDNS.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_DDNS</i> Failed: <i>XML_ResponseStatus</i>

16.11.110 /ISAPI/System/Network/Ehome

Operations about the EHome server access configuration.

Request URI Definition

Table 16-583 GET /ISAPI/System/Network/Ehome

Method	GET
Description	Get the EHome server access parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Ehome</i> Failed: <i>XML_ResponseStatus</i>

Table 16-584 PUT /ISAPI/System/Network/Ehome

Method	PUT
Description	Set the EHome server access parameters.
Query	None.
Request	<i>XML_Ehome</i>
Response	<i>XML_ResponseStatus</i>

16.11.111 /ISAPI/System/Network/Ehome/capabilities

Get the configuration capability of EHome server access.

Request URI Definition

Table 16-585 GET /ISAPI/System/Network/Ehome/capabilities

Method	GET
Description	Get the configuration capability of EHome server access.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_EHome</i> Failed: <i>XML_ResponseStatus</i>

16.11.112 /ISAPI/System/Network/Ehome?centerID=

Operations about the EHome server access configuration.

Request URI Definition

Table 16-586 GET /ISAPI/System/Network/Ehome?centerID=

Method	GET
Description	Get the EHome server access parameters.
Query	centerID : EHome center ID, and if there is only one center, centerID is 1.
Request	None.
Response	Succeeded: <i>XML_Ehome</i> Failed: <i>XML_ResponseStatus</i>

Table 16-587 PUT /ISAPI/System/Network/Ehome?centerID=

Method	PUT
Description	Set the EHome server access parameters.
Query	centerID : EHome center ID, and if there is only one center, centerID is 1.
Request	<i>XML_Ehome</i>
Response	<i>XML_ResponseStatus</i>

16.11.113 /ISAPI/System/Network/EZVIZ

Operations about Hik-Connect access configurations.

Request URI Definition

Table 16-588 GET /ISAPI/System/Network/EZVIZ

Method	GET
Description	Get Hik-Connect parameters.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates

	that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	<i>XML_EZVIZ</i>

Table 16-589 PUT /ISAPI/System/Network/EZVIZ

Method	PUT
Description	Set Hik-Connect parameters.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	<i>XML_EZVIZ</i>
Response	<i>XML_ResponseStatus</i>

16.11.114 /ISAPI/System/Network/EZVIZ/secretKey?format=json

Edit the verification code for Hik-Connect.

Request URI Definition

Table 16-590 PUT /ISAPI/System/Network/EZVIZ/secretKey?format=json

Method	PUT
Description	Edit the verification code for Hik-Connect.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2. format : determine the format of request or response message.

Request *JSON_EZVIZSecretKey*

Response *JSON_ResponseStatus*

16.11.115 /ISAPI/System/Network/ftp

Operations about all FTP configurations.

Request URI Definition

Table 16-591 GET /ISAPI/System/Network/ftp

Method	GET
Description	Get all FTP parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_FTPNotificationList</i> Failed: <i>XML_ResponseStatus</i>

Table 16-592 PUT /ISAPI/System/Network/ftp

Method	PUT
Description	Set all FTP parameters.
Query	None.
Request	<i>XML_FTPNotificationList</i>
Response	<i>XML_ResponseStatus</i>

16.11.116 /ISAPI/System/Network/ftp/<ID>

Operations about the configuration of a specific FTP.

Request URI Definition

Table 16-593 GET /ISAPI/System/Network/ftp/<ID>

Method	GET
Description	Get the parameters of a specific FTP.
Query	None.
Request	None.
Response	Succeeded: <i>XML_FTPNotification</i>

	Failed: <i>XML_ResponseStatus</i>
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Table 16-594 PUT /ISAPI/System/Network/ftp/<ID>

Method	PUT
Description	Set the parameters of a specific FTP.
Query	None.
Request	<i>XML_FTPNotification</i>
Response	<i>XML_ResponseStatus</i>

16.11.117 /ISAPI/System/Network/ftp/capabilities

Get the FTP capability.

Request URI Definition

Table 16-595 GET /ISAPI/System/Network/ftp/capabilities

Method	GET
Description	Get the FTP capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_FTPNotificationList</i> Failed: <i>XML_ResponseStatus</i>

16.11.118 /ISAPI/System/Network/ftp/test

Perform test to check if the FTP server is available.

Request URI Definition

Table 16-596 POST /ISAPI/System/Network/ftp/test

Method	POST
Description	Perform test to check if the FTP server is available.
Query	None.
Request	<i>XML_FTPTestDescription</i>
Response	Succeeded: <i>XML_FTPTestResult</i> Failed: <i>XML_ResponseStatus</i>

16.11.119 /ISAPI/System/Network/interfaces

Get the information of multiple network interfaces.

Request URI Definition

Table 16-597 GET /ISAPI/System/Network/interfaces

Method	GET
Description	Get the information of multiple network interfaces.
Query	None.
Request	None.
Response	Succeeded: <i>XML_NetworkInterfaceList</i> Failed: <i>XML_ResponseStatus</i>

Example

Sample Code of Getting Information of Multiple Network Interfaces

```
GET /ISAPI/System/Network/interfaces HTTP/1.1
```

```
...
```

```
HTTP/1.1 200 OK
```

```
Content-Type: application/xml; charset="UTF-8"
```

```
Content-Length: xxx
```

```
<?xml version="1.0" encoding="UTF-8"?>
<NetworkInterfaceList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <NetworkInterface>
    <id>1</id>
    <IPAddress>
      <ipVersion>v4</ipVersion>
      <addressingType>static</addressingType>
      <ipAddress>172.6.64.7</ipAddress>
      <subnetMask>255.255.255.0</subnetMask>
      <DefaultGateway>
        <ipAddress>172.6.64.1</ipAddress>
      </DefaultGateway>
      <PrimaryDNS>
        <ipAddress>192.0.0.200</ipAddress>
      </PrimaryDNS>
    </IPAddress>
    <Discovery>
      <UPnP>
        <enabled>true</enabled>
      </UPnP>
      <Zeroconf>
        <enabled>true</enabled>
      </Zeroconf>
    </Discovery>
```

```
<Link>
  <MACAddress> 00:40:48:4C:7F:F2</MACAddress>
  <autoNegotiation>true</autoNegotiation>
  <speed>1000</speed>
  <duplex>full</duplex>
  <MTU>1500</MTU>
</Link>
</NetworkInterface>
</NetworkInterfaceList>
```

16.11.120 /ISAPI/System/Network/interfaces/<ID>

Operations about the configuration of a specific network interface.

Request URI Definitions

Table 16-598 GET /ISAPI/System/Network/interfaces/<ID>

Method	GET
Description	Get the information of a specific network interface.
Query	None.
Request	None.
Response	Succeeded: <i>XML_NetworkInterface</i> Failed: <i>XML_ResponseStatus</i>

PUT /ISAPI/System/Network/interfaces/<ID>

Method	PUT
Description	Set a specific network interface.
Query	None.
Request	<i>XML_NetworkInterface</i>
Response	<i>XML_ResponseStatus</i>

16.11.121 /ISAPI/System/Network/interfaces/<ID>/capabilities

Get the capability of a specific network interface.

Request URI Definition

Table 16-599 GET /ISAPI/System/Network/interfaces/<ID>/capabilities

Method	GET
Description	Get the capability of a specific network interface.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_NetworkInterface</i> Failed: <i>XML_ResponseStatus</i>

16.11.122 /ISAPI/System/Network/interfaces/<ID>/discovery

Operations about the network discovery configuration.

Request URI Definition

Table 16-600 GET /ISAPI/System/Network/interfaces/<ID>/discovery

Method	GET
Description	Get the network discovery configuration parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Discovery</i> Failed: <i>XML_ResponseStatus</i>

Table 16-601 PUT /ISAPI/System/Network/interfaces/<ID>/discovery

Method	PUT
Description	Set the network discovery configuration parameters.
Query	None.
Request	<i>XML_Discovery</i>
Response	<i>XML_ResponseStatus</i>

16.11.123 /ISAPI/System/Network/interfaces/<ID>/discovery/capabilities

Get the configuration capability of network discovery.

Request URI Definition

Table 16-602 GET /ISAPI/System/Network/interfaces/<ID>/discovery/capabilities

Method	GET
Description	Get the configuration capability of network discovery.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_Discovery</i> Failed: <i>XML_ResponseStatus</i>

16.11.124 /ISAPI/System/Network/interfaces/<ID>/dynamicHostName?format=json

Get or set dynamic domain name.

Request URI Definition

Table 16-603 GET /ISAPI/System/Network/interfaces/<ID>/dynamicHostName?format=json

Method	GET
Description	Get dynamic domain name.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_DynamicHostName</i> Failed: <i>JSON_ResponseStatus</i>

Table 16-604 PUT /ISAPI/System/Network/interfaces/<ID>/dynamicHostName?format=json

Method	PUT
Description	Set dynamic domain name.
Query	format: determine the format of request or response message.
Request	<i>JSON_DynamicHostName</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID.

16.11.125 /ISAPI/System/Network/interfaces/<ID>/dynamicHostName/capabilities?format=json

Get configuration capability of dynamic domain name.

Request URI Definition

Table 16-605 GET /ISAPI/System/Network/interfaces/<ID>/dynamicHostName/capabilities?format=json

Method	GET
Description	Get configuration capability of dynamic domain name.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_DynamicHostNameCap</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID.

16.11.126 /ISAPI/System/Network/interfaces/<ID>/ieee802.1x

Operations about configuration of IEEE 802.1x protocol access.

Request URI Definition

Table 16-606 GET /ISAPI/System/Network/interfaces/<ID>/ieee802.1x

Method	GET
Description	Get parameters of IEEE 802.1x protocol access.
Query	None.
Request	None.
Response	Succeeded: <i>XML_IEEE802_1x</i> Failed: <i>XML_ResponseStatus</i>

Table 16-607 PUT /ISAPI/System/Network/interfaces/<ID>/ieee802.1x

Method	PUT
Description	Set parameters of IEEE 802.1x protocol access.
Query	None.

Request	<i>XML_IEEE802_1x</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID.

16.11.127 /ISAPI/System/Network/interfaces/<ID>/ieee802.1x/capabilities

Get configuration capability of IEEE802.1x protocol access.

Request URI Definition

Table 16-608 GET /ISAPI/System/Network/interfaces/<ID>/ieee802.1x/capabilities

Method	GET
Description	Get configuration capability of IEEE802.1x protocol access.
Query	None.
Request	None.
Response	<i>XML_Cap_IEEE802_1x</i>

16.11.128 /ISAPI/System/Network/interfaces/<ID>/ipAddress

Operations about IP address configuration of a specific network interface.

Request URI Definition

Table 16-609 GET /ISAPI/System/Network/interfaces/<ID>/ipAddress

Method	GET
Description	Get the IP address of a specific network interface.
Query	None.
Request	None.
Response	Succeeded: <i>XML_IPAddress</i> Failed: <i>XML_ResponseStatus</i>

Table 16-610 PUT /ISAPI/System/Network/interfaces/<ID>/ipAddress

Method	PUT
Description	Set IP address of a specific network interface.
Query	None.

Request	<i>XML_IPAddress</i>
Response	<i>XML_ResponseStatus</i>

Example

Sample Code of Setting IP Address of A Specific Network Interface

```
PUT /ISAPI/System/Network/interfaces/1/ipAddress HTTP/1.1
...
Content-Type: application/xml; charset="UTF-8"
Content-Length: xxx

<?xml version="1.0" encoding="UTF-8"?>
<IPAddress version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ipVersion>v4</ipVersion>
  <addressingType>static</addressingType>
  <ipAddress>172.6.64.16</ipAddress>
  <subnetMask>255.255.255.0</subnetMask>
  <DefaultGateway>
    <ipAddress>172.6.64.1</ipAddress>
  </DefaultGateway>
  <PrimaryDNS>
    <ipAddress>192.0.0.200</ipAddress>
  </PrimaryDNS>
</IPAddress>

HTTP/1.1 200 OK
...
Content-Type: application/xml; charset="UTF-8"
Content-Length:xxx

<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <requestURL>/Network/interfaces/1/ipAddress</requestURL>
  <statusCode>1</statusCode>
  <statusString>OK</statusString>
</ResponseStatus>
```

16.11.129 /ISAPI/System/Network/interfaces/<ID>/ipAddress/capabilities

Get the IP address configuration capability of a specific network interface.

Request URI Definition

Table 16-611 GET /ISAPI/System/Network/interfaces/<ID>/ipAddress/capabilities

Method	GET
Description	Get the IP address configuration capability of a specific network interface.

Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_IpAddress</i> Failed: <i>XML_ResponseStatus</i>

16.11.130 /ISAPI/System/Network/interfaces/<ID>/link

Operations about the connection configuration of a specific network interface.

Request URI Definition

Table 16-612 GET /ISAPI/System/Network/interfaces/<ID>/link

Method	GET
Description	Get the connection parameters of a specific network interface.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Link</i> Failed: <i>XML_ResponseStatus</i>

Table 16-613 PUT /ISAPI/System/Network/interfaces/<ID>/link

Method	PUT
Description	Set the connection parameters of a specific network interface.
Query	None.
Request	<i>XML_Link</i>
Response	<i>XML_ResponseStatus</i>

16.11.131 /ISAPI/System/Network/interfaces/<ID>/link/capabilities

Get the connection configuration capability of a specific network interface.

Request URI Definition

Table 16-614 GET /ISAPI/System/Network/interfaces/<ID>/link/capabilities

Method	GET
Description	Get the connection configuration capability of a specific network interface.

Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_Link</i> Failed: <i>XML_ResponseStatus</i>

16.11.132 /ISAPI/System/Network/interfaces/<ID>/NetworkMode

Operations about the working mode configuration of the network connection.

Request URI Definition

Table 16-615 GET /ISAPI/System/Network/interfaces/<ID>/NetworkMode

Method	GET
Description	Get the working mode parameters of the network connection.
Query	None.
Request	None.
Response	Succeeded: <i>XML_NetWorkMode</i> Failed: <i>XML_ResponseStatus</i>

Table 16-616 PUT /ISAPI/System/Network/interfaces/<ID>/NetworkMode

Method	PUT
Description	Set the working mode parameters of the network connection.
Query	None.
Request	<i>XML_NetWorkMode</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID.

16.11.133 /ISAPI/System/Network/interfaces/<ID>/wireless

Operations about Wi-Fi configuration of a specific network interface.

Request URI Definition

Table 16-617 GET /ISAPI/System/Network/interfaces/<ID>/wireless

Method	GET
Description	Get the Wi-Fi parameters of a specific network interface.
Query	security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	Succeeded: <i>XML_Wireless</i> Failed: <i>XML_ResponseStatus</i>

Table 16-618 PUT /ISAPI/System/Network/interfaces/<ID>/wireless

Method	PUT
Description	Set Wi-Fi parameters of a specific network interface.
Query	security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2.
Request	<i>XML_Wireless</i>
Response	<i>XML_ResponseStatus</i>

16.11.134 /ISAPI/System/Network/interfaces/<ID>/wireless/accessDeviceList/<ID>

Get the information of a specific device connected to the hotspot.

Request URI Definition

Table 16-619 GET /ISAPI/System/Network/interfaces/<ID>/wireless/accessDeviceList/<ID>

Method	GET
Description	Get the information of a specific device connected to the hotspot.
Query	None.
Request	None.
Response	Succeeded: <i>XML_accessDevice</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The first <ID> in the URI refers to the network interface ID, and the second <ID> refers to the device ID.

16.11.135 /ISAPI/System/Network/interfaces/<ID>/wireless/accessDeviceList/capabilities

Get the capability set of the device connected to the hotspot.

Request URI Definition

Table 16-620 GET /ISAPI/System/Network/interfaces/<ID>/wireless/accessDeviceList/capabilities

Method	GET
Description	Get the capability set of the device connected to the hotspot.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_accessDeviceList</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI refers to the network interface ID.

16.11.136 /ISAPI/System/Network/interfaces/<ID>/wireless/accessPointList

Get Wi-Fi hotspot list.

Request URI Definition**Table 16-621 GET /ISAPI/System/Network/interfaces/<ID>/wireless/accessPointList**

Method	GET
Description	Get Wi-Fi hotspot list.
Query	None.
Request	None.
Response	Succeeded: <i>XML_accessPointList</i> Failed: <i>XML_ResponseStatus</i>

16.11.137 /ISAPI/System/Network/interfaces/<ID>/wireless/accessPointList/<ID>

Get a specific Wi-Fi hotspot.

Request URI Definition**Table 16-622 GET /ISAPI/System/Network/interfaces/<ID>/wireless/accessPointList/<ID>**

Method	GET
Description	Get a specific Wi-Fi hotspot.
Query	None.
Request	None.
Response	Succeeded: <i>XML_accessPoint</i> Failed: <i>XML_ResponseStatus</i>

16.11.138 /ISAPI/System/Network/interfaces/<ID>/wireless/capabilities

Get the Wi-Fi configuration capability.

Request URI Definition

Method	GET
Description	Get the Wi-Fi configuration capability.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates

	that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	Succeeded: <i>XML_Cap_Wireless</i> Failed: <i>XML_ResponseStatus</i>

16.11.139 /ISAPI/System/Network/interfaces/<ID>/wirelessServer

Operations about wireless hotspot configuration.

Request URI Definitions

Table 16-623 GET /ISAPI/System/Network/interfaces/<ID>/wirelessServer

Method	GET
Description	Get wireless hotspot parameters.
Query	None.
Request	None.
Response	<i>XML_WirelessServer</i>

Table 16-624 PUT /ISAPI/System/Network/interfaces/<ID>/wirelessServer

Method	PUT
Description	Set wireless hotspot parameters.
Query	None.
Request	<i>XML_WirelessServer</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI refers to the network interface ID.

16.11.140 /ISAPI/System/Network/interfaces/<ID>/wirelessServer/accessDeviceList

Get list of devices that are connected to the hotspot.

Request URI Definition

Table 16-625 GET /ISAPI/System/Network/interfaces/<ID>/wirelessServer/accessDeviceList

Method	GET
Description	Get list of devices that are connected to the hotspot.
Query	None.
Request	None.
Response	Succeeded: <i>XML_accessDeviceList</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI refers to the network interface ID.

16.11.141 /ISAPI/System/Network/interfaces/<ID>/wirelessServer/capabilities

Get configuration capability of wireless hotspot.

Request URI Definition

Table 16-626 GET /ISAPI/System/Network/interfaces/<ID>/wirelessServer/capabilities

Method	GET
Description	Get configuration capability of wireless hotspot.
Query	None.
Request	None.
Response	<i>XML_Cap_WirelessServer</i>

16.11.142 /ISAPI/System/Network/interfaces/<ID>/wirelessServerStatus

Operations about the wireless hotspot status configuration.

Request URI Definition

Table 16-627 GET /ISAPI/System/Network/interfaces/<ID>/wirelessServerStatus

Method	GET
Description	Get the wireless hotspot status parameters.
Query	None.

Request	None.
Response	Succeeded: <i>XML_WirelessServerStatus</i> Failed: <i>XML_ResponseStatus</i>

Table 16-628 PUT /ISAPI/System/Network/interfaces/<ID>/wirelessServerStatus

Method	PUT
Description	Set the wireless hotspot status parameters.
Query	None.
Request	<i>XML_WirelessServerStatus</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID.

16.11.143 /ISAPI/System/Network/interfaces/<ID>/wirelessServerStatus/capabilities

Get the wireless hotspot status capability.

Request URI Definition**Table 16-629 GET /ISAPI/System/Network/interfaces/<ID>/wirelessServerStatus/capabilities**

Method	GET
Description	Get the wireless hotspot status capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_WirelessServerStatus</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID.

16.11.144 /ISAPI/System/Network/interfaces/<ID>/wirelessStatus

Operations about the configuration of wireless network connection status.

Request URI Definition

Table 16-630 GET /ISAPI/System/Network/interfaces/<ID>/wirelessStatus

Method	GET
Description	Get the parameters of wireless network connection status.
Query	None.
Request	None.
Response	Succeeded: <i>XML_WirelessStatus</i> Failed: <i>XML_ResponseStatus</i>

Table 16-631 PUT /ISAPI/System/Network/interfaces/<ID>/wirelessStatus

Method	PUT
Description	Set the parameters of wireless network connection status.
Query	None.
Request	<i>XML_WirelessStatus</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID.

16.11.145 /ISAPI/System/Network/interfaces/<ID>/wirelessStatus/capabilities

Get the configuration capability of wireless network connection status.

Request URI Definition

Table 16-632 GET /ISAPI/System/Network/interfaces/<ID>/wirelessStatus/capabilities

Method	GET
Description	Get the configuration capability of wireless network connection status.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_WirelessStatus</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID.

16.11.146 /ISAPI/System/Network/interfaces/<ID>/WPS/ApPinCode

Operations about the configuration of PIN (Personal Identification Number) code of the wireless access point.

Request URI Definition**Table 16-633 GET /ISAPI/System/Network/interfaces/<ID>/WPS/ApPinCode**

Method	GET
Description	Get the parameters of PIN (Personal Identification Number) code of the wireless access point.
Query	None.
Request	None.
Response	Succeeded: <i>XML_WpsApPincode</i> Failed: <i>XML_ResponseStatus</i>

Table 16-634 PUT /ISAPI/System/Network/interfaces/<ID>/WPS/ApPinCode

Method	PUT
Description	Set the parameters of PIN (Personal Identification Number) code of the wireless access point.
Query	None.
Request	<i>XML_WpsApPincode</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID, and it is set to 2 by default.

16.11.147 /ISAPI/System/Network/interfaces/<ID>/WPS/AutoConnect

Set the configuration of PBC (Push Button Configuration) automatic connection of WPS (Wi-Fi Protected Setup).

Request URI Definition

Table 16-635 PUT /ISAPI/System/Network/interfaces/<ID>/WPS/AutoConnect

Method	PUT
Description	Set the parameters of PBC (Push Button Configuration) automatic connection of WPS (Wi-Fi Protected Setup).
Query	None.
Request	<i>XML_WPS</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID, and it is set to 2 by default.

16.11.148 /ISAPI/System/Network/interfaces/<ID>/WPS/devicePinCode

Get or set the PIN (Personal Identification Number) code of WPS (Wi-Fi Protected Setup) of the device.

Request URI Definition

Table 16-636 GET /ISAPI/System/Network/interfaces/<ID>/WPS/devicePinCode

Method	GET
Description	Get the PIN (Personal Identification Number) code of WPS (Wi-Fi Protected Setup) of the device.
Query	None.
Request	None.
Response	PIN code string.

Table 16-637 PUT /ISAPI/System/Network/interfaces/<ID>/WPS/devicePinCode

Method	PUT
Description	Set the PIN (Personal Identification Number) code of WPS (Wi-Fi Protected Setup) of the device.
Query	None.
Request	PIN code string.
Response	<i>XML_ResponseStatus</i>

16.11.149 /ISAPI/System/Network/ipFilter

Operations about IP address filter configuration.

Request URI Definition

Table 16-638 GET /ISAPI/System/Network/ipFilter

Method	GET
Description	Get the IP filter configuration parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_IPFilter</i> Failed: <i>XML_ResponseStatus</i>

Table 16-639 PUT /ISAPI/System/Network/ipFilter

Method	PUT
Description	Set the IP address filter.
Query	None.
Request	<i>XML_IPFilter</i>
Response	<i>XML_ResponseStatus</i>

16.11.150 /ISAPI/System/Network/ipFilter/capabilities

Get IP address filter configuration capability.

Request URI Definition

Table 16-640 GET /ISAPI/System/Network/ipFilter/capabilities

Method	GET
Description	Get the IP address configuration capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_IPFilter</i> Failed: <i>XML_ResponseStatus</i>

16.11.151 /ISAPI/System/Network/MACFilter

Operations about MAC address filter configuration.

Request URI Definition

Table 16-641 GET /ISAPI/System/Network/MACFilter

Method	GET
Description	Get the MAC address filter configuration parameters.
Query	None.
Request	None.
Response	<i>XML_MACFilter</i>

Table 16-642 PUT /ISAPI/System/Network/MACFilter

Method	PUT
Description	Set the MAC address filter.
Query	None.
Request	<i>XML_MACFilter</i>
Response	<i>XML_ResponseStatus</i>

16.11.152 /ISAPI/System/Network/MACFilter/capabilities

Get the MAC address filter configuration capability.

Request URI Definition

Table 16-643 GET /ISAPI/System/Network/MACFilter/capabilities

Method	GET
Description	Get the MAC address filter configuration capability.
Query	None.
Request	None.
Response	<i>XML_Cap_MACFilter</i>

16.11.153 /ISAPI/System/Network/mailling

Operations about email configuration.

Request URI Definition**Table 16-644 GET /ISAPI/System/Network/mailing**

Method	GET
Description	Get email configuration parameters.
Query	None.
Request	None.
Response	<i>XML_mailingList</i>

Table 16-645 PUT /ISAPI/System/Network/mailing

Method	PUT
Description	Set email parameters.
Query	None.
Request	<i>XML_mailingList</i>
Response	<i>XML_ResponseStatus</i>

16.11.154 /ISAPI/System/Network/mailing/<ID>

Operations about the configuration of a specific email.

Request URI Definition**Table 16-646 GET /ISAPI/System/Network/mailing/<ID>**

Method	GET
Description	Get the configuration parameters of a specific email.
Query	None.
Request	None.
Response	<i>XML_mailing</i>

Table 16-647 PUT /ISAPI/System/Network/mailing/<ID>

Method	PUT
Description	Set the parameters of a specific email.
Query	None.
Request	<i>XML_mailing</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI refers to the email ID.

16.11.155 /ISAPI/System/Network/mailing/capabilities

Get email configuration capability.

Request URI Definition

Table 16-648 GET /ISAPI/System/Network/mailing/capabilities

Method	GET
Description	Get email configuration capability.
Query	None.
Request	None.
Response	<i>XML_Cap_mailingList</i>

16.11.156 /ISAPI/System/Network/mailing/test

Execute the email test to check if the email server and address are available.

Request URI Definition

Table 16-649 POST /ISAPI/System/Network/mailing/test

Method	POST
Description	Execute the email test to check if the email server and address are available.
Query	None.
Request	<i>XML_mailingTestDescription</i>
Response	<i>XML_mailingTestResult</i>

16.11.157 /ISAPI/System/Network/pingtest

Start a ping test and get the test result.

Request URI Definition**Table 16-650 POST /ISAPI/System/Network/pingtest**

Method	POST
Description	Start a ping test and get the test result.
Query	None.
Request	<i>XML_pingTestDescription</i>
Response	<i>XML_pingTestResult</i>

16.11.158 /ISAPI/System/Network/POE/capabilities?format=json

Get capability of PoE port parameters configuration.

Request URI Definition**Table 16-651 GET /ISAPI/System/Network/POE/capabilities?format=json**

Method	GET
Description	Get the capability of PoE port parameters configuration.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_Cap_POE</i>

16.11.159 /ISAPI/System/Network/POE?format=json

Get or set the PoE port configuration parameters.

Request URI Definition**Table 16-652 GET /ISAPI/System/Network/POE?format=json**

Method	GET
Description	Get PoE port configuration parameters.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_POE</i>

Table 16-653 PUT /ISAPI/System/Network/POE?format=json

Method	PUT
Description	Set PoE port configuration parameters.
Query	format : determine the format of request or response message.
Request	<i>JSON_POE</i>
Response	<i>JSON_ResponseStatus</i>

16.11.160 /ISAPI/System/Network/resourceStatistics?format=json

Get the information of network resource statistics.

Request URI Definition**Table 16-654 GET /ISAPI/System/Network/resourceStatistics?format=json**

Method	GET
Description	Get the information of network resource statistics.
Query	None.
Request	None.
Response	Succeeded: <i>JSON_resourceStatistics</i> Failed: <i>JSON_ResponseStatus</i>

16.11.161 /ISAPI/System/Network/SIP

Operations about the configuration of a specific SIP (Session Initiation Protocol) server.

Request URI Definition**Table 16-655 GET /ISAPI/System/Network/SIP**

Method	GET
Description	Get the parameters of a specific SIP (Session Initiation Protocol) server.
Query	None.
Request	None.
Response	Succeeded: <i>XML_SIPServerList</i> Failed: <i>XML_ResponseStatus</i>

Table 16-656 PUT /ISAPI/System/Network/SIP

Method	PUT
Description	Set the parameters of a specific SIP (Session Initiation Protocol) server.
Query	None.
Request	<i>XML_SIPServerList</i>
Response	<i>XML_ResponseStatus</i>

16.11.162 /ISAPI/System/Network/SIP/<ID>

Operations about the configuration of a specific SIP (Session Initiation Protocol) server.

Request URI Definition

Table 16-657 GET /ISAPI/System/Network/SIP/<ID>

Method	GET
Description	Get the parameters of a specific SIP (Session Initiation Protocol) server.
Query	None.
Request	None.
Response	Succeeded: <i>XML_SIPServer</i> Failed: <i>XML_ResponseStatus</i>

Table 16-658 PUT /ISAPI/System/Network/SIP/<ID>

Method	PUT
Description	Set the parameters of a specific SIP (Session Initiation Protocol) server.
Query	None.
Request	<i>XML_SIPServer</i>
Response	<i>XML_ResponseStatus</i>

16.11.163 /ISAPI/System/Network/SIP/<ID>/capabilities

Get the configuration capability of a specific SIP (Session Initiation Protocol) server.

Request URI Definition

Table 16-659 GET /ISAPI/System/Network/SIP/<ID>/capabilities

Method	GET
Description	Get the configuration capability of a specific SIP (Session Initiation Protocol) server.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_SIPServer</i> Failed: <i>XML_ResponseStatus</i>

16.11.164 /ISAPI/System/Network/SIP/<ID>/SIPInfo

Operations about the configuration of a piece of SIP (Session Initiation Protocol) information.

Request URI Definition

Table 16-660 GET /ISAPI/System/Network/SIP/<ID>/SIPInfo

Method	GET
Description	Get the parameters of a piece of SIP (Session Initiation Protocol) information.
Query	None.
Request	None.
Response	Succeeded: <i>XML_SIPInfo</i> Failed: <i>XML_ResponseStatus</i>

Table 16-661 PUT /ISAPI/System/Network/SIP/<ID>/SIPInfo

Method	PUT
Description	Set the parameters of a piece of SIP (Session Initiation Protocol) information.
Query	None.
Request	<i>XML_SIPInfo</i>
Response	<i>XML_ResponseStatus</i>

16.11.165 /ISAPI/System/Network/SIP/<ID>/SIPInfo/capabilities

Get the configuration capability of a specific piece of SIP (Session Initiation Protocol) information.

Request URI Definition

Table 16-662 GET /ISAPI/System/Network/SIP/<ID>/SIPInfo/capabilities

Method	GET
Description	Get the configuration capability of a specific piece of SIP (Session Initiation Protocol) information.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_SIPInfo</i> Failed: <i>XML_ResponseStatus</i>

16.11.166 /ISAPI/System/Network/SIP/<ID>/SIPInfo/multiInfo

Get or set the information list of SIP servers.

Request URI Definition

Table 16-663 GET /ISAPI/System/Network/SIP/<ID>/SIPInfo/multiInfo

Method	GET
Description	Get the information list of SIP servers.
Query	None.
Request	None.
Response	Succeeded: <i>XML_SIPInfoList</i> Failed: <i>XML_ResponseStatus</i>

Table 16-664 PUT /ISAPI/System/Network/SIP/<ID>/SIPInfo/multiInfo

Method	PUT
Description	Set the information list of SIP servers.
Query	None.
Request	<i>XML_SIPInfoList</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the SIP server ID.

16.11.167 /ISAPI/System/Network/SNMP

Get or set batch configuration parameters of SNMP (Simple Network Management Protocol).

Request URI Definition**Table 16-665 GET /ISAPI/System/Network/SNMP**

Method	GET
Description	Get batch configuration parameters of SNMP.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	Succeeded: <i>XML_SNMP</i> Failed: <i>XML_ResponseStatus</i>

Table 16-666 PUT /ISAPI/System/Network/SNMP

Method	PUT
Description	Set batch configuration parameters of SNMP.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	<i>XML_SNMP</i>
Response	<i>XML_ResponseStatus</i>

16.11.168 /ISAPI/System/Network/SNMP/advanced

Get or set advanced parameters of SNMP (Simple Network Management Protocol).

Request URI Definition**Table 16-667 GET /ISAPI/System/Network/SNMP/advanced**

Method	GET
Description	Get advanced parameters of SNMP.
Query	None.
Request	None.
Response	Succeeded: <i>XML_SNMPAAdvanced</i> Failed: <i>XML_ResponseStatus</i>

Table 16-668 PUT /ISAPI/System/Network/SNMP/advanced

Method	PUT
Description	Set advanced parameters of SNMP.
Query	None.
Request	<i>XML_SNMPAAdvanced</i>
Response	<i>XML_ResponseStatus</i>

16.11.169 /ISAPI/System/Network/SNMP/advanced/trapReceivers

Get, set, delete parameters of all trap address, or add a new trap address of SNMP in version 3.

Request URI Definition**Table 16-669 GET /ISAPI/System/Network/SNMP/advanced/trapReceivers**

Method	GET
Description	Get all trap address parameters of SNMP in version 3.
Query	None.
Request	None.
Response	Succeeded: <i>XML_SNMPTrapReceiverList</i> Failed: <i>XML_ResponseStatus</i>

Table 16-670 PUT /ISAPI/System/Network/SNMP/advanced/trapReceivers

Method	PUT
Description	Set parameters for all trap addresses of SNMP in version 3.
Query	None.
Request	<i>XML_SNMPTrapReceiverList</i>
Response	<i>XML_ResponseStatus</i>

Table 16-671 DELETE /ISAPI/System/Network/SNMP/advanced/trapReceivers

Method	DELETE
Description	Delete all trap address parameters of SNMP in version 3.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

Table 16-672 POST /ISAPI/System/Network/SNMP/advanced/trapReceivers

Method	POST
Description	Add a new trap address of SNMP.
Query	None.
Request	<i>XML_SNMPTrapReceiver</i>
Response	<i>XML_ResponseStatus</i>

16.11.170 /ISAPI/System/Network/SNMP/advanced/trapReceivers/<ID>

Get, set or delete parameters for a single trap address.

Request URI Definition

Table 16-673 GET /ISAPI/System/Network/SNMP/advanced/trapReceivers/<ID>

Method	GET
Description	Get trap address parameters of SNMP in version 3.
Query	None.
Request	None.
Response	Succeeded: <i>XML_SNMPTrapReceiver</i> Failed: <i>XML_ResponseStatus</i>

Table 16-674 PUT /ISAPI/System/Network/SNMP/advanced/trapReceivers/<ID>

Method	PUT
Description	Set parameters for all trap addresses of SNMP in version 3.
Query	None.
Request	<i>XML_SNMPTrapReceiver</i>
Response	<i>XML_ResponseStatus</i>

Table 16-675 DELETE /ISAPI/System/Network/SNMP/advanced/trapReceivers/<ID>

Method	DELETE
Description	Delete a trap address of SNMP in version 3.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

16.11.171 /ISAPI/System/Network/socketIP

Get the socket IP of current connection.

Request URI Definition

Table 16-676 GET /ISAPI/System/Network/socketIP

Method	GET
Description	Get the socket IP of current connection.
Query	None.
Request	None.
Response	<i>XML_SocketIP</i>

16.11.172 /ISAPI/System/Network/ssh

Operations about SSH server configurations.

Request URI Definition

Table 16-677 GET /ISAPI/System/Network/ssh

Method	GET
Description	Get SSH server parameters.
Query	None.
Request	None.
Response	<i>XML_SSH</i>

Table 16-678 PUT /ISAPI/System/Network/ssh

Method	PUT
Description	Set SSH server parameters.
Query	None.
Request	<i>XML_SSH</i>
Response	<i>XML_ResponseStatus</i>

16.11.173 /ISAPI/System/Network/StaticRoute

Get, set, or delete the parameters of static route.

Request URI Definition

Table 16-679 GET /ISAPI/System/Network/StaticRoute

Method	GET
Description	Get the parameters of static route.
Query	None.
Request	None.
Response	Succeeded: <i>XML_staticRouteList</i> Failed: <i>XML_ResponseStatus</i>

Table 16-680 PUT /ISAPI/System/Network/StaticRoute

Method	PUT
Description	Set the parameters of static route.
Query	None.

Request	<i>XML_staticRouteList</i>
Response	<i>XML_ResponseStatus</i>

Table 16-681 DELETE /ISAPI/System/Network/StaticRoute

Method	DELETE
Description	Delete the parameters of static route.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

16.11.174 /ISAPI/System/Network/telnetd

Operations about the telnetd service configuration.

Request URI Definition

Table 16-682 GET /ISAPI/System/Network/telnetd

Method	GET
Description	Get the telnetd service parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Telnetd</i> Failed: <i>XML_ResponseStatus</i>

Table 16-683 PUT /ISAPI/System/Network/telnetd

Method	PUT
Description	Set the telnetd service parameters.
Query	None.
Request	<i>XML_Telnetd</i>
Response	<i>XML_ResponseStatus</i>

16.11.175 /ISAPI/System/Network/telnetd/capabilities

Get the telnetd service capability.

Request URI Definition

Table 16-684 GET /ISAPI/System/Network/telnetd/capabilities

Method	GET
Description	Get the telnetd service capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_Telnetd</i> Failed: <i>XML_ResponseStatus</i>

16.11.176 /ISAPI/System/Network/UPnP

Operations about UPnP configuration.

Request URI Definition

Table 16-685 GET /ISAPI/System/Network/UPnP

Method	GET
Description	Get UPnP parameters.
Query	None.
Request	None.
Response	<i>XML_UPnP</i>

Table 16-686 PUT /ISAPI/System/Network/UPnP

Method	PUT
Description	Set UPnP parameters.
Query	None.
Request	<i>XML_UPnP</i>
Response	<i>XML_ResponseStatus</i>

16.11.177 /ISAPI/System/Network/UPnP/ports

Operations about configuration of all UPnP ports.

Request URI Definition

Table 16-687 GET /ISAPI/System/Network/UPnP/ports

Method	GET
Description	Get the parameters of all UPnP ports.
Query	None.
Request	None.
Response	<i>XML_ports</i>

Table 16-688 PUT /ISAPI/System/Network/UPnP/ports

Method	PUT
Description	Set the parameters of all UPnP ports.
Query	None.
Request	<i>XML_ports</i>
Response	<i>XML_ResponseStatus</i>

16.11.178 /ISAPI/System/Network/UPnP/ports/status

Get the mapping statuses of all UPnP ports.

Request URI Definition

Table 16-689 GET /ISAPI/System/Network/UPnP/ports/status

Method	GET
Description	Get the mapping statuses of all UPnP ports.
Query	None.
Request	None.
Response	<i>XML_portsStatus</i>

16.11.179 /ISAPI/System/Network/UPnP/ports/<ID>

Operations about configuration of a specific UPnP port.

Request URI Definition

Table 16-690 GET /ISAPI/System/Network/UPnP/ports/<ID>

Method	GET
Description	Get the parameters of a specific UPnP port.
Query	None.
Request	None.
Response	Succeeded: <i>XML_port</i> Failed: <i>XML_ResponseStatus</i>

Table 16-691 PUT /ISAPI/System/Network/UPnP/ports/<ID>

Method	PUT
Description	Set the parameters of a specific UPnP port.
Query	None.
Request	<i>XML_port</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to UPnP port ID.

16.11.180 /ISAPI/System/Network/UPnP/ports/<ID>/status

Get mapping status of a specific UPnP port.

Request URI Definition

Table 16-692 GET /ISAPI/System/Network/UPnP/ports/<ID>/status

Method	GET
Description	Get mapping status of a specific UPnP port.
Query	None.
Request	None.
Response	Succeeded: <i>XML_portStatus</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to UPnP port ID.

16.11.181 /ISAPI/System/Network/verificationCodeCheck

Verify the encryption key or verification code.

Request URI Definition

Table 16-693 POST /ISAPI/System/Network/verificationCodeCheck

Method	POST
Description	Verify the encryption key or verification code.
Query	None.
Request	<i>XML_CheckInfo</i>
Response	<i>XML_ResponseStatus</i>

16.11.182 /ISAPI/System/Network/WirelessDial/Interfaces

Get or set parameters for all dial network cards.

Request URI Definition

Table 16-694 GET /ISAPI/System/Network/WirelessDial/Interfaces

Method	GET
Description	Get parameters for all dial network cards.
Query	None.
Request	None.
Response	Succeeded: <i>XML_WirelessDialInterface</i> Failed: <i>XML_ResponseStatus</i>

Table 16-695 PUT /ISAPI/System/Network/WirelessDial/Interfaces

Method	PUT
Description	Set parameters for all dial network cards.
Query	None.
Request	<i>XML_WirelessDialInterface</i>
Response	<i>XML_ResponseStatus</i>

16.11.183 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>

Get or set parameters for a single dial network card.

Request URI Definition**Table 16-696 GET /ISAPI/System/Network/WirelessDial/Interfaces/<ID>**

Method	GET
Description	Get parameters for a single dial network card.
Query	None.
Request	None.
Response	Succeeded: <i>XML_WirelessDialInterfaceList</i> Failed: <i>XML_ResponseStatus</i>

Table 16-697 PUT /ISAPI/System/Network/WirelessDial/Interfaces/<ID>

Method	PUT
Description	Set the wireless dial-up parameters.
Query	None.
Request	<i>XML_WirelessDialInterfaceList</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID.

16.11.184 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/dial

Operations about the wireless dial-up configuration.

Request URI Definition**Table 16-698 GET /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/dial**

Method	GET
Description	Get the configuration parameters of wireless dial-up.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates

	that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	Succeeded: <i>XML_Dial</i> Failed: <i>XML_ResponseStatus</i>

Table 16-699 PUT /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/dial

Method	PUT
Description	Set the wireless dial-up parameters.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	<i>XML_Dial</i>
Response	<i>XML_ResponseStatus</i>

16.11.185 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/dial/capabilities

Get the wireless dial-up capability.

Request URI Definition

Table 16-700 GET /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/dial/capabilities

Method	GET
Description	Get the wireless dial-up capability.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.

Request	None.
Response	Succeeded: <i>XML_Cap_Dial</i> Failed: <i>XML_ResponseStatus</i>

16.11.186 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/dialstatus

Operations about the configuration of wireless dial-up connection status.

Request URI Definition

Table 16-701 GET /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/dialstatus

Method	GET
Description	Get the parameters of wireless dial-up connection status.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Dialstatus</i> Failed: <i>XML_ResponseStatus</i>

Table 16-702 PUT /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/dialstatus

Method	PUT
Description	Set the parameters of wireless dial-up connection status.
Query	None.
Request	<i>XML_Dialstatus</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID.

16.11.187 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/keepAlive/capabilities?format=json

Get the configuration capability of network keepalive.

Request URI Definition**Table 16-703 GET /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/keepAlive/capabilities?
format=json**

Method	GET
Description	Get the configuration capability of network keepalive.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_KeepAliveCap</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID.

**16.11.188 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/keepAlive/test?
format=json**

Test network keepalive configuration.

Request URI Definition**Table 16-704 POST /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/keepAlive/test?
format=json**

Method	POST
Description	Test network keepalive configuration.
Query	format: determine the format of request or response message.
Request	<i>JSON_KeepAlive</i>
Response	<i>JSON_TestResult</i>

Remarks

The <ID> in the request URI refers to the network interface ID.

**16.11.189 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/keepAlive?
format=json**

Set network keepalive parameters.

Request URI Definition

Table 16-705 PUT /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/keepAlive?format=json

Method	PUT
Description	Set network keepalive parameters.
Query	format : determine the format of request or response message.
Request	<i>JSON_KeepAlive</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID.

16.11.190 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/messageConfig

Get or set dial-up information parameters.

Request URI Definition

Table 16-706 GET /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/messageConfig

Method	GET
Description	Get dial-up information parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_messageConfig</i> Failed: <i>XML_ResponseStatus</i>

Table 16-707 PUT /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/messageConfig

Method	PUT
Description	Set dial-up information parameters.
Query	None.
Request	<i>XML_messageConfig</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID.

16.11.191 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/messageConfig/messageConfigCap

Get dial-up information capability.

Request URI Definition

Table 16-708 GET /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/messageConfig/messageConfigCap

Method	GET
Description	Get dial-up information capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_messageConfigCap</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID.

16.11.192 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/messageConfig/WhiteList

Get or set all whitelist configuration.

Request URI Definition

Table 16-709 GET /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/messageConfig/WhiteList

Method	GET
Description	Get all whitelist configuration.
Query	None.
Request	None.
Response	Succeeded: <i>XML_SMSWhiteList</i> Failed: <i>XML_ResponseStatus</i>

Table 16-710 PUT /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/messageConfig/WhiteList

Method	PUT
Description	Set all whitelist configuration.
Query	None.
Request	<i>XML_SMSWhiteList</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID.

16.11.193 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/messageConfig/WhiteList/<ID>

Get or set specific whitelist configuration.

Request URI Definition

Table 16-711 GET /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/messageConfig/WhiteList/<ID>

Method	GET
Description	Get specific whitelist configuration.
Query	security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. None. iv: the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	Succeeded: <i>XML_ListMember</i> Failed: <i>XML_ResponseStatus</i>

Table 16-712 PUT /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/messageConfig/WhiteList/<ID>

Method	PUT
Description	Set specific whitelist configuration.

Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. None. iv : the initialization vector, and it is required when security is 1 or 2.
Request	<i>XML_ListMember</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID.

16.11.194 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/schedule

Get or set dial-up schedule parameters.

Request URI Definition

Table 16-713 GET /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/schedule

Method	GET
Description	Get dial-up schedule parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Schedule</i> Failed: <i>XML_ResponseStatus</i>

Table 16-714 PUT /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/schedule

Method	PUT
Description	Set dial-up schedule parameters.
Query	None.
Request	<i>XML_Schedule</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID.

16.11.195 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/trafficMonitor/capabilities?format=json

Get traffic monitoring configuration capability.

Request URI Definition

Table 16-715 GET /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/trafficMonitor/capabilities?format=json

Method	GET
Description	Get traffic monitoring configuration capability.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_TrafficMonitorCap</i> Failed: <i>JSON_ResponseStatus</i>

16.11.196 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/trafficMonitor/status?format=json

Get traffic monitoring status.

Request URI Definition

Table 16-716 GET /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/trafficMonitor/status?format=json

Method	GET
Description	Get traffic monitoring status.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_TrafficMonitorStatus</i> Failed: <i>JSON_ResponseStatus</i>

16.11.197 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/trafficMonitor?format=json

Get traffic monitoring parameters.

Request URI Definition**Table 16-717 GET /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/trafficMonitor?
format=json**

Method	GET
Description	Get traffic monitoring parameters.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_TrafficMonitor</i> Failed: <i>JSON_ResponseStatus</i>

**Table 16-718 PUT /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/trafficMonitor?
format=json**

Method	PUT
Description	Set traffic monitoring parameters.
Query	None.
Request	<i>JSON_TrafficMonitor</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID.

16.11.198 /ISAPI/System/Network/WPS/capabilities

Get the WPS (Wi-Fi Protected Setup) configuration capability.

Request URI Definition**Table 16-719 GET /ISAPI/System/Network/WPS/capabilities**

Method	GET
Description	Get the WPS (Wi-Fi Protected Setup) configuration capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_WPS</i> Failed: <i>XML_ResponseStatus</i>

16.11.199 /ISAPI/System/onlineUpgrade/capabilities

Get the device online upgrade capability.

Request URI Definition

Table 16-720 GET /ISAPI/System/onlineUpgrade/capabilities

Method	GET
Description	Get the device online upgrade capability.
Query	None.
Request	None.
Response	<i>XML_OnlineUpgradeCap</i>

16.11.200 /ISAPI/System/onlineUpgrade/deviceParameter?format=json

Get and set the device online upgrade parameters.

Request URI Definition

Table 16-721 GET /ISAPI/System/onlineUpgrade/deviceParameter?format=json

Method	GET
Description	Get the device online upgrade parameters.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_OnlineUpgradeParameter</i> Failed: <i>JSON_ResponseStatus</i>

Table 16-722 PUT /ISAPI/System/onlineUpgrade/deviceParameter?format=json

Method	PUT
Description	Set the device online upgrade parameters.
Query	format: determine the format of request or response message.
Request	<i>JSON_OnlineUpgradeParameter</i>
Response	<i>JSON_ResponseStatus</i>

16.11.201 /ISAPI/System/onlineUpgrade/downloadPackage/pause?format=json

Pause the upgrade package download.

Request URI Definition

Table 16-723 PUT /ISAPI/System/onlineUpgrade/downloadPackage/pause?format=json

Method	PUT
Description	Pause upgrade package download.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

16.11.202 /ISAPI/System/onlineUpgrade/downloadPackage/resume?format=json

Resume the upgrade package download.

Request URI Definition

Table 16-724 PUT /ISAPI/System/onlineUpgrade/downloadPackage/resume?format=json

Method	PUT
Description	Resume the upgrade package download.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

16.11.203 /ISAPI/System/onlineUpgrade/downloadPackage/status?format=json

Get the upgrade package download progress.

Request URI Definition

Table 16-725 GET /ISAPI/System/onlineUpgrade/downloadPackage/status?format=json

Method	GET
Description	Get the upgrade package download progress.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_DownloadPackageStatus</i> Failed: <i>JSON_ResponseStatus</i>

16.11.204 /ISAPI/System/onlineUpgrade/downloadPackage?format=json

Start or cancel downloading upgrade package to device.

Request URI Definition

Table 16-726 PUT /ISAPI/System/onlineUpgrade/downloadPackage?format=json

Method	PUT
Description	Start to download upgrade package to device.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

Table 16-727 DELETE /ISAPI/System/onlineUpgrade/downloadPackage?format=json

Method	DELETE
Description	Cancel the upgrade package download.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

16.11.205 /ISAPI/System/onlineUpgrade/ignoreCurrentVersion?format=json

Ignore the current upgrade package version.

Request URI Definition

Table 16-728 PUT /ISAPI/System/onlineUpgrade/ignoreCurrentVersion?format=json

Method	PUT
Description	Ignore the current upgrade package version.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

Remarks

If the current upgrade package version is ignored, it will not be found in further detection.

16.11.206 /ISAPI/System/onlineUpgrade/server

Get the online upgrade server status.

Request URI Definition

Table 16-729 GET /ISAPI/System/onlineUpgrade/server

Method	GET
Description	Get the online upgrade server status.
Query	None.
Request	None.
Response	<i>XML_OnlineUpgradeServer</i>

16.11.207 /ISAPI/System/onlineUpgrade/status

Get the online upgrade progress of device.

Request URI Definition

Table 16-730 GET /ISAPI/System/onlineUpgrade/status

Method	GET
Description	Get the online upgrade progress of device.
Query	None.
Request	None.
Response	<i>XML_OnlineUpgradeStatus</i>

16.11.208 /ISAPI/System/onlineUpgrade/upgradeWithoutDownload?format=json

Start device upgrade with existed upgrade package.

Request URI Definition

Table 16-731 PUT /ISAPI/System/onlineUpgrade/upgradeWithoutDownload?format=json

Method	PUT
Description	Start upgrade with existed upgrade package.
Query	format: determine the format of request or response message.

Request None.

Response *JSON_ResponseStatus*

16.11.209 /ISAPI/System/onlineUpgrade/version

Get the information of new upgrade package.

Request URI Definition

Table 16-732 GET /ISAPI/System/onlineUpgrade/version

Method GET

Description Get the information of new upgrade package.

Query check

Request None.

Response *XML_OnlineUpgradeVersion*

16.11.210 /ISAPI/System/PictureServer/capabilities?format=json

Get the picture storage server capability.

Request URI Definition

Table 16-733 GET /ISAPI/System/PictureServer/capabilities?format=json

Method	GET
Description	Get the picture storage server capability.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_Cap_PictureServerInformation</i> Failed: <i>JSON_ResponseStatus</i>

16.11.211 /ISAPI/System/PictureServer?format=json

Operations about the picture storage server configuration parameters.

Request URI Definition

Table 16-734 GET /ISAPI/System/PictureServer?format=json

Method	GET
Description	Get the picture storage server parameters.
Query	format: determine the format of request or response message. security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	Succeeded: <i>JSON_PictureServerInformation</i> Failed: <i>JSON_ResponseStatus</i>

Table 16-735 PUT /ISAPI/System/PictureServer?format=json

Method	PUT
Description	Set the picture storage server parameters.
Query	format: determine the format of request or response message. security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2.
Request	<i>JSON_PictureServerInformation</i>
Response	<i>JSON_ResponseStatus</i>

16.11.212 /ISAPI/System/reboot

Reboot device.

Request URI Definition

Table 16-736 PUT /ISAPI/System/reboot

Method	PUT
Description	Reboot device.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

16.11.213 /ISAPI/System/Serial/authentication/capabilities?format=json

Get capability of serial authentication service.

Request URI Definition

Table 16-737 GET /ISAPI/System/Serial/authentication/capabilities?format=json

Method	GET
Description	Get capability of serial authentication service.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_AuthenticationCap</i> Failed: <i>JSON_ResponseStatus</i>

16.11.214 /ISAPI/System/Serial/authentication?format=json

Get or set parameters of serial authentication service.

Request URI Definition

Table 16-738 GET /ISAPI/System/Serial/authentication?format=json

Method	GET
Description	Get parameters of serial authentication service.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_Authentication</i>

	Failed: <i>JSON_ResponseStatus</i>
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Table 16-739 PUT /ISAPI/System/Serial/authentication?format=json

Method	PUT
Description	Set parameters of serial authentication service.
Query	format : determine the format of request or response message.
Request	<i>JSON_Authentication</i>
Response	<i>JSON_ResponseStatus</i>

16.11.215 /ISAPI/System/Serial/capabilities

Get the serial port capability of the device.

Request URI Definition

Table 16-740 GET /ISAPI/System/Serial/capabilities

Method	GET
Description	Get the serial port capability of the device.
Query	None.
Request	None.
Response	Succeeded: <i>XML_SerialCap</i> Failed: <i>XML_ResponseStatus</i>

16.11.216 /ISAPI/System/Serial/ports

Get the list of serial ports supported by the device.

Request URI Definition

Table 16-741 GET /ISAPI/System/Serial/ports

Method	GET
Description	Get the list of serial ports supported by the device.
Query	None.
Request	None.
Response	Succeeded: <i>XML_SerialPorList</i> Failed: <i>XML_ResponseStatus</i>

16.11.217 /ISAPI/System/Serial/ports/<ID>

Operations about the configuration of a specific serial port supported by the device.

Request URI Definition

Table 16-742 GET /ISAPI/System/Serial/ports/<ID>

Method	GET
Description	Get the parameters of a specific port supported by the device.
Query	None.
Request	None.
Response	Succeeded: <i>XML_SerialPort</i> Failed: <i>XML_ResponseStatus</i>

Table 16-743 PUT /ISAPI/System/Serial/ports/<ID>

Method	PUT
Description	Set the parameters of a specific serial port supported by the device.
Query	None.
Request	<i>XML_SerialPort</i>
Response	<i>XML_ResponseStatus</i>

16.11.218 /ISAPI/System/Serial/ports/<ID>/capabilities

Get the capability of a specific serial port.

Request URI Definition

Table 16-744 GET /ISAPI/System/Serial/ports/<ID>/capabilities

Method	GET
Description	Get the capability of a specific serial port.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_SerialPort</i> Failed: <i>XML_ResponseStatus</i>

16.11.219 /ISAPI/System/Serial/ports/command

Send a command to a serial port.

Request URI Definition

Table 16-745 PUT /ISAPI/System/Serial/ports/command

Method	PUT
Description	Send a command to a serial port.
Query	None.
Request	<i>XML_SerialCommand</i> or query string with raw command in binary format
Response	<i>XML_ResponseStatus</i>

Remarks

- If the device is an encoder that converts analog signals to digital signals and it is connected to analog cameras with PTZ enabled, the encoder will forward the command to the appropriate serial port according to the **<chainNo>** node of the XML message or the query string of the HTTP request.
- If the device is a digital camera with PTZ enabled, the camera will forward the PTZ command to the corresponding serial port.
- The serial port command can be encapsulated in the **<command>** node of the XML message, and the command should be encoded in hexadecimal notation; or the serial port command can be directly uploaded as payload of the HTTP request, and in this case the **Content-Type** should be set to "application/octet-stream", and the query string **chainNo** should be set.

Example

Sample Code of Command in XML Format

```
PUT /ISAPI/System/Serial/ports/999/command HTTP/1.1
Content-Type: application/xml; charset="UTF-8"
Content-Length: xxx

<?xml version="1.0" encoding="UTF-8"?>
<SerialCommand>
  <chainNo>0</chainNo>
  <command>ab45be8778cd</command>
</SerialCommand>
```

Example

Sample Code of Command Using Query String

```
/ISAPI/System/Serial/ports/999/command?chainNo=1 HTTP/1.1
Content-Type: application/octet-stream
```

Content-Length: xxx
(...Raw data of the command should be inserted here...)

16.11.220 /ISAPI/System/Serial/ports/command/capabilities

Get the serial port command capability.

Request URI Definition

Table 16-746 GET /ISAPI/System/Serial/ports/command/capabilities

Method	GET
Description	Get the serial port command capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_SerialCommand</i> Failed: <i>XML_ResponseStatus</i>

16.11.221 /ISAPI/System/serialLogCfg/capabilities?format=json

Get the configuration capability of serial port log redirection.

Request URI Definition

Table 16-747 GET /ISAPI/System/serialLogCfg/capabilities?format=json

Method	GET
Description	Get the configuration capability of serial port log redirection.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_SerialLogCfgCap</i> Failed: <i>JSON_ResponseStatus</i>

16.11.222 /ISAPI/System/serialLogCfg?format=json

Operations about the redirection configuration of serial port log.

Request URI Definition

Table 16-748 GET /ISAPI/System/serialLogCfg?format=json

Method	GET
Description	Get the redirection parameters of serial port log.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_SerialLogCfg</i> Failed: <i>JSON_ResponseStatus</i>

Table 16-749 PUT /ISAPI/System/serialLogCfg?format=json

Method	PUT
Description	Set the redirection parameters of serial port log.
Query	format : determine the format of request or response message.
Request	<i>JSON_SerialLogCfg</i>
Response	<i>JSON_ResponseStatus</i>

16.11.223 /ISAPI/System/shutdown?format=json

Shut down the device.

Request URI Definition

Table 16-750 PUT /ISAPI/System/shutdown?format=json

Method	PUT
Description	Shut down the device.
Query	format : determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

16.11.224 /ISAPI/System/Software/channels/<ID>

Operation about software service configuration.

Request URI Definition

Table 16-751 GET /ISAPI/System/Software/channels/<ID>

Method	GET
Description	Get software service parameters.
Query	None.
Request	None.
Response	<i>XML_SoftwareService</i>

Table 16-752 PUT /ISAPI/System/Software/channels/<ID>

Method	PUT
Description	Set software service parameters.
Query	None.
Request	<i>XML_SoftwareService</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

16.11.225 /ISAPI/System/Software/channels/<ID>/capabilities

Get the configuration capability of software service.

Request URI Definition

Table 16-753 GET /ISAPI/System/Software/channels/<ID>/capabilities

Method	GET
Description	Get the configuration capability of software service.
Query	None.
Request	None.
Response	<i>XML_Cap_SoftwareService</i>

Remarks

The <ID> in the request URI refers to the channel No.

16.11.226 /ISAPI/System/status

Get device status, e.g., CPU, memory, and so on.

Request URI Definition**Table 16-754 GET /ISAPI/System/status**

Method	GET
Description	Get device status, e.g., CPU, memory, and so on.
Query	None.
Request	None.
Response	Succeeded: <i>XML_DeviceStatus</i> Failed: <i>XML_ResponseStatus</i>

16.11.227 /ISAPI/System/syncSignalOutput/<ID>

Operations about the configuration of synchronous signal output.

Request URI Definition**Table 16-755 GET /ISAPI/System/syncSignalOutput/<ID>**

Method	GET
Description	Get the parameters of synchronous signal output.
Query	None.
Request	None.
Response	Succeeded: <i>XML_SyncSignalOutputList</i> Failed: <i>XML_ResponseStatus</i>

Table 16-756 PUT /ISAPI/System/syncSignalOutput/<ID>

Method	PUT
Description	Set the parameters of synchronous signal output.
Query	None.
Request	<i>XML_SyncSignalOutputList</i>
Response	<i>XML_ResponseStatus</i>

16.11.228 /ISAPI/System/syncSignalOutput/<ID>/capabilities

Get the configuration capability of synchronous signal output.

Request URI Definition**Table 16-757 GET /ISAPI/System/syncSignalOutput/<ID>/capabilities**

Method	GET
Description	Get the configuration capability of synchronous signal output.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_SyncSignalOutputList</i> Failed: <i>XML_ResponseStatus</i>

16.11.229 /ISAPI/System/time

Operations about the device time configuration.

Request URI Definition**Table 16-758 GET /ISAPI/System/time**

Method	GET
Description	Get the device time parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Time</i> Failed: <i>XML_ResponseStatus</i>

Table 16-759 PUT /ISAPI/System/time

Method	PUT
Description	Set the device time parameters.
Query	None.
Request	<i>XML_Time</i>
Response	<i>XML_ResponseStatus</i>

16.11.230 /ISAPI/System/time/capabilities

Get the time capability of the device.

Request URI Definition

Table 16-760 GET /ISAPI/System/time/capabilities

Method	GET
Description	Get the time capability of the device.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_Time</i> Failed: <i>XML_ResponseStatus</i>

16.11.231 /ISAPI/System/time/localTime

Operations about the configuration of device local time.

Request URI Definition

Table 16-761 GET /ISAPI/System/time/localTime

Method	GET
Description	Get the configuration parameters of device local time.
Query	None.
Request	None.
Response	String with date and time information (ISO8601 format, e.g., 2018-02-10T17:30:08)

Table 16-762 PUT /ISAPI/System/time/localTime

Method	PUT
Description	Set device local time.
Query	None.
Request	String with date and time information (ISO8601 format, e.g., 2018-02-10T17:30:08)
Response	<i>XML_ResponseStatus</i>

16.11.232 /ISAPI/System/time/ntpServers

Operations about the configurations of multiple NTP servers.

Request URI Definition

Table 16-763 GET /ISAPI/System/time/ntpServers

Method	GET
Description	Get the parameters of multiple NTP servers.
Query	None.
Request	None.
Response	Succeeded: <i>XML_NTPServerList</i> Failed: <i>XML_ResponseStatus</i>

Table 16-764 PUT /ISAPI/System/time/ntpServers

Method	PUT
Description	Set the parameters of multiple NTP servers.
Query	None.
Request	<i>XML_NTPServerList</i>
Response	<i>XML_ResponseStatus</i>

Table 16-765 POST /ISAPI/System/time/ntpServers

Method	POST
Description	Add a NTP server.
Query	None.
Request	<i>XML_NTPServer</i>
Response	<i>XML_ResponseStatus</i>

Table 16-766 DELETE /ISAPI/System/time/ntpServers

Method	DELETE
Description	Delete multiple NTP servers.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

16.11.233 /ISAPI/System/time/ntpServers/<ID>

Operations about the configurations of a NTP server.

Request URI Definition**Table 16-767 GET /ISAPI/System/time/ntpServers/<ID>**

Method	GET
Description	Get the parameters of a NTP server.
Query	None.
Request	None.
Response	Succeeded: <i>XML_NTPServer</i> Failed: <i>XML_ResponseStatus</i>

Table 16-768 PUT /ISAPI/System/time/ntpServers/<ID>

Method	PUT
Description	Set the parameters of a NTP server.
Query	None.
Request	<i>XML_NTPServer</i>
Response	<i>XML_ResponseStatus</i>

Table 16-769 DELETE /ISAPI/System/time/ntpServers/<ID>

Method	DELETE
Description	Delete the configuration parameters of a NTP server.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

16.11.234 /ISAPI/System/time/ntpServers/<ID>/capabilities

Get the configuration capability of a specific NTP server.

Request URI Definition

Table 16-770 GET /ISAPI/System/time/ntpServers/<ID>/capabilities

Method	GET
Description	Get the configuration capability of a specific NTP server.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_NTPServer</i> Failed: <i>XML_ResponseStatus</i>

16.11.235 /ISAPI/System/time/ntpServers/capabilities

Get the NTP service capability.

Request URI Definition

Table 16-771 GET /ISAPI/System/time/ntpServers/capabilities

Method	GET
Description	Get the NTP service capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_NTPServerList</i> Failed: <i>XML_ResponseStatus</i>

16.11.236 /ISAPI/System/time/ntpServers/test

Execute a test to check if the NTP server is available.

Request URI Definition

Table 16-772 POST /ISAPI/System/time/ntpServers/test

Method	POST
Description	Execute a test to check if the NTP server is available.
Query	None.

Request	<i>XML_NTPTestDescription</i>
Response	<i>XML_NTPTestResult</i>

16.11.237 /ISAPI/System/time/timeType?format=json

Operations about device time type configuration.

Request URI Definition

Table 16-773 GET /ISAPI/System/time/timeType?format=json

Method	GET
Description	Get the device time type.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_TimeType</i>

Table 16-774 PUT /ISAPI/System/time/timeType?format=json

Method	PUT
Description	Set the device time type.
Query	format: determine the format of request or response message.
Request	<i>JSON_TimeType</i>
Response	<i>JSON_ResponseStatus</i>

16.11.238 /ISAPI/System/time/timeZone

Operations about the configuration of device time zone.

Request URI Definition

Table 16-775 GET /ISAPI/System/time/timeZone

Method	GET
Description	Get the device time zone parameters.
Query	None.
Request	None.
Response	String containing time zone information

Table 16-776 PUT /ISAPI/System/time/timeZone

Method	PUT
Description	Set the device time zone parameters.
Query	None.
Request	String containing time zone information
Response	<i>XML_ResponseStatus</i>

16.11.239 /ISAPI/System/TwoWayAudio/channels

Get audio parameters of all two-way audio channels.

Request URI Definition

Table 16-777 GET /ISAPI/System/TwoWayAudio/channels

Method	GET
Description	Get parameters of all two-way audio channels.
Query	None.
Request	None.
Response	Succeeded: <i>XML_TwoWayAudioChannelList</i> Failed: <i>XML_ResponseStatus</i>

Example

Sample Code for Getting Parameters of All Two-Way Audio Channels

```
GET /ISAPI/System/TwoWayAudio/channels HTTP/1.1
Host: 10.17.132.49
Connection: Keep-Alive
Authorization: Digest username="admin",
realm="DS-2CD2F12FWD-IWS",
nonce="4e4446464e6a6333516a63365a4445304f47526a4e32553d",
uri="/ISAPI/System/TwoWayAudio/channels",
cnonce="145ef1bc3ab456be09918d39b77e78ae",
nc=00000021,
response="d10bab23689c60e95241230fff3181b0",
qop="auth"
```

```
HTTP/1.1 200 OK
Date: Wed, 15 Mar 2017 09:25:29 GMT
Server: App-webs/
Connection: close
Content-Length: 467
Content-Type: application/xml
```

```
<?xml version="1.0" encoding="UTF-8"?>
<TwoWayAudioChannelList version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <TwoWayAudioChannel version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
    <id>1</id>
    <enabled>>false</enabled>
    <audioCompressionType>G.711ulaw</audioCompressionType>
    <audioInputType>MicIn</audioInputType>
    <speakerVolume>50</speakerVolume>
    <noisereduce>>false</noisereduce>
  </TwoWayAudioChannel>
</TwoWayAudioChannelList>
```

16.11.240 /ISAPI/System/TwoWayAudio/channels/<ID>

Operations about the configuration of a specific two-way audio channel.

Request URI Definition

Table 16-778 GET /ISAPI/System/TwoWayAudio/channels/<ID>

Method	GET
Description	Get the parameters of a specific two-way audio channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_TwoWayAudioChannel</i> Failed: <i>XML_ResponseStatus</i>

Table 16-779 PUT /ISAPI/System/TwoWayAudio/channels/<ID>

Method	PUT
Description	Set the parameters of a specific two-way audio channel.
Query	None.
Request	<i>XML_TwoWayAudioChannel</i>
Response	<i>XML_ResponseStatus</i>

16.11.241 /ISAPI/System/TwoWayAudio/channels/<ID>/audioData

Receive or send audio data from or to a specific two-way audio channel.

Request URI Definition

Table 16-780 GET /ISAPI/System/TwoWayAudio/channels/<ID>/audioData

Method	GET
Description	Receive audio data from a specific two-way audio channel.
Query	sessionId : communication session ID, which is required when the two-way audio is started between multiple channels and a channel; otherwise, the session ID is not required.
Request	None.
Response	Succeeded: audio data Failed: <i>XML_ResponseStatus</i>

Table 16-781 PUT /ISAPI/System/TwoWayAudio/channels/<ID>/audioData

Method	PUT
Description	Send audio data to a specific two-way audio channel.
Query	sessionId : communication session ID, which is required when the two-way audio is started between multiple channels and a channel; otherwise, the session ID is not required.
Request	Audio data.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the two-way audio channel ID.

Example

Sample Code for Sending Audio Data to Device

```
PUT /ISAPI/System/TwoWayAudio/channels/1/audioData HTTP/1.1
HOST: 10.17.132.49
Authorization: Digest username="admin",
realm="DS-2CD2F12FWD-IWS",
nonce="4e3055314e6a64434e7a59365a4445304f545668596a6b3d",
uri="/ISAPI/System/TwoWayAudio/channels/1/audioData",
response="6a03c7e85c17a35bae9ee6f3cfd254b"
Connection: keep-alive
Content-Length: 0
Content-Type: application/octet-stream

HTTP/1.1 200 OK

..U....U.....U.....U..... (Binary audio data)
```

Example

Sample Code for Receiving Audio Data from Device

```
GET /ISAPI/System/TwoWayAudio/channels/1/audioData HTTP/1.1
HOST: 10.17.132.49
Authorization: Digest username="admin",
realm="DS-2CD2F12FWD-IWS",
nonce="4e3055314e6a64434e7a59365a4445304f5456685a44673d",
uri="/ISAPI/System/TwoWayAudio/channels/1/audioData",
response="7ea0ec7bd102b3c4cb225cbf33adb8f8"
Connection: keep-alive
```

```
HTTP/1.1 200 OK
Content-Type: audio/basic
```

```
Q.....U..dzR..\.FfN...q[....z..q.....` (Binary audio data)
```

16.11.242 /ISAPI/System/TwoWayAudio/channels/<ID>/capabilities

Get the capability of a specific two-way audio channel.

Request URI Definition

Table 16-782 GET /ISAPI/System/TwoWayAudio/channels/<ID>/capabilities

Method	GET
Description	Get the capability of a specific two-way audio channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_TwoWayAudioChannelCap</i> Failed: <i>XML_ResponseStatus</i>

16.11.243 /ISAPI/System/TwoWayAudio/channels/<ID>/close

Stop two-way audio of a specific channel.

Request URI Definition

Table 16-783 PUT /ISAPI/System/TwoWayAudio/channels/<ID>/close

Method	PUT
Description	Stop two-way audio of a specific channel.
Query	None.

Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the two-way audio channel ID.

Example

Sample Code for Stopping Two-Way Audio

```
PUT /ISAPI/System/TwoWayAudio/channels/1/close HTTP/1.1
HOST: 10.17.132.49
Authorization: Digest username="admin",
realm="DS-2CD2F12FWD-IWS",
nonce="4e54457a4d7a67354d6a6b365a4445304f5463794d32553d",
uri="/ISAPI/System/TwoWayAudio/channels/1/close",
response="7d5acdc3ccf7c080400802800d7471e3"
Content-Length: 0

HTTP/1.1 200 OK
Date: Wed, 15 Mar 2017 09:25:35 GMT
Server: App-webs/
Connection: close
Content-Length: 298
Content-Type: application/xml

<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <requestURL>/ISAPI/System/TwoWayAudio/channels/1/close</requestURL>
  <statusCode>1</statusCode>
  <statusString>OK</statusString>
  <subStatusCode>ok</subStatusCode>
</ResponseStatus>
```

16.11.244 /ISAPI/System/TwoWayAudio/channels/<ID>/open

Start two-way audio of a specific channel.

Request URI Definition

Table 16-784 PUT /ISAPI/System/TwoWayAudio/channels/<ID>/open

Method	PUT
Description	Start two-way audio of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_TwoWayAudioSession</i>

Failed: *XML_ResponseStatus*

Remarks

The <ID> in the request URI refers to the two-way audio channel ID.

Example

Sample Code for Start Two-Way Audio

```
PUT /ISAPI/System/TwoWayAudio/channels/1/open HTTP/1.1
HOST: 10.17.132.49
Authorization: Digest username="admin",
realm="DS-2CD2F12FWD-IWS",
nonce="4e3055314e6a64434e7a59365a4445304f5456685957453d",
uri="/ISAPI/System/TwoWayAudio/channels/1/open",
response="368dda22535b9783bdccafc3b2ded29a"
Content-Length: 0

HTTP/1.1 200 OK
Date: Wed, 15 Mar 2017 09:25:29 GMT
Server: App-webs/
Connection: close
Content-Length: 183
Content-Type: application/xml

<?xml version="1.0" encoding="UTF-8"?>
<TwoWayAudioSession version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <sessionId>2093716360</sessionId>
</TwoWayAudioSession>
```

16.11.245 /ISAPI/System/unitConfig/capabilities?format=json

Get the capability of unit unified configuration.

Request URI Definition

Table 16-785 GET /ISAPI/System/unitConfig/capabilities?format=json

Method	GET
Description	Get the capability of unit unified configuration.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_unitConfigCap</i> Failed: <i>JSON_ResponseStatus</i>

16.11.246 /ISAPI/System/unitConfig?format=json

Get and set the unit unified configuration parameters.

Request URI Definition

Table 16-786 GET /ISAPI/System/unitConfig?format=json

Method	GET
Description	Get the unit unified configuration parameters.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_unitConfig</i> Failed: <i>JSON_ResponseStatus</i>

Table 16-787 PUT /ISAPI/System/unitConfig?format=json

Method	PUT
Description	Set the unit unified configuration parameters.
Query	format: determine the format of request or response message.
Request	<i>JSON_unitConfig</i>
Response	<i>JSON_ResponseStatus</i>

16.11.247 /ISAPI/System/updateFirmware

Upgrade the device firmware.

Request URI Definition

Table 16-788 PUT or POST /ISAPI/System/updateFirmware

Method	PUT, POST
Description	Upgrade the device firmware.
Query	None.
Request	Opaque data (binary data for PUT method, data in form format for POST method).
Response	<i>XML_ResponseStatus</i>

Remarks

After upgrading the device firmware, the device will automatically reboot according to the response message. If upgrading failed, the following reasons may be returned: "upgrading"-the device is upgrading, "badFlash"-flash error, "badLanguage"-language mismatches.

Example

The URL Interaction in Form Format

```
/ISAPI/System/updateFirmware
Accept: text/html, application/xhtml+xml,
Accept-Language: zh-CN
Content-Type: multipart/form-data; boundary=-----7e13971310878
User-Agent: Mozilla/5.0 (compatible; MSIE 9.0; Windows NT 6.1; WOW64; Trident/5.0)
Accept-Encoding: gzip, deflate
Host: 10.10.36.29:8080
Content-Length: 9907
Connection: Keep-Alive
Cache-Control: no-cache

-----7e13971310878
Content-Disposition: form-data; name="updateFile";
Content-Type: application/octet-stream
Content-Length: 9907

.....`..C.....
..
.....$. ' ",#.(7),01444.'9=82<.342...C. ....
-----7e13971310878--
```

16.11.248 /ISAPI/System/updateFirmware?type=&id=

Upgrade the slave device's firmware.

Request URI Definition

Table 16-789 PUT or POST /ISAPI/System/updateFirmware?type=&id=

Method	PUT, POST
Description	Upgrade the slave device's firmware.
Query	type: device or module type, string type, it can be set to one of the following values: "cardReader"-card reader, "FPModule"-fingerprint module, "securityModule"-secure door control unit, "extendModule"- (IO) extension module, "channelController"-lane controller, "IRModule"-IR module, "lampModule"-indicator module, "elevatorController"-slave elevator controller, and

	"FPAlgorithmProgram"-fingerprint algorithm program of the card reader, "uboot". id : slave device No., e.g., /ISAPI/System/updateFirmware?type=cardReader&id=1 refers to upgrading firmware of card reader 1.
Request	Opaque data (binary data for PUT method, data in form format for POST method).
Response	<i>XML_ResponseStatus</i>

Remarks

After upgrading the slave device's firmware, the device will automatically reboot according to the response message. If upgrading failed, the following reasons may be returned: "upgrading"-the device is upgrading, "badFlash"-flash error, "badLanguage"-language mismatches.

16.11.249 /ISAPI/System/updateFirmware?type=&moduleAddress=

Upgrade the peripheral module's firmware.

Request URI Definition

Table 16-790 PUT or POST /ISAPI/System/updateFirmware?type=&moduleAddress=

Method	PUT, POST
Description	Upgrade the peripheral module's firmware.
Query	type : device or module type, string type, it can be set to one of the following values: "keypad", "wirelessRecv"-wireless receiving module, "wiredZone"-wired zone module. moduleAddress : module address, e.g., /ISAPI/System/updateFirmware?type=wirelessRecv&moduleAddress=1 refers to upgrading firmware of the wireless receiving module with the module address 1.
Request	Opaque data (binary data for PUT method, data in form format for POST method).
Response	<i>XML_ResponseStatus</i>

Remarks

After upgrading the peripheral module's firmware, the device will automatically reboot according to the response message. If upgrading failed, the following reasons may be returned: "upgrading"-the device is upgrading, "badFlash"-flash error, "badLanguage"-language mismatches.

16.11.250 /ISAPI/System/upgradeStatus

Get the device upgrading status and progress.

Request URI Definition

Table 16-791 GET /ISAPI/System/upgradeStatus

Method	GET
Description	Get the device upgrading status and progress.
Query	None.
Request	None.
Response	Succeeded: <i>XML_upgradeStatus</i> Failed: <i>XML_ResponseStatus</i>

16.11.251 /ISAPI/System/usb/capabilities?format=json

Get the USB management capability.

Request URI Definition

Table 16-792 GET /ISAPI/System/usb/capabilities?format=json

Method	GET
Description	Get the USB management capability.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_USBCap</i> Failed: <i>JSON_ResponseStatus</i>

16.11.252 /ISAPI/System/usb?format=json

Get or set the USB parameters.

Request URI Definition

Table 16-793 GET /ISAPI/System/usb?format=json

Method	GET
Description	Get the USB parameters.

Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_USB</i> Failed: <i>JSON_ResponseStatus</i>

Table 16-794 PUT /ISAPI/System/usb?format=json

Method	PUT
Description	Set the USB parameters.
Query	format: determine the format of request or response message.
Request	<i>JSON_USB</i>
Response	<i>JSON_ResponseStatus</i>

16.11.253 /ISAPI/System/userLock/config/capabilities?format=json

Get the configuration capability of locking IP address.

Request URI Definition

Table 16-795 GET /ISAPI/System/userLock/config/capabilities?format=json

Method	GET
Description	Get the configuration capability of locking IP address.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_LockCfgCap</i> Failed: <i>JSON_ResponseStatus</i>

16.11.254 /ISAPI/System/userLock/config?format=json

Operations about IP address locking configuration.

Request URI Definition

Table 16-796 GET /ISAPI/System/userLock/config?format=json

Method	GET
Description	Get the parameters of IP address locking.
Query	format: determine the format of request or response message.

Request	None.
Response	Succeeded: <i>JSON_LockCfg</i> Failed: <i>JSON_ResponseStatus</i>

Table 16-797 PUT /ISAPI/System/userLock/config?format=json

Method	PUT
Description	Set the parameters of IP address locking.
Query	format: determine the format of request or response message.
Request	<i>JSON_LockCfg</i>
Response	<i>JSON_ResponseStatus</i>

16.11.255 /ISAPI/System/userLock/lockedUsers?format=json

Get all locked IP addresses.

Request URI Definition

Table 16-798 GET /ISAPI/System/userLock/lockedUsers?format=json

Method	GET
Description	Get all locked IP addresses.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_List_IPAddress</i> Failed: <i>JSON_ResponseStatus</i>

16.11.256 /ISAPI/System/userLock/unlockUser?format=json

Unlock the IP address.

Request URI Definition

Table 16-799 PUT /ISAPI/System/userLock/unlockUser?format=json

Method	PUT
Description	Unlock the IP address.
Query	format: determine the format of request or response message.

Request	<i>JSON_operType</i>
Response	<i>JSON_ResponseStatus</i>

16.11.257 /ISAPI/System/Video/capabilities

Get video channel capability.

Request URI Definition

Table 16-800 GET /ISAPI/System/Video/capabilities

Method	GET
Description	Get video channel capability.
Query	None
Request	None
Response	Succeeded: <i>XML_VideoCap</i> Failed: <i>XML_ResponseStatus</i>

16.11.258 /ISAPI/System/Video/inputs

Get the video input configuration of the device.

Request URI Definition

Table 16-801 GET /ISAPI/System/Video/inputs

Method	GET
Description	Get the video input configuration parameters of the device.
Query	None.
Request	None.
Response	Succeeded: <i>XML_VideoInput</i> Failed: <i>XML_ResponseStatus</i>

16.11.259 /ISAPI/System/Video/inputs/channels

Get the configuration of all video input channels of the device.

Request URI Definition

Table 16-802 GET /ISAPI/System/Video/inputs/channels

Method	GET
Description	Get the configuration parameters of all video input channels of the device.
Query	None.
Request	None.
Response	Succeeded: <i>XML_VideoInputChannelList</i> Failed: <i>XML_ResponseStatus</i>

16.11.260 /ISAPI/System/Video/inputs/channels/<ID>

Operations about the configuration of a specific video input channel of the device.

Request URI Definition

Table 16-803 GET /ISAPI/System/Video/inputs/channels/<ID>

Method	GET
Description	Get the parameters of a specific video input channel of the device.
Query	None.
Request	None.
Response	Succeeded: <i>XML_VideoInputChannel</i> Failed: <i>XML_ResponseStatus</i>

Table 16-804 PUT /ISAPI/System/Video/inputs/channels/<ID>

Method	PUT
Description	Set the parameters of a specific video input channel of the device.
Query	None.
Request	<i>XML_VideoInputChannel</i>
Response	<i>XML_ResponseStatus</i>

See Also

The <ID> in the request URI refers to the channel No.

16.11.261 /ISAPI/System/Video/inputs/channels?format=json

Get the information of all channels according to the request message.

Request URI Definition

Table 16-805 POST /ISAPI/System/Video/inputs/channels?format=json

Method	POST
Description	Get the information of all channels according to the request message.
Query	format : determine the format of request or response message.
Request	<i>JSON_InputParam</i>
Response	Succeeded: <i>JSON_ChannelInfoList</i> Failed: <i>JSON_ResponseStatus</i>

16.11.262 /ISAPI/System/Video/inputs/channels/<ID>/focus

Perform manual focus on a specific channel.

Request URI Definition

Table 16-806 PUT /ISAPI/System/Video/inputs/channels/<ID>/focus

Method	PUT
Description	Perform manual focus on a specific channel.
Query	None.
Request	<i>XML_FocusData</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to video input channel ID.

16.11.263 /ISAPI/System/Video/inputs/channels/<ID>?format=json

Get the information of a channel according to the request message.

Request URI Definition

Table 16-807 POST /ISAPI/System/Video/inputs/channels/<ID>?format=json

Method	POST
Description	Get the information of a channel according to the request message.
Query	format : determine the format of request or response message.
Request	
Response	Succeeded: <i>JSON_ChannelInfo</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

16.11.264 /ISAPI/System/Video/inputs/channels/<ID>/capabilities

Get the configuration capability of a video input channel.

Request URI Definition

Table 16-808 GET /ISAPI/System/Video/inputs/channels/<ID>/capabilities

Method	GET
Description	Get the configuration capability of a video input channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_VideoInputChannel</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to video input channel ID.

16.11.265 /ISAPI/System/Video/inputs/channels/<ID>/iris

Manually adjust the iris of a specific video input channel.

Request URI Definition

Table 16-809 PUT /ISAPI/System/Video/inputs/channels/<ID>/iris

Method	PUT
Description	Manually adjust the iris of a specific video input channel.
Query	None.
Request	<i>XML_IrisData</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to video input channel ID.

16.11.266 /ISAPI/System/Video/inputs/channels/<ID>/motionDetection

Get or set the motion detection parameters of a specified video input channel.

Request URI Definition

Table 16-810 GET /ISAPI/System/Video/inputs/channels/<ID>/motionDetection

Method	GET
Description	Get the motion detection parameters of a specified video input channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_MotionDetection</i> Failed: <i>XML_ResponseStatus</i>

Table 16-811 PUT /ISAPI/System/Video/inputs/channels/<ID>/motionDetection

Method	PUT
Description	Set the motion detection parameters of a specified video input channel.
Query	None.
Request	<i>XML_MotionDetection</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the video input channel No.

16.11.267 /ISAPI/System/Video/inputs/channels/<ID>/motionDetection/capabilities

Get the motion detection capability of a specified video input channel.

Request URI Definition

Table 16-812 GET /ISAPI/System/Video/inputs/channels/<ID>/motionDetection/capabilities

Method	GET
Description	Get the motion detection capability of a specified video input channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_MotionDetection</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the video input channel No.

16.11.268 /ISAPI/System/Video/inputs/channels/<ID>/motionDetection/layout

Get or set rule region parameters of motion detection.

Request URI Definition

Table 16-813 GET /ISAPI/System/Video/inputs/channels/<ID>/motionDetection/layout

Method	GET
Description	Get rule region parameters of motion detection.
Query	regionType : Detect by region or by grid, string, the available value is "region" or "grid".
Request	None
Response	Succeeded: <i>XML_MotionDetectionLayout</i> Failed: <i>XML_ResponseStatus</i>

Table 16-814 PUT /ISAPI/System/Video/inputs/channels/<ID>/motionDetection/layout

Method	PUT
Description	Set rule region parameters of motion detection.

Query	regionType : Detect by region or by grid, string, the available value is "region" or "grid".
Request	<i>XML_MotionDetectionLayout</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI refers to the video input channel ID.

16.11.269 /ISAPI/System/Video/inputs/channels/<ID>/motionDetection/layout/capabilities

Get rule region configuration capability of motion detection.

Request URI Definition

Table 16-815 GET /ISAPI/System/Video/inputs/channels/<ID>/motionDetection/layout/capabilities

Method	GET
Description	Get rule region configuration capability of motion detection.
Query	None
Request	None
Response	Succeeded: <i>XML_Cap_MotionDetectionLayout</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI refers to the video input channel ID.

16.11.270 /ISAPI/System/Video/inputs/channels/<ID>/motionDetection/layout/gridLayout

Get or set grid layout parameters of motion detection.

Request URI Definition

Table 16-816 GET /ISAPI/System/Video/inputs/channels/<ID>/motionDetection/layout/gridLayout

Method	GET
Description	Get grid layout parameters of motion detection.

Query	regionType : Detect by region or by grid, string, the available value is "region" or "grid".
Request	None
Response	Succeeded: <i>XML_MotionDetectionGridLayout</i> Failed: <i>XML_ResponseStatus</i>

**Table 16-817 PUT /ISAPI/System/Video/inputs/channels/<ID>/motionDetection/layout/
gridLayout**

Method	PUT
Description	Set grid layout parameters of motion detection.
Query	regionType : Detect by region or by grid, string, the available value is "region" or "grid".
Request	<i>XML_MotionDetectionGridLayout</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI refers to the video input channel ID.

16.11.271 /ISAPI/System/Video/inputs/channels/<ID>/overlays

Operations about OSD configuration of a specific video input channel.

Request URI Definition

Table 16-818 GET /ISAPI/System/Video/inputs/channels/<ID>/overlays

Method	GET
Description	Get OSD configuration parameters of a specific video input channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_VideoOverlay</i> Failed: <i>XML_ResponseStatus</i>

Table 16-819 PUT /ISAPI/System/Video/inputs/channels/<ID>/overlays

Method	PUT
Description	Set OSD parameters of a specific video input channel.
Query	None.

Request	<i>XML_VideoOverlay</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the video input channel ID.

Example

Sample Code for Getting OSD Parameters

```
GET /ISAPI/System/Video/inputs/channels/1/overlays HTTP/1.1
Host: 10.17.132.49
Connection: Keep-Alive
Authorization: Digest username="admin",
realm="DS-2CD2F12FWD-IWS",
nonce="4d3046464f454e425254633659325534595749784f546b3d",
uri="/ISAPI/System/Video/inputs/channels/1/overlays",
cnonce="03e455047789a833deaae75e44381653",
nc=00000017,
response="ae141a01b415c1b9ab50ddea7ff467c3",
qop="auth"

HTTP/1.1 200 OK
Date: Tue, 14 Mar 2017 20:38:40 GMT
Server: App-webs/
Connection: close
Content-Length: 1087
Content-Type: application/xml

<?xml version="1.0" encoding="UTF-8"?>
<VideoOverlay version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <normalizedScreenSize>
    <normalizedScreenWidth>704</normalizedScreenWidth>
    <normalizedScreenHeight>576</normalizedScreenHeight>
  </normalizedScreenSize>
  <attribute>
    <transparent>false</transparent>
    <flashing>false</flashing>
  </attribute>
  <fontSize>64*64</fontSize>
  <TextOverlayList size="1">
    <TextOverlay>
      <id>1</id>
      <enabled>true</enabled>
      <positionX>0</positionX>
      <positionY>576</positionY>
      <displayText>222</displayText>
    </TextOverlay>
  </TextOverlayList>
  <DateTimeOverlay>
    <enabled>true</enabled>
    <positionX>0</positionX>
```

```
<positionY>544</positionY>
<dateStyle>MM-DD-YYYY</dateStyle>
<timeStyle>24hour</timeStyle>
<displayWeek>true</displayWeek>
</DateTimeOverlay>
<channelNameOverlay version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <enabled>true</enabled>
  <positionX>512</positionX>
  <positionY>64</positionY>
</channelNameOverlay>
<frontColorMode>auto</frontColorMode>
<frontColor>000000</frontColor>
</VideoOverlay>
```

16.11.272 /ISAPI/System/Video/inputs/channels/<ID>/overlays/BatteryPowerOverlay

Operations about the battery overlay configuration of a specific video input channel.

Request URI Definition

Table 16-820 GET /ISAPI/System/Video/inputs/channels/<ID>/overlays/BatteryPowerOverlay

Method	GET
Description	Get the battery overlay parameters of a specific video input channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_BatteryPowerOverlay</i> Failed: <i>XML_ResponseStatus</i>

Table 16-821 PUT /ISAPI/System/Video/inputs/channels/<ID>/overlays/BatteryPowerOverlay

Method	PUT
Description	Set the battery overlay parameters of a specific video input channel.
Query	None.
Request	<i>XML_BatteryPowerOverlay</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the video input channel No.

16.11.273 /ISAPI/System/Video/inputs/channels/<ID>/overlays/BatteryPowerOverlay/capabilities

Get the configuration capability of battery overlay of a specific video input channel.

Request URI Definition

Table 16-822 GET /ISAPI/System/Video/inputs/channels/<ID>/overlays/BatteryPowerOverlay/capabilities

Method	GET
Description	Get the configuration capability of battery overlay of a specific video input channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_BatteryPowerOverlay</i> Failed: <i>XML_ResponseStatus</i>

16.11.274 /ISAPI/System/Video/inputs/channels/<ID>/overlays/capabilities

Get the OSD capability of a specific video input channel.

Request URI Definition

Table 16-823 GET /ISAPI/System/Video/inputs/channels/<ID>/overlays/capabilities

Method	GET
Description	Get the OSD capability of a specific video input channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_VideoOverlay</i> Failed: <i>XML_ResponseStatus</i>

16.11.275 /ISAPI/System/Video/inputs/channels/<ID>/overlays/channelNameOverlay

Operations about the configuration of channel name overlay of a specific video input channel.

Request URI Definition**Table 16-824 GET /ISAPI/System/Video/inputs/channels/<ID>/overlays/channelNameOverlay**

Method	GET
Description	Get the parameters of channel name overlay of a specific video input channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_channelNameOverlay</i> Failed: <i>XML_ResponseStatus</i>

Table 16-825 PUT /ISAPI/System/Video/inputs/channels/<ID>/overlays/channelNameOverlay

Method	PUT
Description	Set the parameters of channel name overlay of a specific video input channel.
Query	None.
Request	<i>XML_channelNameOverlay</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the video input channel No.

16.11.276 /ISAPI/System/Video/inputs/channels/<ID>/overlays/dateTimeOverlay

Operations about the configuration of date and time information overlay of a specific video input channel.

Request URI Definition**Table 16-826 GET /ISAPI/System/Video/inputs/channels/<ID>/overlays/dateTimeOverlay**

Method	GET
Description	Get the parameters of date and time information overlay of a specific video input channel.
Query	None.
Request	None.
Response	Succeeded:

	Failed: <i>XML_ResponseStatus</i>
--	--

Table 16-827 PUT /ISAPI/System/Video/inputs/channels/<ID>/overlays/dateTimeOverlay

Method	PUT
Description	Set the parameters of date and time information overlay of a specific video input channel.
Query	None.
Request	
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the video input channel No.

16.11.277 /ISAPI/System/Video/inputs/channels/<ID>/overlays/dateTimeOverlay/capabilities

Get the capability of date and time information overlay of a specific video input channel.

Request URI Definition**Table 16-828 GET /ISAPI/System/Video/inputs/channels/<ID>/overlays/dateTimeOverlay/capabilities**

Method	GET
Description	Get the capability of date and time information overlay of a specific video input channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_DatetimeOverlay</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the video input channel No.

16.11.278 /ISAPI/System/Video/inputs/channels/<ID>/overlays/text

Operations about the text overlay configuration of a specific video input channel.

Request URI Definition

Table 16-829 GET /ISAPI/System/Video/inputs/channels/<ID>/overlays/text

Method	GET
Description	Get the text overlay parameters of a specific video input channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_TextOverlayList</i> Failed: <i>XML_ResponseStatus</i>

Table 16-830 PUT /ISAPI/System/Video/inputs/channels/<ID>/overlays/text

Method	PUT
Description	Set the text overlay parameters of a specific video input channel.
Query	None.
Request	<i>XML_TextOverlayList</i>
Response	<i>XML_ResponseStatus</i>

Table 16-831 POST /ISAPI/System/Video/inputs/channels/<ID>/overlays/text

Method	POST
Description	Add a text to overlay on the video for a specific video input channel.
Query	None.
Request	<i>XML_TextOverlay</i>
Response	<i>XML_ResponseStatus</i>

Table 16-832 DELETE /ISAPI/System/Video/inputs/channels/<ID>/overlays/text

Method	DELETE
Description	Delete the text overlay parameters of a specific video input channel.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

16.11.279 /ISAPI/System/Video/inputs/channels/<ID>/overlays/text/<ID>

Operations about the configuration of a specific piece of text overlay information of a specific video input channel.

Request URI Definition

Table 16-833 GET /ISAPI/System/Video/inputs/channels/<ID>/overlays/text/<ID>

Method	GET
Description	Get the parameters of a specific piece of text overlay information of a specific video input channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_TextOverlay</i> Failed: <i>XML_ResponseStatus</i>

Table 16-834 PUT /ISAPI/System/Video/inputs/channels/<ID>/overlays/text/<ID>

Method	PUT
Description	Set the parameters of a specific piece of text overlay information of a specific video input channel.
Query	None.
Request	<i>XML_TextOverlay</i>
Response	<i>XML_ResponseStatus</i>

Table 16-835 DELETE /ISAPI/System/Video/inputs/channels/<ID>/overlays/text/<ID>

Method	DELETE
Description	Delete the parameters of a specific piece of text overlay information of a specific video input channel.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

16.11.280 /ISAPI/System/Video/inputs/channels/<ID>/privacyMask

Get or set privacy mask parameters.

Request URI Definition

Table 16-836 GET /ISAPI/System/Video/inputs/channels/<ID>/privacyMask

Method	GET
Description	Get or set privacy mask parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_PrivacyMask</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the video input channel ID.

16.11.281 /ISAPI/System/Video/inputs/channels/<ID>/privacyMask/privacyMaskCap

Get configuration capability of privacy mask.

Request URI Definition

Table 16-837 GET /ISAPI/System/Video/inputs/channels/<ID>/privacyMask/privacyMaskCap

Method	GET
Description	Get configuration capability of privacy mask.
Query	None.
Request	None.
Response	Succeeded: <i>XML_PrivacyMaskCap</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the video input channel ID.

16.11.282 /ISAPI/System/Video/inputs/channels/<ID>/privacyMask/regions

Get, set, or delete parameters of privacy mask regions, and add a region.

Request URI Definition

Table 16-838 GET /ISAPI/System/Video/inputs/channels/<ID>/privacyMask/regions

Method	GET
Description	Get parameters of privacy mask regions.
Query	None.
Request	None.
Response	Succeeded: <i>XML_PrivacyMaskRegionList</i> Failed: <i>XML_ResponseStatus</i>

Table 16-839 PUT /ISAPI/System/Video/inputs/channels/<ID>/privacyMask/regions

Method	PUT
Description	Set parameters of privacy mask regions.
Query	None.
Request	<i>XML_PrivacyMaskRegionList</i>
Response	<i>XML_ResponseStatus</i>

Table 16-840 DELETE /ISAPI/System/Video/inputs/channels/<ID>/privacyMask/regions

Method	DELETE
Description	Delete parameters of privacy mask regions.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

Table 16-841 POST /ISAPI/System/Video/inputs/channels/<ID>/privacyMask/regions

Method	POST
Description	Add a privacy mask region.
Query	None.
Request	<i>XML_PrivacyMaskRegion</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the video input channel ID.

16.11.283 /ISAPI/System/Video/inputs/channels/<ID>/privacyMask/regions/<ID>

Get, set, or delete parameters of a privacy mask region.

Request URI Definition**Table 16-842 GET /ISAPI/System/Video/inputs/channels/<ID>/privacyMask/regions/<ID>**

Method	GET
Description	Get parameters of a privacy mask region.
Query	None.
Request	None.
Response	Succeeded: <i>XML_PrivacyMaskRegion</i> Failed: <i>XML_ResponseStatus</i>

Table 16-843 PUT /ISAPI/System/Video/inputs/channels/<ID>/privacyMask/regions/<ID>

Method	PUT
Description	Set parameters of a privacy mask region.
Query	None.
Request	<i>XML_PrivacyMaskRegion</i>
Response	<i>XML_ResponseStatus</i>

Table 16-844 DELETE /ISAPI/System/Video/inputs/channels/<ID>/privacyMask/regions/<ID>

Method	DELETE
Description	Delete parameters of a privacy mask region.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The first <ID> in the request URI refers to the video input channel ID; and second <ID> is the privacy mask region ID.

16.11.284 /ISAPI/System/Video/inputs/channels/<ID>/VCAResource

Operation about configuration of intelligent resources switch.

Request URI Definition**Table 16-845 GET /ISAPI/System/Video/inputs/channels/<ID>/VCAResource**

Method	GET
Description	Get parameters of intelligent resources switch.
Query	None.
Request	None.
Response	Succeeded: <i>XML_VCAResource</i> Failed: <i>XML_ResponseStatus</i>

Table 16-846 PUT /ISAPI/System/Video/inputs/channels/<ID>/VCAResource

Method	PUT
Description	Set parameters of intelligent resources switch.
Query	None.
Request	<i>XML_VCAResource</i>
Response	<i>XML_ResponseStatus</i>

Remarks

- The <ID> in the request URI refers to the video input channel ID.
- The intelligent resources switch supports three capture mode, i.e., target capture, face capture, and perimeter capture.
 - For target capture mode, the panoramic camera captures the human body and vehicle pictures, the speed dome tracks and capture the human body for human body or face recognition and comparison. If matched, the thumbnails and original pictures of human body and face will be uploaded; otherwise, only the thumbnail and original picture of face will be uploaded.
 - For face capture mode, the panoramic camera transmits the videos without any intelligent analysis, the speed dome captures face picture in up to 4 scenes.
 - For perimeter capture, the panoramic camera captures vehicle pictures and performs HMS +DGS detection, and it supports detecting and tracking the small target within 40 m; while the speed dome tracks and captures the detected target, and outputs thumbnails and original pictures of human body and face.

16.11.285 /ISAPI/System/Video/inputs/channels/<ID>/VCAResource/capabilities

Get capability of intelligent resources switch.

Request URI Definition

Table 16-847 GET /ISAPI/System/Video/inputs/channels/<ID>/VCAResource/capabilities

Method	GET
Description	Get capability of intelligent resources switch.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_VCAResource</i> Failed: <i>XML_ResponseStatus</i>

16.11.286 /ISAPI/System/Video/inputs/channels/counting/collection/capabilities?format=json

Get the capability of people counting data replenishment.

Request URI Definition

Table 16-848 GET /ISAPI/System/Video/inputs/channels/counting/collection/capabilities?format=json

Method	GET
Description	Get the capability of people counting data replenishment.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_PeopleCounting_CollectionDescriptionCap</i>

16.11.287 /ISAPI/System/Video/inputs/channels/counting/collection?format=json

Perform people counting replenishment.

Request URI Definition

Table 16-849 POST /ISAPI/System/Video/inputs/channels/counting/collection?format=json

Method	POST
Description	Perform people counting replenishment.
Query	format: determine the format of request or response message.

Request	<i>JSON_PeopleCounting_CollectionDescription</i>
Response	<i>JSON_PeopleCounting_CollectionResult</i>

16.11.288 /ISAPI/System/Video/inputs/channels/heatMap/collection/capabilities?format=json

Get the capability of heat map data replenishment.

Request URI Definition

Table 16-850 GET /ISAPI/System/Video/inputs/channels/heatMap/collection/capabilities?format=json

Method	GET
Description	Get the capability of heat map data replenishment.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_HeatMap_CollectionDescriptionCap</i>

16.11.289 /ISAPI/System/Video/inputs/channels/heatMap/collection?format=json

Perform heat map data replenishment.

Request URI Definition

Table 16-851 POST /ISAPI/System/Video/inputs/channels/heatMap/collection?format=json

Method	POST
Description	Perform heat map data replenishment.
Query	format: determine the format of request or response message.
Request	<i>JSON_HeatMap_CollectionDescription</i>
Response	<i>JSON_HeatMap_CollectionResult</i>

16.11.290 /ISAPI/System/Video/inputs/OSDLanguage

Operations about OSD language configuration.

Request URI Definition

Table 16-852 GET /ISAPI/System/Video/inputs/OSDLanguag

Method	GET
Description	Get OSD language parameters.
Query	None.
Request	None.
Response	<i>XML_Language</i>

Table 16-853 PUT /ISAPI/System/Video/inputs/OSDLanguag

Method	PUT
Description	Set OSD language parameters.
Query	None.
Request	<i>XML_Language</i>
Response	<i>XML_ResponseStatus</i>

16.11.291 /ISAPI/System/Video/outputs

Get configuration parameters of video outputs.

Request URI Definition

Table 16-854 GET /ISAPI/System/Video/outputs

Method	GET
Description	Get configuration parameters of all video outputs.
Query	None.
Request	None.
Response	Succeeded: <i>XML_VideoOutput</i> Failed: <i>XML_ResponseStatus</i>

16.11.292 /ISAPI/System/Video/outputs/channels

Get configuration parameters of multiple video outputs.

Request URI Definition

Table 16-855 GET /ISAPI/System/Video/outputs/channels

Method	GET
Description	Get configuration parameters of multiple video outputs.
Query	None.
Request	None.
Response	Succeeded: <i>XML_VideoOutputChannelList</i> Failed: <i>XML_ResponseStatus</i>

16.11.293 /ISAPI/System/Video/outputs/channels/<ID>

Get or set parameters of a video output.

Request URI Definition

Table 16-856 GET /ISAPI/System/Video/outputs/channels/<ID>

Method	GET
Description	Get parameters of a video output.
Query	None.
Request	None.
Response	Succeeded: <i>XML_VideoOutputChannel</i> Failed: <i>XML_ResponseStatus</i>

Table 16-857 PUT /ISAPI/System/Video/outputs/channels/<ID>

Method	PUT
Description	Set parameters of a video output.
Query	None.
Request	<i>XML_VideoOutputChannel</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the video output ID.

16.11.294 /ISAPI/System/Video/outputs/channels/<ID>/capabilities

Get video output configuration capability.

Request URI Definition

Table 16-858 GET /ISAPI/System/Video/outputs/channels/<ID>/capabilities

Method	GET
Description	Get video output configuration capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_VideoOutputChannel</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the video output ID.

16.11.295 /ISAPI/System/Video/outputs/PreviewSwitch/capabilities

Get capability of auto-switch configuration during live view.

Request URI Definition

Table 16-859 GET /ISAPI/System/Video/outputs/PreviewSwitch/capabilities

Method	GET
Description	Get capability of auto-switch configuration during live view.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_PreviewSwitch</i> Failed: <i>XML_ResponseStatus</i>

16.11.296 /ISAPI/System/Video/outputs/PreviewSwitch/capabilities?videoOutType=

Get capability of auto-switch configuration during live view by video output.

Request URI Definition**Table 16-860 GET /ISAPI/System/Video/outputs/PreviewSwitch/capabilities?videoOutType=**

Method	GET
Description	Get capability of auto-switch configuration during live view by video output.
Query	videoOutType: video output type, string type, it can be the following values: CVBS, HDMI, VGA, AUXOutput1, AUXOutput2, AUXOutput3, AUXOutput4, extendHDMI1, extendHDMI2, extendHDMI3, extendHDMI4, noSameSourceVGA1, noSameSourceVGA2, noSameSourceVGA3, noSameSourceVGA4, sameSourceVGA, mainLCD, AUX-VGA, AUX-LCD, and zeroChannel. If there are more than two outputs are homologous, you can combine these outputs by "/", e.g., if HDM1 and VGA1 are homologous outputs, the value of videoOutType equals to HDM1/VGA1.
Request	None.
Response	Succeeded: <i>XML_PreviewSwitchVideoOutCap</i> Failed: <i>XML_ResponseStatus</i>

16.11.297 /ISAPI/System/Video/outputs/PreviewSwitch?groupID=&videoOutType=

Operations about auto-switch configuration in live view.

Request URI Definition**Table 16-861 GET /ISAPI/System/Video/outputs/PreviewSwitch?groupID=&videoOutType=**

Method	GET
Description	Get configuration parameters of auto-switch in live view.
Query	groupNo: group No., integer type, which is used to group the windows for configuration, and up to 64 windows can be added to a group. videoOutType: video output type, string type, it can be the following values: CVBS, HDMI, VGA, AUXOutput1, AUXOutput2, AUXOutput3, AUXOutput4, extendHDMI1, extendHDMI2, extendHDMI3, extendHDMI4, noSameSourceVGA1, noSameSourceVGA2, noSameSourceVGA3, noSameSourceVGA4, sameSourceVGA, mainLCD, AUX-VGA, AUX-LCD, and zeroChannel. If there are more

	than two outputs are homologous, you can combine these outputs by "/", e.g., if HDM1 and VGA1 are homologous outputs, the value of videoOutType equals to HDM1/VGA1.
Request	None.
Response	Succeeded: <i>XML_PreviewSwitch</i> Failed: <i>XML_ResponseStatus</i>

Table 16-862 PUT /ISAPI/System/Video/outputs/PreviewSwitch?groupID=&videoOutType=

Method	PUT
Description	Set parameters for auto-switch in live view.
Query	groupNo: group No., integer type, which is used to group the windows for configuration, and up to 64 windows can be added to a group. videoOutType: video output type, string type, it can be the following values: CVBS, HDMI, VGA, AUXOutput1, AUXOutput2, AUXOutput3, AUXOutput4, extendHDMI1, extendHDMI2, extendHDMI3, extendHDMI4, noSameSourceVGA1, noSameSourceVGA2, noSameSourceVGA3, noSameSourceVGA4, sameSourceVGA, mainLCD, AUX-VGA, AUX-LCD, and zeroChannel. If there are more than two outputs are homologous, you can combine these outputs by "/", e.g., if HDM1 and VGA1 are homologous outputs, the value of videoOutType equals to HDM1/VGA1.
Request	<i>XML_PreviewSwitch</i>
Response	<i>XML_ResponseStatus</i>

16.11.298 /ISAPI/System/Video/outputs/PreviewSwitch? groupID=&videoOutType=&previewFrameNo=

Get configuration parameters of auto-switch in live view.

Request URI Definition

Table 16-863 GET /ISAPI/System/Video/outputs/PreviewSwitch?
groupID=&videoOutType=&previewFrameNo=

Method	GET
Description	Get configuration parameters of auto-switch in live view.

Query	groupNo: group No., integer type, which is used to group the windows for configuration, and up to 64 windows can be added to a group. videoOutType: video output type, string type, it can be the following values: CVBS, HDMI, VGA, AUXOutput1, AUXOutput2, AUXOutput3, AUXOutput4, extendHDMI1, extendHDMI2, extendHDMI3, extendHDMI4, noSameSourceVGA1, noSameSourceVGA2, noSameSourceVGA3, noSameSourceVGA4, sameSourceVGA, mainLCD, AUX-VGA, AUX-LCD, and zeroChannel. If there are more than two outputs are homologous, you can combine these outputs by "/", e.g., if HDM1 and VGA1 are homologous outputs, the value of videoOutType equals to HDM1/VGA1. (Optional) previewFrameNo: number of live view windows, integer string type, and it can be the following values: 1, 4, 6, 8, 9, 25, 32, 36, auto1 (custom window division 1), auto2 (custom window division 2), auto3 (custom window division 3), and auto4 (custom window division 4).
Request	None.
Response	Succeeded: <i>XML_PreviewSwitch</i> Failed: <i>XML_ResponseStatus</i>

16.11.299 /ISAPI/System/wiresharkToUSB/capabilities?format=json

Get the configuration capability of exporting captured packets to USB flash drive.

Request URI Definition

Table 16-864 GET /ISAPI/System/wiresharkToUSB/capabilities?format=json

Method	GET
Description	Get the configuration capability of exporting captured packets to USB flash drive.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_Cap_WiresharkToUSB</i> Failed: <i>JSON_ResponseStatus</i>

16.11.300 /ISAPI/System/wiresharkToUSB/mode?format=json

Set parameters of exporting captured packets to USB flash drive.

Request URI Definition

Table 16-865 PUT /ISAPI/System/wiresharkToUSB/mode?format=json

Method	PUT
Description	Set parameters of exporting captured packets to USB flash drive.
Query	format: determine the format of request or response message.
Request	<i>JSON_WiresharkToUSB</i>
Response	<i>JSON_ResponseStatus</i>

16.11.301 /ISAPI/System/wiresharkToUSB/status?format=json

Get the status of exporting captured packets to USB flash drive.

Request URI Definition

Table 16-866 GET /ISAPI/System/wiresharkToUSB/status?format=json

Method	GET
Description	Get the status of exporting captured packets to USB flash drive.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_WiresharkToUSBStatus</i> Failed: <i>JSON_ResponseStatus</i>

16.11.302 /ISAPI/System/workingstatus/hdStatus?format=json

Get HDD status.

Request URI Definition

Table 16-867 GET /ISAPI/System/workingstatus/hdStatus?format=json

Method	GET
Description	Get all HDDs' status.
Query	format: determine the format of request or response message.

Request	None.
Response	Succeeded: <i>JSON_HDStatus</i> Failed: <i>JSON_ResponseStatus</i>

Table 16-868 POST /ISAPI/System/workingstatus/hdStatus?format=json

Method	POST
Description	Get a specified HDD's status.
Query	format: determine the format of request or response message.
Request	<i>JSON_HDCond</i>
Response	Succeeded: <i>JSON_HDStatus</i> Failed: <i>JSON_ResponseStatus</i>

16.11.303 /ISAPI/System/workingstatus/chanStatus?format=json

Get channel status.

Request URI Definition

Table 16-869 GET /ISAPI/System/workingstatus/chanStatus?format=json

Method	GET
Description	Get all channels' status.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_ChanStatus</i> Failed: <i>JSON_ResponseStatus</i>

Table 16-870 POST /ISAPI/System/workingstatus/chanStatus?format=json

Method	POST
Description	Get a specified channel's status.
Query	format: determine the format of request or response message.
Request	<i>JSON_ChanCond</i>
Response	Succeeded: <i>JSON_ChanStatus</i> Failed: <i>JSON_ResponseStatus</i>

16.11.304 /ISAPI/System/workingstatus/capabilities?format=json

Get capability of getting all statuses of device.

Request URI Definition

Table 16-871 GET /ISAPI/System/workingstatus/capabilities?format=json

Method	GET
Description	Get capability of getting all statuses of device.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_Cap_WorkingStatus</i> Failed: <i>JSON_ResponseStatus</i>

16.11.305 /ISAPI/System/workingstatus?format=json

Get device working status.

Request URI Definition

Table 16-872 GET /ISAPI/System/workingstatus?format=json

Method	GET
Description	Get device working status.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_WorkingStatus</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

Up to 20 bytes are allowed for a device's working status; up to 100 bytes are allowed for a channel's status, a HDD's status, and an IO's status; up to 30 bytes are allowed for a two-way audio channel's status.

16.11.306 /ISAPI/System/workingstatus/IOStatus?format=json

Get device IO status.

Request URI Definition

Table 16-873 GET /ISAPI/System/workingstatus/IOStatus?format=json

Method	GET
Description	Get device IO status.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_IOStatus</i> Failed: <i>JSON_ResponseStatus</i>

16.11.307 /ISAPI/System/zeroBiasCalibration/channels/<ID>?format=json

Calibrate the accelerometer bias.

Request URI Definition

Table 16-874 PUT /ISAPI/System/zeroBiasCalibration/channels/<ID>?format=json

Method	PUT
Description	Calibrate the accelerometer bias.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

16.12 /ISAPI/Thermal

16.12.1 /ISAPI/Thermal/capabilities

Get thermal capability.

Request URI Definition

Table 16-875 GET /ISAPI/Thermal/capabilities

Method	GET
Description	Get the thermal capability.
Query	None.
Request	None.
Response	<i>XML_ThermalCap</i>

16.12.2 /ISAPI/Thermal/temperature/collection/capabilities?format=json

Get the capability of temperature data replenishment.

Request URI Definition

Table 16-876 GET /ISAPI/Thermal/temperature/collection/capabilities?format=json

Method	GET
Description	Get the capability of temperature data replenishment.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_Temperature_CollectionDescriptionCap</i>

16.12.3 /ISAPI/Thermal/temperature/collection?format=json

Perform the temperature data replenishment.

Request URI Definition

Table 16-877 GET /ISAPI/Thermal/temperature/collection?format=json

Method	POST
Description	Perform the temperature data replenishment.
Query	format: determine the format of request or response message.
Request	<i>JSON_CollectionDescription</i>
Response	<i>JSON_CollectionResult</i>

Appendix A. Appendixes

A.1 Request and Response Message

The request and response messages in XML or JSON format of each request URL are listed here for reference. You can search for the parameters by the message name.

A.1.1 JSON Messages

JSON_Adaption

Adaption message in JSON format

```
{
  "Adaption": {
    "enable": ,
    /*required, boolean type, whether to enable network self-adaptive function: true-no, false-yes; this function is bound
    with IntraRefresh encoding strategy*/
    "policy": ,
    /*required,self-adaptive strategy, 0-self-adaptive (adjust the resolution and bit rate, the resolution value should be
    smaller than or equal to the configured value, but the frame rate is constant), 1-resolution in priority (adjust bit rate,
    resolution and frame rate are constant), 2-error correction and retransmission (resolution and frame rate are
    constant)*/
    "percentage":
    /*this node is required when policy is "2", error correction percentage, unit: %, which is between 0 and 100; lager
    percentage represents more corrected data, more higher requirement for bandwidth, and better effect*/
  }
}
```

JSON_AdaptionCap

AdaptionCap message in JSON format

```
{
  "AdaptionCap": {
    "enable": "true,false",
    /*required, boolean type, whether to enable network self-adaptive function: true-no, false-yes; this function is bound
    with IntraRefresh encoding strategy*/
    "policy": {
    /*required,self-adaptive strategy, 0-self-adaptive (adjust the resolution and bit rate, the resolution value should be
    smaller than or equal to the configured value, but the frame rate is constant), 1-resolution in priority (adjust bit rate,
    resolution and frame rate are constant), 2-error correction and retransmission (resolution and frame rate are
    constant)*/
    "@opt": "0,1,2"
    },
  },
}
```

```
"percentage": {  
  /*this node is required when policy is "2", error correction percentage, unit: %, which is between 0 and 100; lager  
  percentage represents more corrected data, more higher requirement for bandwidth, and better effect*/  
  "@min": 0,  
  "@max": 100,  
}  
}  
}
```

JSON_AlarmStatistics

AlarmStatistics message in JSON format

```
{  
  "AlarmStatistics":{  
    /*optional, alarm statistics of added IoT devices*/  
    "totalNumber": "",  
    /*required, the total number of alarms, integer*/  
    "time": "",  
    /*required, statistic time, integer, unit: second*/  
  }  
}
```

JSON_AlgorithmsInfo

AlgorithmsInfo message in JSON format

```
{  
  "AlgorithmsInfo":{  
    "type": "",  
    /*required, string, algorithm library type, e.g, "face"*/  
    "name": "",  
    /*required, string, algorithm library name which is used to distinguish different algorithms, the maximum size is 256  
    bytes*/  
    "versionInfo":{  
      /*required, algorithm library version information*/  
      "major": ,  
      /*required, integer, major version of the algorithm library*/  
      "minor": ,  
      /*required, integer, minor version of the algorithm library*/  
      "revision":  
      /*required, integer, revision version of the algorithm library*/  
    },  
    "platform": "",  
    /*required, string, platform where the algorithm library runs*/  
    "os": "",  
    /*required, string, operating system adaptive to the algorithm library*/  
    "accuracy": "",  
    /*optional, string, computational accuracy of the algorithm library*/  
  }  
}
```

```
"encryption":"","  
/*required, string, encryption method*/  
"buildTime":"","  
/*required, string, compiling date in the format "buildyyyymmdd"*/  
"versionProperties":"","  
/*required, string, version attribute: "alpha", "beta", "release", "custom"*/  
}}  
}
```

JSON_AlgorithmsInfoCap

AlgorithmsInfoCap capability message in JSON format

```
{  
  "AlgorithmsInfoCap":{  
    "type":{  
      /*required, string, algorithm library type, e.g., "face"*/  
      "@opt":["face"]  
    },  
    "maxSize": ,  
    /*required, integer, supported number of algorithm libraries*/  
    "name":{  
      /*required, string, algorithm library name which is used to distinguish different algorithms, the maximum size is 256 bytes*/  
      "@min": ,  
      "@max":  
    },  
    "versionInfo":{  
      /*required, algorithm library version information*/  
      "major":{  
        /*required, integer, major version of the algorithm library*/  
        "@min": ,  
        "@max":  
      },  
      "minor":{  
        /*required, integer, minor version of the algorithm library*/  
        "@min": ,  
        "@max":  
      },  
      "revision":{  
        /*required, integer, revision version of the algorithm library*/  
        "@min": ,  
        "@max":  
      }  
    },  
    "platform":{  
      /*required, string, platform where the algorithm library runs*/  
      "@min": ,  
      "@max":  
    },  
    "accuracy":{
```

```
/*optional, string, computational accuracy of the algorithm library*/
  "@min": ,
  "@max":
},
"os":{
/*required, string, operating system adaptive to the algorithm library*/
  "@min": ,
  "@max":
},
"encryption":{
/*required, string, encryption method*/
  "@min": ,
  "@max":
},
"buildTime":{
/*required, string, compiling date in the format "buildyyyymmdd"*/
  "@min": ,
  "@max":
},
"versionProperties":{
/*required, string, version attribute: "alpha", "beta", "release", "custom"*/
  "@min": ,
  "@max":
}
}
}
```

JSON_AssociatedChannelList

AssociatedChannelList message in JSON format

```
{
  "AssociatedChannelList": [{
/*optional, list of linked channels*/
    "channel": "",
/*required, channel No., integer*/
    "type": "",
/*required, channel type: "video"-video channel, "IoT"-IoT channel, string*/
    "associatedVideoChannels": "",
/*optional, linked video channel No., array, and the unit is integer*/
    "associatedIoTChannels": ""
  }]
}
```

JSON_AudioAlarm

AudioAlarm message in JSON format

```
{
  "AudioAlarm":{
    "audioID": ,
    /*required, alarm sound types, 1-siren, 2-"Warning, this is a restricted area", 3-"Waring, this is a restricted area, please keep away", 4-"Warning, this is a no-parking zone", 5-"Warning, this is a no-parking zone, please keep away", 6-"Attention please. The area is under surveillance", 7-"Welcome, Please notice that the area is under surveillance", 8-"Welcome", 9-"Danger! Please keep away", 10-siren + "Danger, please keep away", 11-Audio Warning, 12-Beep Sound, 13-custom alarm sound 1, 14-custom alarm sound 2, 15-custom alarm sound 3*/
    "audioVolume": ,
    /*required, int, volume, which is between 1 and 100*/
    "alarmTimes": ,
    /*required, int, times of alarm, which is between 1 and 50*/
    "TimeRangeList":[{
    /*optional, alarm output schedule list*/
      "week": ,
      "TimeRange":[{
        "id": ,
        /*required, int, ID of time period for each day*/
        "beginTime": "",
        /*required, string, start time in ISO8601 format*/
        "endTime": ""
        /*required, string, end time in ISO8601 format*/
      }]
    }]
    "audioClass": "",
    /*optional, string, read-only, sound type, "alertAudio"-alarm sound (get the supported alarm sound types from the node AlertAudioTypeListCap), "promptAudio"-prompt sound, "customAudio"-custom sound (configure it according to customAudioID); if this node is not returned, you should get the supported sound types from the node audioTypeListCap*/
    "alertAudioID": ,
    /*dependent, int, read-only, alarm sound ID, this node is valid only when audioClass is alertAudio, 1-"Siren", 2-"Warning, this is a restricted area", 3-"Waring, this is a restricted area, please keep away", 4-"Warning, this is a no-parking zone", 5-"Warning, this is a no-parking zone, please keep away", 6-"Attention please. The area is under surveillance", 7-"Welcome, Please notice that the area is under surveillance", 8-"Welcome", 9-"Danger! Please keep away", 10-"(Siren)&Danger, please keep away", 11-Audio Warning*/
    "customAudioID": ,
    /*dependent, int, read-only, custom sound ID, this node will be returned when audioClass is "customAudio". The alarm sound file can only be uploaded via Hik-Connect in specified format. If there is no value returned in this node, the custom sound (customAudio) will not be returned,either*/
    "alarmType": "",
    /*optional, string, alarm type: "behavior"-behavior analysis, "thermometry"-thermometry, "dynamicFire"-fire detection, "smokingMode"-smoke detection, "noMaskDetection"-no wearing mask detection*/
  }
}
```

Remarks

1. The user can configure the sound file via the node **audioID** only, and when the **audioID** is specified, its value will be automatically synchronized to the nodes of **audioClass**, **alertAudioID**, and **customAudioID**, which are read-only.
2. When the value of **audioID** is from 1 to 11, the corresponding value of **audioClass** is "alertAudio", and the value of **alertAudioID** is also from 1 to 11; when the value of **audioID** is 12, the corresponding value of **audioClass** is "promptAudio"; when the value of **audioID** is from 13 to 15, the corresponding value of **audioClass** is "customAudio", and the value of **customAudioID** is from 1 to 3.

JSON_AudioAlarmCap

JSON message about audible warning configuration capability

```
{
  "AudioAlarmCap":{
    "audioTypeListCap":[{
      /*required, alarm sound types, 1-siren, 2-"Warning, this is a restricted area", 3-"Waring, this is a restricted area,
      please keep away", 4-"Warning, this is a no-parking zone", 5-"Warning, this is a no-parking zone, please keep away",
      6-"Attention please. The area is under surveillance", 7-"Welcome, Please notice that the area is under surveillance",
      8-"Welcome", 9-"Danger! Please keep away", 10-siren + "Danger, please keep away", 11-Audio Warning, 12-Beep
      Sound, 13-custom alarm sound 1, 14-custom alarm sound 2, 15-custom alarm sound 3*/
      "audioID": 1,
      "audioDescription":"siren"
    },
    {
      "audioID": 2,
      "audioDescription":"Warning, this is a restricted area"
    },
    {
      "audioID": 3,
      "audioDescription":"Waring, this is a restricted area, please keep away"
    }
  ]},
  "audioVolume":{
    /*required, int, volume, which is between 1 ad 100*/
    "@min": 1,
    "@max": 100,
    "@def": 100
  },
  "alarmTimes":{
    /*required, int, times of audible warning, which is between 1 and 50*/
    "@min": 1,
    "@max": 50,
    "@def": 5
  },
  "TimeRangeCap":{
    /*optional, audible warning schedule capability*/
    "week":{
```



```

/*required, int, days of the week: 1-Monday, 2-Tuesday, 3-Wednesday, 4-Thursday, 5-Friday,6-Saturday, 7-Sunday*/
"@opt": "1, 2, 3, 4, 5, 6, 7"
},
"id": {
/*required, int, ID of time period for each day*/
"@maxSize": 8
},
"beginTime": {
/*required, string, start time in ISO8601 format*/
"@min": "00:00",
"@max": "24:00"
},
"endTime": {
/*required, string, end time in ISO8601 format*/
"@min": "00:00",
"@max": "24:00"
}
}
"audioClass": {
/*optional, string, read-only, sound type, "alertAudio"-alarm sound (get the supported alarm sound types from the
node AlertAudioTypeListCap), "promptAudio"-prompt sound, "customAudio"-custom sound (configure it according to
customAudioID); if this node is not returned, you should get the supported sound types from the node
audioTypeListCap*/
"@opt": "alertAudio, promptAudio, customAudio",
"@def": "alertAudio"
},
"AlertAudioTypeListCap": [ {
/*dependent, read-only, list of supported alarm sound types, this node will be returned when audioClass is
"alertAudio": 1-"Siren", 2-"Warning, this is a restricted area", 3-"Waring, this is a restricted area, please keep away",
4-"Warning, this is a no-parking zone", 5-"Warning, this is a no-parking zone, please keep away", 6-"Attention please.
The area is under surveillance", 7-"Welcome, Please notice that the area is under surveillance", 8-"Welcome",
9-"Danger! Please keep away", 10-"(Siren)&Danger, please keep away, 11-Audio Warning"*/
"alertAudioID": 1,
"alertAudioDescription": "Siren"
}, {
"alertAudioID": 2,
"alertAudioDescription": "Warning, this is a restricted area"
}, {
"alertAudioID": 3,
"alertAudioDescription": "Waring, this is a restricted area, please keep away"
}
],
"customAudioID": {
/*dependent, int, read-only, custom sound ID, this node will be returned when audioClass is "customAudio". The
alarm sound file can only be uploaded via Hik-Connect in specified format. If there is no value returned in this node,
the custom sound (customAudio) will not be returned,either*/
"@min": 1,
"@max": 3,
"@def": 1
},
"alarmType": {
/*optional, string, alarm type: "behavior"-behavior analysis, "thermometry"-thermometry, "dynamicFire"-fire

```

```
detection, "smokingMode"-smoke detection*/
  "@opt": "behavior,thermometry,dynamicFire,smokingMode",
  "@def": "thermometry"
}
"AlarmBindAudioList": [{
/*optional, alarm type and the list of audio related to alarm type*/
  "alarmType": "behavior",
/*optional, string, alarm type: "behavior"-behavior analysis, "thermometry"-temperature measurement,
"dynamicFire"-fire detection, "smokingMode"-smoke detection, "noMaskDetection"-no wearing mask detection*/
  "audioID": {
/*optional, string, the audible warning type ID corresponding to alarm type, behavior analysis corresponds to 1-12,
thermometry corresponds to 13, fire detection corresponds to 14, smoke detection corresponds to 15*/
    "@opt": "1,2,3,4,5,6,7,8,9,10,11,12"
  }
}]
"isSupportCustomAudio": true,
/*optional, boolean, whether the device supports uploading, downloading, and editing custom alarm sound file*/
"isSupportAudioTest": true,
/*optional, boolean, whether the device supports sound file test*/
"customAudioName": {
/*optional, string, name of custom sound file, the maximum size is 24 bytes*/
  "@min": 1,
  "@max": 24
}
}
```

Remarks

1. The user can configure the sound file via the node **audioID** only, and when the **audioID** is specified, its value will be automatically synchronized to the nodes of **audioClass**, **alertAudioID**, and **customAudioID**, which are read-only.
2. When the value of **audioID** is from 1 to 11, the corresponding value of **audioClass** is "alertAudio", and the value of **alertAudioID** is also from 1 to 11; when the value of **audioID** is 12, the corresponding value of **audioClass** is "promptAudio"; when the value of **audioID** is from 13 to 15, the corresponding value of **audioClass** is "customAudio", and the value of **customAudioID** is from 1 to 3.

JSON_Authentication

JSON message about parameters of serial authentication service

```
{
  "Authentication": {
    "enabled":,
/*required, boolean, whether to enable serial authentication service*/
    "serviceClosurePeriod":
/*dependent, integer, service life, unit: day; it is required when enabled is "false"*/
  }
}
```

JSON_AuthenticationCap

JSON message about serial authentication service

```
{
  "AuthenticationCap":{
    "enabled":{
      /*required, boolean, whether to enable serial authentication service*/
      "@opt":[
        true,
        false
      ]
    },
    "serviceClosurePeriod":{
      /*dependent, integer, service life, unit: day; it is required when enabled is "false"*/
      "@min":1,
      "@max":30,
      "@def":7
    }
  }
}
```

JSON_AutoEagleFocusing

AutoEagleFocusing message in JSON format

```
{
  "AutoEagleFocusing": {
    /*required, auto calibration of rapid focus*/
    "EagleFocusingRegion": [{
      /*optional, string, rapid focus region*/
      "sid": "",
      /*optional, string, scene ID*/
      "spotNum": ,
      /*optional, integer, the number of calibration points*/
      "type": "",
      /*optional, string, region type now only supports "line"*/
      "Region": [{
        /*optional, region list*/
        "x": ,
        "y":
      }],
      "AbsoluteHigh": {
        /*optional, PTZ position*/
        "elevation": ,
        /*optional, xs:integer, tilt angle, range: [-900,2700]*/
        "azimuth": ,
        /*optional, xs:integer, azimuth, range: [0,3600]*/
        "absoluteZoom": 33
      }
    }
  }
}
```

```
/*optional, xs:integer, zoom, range: [1,1000]*/
}
}}
}
}
```

JSON_AutoEagleFocusingCap

AutoEagleFocusingCap message in JSON format

```
{
  "AutoEagleFocusingCap": {
    /*required, auto calibration capability of rapid focus*/
    "EagleFocusingRegion": {
      "sid": {
        /*optional, string, scene ID*/
        "@min": 1,
        "@max": 2
      },
      "spotNum": {
        /*optional, integer, the number of calibration points*/
        "@min": 1,
        "@max": 64
      },
      "type": {
        /*optional, string, region type now only supports "line"*/
        "@opt": "line"
      },
      "RegionCap": {
        "minSize": 3,
        /*required, integer, the minimum number of region edges*/
        "maxSize": 10,
        /*required, integer, the maximum number of region edges*/
        "x": {
          /*required, float, X-coordinate, value range: 0.000 to 1*/
          "@min": 0.000,
          "@max": 1.000,
          "#text": 0.120
        },
        "y": {
          /*required, float, Y-coordinate, value range: 0.000 to 1*/
          "@min": 0.000,
          "@max": 1.000,
          "#text": 0.120
        }
      },
      "AbsoluteHighCap": {
        "elevation": {
          /*optional, xs:integer, tilt angle, range: [-900,2700]*/
          "@min": -900,
          "@max": 2700
        }
      }
    }
  }
}
```

```
    },
    "azimuth": {
/*optional, xs:integer, azimuth, range: [0,3600]*/
        "@min": 0,
        "@max": 3600
    },
    "absoluteZoom": {
/*optional, xs:integer, zoom, range: [1,1000]*/
        "@min": 1,
        "@max": 1000
    }
}
}
```

JSON_AutoGotoCfg

AutoGotoCfg message in JSON format

```
{
  "AutoGotoCfg":{
    "enable": ,
/*optional, boolean type, whether to enable*/
    "autoRecoveryTime":
/*optional, integer type, automatically restored time after the PTZ control is disabled, unit: second*/
  }
}
```

JSON_AutoGotoCfgCap

AutoGotoCfgCap message in JSON format

```
{
  "AutoGotoCfgCap":{
    "enable":"true,false",
/*optional, whether to enable*/
    "autoRecoveryTime":{
/*optional, automatically restored time after the PTZ control is disabled, unit: second*/
      "@min": ,
      "@max":
    }
  }
}
```

JSON_AutoMaintenance

AutoMaintenance message in JSON format

```
{
  "AutoMaintenance":{
    "enabled": true,
    /*required, boolean, enable or not*/
    "dayOfWeek": "",
    /*required, integer, day of the week, ranges from 1 to 7, 1-Monday, 2-Tuesday, and so on*/
    "rebootTime": "",
    /*required, reboot time, corrects to minute, it is 0 by default, ISO8601 format, string*/
  }
}
```

JSON_AutoMaintenanceCap

AutoMaintenanceCap message in JSON format

```
{
  "AutoMaintenanceCap":{
    "enabled": "true,false",
    /*required, boolean, enable or not*/
    "dayOfWeek":{
      /*required, integer, day of the week, ranges from 1 to 7, 1-Monday, 2-Tuesday, and so on*/
      "@min": 1,
      "@max": 7,
      "#text": 1
    },
  }
}
```

JSON_AutoSwitch

AutoSwitch message in JSON format

```
{
  "AutoSwitch":{
    "enabled":true,
    /*required, boolean, whether to enable power on or off*/
    "openTime":"10:00:00",
    /*required, string, power on time, which is accurate to minute with the time format of ISO8601*/
    "closeTime":"10:00:00"
  }
}
```

JSON_AutoSwitchCap

AutoSwitchCap message in JSON format

```
{
  "AutoSwitchCap":{
    "enabled":"true,false"
    /*required, boolean, whether to enable auto power on or off*/
  }
}
```

JSON_BasicParam

BasicParam message in JSON format

```
{
  "BasicParam":{
    /*required, basic parameters of added IoT device*/
    "channelName": ""
    /*required, channel name, string, the max. length is 32*/
  }
}
```

JSON_CalibrationStatus

CalibrationStatus message in JSON format

```
{
  "CalibrationStatus":{
    /*required, string type, installation angle calibration status: "Calibrated"-calibrated, "Uncalibrated"-uncalibrated*/
  }
}
```

JSON_Cap_CalibrationStatus

CalibrationStatus message in JSON format

```
{
  "CalibrationStatus":{
    /*required, string type, installation angle calibration status: "Calibrated"-calibrated, "Uncalibrated"-uncalibrated*/
    "@opt":"Calibrated,Uncalibrated"
  }
}
```

JSON_Cap_CertificateSelect

CertificateSelect capability message in JSON format

```
{
  "CertificateSelect":{
    "Function":{
      /*required, function list*/
      "functionName":"HTTPS",
      /*required, function name, it should be set to one of the following values:
      "HTTPS","WebSocket","SDK_OVER_TLS","SRTP","securityLog","ieee802.1x"*/
      "certificateType":["client", "server", "CA"]
      /*required, certificate type required by this function, it should be set to one or multiple values of the following
      options: "client","server","CA"*/
    },
    "clientCertificate":{
      /*optional, optional client certificate types*/
      "@opt":["name1"]
    },
    "serverCertificate":{
      /*optional, optional server certificate types*/
      "@opt":["name2"]
    },
    "CACertificate":{
      /*optional, optional CA certificate types*/
      "@opt":["name3"]
    },
    "SRTPSecurityAlgorithm":{
      /*optional, string, optional encryption algorithms*/
      "@opt":["AES128", "AES256"]
    }
  }
}
```

JSON_Cap_CleanUpFileList

Message about getting the capability of deleting saved files in a batch in JSON format

```
{
  "CleanUpFileList": {
    "@size": ,
    /*required, int, the maximum number of files that can be deleted in a single time*/
    "trackId":
    /*required, boolean, No. of supported files, read-only*/
  }
}
```


JSON_Cap_ClearingSpaceConfig

JSON message about configuration capability of HDD clearing

```
{
  "ClearingSpaceConfig":{
    /*when the free space of HDD reaches the configured value of insufficientSpace, and if the value of
clearingOriginalData is true, the configured HDD space size will be cleared*/
    "insufficientSpace":{
      /*required, int, HDD free space, which is used as the threshold to start clearing*/
      "@min":0,
      /*minimum value of the range*/
      "@max":100
      /*maximum value of the range*/
    },
    "clearingOriginalData":{
      /*required, boolean, whether to enable clearing the original data*/
      "opt":"true,false"
    },
    "clearingSpace":{
      /*required, int, space size to clear*/
      "@min":0,
      /*minimum value of the range*/
      "@max":100
      /*maximum value of the range*/
    }
  }
}
```

JSON_Cap_CommuMode

CommuMode capability message in JSON format

```
{
  "CommuMode":{
    "mode":{
      /*required, string type, security mode level of the private protocol: "compatibilityMode"-compatibility mode,
      "safeMode"-security mode*/
      "@opt":"compatibilityMode,safeMode"
    }
  }
}
```

JSON_Cap_ExportInfo

ExportInfo capability message in JSON format

```
{
  "ExportInfo":{
    "mediaType":{
      /*required, string, exported data type: "video", "audio", "metadata", "text", "mixed", "other". It should be the same
      as the search type. For pictures, this node should be set to "metadata"*/
      "@opt":["video", "audio", "metadata", "text", "mixed", "other"]
    },
    "playbackURLList":{
      /*required, array, file path*/
      "@size":1
    }
  }
}
```

JSON_Cap_IntelliManagement

IntelliManagement capability message in JSON format

```
{
  "requestURL": "",
  "statusCode": ,
  "statusString": "",
  "subStatusCode": "",
  "errorCode": ,
  "errorMsg": "",
  /*see the description of this node and the above nodes in the message of JSON_ResponseStatus*/
  "isSupportTask":true,
  /*optional, boolean type, whether it supports task: "true"-yes, "false"-no*/
  "isSupportCtrlCenter":true,
  /*optional, boolean type, whether it supports control center: "true"-yes, "false"-no*/
  "isSupportIntelligentSearch":true,
  /*optional, boolean type, whether it supports intelligent search: "true"-yes, "false"-no*/
  "isSupportExecuteControl":true,
  /*optional, boolean type, whether it supports arming: "true"-yes, "false"-no*/
  "isSupportDataStatistics":true,
  /*optional, boolean type, whether it supports statistics: "true"-yes, "false"-no*/
  "isSupportIntelligentStorage":true,
  /*optional, boolean type, whether it supports intelligent storage: "true"-yes, "false"-no*/
  "isSupportServer":true,
  /*optional, boolean type, whether it supports service configuration: "true"-yes, "false"-no*/
  "isSupportAlgorithmModel":true,
  /*optional, boolean type, whether it supports algorithm model configuration: "true"-yes, "false"-no*/
  "isSupportCluster":true,
  /*optional, boolean type, whether it supports cluster: "true"-yes, "false"-no*/
  "isSupportSearchResultDisplay":true,
  /*optional, boolean type, whether it supports display configuration: "true"-yes, "false"-no*/
  "isSupportTranscodeAnalysisUnit":true,
  /*optional, boolean type, whether it supports transcoding analysis unit: "true"-yes, "false"-no*/
  "isSupportFactoryReset":true,
```

```
/*optional, boolean type, whether it supports restoring to factory settings: "true"-yes, "false"-no*/
"isSupportUPS":true,
/*optional, boolean type, whether it supports UPS: "true"-yes, "false"-no*/
"isSupportCity":true,
/*optional, boolean type, whether it supports city: "true"-yes, "false"-no*/
"isSupportClusterEnableStatus":true,
/*optional, boolean type, whether it supports enabling cluster status: "true"-yes, "false"-no*/
"isSupportSensitivity":true,
/*optional, boolean type, whether it supports sensitivity configuration: "true"-yes, "false"-no*/
"isSupportConfidence":true,
/*optional, boolean type, whether it supports confidence configuration: "true"-yes, "false"-no*/
"isSupportEventSearch":true,
/*optional, boolean, whether it supports event search: "true"-yes, "false"-no*/
"isSupportAudioFileUpload":true,
/*optional, boolean type, whether it supports importing audio file: "true"-yes, "false"-no*/
"isSupportIntelligentMode":true,
/*optional, boolean type, whether it supports intelligent mode configuration: "true"-yes, "false"-no*/
"saveCapturedDataMaxDays":90,
/*optional, int*, the captured pictures can be saved in local HDD up to 90 days by default, when the number of days
reaches the limit, the records will be overwritten by day./
"isSupportDataResync":true,
/*optional, boolean, whether it supports data resynchronization: "true"-yes, "false"-no*/
"isSupportCertificateImport":true,
/*optional, boolean, whether it supports importing software license certificate: "true"-yes, "false"-no*/
"isSupportFingerprint":true,
/*optional, boolean, whether it supports exporting hardware fingerprint: "true"-yes, "false"-no*/
"isSupportAuthorization":true,
/*optional, boolean, whether it supports authorization configuration of the algorithm center: "true"-yes, "false"-no*/
"isSupportLicensingDongle":true,
/*optional, boolean, whether it supports dongle: "true"-yes, "false"-no*/
"isSupportAlgorithmUpdate":true,
/*optional, boolean, whether it supports updating algorithm: "true"-yes, "false"-no*/
"isSupportAnalysisResult":true,
/*optional, boolean, whether it supports searching for and counting the number of behavior analysis events: "true"-
yes, "false"-no*/
"isShowCenterComputing":true
/*optional, boolean, whether to display the configuration interface of algorithm resources: "true"-yes, "false"-no. The
configuration interface of algorithm resources will be displayed only when this field is "true" in the single smart
mode*/
}
```

See Also

JSON_ResponseStatus

JSON_Cap_PictureServerInformation

PictureServerInformation capability message in JSON format

```
{
  "PictureServerInformation":{
```

```
"pictureServerType":{
/*required, string type, picture storage server type: "tomact", "VRB", "cloudStorage"-cloud storage, "KMS"*/
"@opt": "tomact,VRB,cloudStorage,KMS",
"#text": ""
},
"addressingFormatType":{
/*required, string type, format type of the picture storage server address: "ipaddress"-IP address (default),
"hostname"-host name*/
"@opt": "ipaddress,hostname",
"#text": ""
},
"hostName":{
/*string type, domain name of the picture storage server, the string length is between 0 and 64. This field is valid
when addressingFormatType is "hostname"*/
"@min": 0,
"@max": 64,
"#text": ""
},
"ipv4Address":{
/*string type, IPv4 address of the picture storage server, the string length is between 0 and 64. This field is valid when
addressingFormatType is "ipaddress"*/
"@min": 0,
"@max": 64,
"#text": ""
},
"ipv6Address":{
/*string type, IPv6 address of the picture storage server, the string length is between 0 and 128. This field is valid
when addressingFormatType is "ipaddress"*/
"@min": 0,
"@max": 128,
"#text": ""
},
"portNo":{
/*required, integer type, port No. of the picture storage server, which is between 1024 and 65535*/
"@min": 1024,
"@max": 65535,
"#text": ""
},
"underlyingProtocol":{
/*optional, string, bottom-level protocol of the picture storage server: "HTTP", "HTTPS". This field is valid when
pictureServerType contains "cloudStorage"*/
"@opt": ["http","https"]
},
"cloudStorage":{
/*parameters of the cloud storage server, which is valid when pictureServerType is "cloudStorage"*/
"cloudManageHttpPort":{
/*required, integer type, HTTP port No. for central management of the cloud storage server, which is between 1024
and 65535*/
"@min": 1024,
"@max": 65535,
"#text": ""
},
},
}
```

```
"cloudTransDataPort":{
/*required, integer type, data transmission port No. of the cloud storage server, which is between 1024 and 65535*/
"@min":1024,
"@max":65535,
"#text":
},
"cloudCmdPort":{
/*required, integer type, signaling port No. of the cloud storage server, which is between 1024 and 65535*/
"@min":1024,
"@max":65535,
"#text":
},
"cloudHeartBeatPort":{
/*required, integer type, heartbeat port No. of the cloud storage server, which is between 1024 and 65535*/
"@min":1024,
"@max":65535,
"#text":
},
"cloudStorageHttpPort":{
/*required, integer type, HTTP port No. of the cloud storage server, which is between 1024 and 65535*/
"@min":1024,
"@max":65535,
"#text":
},
"cloudUsername":{
/*required, string type, user name of the cloud storage server, the string length is between 0 and 32*/
"@min":0,
"@max":32,
"#text":""
},
"cloudPassword":{
/*required, string type, password of the cloud storage server, the string length is between 0 and 32*/
"@min":0,
"@max":32,
"#text":""
},
"cloudPoolId":{
/*required, integer type, cloud storage pool ID, which is between 1 and 4294967295. If this field is not configured by
the upper-level, this field will be set to 1 by default*/
"@min":1,
"@max":4294967295,
"#text":
},
"cloudPoolIdEx":{
/*optional, string, cloud storage pool ID, this node is valid when cloud storage pool ID of type string is supported*/
"@min":0,
"@max":0,
"#text":""
},
"cloudProtocolVersion":{
/*required, string type, protocol version of the cloud storage server, the string length is between 0 and 32*/
"@min":0,
```

```
"@max":32,
"#text":""
},
"cloudAccessKey":{
/*string type, cloud storage server access_key, the string length is between 0 and 64. This field is valid when
cloudProtocolVersion is "V2.0"*/
"@min":0,
"@max":64
},
"cloudSecretKey":{
/*string type, cloud storage server secret_key, the string length is between 0 and 64. This field is valid when
cloudProtocolVersion is "V2.0"*/
"@min":0,
"@max":64
}
}
}
```

JSON_Cap_POE

POE capability message in JSON format

```
{
  "POE": {
    "ipVersion": {
/*optional, supported IP address type, string type*/
"@opt": "IPv4,IPv6",
"#text": "IPv4"
    }
  }
}
```

JSON_Cap_RemarkList

JSON message about capability of adding remarks for videos by video ID

```
{
  "RemarkList": {
    "@size": 100,
/*maximum number of remarks can be added for once, the default value is 100*/
    "trackId": ,
/*whether it supports file No.*/
    "remarks": ,
/*whether it supports adding remarks for file*/
    "recordName":{
/*optional, string, file name of video, audio, or picture*/
"@min":1,
"@max":100
    }
  }
}
```

```
,
"recorderCode":{
/*optional, string, ID of device that collects data, which should be encrypted*/
"@min":1,
"@max":100
},
"policeCode":{
/*optional, string, ID of police that collects the data, which should be encrypted*/
"@min":1,
"@max":100
}
}
}
```

JSON_Cap_ReportCenterCfg

ReportCenterCfg capability message in JSON format

```
{
"ReportCenterCfg":{
"CenterID":{
/*center group No.*/
"@min": ,
"@max":
},
"enable":"true,false",
/*optional, boolean type, whether to enable uploading report*/
"ChanAlarmMode":{
/*alarm channel of the center group*/
"maxSize":1,
"id":{
/*optional, integer type, channel ID: 1-main channel, 2-backup channel 1, 3-backup channel 2, 4-backup channel 3*/
"@min":1,
"@max":2
},
"chanAlarmMode":{
/*optional, string type, alarm channel mode: "T1"-T1 channel, "T2"-T2 channel, "N1"-N1 channel, "N2"-N2 channel,
"G1"-G1 channel, "G2"-G2 channel, "N3"-N3 channel, "N4"-N4 channel*/
"@opt":"T1,T2,N1,N2,G1,G2,N3,N4"
}
}
}
}
```

JSON_Cap_TimeSearchCond

Capability message about searching for recording start and end time in JSON format

```
{
  "TimeSearchCond": {
    "@size": 1024,
    /*the maximum number of search conditions that can be set*/
    "channelID": {
      /*required, int, channel No.*/
      "@min": 1,
      "@max": 256
      /*value range of the channel No.*/
    },
    "streamType": {
      /*optional, string, stream type, "main,sub,adaptive"*/
      "@opt": [
        "main",
        "sub",
        "adaptive"
      ]
    }
  },
  "TimeSearchResult": {
    "@size": 1024,
    /*the maximum number of results that can be returned*/
    "channelID": {
      /*required, int, channel No.*/
      "@min": 1,
      "@max": 256
      /*value range of the channel No.*/
    },
    "streamType": {
      /*optional, string, stream type, "main,sub,adaptive"*/
      "@opt": [
        "main",
        "sub",
        "adaptive"
      ]
    },
    "hasRecord": {
      /*required, boolean, whether the video exists, startTime and endTime are valid only when the value of hasRecord is "true"*/
      "@opt": [
        true,
        false
      ]
    },
    "startTime": {
      /*optional, string, recording start time, e.g., 2010-01-01T00:00:00+08:00, this field is required when the value of hasRecord is "true"*/
      "@min": 0,
      "@max": 64
    },
    "endTime": {
      /*optional, string, recording end time, e.g., 2010-01-01T00:00:00+08:00, this field is required when the value of
```



```
hasRecord is "true"*/
  "@min": 0,
  "@max": 64
}
}
}
```

JSON_Cap_WiresharkToUSB

JSON message about configuration capability of exporting captured packets to USB flash drive

```
{
  "WiresharkToUSB": {
    "switch": "",
    "operateCmd": "",
    "DeviceCondList": [
      {
        "DeviceCond": {
          "id": ,
          "devType": "",
          "devID": ,
          "subSys": ,
        }
      }
    ]
  }
}
```

JSON_Cap_WorkingStatus

WorkingStatus capability message in JSON format

```
{
  "WorkingStatusCap":{
    "devStatus":{
      "@opt": "0,1,2"
      /*required, integer type, value of device status: 0-normal, 1-CPU usage, higher than 85%, 2-hardware error (e.g., serial
      port exception)*/
    },
    "ChanStatus":{
      "chanNo": "",
      /*required, integer type, channel No., which starts from 1*/
      "enable": "",
      /*integer type, it is valid for analog channel only, 0-disable, 1-enable*/
      "online": "",
      /*required, integer type, online status: 0-offline, 1-online*/
      "record":{
        "@opt": "0,1,2",
        /*integer type, recording status: 0-recording, 1-recording exception (HDD exception), 2-recording exception (network
```

```
camera offline), 3-recording exception (other reason)*/
    },
    "signal": "",
    /*required, integer type, signal status: 0-normal, 1-signal loss*/
    "linkNum": "",
    /*required, integer type, number of software clients connected to this channel*/
    "bitRate": ""
    /*required, integer type, channel bit rate, unit: Kpbs*/
    },
    "HDStatus":{
        "hdNo": "",
        /*required, integer type, HDD No., which starts from 1*/
        "enable": "",
        "status":{
            "@opt": "0,1,2",
            /*required, integer type, HDD status: 0-activate, 1-sleep, 2-exception, 3-sleepy HDD error, 4-unformatted, 5-
            disconnected (for network HDD), 6-formatting*/
            },
            "volume": "",
            /*required, integer type, HDD capacity, unit: MB*/
            "linkNum": "",
            /*integer type, *number of connections*/
            "freeSpace": "1"
        }
        /*required, integer type, free space, unit: MD*/
    }
    }
}
```

JSON_CertificateRevocation

CertificateRevocation message in JSON format

```
{
  "CertificateRevocation":{
    "enabled": ,
    /*required, boolean, whether to enable certificate expiry alarm*/
    "expireAlarmTime": ,
    /*required, integer, number of days to prompt in advance before the certificate is expired*/
    "detecteTime": "",
    /*required, time, detection time of certificate expiry alarm*/
    "intervalTime":
    /*required, integer, alarm interval, unit: day*/
  }
}
```

JSON_CertificateRevocationCap

CertificateRevocationCap capability message in JSON format

```
{
  "CertificateRevocationCap":{
    "enabled":{
      /*required, boolean, whether to enable certificate expiry alarm*/
      "@opt":[true, false]
    },
    "expireAlarmTime":{
      /*required, integer, number of days to prompt in advance before the certificate is expired*/
      "@min":1,
      "@max":30,
      "@def":7
    },
    "detecteTime":{
      /*required, time, detection time of certificate expiry alarm*/
      "@min":"00:00",
      "@max":"24:00",
      "@def":"10:00"
    },
    "intervalTime":{
      /*required, integer, alarm interval, unit: day*/
      "@min":1,
      "@max":30,
      "@def":1
    }
  }
}
```

JSON_CertificateSelect

CertificateSelect message in JSON format

```
{
  "CertificateSelect":{
    /*you can choose one field among clientCertificate, serverCertificate, and CACertificate to configure as needed*/
    "clientCertificate": "",
    /*optional, string, client certificate type*/
    "serverCertificate": "",
    /*optional, string, server certificate type*/
    "CACertificate": "",
    /*optional, string, CA certificate type*/
    "SRTPSecurityAlgorithm": ""
    /*optional, string, encryption algorithm*/
  }
}
```

JSON_ChanCond

ChanCond message in JSON format

```
{
  "ChanCond":{
    "chanNo": [...,...]
  /*specify channel No. to get the status*/
  }
}
```

JSON_ChanStatus

ChanStatus message in JSON format

```
{
  "ChanStatus":[{
    "chanNo": ,
    /*required, integer type, channel No., which starts from 1*/
    "enable": ,
    /*integer type, it is valid for analog channel only, 0-disable, 1-enable*/
    "online": ,
    /*required, integer type, online status: 0-offline, 1-online*/
    "record": ,
    /*required, whether the device is recording, 0-no, 1-yes*/
    "recordStatus": ,
    /*integer type, recording status: 0-recording, 1-recording exception (HDD exception), 2-recording exception (network camera offline), 3-recording exception (other reason)*/
    "signal": ,
    /*required, integer type, signal status: 0-normal, 1-signal loss*/
    "linkNum": ,
    /*required, integer type, number of software clients connected to this channel*/
    "bitRate":
    /*required, integer type, channel bit rate, unit: Kpbs*/
  },
  {
    "chanNo": ,
    /*required, integer type, channel No., which starts from 1*/
    "enable": ,
    /*integer type, it is valid for analog channel only, 0-disable, 1-enable*/
    "online": ,
    /*required, integer type, online status: 0-offline, 1-online*/
    "record": ,
    /*required, whether the device is recording, 0-no, 1-yes*/
    "recordStatus": ,
    /*integer type, recording status: 0-recording, 1-recording exception (HDD exception), 2-recording exception (network camera offline), 3-recording exception (other reason)*/
    "signal": ,
    /*required, integer type, signal status: 0-normal, 1-signal loss*/
    "linkNum": ,
    /*required, integer type, number of software clients connected to this channel*/
    "bitRate":
    /*required, integer type, channel bit rate, unit: Kpbs*/
  }
}]
```

```
}}  
}
```

JSON_ChannelInfoList

ChannelInfoList message in JSON format

```
{  
  "ChannelInfoList":{  
    "ChannelInfo":[{  
      "chanNo": ,  
      /*required, integer type, channel No.*/  
      "enable": ,  
      /*integer type, it is valid for analog channel only, 0-disable, 1-enable*/  
      "name": "",  
      /*string type, channel name*/  
      "online": ,  
      /*integer type, online status: 0-offline, 1-online*/  
      "linknum":  
    }]  
  }  
}
```

JSON_ChannelInfo

ChannelInfo message in JSON format

```
{  
  "ChannelInfo": [{  
    "chanNo": ,  
    /*required, integer type, channel No.*/  
    "enable": ,  
    /*integer type, it is valid for analog channel only, 0-disable, 1-enable*/  
    "name": "",  
    /*string type, channel name*/  
    "online": ,  
    /*integer type, online status: 0-offline, 1-online*/  
    "linknum": ,  
    /*number of live view channels*/  
  }]  
}
```

JSON_ChangedStatus

ChangedStatus message in JSON format

```
{
  ChangedStatus: {
    "seq": ,
    /*32-byte integer type, alarm No., for alarm acknowldage*/
    "devStatus": ,
    /*integer type, value of device status: 0-normal, 1-CPU usage, higher than 85%, 2-hardware error (e.g., serial port
    exception)*/
    "devCapHash": "",
    /*global Hash value of device capability, which is generated by MD5, and encrypted by Base64; for NVR, the Hash
    value of network camera should be the calculation source*/
    "zeroChanEnable": ,
    /*boolean type, whether the channel-zero is enabled: true-yes, false-no*/
    "cycleRecordEnable": true,
    /*boolean type, whether to enable recoridng, true-yes, false-no*/
    "chanChanged": [...,...,...]
    /*integer type, channel status changes, e.g., delete, edit, add, the Hash value of this channel changed*/
    "hdChanged": [...,...,...]
    /*integer type, HDD status changes, e.g., delete, edit, add*/
    "ChanStatus": [{
      "chanNo": ,
      /*integer type, channel No., read-only*/
      "enable": 1,
      /*integer type, it is valid for analog channel only, 0-disable, 1-enable*/
      "online": ,
      /*required, integer type, online status: 0-offline, 1-online*/
      "recordStatus": ,
      /*integer type, recording status: 0-recording, 1-recording exception (HDD exception), 2-recording exception (network
      camera offline), 3-recording exception (other reason)*/
      "signal": ,
      /*required, integer type, signal status: 0-normal, 1-signal loss*/
      "arming": ,
      /*integer type, network camera arming status: 0-armed, 1-arming failed*/
    }],
    "HDStatus":[{
      "hdNo": ,
      /*required, integer type, HDD No., which starts from 1*/
      "status": ,
      /*required, integer type, HDD status: 0-activate, 1-sleep, 2-exception, 3-sleepy HDD error, 4-unformatted, 5-
      disconnected (for network HDD), 6-formatting*/
    }],
    "CapChanged": [{
      "name": "",
      /*required, root node of changed capability set*/
      "chan": ,
      /*channel No., it is required when transmitting channel capability; it is optional when transmitting device capability*/
      "url": "/ISAPI/Thermal/channels/1/fireDetection/capabilities",
      /*capability set URL, it is required for capability set based on ISAPI protocol*/
    }],
    {
      "name": "",
      /*required, capability set name, which is usually the root node*/
      "chan": 1,
```

```
/*channel No., it is required when transmitting channel capability; it is optional when transmitting device capability*/
  "abilityType": 0x700,
/*capability type, it is required if the capability is not based on ISAPI protocol*/
  }
}
```

JSON_CleanUpFileList

Message about deleting saved files in a batch in JSON format

```
{
  "CleanUpFileList": [{
    "trackId": ""
/*required, string, file No.*/
  }
]
```

JSON_ClearingSpaceConfig

JSON message about HDD clearing parameters

```
{
  "ClearingSpaceConfig":{
/*when the free space of HDD reaches the configured value of insufficientSpace, and if the value of clearingOriginalData is true, the configured HDD space size will be cleared*/
    "insufficientSpace": ,
/*required, int, HDD free space, which is used as the threshold to start clearing*/
    "clearingOriginalData": ,
/*required, boolean, whether to enable clearing the original data*/
    "clearingSpace":
/*required, int, space size to clear*/
  }
}
```

JSON_CommMode

CommMode message in JSON format

```
{
  "CommMode":{
    "mode":""
/*required, string type, security mode level of the private protocol: "compatibilityMode"-compatibility mode,
"safeMode"-security mode*/
  }
```

```
}  
}
```

JSON_CustomAudioID

JSON message about custom alarm sound ID

```
{  
  "customAudioID":  
  /*required, string, custom alarm sound ID, the value can be 1 to 3*/  
}
```

JSON_CustomAudioInfo

JSON message about custom alarm sound file information

```
{  
  "CustomAudioInfo": {  
    /*required, related information about custom alarm sound file*/  
    "customAudioID": ,  
    /*required, int, custom alarm sound file ID, the value can be 1 to 3*/  
    "customAudioName": ""  
    /*required, string, name of custom sound file, the maximum size is 24 bytes*/  
  }  
}
```

JSON_DeviceCertificate

DeviceCertificate message in JSON format

```
{  
  "DeviceCertificate":{  
    "CertificateInfo":{  
      "issuerDN": "",  
      /*required, string, read-only, name of the authority that issued the certificate*/  
      "subjectDN": "",  
      /*required, string, read-only, certificate holder name*/  
      "startDate": "",  
      /*required, date, read-only, start date of the certificate validity period, it is accurate to day*/  
      "endDate": "",  
      /*required, date, read-only, end date of the certificate validity period, it is accurate to day*/  
      "type": "",  
      /*optional, string, read-only, certificate type: "https", "websockets", "SDKoverTLS", "SRTP", "securityLog",  
      "ieee802.1x". Multiple types should be separated by commas*/  
      "status": "",  
      /*optional, string, read-only, certificate status: "normal", "expired", "exceptional"*/  
      "customID": ""  
    }  
  }  
}
```



```
/*optional, string, read-only, custom certificate ID, it is created when the user imports the certificate and it consists of
digits and characters*/
}
}
}
```

JSON_DeviceCertificateCap

DeviceCertificateCap capability message in JSON format

```
{
  "DeviceCertificateCap":{
    "customID":{
      /*required, string, custom certificate ID, it is created when the user imports the certificate and it consists of digits and
characters*/
      "@min":1,
      "@max":64,
      "status":{
        /*optional, string, certificate status: "normal", "expired", "exceptional"*/
        "@opt":["normal", "expired", "exceptional"]
      }
    }
  }
}
```

JSON_DeviceCertificates

DeviceCertificates message in JSON format

```
{
  "DeviceCertificates":{
    "CertificateInfoList":{
      "CertificateInfo": [{
        "id": ,
        /*required, integer32, read-only, certificate ID, which starts from 1*/
        "issuerDN": "",
        /*required, string, read-only, name of the authority that issued the certificate*/
        "subjectDN": "",
        /*required, string, read-only, certificate holder name*/
        "startDate": "",
        /*required, read-only, start date of certificate expiry period, it is accurate to day*/
        "endDate": "",
        /*required, read-only, end date of certificate expiry period, it is accurate to day*/
        "type": "",
        /*optional, read-only, certificate type: "wpa", "securityLog", "ieee802.1x", "HTTPS", "WebSocketS", "SDK_OVER_TLS",
"SRTP". Multiple types can be separated by commas*/
        "status": "",
        /*optional, string, read-only, certificate status: "normal", "expired", "exceptional"*/
        "customID": ""
      }
    ]
  }
}
```

```
/*optional, string, read-only, custom certificate ID, it is created when the user imports the certificate and it consists of
digits and characters*/
    }}
  }
}
}
```

JSON_DeviceCertificatesCap

DeviceCertificatesCap message in JSON format

```
{
  "DeviceCertificatesCap":{
    "CertificateInfoList":{
      "@size": 64,
      /*required, integer32 type, read-only*/
      "CertificateInfo": [{
        "id": {
          /*required, integer32 type, read-only, certificate ID, which starts from 1*/
          "@min":1,
          "@max":64,
          "#text":
          }
          "issuerDN": "",
          /*required, string type, read-only, certificate issue*/
          "subjectDN": "",
          /*required, string type, read-only, certificate holder name*/
          "startDate": "",
          /*required, read-only, start date of certificate expiry period, it accurate to day*/
          "endDate": ""
          /*required, read-only, end date of certificate expiry period, it accurate to day*/
          "type": {
            /*optional, read-only, certificate type*/
            "@opt": "wpa, securityLog, ieee802.1x"
          }
        }
      ]
    }
  }
}
```

JSON_DeviceCertificateStatus

DeviceCertificateStatus message in JSON format

```
{
  "DeviceCertificateStatus": {
    /*required, status of device certificate*/
    "id": "",
    /*required, read-only, string type, certificate ID, which is generated and returned by device*/
  }
}
```

```
"status": ""
/*required, string type, certificate status: normal, abnormal*/
}
}
```

JSON_DeviceCertificateStatusList

DeviceCertificateStatusList message in JSON format

```
{
  "DeviceCertificateStatusList": [{
    "DeviceCertificateStatus": {
/*required, status of device certificate*/
      "id": "",
/*required, read-only, string type, certificate ID, which is generated and returned by device*/
      "status": ""
/*required, string type, certificate status: normal, abnormal*/
    }
  }]
}
```

JSON_DiagnosisCond

DiagnosisCond message in JSON format

```
{
  "DiagnosisCond":{
    "command": "",
/*required, string type, diagnosis command*/
    "selfdefineCmd": ""
/*string type, custom diagnosis command, it is valid when command is set to "selfdefine"*/
  }
}
```

JSON_DiagnosisCondCap

DiagnosisCondCap capability message in JSON format

```
{
  "DiagnosisCondCap":{
/*diagnosis condition capability*/
    "command":{
/*required, supported diagnosis command*/
      "@opt":[
        "system",
/*diagnose system*/
        "alarm",
```

```
/*diagnose alarm*/
  "peripheral",
/*diagnose peripheral*/
  "wifi",
/*diagnose Wi-Fi network*/
  "cloud",
/*diagnose cloud*/
  "GPRS/3G/4G",
/*diagnose mobile network*/
  "IPC",
/*diagnose network camera*/
  "ARC",
/*diagnose alarm center*/
  "selfdefine"
/*custom diagnosis command*/
  ]
},
  "selfdefineCmd":{
/*length range of the custom diagnosis command, it is valid when "selfdefine" is included in command*/
  "@min": ,
/*required, integer type, the minimum length of the custom diagnosis command*/
  "@max":
/*required, integer type, the maximum length of the custom diagnosis command*/
  }
}
}
```

JSON_DiagnosisResult

DiagnosisResult message in JSON format

```
{
  "DiagnosisResult":{
    "result":""
/*required, string type, diagnosis result*/
  }
}
```

JSON_DoubleVerification

DoubleVerification message in JSON format

```
{
  "DoubleVerification":{
    "enable":""
/*optional, whether enables double verification, boolean*/
  }
}
```

JSON_DownloadPackageStatus

DownloadPackageStatus message in JSON format

```
{
  "DownloadPackageStatus":{
    /*required, upgrade package download status*/
    "status": "notDownload,downloading,pause,finish,incorrectPackage,hdOperationFailed",
    /*required, string, download status: "notDownload"-not download, "downloading"-downloading, "pause"-paused,
    "finish"-completed, "incorrectPackage"-incorrect upgrade package format, "hdOperationFailed"-HD operation failed*/
    "total": "",
    /*optional, float, upgrade package total size, unit: MB, corrects to one decimal place*/
    "remain": "",
    /*optional, float, remaining space, unit: MB, corrects to one decimal place*/
    "speed": ""
    /*optional, float, download speed, unit: KB/s, corrects to two decimal places*/
    "remainTime": "",
    /*optional, integer, estimated remaining time, unit: s*/
    "progress": 0,
    /*required, integer, progress, ranges from 0 to 100*/
  }
}
```

JSON_DynamicHostName

DynamicHostName message in JSON format

```
{
  "DynamicHostName": {
    /*required, configuration parameters of dynamic domain name*/
    "enabled": ,
    /*required, boolean type, whether to enable dynamic domain name*/
    "hostName": ""
    /*required, string type, domain name*/
  }
}
```

JSON_DynamicHostNameCap

DynamicHostNameCap message in JSON format

```
{
  "DynamicHostNameCap": {
    /*required, configuration capability of dynamic domain name*/
    "enabled": "true,false",
    /*required, boolean type, whether to enable dynamic domain name*/
    "hostName": {
      /*required, string type, domain name*/
    }
  }
}
```

```
"@min": 0,  
"@max": 64  
}  
}  
}
```

JSON_EmailCertification

Enter a short description of your reference here (optional).

Enter the syntax information of your reference here (optional).

Enter the actual information in this section (optional).

Example

Enter an example to illustrate your reference here (optional).

JSON_EnableEncryption

EnableEncryption message in JSON format

```
{  
  "enable": true,  
  /*required, boolean type, whether to enable stream encryption, true=yes, false=no*/  
}
```

JSON_EncryptFormat

EncryptFormat message in JSON format

```
{  
  "EncryptFormat":{  
    "password": "",  
    /*required, string, password, it will be encrypted according to the encryption vector*/  
    "encryptFormatType": ""  
    /*optional, string, HDD formatting type: "FAT32", "EXT4"*/  
  }  
}
```

JSON_EncryptionCap

Stream encryption capability in JSON format.

```
{  
  "enable": "true,false",  
  /*required, boolean type, whether to enable stream encryption, true=yes, false=no*/  
  "secretKey":{
```

```
/*required, string type, encryption key*/
"@min": 6,
"@max": 12,
"#text": "user"
}
}
```

JSON_EncryptVerify

EncryptVerify message in JSON format

```
{
  "EncryptVerify":{
    "password":""
  /*required, string, password, it will be encrypted according to the encryption vector*/
  }
}
```

JSON_EPTZMode

EPTZMode message in JSON format

```
{
  "EPTZMode":{
    "mode": "",
    /*required, modes supported by e-PTZ: "cruise"-patrol (PTZ supports preset, patrol, zoom, and eight directions),
    "autoTrack"-automatic tracking*/
    "AutoTrack":{
      /*dependency, auto-tracking mode configuration*/
      "sensitivity": ,
      /*optional, integer, sensitivity*/
      "detectionTarget":["human", "vehicle"],
      /*detection target*/
      "ReginonList":[{
        "id": ,
        /*required, rule ID*/
        "Region":{
          /*required, rule area, the number of edges is between 3 and 10*/
          "x": ,
          /*required, float, X-coordinate, range: [0.000,1]*/
          "y":
          /*required, float, Y-coordinate, range: [0.000,1]*/
        }
      }],
      "FilterSize":{
        /*optional, size filter of rectangle*/
        "MaxTargetRect":{
          "height": ,
          /*required, float, height, range: [0.000,1.000], the value is accurate to three decimal places*/

```

```
    "width": ,
    /*required, float, width, range: [0.000,1.000], the value is accurate to three decimal places*/
    "x": ,
    /*required, float, X-coordinate, range: [0.000,1.000], the value is accurate to three decimal places. The upper-left
    corner is the origin*/
    "y":
    /*required, float, Y-coordinate, range: [0.000,1.000], the value is accurate to three decimal places. The upper-left
    corner is the origin*/
    },
    "MinTargetRect":{
        "height": ,
        /*required, float, height, range: [0.000,1.000], the value is accurate to three decimal places*/
        "width": ,
        /*required, float, width, range: [0.000,1.000], the value is accurate to three decimal places*/
        "x": ,
        /*required, float, X-coordinate, range: [0.000,1.000], the value is accurate to three decimal places. The upper-left
        corner is the origin*/
        "y":
        /*required, float, Y-coordinate, range: [0.000,1.000], the value is accurate to three decimal places. The upper-left
        corner is the origin*/
        }
    }
}
}
```

JSON_EPTZModeCap

EPTZModeCap capability message in JSON format

```
{
  "EPTZModeCap":{
    "mode":{
      /*required, modes supported by e-PTZ: "cruise"-patrol (PTZ supports preset, patrol, zoom, and eight directions),
      "autoTrack"-automatic tracking*/
      "@opt":["cruise", "autoTrack"]
    },
    "AutoTrackCap":{
      /*optional, this node is returned when mode contains "autoTrack"*/
      "sensitivity":{
        /*optional, integer, sensitivity*/
        "@min":0,
        "@max":100
      },
      "detectionTarget":{
        /*detection target*/
        "@opt":["human", "vehicle", "all"]
      },
      "ReginonList":{
        "id":{
          /*required, rule ID*/

```



```
"@min":1,
"@max":4
},
"Region":{
/*required, rule area, the number of edges is between 3 and 10*/
"number":{
/*required, supported number of edges*/
"@min":3,
"@max":10
},
"x":{
/*required, float, X-coordinate, range: [0.000,1]*/
"@min":0,
"@max":1
},
"y":{
/*required, float, Y-coordinate, range: [0.000,1]*/
"@min":0,
"@max":1
}
},
"FilterSize":{
/*optional, size filter of rectangle*/
"MaxTargetRect":{
"height":{
/*required, float, height, range: [0.000,1.000], the value is accurate to three decimal places*/
"@min":0,
"@max":1
},
"width":{
/*required, float, width, range: [0.000,1.000], the value is accurate to three decimal places*/
"@min":0,
"@max":1
},
"x":{
/*required, float, X-coordinate, range: [0.000,1.000], the value is accurate to three decimal places. The upper-left
corner is the origin*/
"@min":0,
"@max":1
},
"y":{
/*required, float, Y-coordinate, range: [0.000,1.000], the value is accurate to three decimal places. The upper-left
corner is the origin*/
"@min":0,
"@max":1
}
},
"MinTargetRect":{
"height":{
/*required, float, height, range: [0.000,1.000], the value is accurate to three decimal places*/
"@min":0,
```

```
        "@max":1
      },
      "width":{
        /*required, float, width, range: [0.000,1.000], the value is accurate to three decimal places*/
        "@min":0,
        "@max":1
      },
      "x":{
        /*required, float, X-coordinate, range: [0.000,1.000], the value is accurate to three decimal places. The upper-left
        corner is the origin*/
        "@min":0,
        "@max":1
      },
      "y":{
        /*required, float, Y-coordinate, range: [0.000,1.000], the value is accurate to three decimal places. The upper-left
        corner is the origin*/
        "@min":0,
        "@max":1
      }
    }
  }
}
```

JSON_EventNotificationAlert_Alarm/EventInfo

EventNotificationAlert message with alarm or event information in JSON format.

```
{
  "ipAddress": "",
  /*required, device IPv4 address , string, the maximum size is 32 bytes*/
  "ipv6Address": "",
  /*optional, device IPv6 address, string, the maximum size is 128 bytes*/
  "portNo": ,
  /*optional, device port No., integer32*/
  "protocol": "",
  /*optional, protocol type, "HTTP, HTTPS", string, the maximum size is 32 bytes*/
  "macAddress": "",
  /*optional, MAC address, string, the maximum size is 32 bytes, e.g., 01:17:24:45:D9:F4*/
  "channelID": "",
  /*optional, device channel No., integer32*/
  "dateTime": "",
  /*optional, string, alarm/event triggered or occurred time based on ISO8601, the maximum size is 32 bytes, e.g.,
  2009-11-14T15:27Z*/
  "activePostCount": "",
  /*required, alarm/event frequency, integer32*/
  "eventType": "",
  /*required, alarm/event type, "captureResult, faceCapture,...", string, the maximum size is 128 bytes*/
  "eventState": "",
  /*required, string, the maximum size is 32 bytes, durative alarm/event status: "active"-valid, "inactive"-invalid*/
}
```

```
"eventDescription": "",
/*required, event description, string, the maximum size is 128 bytes*/
"deviceId":"","
/*string type, device ID*/
"uuid":"","
/*string type, event UUID, which is used to uniquely identify an event, the standard UUID format is xxxxxxxx-xxxx-xxxx-
xxxx-xxxxxxxxxxxx*/
...
/*optional, for different alarm/event types, the nodes are different, see the message examples in different
applications*/
}
```

JSON_EventNotificationAlert_CertificateExpiryAlarmMsg

The certificate expiry alarm details are uploaded in JSON format of EventNotificationAlert message, here shows an example.

```
Content-Type: multipart/form-data; boundary=MIME_boundary
--MIME_boundary
Content-Type: application/json
Content-Length:

{
  "ipAddress":172.6.64.7,
  "ipv6Address":"","
  "portNo":80,
  "protocol":"HTTP",
  "macAddress":"01:17:24:45:D9:F4",
  "channelID":1,
  "dateTime":"2018-11-29T15:32:55+08:00",
  "activePostCount":1,
  "eventType":"certificateRevocation",
  "eventState":"active",
  "eventDescription":"Certificate Revocation Detection",
  /*refer to the message JSON_EventNotificationAlert_Alarm/EventInfo for description details of the above nodes*/
  "channelName":"ABC",
  /*required, channel name (camera name)*/
  "deviceId":"test0123",
  /*optional, device ID, it is also the PUID and it must be returned when passing through ISAPI event by ISUP*/
  "Result":{
    "customID":"","
  /*required, string, custom certificate ID*/
    "certificateType":"CA",
  /*required, string, certificate type: "CA"-CA certificate, "client/server"-client or server certificate*/
    "type":["securityLog"],
  /*optional, function type: "HTTPS", "WebSocketS", "SDK_OVER_TLS", "SRTP", "securityLog", "ieee802.1x"*/
    "status":"expiring",
  /*optional, certificate status: "expiring"-expiring, "expired"-expired, "exceptional"-exceptional*/
    "startDate":"2010-04-17",
  /*required, date, start date of the certificate expiry date, it is accurate to day*/
```

```
"endDate":"2010-04-17"
/*required, date, end date of the certificate expiry date, it is accurate to day*/
}}
}
--MIME_boundary--
```

See Also

JSON_EventNotificationAlert_Alarm/EventInfo

JSON_EventNotificationAlert_HDDBadSectorEventMsg

The event details of the HDD bad sector detection are uploaded in JSON format of EventNotificationAlert message, here shows an example.

```
{
  "ipAddress": "",
  /*required, string type, IPv4 address of the alarm device, the maximum length is 32 bytes*/
  "ipv6Address": "",
  /*optional, string type, IPv6 address of the alarm device, the maximum length is 128 bytes*/
  "portNo": ,
  /*optional, integer32 type, port No. of the alarm device*/
  "protocolType": "",
  /*optional, string type, protocol type: "HTTP", "HTTPS", the maximum length is 32 bytes*/
  "macAddress": "",
  /*optional, string type, MAC address, the maximum length is 32 bytes*/
  "channelID": ,
  /*optional, integer32 type, device channel No. that triggers alarms*/
  "dateTime": "",
  /*required, string type, alarm triggering time in ISO8601 time format, e.g., 2018-03-13T19:42:27+08:00, the maximum
  length is 32 bytes*/
  "activePostCount": ,
  /*required, integer32 type, times that the same alarm has been uploaded*/
  "eventType": "",
  /*required, triggered event type: "hdBadBlock"-HDD bad sector detection event*/
  "eventState": "",
  /*required, string type, event triggering status: "active"-triggered, "inactive"-not triggered (heartbeat data), the
  maximum length is 32 bytes*/
  "eventDescription": "",
  /*required, string type, event description: "HDD bad sector detection event", the maximum length is 128 bytes*/
  "deviceId": "",
  /*optional, device ID, which is the PUID and should be returned for ISUP alarms, e.g., "test0123"*/
  "HDBadBlock":{
    "diskNo":
    /*required, integer32 type, HDD No.*/
  }
}
```

JSON_EventNotificationAlert_HDDHighTemperatureEventMsg

The event details of the HDD high temperature detection are uploaded in JSON format of EventNotificationAlert message, here shows an example.

```
{
  "ipAddress": "",
  /*required, string type, IPv4 address of the alarm device, the maximum length is 32 bytes*/
  "ipv6Address": "",
  /*optional, string type, IPv6 address of the alarm device, the maximum length is 128 bytes*/
  "portNo": ,
  /*optional, integer32 type, port No. of the alarm device*/
  "protocolType": "",
  /*optional, string type, protocol type: "HTTP", "HTTPS", the maximum length is 32 bytes*/
  "macAddress": "",
  /*optional, string type, MAC address, the maximum length is 32 bytes*/
  "channelID": ,
  /*optional, integer32 type, device channel No. that triggers alarms*/
  "dateTime": "",
  /*required, string type, alarm triggering time in ISO8601 time format, e.g., 2018-03-13T19:42:27+08:00, the maximum
length is 32 bytes*/
  "activePostCount": ,
  /*required, integer32 type, times that the same alarm has been uploaded*/
  "eventType": "",
  /*required, triggered event type: "highHDTemperature"-HDD high temperature detection event*/
  "eventState": "",
  /*required, string type, event triggering status: "active"-triggered, "inactive"-not triggered (heartbeat data), the
maximum length is 32 bytes*/
  "eventDescription": "",
  /*required, string type, event description: "HDD high temperature detection event", the maximum length is 128
bytes*/
  "deviceId": "",
  /*optional, device ID, which is the PUID and should be returned for ISUP alarms, e.g., "test0123"*/
  "HighHDTemperature": {
    "diskNo":
  /*required, integer32 type, HDD No.*/
  }
}
```

JSON_EventNotificationAlert_HDDImpactEventMsg

The event details of the HDD impact detection are uploaded in JSON format of EventNotificationAlert message, here shows an example.

```
{
  "ipAddress": "",
  /*required, string type, IPv4 address of the alarm device, the maximum length is 32 bytes*/
  "ipv6Address": "",
  /*optional, string type, IPv6 address of the alarm device, the maximum length is 128 bytes*/
```

```
"portNo": ,
/*optional, integer32 type, port No. of the alarm device*/
"protocolType": "",
/*optional, string type, protocol type: "HTTP", "HTTPS", the maximum length is 32 bytes*/
"macAddress": "",
/*optional, string type, MAC address, the maximum length is 32 bytes*/
"channelID": ,
/*optional, integer32 type, device channel No. that triggers alarms*/
"dateTime": "",
/*required, string type, alarm triggering time in ISO8601 time format, e.g., 2018-03-13T19:42:27+08:00, the maximum
length is 32 bytes*/
"activePostCount": ,
/*required, integer32 type, times that the same alarm has been uploaded*/
"eventType": "",
/*required, triggered event type: "hdlImpact"-HDD impact detection event*/
"eventState": "",
/*required, string type, event triggering status: "active"-triggered, "inactive"-not triggered (heartbeat data), the
maximum length is 32 bytes*/
"eventDescription": "",
/*required, string type, event description: "HDD impact detection event", the maximum length is 128 bytes*/
"deviceId": "",
/*optional, device ID, which is the PUID and should be returned for ISUP alarms, e.g., "test0123"*/
"HdImpact":{
  "diskNo":
/*required, integer32 type, HDD No.*/
}
}
```

JSON_EventNotificationAlert_HDDLowTemperatureEventMsg

The event details of the HDD low temperature detection are uploaded in JSON format of EventNotificationAlert message, here shows an example.

```
{
  "ipAddress": "",
  /*required, string type, IPv4 address of the alarm device, the maximum length is 32 bytes*/
  "ipv6Address": "",
  /*optional, string type, IPv6 address of the alarm device, the maximum length is 128 bytes*/
  "portNo": ,
  /*optional, integer32 type, port No. of the alarm device*/
  "protocolType": "",
  /*optional, string type, protocol type: "HTTP", "HTTPS", the maximum length is 32 bytes*/
  "macAddress": "",
  /*optional, string type, MAC address, the maximum length is 32 bytes*/
  "channelID": ,
  /*optional, integer32 type, device channel No. that triggers alarms*/
  "dateTime": "",
  /*required, string type, alarm triggering time in ISO8601 time format, e.g., 2018-03-13T19:42:27+08:00, the maximum
length is 32 bytes*/
  "activePostCount": ,
```

```
/*required, integer32 type, times that the same alarm has been uploaded*/
"eventType": "",
/*required, triggered event type: "lowHDDTemperature"-HDD low temperature detection event*/
"eventState": "",
/*required, string type, event triggering status: "active"-triggered, "inactive"-not triggered (heartbeat data), the
maximum length is 32 bytes*/
"eventDescription": "",
/*required, string type, event description: "HDD low temperature detection event", the maximum length is 128
bytes*/
"deviceId": "",
/*optional, device ID, which is the PUID and should be returned for ISUP alarms, e.g., "test0123"*/
"LowHDDTemperature": {
  "diskNo":
/*required, integer32 type, HDD No.*/
}
}
```

JSON_EventNotificationAlert_HDDSevereFaultEventMsg

The event details of the HDD severe fault detection are uploaded in JSON format of EventNotificationAlert message, here shows an example.

```
{
  "ipAddress": "",
  /*required, string type, IPv4 address of the alarm device, the maximum length is 32 bytes*/
  "ipv6Address": "",
  /*optional, string type, IPv6 address of the alarm device, the maximum length is 128 bytes*/
  "portNo": ,
  /*optional, integer32 type, port No. of the alarm device*/
  "protocolType": "",
  /*optional, string type, protocol type: "HTTP", "HTTPS", the maximum length is 32 bytes*/
  "macAddress": "",
  /*optional, string type, MAC address, the maximum length is 32 bytes*/
  "channelID": ,
  /*optional, integer32 type, device channel No. that triggers alarms*/
  "dateTime": "",
  /*required, string type, alarm triggering time in ISO8601 time format, e.g., 2018-03-13T19:42:27+08:00, the maximum
length is 32 bytes*/
  "activePostCount": ,
  /*required, integer32 type, times that the same alarm has been uploaded*/
  "eventType": "",
  /*required, triggered event type: "severeHDFailure"-HDD severe fault detection event*/
  "eventState": "",
  /*required, string type, event triggering status: "active"-triggered, "inactive"-not triggered (heartbeat data), the
maximum length is 32 bytes*/
  "eventDescription": "",
  /*required, string type, event description: "HDD severe fault detection event", the maximum length is 128 bytes*/
  "deviceId": "",
  /*optional, device ID, which is the PUID and should be returned for ISUP alarms, e.g., "test0123"*/
  "SevereHDFailure": {
```

```
"diskNo":  
/*required, integer32 type, HDD No.*/  
}  
}
```

JSON_EventNotificationAlert_NetworkTrafficPrealarmMsg

JSON message about details of network traffic pre-alarm.

```
{  
  "ipAddress": "",  
  /*required, string, IPv4 address of security control device, the maximum size is 32 bytes*/  
  "ipv6Address": "",  
  /*optional, string, IPv6 address of security control device, the maximum size is 128 bytes*/  
  "portNo": ,  
  /*optional, integer32, port No. of security control device*/  
  "protocolType": "",  
  /*optional, string, protocol types: "HTTP,HTTPS,Ehome", the maximum size is 32 bytes*/  
  "macAddress": "",  
  /*optional, string, MAC address, the maximum size is 32 bytes*/  
  "channelID": ,  
  /*optional, integer32, device channel No. of the device that triggers alarm*/  
  "dateTime": "",  
  /*required, string, time of alarm trigger with the format of ISO8601, the maximum size is 32 bytes*/  
  "activePostCount": ,  
  /*required, integer32, number of times that an alarm has been uploaded*/  
  "eventType": "",  
  /*required, string, event type, here it should be set to "dataPrealarm"; the maximum size is 128 bytes*/  
  "eventState": "",  
  /*required, event status: "active"-triggered alarm, "inactive"-not triggered alarm, the maximum size is 32 bytes*/  
  "eventDescription": "",  
  /*required, string, event description*/  
  "deviceId": "",  
  /*optional, PUID, e.g., test0123*/  
  "DataPrealarm":{  
    /*req, traffic pre-alarm event*/  
    "interfaceType": "",  
    /*required, string, network card type: "4g,wiredAndWireless"*/  
    "interfaceID": ,  
    /*required, integer, network card ID*/  
    "packageType": "",  
    /*required, string, data plan type: "day"-daily plan, "month"-monthly plan, "year"-annual plan*/  
    "trafficValue": ,  
    /*required, float, data value in the plan, which should be accurate to three decimal places, unit: MB*/  
    "usedTrafficValue":  
  /*required, float, data usage, which should be accurate to three decimal places, unit: MB*/  
  }  
}
```


JSON_EventNotificationAlert_vibrationDetection

Message about alarm details of vibration detection in JSON format

```
{
  "ipAddress": "",
  /*required, string, IPv4 address of alarm device, the maximum size is 32 bytes*/
  "ipv6Address": "",
  /*optional, string, IPv6 address of alarm device, the maximum size is 128 bytes*/
  "portNo": ,
  /*optional, integer32, port No. of alarm device*/
  "protocol": "",
  /*optional, string, protocol type: "HTTP, HTTPS, EHome", it should be the same as the protocolType in /ISAPI/Event/
  notification/httpHosts/<ID>, and the maximum size is 32 bytes. This node should be set to "HTTP" when the JSON
  message is transmitted by SDK; it should be set to "EHome" when the JSON message is transmitted by ISUP; it should
  be set to "HTTPS" when the transmission protocol is HTTPS*/
  "macAddress": "",
  /*optional, string, MAC address, the maximum size is 32 bytes*/
  "channelID": ,
  /*optional, integer32, channel ID of the device that triggers alarm*/
  "dateTime": "",
  /*required, string, alarm triggering time in ISO8601 format, the maximum size is 32 bytes*/
  "activePostCount": ,
  /*required, integer32, times of the same alarm being uploaded*/
  "eventType": "",
  /*required, string, event type that triggers alarm, here it should be set to "vibrationDetection" (vibration detection),
  the maximum size is 128 bytes*/
  "eventState": "",
  /*required, string, event triggering status: "active"-triggered, "inactive"-not triggered (heartbeat data), the maximum
  size is 32 bytes*/
  "eventDescription": "",
  /*required, string, event description (e.g., vibration detection), the maximum size is 128 bytes*/
  "channelName": " ",
  /*optional, string, channel name (camera name), which should be the same as the channelName in /ISAPI/Streaming/
  channels/<ID>*/
  "deviceId": " ",
  /*optional, device ID which is also the PUID, which should be returned in ISUP alarm and should be the same as the
  deviceId in /ISAPI/System/Network/Ehome*/
  "VibrationDetection": {
    "sensitivity":
    /*optional, int, sensitivity, value range: [0-100]*/
  }
}
```

JSON_EventNotificationAlert_voltageinstable

The supply voltage exception alarm is uploaded in the JSON format of EventNotificationAlert message.

```
{
  "ipAddress": "",
  /*required, device IPv4 address , string, the maximum size is 32 bytes*/
  "ipv6Address": "",
  /*optional, device IPv6 address, string, the maximum size is 128 bytes*/
  "portNo":,
  /*optional, device port No., integer32*/
  "protocol": "",
  /*optional, protocol type, "HTTP, HTTPS", string, the maximum size is 32 bytes*/
  "macAddress": "",
  /*optional, MAC address, string, the maximum size is 32 bytes, e.g., 01:17:24:45:D9:F4*/
  "channelID": "",
  /*optional, device channel No., integer32*/
  "dateTime": "",
  /*optional, string, alarm/event triggered or occurred time based on ISO8601, the maximum size is 32 bytes, e.g.,
  2009-11-14T15:27Z*/
  "activePostCount":,
  /*required, alarm/event frequency, integer32*/
  "eventType": "voltageinstable",
  /*required, alarm/event type, string, the maximum size is 128 bytes*/
  "eventState": "",
  /*required, string, the maximum size is 32 bytes, durative alarm/event status: "active"-valid, "inactive"-invalid*/
  "eventDescription": "",
  /*required, event description, string, the maximum size is 128 bytes, "overvoltage"-high supply voltage,
  "undervoltage"-low supply voltage*/
  "voltageValue": ""
  /*optional, supply voltage value, float type, unit: V, corrects to one decimal place*/
}
```

JSON_EventSearchCap

EventSearchCap message in JSON format

```
{
  "startTime": "2004-05-03T17:30:08Z",
  /*required, start time, ISO8601_time, string*/
  "endTime": "2004-05-03T17:30:08Z",
  /*required, end time, ISO8601_time, string*/
  "resultMaxNum": 100,
  /*required, supported maximum number of searching, int*/
  "channelIDLen":{
    /*optional, channel ID length*/
    "@min":,
    "@max":
  },
  "eventType":{
    /*event type: studentStoodUp-student stands up, accessController-access controller, videoIntercom-video intercom,
    OPTEX-OPTEX security control panel, Luminite-Luminite security control panel, GJD-GJD security control panel,
    cameraDetector-detector, securityControlPanel-security control panel, multiple selections are separated by comma,
    string*/
  }
```

```
"@opt": "studentStoodUp, accessController,videoIntercom, OPTEX,Luminite,GJD, cameraDetector,
securityControlPanel",
"#text": "studentStoodUp"
},
"type":{
/*optional, channel type: video-video channel, IOT-IOT channel, if this node does not exist, it indicates video channel,
string*/
"@opt": "video,IOT",
"#text": "video"
},
"AccessController":{
/*optional*/
"eventType":{
/*optional, event type, if this node does not exist, it indicates searching all supported events, string*/
"@opt": "",
"#text": ""
},
"nameLen":{
/*optional, name length*/
"@min": ,
"@max":
},
"cardNoLen":{
/*optional, card No. length*/
"@min": ,
"@max":
},
},
"VideoIntercom":{
/*optional*/
"eventType":{
/*optional, optional, event type, if this node does not exist, it indicates searching all supported events, string*/
"@opt": "",
"#text": ""
},
"nameLen":{
/*optional, name length*/
"@min": ,
"@max":
},
"cardNoLen":{
/*optional, card No. length*/
"@min": ,
"@max":
},
},
"OPTEX":{
/*optional*/
"eventType":{
/*optional, optional, event type, if this node does not exist, it indicates searching all supported events, string*/
"@opt": "",
"#text": ""
```

```
,
},
"Luminite":{
/*optional*/
"eventType":{
/*optional, optional, event type, if this node does not exist, it indicates searching all supported events, string*/
"@opt": "",
"#text": ""
},
},
"GJD":{
/*optional*/
"eventType":{
/*optional, optional, event type, if this node does not exist, it indicates searching all supported events, string*/
"@opt": "",
"#text": ""
},
},
"CameraDetector":{
/*optional*/
"eventType":{
/*optional, optional, event type, if this node does not exist, it indicates searching all supported events, string*/
"@opt": "",
"#text": ""
},
},
"SecurityControlPanel":{
/*optional, security control panel event*/
"eventType":{
/*optional, optional, event type, if this node does not exist, it indicates searching all supported events, string*/
"@opt": ""
},
"zoneNo":{
/*required, zone No., integer*/
"size": 1,
/*required, the maximum number of items, integer*/
"@min": 1,
"@max": 16,
"#text": 1
},
}
}
```

JSON_EventSearchCond

EventSearchCond message in JSON format

```
{
"searchID": "",
/*required, string, unique record search ID.*/
"searchResultPosition": "",
```

```
/*required, initial position of search result in the list, integer, When there are multiple records, and cannot get all
records in one time searching, you can search the records followed specified position for next search*/
"maxResults": "",
/*required, the max. number of results for current search, int*/
"startTime": "",
/*required, start time, ISO8601_time, string*/
"endTime": "",
/*required, end time, ISO8601_time, string*/
"choiceChannel": [{
/*optional*/
"channelID": "",
/*optional, channel ID,string*/
}],
"eventType": "",
/*required, event type: studentStoodUp-student stands up, accessController-access controller, videoIntercom-video
intercom, OPTEX-OPTEX security control panel, Luminite-Luminite security control panel, GJD-GJD security control
panel, cameraDetector-detector, securityControlPanel-security control panel, multiple selections are separated by
comma, string*/
"type": "video,IOT",
/*optional, channel type: video-video channel, IOT-IOT channel, if this node does not exist, it indicates video channel,
string*/
"AccessController":{
"eventType": "",
/*optional, IoT event type, if this node does not exist, it indicates searching all supported events, string*/
"name": "",
/*optional, name*/
"cardNo": "",
/*optional, card No.*/
},
"VideoIntercom":{
"eventType": "",
/*optional, IoT event type, if this node does not exist, it indicates searching all supported events, string*/
"name": "",
/*optional, name*/
"cardNo": "",
/*optional, card No.*/
},
"OPTEX":{
"eventType": "",
/*optional, IoT event type, if this node does not exist, it indicates searching all supported events, string*/
},
"Luminite":{
"eventType": "",
/*optional, IoT event type, if this node does not exist, it indicates searching all supported events, string*/
},
"GJD":{
"eventType": "",
/*optional, IoT event type, if this node does not exist, it indicates searching all supported events, string*/
},
"CameraDetector":{
"eventType": "",
/*optional, IoT event type, if this node does not exist, it indicates searching all supported events, string*/
```

```
}
"SecurityControlPanel":{
  "eventType": "",
  /*optional, IoT event type, if this node does not exist, it indicates searching all supported events, string*/
  "zoneNo": [1,2]
  /*optional, zone No., it is valid when eventType values "zone", array, the sub type is integer*/
}
}
```

JSON_EventSearchResult

EventSearchResult message in JSON format

```
{
  "responseStatusStr": "",
  /*optional, searching status: OK- Searching ended, NO MATCHES-No matched data found, MORE-Search again for
  more results, string, the max. length is 32, {dep if errcode == 1 && errMsg == ok}*/
  "numOfMatches": "",
  /*optional, returned number of results for current search, integer32, {dep if errcode == 1 && errMsg == ok}*/
  "totalMatches": "",
  /*optional, total number of matched results, integer32, {dep if errcode == 1 && errMsg == ok}*/
  "eventInfo": [{
    "captureTime": "",
    /*required, capture time, ISO8601_time, string*/
    "channelID": "",
    /*optional, camera ID, string*/
    "channelName": "",
    /*optional, camera name*/
    "picUrl": "",
    /*optional, picture URL, string*/
    "subPicUrl": "",
    /*optional, thumbnail URL, string */
    "eventType": "",
    /*required, event type: studentStoodU-student stands up, videoIntercom-video intercom, string*/
    "studentStoodUp":{
      /*optional, student stand up event information*/
      "studentsStoodUp_number": ""
    }
    /* optional, xs:integer, the number of students, who are standing up*/
  },
  "AccessController":{
    "eventType": "",
    /*required, event type, string*/
    "name": "",
    /*optional, name*/
    "cardNo": "",
    /*optional, card No.*/
    "cardType": "",
    /*optional, card type: invalid-invalid, ordinary-normal card, disabled-card for the disabled,blacklist-card in blacklist,
    patrol-patrol card, super-super card, guest-visitor card, remove-removed card*/
    "videoChannelInfo": {
      /*optional, video channel information*/
```

```
"videoChannel": "",
/*required, video channel No., integer*/
"picUrl": "",
/*optional, picture URL, string*/
}}
},
"VideoIntercom":{
  "eventType ": "",
/*required, event type, string*/
  "name": "",
/*optional, name*/
  "cardNo": "",
/*optional, card No.*/
  "cardType": "invalid,ordinary,disabled,blacklist,patrol,super,guest,remove",
/*optional, card type: invalid-invalid, ordinary-normal card, disabled-card for the disabled,blacklist-card in blacklist,
patrol-patrol card, super-super card, guest-visitor card, remove-removed card*/
  "videoChannelInfo": [{
/*optional, video channel information*/
    "videoChannel": "",
/*required, video channel No., integer*/
    "picUrl": "",
/*optional, picture URL, string*/
  }]
},
"OPTEx":{
  "eventType ": "",
/*required, event type, string*/
  "videoChannelInfo": [{
/*optional, video channel information*/
    "videoChannel": "",
/*required, video channel No., integer*/
    "picUrl": "",
/*optional, picture URL, string*/
  }]
},
"Luminite":{
  "eventType ": "",
/*required, event type, string*/
  "videoChannelInfo": [{
/*optional, video channel information*/
    "videoChannel": "",
/*required, video channel No., integer*/
    "picUrl": "",
/*optional, picture URL, string*/
  }]
},
"GJD":{
  "eventType ": "",
/*required, event type, string*/
  "videoChannelInfo": [{
/*optional, video channel information*/
    "videoChannel": "",
```

```
/*required, video channel No., integer*/
  "picUrl": "",
/*optional, picture URL, string*/
  },
  "CameraDetector":{
    "eventType ": "",
/*required, event type, string*/
    "videoChannelInfo": [{
/*optional, video channel information*/
      "videoChannel": "",
/*required, video channel No., integer*/
      "picUrl": "",
/*optional, picture URL, string*/
    }]
  },
  "SecurityControlPanel":{
    "eventType ": "",
/*required, event type, string*/
    "videoChannelInfo": [{
/*optional, video channel information*/
      "videoChannel": "",
/*required, video channel No., integer*/
      "picUrl": "",
/*optional, picture URL, string*/
    }]
    "zoneNo": "",
/*optional, zone No., it is valid when eventType values "zone", integer*/
    "code":1103,
/*optional, CID event No., it is valid when eventType values "zone", integer*/
    "zoneName": ""
  }
}]
}
```

JSON_ExportInfo

ExportInfo message in JSON format

```
{
  "ExportInfo":{
    "mediaType":"","
/*required, string, exported data type: "video", "audio", "metadata", "text", "mixed", "other". It should be the same
as the search type. For pictures, this node should be set to "metadata"*/
    "playbackURLList":
/*required, array, file path*/
  }
}
```


JSON_ExportLogToUSB

JSON message about parameters of exporting logs to USB flash drive

```
{
  "ExportLogToUSB": {
    "switch": "",
    "DeviceCondList": [
      {
        "DeviceCond": {
          "id": ,
          "devType": "",
          "devID": ,
          "subSys": ,
        }
      }
    ]
  }
}
```

JSON_ExportLogToUSBCap

JSON message about capability of exporting logs to USB flash drive

```
{
  "ExportLogToUSBCap": {
    "switch": {
      "@opt": [
        "stop",
        "start"
      ]
    },
    "DeviceCondList": [
      {
        "DeviceCond": {
          "id": {
            "@min": 1,
            "@max": ,
          },
          "devType": {
            "@opt": [
              "mainDev",
              "subDev"
            ]
          },
          "devID": {
            "@min": ,
            "@max":
          },
        }
      }
    ]
  }
}
```

```
"subSys": {  
  "@min": ,  
  "@max":  
},  
}  
}  
]  
}  
}
```

JSON_ExportLogToUSBStatus

JSON message about status of exporting logs to USB flash drive

```
{  
  "ExportLogToUSBStatus":{  
    "DeviceLogExportList":[{  
      "DeviceLogExport":{  
        "devType": "",  
        "devID": ,  
        "subSys": ,  
        "switch": "",  
        "exportStatus": "",  
        "exceptionMsg": ""  
      }  
    }  
  ]  
}  
}
```

JSON_ExporttoUSB_TaskInfo

TaskInfo message in JSON format

```
{  
  "TaskInfo":{  
    "taskId": ""  
    /*required, string, task ID*/  
  }  
}
```

JSON_EZVIZSecretKey

EZVIZSecretKey message in JSON format

```
{  
  "EZVIZSecretKey":{  
    /*required, verification code for Hik-Connect, string, sensitive information should be encrypted, the string length
```

```
ranges from 0 to 64. E.g., 34020000002000000001*/
"secretKey": ""
}
}
```

JSON_FileExport

FileExport message in JSON format

```
{
  "FileExport":{
    "fileType":"","
    /*required, string type, file type*/
    "selfDefineCmd":""
    /*string type, custom file type to be exported, it is valid when fileType is set to "selfDefine"*/
  }
}
```

JSON_FileExportCap

FileExportCap capability message in JSON format

```
{
  "FileExportCap":{
    "fileType":{
      "@opt":["serialLog","coreDump","debug","selfDefine"]
      /*required, string type, file type: "serialLog"-serial port log, "coreDump"-core dump log, "debug"-debug log,
      "selfDefine"-custom file type to be exported*/
    },
    "selfDefineCmd":{
      /*string type, custom file type to be exported, it is valid when fileType is set to "selfDefine"*/
      "@min": ,
      /*required, integer type, the minimum length of the custom file type to be exported*/
      "@max":
      /*required, integer type, the maximum length of the custom file type to be exported*/
    }
  }
}
```

JSON_FileExportResult

FileExportResult message in JSON format

```
{
  "FileExportResult":{
    "url":""
    /*required, string type, file URL, e.g., "http://ip:port//ISAPI/ContentMgmt/logSearch/dataPackage/data?identifier"*/
  }
}
```

```
}  
}
```

JSON_GetCustomAudio

JSON Message about getting the custom alarm sound file

```
{  
  "CustomAudioInfo": {  
    "CustomAudioList": [{  
      /*required, the file list of custom alarm sound*/  
      "customAudioID": ,  
      /*required, int, custom alarm sound ID, the value can be 1 to 3*/  
      "customAudioName": ""  
    } /*required, string, name of the custom alarm sound, the maximum size is 24 bytes*/  
  ],  
}  
}
```

JSON_GPSCalibration

Message about configuration parameters of a single GPS calibration point in JSON format

```
{  
  "GPSCalibration": {  
    "id": ,  
    /*optional, int, calibration point ID, which is used for parsing in upper layer and cannot be configured*/  
    "calibrated": ,  
    /*optional, boolean, calibration status*/  
    "longitude": ,  
    /*required, float, longitude, accurate to 6 decimal places*/  
    "latitude": ,  
    /*required, float, latitude, accurate to 6 decimal places*/  
    "x": ,  
    /*required, float, x-coordinate, accurate to 6 decimal places*/  
    "y": ,  
    /*required, float, y-coordinate, accurate to 6 decimal places*/  
  }  
}
```

JSON_GPSCalibrationCap

Message about GPS calibration capability in JSON format

```
{  
  "GPSCalibrationCap": {
```

```
/*required, GPS calibration configuration capability*/
  "enabled": "true,false",
/*required, boolean, whether the channel supports GPS calibration configuration*/
  "id": {
/*required, int, calibration No.*/
    "@min":1,
    "@max":64,
    "@text":1
  },
  "Longitude":{
/*required, float, longitude, accurate to 6 decimal places*/
    "@min":-180.000000,
    "@max":180.000000,
    "@text":56.000000
  },
  "Latitude":{
/*required, float, latitude, accurate to 6 decimal places*/
    "@min":-90.000000,
    "@max":90.000000,
    "@text":32.000000
  },
  "x":{
/*required, float, x-coordinate, accurate to 6 decimal places*/
    "@min":0.000000,
    "@max":1.000000,
    "@text":1.000000
  },
  "y":{
/*required, float, y-coordinate, accurate to 6 decimal places*/
    "@min":0.000000,
    "@max":1.000000,
    "@text":1.000000
  },
  "showEnabled": "true,false",
/*optional, boolean, whether to enable calibration stream overlay, if it is set to "true", calibration identification will be
displayed on the live view screen*/
  "calibratated": "true,false",
/*optional, boolean, calibration status*/
  "isSupportCalibrationStatus":true
/*optional, boolean, whether it supports getting all the calibration status (related URI: /ISAPI/System/
GPSCalibratation/channels/<ID>/points/status?format=json)*/
  }
}
```

JSON_GPSCalibratationList

Message about configuration parameters of multiple GPS calibration points in JSON format

```
{
  "GPSCalibratationList":[{
    "id": ,
```

```
/*required, int, calibration point ID*/
"calibrated":,
/*optional, boolean, calibration status*/
"longitude":,
/*required, float, longitude, accurate to 6 decimal places*/
"latitude":,
/*required, float, latitude, accurate to 6 decimal places*/
"x":,
/*required, float, x-coordinate, accurate to 6 decimal places*/
"y":
/*required, float, y-coordinate, accurate to 6 decimal places*/
}}
}
```

JSON_GPSCalibrationStatus

Message about GPS calibration status in JSON format

```
{
  "GPSCalibrationStatus": {
    /*required, GPS calibration status*/
    "status": ""
    /*required, string, GPS calibration status, calib-calibrated already,noCalib-Not been calibrated yet*/
  }
}
```

JSON_GuideConfig

GuideConfig message in JSON format

```
{
  "GuideConfig": {
    "GuideStep": [{
      /*required, configuration steps of quick setup instruction*/
      "stepIndex": ,
      /*required, integer, the maximum supported index*/
      "StepFunction": {
        /*required, string, instruction functions*/
        "channelID": ,
        /*optional, integer, channel number*/
        "type": ""
        /*required, "masterSlaveTracking" (linkage calibration, corresponding URL: /ISAPI/MasterSlaveTracking/channels/<ID>/
        slaveCameraCalibrating/capabilities), "mixedTargetDetection" (multi-target type detection, corresponding URL : /ISAPI/
        Intelligent/channels/<ID>/mixedTargetDetection/capabilities?format=json), eagleFocusing (rapid focus, corresponding
        URL: /ISAPI/PTZCtrl/channels/<ID>/EagleFocusing/capabilities)*/
      },
      "status": "unfinished"
    /*required, string, instruction status, whether the single configuration has completed*/
    }],
  }
```

```
"guideStatus": "unfinished"
/*required, string, instruction overall status, whether all configurations have completed*/
}
}
```

JSON_GuideConfigCap

GuideConfigCap message in JSON format

```
{
  "GuideConfigCap": {
    "GuideStep": {
      /*required, configuration steps of quick setup instruction*/
      "stepIndexNum": ,
      /*required, integer, the maximum supported index*/
      "StepFunction": {
        /*required, string, instruction functions*/
        "channels": ,
        /*optional, integer, channel number*/
        "type": {
          "@opt": ["masterSlaveTracking", "mixedTargetDetection", "eagleFocusing"]
          /*required, "masterSlaveTracking" (linkage calibration, corresponding URL: /ISAPI/MasterSlaveTracking/channels/<ID>/
          slaveCameraCalibrating/capabilities), "mixedTargetDetection" (multi-target type detection, corresponding URL : /ISAPI/
          Intelligent/channels/<ID>/mixedTargetDetection/capabilities?format=json), eagleFocusing (rapid focus, corresponding
          URL: /ISAPI/PTZCtrl/channels/<ID>/EagleFocusing/capabilities)*/
        },
        "guideStatus": {
          /*required, string, instruction status, whether the instruction configuration has completed*/
          "@opt": ["finished", "unfinished"]
        }
      },
      "MasterSlaveTrackingCap": {
        /*optional, dep:StepFunction, linkage calibration capability*/
        "enabled": {
          /*required, boolean, whether to enable, the default value is "true"*/
          "@opt": "true,false",
          "@def": "true"
        },
        "CalibratingMode": {
          /*required, string, calibration mode*/
          "@opt": "auto",
          "@def": "auto"
        }
      },
      "MixedTargetDetectionCap": {
        /*optional, dep:StepFunction, capability of multi-target type detection*/
        "enabled": {
          /*required, boolean, whether to enable, the default value is "true"*/
          "@opt": "true,false",
          "@def": "true"
        }
      }
    }
  }
}
```

```
"RuleInfoCap": {
/*required, rule information*/
  "maxSize": 1,
/*required, integer, number of supported rules*/
  "ruleID": {
/*required, integer, rule ID*/
    "@min": 1,
    "@max": 1
  },
  "RegionCap": {
/*required, rule region capability*/
    "minSize": 3,
/*required, integer, the minimum region edges*/
    "maxSize": 10,
/*required, integer, the maximum region edges*/
    "x": {
/*required, float, X-coordinate, range: from 0.000 to 1*/
      "@min": 0.000,
      "@max": 1.000,
      "#text": 0.120
    },
    "y": {
/*required, float, Y-coordinate, range: from 0.000 to 1*/
      "@min": 0.000,
      "@max": 1.000,
      "#text": 0.120
    }
  }
},
  "EagleFocusingCap": {
/*optional, dep:StepFunction, rapid focus capability*/
    "focusMode": {
/*required, string, focus mode*/
      "@opt": "auto",
      "@def": "auto"
    },
    "controlEnabled": {
/*required, boolean, whether to enable rapid focus, corresponding URL: /ISAPI/PTZCtrl/channels/<ID>/EagleFocusing/control/capabilities*/
      "@opt": "true,false",
      "@def": "true"
    }
  }
}
```


JSON_HeatMap_CollectionDescription

CollectionDescription message in JSON format

```
{
  "CollectionDescription":{
    "searchID": "",
    /*required, string, unique search ID, the content of searchID remains unchanged when search condition remains
    unchanged.*/
    "searchResultPosition": "",
    /*required, initial position of search result list, integer type. When there are multiple records, and cannot get all
    records in one time searching, you can search the records followed specified position for next search*/
    "maxResults": "",
    /*required, number of matched records per search, integer type*/
    "timeSpanList": [{
    /*required, time interval list*/
      "startTime": "",
      /*required, start time, ISO8601_time, string*/
      "endTime": "",
      /*required, end time, ISO8601_time, string*/
    }],
    "channels": "",
    /*optional, sub type, array, integer32, related channel No., array. If the channel information is not specified, it
    indicates searching all channels*/
    "statisticsType": "PDC",
    /*required, string, statistic type: OLD-heat map, PDC-people counting, DURATION-people dwell time, INTERSECTION-
    people counting of intersection*/
  }
}
```

JSON_HeatMap_CollectionDescriptionCap

CollectionDescriptionCap message in JSON format

```
{
  "CollectionDescriptionCap":{
    "timeRange": "",
    /*required, supported time range for replenishment, unit: day, integer*/
    "timeSpanMaxNum": "",
    /*required, the maximum number of time buckets supported for one time searching, integer*/
    "resultMaxNum": "",
    /*required, the maximum number of items can be searched, integer*/
    "channelMaxNum": "",
    /*required, the maximum number of channels supported for one time search, integer*/
    "statisticsType":{
    /*required, string, statistics type: OLD-heat map, PDC-people counting statistics, DURATION-people staying time
    duration, INTERSECTION-people counting of intersection*/
      "@opt": "OLD,PDC,DURATION,INTERSECTION"
    }
  }
}
```

```
}  
}
```

JSON_HeatMap_CollectionResult

CollectionResult message in JSON format

```
{  
  "CollectionResult":{  
    "responseStatusStrg": "",  
    /*required, searching status: OK- Searching ended, NO MATCHES-No matched data found, MORE-Search again for  
    more results, string, the max. length is 32, {dep if errcode == 1 && errMsg == ok}*/  
    "numOfMatches": "",  
    /*required, returned number of results for current search, integer32, {dep if errcode == 1 && errMsg == ok}*/  
    "totalMatches": "",  
    /*required, total number of matched results, integer32, {dep if errcode == 1 && errMsg == ok}*/  
    "targets": [{  
    /*optional, people gathering result*/  
      "startTime": "",  
    /*required, start time*/  
      "endTime": "",  
    /*required, end time*/  
      "channel": "",  
    /*required, integer, channel No.*/  
      "statisticsType": "",  
    /*required, string, statistic type: OLD-heat map, PDC-people counting, DURATION-people staying time, INTERSECTION-  
    people counting of intersection*/  
      "HeatMapValue": [{  
        "heatmapDataType": "",  
    /*optional, string, heat map data type, it is valid only when statisticsType values "PDC". "PDC_stayNum"-the number  
    of people dwell in the image, "PDC_leaveNum"-the number of exiting people in the image*/  
        "maxHeatMapValue": ,  
    /*required, integer, the max. heat value, 4-byte*/  
        "minHeatMapValue": ,  
    /*required, integer, the min. heat value, 4-byte*/  
        "timeHeatMapValue": ,  
    /*required, integer, the average heat value, 4-byte*/  
        "lineValue": ,  
    /*required, integer, row value, 4-byte*/  
        "columnValue": ,  
    /*required, integer, column value, 4-byte*/  
        "curNumber": ,  
    /*optional, integer, the number of current people, it is valid only when statisticsType values "PDC"*/  
        "leaveNumber": ,  
    /*optional, integer, the number of exiting people, it is valid only when statisticsType values "PDC"*/  
        "totalTime": ,  
    /*optional, integer, total dwell time, it is valid only when statisticsType values "DURATION"*/  
      }],  
      "heatmapURL": "",  
    /*optional, string, URL of pixel data for heat map image, it is valid when statisticsType values "OLD", "DURATION" or  
    "INTERSECTION"*/  
    }  
  }  
}
```

```
"PDC_stayURL": "",
/*optional, string, URL of pixel data for dwell people in heat map image, it is valid when statisticsType values "PDC"*/
"PDC_leaveURL": "",
/*optional, string, URL of pixel data for exiting people, it is valid when statisticsType values "PDC"*/
"arrayUnitType": ""
/*optional, string, matrix unit data type (each pixel data type of matrix information): "byte"-a byte, "short"-2 bytes,
"int"-four bytes*/
}}
}
```

JSON_HDCond

HDCond message in JSON format

```
{
  "HDCond":{
    "hdNo": [...,...]
/*specify the channel No. to get the corresponding HDD status*/
  }
}
```

JSON_HddFormatList

HddFormatList message in JSON format

```
{
  "HddFormatList": [{
/*required, HDD list to be formatted*/
    "HddFormat": {
/*required, initialization parameter of a HDD*/
      "id": ,
/*required, string type, ID of HDDs that specified to be formatted*/
      "formatType": ""
/*optional, string type, formatting type: FAT32, EXT4; this node is only available for SD card; if this node does not exist,
the formatting type is the default type "FAT32"*/
    }
  }]
}
```

JSON_HDStatus

HDStatus message in JSON format

```
{
  "HDStatus": [{
    "hdNo": ,
```

```
/*required, integer type, HDD No., which starts from 1*/
"status": ,
/*required, integer type, HDD status: 0-activate, 1-sleep, 2-exception, 3-sleepy HDD error, 4-unformatted, 5-
disconnected (for network HDD), 6-formatting*/
"volume": ,
/*required, integer type, HDD capacity, unit: MB*/
"freeSpace":
/*required, integer type, free space, unit: MD*/
}}
}
```

JSON_id

id message in JSON format

```
{
  "id": "",
  /*optional, integer, user ID, it returns when double verification user is added*/
}
```

JSON_ImportCustomAudio

JSON message about custom alarm sound file information

```
{
  "CustomAudioInfo": {
    /*required, related information about custom alarm sound file*/
    "customAudioName": ""
  }
  /*required, string, custom alarm sound file name, the maximum size is 24 bytes*/
}
```

JSON_InputParam

InputParam message in JSON format

```
{
  "InputParam":{
    "condType": ["name", "enable", "online", "linknum"]
  }
  /*enter channel name, enable or disable channel, online status, and number of connections*/
}
```

JSON_IntelligentSearchCap

JSON message about intelligent search capability

```
{
  "requestURL": "",
  /*optional, string type, request URL*/
  "statusCode": 1,
  /*required, int type, status code*/
  "statusString": "",
  /*required, string type, status description*/
  "subStatusCode": "",
  /*required, string type, sub status code*/
  "errorCode": 1,
  /*optional, int type, error code, this node is required when statusCode is not 1 and it corresponds to subStatusCode*/
  "errorMsg": "ok",
  /*optional, string type, error details, this node is required, when statusCode is not 1 and it can be error details about a specific parameter*/
  "startTime": "2004-05-03T17:30:08+08:00",
  /*required, string type, start time in ISO8601 time format. This node only indicates that the device supports this field*/
  "endTime": "2004-05-03T17:30:08+08:00",
  /*required, string type, end time in ISO8601 time format. This node only indicates that the device supports this field*/
  "resultMaxNum": 100,
  /*required, int type, supported maximum number of results that can be searched*/
  "channelIDLen": {
    /*optional, camera ID length*/
    "min": 1,
    "max": 32
  },
  "channelNameLen": {
    /*optional, camera name length*/
    "min": 1,
    "max": 32
  },
  "targetRect": {
    /*required, target frame*/
    "height": {
      "min": 1,
      "max": 100
    },
    "width": {
      "min": 1,
      "max": 100
    },
    "x": {
      "min": 1,
      "max": 100
    },
    "y": {
      "min": 1,
      "max": 100
    }
  }
}
```

```

    }
  },
  "picUrlLen":{
    /*required, URL length of the large picture*/
    "min":1,
    "max":128
  },
  "targetType":"vehicle",
  /*string type, search object type*/
  "isSupportLicense":true,
  /*required, boolean type, whether it supports license plate number*/
  "plateType":"unknown,92TypeCivil,arm,upDownMilitay,92TypeArm,leftRightMilitay,
02TypePersonalized,yellowTwoLine,
04NewMilitay,embassy,oneLineArm,twoLineArm,yellow1225FarmVehicle,green1325FarmVehicle,yellow1325FarmVehi
cle,motorola,coach,tempTravl,trailer,consulate,hongKongMacao,tempEntry,civilAviation,newEnergy",
  /*optional, string type, license plate type: license plate type: "unknown", "92TypeCivil"-92-style civil vehicle, "arm"-
police vehicle, "upDownMilitay"-military vehicle (top-bottom type), "92TypeArm"-92-style police vehicle,
"leftRightMilitay"-military vehicle (left-right type), "02TypePersonalized"-02-style customized vehicle, "yellowTwoLine"-
yellow two-line rear license plate, "04NewMilitay"-04-style new military vehicle, "embassy"-embassy vehicle,
"oneLineArm"-new armed police vehicle (one-line), "twoLineArm"-new armed police vehicle (two-line),
"yellow1225FarmVehicle"-yellow agricultural vehicle with 1225 structure, "green1325FarmVehicle"-green agricultural
vehicle with 1325 structure, "yellow1325FarmVehicle"-yellow agricultural vehicle with 1325 structure, "motorola"-
motorcycle, "coach"-driver-training vehicle, "tempTravl"-vehicle with temporary license plate, "trailer"-trailer,
"consulate"-consular vehicle, "hongKongMacao"-vehicle entering and leaving Hong Kong/Macao, "tempEntry"-
temporary entry vehicle, "civilAviation"-civil aviation license plate, "newEnergy"-new energy license plate. Multiple
types should be separated by commas*/

  "vehicleColor":"white,silver,gray,black,red,deepBlue,blue,yellow,green,brown,pink,purple,deepGray,cyan,orange,unkn
own",
  /*optional, string type, vehicle color: "unknown", "white", "silver"-silvery, "gray", "black", "red", "deepBlue"-dark
blue, "blue", "yellow", "green", "brown", "pink", "purple", "deepGray"-dark gray, "cyan", "orange". Multiple colors
should be separated by commas*/
  "plateColor":"white,yellow,blue,black,green,civilAviationBlack,civilAviationGreen,other",
  /*optional, string type, license plate color: "white", "yellow", "blue", "black", "green", "civilAviationBlack"-civil
aviation black, "civilAviationGreen"-civil aviation green, "0xff" or "other"-other color, "unknown"-unrecognized.
Multiple colors should be separated by commas*/

  "vehicleType":"largeBus,truck,vehicle,van,buggy,pedestrian,twoWheelVehicle,threeWheelVehicle,SUVMPV,mediumBus
,motorVehicle,nonmotorVehicle,smallCar,miniCar,pickupTruck,unknown",
  /*optional, string type, vehicle type: "largeBus"-large-sized bus, "truck"-truck, "vehicle"-salon car, "van"-minivan,
"buggy"-light truck, "pedestrian", "twoWheelVehicle"-two wheeler, "threeWheelVehicle"-tricycle, "SUVMPV"-SUV/
MPV, "mediumBus"-middle-sized bus, "motorVehicle"-motor vehicle (it will be transformed to "vehicle" (salon car) in
the platform), "nonmotorVehicle"-non-motor vehicle (it will be transformed to "threeWheelVehicle" (tricycle) in the
platform), "smallCar"-small sedan (it will be transformed to "vehicle" (salon car) in the platform), "miniCar"-mini
sedan (it will be transformed to "vehicle" (salon car) in the platform), "pickupTruck"-pick-up truck, "unknown".
Multiple types should be separated by commas*/
  "isSupportVehicleLogo":true,
  /*optional, boolean type, whether it supports vehicle parent brand*/
  "isSupportVehicleSubLogo":true,
  /*optional, boolean type, whether it supports vehicle sub brand*/
  "isSupportVehicleModel":true,
  /*optional, boolean type, whether it supports vehicle model year*/

```

```
"channelMaxNum":1,
/*required, integer type, maximum number of channels that support simultaneous search*/
"supportSearchByUTC":true
/*optional, boolean type, whether it supports searching by UTC*/
"linkFaceBodyId": {
  "@min": 0,
  "@max": 64
},
/*optional, linkage ID between human face and body, the maximum length is 64 bytes*/
"sortType": {
  "@opt":["similarity", "time"]
}
/*optional, string, sorting type: similarity-by similarity(by default), time-by time, the maximum length is 32 bytes*/
}
```

JSON_IntelligentSearchCondition

JSON message about intelligent search conditions

```
{
  "searchID":"","
/*required, string type, search ID. The content of searchID remains unchanged if the search conditions remain
unchanged. This node is used to check the same search. When the device performance is limited, you can search
asynchronously by applying conditions with the same searchID several times and getting the search progress*/
  "searchResultPosition": ,
/*required, int type, the start position of the search result in the result list. When there are multiple records and you
cannot get all search results at a time, you can search for the records after the specified position next time*/
  "maxResults": ,
/*required, int type, maximum number of search results obtained this time*/
  "startTime":"","
/*required, string type, start time in ISO8601 time format*/
  "endTime":"","
/*required, string type, end time in ISO8601 time format*/
  "choiceChannel":[{"
/*optional*/
    "channelID":"","
/*optional, string type, camera No.*/
    "plateRecogRegionID":
/*optional, integer type, license plate recognition area No. If this node is not configured, it refers to all areas*/
  }],
  "targetType":"","
/*optional, string type, search object type. If this node is not returned, it indicates no limit*/
  "sortType": "similarity",
/*optional, string, sorting type: similarity-by similarity(by default), time-by time, the maximum length is 32 bytes*/
  "humanInfo":{"
/*dependent, this node is valid only when the value of targetType is human*/
    "ageGroup":"child",
/*optional, string, age*/
    "gender":"male",
/*optional, string, gender, string, gender*/
    "glass":"yes",
```

```
/*optional, string, wear glasses or not*/
  "bag": "yes",
/*optional, string, whether with bag or not*/
  "hat": "yes",
/*optional, string, whether with hat or not*/
  "mask": "yes",
/*optional, string, whether with mask or not*/
  "jacketType": "shortSleeve",
/*optional, string, jacket type*/
  "trousersType": "shortTrousers",
/*optional, string, trousers type*/
  "hairStyle": "shortHair",
/*optional, string, hair style*/
  "jacketColor": "white,black",
/*optional, string, jacket color*/
  "trousersColor": "white,black",
/*optional, string, trousers color*/
  "ride": "yes",
/*optional, string, ride a bike or not*/
  "things": "yes",
/*optional, string, whether with handbag or not*/
  "cyclingType": "",
/*optional, string, vehicle type*/
  "cyclingPersonNumber": "",
/*optional, string, number of passengers*/
  "hID": "F4F665D6A18E41308CE9934DCDDD2222",
/*required, string, person ID, the maximum length is 32 bytes*/
  "name": "小王",
/*required, string, name of the person, the maximum length is 32 bytes*/
  "linkFaceBodyId": "XXX"
/*optional, linkage ID between human face and body, the maximum length is 64 bytes*/
  "vehicleInfo": {
/*this node is valid when targetType is "vehicle"*/
    "license": "",
/*optional, string type, license plate number*/
    "plateType": "",
/*optional, string type, license plate type: license plate type: "unknown", "92TypeCivil"-92-style civil vehicle, "arm"-
police vehicle, "upDownMilitary"-military vehicle (top-bottom type), "92TypeArm"-92-style police vehicle,
"leftRightMilitary"-military vehicle (left-right type), "02TypePersonalized"-02-style customized vehicle, "yellowTwoLine"-
yellow two-line rear license plate, "04NewMilitary"-04-style new military vehicle, "embassy"-embassy vehicle,
"oneLineArm"-new armed police vehicle (one-line), "twoLineArm"-new armed police vehicle (two-line),
"yellow1225FarmVehicle"-yellow agricultural vehicle with 1225 structure, "green1325FarmVehicle"-green agricultural
vehicle with 1325 structure, "yellow1325FarmVehicle"-yellow agricultural vehicle with 1325 structure, "motorola"-
motorcycle, "coach"-driver-training vehicle, "tempTravel"-vehicle with temporary license plate, "trailer"-trailer,
"consulate"-consular vehicle, "hongKongMacao"-vehicle entering and leaving Hong Kong/Macao, "tempEntry"-
temporary entry vehicle, "civilAviation"-civil aviation license plate, "newEnergy"-new energy license plate. If this node
is not returned, it indicates no limit*/
    "vehicleColor": "",
/*optional, string type, vehicle color: "unknown", "white", "silver"-silvery, "gray", "black", "red", "deepBlue"-dark
blue, "blue", "yellow", "green", "brown", "pink", "purple", "deepGray"-dark gray, "cyan", "orange". Multiple colors
should be separated by commas. If this node is not returned, it indicates no limit*/
    "vehicleType": ""
```



```
/*optional, string type, vehicle type: "largeBus"-large-sized bus, "truck"-truck, "vehicle"-salon car, "van"-minivan,
"buggy"-light truck, "pedestrian", "twoWheelVehicle"-two wheeler, "threeWheelVehicle"-tricycle, "SUVMPV"-SUV/
MPV, "mediumBus"-middle-sized bus, "unknown". If this node is not returned, it indicates no limit*/
"vehicleLogo": ,
/*optional, int type, vehicle parent brand. If this node is not returned, it indicates no limit*/
"vehicleSubLogo": ,
/*optional, int type, vehicle sub brand. If this node is not returned, it indicates no limit*/
"vehicleModel": ,
/*optional, int type, vehicle model year. If this node is not returned, it indicates no limit*/
"plateColor": "",
/*optional, string type, license plate color: "white", "yellow", "blue", "black", "green", "civilAviationBlack"-civil
aviation black, "civilAviationGreen"-civil aviation green, "0xff" or "other"-other color. If this node is not returned, it
indicates no limit*/
"pilotSafebelt": "",
/*optional, string type, whether the driver buckled up. If this node is not returned, it indicates no limit*/
"pilotSunvisor": "",
/*optional, string type, whether the driver's sun visor is pulled down. If this node is not returned, it indicates no limit*/
"vicePilotSafebelt": "",
/*optional, string type, whether the front passenger buckled up*/
"vicePilotSunvisor": "",
/*optional, string type, whether the front passenger's sun visor is pulled down*/
"uphone": ""
/*optional, string type, whether the person is making a call*/
}
}
```

JSON_IntelligentSearchResult

JSON message about intelligent search results

```
{
  "requestURL": "",
  /*required, string type, request URL*/
  "statusCode": ,
  /*required, int type, status code*/
  "statusString": "",
  /*required, string type,status description*/
  "subStatusCode": "",
  /*required, string type, sub status code*/
  "errorCode": ,
  /*optional, int type, error code, this node is required when statusCode is not 1 and it corresponds to subStatusCode*/
  "errorMsg": "",
  /*optional, string type, error details, this node is required, when statusCode is not 1 and it can be error details about a
specific parameter*/
  "responseStatusStr": "",
  /*required, string type, searching status: "OK"-searching completed, "NO MATCH"-no matched results, "MORE"-
searching for more results. The maximum length is 32 bytes. This node is valid when errorCode is 1 and errorMsg is
"ok"*/
  "numOfMatches": ,
  /*required, integer32, number of results returned this time. This node is valid when errorCode is 1 and errorMsg is
"ok"*/
}
```

```
"totalMatches": ,
/*required, integer32, total number of matched results. This node is valid when errorCode is 1 and errorMsg is "ok"*/
"progress": ,
/*optional, integer32 type, VCA search progress*/
"targetInfo":{
/*this node is valid when progress is 100*/
  "captureTime": "",
/*required, string type, capture time in ISO8601 time format*/
  "channelID": "",
/*optional, string type, camera No.*/
  "channelName": "",
/*optional, string type, camera name*/
  "streamType": "",
/*optional, string type, streaming type: "realtime,historyvideo,localvideo"*/
  "targetRect":{
/*required, target frame*/
    "height": ,
    "width": ,
    "x": ,
    "y":
  },
  "picUrl": "",
/*optional, string type, large picture URL*/
  "targetType": "",
/*optional, string type, search object type*/
  "humanInfo":{
/*dependent, this node is valid only when the value of targetType is human*/
    "ageGroup": "child",
/*optional, string, age*/
    "gender": "male",
/*optional, string, gender, string, gender*/
    "glass": "yes",
/*optional, string, wear glasses or not*/
    "bag": "yes",
/*optional, string, whether with bag or not*/
    "hat": "yes",
/*optional, string, whether with hat or not*/
    "mask": "yes",
/*optional, string, whether with mask or not*/
    "jacketType": "shortSleeve",
/*optional, string, jacket type*/
    "trousersType": "shortTrousers",
/*optional, string, trousers type*/
    "hairStyle": "shortHair",
/*optional, string, hair style*/
    "jacketColor": "white,black",
/*optional, string, jacket color*/
    "trousersColor": "white,black",
/*optional, string, trousers color*/
    "ride": "yes",
/*optional, string, ride a bike or not*/
    "things": "yes",
```

```
/*optional, string, whether with handbag or not*/
  "cyclingType":"","
/*optional, string, vehicle type*/
  "cyclingPersonNumber":"","
/*optional, string, number of passengers*/
  "posture":{
/*optional, posture*/
  "confidence":99.9999,
/*optional, float, confidence*/
  "value":""
/*optional, string, posture type*/
  },
  "listening":{
/*optional, listening*/
  "confidence":99.9999,
/*optional, float, listening*/
  "value":""
/*optional, string, listening type*/
  },
  "reading":{
/*optional, reading*/
  "confidence":99.9999,
/*optional, float, confidence*/
  "value":""
/*optional, string, reading type*/
  },
  "writing":{
/*optional, writing*/
  "confidence":99.9999,
/*optional, float, confidence*/
  "value":""
/*optional, string, writing type*/
  },
  "raiseHand":{
/*optional, raise hand*/
  "confidence":99.9999,
/*optional, float, confidence*/
  "value":""
/*optional, string, hand raising type*/
  },
  "lyingOnTable":{
/*optional, lying on the table*/
  "confidence":99.9999,
/*optional, float, confidence*/
  "value":""
/*optional, string, types of lying on the table*/
  },
  "playPhone":{
/*optional, playing phone*/
  "confidence":99.9999,
/*optional, float, confidence*/
  "value":""
```

```
/*optional, string , types of playing phone*/
},
"orientation":{
/*optional, orientation*/
"confidence":99.9999,
/*optional, float, confidence*/
"value":""
/*optional, string, types of orientation*/
},
"discuss":{
/*optional, discussion*/
"confidence":99.9999,
/*optional, string, confidence*/
"value":""
/*optional, string, types of discussion: unknown, no discussion, have discussion*/
},
"linkFaceBodyId":"XXX"
/*optional, linkage ID between human body and face, the maximum length is 64 bytes*/
}
"vehicleInfo ":{
/*this node is valid when targetType is "vehicle"*/
"license": "",
/*optional, string type, license plate number*/
"plateType": "",
/*optional, string type, license plate type: license plate type: "unknown", "92TypeCivil"-92-style civil vehicle, "arm"-
police vehicle, "upDownMilitay"-military vehicle (top-bottom type), "92TypeArm"-92-style police vehicle,
"leftRightMilitay"-military vehicle (left-right type), "02TypePersonalized"-02-style customized vehicle, "yellowTwoLine"-
yellow two-line rear license plate, "04NewMilitay"-04-style new military vehicle, "embassy"-embassy vehicle,
"oneLineArm"-new armed police vehicle (one-line), "twoLineArm"-new armed police vehicle (two-line),
"yellow1225FarmVehicle"-yellow agricultural vehicle with 1225 structure, "green1325FarmVehicle"-green agricultural
vehicle with 1325 structure, "yellow1325FarmVehicle"-yellow agricultural vehicle with 1325 structure, "motorola"-
motorcycle, "coach"-driver-training vehicle, "tempTravl"-vehicle with temporary license plate, "trailer"-trailer,
"consulate"-consular vehicle, "hongKongMacao"-vehicle entering and leaving Hong Kong/Macao, "tempEntry"-
temporary entry vehicle, "civilAviation"-civil aviation license plate, "newEnergy"-new energy license plate*/
"vehicleColor": "",
/*optional, string type, vehicle color: "unknown", "white", "silver"-silvery, "gray", "black", "red", "deepBlue"-dark
blue, "blue", "yellow", "green", "brown", "pink", "purple", "deepGray"-dark gray, "cyan", "orange". Multiple colors
should be separated by commas*/
"vehicleType": "",
/*optional, string type, vehicle type: "largeBus"-large-sized bus, "truck"-truck, "vehicle"-salon car, "van"-minivan,
"buggy"-light truck, "pedestrian", "twoWheelVehicle"-two wheeler, "threeWheelVehicle"-tricycle, "SUVMPV"-SUV/
MPV, "mediumBus"-middle-sized bus, "unknown"*/
"vehicleLogo": ,
/*optional, int type, vehicle parent brand*/
"vehicleSubLogo": ,
/*optional, int type, vehicle sub brand*/
"vehicleModel": ,
/*optional, int type, vehicle model year*/
"plateColor": "",
/*optional, string type, license plate color: "white", "yellow", "blue", "black", "green", "civilAviationBlack"-civil
aviation black, "civilAviationGreen"-civil aviation green, "0xff" or "other"-other color.*/
"pilotSafebelt": "",
```

```
/*optional, string type, whether the driver buckled up*/
  "pilotSunvisor":"","
/*optional, string type, whether the driver's sun visor is pulled down*/
  "vicePilotSafebelt":"","
/*optional, string type, whether the front passenger buckled up*/
  "vicePilotSunvisor":"","
/*optional, string type, whether the front passenger's sun visor is pulled down*/
  "uphone":"","
/*optional, string type, whether the person is making a call*/
  "region":"","
/*optional, string type, region that the license plate belongs to: "EU"-Europe, "ER"-Russian region, "EUandCIS"-Europe
and Russia, "ME"-the Middle East, "All"-all regions, "other"*/
  "country":"","
/*optional, string type, country that the license plate belongs to: "EU"-Europe, "ER"-Russian region, "EUandCIS"-
Europe and Russia, "ME"-the Middle East, "All"-all regions, "other"*/
}
}}
}
```

JSON_IOStatus

IOStatus message in JSON format

```
{
  "IOStatus": {
    "IOInTrig": [...,...,...],
    /*integer type, triggered alarm input No., two places and below: analog alarm input No., two places and above: digital
    alarm input No.; and the low 2-bit is the alarm input No., the 3-bit or above is the digital channel No., e.g., 3201-alarm
    input No.1 of digital channel No.32*/
    "IOOutTrig": [...,...,...]
    /*integer type, triggered alarm output No., two places and below: analog alarm output No., two places and above:
    digital alarm output No.; and the lower 2 bits are the alarm output No., the 3-bit or above is the digital channel No.,
    e.g., 3201-alarm output No.1 of digital channel No.32*/
  }
}
```

JSON_IOT_ChannelInfo

ChannelInfo message in JSON format

```
{
  "ChannelInfo":{
    "channel": "",
    /*required, channel No., it should be returned when the channel is added, integer*/
  }
}
```

JSON_IOT_ChannelInfoList

ChannelInfoList message in JSON format

```
{
  "ChannelInfoList": [{
    /*optional, channel information, if this node does not exist, it indicates all channels, array*/
    "channel": "",
    /*required, channel No., integer*/
    "type": ""
  }]
}
```

JSON_IOT_ErrorList

ErrorList message in JSON format

```
{
  "ErrorList": [{
    /*dep, detailed error information, it is valid when subStatusCode values "badParameters"*/
    "errorRowNo": "",
    /*required, error row No., integer*/
    "errorType": "",
    /*required, error type, string, channelNoInvalid-invalid channel No., channelNoConflict-conflicted channel No.,
    channel IP/Domain invalid-invalid channel IP or domain name, channel IP/Domain conflict-conflicted channel IP or
    domain name, "channel IP conflict with local IP"-channel IP is conflicted with local IP, protocolError-incorrect protocol,
    adminPortError-incorrect admin port, channelError-incorrect channel No., UserNameInvalid-invalid user name,
    passwordInvalid-invalid password, transProtocalError-incorrect transmission protocol, inductiveTypeInvalid-invalid
    inductive type*/
  }]
}
```

JSON_IOTChannel

IOTChannel message in JSON format

```
{
  "IOTChannel":{
    /*optional, added IoT devices information*/
    "channel": "",
    /*optional, channel No., if this node does not exist, it indicates any free channel, integer*/
    "IOTProtocolType": "",
    /*required, IOT protocol type: "HIKVISION"-Hikvision device, "OPTEX"-OPTEX security control panel, "Luminite"-
    Luminite security control panel, "GJD"-GJD security control panel, "SIA-CID"-SIA-CID protocol, string, the max. length is
    32*/
    "addressingFormatType": "",
    /*optional, address type, "ipaddress"-ip address, "hostname"-host name, string, the max. length is 32*/
  }
}
```

```
"hostName": "",
/*optional, host name, it is required when addressingFormatType values "hostname", string, the max. length is 64*/
"ipAddress": "",
/*optional, device IPV4 address, it is required when addressingFormatType values "ipaddress", string, the max. length
is 32*/
"ipv6Address": "",
/*optional, device IPV6 address, it is required when addressingFormatType values "ipaddress", string, the max. length
is 128*/
"protocol": "",
/*optional, protocol type: "tcp"-TCP protocol, "udp"-UDP protocol, "multicast"-Multicast protocol, string, the max.
length is 32*/
"portNo": "",
/*optional, device port No., integer*/
"userName": "",
/*optional, user name, string, the sensitive information is encrypted, the max. length is 32*/
"password": "",
/*read-only, password, string, the sensitive information is encrypted, the max. length is 16. It will be applied only
when configuring, and it is not returned when getting information*/
"deviceChannel": "",
/*required, added IoT channel No., integer*/
"videoChannels": [1, 2],
/*optional, added video channel No., including the cameras and IoT devices, array, and the array unit is integer*/
"addAll": "",
/*optional, whether adds all channels (including video and IoT channel) of device, boolean*/
"inductiveDeviceType": "",
/*optional, inductive device type, "inductiveType1"-inductive type 1, "inductiveType2"-inductive type 2. When getting
added channel information according inductive type, if this node does not exist, all added device channels will be
returned, string*/
"zoneNum": ""
/*optional, the number of added zones, it is valid when IOTProtocolType values "SIA-CID", integer*/
}
}
```

Remarks



Note

For different IoT devices, the inputted parameters in IOTChannel message are different.

Example

IOTChannel Message of IoT Devices via Private Protocol

```
"IOTChannel":{
  "IOTProtocolType": "HIKVISION",
  "addressingFormatType": "ipaddress",
  "ipAddress": "10.11.12.13",
  "protocol": "tcp",
  "portNo": 80,
  "userName": "admin",
  "password": "12345",
  "deviceChannel": 1,
```

```
"videoChannels": [1, 2]
}
```

Example

IOTChannel Message of IoT Devices via SIA-CID Protocol

```
"IOTChannel":{
  "IOTProtocolType": "HIKVISION",
  "addressingFormatType": "ipaddress",
  "ipAddress": "10.11.12.13",
  "protocol": "tcp",
  "portNo": 80,
  "userName": "admin",
  "password": "12345",
  "deviceChannel": 1,
  "videoChannels": [1, 2],
  "zoneNum": 1
}
```

Example

IOTChannel Message of OPTEX IoT Devices

```
"IOTChannel":{
  "IOTProtocolType": "OPTEX",
  "addressingFormatType": "ipaddress",
  "ipAddress": "10.11.12.13",
  "protocol": "tcp",
  "deviceChannel": 1
}
```

Example

IOTChannel Message of Luminite IoT Devices

```
"IOTChannel":{
  "IOTProtocolType": "Luminite",
  "addressingFormatType": "ipaddress",
  "ipAddress": "10.11.12.13",
  "protocol": "tcp",
  "portNo": 80,
  "password": "12345",
  "deviceChannel": 1
}
```

Example

IOTChannel Message of GJD IoT Devices

```
"IOTChannel":{
  "IOTProtocolType": "GJD",
  "addressingFormatType": "ipaddress",
  "ipAddress": "10.11.12.13",
  "deviceChannel": 1
}
```


JSON_IOTChannelEventCap

IOTChannelEventCap message in JSON format

```
{
  "IOTChannelEventCap":{
    /*required, event capability*/
    "AccessController":{
      /*optional, access controller event*/
      "eventType":{
        /*required, event type, array, the sub type is string*/
        "@opt": ["authenticated", "openingDoor"]
      },
    },
    "VideoIntercom":{
      /*optional, video intercom event*/
      "eventType":{
        /*required, event type, string*/
        "@opt": ["tampering", "duressAlarm"]
      },
    },
    "GJD":{
      /*optional, GJD security control panel event*/
      "eventType":{
        /*required, event type, string*/
        "@opt": ["pirDetection", "tampering"]
      },
    },
    "Luminite":{
      /*optional, Luminite security control panel*/
      "eventType":{
        /*required, event type, string*/
        "@opt": ["pirDetection", "pirTamper"]
      },
    },
    "OPTEx":{
      /*optional, OPTEx security control panel*/
      "eventType":{
        /*required, event type, string*/
        "@opt": ["pcFn", "pcFr"]
      },
    },
    "cameraDetector":{
      /*optional, detector evnet*/
      "eventType":{
        /*required, event type, string*/
        "@opt": ["wirelessTemperature", "wirelessPIR"]
      },
    },
    "SecurityControlPanel":{
      /*optional, security control panel*/
    }
  }
}
```

```
"eventType":{
/*required, event type, string*/
  "@opt": ["zone","host"]
},
"zoneNum":{
/*required, the number of zones, integer*/
  "@min": 1,
  "@max": 16,
  "#text": 1
},
}
}
```

JSON_IOTChannelList

IOTChannelList message in JSON format

```
{
  "IOTChannelList": [{
    "IOTChannel"
/*optional, added IoT devices information, see details in
    JSON_IOTChannel
*/
  }]
}
```

See Also

JSON_IOTChannel

JSON_IOTChannelStatus

IOTChannelStatus message in JSON format

```
{
  "IOTChannelStatus":{
/*optional, added IoT device status*/
    "IOTChannel":{
/*optional, added IoT device information, see details in
        JSON_IOTChannel
*/
    },
    "channelName": "",
/*required, channel name, string, type, the max. length is 32*/
    "onlineStatus": "",
/*required, online status, boolean*/
    "channelDetectResult": "connecting,networkUnreachable,accessAbnormal,normal,gaurdFailed",
/*optional, channel detection status: "connecting"-connecting, "networkUnreachable"-unreachable network,
"accessAbnormal"-abnormal access, "normal"-normal, "guardFailed"-arming failed, string*/
    "SecurityStatus":{
```

```
/*optional, security status*/
  "passwordStatus": ""
/*optional, password status: notActivated-device is inactivated, risk-risky password, weak-weak password, medium-
medium password, strong-strong password, invalid-invalid status, string*/
},
"DeviceInfo":{
/*optional, device information*/
  "deviceName": "",
/*optional, device name, string, the max. length is 32*/
  "deviceType": "",
/*required, device type: "accessController"-access controller, "videoIntercom"-video intercom, "OPTEX"-OPTEX
security control panel, Luminite-Luminite security control panel, GJD-GJD security control panel, securityControlPanel-
security control panel, string*/
  "model": "",
/*optional, device model, string, the max. length is 128*/
  "firmwareVersion": "",
/*optional, firmware version, string, the max. length is 128*/
  "firmwareReleasedDate": ""
/*optional, firmware compiled date, string, the max. length is 128*/
}
}
}
```

JSON_IOTChannelStatusList

IOTChannelStatusList message in JSON format

```
{
  "IOTChannelStatusList": [{
    "IOTChannelStatus":
/*optional, added IoT device status, see details in JSON_IOTChannelStatus*/
  }]
}
```

See Also

JSON_IOTChannelStatus

JSON_IOTSourceDescription

IOTSourceDescription message in JSON format

```
{
  "IOTSourceDescription":{
/*optional, protocol information*/
  "IOTProtocolType": "",
/*required, IOT protocol type: "HIKVISION"-Hikvision device, "OPTEX"-OPTEX security control panel, "Luminite"-
Luminite security control panel, "GJD"-GJD security control panel, "SIA-CID"-SIA-CID protocol, string, the max. length is
32*/
  "addressingFormatType": "",
```

```
/*optional, address type, "ipaddress"-ip address, "hostname"-host name, string, the max. length is 32*/
  "hostName": "",
/*optional, host name, it is required when addressingFormatType values "hostname", string, the max. length is 64*/
  "ipAddress": "",
/*optional, device IPV4 address, it is required when addressingFormatType values "ipaddress", string, the max. length
is 32*/
  "ipv6Address": "",
/*optional, device IPV6 address, it is required when addressingFormatType values "ipaddress", string, the max. length
is 128*/
  "protocol": "",
/*optional, ptocol type: "tcp"-TCP protocol, "ud"-UDP protocol, "muticast"-Multicast protocol, string, the max.
length is 32*/
  "portNo": "",
/*optional, device port No., integer*/
  "userName": "",
/*optional, user name, string, the sensitive information is encrypted, the max. length is 32*/
  "password": ""
/*optional, password, string, the sensitive information is encrypted, the max. length is 16*/
}
}
```

JSON_IOTSourceList

IOTSourceList message in JSON format

```
{
  "IOTSourceList": [{
/*optional, source informaiton, array*/
    "IOTProtocolType": "",
/*required, IOT protocol type: "HIKVISION"-Hikvision device, "OPTEX"-OPTEX security control panel, "Luminite"-
Luminite security control panel, "GJD"-GJD security control panel, "SIA-CID"-SIA-CID protocol, string, the max. length is
32*/
    "addressingFormatType": "",
/*optional, address type, "ipaddress"-ip address, "hostname"-host name, string, the max. length is 32*/
    "hostName": "",
/*optional, host name, it is required when addressingFormatType values "hostname", string, the max. length is 64*/
    "ipAddress": "",
/*optional, device IPV4 address, it is required when addressingFormatType values "ipaddress", string, the max. length
is 32*/
    "ipv6Address": "",
/*optional, device IPV6 address, it is required when addressingFormatType values "ipaddress", string, the max. length
is 128*/
    "portNo": "",
/*optional, device port No., integer*/
    "userName": "",
/*optional, user name, string, sensitive information will be encrypted, the max. length is 32*/
    "password": "",
/*optional, password, string, sensitive information will be encrypted, the max. length is 16*/
    "activeStatus": "",
/*required, active or not, boolean*/
    "deviceType": "",
```

```
/*optional, string, devive type: accessController-access controller, videoIntercom-video intercom, OPTEX-OPTEX
security control panel, Luminite-Luminite security control panel, GJD-GJD security control panel, securityControlPanel-
security control panel*/
"macAddress": "",
/*optional, device MAC address, string, the max. length is 48*/
"serialNumber": "",
/*optional, device serial No., string, the max. length is 48*/
"firmwareVersion": "",
/*optional, firmware version (including compiled date), string, the max. length is 128*/
}}
}
```

JSON_IOTSourceSupport

IOTSourceSupport message in JSON format

```
{
  "IOTSourceSupport":{
    /*required, supported source information*/
    "sourceNum": "",
    /*required, the number of protocols, integer*/
    "IOTSourceDescriptions": [{
    /*optional, protocol informaiton, array*/
      "IOTProtocolType": "",
      /*required, IOT protocol type: "HIKVISION"-Hikvision device, "OPTEX"-OPTEX security control panel, "Luminite"-
      Luminite security control panel, "GJD"-GJD security control panel, "SIA-CID"-SIA-CID protocol, string, the max. length is
      32*/
      "addressingFormatType":{
        /*optional, address type, "ipaddress"-ip address, "hostname"-host name, string, the max. length is 32*/
        "@opt": "ipaddress,hostname",
        "#text": "ipaddress"
      },
      "protocol":{
        /*optional, protocol type, "tcp, udp, muticast", string, the max. length is 32*/
        "@opt": "tcp,udp,multicast",
        "#text": "tcp"
      },
      "portNo":{
        /*optional, device port No., integer*/
        "@min": 0,
        "@max": 65535,
        "#text": 80
      },
      "userName ":{
        /*optional, user name, string, the max. length is 32*/
        "@min": 0,
        "@max": 32,
        "#text": "admin"
      },
      "password":{
        /*optional, password, string, the max. length is 16*/
```

```
"@min": 0,
"@max": 16,
"#text": "12345"
},
"maxVideoChannelNum": "",
/*optional, the max. number of supported video channel, integer*/
"zoneNum":{
/*optional, number of connected zones, integer, it is valid when the IOTProtocolType values "SIA-CID"*/
"@min": 1,
"@max": 256,
"#text": 1
},
}},
"DeviceInductiveType":{
/*optional, device inductive type*/
"inductiveType1": ["accessController", "videoIntercom"],
/*optional, array, inductive type 1: accessController-access controller, videoIntercom-video intercom, OPTEX-OPTEX
security control panel, Luminite-Luminite security control panel, GJD-GJD security control panel, securityControlPanel-
security control panel, the sub type is string*/
"inductiveType2": ["OPTEX", "Luminite", "GJD", "securityControlPanel"]
/*optional, array, inductive type 2: accessController-access controller, videoIntercom-video intercom, OPTEX-OPTEX
security control panel, Luminite-Luminite security control panel, GJD-GJD security control panel,securityControlPanel-
security control panel, the sub type is string*/
}
}
}
```

JSON_GPSInfo

JSON message about GPS information

```
{
  "GPSInfo":{
/*required, actual GPS information*/
    "longitude": ,
/*required, float, longitude, range: [-180.000000,180.000000], corrects to 6 decimal places*/
    "latitude":
/*required, float latitude, range: [-90.000000,90.000000], corrects to 6 decimal places*/
  }
}
```

JSON_GPSVerification_ResultInfo

JSON message about calibration verification result information

```
{
  "ResultInfo":{
/*required, calibration verification result information*/
    "id": ,
  }
}
```

```
/*required, int, verification point ID; PanoVu series camera supports up to 16 points, and speed dome supports up to 64 points*/  
  "longitude": ,  
/*required, float, longitude, corrects to 6 decimal places, range:[-180.000000,180.000000]*/  
  "latitude":  
/*required, float, latitude, corrects to 6 decimal places, range: [-90.000000,90.000000]*/  
}  
}
```

JSON_GPSVerification_ResultInfoList

JSON message about calibration verification results

```
{  
  "ResultInfoList":[{  
/*required, verification results information*/  
    "id": ,  
/*required, int, verification point ID; PanoVu series camera supports up to 16 points, and speed dome supports up to 64 points*/  
    "longitude": ,  
/*required, float, longitude which is calculated based on the pan and tilt, corrects to 6 decimal places, range: [-180.000000,180.000000]*/  
    "latitude":  
/*required, float, latitude which is calculated based on the pan and tilt, corrects to 6 decimal places, range: [-90.000000,90.000000]*/  
  }  
}]  
}
```

JSON_GyroVerify

JSON message about gyroscope calibration parameters

```
{  
  "GyroVerify": {  
/*required, gyroscope calibration*/  
    "enabled":  
/*required, whether to enable gyroscope calibration: true=yes, false=no*/  
  }  
}
```

JSON_GyroVerifyCap

JSON message about gyroscope calibration capability.

```
{  
  "GyroVerifyCap":{  
    "enabled":{
```

```
/*required, whether to enable gyroscope calibration*/
  "@opt": [true,false],
  "@def":false
}
}
}
```

JSON_KeepAlive

KeepAlive message in JSON format

```
{
  "KeepAlive":{
    "enabled":"","
/*required, boolean, whether to enable network keepalive*/
    "interval":"","
/*required, integer, heartbeat interval, unit: second*/
    "addressingFormatType":"","
/*required, string, address types: "ipaddress"-IP address, "hostname"-host name, the max. length is 32*/
    "hostName": "",
/*optional, string, host name, which is required when addressingFormatType is "hostname", the max. length is 64*/
    "ipAddress": "",
/*optional, string, IPv4 address, which is required when addressingFormatType is "ipaddress", the max. length is 32*/
    "ipv6Address": "",
/*optional, string, IPv6 address, which is valid when addressingFormatType is "ipaddress", the max. length is 128*/
  }
}
```

JSON_KeepAliveCap

KeepAliveCap message in JSON format

```
{
  "KeepAliveCap":{
    "enabled": "true,false",
/*required, boolean, whether to enable network keepalive*/
    "interval":{
      "@min": 60,
      "@max": 300,
      "@def": 300
    },
/*required, integer, heart beat interval which is 300s by default, unit: second*/
    "addressingFormatType":{
      "@opt": ["ipaddress", "hostname"],
    },
/*required, string, address types: "ipaddress"-IP address, "hostname"-host name, the max. length is 32*/
    "supportServerTest": true,
  }
}
```



```
/*optional, boolean, whether to support server test*/  
}  
}
```

JSON_LaserOpticalAxis

JSON message about laser optical axis parameters.

```
{  
  "LaserOpticalAxis": {  
    "adjustmentEnabled": ,  
    /*required, boolean, whether it enables adjusting laser optical axis*/  
    "sensitivity":  
    /*required, int, sensitivity*/  
  }  
}
```

JSON_LaserOpticalAxis_direction

JSON message about direction of adjusting laser optical axis

```
{  
  "direction": ""  
  /*required, string, direction of adjusting laser optical axis: "up"-up, "down"-down, "left"-left, "right"-right*/  
}
```

JSON_LaserOpticalAxisCap

JSON message about capability of configuring laser optical axis

```
{  
  "LaserOpticalAxisCap": {  
    "adjustmentEnabled": {  
    /*required, boolean, whether it enables adjusting laser optical axis*/  
      "@opt": [true, false]  
    },  
    "sensitivity": {  
    /*required, int, sensitivity*/  
      "@min": 1,  
      "@max": 100  
    }  
    "direction": {  
    /*required, string, direction of adjusting laser optical axis: "up"-up, "down"-down, "left"-left, "right"-right*/  
      "@opt": ["up", "down", "left", "right"],  
      "#text": "up"  
    }  
  }  
}
```

```
}  
}
```

JSON_LensCorrection

LensCorrection in JSON format

```
{  
  "LensCorrection":{  
    "enabled": "",  
    /*required, enable or not*/  
    "TimeTaskList": [{  
      "TimeTaskBlock":{  
        "dayOfWeek": "",  
        /*required, integer32, ISO8601 weekday number, 1=Monday*/  
        "TimeTaskRange": [{  
          "taskID": "",  
          /*required, integer32, task ID, starts from 1*/  
          "beginTime": "",  
          /*required, begin time, accurate to second, string type*/  
          "endTime": ""  
        /*required, end time, accurate to second, string type*/  
        }]  
      }  
    }  
  }  
}
```

JSON_LensCorrectionCap

LensCorrectionCap message in JSON format

```
{  
  "LensCorrectionCap":{  
    "enabled": "true,false",  
    /*required, enable or not*/  
    "TimeTaskList":{  
      "maxSize": ,  
      "TimeTaskBlock":{  
        "dayOfWeek": {  
          /*required, integer32, ISO8601 weekday number, 1=Monday*/  
          "@min": "",  
          "@max": "",  
        },  
        "TimeTaskRange": {  
          "maxSize": ,  
          "taskID": {  
            /*required, integer32, task ID, starts from 1*/  
            "@min": ,
```

```
        "@max": ,
    },
    "beginTime": "01:00:00",
    /*required, begin time, corrects to second, string*/
    "endTime": "02:00:00"
    /*required, end time, corrects to second, string*/
  }
}
}
}
```

JSON_LinkageChansCond

LinkageChansCond message in JSON format

```
{
  "LinkageChansCond":{
    "eventType": "",
    /*required, string, for access controller or video intercom, the EventType == AcsEvent_major_minor, for security control panel, the EventType == AlarmHost_zoneNo*/
    "monitorId": "",
    /*dep, string, camera ID, strlen=[32,64]. For access controller or video intercom, the format is: device serial No._DOOR_door No., for security control panel, the format is: device serial No._ALARM_security control panel No.*/
    "unlockType": ""
    /*optional, unlock type, string, password-unlock by password, hijacking-hijacking unlock, card-unlock by swiping card, householder-unlock by householder, centerplatform-unlock by center platform, bluetooth-bluetooth unlock, qrcode-unlock by QR code, face-unlock by face, fingerprint-unlock by fingerprint*/
  }
}
```

JSON_LinkedTracking

JSON message about advanced parameters of linked tracking capture

```
{
  "LinkedTracking":{
    /*required, advanced parameters of linked tracking capture*/
    "trackingTakeoverEnabled": ,
    /*optional, boolean, whether to enable tracking takeover, it is disabled by default; the function is supported only when you select VCA resource as Smart Event*/
    "linkedTrackingCaptureEnabled": ,
    /*optional, boolean, whether to enable linked tracking capture, it is disabled by default; the function is supported only when you select VCA resource as Smart Event*/
    "captureThreshold": ,
    /*optional, int, capture threshold, it refers for the quality of face to trigger capture and alarm, range: [0,100], default value: 20*/
    "Exposure":{
      /*optional, face exposure*/
    }
  }
}
```

```
"exposureEnabled": ,
/*optional, boolean, whether to enable face exposure, it is disabled by default*/
"brightRef": ,
/*optional, int, reference brightness of a face in the face exposure mode, range: [0,100]*/
"expDurationTime":
/*optional, int, minimum duration, it is the extra time the device keeps the face exposure lever after the face
disappears from the scene; range: [0,60], unit: minute, default value: 5m*/
},
"RemoveDuplicate":{
/*optional, remove duplicates of captured face pictures*/
"enabled": ,
/*optional, whether to enable removing duplicates of captured face pictures*/
"threshold": ,
/*optional, int, threshold of removing duplicates, when the picture similarity is larger than the value, the picture is
duplicated*/
"faceScore": ,
/*optional, int, the face grading threshold for removing duplicates, only when the face grading is larger than the
configured threshold, the face picture will be compared for removing duplicates, in case that low quality face pictures
influence the comparison effect*/
"updateTime":
/*optional, int, face modal data updating time interval (the time interval between the modal data being added into
the library and being deleted)*/
}
}
}
```

JSON_LinkedTrackingCap

JSON message about configuration capability of advanced parameters of linked tracking capture

```
{
  "LinkedTrackingCap":{
    /*required, advanced parameters capability of linked tracking capture*/
    "trackingTakeoverEnabled":{
      /*optional, boolean, whether to enable tracking takeover, it is disabled by default; the function is supported only
      when you select VCA resource as Smart Event*/
      "@opt": [true, false],
      "@def": false
    },
    "linkedTrackingCaptureEnabled":{
      /*optional, boolean, whether to enable linked tracking capture, it is disabled by default; the function is supported only
      when you select VCA resource as Smart Event*/
      "@opt": [true, false],
      "@def": false
    },
    "captureThreshold":{
      /*optional, int, capture threshold, it refers for the quality of face to trigger capture and alarm, range: [0,100], default
      value: 20*/
      "@min": 0,
      "@max": 100,
      "@def": 20
    }
  }
}
```

```
,
  "ExposureCap":{
/*optional, face exposure capability*/
    "exposureEnabled":{
/*optional, boolean, whether to enable face exposure, it is disabled by default*/
        "@opt": [true, false],
        "@def":false
    },
    "brightRef":{
/*optional, int, reference brightness of a face in the face exposure mode, range: [0,100]*/
        "@min":0,
        "@max":100,
        "@def":50
    },
    "expDurationTime":{
/*optional, int, minimum duration, it is the extra time the device keeps the face exposure lever after the face
disappears from the scene; range: [0,60], unit: minute, default value: 5m*/
        "@min":1,
        "@max":60,
        "@def":5
    }
  },
  "RemoveDuplicateCap":{
/*optional, capability of removing duplicates of captured face pictures*/
    "enabled":{
/*optional, whether to enable removing duplicates of captured face pictures*/
        "@opt": [true, false],
        "@def":false
    },
    "threshold":{
/*optional, int, threshold of removing duplicates, when the picture similarity is larger than the value, the picture is
duplicated*/
        "@min":70,
        "@max":100,
        "@def":88
    },
    "faceScore":{
/*optional, int, the face grading threshold for removing duplicates, only when the face grading is larger than the
configured threshold, the face picture will be compared for removing duplicates, in case that low quality face pictures
influence the comparison effect*/
        "@min":20,
        "@max":60,
        "@def":20
    },
    "updateTime":{
/*optional, int, face modal data updating time interval (the time interval between the modal data being added into
the library and being deleted)*/
        "@min":0,
        "@max":300,
        "@def":30
    }
  }
}
```

```
}  
}
```

Remarks

Tracking takeover and linked tracking capture are two mutually exclusive modes for panorama tracking, they cannot be enabled at the same time.

JSON_List_IPAddress

List_IPAddress message in JSON format

```
{  
  "List":[{  
    "IPAddress":{  
      "ipVersion":"","  
/*required, string type, version information of IP address: "v4"-IPv4, "v6"-IPv6*/  
      "ipAddress":"","  
/*required, string type, IP address*/  
    }  
  }  
}
```

JSON_LockCfg

LockCfg message in JSON format

```
{  
  "LockCfg":{  
    "maxTryTimes": ,  
/*optional, integer, the maximum attempts of logging in by the IP address*/  
    "lockedTime":  
/*optional, integer, time range of locking the IP address, unit: second*/  
  }  
}
```

JSON_LockCfgCap

LockCfgCap capability message in JSON format

```
{  
  "LockCfgCap":{  
    "maxTryTimes":{  
/*optional, the maximum attempts of logging in by the IP address*/  
      "@opt":"5,..."  
    },  
    "lockedTime":{  
/*optional, time range of locking the IP address, unit: second*/
```

```
"@min": ,
"@max":
}
}
}
```

JSON_LoginLinkNum

LoginLinkNum message in JSON format

```
{
  "LoginLinkNum": {
    "maxLinkNum": ,
    /*required, integer type, maximum number of logged in accounts*/
    "adminPassword":""
    /*required, string type, confirm the password of admin user, this node is valid only when setting maximum number
    (PUT), otherwise, this node does not exits*/
  }
}
```

JSON_logProTestResult

JSON message about preverification results of logging HDD.

```
{
  "requestURL": "",
  /*optional, string type, request URL*/
  "statusCode": ,
  /*required, integer type, status code*/
  "statusString": "",
  /*required, string type, status description*/
  "subStatusCode": "",
  /*required, string type, sub status code*/
  "errorCode": ,
  /*optional, integer type, error code, which corresponds to subStatusCode, this field is required when statusCode is
  not 1. The returned value is the transformed decimal number*/
  "errorMsg": "",
  /*optional, string type, error details, this field is required when statusCode is not 1*/
  "logProTestResult": {
    /*preverification results*/
    "enough": ,
    /*required, boolean, whether there is enough storage space in the HDD: "true"-enough, "false"-not enough*/
    "CoverDate": {
      /*dependent, it is returned when the field enough is "false"*/
      "beginTime": "",
      /*required, string, start time of HDD coverage, which should in ISO 8601 format,the maximum size is 32 bytes*/
      "endTime": "",
      /*required, string, end time of HDD coverage, which should in ISO 8601 format,the maximum size is 32 bytes*/
      "coverSpace":
```

```
/**required, integer, size of the covered HDD, unit: GB*/
}
}
}
```

JSON_MatchList

JSON Message about log search results

```
{
  "requestURL": "",
  /*optional, string, request URL*/
  "statusCode": ,
  /*required, int, status code*/
  "statusString": "",
  /*required, string, status description*/
  "subStatusCode": "",
  /*required, string, sub status code*/
  "errorCode": ,
  /*optional, int, error code, which corresponds to the subStatusCode; this node is required when the value of
statusCode is not 1*/
  "errorMsg": "",
  /*optional, string, error description, which corresponds to errorCode*/
  "searchID": "",
  /*required, string, search ID, which is used to check whether the current search requester is the same as the previous
  one. If they are the same, the search record will be stored in the device to speed up the next search, the maximum
  length is 32 bytes*/
  "responseStatusStr": "",
  /*required, string, search status: "OK"-search completed, "FAILED"-search failed, "MORE"-more data waiting to be
  searched, "PARAM ERROR"-incorrect parameters, "INVALID TIME"-invalid time, "TIMEOUT"-timeout, "NO MATCHES"-no
  matched data, the maximum length is 32 bytes*/
  "numOfMatches": ,
  /*required, integer32, number of matched data records*/
  "matchList": [{
  /*optional, array, list of matched data records*/
    "searchMatchItem": {
  /*optional, single matched record*/
    "logDescriptor": {
  /*required, log description*/
    "metaID": "",
  /*required, string, log type string, log type includes "Alarm"-alarm log, "Exception"-exception log, "Operation"-
  operation log, and "Information"-additional information; e.g., "log.std-cgi.com/Alarm", the maximum length is 64
  bytes*/
    "startTime": "",
  /*required, string, log time, ISO8601 time format, the maximum length is 32 bytes*/
    "localId": "",
  /*optional, string, local channel, different logs are corresponding to different local channel types, the maximum length
  is 32 bytes*/
    "paraType": "",
  /*optional, string, function modules that generate the logs: "videoout, image, encode, network, alarm, exception,
  decoder, rs232, preview, security, datatype, frametype"; the maximum length is 32 bytes*/
```



```
"userName": "",
/*optional, string, user name that performs current search task, the maximum length is 32 bytes*/
"ipAddress": "",
/*optional, string, device IP address, the maximum length is 32 bytes*/
"logInfo": ""
/*optional, string, log information, the maximum length is 32 bytes*/
"operationType":,
/*optional, operation type, it is valid only when metaID is set to "Operation": 0-all, 1-software started, 2-software
stopped, 3-configuration page entered, 4-parameters edited, 5-file search, 6-device connected, 7-information
collection started, 8-information collection ended, 9-devices disconnected, 10-password changed, 11-disk space
checked, 12-network status checked*/
"remarks":""
/*optional, operation remarks*/
}
}
}}
}
```

JSON_MoveAutoTracking

JSON message about status of moving object tracking

```
{
  "MoveAutoTracking":{
    "enabled":
/*required, boolean, whether to enable moving object tracking*/
  }
}
```

JSON_OnlineUpgradeParameter

OnlineUpgradeParameter message in JSON format

```
{
  "OnlineUpgradeParameter":{
/*required, online upgrade parameters*/
    "autoDownloadPackage": "",
/*required, whether enables automatic download of upgrade package, boolean, it values "false" by default*/
    "timingUpgrade": "",
/*optional, whether enable scheduled upgrade, boolean, it values "false" by default*/
    "upgradeTime": "",
/*optional, upgrade time, corrects to minute, and second is 0 by default, ISO8601 format, string type, it is valid when
timingUpgrade values "true"*/
  }
}
```

JSON_operType

operType message in JSON format

```
{
  "operType": "",
  /*required, string type, operation type: "unlock"-unlock one or multiple IP addresses, "unlockAll"-unlock all IP addresses*/
  "List": [{
    "IPAddress": {
      "ipVersion": "",
      /*required, string, IP address version information: "v4"-IPv4, "v6"-IPv6. This node is valid only when openType is "unlock"*/
      "ipAddress": ""
    }
  /*required, string, IP address, this node is valid only when openType is "unlock"*/
  }
  ]
}
```

JSON_OSD

OSD message in JSON format

```
{
  "OSD": {
    /*required, OSD information*/
    "overlayItem": "",
    /*required, string, overlaid item, supports multiple items, each item is separated by comma; deviceName-device name, cardNo-card number, eventName-event name, name-name, identityCard-ID card, privacyInfo-private information*/
    "characterEncoding": "",
    /*required, character encoding, string*/
    "fontSize": "",
    /*required, string, font size: big-large, medium-medium, small-small*/
    "fontColor": "",
    /*required, font color, hexBinary*/
    "overlayType": "",
    /*required, overlay type: "flip, scroll", string*/
    "displayTime": "",
    /*required, display time, unit; second, integer*/
    "PrivacyInfo": [ {
      /*optional, private information list, it is required when the value of overlayItem contains "privacyInfo", array*/
      "content": "",
      /*required, private information, string, the max. length is 16*/
    } ],
    "normalizedScreenSize": {
      /*required, read-only, normalized coordinates*/
      "normalizedScreenWidth": "",
      /*required, normalized width, integer*/
    }
  }
}
```

```
"normalizedScreenHeight": ""
/*required, normalized height, integer*/
},
"Rect":{
/*required, OSD position*/
"height": "",
/*required, height, float*/
"width": "",
/*required, width, float*/
"x": "",
/*required, X-coordinate, float*/
"y": "",
/*required, Y-coordinate, float*/
}
}
}
```

JSON_OSDCap

OSDCap message in JSON format

```
{
  "OSDCap":{
    /*required, OSD information*/
    "overlayItem":{
      /*required, string, overlaid item, supports multiple items, each item is separated by comma; deviceName-device
      name, cardNo-card number, eventName-event name, name-name, identityCard-ID card, privacyInfo-private
      information*/
      "@opt": "deviceName,cardNo,eventName,name,identityCard,privacyInfo",
      "#text": "deviceName"
    },
    "characterEncoding":{
      /*required, character encoding, string*/
      "@opt": "GB2312,Latin-1",
      "#text": "GB2312"
    },
    "fontSize":{
      /*required, string, font size: big-large, medium-medium, small-small*/
      "@opt": "big,medium,small",
      "#text": "big"
    },
    "fontColor":{
      /*required, font color, hexBinary*/
      "@opt": "FFFFFF",
    },
    "overlayType":{
      /*required, overlay type: "flip, scroll", string*/
      "@opt": "flip,scroll",
      "#text": "flip"
    },
    "displayTime":{
```

```
/*required, display time, unit; second, integer*/
"@min": 1,
"@max": 120,
"#text": 15
},
"PrivacyInfo": [{
/*optional, private information list, it is required when the value of overlayItem contains "privacyInfo", array*/
"size": 3,
/*required, the max. number of items, integer*/
"content":{
/*required, private information, string, the max. length is 16*/
"@min": 1,
"@max": 16,
"#text": "12345"
},
}],
"Rect":{
/*required, OSD position*/
"height":{
/*required, height, float*/
"@min": 0.000,
"@max": 1.000,
"#text": 0.311
},
"width":{
/*required, width, float*/
"@min": 0.000,
"@max": 1.000,
"#text": 0.311
},
"x":{
/*required, X-coordinate, float*/
"@min": 0.000,
"@max": 1.000,
"#text": 0.311
},
"y":{
/*required, Y-coordinate, float*/
"@min": 0.000,
"@max": 1.000,
"#text": 0.311
},
}
}
}
```

JSON_PeopleCounting_CollectionDescription

CollectionDescription message in JSON format

```
{
  "CollectionDescription":{
    "searchID": "",
    /*required, string, unique search ID, the content of searchID remains unchanged when search condition remains
    unchanged.*/
    "searchResultPosition": "",
    /*required, initial position of search result list, integer type. When there are multiple records, and cannot get all
    records in one time searching, you can search the records followed specified position for next search*/
    "maxResults": "",
    /*required, number of matched records per search, integer type*/
    "timeSpanList": [{
    /*required, time interval list*/
      "startTime": "",
    /*required, start time, ISO8601_time, string*/
      "endTime": "",
    /*required, end time, ISO8601_time, string*/
    }],
    "channels": "",
    /*optional, sub type, array, integer32, related channel No., array. If the channel information is not specified, it
    indicates searching all channels*/
  }
}
```

JSON_PeopleCounting_CollectionDescriptionCap

CollectionDescriptionCap message in JSON format.

```
{
  "CollectionDescriptionCap":{
    "timeRange": "",
    /*required, supported time range for people counting replenishment, unit: day, integer*/
    "timeSpanMaxNum": "",
    /*required, the maximum number of time buckets supported for one time search integer*/
    "resultMaxNum": "",
    /*required, the maximum number of items can be searched, integer*/
    "channelMaxNum": ""
  }
}
```

JSON_PeopleCounting_CollectionResult

CollectionResult message in JSON format

```
{
  "CollectionResult":{
    "responseStatusStrg": "",
    /*required, searching status: OK- Searching ended, NO MATCHES-No matched data found, MORE-Search again for
```

```
more results, string, the max. length is 32, {dep if errcode == 1 && errMsg == ok}*/
  "numOfMatches": "",
  /*required, returned number of results for current search, integer32, {dep if errcode == 1 && errMsg == ok}*/
  "totalMatches": "",
  /*required, total number of matched results, integer32, {dep if errcode == 1 && errMsg == ok}*/
  "targets": [{
    /*optional, people gathering result*/
    "startTime": "",
    /*required, start time of people gathering*/
    "endTime": "",
    /*required, end time of people gathering*/
    "channel": "",
    /*required, integer, channel No.*/
    "enter": "",
    /*required, integer, the number of people entered*/
    "exit": "",
    /*required, integer, the number of people exited*/
    "pass": "",
    /*required, integer, the number of people passed by*/
  }]
}
```

JSON_PictureServerInformation

PictureServerInformation message in JSON format

```
{
  "PictureServerInformation":{
    "pictureServerType": "",
    /*required, string type, picture storage server type: "tomact,VRB,cloudStorage,KMS"*/
    "addressingFormatType": "",
    /*required, string type, format type of the picture storage server address: "ipaddress"-IP address (default),
    "hostname"-host name*/
    "hostName": "",
    /*string type, domain name of the picture storage server, the string length is between 0 and 64. This field is valid
    when addressingFormatType is "hostname"*/
    "ipv4Address": "",
    /*string type, IPv4 address of the picture storage server, the string length is between 0 and 64. This field is valid when
addressingFormatType is "ipaddress"*/
    "ipv6Address": "",
    /*string type, IPv6 address of the picture storage server, the string length is between 0 and 128. This field is valid
    when addressingFormatType is "ipaddress"*/
    "portNo": ,
    /*required, integer type, port No. of the picture storage server, which is between 1024 and 65535*/
    "underlyingProtocol": "",
    /*optional, string, bottom-level protocol of the picture storage server: "HTTP", "HTTPS". This field is valid when
pictureServerType contains "cloudStorage". If this field does not exist, the default bottom-level protocol is HTTP*/
    "cloudStorage":{
      /*parameters of the cloud storage server, which is valid when pictureServerType is "cloudStorage"*/
      "cloudManageHttpPort": ,
```

```
/*required, integer type, HTTP port No. for central management of the cloud storage server, which is between 1024 and 65535*/
"cloudTransDataPort": ,
/*required, integer type, data transmission port No. of the cloud storage server, which is between 1024 and 65535. This field is not supported by access control devices*/
"cloudCmdPort": ,
/*required, integer type, signaling port No. of the cloud storage server, which is between 1024 and 65535*/
"cloudHeartBeatPort": ,
/*required, integer type, heartbeat port No. of the cloud storage server, which is between 1024 and 65535. This field is not supported by access control devices*/
"cloudStorageHttpPort": ,
/*required, integer type, HTTP port No. of the cloud storage server, which is between 1024 and 65535. This field is not supported by access control devices*/
"cloudUsername": "",
/*required, string type, user name of the cloud storage server, the string length is between 0 and 32. This field is not supported by access control devices*/
"cloudPassword": "",
/*required, string type, password of the cloud storage server, the string length is between 0 and 32. This field is not supported by access control devices*/
"cloudPoolId": ,
/*required, integer type, cloud storage pool ID, which is between 1 and 4294967295. If this field is not configured by the upper-level, this field will be set to 1 by default*/
"cloudPoolIdEx": "",
/*optional, string type, cloud storage pool ID, this node is valid when cloud storage pool ID of type string (cloud storage protocol in version 3.0) is supported*/
"cloudProtocolVersion": "",
/*required, string type, protocol version of the cloud storage server, the string length is between 0 and 32*/
"cloudAccessKey": "",
/*string type, cloud storage server access_key, the string length is between 0 and 64. This field is valid when cloudProtocolVersion is "V2.0"*/
"cloudSecretKey": ""
/*string type, cloud storage server secret_key, the string length is between 0 and 64. This field is valid when cloudProtocolVersion is "V2.0"*/
}
}
}
```

JSON_POE

POE message in JSON format

```
{
  "POE": {
    "@opt": "IPv4,IPv6",
    /*optional, IPv4 address, string type, the max. length is 32 bytes*/
    "#text": "IPv4"
    /*optional, IPv6 address, string type, the max. length is 128*/
  }
}
```

JSON_ProgressInfo

ProgressInfo message in JSON format

```
{
  "ProgressInfo":{
    /*required, progress information*/
    "status": "",
    /*required, string, exporting status: "exporting", "failed", "success"*/
    "progress": ""
    /*required, string, exporting progress, format example: 2/101 (there are 101 files in total to be exported, and 2 files
    have been exported)*/
  }
}
```

JSON_RemarkList

JSON message about remark information list of videos

```
{
  "RemarkList": [{
    "trackId": "",
    /*required, file No.*/
    "remarks": "",
    /*required, string, file remarks, which should be encrypted*/
    "recordName": "",
    /*optional, string, file name of video, audio, or picture*/
    "recorderCode": "",
    /*optional, string, ID of device that collects data, which should be encrypted*/
    "policeCode": ""
    /*optional, string, ID of police that collects data, which should be encrypted*/
  }]
}
```

JSON_ReportCenterCfg

ReportCenterCfg message in JSON format

```
{
  "ReportCenterCfg":{
    "enable": ,
    /*optional, boolean type, whether to enable uploading report*/
    "ChanAlarmMode":{
    /*alarm channel of the center group*/
      "id": ,
      /*optional, integer type, channel ID: 1-main channel, 2-backup channel 1, 3-backup channel 2, 4-backup channel 3*/
      "chanAlarmMode": ""
    }
    /*optional, string type, alarm channel mode: "T1"-T1 channel, "T2"-T2 channel, "N1"-N1 channel, "N2"-N2 channel,
```



```
"G1"-G1 channel, "G2"-G2 channel, "N3"-N3 channel, "N4"-N4 channel*/
  }}
}
}
```

JSON_ResponseStatus

ResponseStatus message in JSON format.

```
{
  "requestURL": "",
  /*optional, string type, request URL*/
  "statusCode": ,
  /*required, integer type, status code*/
  "statusString": "",
  /*required, string type, status description*/
  "subStatusCode": "",
  /*required, string type, sub status code*/
  "errorCode": ,
  /*optional, integer type, error code, which corresponds to subStatusCode, this field is required when statusCode is
  not 1*/
  "errorMsg": "",
  /*optional, string type, error details, this field is required when statusCode is not 1*/
}
```



Note

See **ISAPI Response Codes** for details about the status codes, sub status codes, error codes, and error descriptions.

JSON_resourceStatistics

resourceStatistics message in JSON format

```
{
  "inputBandwidth": ,
  /*required, input bandwidth, unit: kbps, float type, corrects to one decimal*/
  "outputBandwidth": ,
  /*required, output bandwidth, unit: kbps, float type, corrects to one decimal*/
  "StreamingInformation": [{
  /*optional, stream information, array*/
    "moduleType": "",
    /*required, module type: "remoteStreaming"-remote streaming, "remotePlayBack"-remote playback, string type, the
    maximum length is 32 bits*/
    "ipAddress": "",
    /*required, device IPv4 address, string type, the maximum length is 32 bits*/
    "ipv6Address": "",
    /*optional, device IPv6 address, string type, the maximum length is 128 bits*/
    "bandwidth":
```

```
/*required, bandwidth, unit: kbps, float type, corrects to one decimal*/
}},
"inputPictureBandwidth": ,
/*optional, picture input bandwidth (including all interfaces of inputting binary picture data), unit: kbps, float type,
corrects to one decimal*/
"outputPictureBandwidth": ,
/*optional, picture output bandwidth (including all interfaces of returning binary picture data), unit: kbps, float type,
corrects to one decimal*/
}
```

JSON_Result

Result message in JSON format

```
{
  "Result":{
    "channels": [1, 2],
    /*optional, integer, linked video channel No.*/
    "IoTChannels": [1, 2]
  }
}
```

JSON_SearchCondition

JSON message about condition information of log search

```
{
  "searchID": "",
  /*required, string, Search ID, which is used to check whether the current search requester is the same as the previous
  one. If they are the same, the search record will be stored in the device to speed up the next search, the maximum
  length is 32 bytes*/
  "timeSpanList": {
    /*required, list of time periods*/
    "timeSpan": {
      /*required, time periods*/
      "startTime": "",
      /*required, string, start time of search, ISO8601 time format, the maximum length is 32 bytes*/
      "endTime": ""
    }
  },
  /*required, string, end time of search, ISO8601 time format, the maximum length is 32 bytes*/
  "metaID": "",
  /*required, string, log type string, log type includes "Alarm"-alarm log, "Exception"-exception log, "Operation"-
  operation log, and "Information"-additional information; e.g., "log.std-cgi.com/Alarm", the maximum length is 64
  bytes*/
  "searchResultPostion": ,
  /*required, integer32, the start position of search result in result list*/
  "maxResults": ,
}
```

```
/*required, integer32, the maximum number of records supported in this search*/
"wildcard ": "",
/*optional, string, wildcard, which is used for fuzzy search, the maximum length is 48 bytes*/
"operationType": ,
/*optional, operation type, it is valid only when metaID is set to "Operation": 0-all, 1-software started, 2-software
stopped, 3-configuration page entered, 4-parameters edited, 5-file search, 6-device connected, 7-information
collection started, 8-information collection ended, 9-devices disconnected, 10-password changed, 11-disk space
checked, 12-network status checked*/
}
```

JSON_secretKey

secretKey message in JSON format

```
{
  "secretKey":"","
/*required, string type, encryption key*/
}
```

JSON_SecurityAdvanced

SecurityAdvanced message in JSON format

```
{
  "SecurityAdvanced ": {
    "securityEnhanced": ,
/*optional, boolean type, whether to enable security reinforcement: true*/
    "noOperationEnabled": ,
/*optional, boolean type, whether to enable control timeout: true*/
    "noOperationTime": ,
/*optional, integer type, control timeout, value range: [1,60], default value: 15, unit: minute*/
    "digestStatus":"SHA256"
/*read-only, string type, digest algorithm: SHA256, MD5*/
  }
}
```

JSON_SecurityEmail

SecurityEmail message in JSON format

```
{
  "SecurityEmail":{
    "SecurityInformation": [{
/*required, recovery email information*/
      "emailAddress": "",
/*required, string type, email address, the sensitive information should be encrypted*/
    }],
}
```

```
},  
}
```

JSON_SecurityEmailCap

SecurityEmailCap message in JSON format

```
{  
  "SecurityEmailCap":  
    /*required, recovery email configuration capability*/  
    {  
      "emailAddressNum": ,  
      /*required, number of recovery emails*/  
      "emailAddress":  
        /*required, string type, recovery email information*/  
        {  
          "@min": 1,  
          "@max": 128,  
        },  
    },  
}
```

JSON_SecurityEmailQrCode

SecurityEmailQrCode message in JSON format

```
{  
  "SecurityEmailQrCode":  
    /*optional, QR code of recovery email*/  
    {  
      "SecurityInformation": [{  
        /*required, security information, the sensitive information should be encrypted*/  
        "emailAddress": "",  
        /*required, string type, email address, the sensitive information should be encrypted*/  
        "qrCode": "",  
        /*required, string type, QR code, the maximum size is 1024 bytes, the sensitive information should be encrypted*/  
      }],  
      "serverEmailAddress": "",  
      /*optional, string type, manufacturer service email address, the sensitive information should be encrypted*/  
      "isDisclaimerDisplay":  
        /*optional, boolean type, whether to display disclaimer*/  
      },  
    }  
}
```

JSON_SerialLogCfg

SerialLogCfg message in JSON format

```
{
  "SerialLogCfg":{
    "enabled":
/*required, boolean type, whether to enable serial port log redirection: "true"-yes, "false"-no*/
  }
}
```

JSON_SerialLogCfgCap

SerialLogCfgCap capability message in JSON format

```
{
  "SerialLogCfgCap":{
    "enabled":"true, false"
/*required, boolean type, whether to enable serial port log redirection: "true"-yes, "false"-no*/
  }
}
```

JSON_ServerCertificateCap

ServerCertificateCap capability message in JSON format

```
{
  "ServerCertificateCap":{
    "customID":{
/*required, string, custom certificate ID, it is created when the user imports the certificate and it consists of digits and characters*/
      "@min":1,
      "@max":64
    },
    "status":{
/*optional, string, certificate status: "normal", "expired", "exceptional"*/
      "@opt":["normal", "expired", "exceptional"]
    },
    "RSAKeyLength":{
/*optional, integer, RSA key length*/
      "@opt":[1024, 2048]
    },
    "countryName":{
/*optional, string, country*/
      "@min":1,
      "@max":32
    },
    "commonName":{
/*optional, string, domain name or IP address*/
      "@min":1,
      "@max":32
    },
  },
}
```

```
"validity":{
/*optional, integer, valid days*/
"@min":1
},
"passwd":{
/*optional, string, password*/
"@min":1,
"@max":64
},
"stateOrProvinceName":{
/*optional, string, province or state*/
"@min":1,
"@max":32
},
"localityName":{
/*optional, string, region*/
"@min":1,
"@max":32
},
"organizationName":{
/*optional, string, organization*/
"@min":1,
"@max":32
},
"organizationUnitName":{
/*optional, string, organization unit*/
"@min":1,
"@max":32
},
"email":{
/*optional, string, email*/
"@min":1,
"@max":32
}
}
```

JSON_ServerCertificates

ServerCertificates message in JSON format

```
{
  "ServerCertificates":{
    "CertificateInfo":[{
      "issuerDN": "",
      /*required, string, read-only, name of the authority that issued the certificate*/
      "subjectDN": "",
      /*required, string, read-only, certificate holder name*/
      "startDate": "",
      /*required, date, read-only, start date of the certificate validity period, it is accurate to day*/
      "endDate": ""
    }
  ]
}
```

```
/*required, date, read-only, end date of the certificate validity period, it is accurate to day*/
  "type":"","
/*optional, string, read-only, certificate type: "HTTPS", "WebSocketS", "SDK_OVER_TLS", "SRTP", "securityLog",
"ieee802.1x". Multiple types should be separated by commas*/
  "status":"","
/*optional, string, read-only, certificate status: "normal", "expired", "exceptional"*/
  "customID":""
/*optional, string, read-only, custom certificate ID, it is created when the user imports the certificate and it consists of
digits and characters*/
  }}
}
}
```

JSON_SmartOverlap

SmartOverlap message in JSON format

```
{
  "SmartOverlap":{
    "enabled": ,
/*required, boolean, whether to enable the stream*/
    "streamType": ,
/*required, integer, stream type: 1-main stream, 2-sub-stream, 3-third stream*/
    "bkgImageOverlapEnabled":
/*required, boolean, whether to overlay rule frame and target frame on the background picture*/
  }
}
```

JSON_SmartOverlapCap

SmartOverlapCap capability message in JSON format

```
{
  "SmartOverlapCap":{
    "enabled":{
/*required, boolean, whether to enable the stream*/
      "@opt":[ true, false]
    },
    "streamType":{
/*required, integer, stream type: 0-disable, 1-main stream, 2-sub-stream, 3-third stream*/
      "@opt":[2]
    },
    "bkgImageOverlapEnabled":{
/*required, boolean, whether to overlay rule frame and target frame on the background picture*/
      "@opt":[true, false]
    },
    "supportEventType":{
/*required, string, supported event type: "fieldDetection"-intrusion, "attendedBaggage"-object removal,
"unattendedBaggage"-unattended baggage, "regionExiting"-region exiting, "regionEntrance"-region entrance,
```

```
"lineDetection"-line crossing*/
  "@opt":["fieldDetection", "attendedBaggage", "unattendedBaggage", "regionExiting", "regionEntrance",
"lineDetection"]
  }
}
}
```

JSON_SourceCapabilities

SourceCapabilities message in JSON format

```
{
  "SourceCapabilities":{
/*required, the supported source capabilities*/
    "IOTChannelNum": "",
/*required, the number of IoT channels, integer*/
    "videoChannelNum": "",
/*optional, the number of video channels, integer*/
    "accessChannels": [{
/*optional, array, added channel No.*/
      "channel": "",
/*required, channel No., integer*/
      "type": "",
/*required, channel type: "video"-video channel, "IoT"-IoT channel, string*/
    }]
  }
}
```

JSON_CollectionDescription

CollectionDescription message in JSON format

```
{
  "CollectionDescription":{
    "searchID": "",
/*required, string, unique search ID, the content of searchID remains unchanged when search condition remains
unchanged.*/
    "searchResultPosition": "",
/*required, initial position of search result list, integer type. When there are multiple records, and cannot get all
records in one time searching, you can search the records followed specified position for next search*/
    "maxResults": "",
/*required, number of matched records per search, integer type*/
    "timeSpanList": [{
/*required, time interval list*/
      "startTime": "",
/*required, start time, ISO8601_time, string*/
      "endTime": "",
/*required, end time, ISO8601_time, string*/
    }],
  },
}
```



```
"channels": "",
/*optional, sub type, array, integer32, related channel No., array. If the channel information is not specified, it
indicates searching all channels*/
"alarmLevel": "TMA",
/*required, string, alarm level: TMA--thermometry alarm, TMPA-thermometry pre-alarm*/
}
}
```

JSON_Temperature_CollectionDescriptionCap

CollectionDescriptionCap message in JSON format

```
{
  "CollectionDescriptionCap":{
    "timeRange": 15,
    /*required, supported time range for replenishment, unit: day, integer */
    "timeSpanMaxNum": 1,
    /*required, the maximum number of time buckets supported for one time search, integer*/
    "resultMaxNum": 100,
    /*required, the maximum number of items can be searched, integer*/
    "channelMaxNum": 1,
    /*required, the maximum number of channels supported for one time search, integer*/
    "alarmLevel":{
      /*required, string, alarm level: TMA--thermometry alarm, TMPA-thermometry pre-alarm*/
      "@opt": "TMA,TMPA"
    }
  }
}
```

JSON_CollectionResult

CollectionResult message in JSON format

```
{
  "CollectionResult":{
    "responseStatusStrg": "",
    /*required, searching status: OK- Searching ended, NO MATCHES-No matched data found, MORE-Search again for
more results, string, the max. length is 32, {dep if errcode == 1 && errMsg == ok}*/
    "numOfMatches": "",
    /*required, returned number of results for current search, integer32, {dep if errcode == 1 && errMsg == ok}*/
    "totalMatches": "",
    /*required, total number of matched results, integer32, {dep if errcode == 1 && errMsg == ok}*/
    "targets": [{
      /*optional, statistics result*/
      "time": "2017-08-25T11:34:59+08:00",
      /*required, statistic time*/
      "channel": "",
      /*required, integer, channel No.*/
      "Region": [{
```

```
/*required, float, X-coordinate, ranges from 0.000 to 1*/
  "x": "",
/*required, float, Y-coordinate, ranges from 0.000 to 1*/
  "y": ""
}},
  "thermometryUnit": "",
/*required, string, temperature unit: celsius, fahrenheit, kelvin*/
  "ruleTemperature": ,
/*required, float, themometry rule*/
  "currTemperature": ,
/*required, float, current temperature*/
  "ruleCalibType": "",
/*required, string, rule calibration type: point,line,region*/
  "ruleType": "",
/*required, string, rule: "highest temp is higher than"-the maximum temperature is higher than, "lowest temp is
higher than"-the minimum temperature is higher than, "average temp is higher than"-the average temperature is
higher than, "temp diff is higher than"-temperature difference is higher than, "highest temp is lower than"-the
maximum temperature is lower than, "lowest temp is lower than"-the minimum temperature is lower than, "average
temp is lower than"-the average temperature is lower than, "temp diff is lower than"-temperature difference is lower
than*/
  "MaximumTemperaturePoint":{
/*optional, the maximum temperature point*/
    "x": "",
/*required, float, X-coordinate, ranges from 0.000 to 1*/
    "y": ""
  },
/*required, float, Y-coordinate, ranges from 0.000 to 1*/
  },
  "AbsoluteHigh":{
/*optional, absolute height*/
    "elevation": "",
/*required, float, elevation*/
    "azimuth": "",
/*required, float, azimuth*/
    "absoluteZoom": ""
  },
/*required, float, absolute zoom*/
  },
  "ruleID": "",
/*required, integer, rule ID*/
  "presetNo": "",
/*required, integer, preset ID*/
  "visibleLightURL": "",
/*optional, string, visible light picture URL*/
  "thermalURL": "",
/*optional, string, thermal picture URL*/
  "thermalInfoURL": "",
/*optional, string, thermal additional information URL*/
  }}
}
}
```

JSON_TestResult

TestResult message in JSON format

```
{
  "TestResult":{
    "status":"ok",
    /*required, string, server status: "ok"-normal, "connectServerFail"-server connection failed
  }
}
```

JSON_TimeSearchCond

Message about recording time search condition in JSON format

```
{
  "TimeSearchCond": {
    "channelID": ,
    /*required, int, channel No.*/
    "streamType": ""
    /*required, string, stream type, "main,sub,adaptive"*/
  }
}
```

JSON_TimeSearchResult

Message about recording time search result in JSON format

```
{
  "TimeSearchResult": {
    "channelID": ,
    /*required, int, channel No.*/
    "streamType": "",
    /*required, string, stream type, "main,sub,adaptive"*/
    "hasRecord": ,
    /*required, boolean, whether video exists, startTime and endTime are valid only when the value of hasRecord is
    "true"*/
    "startTime": "",
    /*optional, string, recording start time, e.g., 2010-01-01T00:00:00+08:00, this field is required when the value of
    hasRecord is "true"*/
    "endTime": "",
    /*optional, string, recording end time, e.g., 2010-01-01T00:00:00+08:00, this field is required when the value of
    hasRecord is "true"*/
  }
}
```

JSON_TimeType

TimeType message in JSON format

```
{
  "TimeType":{
    "type": ""
  /*required, value: local, UTC*/
  }
}
```

JSON_TrackingRatio

JSON message about tracking parameters

```
{
  "TrackingRatio":{
  /*required, tracking parameters of the speed dome*/
    "coefficient": ,
  /*required, int, tracking zoom ratio, it is between 1 and 100 and the default value is 50. The tracking zoom ratio is the
  ratio of the target to the image (unit: percentage). The radar will send the target's distance to the application during
  tracking. For human bodies, the ratio of a person's height to the image will be calculated assuming that the person is
  1.7 meters tall; for vehicles, the ratio of the vehicle's width to the image will be calculated assuming that the vehicle is
  2 meters wide*/
    "durationTime": ,
  /*required, int, tracking duration, it is between 5 and 60 and the default value is 20, unit: second*/
    "whiteLight":
  /*optional, boolean, whether to support the flashing light*/
    "trackingTakeoverEnabled":
  /*required, boolean, whether to enable tracking takeover*/
  }
}
```

JSON_TrackingRatioCap

JSON message about the configuration capability of tracking parameters

```
{
  "TrackingRatioCap":{
  /*required, configuration capability of the speed dome's tracking parameters*/
    "coefficient":{
  /*required, int, tracking zoom ratio, it is between 1 and 100 and the default value is 50. The tracking zoom ratio is the
  ratio of the target to the image (unit: percentage). The radar will send the target's distance to the application during
  tracking. For a human body, the ratio of its height to the image will be calculated assuming that the person is 1.7
  meters tall; for a vehicle, the ratio of its width to the image will be calculated assuming that the vehicle is 2 meters
  wide*/
    "@min":1,
    "@max":100,
```

```
"@def":50
},
"durationTime":{
/*required, int, tracking duration, it is between 5 and 60 and the default value is 20, unit: second*/
"@min":5,
"@max":60,
"@def":20
},
"whiteLight":{
/*optional, boolean, whether to support the flashing light*/
"@opt":[true, false]
}
"trackingTakeoverEnabled": {
/*optional, boolean, whether to enable tracking takeover; by default, it is false*/
"@opt": [true, false]
}
}
}
```

JSON_TrafficMonitor

TrafficMonitor message in JSON format

```
{
  "TrafficMonitor":{
    "enabled":,
    /*required, boolean, whether to enable traffic monitoring*/
    "packageType": "",
    /*required, string, data plan types: "day"-daily plan, "month"-monthly plan, "year"-annual plan */
    "trafficValue":,
    /*required, float, data value, which should be accurate to three decimal places, unit:MB */
    "overrunOperation": "",
    /*optional, string, operations when the data exceeds: "brokenNetworkAndAlert"-turn off the cellular network and
    alert users, "alert"-notify users */
    "alertType":[
      "beep",
      "email",
      "SMS"
    ],
    /*required, string, notification methods: "beep"-audio warning, "email"-send email, "SMS"-send message, "center"-
    upload to center*/
    "alertValue":,
    /*required, integer, alarm value(%)*/
    "startDay":
    /*optional, integer, the start day of the monthly plan, which is valid only when packageTypeis "month"*/
  }
}
```

JSON_TrafficMonitorCap

TrafficMonitorCap message in JSON format

```
{
  "TrafficMonitorCap":{
    "enabled":"","
    /*required, boolean, whether to enable traffic monitoring*/
    "packageType":{
      "@opt":[
        "day",
        "month",
        "year"
      ]
    },
    /*required, string, data plan types: "day"-daily plan, "month"-monthly plan, "year"-annual plan*/
    "overrunOperation":{
      "@opt":[
        "brokenNetworkAndAlert",
        "alert"
      ]
    },
    /*optional, string, operations after the data exceeds: "brokenNetworkAndAlert"-turn off the cellular network and alert users, "alert"-notify users */
    },
    "alertType":{
      "@opt":[
        "beep",
        "email",
        "SMS",
        "center"
      ]
    },
    },
    /*required, string, notification methods: "beep"-audio warning, "email"-send email, "SMS"-send message, "center"-upload to center*/
    "alertValue":{
      "@min":,
      "@max":
    },
    },
    /*required, integer, alarm value(%)*/
    "supportTrafficMonitorStatus":
    /*optional, boolean, whether it supports getting the current network data usage*/
  }
}
```

JSON_TrafficMonitorStatus

TrafficMonitorStatus message in JSON format

```
{
  "TrafficMonitorStatus":{
    "remainTrafficValue":,
    /*required, float, remaining data, which should be accurate to three decimal places, unit:MB*/
    "trafficValue":,
    /*required, float, data value, which should be accurate to three decimal places, unit:MB*/
    "todayUsedTrafficValue":
    /*required, float, today's data usage, which should be accurate to three decimal places, unit:MB*/
    "todayUsedUpstreamTrafficValue":,
    /*optional, float, today's upstream data usage, which should be accurate to three decimal places, unit:MB*/
    "todayUsedDownstreamTrafficValue": ,
    /*optional, float, today's downstream data usage, which should be accurate to three decimal places, unit:MB*/
    "monthUsedTrafficValue": ,
    /*optional, float, data usage of this month, which should be accurate to three decimal places, unit:MB*/
    "monthUsedUpstreamTrafficValue": ,
    /*optional, float, upstream data usage of this month, which should be accurate to three decimal places, unit:MB*/
    "monthUsedDownstreamTrafficValue": ,
    /*optional, float, downstream data usage of this month, which should be accurate to three decimal places, unit:MB*/
    "yearUsedTrafficValue": ,
    /*optional, float, data usage of this year, which should be accurate to three decimal places, unit:MB*/
    "yearUsedUpstreamTrafficValue": ,
    /*optional, float, upstream data usage of this year, which should be accurate to three decimal places, unit:MB*/
    "yearUsedDownstreamTrafficValue": ,
    /*optional, float, downstream data usage of this year, which should be accurate to three decimal places, unit:MB*/
    "enabled": ,
    /*optional, boolean, whether traffic monitoring is enabled*/
    "packageType": "",
    /*optional, string, data plan types: "day"-daily plan, "month"-monthly plan, "year"-annual plan*/
    "alertValue":
    /*optional, integer, alarm value(%)*/
  }
}
```

JSON_unitConfig

unitConfig message in JSON format

```
{
  "enabled": "",
  /*required, enable unit unifying configuration or not, boolean type*/
  "temperatureRange": "",
  /*optional, string type, temperature unit: degreeCentigrade-Centigrade (°C), degreeFahrenheit-Fahrenheit (°F) ,
  degreeKelvin-Kelvin (K), the default value is "degreeCentigrade"*/
  "distanceUnit": ""
  /*optional, string type, distance unit: centimeter, meter, feet, the default value is "centimeter"*/
}
```

Remarks

- When the **enabled** values "false", it indicates the unit unifying function is disabled, the unit can be configured in thermometry basic settings, and the configured unit has no influence on the unit in system settings.
- When the **enabled** values "false", the thermometry unit configuration in applied URL remains unchanged and takes effect; while when the **enabled** values "true" and the thermometry unit is different with the system unit, the thermometry unit configuration in applied URL takes no effect and the error code will be returned.

JSON_unitConfigCap

unitConfigCap message in JSON format

```
{
  "enabled":"true,false",
  /*required, enable unit unified configuration or not, boolean type*/
  "temperatureRange":{
    /*optional, string type, temperature unit: degreeCentigrade-Centigrade (°C), degreeFahrenheit-Fahrenheit (°F) ,
    degreeKelvin-Kelvin (K), the default value is "degreeCentigrade"*/
    "@opt":"degreeCentigrade,degreeFahrenheit,degreeKelvin"
  },
  "distanceUnit":{
    /*optional, string type, distance unit: centimeter, meter, feet, the default value is "centimeter"*/
    "@opt":"centimeter,meter,feet"
  }
}
```

JSON_USB

JSON message about USB parameters

```
{
  "USB": {
    "usbMode": ""
  }
  /*optional, string, USB mode: "UVC", "NCM"*/
}
```

JSON_USBCap

JSON message about USB management capability

```
{
  "USBCap": {
    "usbMode": {
```



```
/*optional, string, USB mode: "UVC", "NCM"*/
"@opt":["UVC", "NCM"],
"#text":"UVC"
}
}
}
```

JSON_User

User message in JSON format

```
{
  "User":{
    "userName": "",
    /*required, string, double verification user name*/
    "password": ""
    /*required, string, double verification password*/
  }
}
```

JSON_UserCap

UserCap message in JSON format

```
{
  "UserCap":{
    "userMaxNumber": "",
    /*required, the maximum number of supported double verification users*/
    "userNameLen":{
      /*required, the length of double verification user name*/
      "@min": "",
      "@max": "",
    },
    "passwordLen":{
      /*required, the length of double verification user password*/
      "@min": ,
      "@max": ,
    }
  }
}
```

JSON_UserList

UserList message in JSON format

```
{
  "UserList": [{
```

```
"id": ,
/*required, integer, user ID*/
"userName":"","
/*required, string, double verification user name*/
}}
}
```

JSON_UserPermission

UserPermission message in JSON format

```
{
  "UserPermission":{
    "localPlayback":{
      /*optional, local playback permission*/
      "enable": "",
      /*required, enables permission or not, boolean*/
      "channelPermission": [{
        /*optional, channel permission*/
        "channelID": ,
        /*required, channel ID, integer*/
        "enable": ""
      }
      /*required, enables permission or not, boolean*/
    }
  },
  "localBackup":{
    /*optional, local backup permission*/
    "enable": "",
    /*required, enables permission or not, boolean*/
    "channelPermission": [{
      /*optional, channel permission*/
      "channelID": ,
      /*required, channel ID, integer*/
      "enable": ""
    }
    /*required, enables permission or not, boolean*/
  },
  "remotePlayback":{
    /*optional, remote playback and download permission*/
    "enable": "",
    /*required, enables permission or not, boolean*/
    "channelPermission": [{
      /*optional, channel permission*/
      "channelID": ,
      /*required, channel ID, integer*/
      "enable": ""
    }
    /*required, enables permission or not, boolean*/
  }
}
```

```
}  
}
```

JSON_UserPermissionCap

UserPermissionCap message in JSON format

```
{  
  "UserPermissionCap":{  
    "localPlayback":{  
      /*required, local playback permission*/  
      "channelID":{  
        /*required, the channel which supports configuring local playback permission*/  
        "@min": "",  
        "@max": ,  
      },  
      "enable": "true,false"  
      /*required, enables permission or not, boolean*/  
    },  
    "localBackup":{  
      /*required, local backup permission*/  
      "channelID":{  
        /*required, the channel which supports configuring local backup permission*/  
        "@min": "",  
        "@max": ,  
      },  
      "enable": "true,false"  
      /*required, enables permission or not, boolean*/  
    },  
    "remotePlayback":{  
      /*required, remote playback permission*/  
      "channelID":{  
        /*required, the channel which supports configuring remote playback permission*/  
        "@min": "",  
        "@max": "",  
        "#text": ""  
      },  
      "enable": "true,false"  
      /*required, enables permission or not, boolean*/  
    }  
  }  
}
```

JSON_VerificationDisplay

JSON message about parameters of calibration verification result display

```
{  
  "VerificationDisplay": {
```

```
/*required, display calibration verification result*/
  "enabled":
/*required, boolean, whether to display the calculated point position and verification point position on video: true-yes*/
}
}
```

JSON_VerificationPoint

JSON message about calibration verification point information

```
{
  "VerificationPoint":[
/*required, calibration verification point*/
    "pan": ,
/*optional, float, pan, range: [0.00,360.00]*/
    "tilt": ,
/*optional, float, tilt, range: [-90.00,90.00]*/
    "zoom": ,
/*optional, float, zoom, range: [0.00,90.00]*/
    "longitude": ,
/*required, float, longitude which is calculated based on the pan and tilt, corrects to 6 decimal places, range: [-180.000000,180.000000]*/
    "latitude": ,
/*required, float, latitude which is calculated based on the pan and tilt, corrects to 6 decimal places, range: [-90.000000,90.000000]*/
    "x": ,
/*optional, float, x-coordinate, [0.000,1.000]*/
    "y":
/*optional, float, y-coordinate, [0.000,1.000]*/
  ]
}
```

JSON_VerificationPointList

JSON message about verification points information

```
{
  "VerificationPointList":[{
/*required, calibration verification points*/
    "id": ,
/*required, int, verification point ID; PanoVu series camera supports up to 16 points, and speed dome supports up to 64 points*/
    "pan": ,
/*optional, float, pan, range: [0.00,360.00]*/
    "tilt": ,
/*optional, float, tilt, range: [-90.00,90.00]*/
    "zoom": ,
/*optional, float, zoom, range: [0.00,90.00]*/
  }
]
```

```
"longitude": ,
/*required, float, longitude, corrects to 6 decimal places, range:[-180.000000,180.000000]*/
"latitude": ,
/*required, float, latitude, corrects to 6 decimal places, range: [-90.000000,90.000000]*/
"x": ,
/*optional, float, x-coordinate, [0.000,1.000]*/
"y":
/*optional, float, y-coordinate, [0.000,1.000]*/
}}
}
```

JSON_VerificationPointsCap

JSON message about capability of verify calibration

```
{
  "VerificationPointsCap":{
    /*required, calibration verification capability*/
    "id":{
      /*required, int, verification point ID; PanoVu series camera supports up to 16 points, and speed dome supports up to 64 points*/
      "@min":1,
      "@max":64
    },
    "pan":{
      /*optional, float, pan*/
      "@min":0,
      "@max":360
    },
    "tilt":{
      /*optional, float, tilt*/
      "@min":-90,
      "@max":90
    },
    "zoom":{
      /*optional, float, zoom*/
      "@min":0,
      "@max":90
    },
    "longitude":{
      /*required, float, longitude, corrects to 6 decimal places*/
      "@min":-180,
      "@max":180
    },
    "latitude":{
      /*required, float, latitude, corrects to 6 decimal places*/
      "@min":-90,
      "@max":90
    },
    "x":{
      /*optional, float, x-coordinate*/
```

```
"@min":0,
"@max":1
},
"y":{
/*optional, float, y-coordinate*/
"@min":0,
"@max":1
},
"isSupportDisplay":
/*optional, boolean, whether device supports display the verification result (related URI: /ISAPI/System/
GPSVerification/channels/<ID>/display?format=json)*/
}
}
```

JSON_VibrationDetection

Vibration detection message in JSON format

```
{
  "VibrationDetection": {
/*vibration detection*/
    "enabled": ,
/*required, boolean, whether to enable vibration detection, "true"-enable, "false"-disable*/
    "sensitivity":
/*required, int, sensitivity, value range: [0-100], by default: 50*/
  }
}
```

JSON_VibrationDetectionCap

Message about vibration detection capability in JSON format

```
{
  "VibrationDetectionCap": {
/*vibration detection capability*/
    "enabled": {
/*required, boolean, whether to enable vibration detection, "true"-enable, "false"-disable*/
      "@opt": [true, false]
    },
    "sensitivity": {
/*required, int, sensitivity, value range: [0-100]*/
      "@min": 0,
      "@max": 100,
      "@def": 50
    }
  }
}
```

JSON_WhiteLightAlarm

WhiteLightAlarm message in JSON format

```
{
  "WhiteLightAlarm":{
    "durationTime": ,
    /*required, int, flashing duration of supplement light, which is between 1 and 60 s*/
    "frequency": "",
    /*required, string, flashing frequency of supplement light: "high"-flashing 0.5s per second, "medium"-flashing 0.8s per
    1.6s, "low"-flashing 1s per 2s*/
    "brightness": ,
    /*optional, int, supplement light brightness, which is between 1 and 100*/
    "TimeRangeList": [{
    /*optional, alarm output schedule list*/
      "week": ,
      /*required, int, days of the week: 1-Monday, 2-Tuesday, 3-Wednesday, 4-Thursday, 5-Friday, 6-Saturday, 7-Sunday*/
      "TimeRange": [{
        "id": ,
        /*required, int, ID of time period for each day*/
        "beginTime": "",
        /*required, string, start time in ISO8601 format*/
        "endTime": ""
        /*required, string, end time in ISO8601 format*/
      }]
    }]
  }
}
```

JSON_WhiteLightAlarmCap

WhiteLightAlarmCap message in JSON format

```
{
  "WhiteLightAlarmCap":{
    "durationTime":{
    /*required, int, flashing duration of supplement light, which is between 1 and 60 s*/
      "@min": 1,
      "@max": 60,
      "@def": 15
    },
    "frequency":{
    /*required, string, flashing frequency of supplement light: "high"-flashing 0.5s per second, "medium"-flashing 0.8s per
    1.6s, "low"-flashing 1s per 2s*/
      "@opt": "high,medium,low,normallyOn",
      "@def": "high"
    },
    "brightness":{
    /*optional, int, supplement light brightness, which is between 1 and 100*/
    }
  }
}
```

```
"@min": 1,
"@max": 100,
"@def": 50
},
"TimeRangeCap":{
/*optional, alarm output schedule capability*/
"week":{
/*required, int, days of the week: 1-Monday, 2-Tuesday, 3-Wednesday, 4-Thursday, 5-Friday,6-Saturday, 7-Sunday*/
"@opt":"1,2,3,4,5,6,7"
},
"id":{
/*required, int, ID of time period for each day*/
"@maxSize": 8
},
"beginTime":{
/*required, string, start time in ISO8601 format*/
"@min":"00:00",
"@max":"24:00"
},
"endTime":{
/*required, string, end time in ISO8601 format*/
"@min":"00:00",
"@max":"24:00"
}
},
"whiteLightMode":{
/*optional, string, supplement light mode: "keepOn"-always on, "flashing"-flashing*/
"@opt":"keepOn,flashing",
"@def":"keepOn"
}
}
}
```

JSON_WiresharkToUSB

JSON message about parameters of exporting captured packets to USB flash drive

```
{
  "WiresharkToUSB": {
    "switch": "",
    "operateCmd": "",
    "DeviceCondList": [
      {
        "DeviceCond": {
          "id": ,
          "devType": "",
          "devID": ,
          "subSys": ,
        }
      }
    ]
  }
}
```



```
}  
}
```

JSON_WiresharkToUSBStatus

JSON message about status of exporting captured packets to USB flash drive

```
{  
  "WiresharkToUSBStatus":{  
    "WiresharkExportList":[{  
      "WiresharkExport":{  
        "devType": "",  
        "devID": ,  
        "subSys": ,  
        "switch": "",  
        "operateCmd": "",  
        "exportStatus": "",  
        "exceptionMsg": ""  
      }  
    }  
  }  
}
```

JSON_WorkingStatus

WorkingStatus message in JSON format

```
{  
  "WorkingStatus":{  
    "devStatus": ,  
    /*required, integer type, value of device status: 0-normal, 1-CPU usage, higher than 85%, 2-hardware error (e.g., serial  
    port exception)*/  
    "ChanStatus":[{  
      "chanNo": ,  
      /*required, integer type, channel No., which starts from 1*/  
      "enable": ,  
      /*integer type, it is valid for analog channel only, 0-disable, 1-enable*/  
      "online": ,  
      /*required, integer type, online status: 0-offline, 1-online*/  
      "record": ,  
      /*required, whether the device is recording, 0-no, 1-yes*/  
      "recordStatus": ,  
      /*integer type, recording status: 0-recording, 1-recording exception (HDD exception), 2-recording exception (network  
      camera offline), 3-recording exception (other reason)*/  
      "signal": ,  
      /*required, integer type, signal status: 0-normal, 1-signal loss*/  
      "linkNum": ,  
      /*required, integer type, number of software clients connected to this channel*/  
      "bitRate":
```

```
/*required, integer type, channel bit rate, unit: Kpbs*/
  },
  "HDStatus":{
    "hdNo": ,
/*required, integer type, HDD No., which starts from 1*/
    "status": ,
/*required, integer type, HDD status: 0-activate, 1-sleep, 2-exception, 3-sleepy HDD error, 4-unformatted, 5-
disconnected (for network HDD), 6-formatting*/
    "volume": ,
/*required, integer type, HDD capacity, unit: MB*/
    "freeSpace":
/*required, integer type, free space, unit: MD*/
  },
  "IOStatus":{
    "IOInTrig":[...,...,...],
/*integer type, triggered alarm input No., two places and below: analog alarm input No., two places and above: digital
alarm input No.; and the low 2-bit is the alarm input No., the 3-bit or above is the digital channel No., e.g., 3201-alarm
input No.1 of digital channel No.32*/
    "IOOutTrig":[...,...,...]
/*integer type, triggered alarm output No., two places and below: analog alarm output No., two places and above:
digital alarm output No.; and the lower 2 bits are the alarm output No., the 3-bit or above is the digital channel No.,
e.g., 3201-alarm output No.1 of digital channel No.32*/
  }
}
}
```

JSON_XX

XX message in JSON format

```
{
  "XX":{
/*required, event/alarm configuration information*/
    "enabled": "",
/*required, enable or not, boolean*/
    "zoneName": ""
/*optional, zone name, it is valid only when event type is zone, string*/
  }
}
```

Remarks

The XX in the parameter "**XX**" corresponds to detailed event type. E.g., if the event type is humanRecognition, then the returned node is "HumanRecognition".

JSON_XXCap

XXCap message in JSON format

```
{
  "XXCap":{
    /*required, event configuration information*/
    "enabled": "true,false",
    /*required, enable or not, boolean*/
    "zoneName":{
      /*optional, zone name, it is valid only when event type is zone, string*/
      "@min": 1,
      "@max": 64,
    },
  }
}
```

Remarks

The XX in the parameter "**XXCap**" corresponds to detailed event type. E.g., if the event type is humanRecognition, then the returned node is "HumanRecognitionCap".

A.1.2 XML Messages

XML_accessDevice

accessDevice message in XML format

```
<accessDevice version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:integer--></id>
  <MACAddress><!--opt, xs:string--></MACAddress>
  <ipV4Address><!--dep, xs:string--></ipV4Address>
  <accessTime><!--req, xs:time, ISO8601 data--></accessTime>
</accessDevice>
```

XML_accessDeviceList

accessDeviceList message in XML format

```
<accessDeviceList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <accessDevice/><!--see details in the message of XML_accessDevice-->
</accessDeviceList>
```

See Also

XML_accessDevice

XML_accessPoint

accessPoint message in XML format

```
<accessPoint version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:integer--></id>
  <networkMode>
    <!--opt, xs:string, "infrastructure, adhoc"-->
  </networkMode>
  <channel><!--opt, xs:string, "1 to 14, auto"--></channel>
  <ssid><!--req, xs:string--></ssid>
  <speed><!--opt, xs:integer, unit: Mbps--></speed>
  <signalStrength><!--opt, xs:integer, signal strength, which is between 0 and 100--></signalStrength>
  <securityMode>
    <!--req, xs:string, "disable,WEP,WPA-personal,WPA2-personal,WPA-RADIUS,WPA-enterprise,WPA2-enterprise"-->
  </securityMode>
  <connected><!--opt, xs:boolean--></connected>
  <connecting><!--opt, xs:boolean, whether the device is connecting to the Wi-Fi--></connecting>
</accessPoint>
```

XML_accessPointList

accessPointList message in XML format

```
<accessPointList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <accessPoint/><!--see details in the message of XML_accessPoint>
</accessPointList>
```

See Also

XML_accessPoint

XML_ActivateInfo

ActivateInfo message in XML format.

```
<ActivateInfo version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <password><!--req, xs:string--></password>
</ActivateInfo>
```

XML_AdminAccessProtocol

AdminAccessProtocol message in XML format

```
<AdminAccessProtocol version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--required, xs:string, ID--></id>
  <enabled><!--optional, xs:boolean, "true,false"--></enabled>
  <protocol><!--required, xs:string, protocol name:
  "HTTP,HTTPS,RTSP,DEV_MANAGE,IOT,WebSocket,WebSocketS,SDK_OVER_TLS,TLS1_1Enable,TLS1_2Enable,SRTP,MATRI
  X_GATEWAY,Bonjour"--></protocol>
  <portNo><!--required, xs:integer--></portNo>
```

```
<redirectToHttps>
  <!--opt, xs: boolean, whether to automatically go to HTTPS when connecting to HTTP port and HTTPS is enabled-->
</redirectToHttps>
<streamOverTls>
  <!--opt, xs: boolean, whether to enable TLS link encryption when the streaming mode is SDK_OVER_TLS, true=yes,
false=no-->
</streamOverTls>
<TLS1_1Enable><!--dep, xs:boolean--></TLS1_1Enable>
<TLS1_2Enable><!--dep, xs:boolean--></TLS1_2Enable>
</AdminAccessProtocol>
```

Remarks

"TLS1_1Enable" and "TLS1_2Enable" are not available for port configuration.

XML_AdminAccessProtocolList

AdminAccessProtocolList message in XML format

```
<AdminAccessProtocolList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <AdminAccessProtocol/><!--see details in the message of XML_AdminAccessProtocol-->
</AdminAccessProtocolList>
```

See Also

XML_AdminAccessProtocol

XML_AlgorithmsVersion

Message about version information of algorithm library in XML format

```
<AlgorithmsVersion version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <AlgorithmsVersionInfoList><!--optional-->
    <AlgorithmsVersionInfo>
      <majorVersion><!--required, xs:integer, major version No.--></majorVersion>
      <minorVersion><!--required, xs:integer, minor version No.--></minorVersion>
      <revisionNumber><!--required, xs:integer, revision No.--></revisionNumber>
      <buildNumber><!--required, xs:integer, build No.--></buildNumber>
      <versionYear><!--required, xs:integer, date: year--></versionYear>
      <versionMonth><!--required, xs:integer, date: month--></versionMonth>
      <versionDay><!--required, xs:integer, date: day--></versionDay>
      <name>
        <!--required, xs:string, algorithm library name, "faceProperties"-face attributes, "behaviorAnalysis"-behavior
analysis, "humanAttribute"-human body attributes, "faceSnap"-face capture, "faceRecognition"-face recognition,
"faceContrast"-face picture comparison, "peopleCounting"-people counting, "faceScore"-face picture score,
"personQueueDetection"-queue management, "depthMap"-depth map, "safetyHelmetDetection"-hard hat detection,
"streetBehavior"-street behavior, "workBehavior"-work behavior, "cityManagement"-city management,
"fireEscapeDetection"-fire engine access detection, "takingElevatorDetection"-elevator detection-->
        <name>
      </AlgorithmsVersionInfo>
```

```
</AlgorithmsVersionInfoList>
</AlgorithmsVersion>
```

Example

AlgorithmsVersion Message Example

```
<AlgorithmsVersion version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <AlgorithmsVersionInfoList>
    <AlgorithmsVersionInfo>
      <majorVersion>1</majorVersion>
      <minorVersion>0</minorVersion>
      <revisionNumber>1</revisionNumber>
      <versionYear>17</versionYear>
      <versionMonth>12</versionMonth>
      <versionDay>12</versionDay>
      <name>faceSnap</name>
    </AlgorithmsVersionInfo>
    <AlgorithmsVersionInfo>
      <majorVersion>1</majorVersion>
      <minorVersion>0</minorVersion>
      <revisionNumber>0</revisionNumber>
      <versionYear>17</versionYear>
      <versionMonth>10</versionMonth>
      <versionDay>31</versionDay>
      <name>faceScore</name>
    </AlgorithmsVersionInfo>
    <AlgorithmsVersionInfo>
      <majorVersion>3</majorVersion>
      <minorVersion>4</minorVersion>
      <revisionNumber>1</revisionNumber>
      <versionYear>17</versionYear>
      <versionMonth>12</versionMonth>
      <versionDay>25</versionDay>
      <name>faceContrast</name>
    </AlgorithmsVersionInfo>
  </AlgorithmsVersionInfoList>
</AlgorithmsVersion>
```

XML_anonymouslogin

XML message about parameters of managing anonymous user permission

```
<?xml version="1.0" encoding="utf-8"?>
<anonymouslogin version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--required, xs:boolean--></enabled>
</anonymouslogin>
```

XML_AudioCap

AudioCap capability message in XML format

```
<AudioCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <audioInputNums><!--req, xs:integer--></audioInputNums>
  <audioOutputNums><!--req, xs:integer--></audioOutputNums>
</AudioCap>
```

XML_AudioChannel

AudioChannel message in XML format

```
<AudioChannel version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:string--></id>
  <enabled><!--req, xs:boolean--></enabled>
</AudioChannel>
```

XML_AudioChannelList

AudioChannelList message in XML format

```
<AudioChannelList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <AudioChannel/><!--opt, see details in the message of XML_AudioChannel-->
</AudioChannelList>
```

See Also

XML_AudioChannel

XML_AudioDescriptor

AudioDescriptor message in XML format

```
<AudioDescriptor version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <audioCompressionType>
    <!--opt, xs: string, audio encoding type: "G.711alaw,G.711ulaw,G.726,G.729,G.729a,G.
729b,PCM,MP3,AC3,AAC,ADPCM,MP2L2"-->
  </audioCompressionType>
  <audioSamplingRate><!--opt, xs: float, kHz--></audioSamplingRate>
</AudioDescriptor>
```

XML_AudiIn

AudiIn message in XML format

```
<AudiIn version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id>
    <!--req, xs:string-->
  </id>
  <MixAudiIn>
    <!--req, ro-->
    <enabled>
      <!--req, xs:Boolean-->
    </enabled>
    <audioInputType>
      <!--opt, xs:string-->
    </audioInputType>
    <highPassFilter>
      <!--req, xs:integer, unit: Hz-->
    </highPassFilter>
    <noiseMargin>
      <!--req, xs:integer, unit: dB-->
    </noiseMargin>
    <AutoLimitWave>
      <!--dep-->
      <FBCEnable>
        <!--req, xs:Boolean-->
      </FBCEnable>
      <mode>
        <!--req, xs:string-->
      </mode>
      <filterQValue>
        <!--req, xs:string-->
      </filterQValue>
      <staticFilterNum>
        <!--req, xs:integer-->
      </staticFilterNum>
    </AutoLimitWave>
  </MixAudiIn>
  <AudiInVolumelist>
    <AudiInVlome>
      <type>
        <!--req, xs:string, "audioInput,audioEncode"-->
      </type>
      <volume>
        <!--req, xs:integer-->
      </volume>
    </AudiInVlome>
  </AudiInVolumelist>
</AudiIn>
```


XML_AudioInCap

AudioInCap capability message in XML format

```
<AudioInCap version="2.0" xmlns="http://www.wisapi.org/ver20/XMLSchema">
  <id>
    <!--req, xs:string-->
  </id>
  <MixAudioIn>
    <!--req, ro-->
    <enabled opt="true,false">
      <!--req, xs:Boolean-->
    </enabled>
    <audioInputType opt="micIn, lineIn">
      <!--opt, xs:string-->
    </audioInputType>
    <highPassFilter
      opt="0,8,16,24,31,39,47,55,63,71,79,87,94,102,110,118,126,134,142,150,157,165,173,181,189,197,205,213,220,228,
      236,244,252,260,268,276,283,291,299,307,315,323,331,339,346,354,362,370,378,386,394,402,409,417,425,433,441,
      449,457,465,472,480,488,496,504,512,520,528,535,543,551,559,567,575,583,591,598,606,614,622,630,638,646,654,
      661,669,677,685,693,701,709,717,724,732,740,748,756,764,772,780,787,795,803,811,819,827,835,843,850,858,866,
      874,882,890,898,906,913,921,929,937,945,953,961,969,976,984,992,1000" default="30">
      <!--req, xs:integer, unit: Hz-->
    </highPassFilter>
    <noiseMargin
      opt="1000,660,657,653,650,647,643,640,637,633,-630,627,623,620,617,613,610,607,603,600,597,593,590,587,583,5
      80,577,573,570,567,563,560,557,553,550,547,543,540,537,533,530,527,523,520,517,513,510,507,503,500,497,493,4
      90,487,483,480,477,473,470,467,463,460,457,453,450,447,443,440,437,433,430,427,423,420,417,413,410,407,403,4
      00,397,393,390,387,383,380,377,373,370,367,363,360,357,353,350,347,343,340,337,333,330,327,323,320,317,313,3
      10,307,303,300,297,293,290,287,283,280,277,273,270,267,263,260,257,253,250,247,243,240" default="30">
      <!--req, xs:integer, unit: dB-->
    </noiseMargin>
    <AutoLimitWave>
      <!--dep, audioInputType-->
      <FBCEnable opt="true,false">
        <!--req, xs:Boolean-->
      </FBCEnable>
      <mode opt="fast,general,slow" default="general">
        <!--req, xs:string-->
      </mode>
      <filterQValue opt="40,10" default="40">
        <!--req, xs:integer-->
      </filterQValue>
      <staticFilterNum min="0" max="12" default="0">
        <!--req, xs:integer-->
      </staticFilterNum>
    </AutoLimitWave>
  </MixAudioIn>
  <AudioInVolumelist>
    <AudioInVlome>
      <type>
```

```
<!--req, xs:string, "audioOutput,audioEncode"-->
</type>
<volume min="0" max="127" default="50">
  <!--req, xs:integer-->
</volume>
</AudioInVlome>
</AudioInVolumelist>
</AudioInCap>
```

XML_AudioOut

AudioOut message in XML format

```
<AudioOut version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:string--></id>
  <MixAudioOut>
    <enabled><!--req, xs:Boolean--></enabled>
    <modulatorEnable><!--req, xs:Boolean--></modulatorEnable>
    <postFilter><!--req, xs:Boolean--></postFilter>
    <limitPressure><!--req, xs:Boolean--></limitPressure>
    <modulatorValue><!--req, xs:integer--></modulatorValue>
    <triggerTime><!--req, xs:integer, unit: ms--></triggerTime>
    <freeTime><!--req, xs:integer, unit: ms--></freeTime>
    <compressThreshold><!--req, xs:integer--></compressThreshold>
    <compressMode><!--req, xs:string--></compressMode>
    <compressRate><!--req, xs:integer--></compressRate>
    <recoveryGain><!--req, xs:integer--></recoveryGain>
    <outputGain><!--req, xs:integer --></outputGain>
  </MixAudioOut>
  <AudioOutVolumelist>
    <AudioOutVlome>
      <type><!--req, xs:string, "audioOutput,audioDecode"--></type>
      <volume><!--req, xs:integer--></volume>
    </AudioOutVlome>
  </AudioOutVolumelist>
</AudioOut>
```

XML_AudioOutCap

AudioOutCap capability message in XML format

```
<AudioOutCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0" >
  <id>
    <!--req, xs:string-->
  </id>
  <MixAudioOut>
    <enabled opt="true,false" >
      <!--req, xs:Boolean-->
    </enabled>
```

```
<modulatorEnbale opt="true,false" >  
  <!--req, xs:Boolean-->  
</modulatorEnbale>  
<postFilter opt="false,true" default="true" >  
  <!--req, xs:Boolean-->  
</postFilter>  
<limitPressure opt="false,true" default="true" >  
  <!--req, xs:Boolean-->  
</limitPressure>  
<modulatorValue min="0" max="127" default="10" >  
  <!--req, xs:integer-->  
</modulatorValue>  
<triggerTime  
opt="1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,2,2,2,2,2,2,2,2,2,2,3,3,3,3,3,3,3,3,4,4,4,4,4,5,5,5,5,6,6,6,7,7,7,8,8,9,9,10,  
10,11,11,12,12,13,14,15,15,16,17,18,19,20,21,23,24,25,27,29,30,32,34,36,38,41,43,46,49,52,55,59,63,67,72,76,82,87,  
93,10,17,114,123,132,141,152,163,175,188,23,218,235,254,274,296,320,347,375,47,441,479,520,566,616,671,731,79  
8,872" default="4" >  
  <!--req, xs:integer, unit: ms-->  
</triggerTime>  
<freeTime  
opt="98,11,15,18,112,116,120,124,129,133,138,143,148,154,159,165,171,177,184,191,198,26,213,222,230,239,248,2  
58,268,278,289,31,313,325,338,352,366,381,397,413,430,448,467,486,57,528,551,575,60,626,653,682,712,743,776,8  
11,848,887,927,970,1014,1062,1111,1163,1218,1276,1337,142,1470,1541,1617,1697,1781,1870,1964,2063,2168,227  
9,2397,2521,2653,2792,2940,3097,3263,3439,3626,3825,4035,4259,4497,4750,5020,536,5611,5936,6282,6651,7045,  
7465,7914,8393,895,9452,10037,10664,11335,12053,12824,13650,14537,15489,16511,17610,18793,20065,21435,22  
911,2452,26219,28073,30075,32239,34580,37114,39859,42834,46062" default="42" >  
  <!--req, xs:integer, unit: ms-->  
</freeTime>  
<compressThreshold opt="1,2,4,8" defalut="2" >  
  <!--req, xs:integer-->  
</compressThreshold>  
<compressMode opt="soft,hard" default="soft" >  
  <!--req, xs:string-->  
</compressMode>  
<compressRate min="0" max="127" >  
  <!--req, xs:integer-->  
</compressRate>  
<recoveryGain opt="1,2,4,8" defalut="2" >  
  <!--req, xs:integer-->  
</recoveryGain>  
<outputGain  
opt="100,421,361,325,30,281,265,252,240,230,221,212,25,198,192,186,180,175,170,165,161,156,152,148,145,141,1  
38,134,131,128,125,122,120,117,114,112,110,17,15,13,10,98,96,94,92,90,88,86,85,83,81,79,78,76,74,73,71,70,68,67,  
65,64,62,61,60,58,57,56,54,53,52,51,49,48,47,46,45,43,42,41,40,39,38,37,36,35,34,33,32,31,30,29,28,27,26,25,24,23,  
23,22,21,20,19,18,17,17,16,15,14,13,12,12,11,10,9,9,8,7,6,6,5,4,3,3,2,1,1,0" >  
  <!--req, xs:integer-->  
</outputGain>  
</MixAudioOut>  
<AudioOutVolumelist>  
<AudioOutVlome>  
<type>  
  <!--req, xs:string, "audioOutput,audioEncode"-->
```

```
</type>
<volume min="0" max="127" default="50" >
  <!--req, xs:integer-->
</volume>
</AudioOutVolume>
</AudioOutVolumeList>
</AudioOutCap>
```

XML_BadSectorsTest

BadSectorsTest message in XML format

```
<BadSectorsTest version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <testType><!--opt, xs: string, "full,metadata"--></testType>
</BadSectorsTest>
```

XML_BadSectorsTestStatus

BadSectorsTestStatus message in XML format

```
<BadSectorsTestStatus version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <diskID><!--req, xs: string--></diskID>
  <MaskAreaList><!--req-->
    <MaskArea><!--list-->
      <maskAreaID><!--req, xs: integer--></maskAreaID>
      <startLBA><!--opt,xs:integer--></startLBA>
      <endLBA><!--opt, xs: integer--></endLBA>
    </MaskArea>
  </MaskAreaList>
  <BlockAreaTestStatus><!--opt-->
    <testType><!--opt, xs: string, "full,metadata"--></testType>
    <testStatus><!--req, xs: string, "none,running,pause,complete,exceed,abort"--></testStatus>
    <firstBlock><!--opt, xs: integer--></firstBlock>
    <lastBlock><!--opt, xs: integer--></lastBlock>
    <currentBlock><!--opt, xs: integer--></currentBlock>
    <BadSectorsList><!--opt-->
      <BadSectors><!--list-->
        <id><!--req, xs: integer--></id>
        <block><!--opt, xs: integer--></block>
      </BadSectors>
    </BadSectorsList>
  </BlockAreaTestStatus>
</BadSectorsTestStatus>
```

XML_BatteryPowerOverlay

BatteryPowerOverlay message in XML format

```
<BatteryPowerOverlay version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--req, xs:boolean--></enabled>
  <positionX><!--req, xs:integer, X-coordinate--></positionX>
  <positionY><!--req, xs:integer, Y-coordinate--></positionY>
</BatteryPowerOverlay>
```

XML_BLC

BLC message in XML format

```
<BLC version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <enabled><!--req, xs:boolean-->
  <BLCMode><!--opt, xs:string, BLC mode: "UP, DOWN, LEFT, RIGHT, CENTER, MULTI-AREA, Region"-->
  <BLCLevel><!--opt, xs:integer--></BLCLevel>
  <BLCRegionList><!--dep-->
    <BLCRegion>
      <id><!--req, xs:integer--></id>
      <RegionCoordinatesList/>
    </BLCRegion>
  </BLCRegionList>
</BLC>
```

XML_BluetoothCap

BluetoothCap capability message in XML format

```
<BluetoothCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <isSupportIbeacon>
    <!--opt, xs:boolean, whether to support configuring ibeacon bluetooth parameters-->
  </isSupportIbeacon>
</BluetoothCap>
```

XML_BluetoothStatus

BluetoothStatus message in XML format

```
<BluetoothStatus version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <connectStatus><!--opt, xs:boolean--></connectStatus>
  <connectDeviceName><!--opt, xs:string--></connectDeviceName>
</BluetoothStatus>
```

XML_Bond

Bond message in XML format

```
<Bond version="2.0" xmlns="urn:selfextension:ISAPIext-ver10-xsd">
  <id>
    <!--req, xs:integer-->
  </id>
  <enabled>
    <!--req, xs:boolean-->
  </enabled>
  <workMode>
    <!--req, xs:string, working mode: "balance-rr"-round-robin, "active-backup"-->
  </workMode>
  <primaryIf>
    <!--req, xs:string, default route ID-->
  </primaryIf>
  <slaveIfList>
    <!--req-->
    <ethernetIfId>
      <!--req, xs:string, ID-->
    </ethernetIfId>
  </slaveIfList>
  <IPAddress>
    <ipVersion>
      <!--req, xs:string, "v4,v6,dual"-->
    </ipVersion>
    <addressingType>
      <!--req, xs:string, "static,dynamic,apiipa"-->
    </addressingType>
    <ipAddress>
      <!--dep, xs:string-->
    </ipAddress>
    <subnetMask>
      <!--dep, xs:string, subnet mask for IPv4 address-->
    </subnetMask>
    <ipv6Address>
      <!--dep, xs:string-->
    </ipv6Address>
    <bitMask>
      <!--dep, xs:integer, bitmask for IPv6 address-->
    </bitMask>
  <DefaultGateway>
    <!--dep-->
    <ipAddress>
      <!--dep, xs:string-->
    </ipAddress>
    <ipv6Address>
      <!--dep, xs:string-->
    </ipv6Address>
  </DefaultGateway>
  <PrimaryDNS>
    <!--dep-->
    <ipAddress>
      <!--dep, xs:string-->
    </ipAddress>
```

```
<ipv6Address>
  <!--dep, xs:string-->
</ipv6Address>
</PrimaryDNS>
<SecondaryDNS>
  <!--dep-->
  <ipAddress>
    <!--dep, xs:string-->
  </ipAddress>
  <ipv6Address>
    <!--dep, xs:string-->
  </ipv6Address>
</SecondaryDNS>
</IPAddress>
</Bond>
```

XML_BondList

BondList message in XML format

```
<BondList version="2.0" xmlns="urn:selfextension:ISAPIext-ver10-xsd">
  <Bond/><!--see details in the message of XML_Bond-->
</BondList>
```

See Also

XML_Bond

XML_Cap_accessDeviceList

accessDeviceList capability message in XML format

```
<accessDeviceList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <accessDevice size="4">
    <id><!--req, xs:integer--></id>
    <MACAddress><!--opt, xs:string--></MACAddress>
    <ipV4Address><!--opt, xs:string--></ipV4Address>
    <accessTime><!--req, xs:time, ISO8601 data--></accessTime>
  </accessDevice>
</accessDeviceList>
```

XML_Cap_AcsUpdate

AcsUpdate capability message in XML format

```
<AcsUpdate version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <type
    opt="cardReader,FPMModule,securityModule,extendModule,channelController,IRModule,lampModule,elevatorControl
```

```
er,FPAAlgorithmProgram,uboot,keypad,wirelessRecv,wiredZone">
  <!--opt, upgrading type: "cardReader"-card reader, "FPModule"-fingerprint module, "securityModule"-secure door
control unit, "extendModule"-(IO) extension module, "channelController"-lane controller, "IRModule"-IR module,
"lampModule"-indicator module, "elevatorController"-slave elevator controller, "FPAAlgorithmProgram"-fingerprint
algorithm program of the card reader, "uboot", "keypad", "wirelessRecv"-wireless receiving module, "wiredZone"-
wired zone module-->
</type>
<cardReaderNo min="" max="">
  <!--opt, range of No. of the card reader, this node is valid only when <type> contains "cardReader"-->
</cardReaderNo>
<FPModuleNo min="" max="">
  <!--opt, range of No. of the fingerprint module, this node is valid only when <type> contains "FPModule"-->
</FPModuleNo>
<securityModuleNo min="" max="">
  <!--opt, range of No. of the secure door control module, this node is valid only when <type> contains
"securityModule"-->
</securityModuleNo>
<extendModuleNo min="" max="">
  <!--opt, range of No. of the (IO) extension module, this node is valid only when <type> contains "extendModule"-->
</extendModuleNo>
<channelControllerNo min="" max="">
  <!--opt, range of No. of the lane controller, this node is valid only when <type> contains "channelController"-->
</channelControllerNo>
<IRModuleNo min="" max="">
  <!--opt, range of No. of the IR module, this node is valid only when <type> contains "IRModule"-->
</IRModuleNo>
<lampModuleNo min="" max="">
  <!--opt, range of No. of the indicator module, this node is valid only when <type> contains "lampModule"-->
</lampModuleNo>
<elevatorControllerNo min="" max="">
  <!--opt,range of No. of the slave elevator controller, this node is valid only when <type> contains
"elevatorController"-->
</elevatorControllerNo>
<FPAAlgorithmProgramNo min="" max="">
  <!--opt, range of No. of the fingerprint algorithm program of the card reader, this node is valid only when <type>
contains "FPAAlgorithmProgram"-->
</FPAAlgorithmProgramNo>
<keypadAddress opt="1,3,5">
  <!--opt, range of keypad module address, this node is valid when <type> contains "keypad"-->
</keypadAddress>
<wirelessRecvAddress opt="1,3,5">
  <!--opt, range of wireless receiving module address, this node is valid when <type> contains "wirelessRecv"-->
</wirelessRecvAddress>
<wiredZoneAddress opt="1,3,5">
  <!--opt, range of wired zone module address, this node is valid when <type> contains "wiredZone"-->
</wiredZoneAddress>
</AcsUpdate>
```


XML_Cap_AdminAccessProtocolList

AdminAccessProtocol capability message in XML format

```
<AdminAccessProtocolList version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <AdminAccessProtocol><!--multiple <AdminAccessProtocol> nodes are allowed-->
    <id><!--req, xs: string, ID--></id>
    <enabled opt="true,false"><!--opt, xs: boolean--></enabled>
    <protocol
opt="HTTP,HTTPS,SDK,RTSP,DEV_MANAGE,WebSocket,WebSocketS,SDK_OVER_TLS,SRTP,MATRIX_GATEWAY,IOT">
      <!--req, xs: string-->
    </protocol>
    <portNo min="2000" max="65535" default="8443"><!--req, xs:integer --></portNo>
    <redirectToHttps opt="true,false">
      <!--opt, xs: boolean, whether to automatically go to HTTPS when connecting to HTTP port and HTTPS is enabled.
This node is valid when protocol is "HTTPS"-->
    </redirectToHttps>
    <streamOverTls opt="true,false">
      <!--opt, xs: boolean, whether to enable TLS link encryption when the streaming mode is SDK_OVER_TLS, true=yes,
false-no-->
    </streamOverTls>
    <TLS1_1Enable opt="true,false"><!--dep, xs: boolean, whether to enable TLS version 1.1, it is valid when protocol is
"HTTPS", "true"-yes, "false"-no--></TLS1_1Enable>
    <TLS1_2Enable opt="true,false"><!--dep, xs: boolean, whether to enable TLS version 1.2, it is valid when protocol is
"HTTPS", "true"-yes, "false"-no--></TLS1_2Enable>
  </AdminAccessProtocol>
  <TLS1_0Enable opt="true,false">
    <!--dep, xs: boolean, whether to enable TLS version 1.0, it is valid when protocol is "HTTPS", "true"-yes, "false"-no-->
  </TLS1_0Enable>
  <TLS1_1Enable opt="true,false">
    <!--dep, xs: boolean, whether to enable TLS version 1.1, it is valid when protocol is "HTTPS", "true"-yes, "false"-no-->
  </TLS1_1Enable>
  <TLS1_2Enable opt="true,false">
    <!--dep, xs: boolean, whether to enable TLS version 1.2, it is valid when protocol is "HTTPS", "true"-yes, "false"-no-->
  </TLS1_2Enable>
</AdminAccessProtocolList>
```

XML_Cap_BatteryPowerOverlay

BatteryPowerOverlay capability message in XML format

```
<BatteryPowerOverlay version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled>
    <!--req, xs:boolean-->
  </enabled>
  <positionX>
    <!--req, xs:integer, X-coordinate-->
  </positionX>
  <positionY>
```

```
<!--req, xs:integer, Y-coordinate-->
</positionY>
</BatteryPowerOverlay>
```

XML_Cap_Color

Color capability message in XML format

```
<?xml version="1.0" encoding="utf-8"?>
<Color version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <brightnessLevel min="" max=""><!--opt,xs:integer, brightness--></brightnessLevel>
  <contrastLevel min="" max=""><!--opt,xs:integer, contrast--></contrastLevel>
  <saturationLevel min="" max=""><!--opt,xs:integer, saturation--></saturationLevel>
  <hueLevel min="" max=""><!--opt, xs:integer, hue--></hueLevel>
  <grayScale><!--opt, gray scale-->
    <grayScaleMode><!--opt,xs:string, gray scale mode: "indoor,outdoor"--></grayScaleMode>
  </grayScale>
  <nightMode opt="true,false">
    <!--opt, xs:boolean, enable night mode, when its value is "true", the saturation can be adjusted, otherwise, the
saturation cannot be adjusted-->
  </nightMode>
</Color>
```

XML_Cap_DatetimeOverlay

DatetimeOverlay capability message in XML format

```
<DateTimeOverlay version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled>
    <!--req, xs:boolean, whether to enable date and time overlay-->
  </enabled>
  <positionX min="0" max="1000">
    <!--req, xs:integer, X-coordinate-->
  </positionX>
  <positionY min="0" max="1000">
    <!--req, xs:integer, Y-coordinate-->
  </positionY>
  <dateStyle>
    <!--opt, xs:string, date format: "YYYY-MM-DD, MM-DD-YYYY, DD-MM-YYYY, CHR-YYYY-MM-DD, CHR-MM-DD-YYYY,
CHR-DD-MM-YYYY"-->
  </dateStyle>
  <timeStyle>
    <!--opt, xs:string, "12hour, 24hour"-->
  </timeStyle>
  <displayWeek>
    <!--opt, xs:boolean, whether to display week information-->
  </displayWeek>
  <displayMilliSecond>
    <!--opt, xs:boolean, whether to display millisecond-->
```

```
</displayMilliSecond>  
</DateTimeOverlay>
```

XML_Cap_DDNS

DDNS capability message in XML format

```
<DDNS version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">  
  <id>  
    <!--req, xs:string-->  
  </id>  
  <enabled>  
    <!--req, xs:boolean, "true,false"-->  
  </enabled>  
  <ethernetIfId opt="1,2,3...">  
    <!--opt, xs:string, ID-->  
  </ethernetIfId>  
  <provider opt="IPServer,DynDNS,PeanutHall,NO-IP, HiDDNS">  
    <!--req, xs:string-->  
  </provider>  
  <serverAddress>  
    <addressingFormatType opt="ipaddress,hostname">  
      <!--req, xs:string-->  
    </addressingFormatType>  
    <hostname min="0" max="260">  
      <!--dep, xs:string-->  
    </hostName>  
    <ipAddress min="0" max="64">  
      <!--dep, xs:string-->  
    </ipAddress>  
    <ipv6Address min="0" max="64">  
      <!--dep, xs:string-->  
    </ipv6Address>  
  </serverAddress>  
  <portNo min="0" max="65535">  
    <!--opt, xs:integer-->  
  </portNo>  
  <deviceDomainName min="0" max="64">  
    <!--dep, xs:string-->  
  </deviceDomainName>  
  <username min="0" max="32">  
    <!--dep, xs:string-->  
  </userName>  
  <password min="0" max="16">  
    <!--wo, dep, xs:string-->  
  </password>  
</DDNS>
```

XML_Cap_DeviceInfo

DeviceInfo capability message in XML format

```
<DeviceInfo version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <deviceName min="" max="">
    <!--req, xs:string, device name-->
  </deviceName>
  <DockStation>
    <!--opt, dock station configuration-->
    <Platform>
      <!--opt, platform configuration-->
      <type opt="none, 9533, 8618, ISAPI"><!--req, xs:string, platform type--></type>
      <ip><!--opt, xs:string, IP address --></ip>
      <port><!--opt, xs:integer, communication port--></port>
      <userName><!--req, xs:string, user name, which is used for the dock station to log in to platform--></userName>
      <password><!--req, xs:string, password, which is used for the dock station to log in to platform, it should be
encrypted--></password>
    </Platform>
    <centralStorageBackupEnabled opt="true, false"><!--opt, xs:boolean, whether to enable central storage backup--></
centralStorageBackupEnabled>
  </DockStation>
  <languageType opt="chinese,english,spanish,portuguese,italian,french,russian,german,polish"/><!--opt, xs:string-->
  <deviceID min="0" max="128">
    <!--ro, req, xs:string, uuid-->
  </deviceID>
  <deviceDescription min="0" max="16">
    <!--opt, xs:string-->
  </deviceDescription>
  <deviceLocation opt="STD-CGI, hangzhou">
    <!--opt, xs:string-->
  </deviceLocation>
  <systemContact opt="STD-CGI, Hikvision.China">
    <!--opt, req, xs:string-->
  </systemContact>
  <model min="0" max="64">
    <!--ro, req, xs:string-->
  </model>
  <serialNumber min="0" max="48">
    <!--ro, req, xs:string-->
  </serialNumber>
  <macAddress min="0" max="64">
    <!--ro, req, xs:string-->
  </macAddress>
  <firmwareVersion min="0" max="64">
    <!--ro, req, xs:string-->
  </firmwareVersion>
  <firmwareReleasedDate min="0" max="64">
    <!--ro, opt, xs:string-->
  </firmwareReleasedDate>
  <bootVersion min="0" max="16">
```

```
<!--ro, opt, xs:string-->
</bootVersion>
<bootReleasedDate min="0" max="16">
  <!--ro, opt, xs:string-->
</bootReleasedDate>
<hardwareVersion min="0" max="16">
  <!--ro, opt, xs:string-->
</hardwareVersion>
<encoderVersion min="0" max="64">
  <!--ro, opt, xs:string-->
</encoderVersion>
<encoderReleasedDate min="0" max="64">
  <!--ro, opt, xs:string-->
</encoderReleasedDate>
<decoderVersion min="0" max="64">
  <!--ro, opt, xs:string-->
</decoderVersion>
<decoderReleasedDate min="0" max="64">
  <!--ro, opt, xs:string-->
</decoderReleasedDate>
<deviceType opt="IPCamera, IPDome, DVR, HybirdNVR, NVR, DVS, IPZoom">
  <!--ro, req, xs:string-->
</deviceType>
<telecontrolID min="1" max="255">
  <!--opt, xs:integer, "1-255"-->
</telecontrolID>
<supportBeep>
  <!--opt, xs:boolean: "true,false"-->
</supportBeep>
<firmwareVersionInfo>
  <!--ro, opt, xs:string-->
</firmwareVersionInfo>
<subChannelEnabled>
  <!--opt, xs:boolean: "true,false"-->
</subChannelEnabled>
<thrChannelEnabled>
  <!--opt, xs:boolean: "true,false"-->
</thrChannelEnabled>
<actualFloorNum>
  <!--req, xs:integer, "1-128"-->
</actualFloorNum>
<radarVersion>
  <!--opt, xs:string, radar version-->
</radarVersion>
<powerOnMode opt="button,adapter" def="button">
  <!--optional, xs:string, device startup mode: "button"-press button to power on (default), "adapter"-connect
  adapter to power on-->
</powerOnMode>
</DeviceInfo>
```

XML_Cap_DefaultParam

DefaultParam capability message in XML format

```
<DefaultParam version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  < DialNum min="1" max="32"><!--opt, xs:string--></DialNum>
  < Username min="1" max="32"><!--opt, xs:string--></Username>
  < Password min="1" max="32"><!--opt, xs:string--></Password>
  < APNname min="1" max="32"><!--opt, xs:string--></APNname>
  < VerifyProto opt="auto,CHAP,PAP"><!--req, xs:string, verification protocol: "auto, CHAP, PAP"--></VerifyProto>
</DefaultParam>
```

XML_Cap_DeviceLanguage

DeviceLanguage capability message in XML format

```
<DeviceLanguage version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  < language
    opt="SimChinese,TraChinese,English,Russian,Bulgarian,Hungarian,Greek,German,Italian,Czech,Slovakia,French,Polish,D
    utch,Portuguese,Spanish,Romanian,Turkish,Japanese,Danish,Swedish,Norwegian,Finnish,Korean,Thai,Estonia,Vietname
    se,Hebrew,Latvian,Arabic,Sovenian,Croatian,Lithuanian,Serbian,BrazilianPortuguese,Indonesian,Ukrainian">
    <!--req, xs:string, language supported by the device: "SimChinese"-simplified Chinese, "TraChinese"-traditional
    Chinese, "English", "Russian", "Bulgarian", "Hungarian", "Greek", "German", "Italian", "Czech", "Slovakia", "French",
    "Polish", "Dutch", "Portuguese", "Spanish", "Romanian", "Turkish", "Japanese", "Danish", "Swedish", "Norwegian",
    "Finnish", "Korean", "Thai", "Estonia", "Vietnamese", "Hebrew", "Latvian", "Arabic", "Sovenian"-Slovenian, "Croatian",
    "Lithuanian", "Serbian", "BrazilianPortuguese"-Brazilian Portuguese, "Indonesian", "Ukrainian"-->
  </language>
</DeviceLanguage>
```

XML_Cap_DiagnosedDataParameter

XML message about capability of exporting diagnose information

```
<?xml version="1.0" encoding="utf-8"?>
<DiagnosedDataParameter version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled opt="true,false">
    <!--required, xs:boolean, whether to enable exporting diagnose information: true=yes, false=no-->
  </enabled>
  <HardwareInfo opt="true,false">
    <!--optional, xs:boolean,opt="true,false", whether to enable exporting hardware information, such as running
    information of motor, fan, heater, compressor, supplement light, wiper, and so on-->
  </HardwareInfo>
  <SoftwareInfo opt="true,false">
    <!--optional, xs:boolean, whether to enable exporting software information, such as software status, running log,
    system information-->
  </SoftwareInfo>
  <logInfo opt="true,false">
```

```
<!--optional, xs:boolean, whether to enable exporting startup log--></logInfo>
</DiagnosedDataParameter>
```

XML_Cap_Dial

Dial capability message in XML format

```
<Dial version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled opt="true,false" def="false"><!--req, xs:boolean--></enabled>
  <DialMethod opt="auto,manual">
    <!--req, xs:string, "auto, manual", set it to manual when accessing the private network, and you can configure dial-
up parameters(including access number, user name, password, APN, and verification protocol-->
  </DialMethod>
  <SwitchMethod opt="auto,4GFirst,3GFirst,manualto2G,manualto3G,manualto4G,cableFirst">
    <!--req, xs:string, "auto,4GFirst,3GFirst, manualto2G, manualto3G,manualto4G,cableFirst"-->
  </SwitchMethod>
  <OfflineTime min="30" max="65535"><!--opt, xs:integer, unit: second--></OfflineTime>
  <UIMCardNum min="1" max="32"><!--opt, xs:string--></UIMCardNum>
  <DialNum min="1" max="32"><!--opt, xs:string--></DialNum>
  <Username min="1" max="32"><!--opt, xs:string--></Username>
  <Password min="1" max="32"><!--opt, xs:string--></Password>
  <APNname min="1" max="32"><!--opt, xs:string--></APNname>
  <SIMNum min="" max=""><!--opt, xs:string, SIM card No. (mobile phone number)--></SIMNum>
  <MTU min="100" max="1500"><!--opt, xs: integer--></MTU>
  <VerifyProto opt="auto,CHAP,PAP"><!--req, xs:string, verification protocol: "auto, CHAP, PAP"--></VerifyProto>
  <DefaultParam/><!--opt, default parameters, see details in the message of XML_Cap_DefaultParam-->
  <netAPN min="" max=""><!--opt, xs:string, APN configuration for the private network-->
  <Flow><!--opt, flow configuration-->
    <limitEnabled opt="true,false"><!--opt, xs:boolean, whether to enable flow limitation--></limitEnabled>
    <consumeFlow opt="true,false"><!--opt, xs:boolean, whether supports displaying flow usage, unit: MB--></
consumeFlow>
    <threshold min="" max=""><!--opt, threshold of flow, unit: MB--></threshold>
  </Flow>
  <enabled4G opt="true,false" def="true"><!-- opt, xs:boolean, whether to enable 4G--></enabled4G>
  <isSupportNetworkKeepAlive><!--opt, xs:boolean, whether it supports network keepalive --></
isSupportNetworkKeepAlive>
  <enabledDNS opt="true,false" def="true"><!-- opt, xs:boolean, whether to enable DNS manual configuration--></
enabledDNS>
</Dial>
```

See Also

XML_Cap_DefaultParam

XML_Cap_Discovery

Discovery capability message in XML format

```
<Discovery version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <UPnP><!--req-->
```

```
<enabled><!--req, xs:boolean, "true, false"--></enabled>
</UPnP>
<ZeroConf><!--opt, channel-zero configuration-->
  <enabled><!--req, xs:boolean, "true, false"--></enabled>
</ZeroConf>
</Discovery>
```

XML_Cap_EagleFocusing

EagleFocusing capability message in XML format

```
<?xml version="1.0" encoding="utf-8"?>
<EagleFocusing version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req,xs:string--></id>
  <enabled opt="true,false"><!--req, xs:boolean, "true"-start calibration, "false"-stop calibration--></enabled>
  <normalizedScreenSize><!--req,ro-->
    <normalizedScreenWidth><!--req,ro,xs:integer--></normalizedScreenWidth>
    <normalizedScreenHeight><!--req,ro,xs:integer--></normalizedScreenHeight>
  </normalizedScreenSize>
  <EagleFocusingRegionList><!--opt-->
    <EagleFocusingRegion><!--list-->
      <id><!--req, xs:string--></id>
      <sid><!--opt,xs:string, scene ID--></sid>
      <rate><!--opt,xs:integer, rate--></rate>
      <spotNum><!--opt,xs:integer, the number of calibration points--></spotNum>
      <type><!--req,xs:string, "line", region type, now only supports "line"--></type>
      <Region><!--dep, depends on type-->
        <RegionCoordinatesList size="4"><!--opt-->
          <RegionCoordinates><!--list-->
            <positionX><!--req,xs:integer,coordinate--></positionX>
            <positionY><!--req,xs:integer,coordinate--></positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </Region>
    </EagleFocusingRegion>
  </EagleFocusingRegionList>
  <sceneNum min="" max=""><!--opt,xs:integer, the maximum number of supported scenes--></sceneNum>
  <isSupportSceneGoto opt="true,false"><!--opt,xs:boolean, whether device supports turning to specified scene--></isSupportSceneGoto>
  <rate min="" max=""><!--opt,xs:integer, rate--></rate>
  <spotNum min="" max=""><!--opt,xs:integer, the number of calibration points on line--></spotNum>
  <onlyReadParam opt="rate,spotNum"><!--opt,xs:string, "rate,spotNum"--></onlyReadParam>
  <isSupportAuto opt="true,false"><!--req,xs:boolean, whether device supports automatic calibration of rapid focus--></isSupportAuto>
</EagleFocusing>
```


XML_Cap_EHome

EHome capability message in XML format

```
<Ehome version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id min="1" max="1"><!--opt, xs:string, EHome center ID--></id>
  <GPRSAddressingFormatType opt="ipaddress"><!--opt, xs:string, GPRS address type. If this node is not returned, the
address type will be determined by addressingFormatType which indicates that the address types configured by
different networks do not need to be distinguished--></GPRSAddressingFormatType>
  <addressingFormatType opt="ipaddress,hostname"><!--req, xs:string--></addressingFormatType>
  <hostName min="" max=""><!--dep, xs:string, domain name--></hostName>
  <ipAddress min="" max=""><!--dep, xs:string--></ipAddress>
  <ipv6Address min="" max=""><!--dep, xs:string--></ipv6Address>
  <portNo min="" max=""><!--opt, xs:integer--></portNo>
  <deviceId min="" max=""><!--req, xs:string--></deviceId>
  <registerStatus min="" max=""><!--ro, xs:boolean--></registerStatus>
  <key min="" max=""><!--opt, xs:string, encryption key of EHome version 5.0--></key>
  <version min="" max=""><!--ro, xs:string--></version>
  <netWork opt="0,1,2,3,4"/><!--opt,xs:integer, 0-make no sense, 1-automatic, 2-wired network preferred, 3-wired
network, 4-3G/4G/GPRS-->
  <voiceDeviceType opt="bluetooth,client,local"><!--opt, xs:string, two-way audio device type: "bluetooth"-bluetooth
device, "client", "local"--></voiceDeviceType>
  <protocolVersion opt="v2.0,v2.6,v4.0,v5.0"><!--opt, xs:string, protocol version: "v2.0,v2.6,v4.0,v5.0". If this node is
set to "v2.0", the device can only use protocol v2.0 to register; if this node is set to "v2.6", "v4.0" or "v5.0", the device
will firstly use this protocol to register. If this node is not returned, the protocol version will be determined by
<version>--></protocolVersion>
</Ehome>
```

XML_Cap_ExternalDevice

XML message about external device configuration capability

```
<?xml version="1.0" encoding="utf-8"?>
<ExternalDevice version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <SupplementLight/><!--opt, whether device supports configuring supplement light, see XML_Cap_SupplementLight
for details-->
  <THScreen/><!--opt, capability of peripheral screen configuration, corresponding url: /ISAPI/System/externalDevice/
THScreen-->
  <isSupportSupplementLightChannelSetting>
    <!--opt, xs:boolean, whether device supports configuring peripheral supplement light of specified channel,
corresponding url: /ISAPI/System/channel/<ID>/externalDevice/supplementLight-->
  </isSupportSupplementLightChannelSetting>
</ExternalDevice>
```

See Also

XML_Cap_SupplementLight

XML_Cap_ExtraInfo

ExtraInfo capability message in XML format

```
<ExtraInfo version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <diskMode opt="singleDisk,RAID"><!--req, xs:string, disk mode: "singleDisk"-single disk, "RAID". The device needs to
reboot after changing the disk mode--></diskMode>
</ExtraInfo>
```

XML_Cap_FocusConfiguration

FocusConfiguration capability message in XML format

```
<FocusConfiguration>
  <focusStyle opt="AUTO,MANUAL,SEMIAUTOMATIC"><!--req, xs:string--></focusStyle>
  <focusLimited opt="50,100,300,600,1000,2000,5000,10000,15000" def="1000"><!--opt, xs:integer--></focusLimited>
  <focusPosition/><!--dep, xs:integer, depends on FocusStyle-->
  <focusSpeed><!--opt, xs:integer--></focusSpeed>
  <focusSensitivity min="0" max="2" def="1"><!--opt, xs:intger, sensitivity of focus, ranging from 0 to 2. It is valid when
the focus mode is automatic or semi-automatic--></focusSensitivity>
  <temperatureChangeAdaptEnabled opt="true,false"--><!--opt, xs:boolean--></temperatureChangeAdaptEnabled>
  <relativeFocusPos min="0" max="4000" def=""><!--opt, xs:intger--></relativeFocusPos>
  <highTemperaturePriority opt="true,false"><!--opt, xs:boolean, enable high temperature priority mode--></
highTemperaturePriority>
</FocusConfiguration>
```

XML_Cap_FTPNotification

FTPNotification capability message in XML format

```
<FTPNotification version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id>
    <!--req, xs:string, FTP ID-->
  </id>
  <enabled>
    <!--req, xs:boolean, whether to enable the FTP server-->
  </enabled>
  <useSSL>
    <!--opt, xs:boolean-->
  </useSSL>
  <addressingFormatType opt="ipaddress,hostname">
    <!--req, xs:string, this node can only be set to "ipaddress"-->
  </addressingFormatType>
  <hostName>
    <!--dep, xs:string-->
  </hostName>
  <ipAddress>
    <!--dep, xs:string-->
  </ipAddress>
```

```
</ipAddress>
<ipv6Address>
  <!--dep, xs:string-->
</ipv6Address>
<portNo>
  <!--opt, xs:integer, FTP port No.-->
</portNo>
<userName>
  <!--req, xs:string, user name-->
</userName>
<password>
  <!--wo, xs:string, password-->
</password>
<passiveModeEnabled>
  <!--opt, xs:boolean-->
</passiveModeEnabled>
<annoyftp>
  <!--opt, xs:boolean-->
</annoyftp>
<uploadPicture>
  <!--opt, xs:boolean-->
</uploadPicture>
<uploadVideoClip>
  <!--opt, xs:boolean-->
</uploadVideoClip>
<uploadPath>
  <!--req-->
  <pathDepth>
    <!--req, xs:integer, directory levels, up to 16 levels can be supported-->
  </pathDepth>
  <topDirNameRule opt="none, devName, devId, devIp, positionInfo, time_month, time_date, illegalType, direction,
site, chanName, channelId, lanId, customize ,time, buildUnitNo">
    <!--dep, xs:string, parent directory name: "none", "devName"-device name, "devId"-device ID, "devIp"-device IP
address, "positionInfo"-camera 1, "time_month"-usage date (YYYY-MM), "time_date"-usage date (YYYY-MM-DD),
"illegalType"-violation type, "direction"-direction, "site"-place, "chanName"-channel name, "chanId"-channel No.,
"lanId"-lane No., "customize"-custom, "time", "buildUnitNo"-building No. and unit No. This node is set to NULL by
default-->
  </topDirNameRule>
  <topDirName min="" max="">
    <!--dep, xs:string, custom string for parent directory, the maximum string length is 32 bytes. This node is valid
when <topDirNameRule> is "customize". If attributes of "min" and "max" are not obtained after parsing, the default
value will be returned without error message-->
    <topDirName/><!--dep, xs:string, custom string for parent directory, the maximum string length is 32 bytes. This
node is valid when <topDirNameRule> is "customize"-->
    <subDirNameRule opt="none, devName, devId, devIp, positionInfo, time_month, time_date, illegalType, direction,
site, chanName, channelId, lanId, customize ,time, buildUnitNo">
      <!--dep, xs:string, child directory name-->
    </subDirNameRule>
    <subDirName min="" max="">
      <!--dep, xs:string, custom string for child directory, the maximum string length is 32 bytes. This node is valid when
<subDirNameRule> is "customize". If attributes of "min" and "max" are not obtained after parsing, the default value
will be returned without error message-->
```

```
<subDirName/><!--dep, xs:string, custom string for child directory, the maximum string length is 32 bytes. This node
is valid when <subDirNameRule> is "customize"-->
  <threeDirNameRule opt="none, devName, devId, devIp, positionInfo, time_month, time_date, illegalType, direction,
site, chanName, chanId, lanId, customize ,time, buildUnitNo">
    <!--dep, xs:string, the third directory name-->
    </threeDirNameRule>
    <threeDirName/><!--req, xs:string, custom string for the third directory, the maximum string length is 32 bytes. This
node is valid when <threeDirNameRule> is "customize"-->
    <fourDirNameRule opt="none, devName, devId, devIp, positionInfo, time_month, time_date, illegalType, direction,
site, chanName, chanId, lanId, customize ,time, buildUnitNo">
    <!--dep, xs:string, the fourth directory name-->
    </fourDirNameRule>
    <fourDirName/><!--req, xs:string, custom string for the fourth directory, the maximum string length is 32 bytes. This
node is valid when <fourDirNameRule> is "customize"-->
  </uploadPath>
  <picArchivingInterval min="" max="">
    <!--opt, xs:integer, the value is between 1 and 30, 0-close-->
  </picArchivingInterval>
  <picNameRuleType opt="default,prefix">
    <!--opt, xs:string-->
  </picNameRuleType>
  <picNamePrefix min="0" max="32">
    <!--dep, xs:string-->
  </picNamePrefix>
  <ftpPicNameRuleType opt="videoIntercom,ITC">
    <!--req, xs:string, type of FTP picture name rule: "videoIntercom"-rule used by video intercom products, "ITC"-rule
used by traffic cameras-->
  </ftpPicNameRuleType>
  <FTPPicNameRule>
    <!--dep, picture name rule of a specific FTP-->
    <ItemList/><!--req, see details in the message of XML_Cap_ItemList-->
    <delimiter>
      <!--req, xs:string, delimiter, which is a single character and the default value is "_"-->
    </delimiter>
    <customStr min="1" max="128">
      <!--req, xs:string, custom string-->
    </customStr>
  </FTPPicNameRule>
  <upDataType opt="0,1,2">
    <!-- opt, xs:integer, picture uploading type: 0-all, 1-checkpoint, 2-violation. When only one FTP server is enabled,
this node can only be set to 0. When two FTP servers are both enabled, you should set 1 for one FTP server and set 2
for another FTP server, which means that two FTP servers cannot be set to the same type-->
  </upDataType>
  <uploadPlateEnable>
    <!--opt, xs:boolean, whether to enable uploading license plate thumbnail-->
  </uploadPlateEnable>
  <site min="1" max="128">
    <!--req, xs:string, place, the maximum string length is 128 bytes-->
  </site>
  <roadNum min="1" max="32">
    <!--req, xs:string, intersection No., the maximum string length is 32 bytes-->
  </roadNum>
```

```
<instrumentNum min="1" max="32">
  <!--req, xs:string, device No., the maximum string length is 32 bytes-->
</instrumentNum>
<direction min="1" max="32">
  <!--req, xs:string, direction No., the maximum string length is 32 bytes-->
</direction>
<directionDesc min="1" max="32">
  <!--req, xs:string, direction description, the maximum string length is 32 bytes-->
</directionDesc>
<monitoringInfo1 min="1" max="44">
  <!--req, xs:string, camera 1 information, the maximum string length is 44 bytes-->
</monitoringInfo1>
<uploadAttachedInfomation>
  <!--req, xs:boolean, whether to upload additional information-->
</uploadAttachedInfomation>
<BrokenNetHttp><!--opt, whether it supports ANR (automatic network replenishment)-->
  <enabled opt="true,false"><!--opt, xs:boolean, whether to enable ANR (automatic network replenishment)--></
enabled>
  <supportEventType opt="personQueueCounting"/><!--opt, xs:string, event types supporting FTP ANR:
"personQueueCounting"-person queue counting detection, "personQueueTime"-person queue time detection,
"personQueueRealTimeData"-upload real-time data of person queue counting detection, "faceCapture"-face capture
and recognition, "fieldDetection"-intrusion, "attendedBaggage"-object removal, "unattendedBaggage"-unattended
baggage, "regionExiting"-region exiting, "regionEntrance"-region entrance, "lineDetection"-line crossing detection-->
</BrokenNetHttp>
</FTPNotification>
```

See Also

XML_Cap_ItemList

XML_Cap_FTPNotificationList

FTPNotificationList capability message in XML format

```
<FTPNotificationList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <FTPNotification/><!--opt, see details in the message of XML_Cap_FTPNotification-->
</FTPNotificationList>
```

See Also

XML_Cap_FTPNotification

XML_Cap_GuardAgainstTheft

GuardAgainstTheft capability message in XML format

```
<GuardAgainstTheft version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled opt="true,false"><!--required, xs:boolean, whether it supports enabling device anti-theft--></enabled>
  <SMSEnabled opt="true,false"><!--required, xs:boolean, whether it supports enabling SMS--></SMSEnabled>
  <phoneNum min="0" max="32"><!--dependency, xs:string, phone No.--></phoneNum>
```

```
<longitudeLatitudeEnabled opt="true,false"><!--dependency, xs:boolean, whether it supports enabling longitude and latitude--></longitudeLatitudeEnabled>
<siteLocationEnabled opt="true,false"><!--dependency, xs:boolean, whether it supports enabling site location--></siteLocationEnabled>
<siteLocationInfo min="0" max="128"><!--dependency, xs:string, site location information--></siteLocationInfo>
<timeEnabled opt="true,false"><!--dependency, xs:boolean, whether it supports enabling time--></timeEnabled>
<customInfo min="0" max="128"><!--dependency, xs:string, custom information--></customInfo>
</GuardAgainstTheft>
```

XML_Cap_HardwareService

HardwareService capability message in XML format

```
<HardwareService version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <IrLightSwitch><!--opt-->
    <mode opt="open,close"><!--req, xs:string --></mode>
  </IrLightSwitch>
  <ABF><!--opt-->
    <enabled><!--req, xs:boolean --></enabled>
  </ABF>
  <LED><!--opt-->
    <enabled><!--req, xs:boolean --></enabled>
  </LED>
  <Defog><!--opt-->
    <enabled><!--req, xs:boolean --></enabled>
  </Defog>
  <SupplementLight><!--opt-->
    <enabled><!--req, xs:boolean --></enabled>
    <isSupportFireLaserLight opt="true,false"><!--opt,xs:boolean--></isSupportFireLaserLight>
    <isSupportSupplementLightWord opt="true,false"><!--opt,xs:boolean--></isSupportSupplementLightWord>
    <captureWithSupplimentLightEnabled opt="true,false">
      <!--opt, xs:boolean, enable snapshot supplement light or not-->
    </captureWithSupplimentLightEnabled>
  </SupplementLight>
  <Deicing><!--opt-->
    <enabled><!--req, xs:boolean --></enabled>
  </Deicing>
  <ManualDeicing><!--opt-->
    <enabled><!--req, xs:boolean --></enabled>
  </ManualDeicing>
  <mutexAbility opt="laserLight, deicing">
    <!--req, mutual exclusion ability, the laser light and deicing (including manual deicing and automatic deicing) are mutual exclusive-->
  </mutexAbility>
  <HighTemperatureProtection><!--opt-->
    <enabled><!--req, xs:boolean --></enabled>
    <temperatureType opt="90,100,110" def="90"><!--dep, xs:string,unit:°C --></temperatureType>
  </HighTemperatureProtection>
</HardwareService>
```

XML_Cap_hddList

hddList capability message in XML format

```
<hddList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema" size="">
  <hdd><!--list-->
    <id><!--read-only, required, xs: string; ID--></id>
    <hddName><!--read-only, required, xs: string--></hddName>
    <hddPath><!--read-only, optional, xs: string--></hddPath>
    <hddType opt="IDE,SATA,eSATA,NFS,iSCSI,Virtual Disk"><!--read-only, required, xs: string--></hddType>
    <status
      opt="ok,unformatted,error,idle,mismatch,offline,smartFailed,reparing,formatting,notexist,unRecordHostFormatted">
      <!--read-only, required, xs: string, unRecordHostFormatted--unformatted in education sharing system-->
    </status>
    <capacity><!--read-only, required, xs: float, unit: MB--></capacity>
    <freeSpace><!--read-only, required, xs: float, unit: MB--></freeSpace>
    <property opt="RW,read-only,Redund"><!--required, xs: string, HDD properties--></property>
    <group><!--optional, xs: string; HDD group ID--></group>
    <reservedDayCfg min="0" max="1000"><!--optional, xs:integer, video saving duration, it is between 0 and 1000,
unit: day--></reservedTimeCfg>
    <DataModeList><!--optional, read-only, current HDD allocation mode-->
      <DataMode>
        <type opt="recordStorage,pictureCloudStorage,fileStorage">
          <!--required, xs: string, storage application type, recordStorage-video storage, pictureCloudStorage-picture to be
saved in cloud storage, fileStorage-file-storage-->
        </type>
        <occupancyRate><!--required, xs: integer, HDD usage, range: [0,100]--></occupancyRate>
      </DataMode>
    </DataModeList>
    <formatType opt="FAT32,EXT4" def="FAT32">
      <!--read-only, optional, xs: string, formatting type, this node is only available for SD card; if this node does not
exist, the default formatting type is FAT32-->
    </formatType>
    <Encryption>
      <passwordLen min="6" max="64"/>
      <encryptionStatus opt="unencrypted,encrypted,verifyFailed"><!--read-only, optional, xs:string, encryption status:
"unencrypted", "encrypted", "verifyFailed"-verification failed--></encryptionStatus>
      <encryptFormatType opt="FAT32,EXT4"><!--read-only, optional, xs:string--></encryptFormatType>
    </Encryption>
    <isSupportLogStorage><!--optional, xs:boolean, whether it can be used as a log HDD--></isSupportLogStorage>
  </hdd>
</hddList>
```

XML_Cap_ibeaconParam

IbeaconParam capability message in XML format

```
<IbeaconParam version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <UUID min="" max="">
```

```
<!--req, xs:string, parking lot ID, which supports using digits, letters and hyphen as the tag-->
</UUID>
<major min="" max="">
  <!--req, xs:integer, number of floors in the parking lot, which cannot be 0x00 for all-->
</major>
<minor min="" max="">
  <!--req, xs:integer, number of parking spaces on each floor, which cannot be 0x00 for all-->
</minor>
<sendPower min="" max="">
  <!--req, xs:integer, transmitted power: 01-0 dbm, 02- -6 dbm, 03- -23 dbm-->
</sendPower>
<frequency min="" max="">
  <!--req, xs:integer, transmitted frequency, which is the broadcast time interval, unit: 625 μs. The broadcast time
interval is between 32 (20 ms) to 8000 (5s), and the default value is 160 (100 ms)-->
</frequency>
<measurePower min="" max="">
  <!--req, xs:integer, rated power, which is the RSSI (Received Signal Strength Indication) in the position of 1 meter
away, unit: dbm-->
</measurePower>
</lbeaconParam>
```

XML_Cap_IEEE802_1x

IEEE802_1x capability message in XML format

```
<IEEE802_1x version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--req, xs: boolean--></enabled>
  <authenticationProtocolType opt="EAP-TLS,EAP-TTLS,EAP-PEAP,EAP-LEAP,EAP-FAST,EAP-MD5">
    <!--req, xs: string-->
  </authenticationProtocolType>
  <innerTTLSAuthenticationMethod opt="MS-CHAP,MS-CHAPv2,PAP,EAP-MD5">
    <!--dep, xs: string, this node is required when <authenticationProtocolType> is "EAP-TLS"-->
  </innerTTLSAuthenticationMethod>
  <innerEAPProtocolType opt="EAP-POTP,MS-CHAPv2">
    <!--dep, xs: string, this node is required when <authenticationProtocolType> is "EAP-PEAP" or "EAP-FAST"-->
  </innerEAPProtocolType>
  <validateServerEnabled><!--dep, xs: boolean--></validateServerEnabled>
  <userName><!--dep, xs: string--></userName>
  <password><!--dep, xs: string--></password>
  <anonymousID><!--opt, xs: string--></anonymousID>
  <autoPACProvisioningEnabled>
    <!--dep, xs: boolean, this node is required when <authenticationProtocolType> is "EAP-FAST"-->
  </autoPACProvisioningEnabled>
  <Extensions><!--opt-->
    <EAPOLVersion><!--opt, xs: string, "1, 2"--></EAPOLVersion>
  </Extensions>
</IEEE802_1x>
```


XML_Cap_ImageChannel

XML message about image channel capability

```
<ImageChannel version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:integer--></id>
  <enabled><!--req, xs:boolean--></enabled>
  <videoInputID><!--req, xs:integer--></videoInputID>
  <FocusConfiguration/><!--opt, refer to the message XML_Cap_FocusConfiguration for details-->
  <LensInitialization/><!--opt-->
  <ImageFlip/><!--opt-->
  <ImageFreeze/><!--opt-->
  <proportionalpan/><!--opt-->
  <WDR>
    <mode opt="close,open,auto"><!--req, xs: string, WDR mode: "open,close,auto"--></mode>
    <WDRLevel max="100" min="0"><!--opt, xs: integer--></WDRLevel>
    <WDRContrastLevel max="" min="0"><!--opt, xs:integer--></WDRContrastLevel>
    < WDRLevel1 max="" min="0"><!--opt,xs:integer-->< /WDRLevel1>
  </WDR>
  <BLC/><!--opt-->
  <NoiseReduce/><!--opt-->
  <ImageEnhancement/><!--opt-->
  <SlowShutter/><!--opt-->
  <DSS/><!--opt-->
  <WhiteBlance/><!--opt-->
  <Exposure/><!--opt, exposure configuration parameters, refer to the message XML_Exposure for details-->
  <DayNightGate/><!--opt-->
  <BrightEnhance/><!--opt-->
  <Sharpness>
    <SharpnessLevel max="100" min="0"><!--req, xs:integer--></SharpnessLevel>
  </Sharpness>
  <gammaCorrection/><!--opt-->
  <powerLineFrequency/><!--opt-->
  <Color><!--opt, image adjustment capability-->
    <brightnessLevel max="100" min="0"><!--opt, xs:integer, brightness--></brightnessLevel>
    <contrastLevel max="100" min="0"><!--opt,xs:integer, contrast--></contrastLevel>
    <saturationLevel max="100" min="0"><!--opt, xs:integer, saturation--></saturationLevel>
    <grayScale>
      <grayScaleMode opt="outdoor,indoor"><!--opt,xs:string, gray scale mode: "indoor,outdoor"--></grayScaleMode>
    </grayScale>
  </Color>
  <IrcutFilter/><!--opt, configuration parameters of day/night auto-switch, refer to the message XML_IrcutFilter for details-->
  <Scene/><!--opt-->
  <EPTZ/ ><!--opt-->
  <EIS/><!--opt-->
  <HLC/> <!--opt-->
  <ZoomLimit/> <!--opt-->
  <corridor/><!--opt-->
  <Dehaze/><!--opt-->
  <ImageMode opt="standard, indoor, outdoor, dimLight"/><!--opt, xs:string-->
</ImageChannel>
```

```

<enableImageLossDetection><!--opt, boolean--></enableImageLossDetection>
<CaptureMode/><!--opt-->
<IrLight/><!--opt-->
<LensDistortionCorrection/><!--opt-->
<SupplementLight/><!--opt, supplement light configuration capability, refer to the message
XML_Cap_SupplementLight for details-->
<OpticalDehaze/><!--opt-->
<ManualRanging/><!--opt-->
<OIS/><!--opt-->
<isSupportlaserSpotManual><!--opt, boolean--></isSupportlaserSpotManual>
<isSupportLaserSpotAdjustment><!--opt, boolean--></isSupportLaserSpotAdjustment>
<DigitalZoom>
  <ZoomRatio opt="1x,2x,4x,8x,16x,32x"><!--req, xs:string--></ZoomRatio>
</DigitalZoom>
<Palettes>
  <mode
opt="WhiteHot,BlackHot,Fusion1,Rainbow,Fusion2,Ironbow1,Ironbow2,Sepia,Color1,Color2,IceFire,Rain,RedHot,Green
Hot,DeepBlue,Color3"><!--
opt,xs:string,"WhiteHot,BlackHot,Fusion1,Rainbow,Fusion2,Ironbow1,Ironbow2,Sepia,Color1,Color2,IceFire,Rain,RedH
ot,GreenHot,DeepBlue,Color3"--></mode>
  <ColorateTarget><!--dep,mode="WhiteHot","target coloration"-->
  <ColorateTargetModeList>
    <ColorateTargetMode><!--list-->
      <id><!--req,xs:integer "index,which starts from 1"--></id>
      <mode opt="colorateHotAreae,colorateIntervalArea,colorateColdArea"><!--
req,xs:string,"colorateHotAreae,colorateIntervalArea,colorateColdArea"--></mode>
      <enabled opt="true,false"><!--req,xs:boolean,"true,false"--></enabled>
      <TemperatureLimit><!--req "temperature range"-->
        <minTemperature><!--dep <mode> is colorateHotAreae/colorateIntervalArea,xs:float--></minTemperature>
        <maxTemperature><!--dep <mode> is colorateColdArea/colorateIntervalArea,xs:float--></maxTemperature>
      </TemperatureLimit>
      <Color><!--req,"area color"-->
        <R><!--req,xs:integer--></R>
        <G><!--req,xs:integer--></G>
        <B><!--req,xs:integer--></B>
      </Color>
    </ColorateTargetMode>
  </ColorateTargetModeList>
</ColorateTarget>
  <supportColorateTargetMode opt="WhiteHot,BlackHot"><!--opt,xs:string, palette modes that support target
coloration--></supportColorateTargetMode>
</Palettes>
<ExposureSync opt="true,false" def="false">
  <enabled><!--req, xs:boolean--></enabled>
</ExposureSync>
<BrightnessSuddenChangeSuppressionCap/><!--opt-->
<isSupportIcr><!--opt, boolean--></isSupportIcr>
<isSupportMultishut><!--opt, boolean--></isSupportMultishut>
<isSupportPlateBright><!--opt, boolean--></isSupportPlateBright>
<isSupportJPEGParam><!--opt, boolean--></isSupportJPEGParam>
<isSupportDarkEnhance><!--opt, boolean--></isSupportDarkEnhance>
<isSupportHdr><!--opt, boolean--></isSupportHdr>

```

```

<isSupportLse><!--opt, boolean--></isSupportLse>
<isSupportMce><!--opt, boolean--></isSupportMce>
<isSupportScve><!--opt, boolean--></isSupportScve>
<isSupportSectionCtrl><!--opt, boolean--></isSupportSectionCtrl>
<isSupportAutoContrast><!--opt, boolean--></isSupportAutoContrast>
<isSupportGrayRange><!--opt, boolean--></isSupportGrayRange>
<isSupportLSEDetail><!--opt, boolean--></isSupportLSEDetail>
<isSupportCapture><!--opt, boolean--></isSupportCapture>
<isSupportBrightEnhance><!--opt, boolean--></isSupportBrightEnhance>
<isSupportRecord><!--opt, xs:boolean--></isSupportRecord>
<isSupportDefog><!--opt, xs:boolean--></isSupportDefog>
<isSupportGeneral><!--opt, xs:boolean, whether to display general configuration page on the interface--></
isSupportGeneral>
<isSupportIa><!--opt, xs:boolean, whether to display video configuration page on the interface--></isSupportIa>
<NoiseReduce2D>
  <noiseReduce2DEnable><!--req, xs:boolean--></noiseReduce2DEnable>
  <noiseReduce2DLevel min="0" max="100"><!--dep,xs:integer--></noiseReduce2DLevel>
</NoiseReduce2D>
<Shutter>
  <ShutterLevel min="100" max="40000"><!--req, xs:integer--></ShutterLevel>
</Shutter>
<Gain>
  <GainLevel min="0" max="100"><!--req, xs:integer--></GainLevel>
</Gain>
<TempRange/><!--opt, temperature range capability, see XML_Cap_tempRange for details-->
</ImageChannel>

```

Remarks

The nodes **<DigitalZoom>** and **<ExposureSync>** are not supported by thermographic automation thermal camera (DS-2TA03-15SVI, DS2TA06-25SVI).

XML_Cap_InputProxyChannel

InputProxyChannel capability message in XML format

```

<InputProxyChannel version="1.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <id min="" max=""></id>
  <name min="" max=""></name>
  <sourceInputPortDescriptor>
    <adminProtocol opt="HIKVISION,SONY,ISAPI,ONVIF"></adminProtocol>
    <addressingFormatType opt="ipaddress,hostname"></addressingFormatType>
    <hostName min="" max=""></hostName>
    <ipAddress></ipAddress>
    <ipv6Address></ipv6Address>
    <adminPortNo min="2000" max="65535"></adminPortNo>
    <srcInputPort min="" max=""></srcInputPort>
    <userName min="" max=""><!--user name, which should be encrypted--></userName>
    <password min="" max=""><!--password, which should be encrypted--></password>
    <streamType opt="auto,tcp,udp"></streamType>
    <deviceId min="" max=""></deviceId>
  </sourceInputPortDescriptor>
</InputProxyChannel>

```

```
</sourceInputPortDescriptor>
<enableAnr opt="true,false"></enableAnr>
<NVRInfo>
  <ipAddressNVR>
    <!--opt, xs:string, IP address of NVR-->
  </ipAddressNVR>
  <portNVR min="2000" max="65535">
    <!--opt, xs:integer, port No. of NVR-->
  </portNVR>
  <ipcChannelNo min="" max="">
    <!--opt, xs:integer, channel No. of the network camera in NVR-->
  </ipcChannelNo>
</NVRInfo>
</InputProxyChannel>
```

XML_Cap_IOInputPortList

IOInputPortList capability message in XML format

```
<IOInputPortList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <IOInputPort><!--req-->
    <id><!--req, xs:string--></id>
    <triggeringType opt="high,low"><!--req, xs:string--></triggeringType>
    <name min="" max=""><!--req, xs:string--></name>
    <inputType opt="switch,semaphore">
      <!--opt, xs:string, switch, semaphore-->
    </inputType>
    <CombinationAlarmCap><!--opt, composite alarm capability-->
      <channel min="" max="">
        <!--req, xs:integer, channel No. range-->
      </channel>
      <EventTypeList size=""><!--req, event type list, the "size" indicates the max. number of supported events for composition-->
        <eventType>
          <!--req, xs:string, event type-->
        </eventType>
      </EventTypeList>
    </CombinationAlarmCap>
  </IOInputPort>
</IOInputPortList>
```

XML_Cap_IOOutputPortList

IOOutputPortList capability message in XML format

```
<IOOutputPortList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <IOOutputPort>
    <id min="" max=""><!-- req, xs:integer, "2" --></id>
    <PowerOnState><!--req, output port configuration parameters when the device is powered on>
```

```
<defaultState opt="high,low">
  <!--ro, req, xs:string,default output port signal when it is not triggered-->
</defaultState>
<outputState opt="high,low,pulse">
  <!--ro, req, xs:string, output port signal when it is being triggered-->
</outputState>
<pulseDuration min="" max="">
  <!--dep, xs:integer, duration of a output port signal when it is being triggered, it is valid when outputState is
"pulse", unit: milliseconds-->
</pulseDuration>
</PowerOnState>
<name><!--opt, xs:string--></name>
<IOUseType opt="disable,electricLock,custom"><!--opt, xs:string--></IOUseType>
<normalStatus opt="open, close"><!--opt, xs:string,normal status: open-remain open, close-remain closed--></
normalStatus>
</IOOutputPort>
</IOOutputPortList>
```

XML_Cap_IpAddress

IpAddress capability message in XML format

```
<IpAddress version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ipVersion opt="IPv4,IPv6,dual">
    <!--req, xs:string-->
  </ipVersion>
  <addressingType opt="static,dynamic,apiipa">
    <!--req, xs:string-->
  </addressingType>
  <ipAddress min="0" max="40">
    <!--dep, xs:string-->
  </ipAddress>
  <subnetMask min="0" max="15">
    <!--dep, xs:string-->
  </subnetMask>
  <ipv6Address min="0" max="40">
    <!--dep, xs:string-->
  </ipv6Address>
  <bitMask ask min="0" max="128">
    <!--dep, xs:integer-->
  </bitMask>
  <DefaultGateway>
    <!--dep-->
    <ipAddress min="0" max="40">
      <!--dep, xs:string-->
    </ipAddress>
    <ipv6Address min="0" max="40">
      <!--dep, xs:string-->
    </ipv6Address>
  </DefaultGateway>
  <PrimaryDNS>
```

```
<!--dep-->
<ipAddress min="0" max="40">
  <!--dep, xs:string-->
</ipAddress>
<ipv6Address min="0" max="40">
  <!--dep, xs:string-->
</ipv6Address>
</PrimaryDNS>
<SecondaryDNS>
  <!--dep-->
  <ipAddress min="0" max="40">
    <!--dep, xs:string-->
  </ipAddress>
  <ipv6Address min="0" max="40">
    <!--dep, xs:string-->
  </ipv6Address>
</SecondaryDNS>
<Ipv6Mode><!--opt-->
  <ipV6AddressingType opt="router,ra,manual,dhcp">
    <!--dep, xs:string-->
  </ipV6AddressingType>
  <ipv6AddressList>
    <v6Address>
      <id min="0" max="255">
        <!--dep, xs:string-->
      </id>
      <type opt="router,ra,manual,dhcp">
        <!--dep, xs:string-->
      </type>
      <address min="0" max="40">
        <!--dep, xs:string-->
      </address>
      <bitmask min="0" max="255">
        <!--dep, xs:integer-->
      </bitMask>
    </v6Address>
  </ipv6AddressList>
</Ipv6Mode>
</IpAddress>
```

XML_Cap_IPFilter

IPFilter capability message in XML format

```
<IPFilter version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled opt="true, false"><!--req, xs:boolean--></enabled>
  <permissionType opt="deny, allow">
    <!--req, xs:string, IP address filter status: "deny, allow". If this node is configured, it will overwrite value of all
<permissionType> in <IPFilterAddress>-->
  </permissionType>
  <IPFilterAddressList size="32"><!--opt, the attribute "size" refers to the maximum number of IP addresses supported
```

by the device. If the attribute "size" is not returned, the default supported number of IP addresses is 48-->

```
<IPFilterAddress>
  <id min="" max=""><!--req, xs:string, ID--></id>
  <permissionType opt="deny, allow"><!--dep, ro, xs:string--></permissionType>
  <addressFilterType opt="mask,range">
    <!--dep, ro, xs:string, filter type: "mask"-single address, "range"-address range-->
  </addressFilterType>
  <AddressRange><!--dep, this node is valid when <addressFilterType> contains "range"-->
    <startIPAddress min="" max=""><!--dep, xs:string--></startIPAddress>
    <endIPAddress min="" max=""><!--dep, xs:string--></endIPAddress>
    <startIPv6Address min="" max=""><!--dep, xs:string--></startIPv6Address>
    <endIPv6Address min="" max=""><!--dep, xs:string--></endIPv6Address>
  </AddressRange>
  <AddressMask><!--dep, it is valid when <addressFilterType> contains "mask"-->
    <ipAddress min="" max=""><!--dep, xs:string--></ipAddress>
    <ipv6Address min="" max=""><!--dep, xs:string--></ipv6Address>
    <bitMask min="" max=""><!--opt, xs:string--></bitMask>
    <bitMaskIPv6 min="" max=""><!--req, xs:string, IPv6 prefix length--></bitMaskIPv6>
  </AddressMask>
  <describeMsg min="" max=""><!--opt, xs:string--></describeMsg>
</IPFilterAddress>
</IPFilterAddressList>
</IPFilter>
```

XML_Cap_ItemList

ItemList capability message in XML format

```
<ItemList size="15">
  <Item>
    <itemID min="1" max="15">
      <!--req, xs:string, item ID, which is between 1 and 15-->
    </itemID>
    <itemOrder
      opt="none,devIp,time,buildUnitNo,outDoorDevNo,unlockType,devName,deviceNo,channelName,channelNo,
      plateNo,plateColor,laneNo,carSpeed,
      positionInfo1,pictureNo,CarNo,speedLimit,illegalCode,siteNo,directionNo,carColor,platePosition,carType,illegalType,cu
      stom">
      <!--req, xs:string, name element: "none", "devIp"-device IP address, "time", "buildUnitNo"-building No. and unit
      No., "outDoorDevNo"-door station, "unlockType"-unlocking type, "devName"-device name, "deviceNo"-device No.,
      "channelName"-channel name, "channelNo"-channel No., "plateNo"-license plate number, "plateColor"-license plate
      color, "laneNo"-lane No., "carSpeed"-vehicle speed, "positionInfo1"-camera 1, "pictureNo"-picture No., "CarNo"-
      vehicle No., "speedLimit"-speed limit, "illegalCode"-violation code, "siteNo"-intersection No., "directionNo"-direction
      No., "carColor"-vehicle color, "platePosition"-license plate coordinates, "carType"-vehicle type, "illegalType"-violation
      type, "custom"-->
    </itemOrder>
    <itemCustomStr min="1" max="32">
      <!--req, xs:string, element custom string, which is between 1 and 32, unit: bytes. This node is valid only when
      <itemOrder> is "custom". Currently traffic cameras only support one custom name-->
    </itemCustomStr>
```

```
</Item>
</ItemList>
```

XML_Cap_LensDistortionCorrection

LensDistortionCorrection capability message in XML format

```
<LensDistortionCorrection version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--req, xs: boolean--></enabled>
  <level opt="level1,level2,level3,custom">
    <!--opt, xs: string, distortion correction level: level 1, level 2, level 3, custom-->
  </level>
  <accurateLevel min="0" max="100">
    <!--opt, xs: integer, correction strength, value range: [0,100]-->
  </accurateLevel>
  <zoomedInDistantViewLevel min="0" max="100">
    <!--opt, xs: integer, zooming in range, value range: [0-100], this node is valid when distortion correction is enabled-->
  </zoomedInDistantViewLevel>
  <horizontalFOV min="0" max="100"><!--opt, xs: integer, horizontal FOV, value range: [0,100]--></horizontalFOV>
  <verticalFOV min="0" max="100"><!--opt, xs: integer, vertical FOV, value range: [0,100]--></verticalFOV>
</LensDistortionCorrection>
```

XML_Cap_Link

Link capability message in XML format.

```
<Link version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <MACAddress><!--req, xs:string--></MACAddress>
  <autoNegotiation><!--req, xs:boolean, "true, false"--></autoNegotiation>
  <speed><!--req, xs:integer, "10.100, 1000"--></speed>
  <duplex><!--req, xs:string, "half, full"--></duplex>
  <MTU><!--req, xs:integer, "500-1500"--></MTU>
</Link>
```

XML_Cap_LogConfig

LogConfig capability message in XML format.

```
<LogConfig version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--required, boolean type, whether to enable log--></enabled>
  <level opt="none,debug,info,error,fault,all">
    <!--required, string type, log types, multiple type can be selected, and each type should be separated by comma-->
  </level>
</LogConfig>
```


XML_Cap_MACFilter

MACFilter capability message in XML format

```
<MACFilter version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled opt="true,false"><!--req, xs:boolean--></enabled>
  <permissionType opt="deny, allow"><!-- req, xs:string,"deny, allow"--></permissionType>
  <MACFilterAddressList size="">
    <MACFilterAddress>
      <id> <!--req, xs:string;id--></id>
      <MACAddress max=""><!--req, xs:string--></MACAddress>
    </MACFilterAddress>
  </MACFilterAddressList>
</MACFilter>
```

XML_Cap_mailingList

mailingList capability message in XML format.

```
<mailingList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <mailing><!--opt,xs:string--></mailing>
</mailingList>
```

XML_Cap_MaxElevation

MaxElevation capability message in XML format

```
<MaxElevation version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <mElevation min="-20" max="0"><!--req, xs:integer, the lower limit of max. tilt-angle--> </mElevation>
  <mVerElevation><!--req, xs:integer, the upper limit of max. tilt-angle, normally it is 90°--></mVerElevation>
</MaxElevation>
```

XML_Cap_MotionDetection

XML message about motion detection capability

```
<MotionDetection version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled opt="true,false"><!--required, xs:boolean, whether to enable motion detection--></enabled>
  <enableHighlight opt="true,false"><!--optional, xs:boolean, whether to enable highlight--></enableHighlight>
  <samplingInterval><!--optional, xs:integer, number of frames--></samplingInterval>
  <startTriggerTime><!--optional, xs:integer, unit: millisecond--></startTriggerTime>
  <endTriggerTime><!--opt, xs:integer, unit: millisecond--></endTriggerTime>
  <regionType opt="grid,roi,none, region"><!--required, read-only, xs:string, region type: "grid", "roi", "none",
"region"--></regionType>
  <Grid><!--dependency-->
    <rowGranularity><!--required, read-only, xs:integer--></rowGranularity>
```

```

<columnGranularity><!--required, read-only, xs:integer--></columnGranularity>
</Grid>
<ROI><!--dependency-->
  <normalizedScreenWidth><!--required, read-only, xs:integer--></normalizedScreenWidth>
  <normalizedScreenHeight><!--required, read-only, xs:integer--></normalizedScreenHeight>
</ROI>
<MotionDetectionLayout><!--required, rule region configuration of motion detection-->
  <sensitivityLevel min="0" max="100" step="20"><!--required, xs:integer, sensitivity--></sensitivityLevel>
  <layout>
    <gridMap>
      <!--dependent, xs:string, the image is divided in to N × N grids, each grid can be enabled (value: 1) or disabled
      (value 0) motion detection, each four horizontal grids consist of a hexadecimal number, e.g., the corresponding
      hexadecimal number of "0011" is 3-->
    </gridMap>
    <roiMap><!--dependent, xs:string--></roiMap>
    <RegionList size="4"><!--this node is available when the regionType is set to "region"-->
      <Region><!--list-->
        <id><!--required, xs:string--></id>
        <RegionCoordinatesList><!--opt-->
          <RegionCoordinates><!--list-->
            <positionX><!--required, xs:integer; X-coordinate--></positionX>
            <positionY><!--required, xs:integer; Y-coordinate--></positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </Region>
    </RegionList>
  </layout>
  <targetType opt="human,vehicle"><!--optional, xs:string, target type, "human", "vehicle"--></targetType>
</MotionDetectionLayout>
<enableWithMoving opt="true,false"><!--optional, xs:boolean--></enableWithMoving>
</MotionDetection>

```

XML_Cap_MotionDetectionLayout

XML message about rule region parameters of motion detection

```

<?xml version="1.0" encoding="utf-8"?>
<MotionDetectionLayout version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <sensitivityLevel min="0" max="100" step="20"><!--required, xs:integer, sensitivity--></sensitivityLevel>
  <layout>
    <gridMap>
      <!--dependent, xs:string, the image is divided in to N × N grids, each grid can be enabled (value: 1) or disabled
      (value 0) motion detection, each four horizontal grids consist of a hexadecimal number, e.g., the corresponding
      hexadecimal number of "0011" is 3-->
    </gridMap>
    <roiMap><!--dependent, xs:string--></roiMap>
    <RegionList size="4"><!--this node is available when the regionType is set to "region"-->
      <Region><!--list-->
        <id><!--required, xs:string--></id>
        <RegionCoordinatesList><!--opt-->
          <RegionCoordinates><!--list-->

```

```
<positionX><!--required, xs:integer; X-coordinate--></positionX>
<positionY><!--required, xs:integer; Y-coordinate--></positionY>
</RegionCoordinates>
</RegionCoordinatesList>
</Region>
</RegionList>
</layout>
<targetType opt="human,vehicle"><!--optional, xs:string, target type, "human", "vehicle"--></targetType>
</MotionDetectionLayout>
```

XML_Cap_MountingScenario

MountingScenario capability message in XML format

```
<MountingScenario version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <mode
    opt="indoor,outdoor,day,night,morning,nightfall,mode1,mode2,mode3,mode4,street,lowillumination,custom1,custom
    2,normal,road,faceSnap,highway,road2,backlight,frontlight">
    <!--req, xs:string, "lowillumination"-low illumination, "backlight"-back light, "frontlight"-front light, "faceSnap"-face
    picture capture-->
  </mode>
</MountingScenario>
```

XML_Cap_NetworkInterface

NetworkInterface capability message in XML format

```
<NetworkInterface version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:string--></id>
  <IPAddress><!--req-->
  <Wireless><!--opt-->
  <Discovery><!--opt-->
  <Link><!--opt-->
  <defaultConnection><!--opt, xs:boolean--></defaultConnection>
  <macAddress min="" max=""><!--opt, xs:string--></macAddress>
  <EthernetPortList size="4"><!--opt, network interface information-->
    <EthernetPort><!--opt -->
      <id min="1",max="4"><!--req, xs: integer--></id>
      <MACAddress><!--req, xs:string--></MACAddress>
      <status opt="connected, disconnect"><!--opt, xs:string--></status>
      <speed><!--req, xs:integer, "10, 100, 1000, 10000"--></speed>
    </EthernetPort>
  </EthernetPortList>
</NetworkInterface>
```

XML_Cap_NTPServer

NTPServer capability message in XML format

```
<NTPServer version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id min="1" max=""><!--req, xs:string, ID--></id>
  <addressingFormatType opt="ipaddress,hostname"><!--req, xs:string, which field will be used to locate the NTP
server: "ipaddress, hostname"--></addressingFormatType>
  <hostname min="0" max="64"><!--dep, xs:string--></hostname>
  <ipAddress min="0" max="64"><!--dep, xs:string--></ipAddress>
  <ipv6Address min="0" max="64"><!--dep, xs:string--></ipv6Address>
  <portNo min="1" max="65535"><!--opt, xs:integer--></portNo>
  <synchronizeInterval min="0" max="65535"><!--opt, xs:integer, NTP time synchronization interval, unit: minute--></
synchronizeInterval>
</NTPServer>
```

XML_Cap_ParkAction

ParkAction capability message in XML format

```
<ParkAction version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--required, xs:boolean, whether to enable parking--></enabled>
  <Parktime min="" max=""><!--required, xs:integer, parking time, unit: second--></Parktime>
  <Action><!--parking actions-->
    <ActionType opt="atuoscan,framescan,randomscan,panoramascan,patrol,pattern,preset,areaScan"><!--required,
xs:string, action type: "atuoscan,framescan,randomscan,panoramascan,patrol,pattern,preset,areaScan"--></
ActionType>
    <ActionNum min="0" max="255"><!--required, xs:integer, parking No. This node will be used when <ActionType> is
"patrol", "pattern", or "preset". For other parking action types, this node should be set to 0--></ActionNum>
  </Action>
</ParkAction>
```

XML_Cap_PreviewSwitch

PreviewSwitch message in XML format

```
<PreviewSwitch version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <groupNo min="0" max="15"><!--opt, xs: integer, group No.--></groupNo>
  <videoOutType
opt="CVBS,HDMI,VGA,AUXOutput1,AUXOutput2,AUXOutput3,AUXOutput4,extendHDMI1,extendHDMI2,extendHDMI3
,extendHDMI4,noSameSourceVGA1,noSameSourceVGA2,noSameSourceVGA3,noSameSourceVGA4,sameSourceVGA,m
ainLCD,AUX-VGA,AUX-LCD,zeroChannel">
    <!--opt, xs: string, video output types, if there are more than two outputs are homologous, you can combine these
outputs by "/" to be returned, e.g., if HDM1 and VGA1 are homologous outputs, "HDM1/VGA1" will be returned-->
  </videoOutType>
  <DisplayWindowList size="64">
    <DisplayWindow>
      <id><!--req, xs: string, window No., which equals to id × groupNo--></id>
```

```
<displayChannelNo min="0" max="64"><!--req, xs: string, 0-not display, 1 to 64-displayed channel No.--></displayChannelNo>
</DisplayWindow>
</DisplayWindowList>
<previewFrameNo opt="1,4,6,8,9,25,32,36,auto1,auto2,auto3,auto4">
  <!--req, xs: string, number of live view windows, auto1-custom window division 1, auto2-custome window division 2, auto3-custome window division 3, auto4-custome window division 4-->
</previewFrameNo>
<sound><!--req, xs: boolean, whether to turn on audio during live view, true-yes--></sound>
<switchTime opt="0,5,10,20,30,60,120,300"><!--req, xs: string, switching interval, unit: s--></switchTime>
<sameSource>
  <!--dep, xs: boolean, whether the output is homologous, true-yes, this node is valid only when VideoOutType is "noSameSourceVGA1", "noSameSourceVGA2", "noSameSourceVGA3", and "noSameSourceVGA4"-->
</sameSource>
<supportGetByPreviewNum>
  <!--opt, whether supports setting number of live view windows in the URL, true-yes-->
</supportGetByPreviewNum>
<supportGetByVideoOutType>
  <!--opt, xs: boolean, whether supports getting capability by output-->
</supportGetByVideoOutType>
</PreviewSwitch>
```

XML_Cap_RegionClip

RegionClip capability message in XML format

```
<RegionClip version="2.0" xmlns="http://www.std-cgi.org/ver20/XMLSchema">
  <id><!--req, xs: string--></id>
  <enabled><!--req, xs: boolean--></enabled>
  <normalizedScreenSize><!--req, read-only-->
    <normalizedScreenWidth><!--req, read-only, xs: integer--></normalizedScreenWidth>
    <normalizedScreenHeight><!--req, read-only, xs: integer--></normalizedScreenHeight>
  </normalizedScreenSize>
  <regionType opt="rectangle,convexPolygon,concavePolygon"><!--req, ro, xs:string--></regionType>
  <videoResolutionWidth opt="704"></videoResolutionWidth>
  <videoResolutionHeight opt="576"></videoResolutionHeight>
  <ClipRegionList>
    <ClipRegion>
      <RegionCoordinatesList size="1">
        <RegionCoordinates><!--req-->
          <positionX><!--req, xs: integer; x-coordinate--></positionX>
          <positionY><!--req, xs: integer; y-coordinate--></positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </ClipRegion>
  </ClipRegionList>
</RegionClip>
```

XML_Cap_RuleInfo

Capability message of setting behavior analysis in XML format.

```
<RuleInfo version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <ruleId opt="1,2,3,4,5,6,7,8"><!--required, xs:string--></ruleId>
  <ruleName min="" max=""><!--required, xs:string--></ruleName>
  <combinedRuleId opt="15,16"><!--required, xs:string--></combinedRuleId>
  <enabled opt="true,false"><!--required, xs:boolean--></enabled>
  <ruleType opt="region,line"><!--required, xs:string--></ruleType>
  <LineDetectionParam>
    <detectionTarget opt="all,human,vehicle, human_vehicle "/>
    <!--detection target, node with underline indicates that it supports multiple selections, e.g., human_vehicle
    indicates human and vehicle-->
    <directionSensitivity opt="left-right,right-left,any" def="any"><!--required, xs:string--></directionSensitivity>
    <isSupportHumanMisinfoFilter>
    <!--optional, xs:boolean, whether supports false alarm filter of human: true-supports, false or no return-not
    support-->
    </isSupportHumanMisinfoFilter>
    <isSupportVehicleMisinfoFilter>
    <!--optional, xs:boolean, whether supports false alarm filter of vehicle: true-supports, false or no return-not
    support-->
    </isSupportVehicleMisinfoFilter>
    <sensitivity min="1" max="100" default="50"><!--required, xs:integer, sensitivity, from 1 to 100--></sensitivity>
  </LineDetectionParam>
  <FieldDetectionParam>
    <durationTime min="1" max="100" def="5"><!--required,xs:integer--></durationTime>
    <sensitivityLevel min="1" max="100"><!--required, xs:integer, sensitivity from 0 to 100--></sensitivityLevel>
    <detectionTarget opt="all,human,vehicle, human_vehicle "/>
    <!--detection target, node with underline indicates that it supports multiple selections, e.g., human_vehicle
    indicates human and vehicle-->
    <isSupportHumanMisinfoFilter>
    <!--optional, xs:boolean, whether supports false alarm filter of human: true-supports, false or no return-not
    support-->
    </isSupportHumanMisinfoFilter>
    <isSupportVehicleMisinfoFilter>
    <!--optional, xs:boolean, whether supports false alarm filter of vehicle: true-supports, false or no return-not
    support-->
    </isSupportVehicleMisinfoFilter>
  </FieldDetectionParam>
  <RegionEntranceParam>
    <detectionTarget opt="all,human,vehicle, human_vehicle "/>
    <!--detection target, node with underline indicates that it supports multiple selections, e.g., human_vehicle
    indicates human and vehicle-->
    <isSupportHumanMisinfoFilter>
    <!--optional, xs:boolean, whether supports false alarm filter of human: true-supports, false or no return-not
    support-->
    </isSupportHumanMisinfoFilter>
    <isSupportVehicleMisinfoFilter>
    <!--optional, xs:boolean, whether supports false alarm filter of vehicle: true-supports, false or no return-not
    support-->
    </isSupportVehicleMisinfoFilter>
  </RegionEntranceParam>
</RuleInfo>
```

```

</isSupportVehicleMisinfoFilter>
</RegionEntranceParam>
<RegionExitingParam>
  <detectTarget opt="all,human,vehicle, human_vehicle "/><!-- detection target, node with underline indicates
that it supports multiple selections, e.g., human_vehicle indicates human and vehicle-->
  <isSupportHumanMisinfoFilter>
    <!--optional, xs:boolean, whether supports false alarm filter of human, true-supports, false or no return-not
support-->
  </isSupportHumanMisinfoFilter>
  <isSupportVehicleMisinfoFilter>
    <!--optional, xs:boolean, whether supports false alarm filter of vehicle, true-supports, false or no return-not
support-->
  </isSupportVehicleMisinfoFilter>
</RegionExitingParam>
<PeopleNumChangeParam>
  <peopleNumThreshold min="" max="" default=""><!--optional, xs:integer--></peopleNumThreshold>
  <detectMode opt="greaterthan,lessThan,equal,notEqual"><!--optional, xs:string--></detectMode>
  <noneStateEffectiveEnabled><!--optional,xs:boolean--></noneStateEffectiveEnabled>
  <durationTime min="" max="" default=""><!--optional, xs:integer--></durationTime>
</PeopleNumChangeParam>
<SpacingChangeParam>
  <spacingThreshold min="" max="" default=""><!--optional, xs: integer--></spacingThreshold>
  <detectMode><!--optional,xs:string, "greaterthan,lessThan"--></detectMode>
  <durationTime min="" max="" default=""><!--optional, xs:integer--></durationTime>
</SpacingChangeParam>
<ViolentMotionParam>
  <sensitivity min="" max="" default=""><!--optional, xs:integer--></sensitivity>
</ViolentMotionParam>
<LeavePositionParam>
  <leaveDelay min="" max="" default=""><!--optional, xs:integer--></leaveDelay>
  <onPosition min="" max="" default=""><!--optional, xs:integer--></onPosition>
</LeavePositionParam>
<FailDownParam>
  <durationTime min="" max="" default=""><!--optional, xs:integer--></durationTime>
  <sensitivity min="" max="" def=""><!--optional, xs:integer, sensitivity--></sensitivity>
  <heightThreshold min="" max="" default=""><!--optional, xs:integer--></heightThreshold>
  <stackTarget opt="true,false"><!--optional, xs:boolean, whether to overlay alarm target frame--></stackTarget>
  <stackRule opt="true,false"><!--optional, xs:boolean, whether to overlay alarm rule frame--></stackRule>
</FailDownParam>
<RunningParam><!--running detection-->
  <speed min="" max="" default=""><!--optional, xs:integer--></speed>
  <durationTime min="" max="" def=""><!--optional, xs:integer, duration time--></durationTime>
  <sensitivity min="" max="" def=""><!--optional, xs:integer, sensitivity--></sensitivity>
  <stackTarget opt="true,false"><!--optional, xs:boolean, whether to overlay alarm target frame--></stackTarget>
  <stackRule opt="true,false"><!--optional, xs:boolean, whether to overlay alarm rule frame--></stackRule>
  <mode opt="single,many"><!--optional, xs:string, running mode, "single"-single person running, "many"-multiple
persons running--></mode>
</RunningParam>
<RetentionParam><!--dependent-->
  <durationTime min="" max="" default="">
    <!--required, xs:integer, duration time, from 60 seconds to 3600 seconds, default value: 1800 seconds-->
  </RetentionParam>

```

```

<ParkingParam>
  <durationTime min="5" max="100"><!-- required,xs:integer--> </durationTime>
</ParkingParam>
<eventType
opt="none,lineDetection,fieldDetection,regionEntrance,regionExiting,parking,peopleNumChange,spacingChange,viole
ntMotion,leavePosition,failDown,running,retentionParam,playCellphone">
  <!--required, xs:string-->
</eventType>
<SizeFilter>
  <enabled opt="true,false"><!--required, xs:boolean--></enabled>
  <mode opt="pixels,actualSize"><!--required, xs:string, filter mode: "pixels"-filter according to pixel, "actualSize"-
filter according to actual size--></mode>
  <ObjectSizeList size="4">
    <ObjectSize><!--only return the default value of supported detection type-->
      <detectionTarget><!--optional,xs:string, detection target, "all"-all, "human"-human, "vehicle"-vehicle,
"human_vehicle"-human and vehicle--></detectionTarget>
      <MaxObjectSize><!--maximum size:float-->
        <positionX><!--required, xs:integer;coordinate--></positionX>
        <positionY><!--required, xs:integer;coordinate--></positionY>
        <width><!--required, xs:integer, default width value: when detectionTarget values "all", it is 500, "human"-200,
"vehicle"-500, "human_vehicle"-500, unit: cm--></width>
        <height><!--required, xs:integer, default height value: when detectionTarget values "all", it is 100, "human"-300,
"vehicle"-100, "human_vehicle"-100, unit: cm --></height>
      </MaxObjectSize>
      <MinObjectSize><!--minimum size:float-->
        <positionX><!--required, xs:integer;coordinate--></positionX>
        <positionY><!--required, xs:integer;coordinate--></positionY>
        <width><!--required, xs:integer, default width value: when detectionTarget values "all", it is 60, "human"-60,
"vehicle"-180, "human_vehicle"-60, unit: cm--></width>
        <height><!--required, xs:integer, default height value: when detectionTarget values "all", it is 10, "human"-10,
"vehicle"-140, "human_vehicle"-10, unit: cm--></height>
      </MinObjectSize>
    </ObjectSize>
  </ObjectSizeList>
</SizeFilter>
<RuleRegion>
  <RegionCoordinatesList size="4">
    <RegionCoordinates>
      <positionX min="" max=""><!--required, xs:integer--></positionX>
      <positionY min="" max=""><!--required, xs:integer--></positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
</RuleRegion>
<backgroundSuppression opt="open,close,selfAdapt"><!--optional, xs:string, background suppression--></
backgroundSuppression>
<PlayCellphoneParam><!--optional, playing mobile phone-->
  <durationTime min="1" max="3600" def=""><!--optional, xs:integer, duration time--></durationTime>
  <stackTarget opt="true,false"><!--optional, xs:boolean, whether to overlay alarm target frame--></stackTarget>
  <stackRule opt="true,false"><!--optional, xs:boolean, whether to overlay alarm rule frame--></stackRule>
</PlayCellphoneParam>
<streetBehavior opt="running,group,violentMotion,failDown">
  <!--optional, xs:string, street behavior capability: "running"-fast moving, "group"-people gathering, "violentMotion"-

```



```
violent motion, "failDown"-failing down-->
</streetBehavior>
<workBehavior opt="playCellphone,peopleNumChange,leavePosition,retention">
  <!--optional, xs:string, working behavior capability: "playCellphone"-playing mobile phone, "peopleNumChange"-
number of persons changed, "leavePosition"-absent, "retention"-loitering-->
</workBehavior>
<isSupportMultiRules><!--optional, xs:boolean, whether it supports multiple rules for one event; by default, only one
rule can be configured for one event--></isSupportMultiRules>
<isSupportSchedulesEventType><!--optional, xs:boolean, whether to configure arming schedule by event type; by
default, it is configured by rule ID--></isSupportSchedulesEventType>
<isSupportTriggerEventType><!--optional, xs:boolean, whether to configure alarm linkage by event type; by default, it
is configured by rule ID--></isSupportTriggerEventType>
<analysisMode opt="realTimeMode,autoSwitchMode"><!--optional, xs:string, analysis mode, the default mode is
"realTimeMode"--></analysisMode>
</RuleInfo>
```

Remarks

When the **mode** values "actualSize", the input value of **positionX** and **positionY** is 0.

XML_Cap_SalveCamera

SalveCamera capability message in XML format

```
<SlaveCamera version="2.0" xmlns="http://www.std-cgi.org/ver20/XMLSchema">
  <SlaveCameraInfoList>
    <SlaveCameraInfo>
      <id min="1" max="4">
        <!--req, xs:integer-->
      </id>
      <serverAddress>
        <addressingFormatType opt="ipaddress,hostname">
          <!--req, xs:string, "ipaddress,hostname"-->
        </addressingFormatType>
        <hostName max="">
          <!--dep, xs:string-->
        </hostName>
        <ipAddress>
          <!--dep, xs:string-->
        </ipAddress>
        <ipv6Address>
          <!--dep, xs:string-->
        </ipv6Address>
      </serverAddress>
      <userName max="">
        <!--req, xs:string-->
      </userName>
      <passWord max="">
        <!--req, xs:string-->
      </passWord>
      <portNo min="" max="">
```

```
<!--req, xs:integer-->
</portNo>
<loginStatus opt="login,logout">
  <!--req, xs:string-->
</loginStatus>
</SlaveCameraInfo>
</SlaveCameraInfoList>
</SlaveCamera>
```

XML_Cap_SerialCommand

SerialCommand capability message in XML format

```
<SerialCommand version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <chainNo min="1" max="16">
    <!--opt, xs:integer-->
  </chainNo>
  <command>
    <!--req, xs:hexBinary-->
  </command>
</SerialCommand>
```

XML_Cap_SerialPort

SerialPort capability message in XML format

```
<SerialPort version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id min="0" max="255">
    <!--req, xs:string, serial port ID-->
  </id>
  <enabled>
    <!--req, xs:boolean, whether to enable the serial port: "true, false"-->
  </enabled>
  <serialNumber opt="1,2,3,4,5">
    <!--opt, xs:integer, serial port No., which is between 1 and 5-->
  </serialNumber>
  <serialPortType opt="RS485,RS422,RS232">
    <!--req, xs:string, serial port type: "RS485", "RS422", "RS232"-->
  </serialPortType>
  <duplexMode opt="half,full">
    <!--req, xs:string, duplex mode of the serial port: "half", "full"-->
  </duplexMode>
  <direction opt="monodirectional,bidirectional">
    <!--req, xs:string-->
  </direction>
  <baudRate range="600,1200,2400,4800,9600,19200,38400,57600,76800,115200">
    <!--req, xs:integer-->
  </baudRate>
  <dataBits min="5" max="8">
```

```
<!--req, xs:integer-->
</dataBits>
<parityType opt="none,even,odd,mark,space">
  <!--req, xs:string-->
</parityType>
<stopBits range="1,2">
  <!--req, xs:string, stop bit: "1,1.5,2"-->
</stopBits>
<workMode opt="console,transparent,narrowband,audiomixer,stairsControl,cardReader,disabled,custom">
  <!--dep, xs:string, working mode: "console","transparent","narrowband","audiomixer","stairsControl"-elevator
control,"cardReader"-card reader,"disabled","custom". This node is required only when <serialPortType> is set to
"RS232"-->
</workMode>
<flowCtrl opt="none,software,hardware">
  <!--req, xs:string-->
</flowCtrl>
<rs485WorkMode opt="Led, CaptureTrigger">
  <!--opt, xs:string, working mode of RS-485 serial port, which is used for LED display or triggering transmission of
captured pictures. This node is valid only when <serialPortType> is "RS485"-->
</rs485WorkMode>
<audiomixerProtocolType opt="HIKVISION">
  <!--opt, xs:string, "HIKVISION"-->
</audiomixerProtocolType>
<deviceName min="0" max="32"><!--dependent, xs:string, this node exists only when <serialPortType> is "RS485"-->
</deviceName>
<deviceType min="" max=""><!--dependent, xs:integer, this node exists only when <serialPortType> is "RS485"-->
</deviceType>
<deviceProtocol min="" max=""><!--dependent, xs:integer, this node exists only when <serialPortType> is "RS485"-->
</deviceProtocol>
<mode opt="readerMode,clientMode,externMode,accessControlHost, disabled "><!--dependent, xs:string, working
mode, this node exists only when <serialPortType> is "RS485"-->
</mode>
</SerialPort>
```

XML_Cap_SIPInfo

SIPInfo capability message in XML format

```
<SIPInfo version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <videoID opt="1,2,3">
    <!--req, xs:string-->
  </videoID>
  <AlarmInList>
    <AlarmIn>
      <id min="0" max="8">
        <!--req, xs:integer-->
      </id>
      <alarmInID min="0" max="64">
        <!--req, xs:string-->
      </alarmInID>
    </AlarmIn>
  </AlarmInList>
</SIPInfo>
```

```
</AlarmInList>
</SIPInfo>
```

XML_Cap_SIPServer

SIPServer capability message in XML format

```
<SIPServer version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id min="1" max="">
    <!--req, xs:integer, the maximum ID depends on the number of channels-->
  </id>
  <localPort min="0" max="65535">
    <!--req, xs:integer-->
  </localPort>
  <streamID min="1" max="255">
    <!--req, xs:integer, stream type: 1-main stream, 2-sub-stream, 3-third stream-->
  </streamID>
  <Standard>
    <!--opt-->
    <registerStatus>
      <!--ro, req, xs:boolean, registration status: "false"-unregistered, "true"-registered-->
    </registerStatus>
    <enabled >
      <!--req, xs:string, "true"-log in, "false"-log out-->
    </enabled>
    <registrar min="0" max="128">
      <!--req, xs:string-->
    </registrar>
    <registrarPort min="1024" max="65535">
      <!--req, xs:integer-->
    </registrarPort>
    <proxy min="0" max="128">
      <!--req, xs:string-->
    </proxy>
    <proxyPort min="0" max="65535">
      <!--req, xs:integer-->
    </proxyPort>
    <displayName min="0" max="64">
      <!--req, xs:string-->
    </displayName>
    <username min="0" max="64">
      <!--req, xs:string-->
    </userName>
    <authID min="0" max="64">
      <!--req, xs:string-->
    </authID>
    <password min="0" max="32">
      <!--wo, req, xs:string-->
    </password>
    <expires min="0" max="100000">
      <!--req, xs:integer-->
```

```
</expires>
</Standard>
<GB28181>
  <!--opt-->
    <registerStatus>
      <!--req, xs:boolean, "true,false"-->
    </registerStatus>
    <enabled>
      <!--req, xs:boolean, "true,false"-->
    </enabled>
    <registrar min="0" max="128">
      <!--req, xs:string-->
    </registrar>
    <registrarPort min="1024" max="65535">
      <!--req, xs:integer-->
    </registrarPort>
    <serverId min="0" max="64">
      <!--req, xs:string-->
    </serverId>
    <serverDomain min="0" max="128">
      <!--req, xs:string-->
    </serverDomain>
    <username min="0" max="64">
      <!--req, xs:string-->
    </userName>
    <authID min="0" max="64">
      <!--req, xs:string-->
    </authID>
    <password min="0" max="32">
      <!--wo, req, xs:string-->
    </password>
    <expires min="30" max="100000">
      <!--req, xs:integer-->
    </expires>
    <liveTime min="30" max="100000">
      <!--req, xs:integer-->
    </liveTime>
    <heartbeatTime min="0" max="255">
      <!--req, xs:integer-->
    </heartbeatTime>
    <heartbeatCount min="3" max="255">
      <!--req, xs:integer-->
    </heartbeatCount>
  </GB28181>
</SIPServer>
```

XML_Cap_SmartSearchDescription

SmartSearchDescription capability message in XML format

```
<SmartSearchDescription version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <searchID>
    <!--req, xs:string, search ID, which is used to confirm the upper-level platform or system. If the platform or system
are same during two times of search, the search history will be recorded in the memory for next fast search-->
  </searchID>
  <searchResultPosition><!--req, xs: integer--></searchResultPosition>
  <maxResults><!--req, xs: integer--></maxResults>
  <trackID><!--req, xs: integer, channel ID, e.g., 101-main stream if first channel--></trackID>
  <startTime><!--req, xs: datetime, start time of validity period, e.g., 2013-06-10T 12:00:00Z--></startTime>
  <endTime><!--req, xs: datetime, end time of validity period, e.g., 2013-06-10T 13:00:00Z--></endTime>
  <type opt="motionDetection, traversePlaneDetection,FieldDetection, faceDetection">
    <!--req, xs: string; VCA event type-->
  </type>
  <MotionDetection><!--dep, it is valid when type is "motionDetection"-->
    <Grid>
      <rowGranularity></rowGranularity>
      <columnGranularity></columnGranularity>
    </Grid>
    <MotionDetectionLayout>
      <layout>
        <gridMap></gridMap>
      </layout>
    </MotionDetectionLayout>
    <sensitivity min="1", max="100"><!--opt, xs: integer--></sensitivity>
  </MotionDetection>
  <TraversePlane><!--dep, it is valid when type is "traversePlaneDetection"-->
    <TraversePlaneParam>
      <LineList>
        <Line>
          <StartPoint>
            <x><!--req, xs: float--></x>
            <y><!--req, xs: float--></y>
          </StartPoint>
          <EndPoint>
            <x><!--req, xs: float--></x>
            <y><!--req, xs: float--></y>
          </EndPoint>
        </Line>
      </LineList>
      <crossDirection opt="bothDirection, leftToRight, rightToLeft"><!--opt, xs: string, crossing direction--
    </crossDirection>
    <sensitivity min="1", max="100"><!-- opt, xs: integer--></sensitivity>
    <planeHeight><!--opt, xs: integer--></planeHeight>
    <detectionTarget opt="all, human, vehicle"><!--req, xs: string--><detectionTarget>
  </TraversePlaneParam>
  <dwPreTime><!--opt, xs: integer, pre-recording time--></dwPreTime>
  <dwDelayTime><!--opt, xs: integer,post-recording time--></dwDelayTime>
  <PTZPos>
    <P><!--opt, xs: integer--></P>
    <T><!--opt, xs: integer--></T>
    <Z><!--opt, xs: integer--></Z>
  </PTZPos>
```

```

<advancedType opt="humanFeature, objectFeature"><!--opt, xs: string--></advancedType>
<advance><!--dep, it is valid when type is "motionDetection"-->
  <HumanFeature>
    <age min="0" max="100"><!--opt, xs: integer--></age>
    <sex opt="male,female"><!--opt, xs: string--></sex>
    <eyeGlass><!--opt, xs: boolean--></eyeGlass>
  </HumanFeature><!--dep, it is valid when type is "humanFeature"-->
  <ObjectFeature>
    <colorRatel><!--opt, xs: integer--></colorRatel>
    <R><!--opt, xs: integer--></R>
    <G><!--opt, xs: integer--></G>
    <B><!--opt, xs: integer--></B>
  </ObjectFeature><!--dep, it is valid when type is "objectFeature"-->
</advance>
</TraversePlane>
<FieldDetection><!--dep, it is valid when type is "FieldDetection"-->
  <normalizedScreenSize>
    <normalizedScreenWidth><!--opt, xs: integer--></normalizedScreenWidth>
    <normalizedScreenHeight><!--opt, xs: integer--></normalizedScreenHeight>
  </normalizedScreenSize>
  <param>
    <region>
      <pointList>
        <point>
          <x><!--opt, xs: integer--></x>
          <y><!--opt, xs: integer--></y>
        </point>
      </pointList>
    </region>
  </param>
  <duration min="1", max="100"><!--opt, xs: integer--></duration>
  <sensitivity min="1", max="100"><!--opt, xs: integer--></sensitivity>
  <rate min="1", max="100"><!--opt, xs: integer--></rate>
  <detectionTarget opt="all, human, vehicle"><!--req, xs: string--><detectionTarget>
  <dwPreTime><!--opt, xs: integer, pre-recording time--></dwPreTime>
  <dwDelayTime><!--opt, xs: integer, post-recording time--></dwDelayTime>
  <PTZPos>
    <P><!--opt, xs: integer--></P>
    <T><!--opt, xs: integer--></T>
    <Z><!--opt, xs: integer--></Z>
  </PTZPos>
  <advancedType opt="humanFeature, objectFeature"><!--opt, xs: string--></advancedType>
  <advance><!--dep, it is valid when type is "motionDetection"-->
    <HumanFeature>
      <age min="0" max="100"><!--opt, xs: integer--></age>
      <sex opt="male,female"><!--opt, xs: string--></sex>
      <eyeGlass><!--opt, xs: boolean--></eyeGlass>
    </HumanFeature><!--dep, it is valid when type is "humanFeature"-->
    <ObjectFeature>
      <colorRatel><!--opt, xs: integer--></colorRatel>
      <R></R>
      <G></G>

```

```
<B></B>
</ObjectFeature><!--dep, it is valid when type is "objectFeature"-->
</advance>
</FieldDetection>
<FaceDetection><!--dep, it is valid when type is "faceDetection"-->
  <param>
    <region>
      <pointList>
        <point>
          <x/>
          <y/>
        </point>
      </pointList>
    </region>
  </param>
  <duration min="1", max="100"><!--opt, xs: integer--></duration>
  <sensitivity min="1", max="100"><!--opt, xs: integer--></sensitivity>
  <rate min="1", max="100"><!--opt, xs: integer--></rate>
  <detectionTarget opt="all, human, vehicle"><!--req, xs: string--><detectionTarget>
  <dwPreTime><!--opt, xs: integer, pre-recording time--></dwPreTime>
  <dwDelayTime><!--opt, xs: integer, post-recording time--></dwDelayTime>
  <PTZPos>
    <P><!--opt, xs: integer--></P>
    <T><!--opt, xs: integer--></T>
    <Z><!--opt, xs: integer--></Z>
  </PTZPos>
  <advancedType opt="humanFeature, objectFeature"><!--opt, xs: string--></advancedType>
  <advance><!--dep, it is valid when type is "motionDetection"-->
  <HumanFeature>
    <colorRatel><!--opt, xs: integer--></colorRatel>
    <age min="0" max="100"><!--opt, xs: integer--></age>
    <sex opt="male,female"><!--opt, xs: string--></sex>
    <eyeGlass><!--opt, xs: boolean--></eyeGlass>
  </HumanFeature> <!--dep, it is valid when type is "humanFeature"-->
  <ObjectFeature>
    <colorRatel><!--opt, xs: integer--></colorRatel>
    <R></R>
    <G></G>
    <B></B>
  </ObjectFeature><!--dep, it is valid when type is "objectFeature"-->
</advance>
</FaceDetection>
</SmartSearchDescription>
```

XML_Cap_SoftwareService

SoftwareService capability message in XML format

```
<SoftwareService version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ThirdStream><!--opt, third stream configuration-->
    <enabled opt="true,false"><!--req, xs: boolean, whether to enable third stream configuration--></enabled>
```



```
<thirdStreamPrompt opt="prompt5">
  <!--req, xs: string, set audible prompt for third stream, prompt5-"When third stream is enabled, functions such as
intrusion detection, line crossing detection, region entrance detection, region exiting detection, face detection, HDMI,
H.264+ and H265+, are not supported. New settings will take effect after a reboot."-->
</thirdStreamPrompt>
</ThirdStream>
<PanoramaDisplay><!--opt, display the image range on panorama view-->
  <enabled opt="true,false" default="false">
    <!--req, xs: boolean, whether to enable image range display on panorama view-->
  </enabled>
</PanoramaDisplay>
<MotionDetect>
  <enabled opt="true,false"><!--whether to enable motion detection--></enabled>
  <MotionDetectPrompt opt="prompt1">
    <!--req, xs: string, motion detection audible prompt, prompt1-"When motion detection is enabled, functions such
as intrusion detection, line crossing detection, region entrance detection, and region exiting detection, are not
supported. New settings will take effect after a reboot."-->
  </MotionDetectPrompt>
</MotionDetect>
</SoftwareService>
```

XML_Cap_storageExtension

XML message about storage strategy capability.

```
<storageExtension xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0" >
  <LoopEnable,opt="true,false">
    <!--optional, xs:boolean, whether to enable overwriting-->
  </LoopEnable>
  <enableDormant>
    <!--optional, xs:boolean-->
  </enableDormant>
  <packDuration max="300" min="1">
    <!--optional, xs:integer, unit: minute-->
  </packDuration>
  <logStorageMode opt="system,custom">
    <!--optional, xs:string, log storage mode: "system"-system default mode, "custom"-custom mode-->
  </logStorageMode>
  <logStorageId>
    <!--dependent, xs:integer, disk ID (saving logs in a disk) or array number (saving logs in disk arrays), which are valid
only when logStorageMode is "custom"; -->
  </logStorageId>
  <logStorageCycle min="0" max="90" def="90">
    <!--dependent, xs:integer, log storage period, which is valid only when logStorageMode is "custom"-->
  </logStorageCycle>
</storageExtension>
```

XML_Cap_StreamingChannel

StreamingChannel capability message in XML format

```
<StreamingChannel version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id opt="111,222,333,444"><!--req, xs:string, ID--></id>
  <channelName min="0" max="64"><!--req, xs:string, channel name--></channelName>
  <enabled opt="true,false" def="true"><!--req, xs:boolean--></enabled>
  <Transport><!--req-->
    <rtspPortNo min="0" max="65535" def="554"></rtspPortNo>
    <maxPacketSize min="0" max="1500"><!--opt, xs:integer--></maxPacketSize>
    <audioPacketLength min="0" max="5000"/>
    <audioInboundPacketLength min="0" max="5000"/><!--opt, xs:integer-->
    <audioInboundPortNo min="0" max="65535"/><!--opt, xs:integer-->
    <videoSourcePortNo min="0" max="65535"/><!--opt, xs:integer-->
    <audioSourcePortNo min="0" max="65535"/><!--opt, xs:integer-->
    <ControlProtocolList><!--req-->
      <ControlProtocol><!--list-->
        <streamingTransport opt="RTSP/RTP,HTTP"><!--req, xs:string, "HTTP,RTSP,SHTTP,SRTP"--></streamingTransport>
      </ControlProtocol>
    </ControlProtocolList>
    <Unicast><!--opt-->
      <enabled opt="true,false" def="false"/><!--req, xs:boolean-->
      <rtpTransportType opt="RTP/UDP,RTP/TCP"/><!--opt, xs:string, "RTP/UDP,RTP/TCP"-->
    </Unicast>
    <Multicast><!--opt-->
      <enabled opt="true,false" def="false"/><!--req, xs:boolean-->
      <userTriggerThreshold/><!--opt, xs:integer-->
      <videoDestPortNo min="1" max="65535" def="8860"/><!--opt, xs:integer-->
      <audioDestPortNo min="1" max="65535" def="8860"/><!--opt, xs:integer-->
      <destIPAddress min="8" max="16"/><!--dep, xs:string-->
      <destIPv6Address min="15" max="39"/><!--dep, xs:string-->
      <ttl min="0" max="127" def="1"/><!--opt, xs:integer-->
      <activeMulticastEnabled>
        <!--opt, xs: boolean, "true, false", whether to enable active multicast, which is mutually exclusive with passive
multicast-->
      </activeMulticastEnabled>
      <packagingFormat opt="RTP,TS,ES,PS"><!--opt, xs: string, encapsulation format--></packagingFormat>
      <FecInfo><!--opt-->
        <fecRatio min="0" max="100">
          <!--req, read-only, xs: integer, extra bandwidth occupation ratio of forward error correction (FEC) data, the value
is between 0 and 100, the default value is 0-->
        </fecRatio>
        <fecDestPortNo>
          <!--opt, xs: integer, port No. of FEC multicast, the default port No. is specified by device-->
        </fecDestPortNo>
      </FecInfo>
    </Multicast>
    <Security><!--opt-->
      <enabled opt="true,false" def="false"/><!--req, xs:boolean-->
      <certificateType opt="digest,digest/baisc" def="digest"><!--req, xs:string--></certificateType>

```

```

    <SecurityAlgorithm><!--dependency, this node is valid when certificateType is "digest" or "digest/basic"-->
      <algorithmType opt="MD5,SHA256,MD5/SHA256"><!--optional, xs:string, algorithm type: "MD5", "SHA256",
"MD5/SHA256"--></algorithmType>
    </SecurityAlgorithm>
  </Security>
  <SRTPMulticast><!--opt-->
    <SRTPVideoDestPortNo min="" max=""><!--opt, xs:integer--></SRTPVideoDestPortNo>
    <SRTPAudioDestPortNo min="" max=""><!--opt, xs:integer--></SRTPAudioDestPortNo>
  </SRTPMulticast>
</Transport>
<Video>
  <enabled opt="true,false">true</enabled>
  <videoInputChannelID opt="1,2,3,4">2</videoInputChannelID>
  <videoCodecType opt="MJPEG,MPEG4">MPEG4</videoCodecType>
  <videoScanType opt="interlaced,progressive">progressive</videoScanType>
  <videoResolutionWidth min="0" max="640">640</videoResolutionWidth>
  <videoResolutionHeight min="0" max="480">480</videoResolutionHeight>
  <videoDiffResolutionList><!--opt, list, if this node exists, it indicates that the resolution depends on the node
<videoCodecType>, and <videoResolutionWidth> and <videoResolutionHeight> are invalid. For forward
compatibility, the maximum set of <videoCodecType>, <videoResolutionWidth>, and <videoResolutionHeight>-->
    <videoDiffResolution>
      <videoCodecType>
        <!--req, xs:string, "MPEG4,MJPEG,3GP,H.264,HK.264,MPNG,SVAC"-->
      </videoCodecType>
      <videoResolutionWidth><!--req, xs:integer--></videoResolutionWidth>
      <videoResolutionHeight><!--req, xs:integer--></videoResolutionHeight>
    </videoDiffResolution>
  </videoDiffResolutionList>
  <videoPositionX min="0" max="640">0</videoPositionX>
  <videoPositionY min="0" max="480">0</videoPositionY>
  <videoQualityControlType opt="CBR,VBR">CBR</videoQualityControlType>
  <constantBitRate min="50" max="4000" dynamic="true">2000</constantBitRate>
  <maxFrameRate opt="2500,1250,625,312,156,78, 830">2500</maxFrameRate>
  <keyFrameInterval min="0", max="10000">1000</keyFrameInterval>
  <rotationDegree opt="0,90,180,270" def="0">0</rotationDegree>
  <mirrorEnabled opt="true,false" def="false">false</mirrorEnabled>
  <snapshotImageType opt="JPEG" def="JPEG">JPEG</snapshotImageType>
  <IntelligentInfoDisplayMethod opt="player,non-player"><!--opt, xs:string, method of displaying intelligent
information: "player", "non-player"--></IntelligentInfoDisplayMethod>
  <minimumResolutionSupportedBySmartCode>
    <!--opt, xs:string, the minimum resolution supported by smart coding (smart264 and smart265), e.g., "640*512". If
this node is returned, it indicates the smart code of current device (H7, H5) is limited by the minimum resolution, and
the upper layer can get the supported minimum resolution for smart coding of current device. For the resolution
lower than the minimum resolution does not support smart coding. If this node is not returned, it indicates the smart
coding of current device (H3) is not limited by minimum resolution-->
  </minimumResolutionSupportedBySmartCode>
</Video>
<Audio>
  <enabled opt="true,false" def="false">false</enabled>
  <audioInputChannelID opt="1,2,3,4">2</audioInputChannelID>
  <audioCompressionType opt="G.726,G.711ulaw" def="G.726">G.726</audioCompressionType>
  <audioBitRate opt="16,24,32,40" def="32" dynamic="true">24</audioBitRate>

```

```
<audioSamplingRate opt="8" dynamic="true">8</audioSamplingRate>
<audioResolution opt="3,4,5,6" dynamic="true"/>
</Audio>
<isSupportDynamicCapWithCondition><!--opt, xs:boolean, whether to support dynamic capability with conditions--></isSupportDynamicCapWithCondition>
<enableCABAC><!--opt, xs:boolean, whether it supports compressing stream to improve performance--></enableCABAC>
<subStreamRecStatus><!--opt, xs:boolean--></subStreamRecStatus>
<isSupportRefreshFrame><!--opt, xs:boolean, whether it supports refreshing frames when Smart264 is enabled--></isSupportRefreshFrame>
<isSupportBareDataOverlay><!--opt, xs:boolean--></isSupportBareDataOverlay>
<isSupportRTCPCfg><!--opt, xs:boolean--></isSupportRTCPCfg>
<customStreamEnable><!--opt, xs:boolean, whether the stream is custom stream: "true"-yes, this node is not returned-no--></customStreamEnable>
</StreamingChannel>
```

XML_Cap_SupplementLight

SupplementLight capability message in XML format

```
<SupplementLight><!--opt-->
  <mode opt="schedule,off,on,auto"><!--opt, xs: string, adjustment mode of supplement light: "schedule,off,on,auto"--></mode>
  <Schedule>
    <TimeRange><!--req-->
      <beginTime><!--req, xs: time, ISO8601 time--></beginTime>
      <endTime><!-- req, xs: time, ISO8601 time--></endTime>
    </TimeRange>
  </Schedule>
  <brightnessLimit min="0" max="100">
    <!--dep, xs: integer, brightness of supplement light, which is between 0 and 100; the brightness can be adjust when <bmode> is set to "on"-->
  </brightnessLimit>
  <supplementLightMode opt="mixed,whiteLight,close">
    <!--opt, xs: string, illumination mode: "mixed"-hybrid, "whitelight"-white light, "close"-disabled-->
  </supplementLightMode>
  <irLightBrightness min="0" max="100">
    <!--dep, xs: integer, brightness of IR supplement light, which is between 0 and 100; this node is valid only when <supplementLightMode> is set to "mixed" and <mixedLightBrightnessRegulatMode> is set to "manual"-->
  </irLightBrightness>
  <mixedLightBrightnessRegulatMode opt="manual,auto">
    <!--dep, xs: string, brightness adjustment mode of hybrid supplement light; this node is valid only when <supplementLightMode> is set to "mixed"-->
  </mixedLightBrightnessRegulatMode>
  <mixedModeSupportType opt="irLight,whiteLight">
    <!--dep, xs: string, light type of hybrid illumination mode, "irLight"-IR light, "whiteLight"-white light-->
  </mixedModeSupportType>
  <highIrLightBrightness min="0" max="100">
    <!--dep, xs: integer, brightness of far IR light, which is between 0 and 100; this node cannot be configured together with node <irLightBrightness>-->
  </highIrLightBrightness>
```

```
<highWhiteLightBrightness min="0" max="100">
  <!--dep, xs: integer, brightness of far white light, which is between 0 and 100; this node cannot be configured
together with node <brightnessLimit>-->
</highWhiteLightBrightness>
<lowIrLightBrightness min="0" max="100">
  <!--dep, xs: integer, brightness of near IR light, which is between 0 and 100; this node cannot be configured together
with node <irLightBrightness>-->
</lowIrLightBrightness>
<lowWhiteLightBrightness min="0" max="100">
  <!--dep, xs: integer, brightness of near white light, which is between 0 and 100; this node cannot be configured
together with node <brightnessLimit>-->
</lowWhiteLightBrightness>
<whiteLightBrightness min="0" max="100"><!--dep, xs: integer, white light brightness, which is between 0 and 100-->
</whiteLightBrightness>
<irLightbrightnessLimit min="0" max="100"><!--dep, xs:integer, IR light brightness limit, range: [0,100]; this node is
valid when the value of mode "auto"--></irLightbrightnessLimit>
<whiteLightbrightnessLimit min="0" max="100"><!--dep, xs:integer, white light brightness limit, range: [0,100]; this
node is valid when the value of mode "auto"--></whiteLightbrightnessLimit>
<unscheduledBrightness min="" max=""><!--optional, xs:integer, brightness in the unscheduled time, by default it is
0 (the supplement light is disabled) and the value is between 0 and 100--></unscheduledBrightness>
</SupplementLight>
```

XML_Cap_SyncSignalOutputList

SyncSignalOutputList capability message in XML format

```
<SyncSignalOutputList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <IOOutNum min="3" max="8">
    <!--req, xs:string, number of IO outputs-->
  </IOOutNum>
  <SyncSignalOutput>
    <id min="1" max="8">
      <!--req, xs:integer, ID, which is between 1 and 8-->
    </id>
    <IOWorkMode opt="flashLight,polarizer,continuousLight,flashLightSwitch">
      <!--req, xs:string, IO output mode: "flashLight"-strobe light control mode, "polarizer"-polarizer control mode,
"continuousLight"-solid light control mode, "flashLightSwitch"-strobe light switching mode (used to switch between
the strobe supplement light mode and IR mode). When this node is set to "polarizer", only <outputStatus>,
<detectBrightnessEnable>, <brightnessThreshold>, <flashEnabled>, <startHour>, <startMinute>, <endHour>, and
<endMinute> are valid, and the <outputStatus> can only be set to "high" or "low". When this node is set to
"continuousLight", only <detectBrightnessEnable>, <brightnessThreshold>, <flashEnabled>, <startHour>,
<startMinute>, <endHour>, and <endMinute> are valid-->
    </IOWorkMode>
    <defaultStatus opt="hige,low">
      <!--req, xs:string, effective status-->
    </defaultStatus>
    <outputStatus opt="high,low,pulse">
      <!--req, xs:string, default status-->
    </outputStatus>
    <aheadTime min="" max="">
      <!--req, xs:integer, pre-output time-->
    </aheadTime>
  </SyncSignalOutput>
</SyncSignalOutputList>
```

```
</aheadTime>
<timeDelay min="" max="">
  <!--req, xs:integer, output duration-->
</timeDelay>
<freqMultiplyulti min="1" max="15">
  <!--req, xs:integer, frequency multiplication, which is between 1 and 15-->
</freqMultiplyulti>
<dutyRate min="0" max="40">
  <!--req, xs:integer, duty ratio, which is between 0 and 40-->
</dutyRate>
<postFlashEnable>
  <!--req, xs:boolean-->
</postFlashEnable>
<illegalFlashEnable>
  <!--req, xs:boolean-->
</illegalFlashEnable>
<videoFlashEnable>
  <!--req, xs:boolean-->
</videoFlashEnable>
<detectBrightnessEnable>
  <!--req, xs:boolean-->
</detectBrightnessEnable>
<brightnessThreshold min="0" max="100">
  <!--dep, xs:integer, brightness threshold, which is between 0 and 100-->
</brightnessThreshold>
<flashEnabled>
  <!--req, xs:boolean-->
</flashEnabled>
<startHour min="" max="">
  <!--dep, xs:integer-->
</startHour>
<startMinute min="" max="">
  <!--dep, xs:integer-->
</startMinute>
<endHour min="" max="">
  <!--dep, xs:integer-->
</endHour>
<endMinute min="" max="">
  <!--dep, xs:integer-->
</endMinute>
<plateBrightness>
  <!--req, xs:boolean-->
</plateBrightness>
<incrBrightEnable>
  <!--opt, xs:boolean, whether to enable brightness enhancement mode (for solid light mode)-->
</incrBrightEnable>
<incrBrightTime min="0" max="10000">
  <!--req, xs:integer, brightness enhancement duration, which is between 0 and 10000, unit: millisecond. This node is
valid only when <incrBrightEnable> is "true"-->
</incrBrightTime>
<incrBrightPercent min="0" max="100">
  <!--req, xs:integer, percentage of brightness enhancement, which is between 0 and 100. This node is valid only
```

```
when <incrBrightEnable> is "true"-->
  </incrBrightPercent>
  <brightness min="0" max="100">
    <!--dep, xs:integer, solid light brightness (for solid light mode), which is between 0 and 100-->
  </brightness>
  <delayCaptureTime min="1" max="1000">
    <!--dep, xs:integer, delayed capture time, which is between 1 and 1000, unit: millisecond. This node is valid only
when <incrBrightEnable> is "true"-->
  </delayCaptureTime>
  <manualBrightnessEnable>
    <!--req, xs:boolean, whether to enable adjusting brightness manually-->
  </manualBrightnessEnable>
  <manualBrightness>
    <!--dep, xs:integer, brightness adjusted manually, which is between 0 and 100-->
  </manualBrightness>
</SyncSignalOutput>
</SyncSignalOutputList>
```

XML_Cap_Telnetd

Telnetd capability message in XML format

```
<Telnetd version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled>
    <!--req, xs:boolean, whether to enable telnet: "true, false"-->
  </enabled>
  <radarMsgEnabled>
    <!--req, xs:boolean, whether to enable uploading radar debugging information: "true, false"-->
  </radarMsgEnabled>
  <illegalLoginLock>
    <!--req, xs:boolean, "true, false"-->
  </illegalLoginLock>
</Telnetd>
```

XML_Cap_tempRange

tempRange capability message in XML format

```
<?xml version="1.0" encoding="utf-8"?>
<TempRange version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <mode opt="automatic,manual">
    <!--req, xs:string, temperature range mode: "automatic"-automatic mode, "manual"-manual mode-->
  </mode>
  <temperatureUpperLimit min="-20" max="550" def="150">
    <!--dep, xs:float, maximum temperature, corrects to one decimal, the minimum value is "-20", the maximum value
is "550", unit: Celsius degree; its value should be larger than temperatureLowerLimit; it is valid when the value of
mode is "manual"-->
  </temperatureUpperLimit>
  <temperatureLowerLimit min="-20" max="550" def="0">
```

```
<!--dep, xs:float, minimum temperature, corrects to one decimal, the minimum value is "-20", the maximum value is
"550", unit: Celsius degree; its value should be smaller than temperatureUpperLimit; it is valid when the value of
mode is "manual"-->
</temperatureLowerLimit>
</TempRange>
```

XML_Cap_Time

Time capability message in XML format

```
<Time version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <timeMode opt="NTP>manual, local,satellite,timecorrect" def="satellite"><!--req, xs:string--></timeMode>
  <localTime min="0" max="256">
    <!--dep, xs:datetime, this node is required when <timemode> is "manual" or "local"-->
  </localTime>
  <timeZone min="0" max="256">
    <!--dep, xs:string, POSIX time zone string, this node is required when <timemode> is "manual", "local" or "NTP"-->
  </timeZone>
  <satelliteInterval min="" max="">
    <!--dep, xs:integer, minutes, time synchronization interval of locating by satellite, this node is valid only when
<timemode> is "satellite"-->
  </satelliteInterval>
  <timeType opt="local, UTC">
    <!--opt, xs: string, time type, it can be local, or UTC, or local and UTC. If this node is not returned, it indicates that
device does not support getting or setting time type-->
  </timeType>
</Time>
```

XML_Cap_Track

Track capability message in XML format

```
<Track version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--xs:integer, the value of <id> and <Channel> is the same. For example, 201 refers to the main stream of
channel 1, 202 refers to the sub-stream of channel 2, and so on--></id>
  <Channel><!--req, xs:integer--></Channel>
  <Enable><!--req, xs:boolean--></Enable>
  <Description><!--req, xs:string, e.g., trackType=standard,sourceTag=AXIS210a,contentType=video,codecType=MPEG4-
SP,resolution=640x480,frameRate=20 fps,bitrate=6000 kbps--></Description>
  <TrackGUID><!--req, xs:string, GUID generated by the client, e.g., A01AAAAA-BBBB-CCCC-DDDD-033595353625--></
TrackGUID>
  <Size><!--opt, xs:integer--></Size>
  <Duration min="0" max="750"><!--opt, xs:string, video expiry date, e.g., P10DT15H indicates that the video will
expire after 10 days and 15 hours--></Duration>
  <DefaultRecordingMode opt="CMR,MOTION,ALARM,EDR,ALARMANDMOTION,AllEvent,POS"><!--req, xs:string,
default video type--></DefaultRecordingMode>
  <LoopEnable><!--opt, xs:boolean, whether to support recurrently overwriting--></LoopEnable>
  <SrcDescriptor><!--video source description-->
  <SrcGUID><!--req, xs:string, GUID generated by the device, e.g., E800A543-9D53-4520-8BB8-9509062C692D--></
```



```
SrcGUID>
  <SrcChannel><!--req, xs:integer, source channel--></SrcChannel>
  <StreamHint><!--req, xs:string, e.g., "video, mp4, 640x480, 20 fps, 6000 kbps"--></StreamHint>
  <SrcDriver><!--req, xs:string, stream executable driver name, e.g., RTP/RTSP--></SrcDriver>
  <SrcType><!--opt, xs:string, source type, e.g., "mp4", "video"--></SrcType>
  <SrcUrl opt="rtsp://localhost/PSIA/Streaming/channels/101,rtsp://localhost/PSIA/Streaming/channels/102,rtsp://localhost/PSIA/Streaming/channels/103,,rtsp://localhost/PSIA/Streaming/channels/107"><!--req, xs:string, e.g., rtsp://10.3.2.26/mpeg4/media.amp--></SrcUrl>
  <SrcUrlMethods><!--req, xs:string, methods supported by the source: "DESCRIBE,SETUP,PLAY,TEARDOWN"--></SrcUrlMethods>
SrcUrlMethods>
  <SrcLogin><!--req, xs:string, login source password--></SrcLogin>
</SrcDescriptor>
<TrackSchedule><!--recording schedule-->
  <ScheduleBlockList>
    <ScheduleBlock><!--list, recording schedule list-->
      <ScheduleBlockGUID><!--xs:string, schedule GUID, e.g., ABC12345-CDEF-4520-8BB8-7135789C8790--></ScheduleBlockGUID>
ScheduleBlockGUID>
  <ScheduleBlockType><!--req, xs:string--></ScheduleBlockType>
  <ScheduleAction><!--list-->
    <id><!--req, xs:integer--></id>
    <ScheduleActionStartTime><!--start time of the schedule-->
      <DayOfWeek><!--req, xs:string, "Monday,Tuesday,Wednesday,Thursday,Friday,Saturday,Sunday"--></DayOfWeek>
DayOfWeek><!--inclusive-->
    <TimeOfDay><!--xs:time, 00:00:00--></TimeOfDay>
    </ScheduleActionStartTime>
    <ScheduleActionEndTime><!--end time of the schedule-->
      <DayOfWeek><!--req, xs:string, "Monday,Tuesday,Wednesday,Thursday,Friday,Saturday,Sunday"--></DayOfWeek>
DayOfWeek><!--exclusive-->
    <TimeOfDay><!--xs:time, 08:00:00--></TimeOfDay>
    </ScheduleActionEndTime>
    <ScheduleDSTEnable><!--req, xs:boolean, whether to enable DST (daylight saving time)--></ScheduleDSTEnable>
    <Description><!--req, xs:string, PreMorning (midnight to 8 a.m., local time)--></Description>
    <Actions><!--alarm or motion detection that triggers recording-->
      <Record><!--opt, xs:boolean--></Record>
      <Log><!--opt, xs:boolean--></Log>
      <SaveImg><!--opt, xs:boolean--></SaveImg>
      <ActionRecordingMode><!--req, xs:string, recording schedule type: "CMR,EDR,ALARM,MOTION,ALARMANDMOTION,COMMAND,SMART"--></ActionRecordingMode>
      <PreRecordTimeSeconds><!--opt, xs:integer, pre-record time, it is between 0 and 10 and the default value is 5, unit: second--></PreRecordTimeSeconds>
      <PostRecordTimeSeconds><!--opt, xs:integer, post-record time, it is between 0 and 20 and the default value is 5, unit: second--></PostRecordTimeSeconds>
    </Actions>
  </ScheduleAction>
</ScheduleBlock>
</ScheduleBlockList>
</TrackSchedule>
<CustomExtensionList>
  <CustomExtension>
    <CustomExtensionName><!--opt, xs:string, example: www.isapi.com/RaCM/trackExt/ver10--></CustomExtensionName>
CustomExtensionName>
    <enableSchedule><!--opt, xs:boolean, whether to enable recording schedule configuration--></enableSchedule>
```

```
<SaveAudio><!--opt, xs:boolean, whether to enable recording audio--></SaveAudio>
<PreRecordTimeSeconds><!--opt, xs:integer, pre-record time, unit: second--></PreRecordTimeSeconds>
<PostRecordTimeSeconds><!--opt, xs:integer, post-record time, unit: second--></PostRecordTimeSeconds>
<HolidaySchedule>
  <ScheduleBlock>
    <ScheduleBlockGUID><!--req, xs:string, example: 00000000-0000-0000-0000-000000000000--></
ScheduleBlockGUID>
    <ScheduleBlockType><!--opt, xs:string, e.g., www.isapi.com/racm/schedule/ver10--></ScheduleBlockType>
  </ScheduleBlock>
</HolidaySchedule>
</CustomExtension>
</CustomExtensionList>
<IntelligentRecord><!--opt, xs:boolean, whether to enable VCA recording function: 0-no, 1-yes--></IntelligentRecord>
<delayTime opt="0,3,4,5,10,30,60,120,300"><!--opt, xs:integer, capture delay time, unit: second--></delayTime>
<durationEnabled opt="true,false"><!--opt, xs:boolean, whether to enable video expiry time. If this function is not
supported, this node will not be returned. If this function is supported, the video expiry date will be set by the node
<Duration>--></durationEnabled>
</Track>
```

XML_Cap_Tracking

Tracking capability message in XML format

```
<Tracking version="2.0" xmlns="http://www.std-cgi.org/ver20/XMLSchema">
  <enabled>
    <!--req, xs:boolean-->
  </enabled>
  <mode opt="auto,manual">
    <!--dep, xs:string, "auto,manual"-->
  </mode>
  <trackingTime min="0" max="60" def="5">
    <!--dep, xs:integer, tracking time-->
  </trackingTime>
  <normalizedScreenSize>
    <!--req, ro-->
    <normalizedScreenWidth>
      <!--req, ro, xs:integer-->
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, ro, xs:integer-->
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <ManualRegionCoordinatesList min="4" size="4">
    <!--dep, tracking area-->
    <RegionCoordinates>
      <!--opt-->
      <positionX>
        <!--req, xs:integer, X-coordinate-->
      </positionX>
      <positionY>
        <!--req, xs:integer, Y-coordinate-->
      </positionY>
    </RegionCoordinates>
  </ManualRegionCoordinatesList>
</Tracking>
```

```
</positionY>
</RegionCoordinates>
</ManualRegionCoordinatesList>
<isNotSupportTrackingDisplay>
  <!--req, xs:boolean, whether to support displaying master-slave tracking on the interface: "true"-yes, "false"-no-->
</isNotSupportTrackingDisplay>
<Schedule>
  <!--opt,dep, schedule, this node is valid only when enabled is "true". By default the schedule is from 00:00 to 24:00
if no schedule is configured-->
  <TimeRange >
    <beginTime>
      <!--UTC time, e.g.: "2018-03-13T19:42:27+08:00"-->
    </beginTime>
    <endTime>
      <!--UTC time, e.g.: "2018-03-13T19:42:27+08:00"--->
    </endTime>
  </TimeRange >
</Schedule >
</Tracking>
```

XML_Cap_VCResource

Capability message of intelligent resources switch in XML format

```
<VCResource version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <type
opt="basicBehavior,fullBehavior,facesnapBehavior,facesnap,TFS,smartVehicleDetection,smartHVTDetection,smart,judic
ial,smart264AndRoadDetection,smart264AndFaceDetection,smart264AndHeatMap,smartVehicleIllegalParkingDetectio
n,smartIntelligentMonitor,smartTrafficDataCollection, roadDetection,humanRecognition, perimeterCapture,
vehicleDetection,HVTDetection,mixedTargetDetection,trackingCaptureMode,nonTrackingCaptureMode,close,faceHum
anModelingContrast,cityManagement,teacherBehavior,
12MPLiveView,personQueueDetection,verticalPeopleCounting,safetyHelmet,faceCounting">
  <!--required, xs: string, intelligent resource types, "trackingCaptureMode"-panoramic capture mode,
"nonTrackingCaptureMode"-single target capture mode, "faceHumanModelingContrast"-resources of face/human
body+face modeling+face picture comparison, "verticalPeopleCounting"-vertical people counting, "safetyHelmet"-
hard hat detection, "faceCounting"-face counting-->
</type>
<PromptList><!--optional, prompt for switching intelligent resources-->
  <Prompt><!--list-->
    <resourceType opt="roadDetection">
      <!--optional, xs: string, intelligent resource to prompt, currently, only "roadDetection" is supported-->
    </resourceType>
    <resourcePrompt opt="smartCode">
      <!--optional, xs: string, prompt contents-->
    </resourcePrompt>
  </Prompt>
</PromptList>
</VCResource>
```

XML_Cap_VideoInputChannel

VideoInputChannel capability message in XML format

```
<VideoInputChannel version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs: string--></id>
  <inputPort><!--req, xs: string--></inputPort>
  <videoInputEnabled><!--opt, xs: boolean--></videoInputEnabled>
  <name><!--opt, xs: string--></name>
  <videoFormat opt="PAL,NTSC"><!--opt, xs: string, video standards--></videoFormat>
  <portType opt="SDI,OPT,VGA,HDMI,YPbPr"><!--opt, xs: string, port type--></portType>
  <resDesc><!--opt, xs: string--></resDesc>
</VideoInputChannel>
```

XML_Cap_VideoOutputChannel

VideoOutputChannel capability message in XML format

```
<VideoOutputChannel version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs: string; ID--></id>
  <type opt="VGA,CVBS,HDMI,Spot,SDI,LCD"><!--req, xs: string, video output type--></type>
  <menu><!--dep, read-only-->
    <mirrorMenu><!--req, xs: boolean--></mirrorMenu>
  </menu>
  <mode opt="close,clip,scale,open,SDI_1080P25,...,HDMI_1080P,HDMI_720P,HDMI_2160P">
    <!--opt, xs: string, video output mode-->
  </mode>
  <resolution opt="1920*1080/60HZ,1280*720/50HZ,..."><!--opt, xs: string, video resolution--></resolution>
</VideoOutputChannel>
```

XML_Cap_VideoOverlay

VideoOverlay capability message in XML format

```
<VideoOverlay version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <normalizedScreenSize>
    <!--req-->
    <normalizedScreenWidth>
      <!--ro, req, xs:integer-->
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, xs:integer-->
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <attribute>
    <!--opt-->
    <transparent>
      <!--req, xs:boolean-->
    </transparent>
  </attribute>
</VideoOverlay>
```

```
</transparent>
<flashing>
  <!--req, xs:boolean-->
</flashing>
</attribute>
<TextOverlayList/><!--opt, see details in the message of XML_TextOverlayList-->
<DateTimeOverlay/><!--opt, see details in the message of XML_DateTimeOverlay-->
<channelNameOverlay/><!--opt, see details in the message of XML_channelNameOverlay-->
<fontSize>
  <!--opt, xs:string, "adaptive,16*16,32*32,48*48,64*64,80*80,96*96,112*112,128*128"-->
</fontSize>
<frontColorMode>
  <!--opt, string, "auto,customize"-->
</frontColorMode>
<frontColor>
  <!--dep, xs:hexBinary;color-->
</frontColor>
<BatteryPowerOverlay/><!--opt, see details in the message of XML_BatteryPowerOverlay-->
<alignment opt="customize,alignRight,alignLeft">
  <!--opt, xs:string, "customize,alignRight,alignLeft"-->
</alignment>
<publicSecurity>
  <!--req, xs:boolean-->
</publicSecurity>
<DeviceStatusDisplay><!--opt-->
  <batteryCapacityDisplay>
    <!--opt, xs:boolean-->
  </batteryCapacityDisplay>
  <chargingStatusDisplay>
    <!--opt, xs:boolean-->
  </chargingStatusDisplay>
  <bluetoothStatusDisplay>
    <!--opt, xs:boolean-->
  </bluetoothStatusDisplay>
  <dialStatusDisplay>
    <!--opt, xs:boolean-->
  </dialStatusDisplay>
</DeviceStatusDisplay>
<boundary>
  <!--opt, xs:integer-->
</boundary>
</VideoOverlay>
```

See Also

XML_TextOverlayList

XML_DateTimeOverlay

XML_channelNameOverlay

XML_BatteryPowerOverlay

XML_Cap_Wireless

Wireless capability message in XML format

```
<Wireless version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--req, xs:boolean--></enabled>
  <wirelessNetworkMode opt="infrastructure,adhoc"><!--opt, xs:string--></wirelessNetworkMode>
  <channel opt="1,2,3,4,5,6,7,8,9,10,11,12,13,14,auto"><!--opt, xs:string--></channel>
  <ssid min="" max=""><!--opt, xs:string--></ssid>
  <wmmEnabled><!--opt, xs:boolean--></wmmEnabled>
  <WirelessSecurity><!--opt-->
    <securityMode opt="disable,WEP,WPA-personal,WPA2-personal,WPA-RADIUS,WPA-enterprise,WPA2-
enterprise,WPA/WPA2">
      <!--opt, xs:string-->
    </securityMode>
    <WEP><!--dep, this node depends on <securityMode-->
      <authenticationType opt="open,sharedkey,auto"><!--req, xs:string--></authenticationType>
      <defaultTransmitKeyIndex min="" max=""><!--req, xs:integer--></defaultTransmitKeyIndex>
      <wepKeyLength opt="64,128"><!--opt, xs:integer--></wepKeyLength>
      <EncryptionKeyList>
        <encryptionKey>
          <!--req, xs:hexBinary, WEP encryption key in hexadecimal format-->
        </encryptionKey>
      </EncryptionKeyList>
    </WEP>
    <WPA><!--dep, depends on <securityMode-->
      <algorithmType opt="TKIP,AES,TKIP/AES"><!--req, xs:string--></algorithmType>
      <sharedKey><!-- req, xs:string, pre-shared key used in WPA--></sharedKey>
      <wpaKeyLength min="8" max="64"><!--req, xs: integer--></wpaKeyLength>
    </WPA>
    <support64bitKey opt="WPA-personal, WPA2-personal"/><!--opt, xs:string--></support64bitKey>
  </WirelessSecurity>
  <AccessPointList>
    <InterfaceDisplay><!--opt-->
      <TableElementList>
        <TableElement>
          <!--opt,xs:string,opt="SSID,workingMode,securityMode,channel,signalStrength, speed,connectionStatus"-->
        </TableElement>
      </TableElementList>
    </InterfaceDisplay>
  </AccessPointList>
  <isSupportConnectStatus>
    <!-- opt, xs:boolean, whether supports Wi-Fi connection status-->
  </isSupportConnectStatus>
  <workScene opt="computerRoom,monitorTerminal"><!--opt, xs:string, working scenario--></workScene>
  <protocol opt="802.11ac"><!--req,xs:string, protocol mode--></protocol>
  <hideSsid><!--opt, xs:boolean--></hideSsid>
  <ChannelConfig><!--opt-->
    <transmitPower opt="9,12,15,18,21,24,27"><!--opt, xs:integer, unit: dBm--></transmitPower>
    <ChannelCountryList><!--opt>
      <Country>
```

```
<countryID><!--req, xs:integer, country code--></countryID>
<ChannelList>
  <Node>
    <width opt="auto,20,40,80"><!--req, xs:string, bandwidth, unit: MHz--></width>
    <channel><!--req,dep, xs:string, unit: MHz></channel>
  </Node>
</ChannelList>
</Country>
</ChannelCountryList>
</ChannelConfig>
<isSupportNullSsid>
  <!--opt, xs: boolean, whether supports setting SSID to null, if supports, return "true", otherwise, this node will not
  be returned. If the SSID is empty, the device will filter the SSID configuration when configuring Wi-Fi-->
</isSupportNullSsid>
</Wireless>
```

XML_Cap_WirelessServer

WirelessServer capability message in XML format

```
<WirelessServer version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <wifiApEnabled opt="true,false"><!--opt,xs: boolean,--></wifiApEnabled>
  <broadcastEnabled opt="true,false"><!--opt,xs: boolean--></broadcastEnabled>
  <wlanShareEnabled opt="true,false"><!--opt,xs: boolean--></wlanShareEnabled>
  <ssid min="" max=""><!--opt, xs: string--></ssid>
  <WirelessSecurity><!--req-->
    <securityMode opt="disable,WEP,WPA-personal,WPA2-personal,WPA-RADIUS,WPA-enterprise,WPA2-enterprise">
      <!--opt, xs: string-->
    </securityMode>
    <WEP><!--dep, depends on <securityMode>-->
      <authenticationType opt="open,sharedkey,auto">
        <!--req, xs: string-->
      </authenticationType>
      <defaultTransmitKeyIndex min="" max="">
        <!--req, xs: integer-->
      </defaultTransmitKeyIndex>
      <wepKeyLength opt="64,128"><!--opt, xs: integer--></wepKeyLength>
      <EncryptionKeyList size="">
        <encryptionKey>
          <!--req, xs: hexBinary, WEP encryption key in hexadecimal format-->
        </encryptionKey>
      </EncryptionKeyList>
    </WEP>
    <WPA><!--dep, depends on <securityMode>-->
      <algorithmType opt="TKIP,AES,TKIP/AES">
        <!--req, xs: string, "TKIP,AES,TKIP/AES"-->
      </algorithmType>
      <sharedKey><!--opt, xs: string, shared key used in WPA--></sharedKey>
      <wpaKeyLength min="" max=""><!--opt, xs: integer, the key length is between 8 and 63--></wpaKeyLength>
      <defaultPassword><!--opt, xs: boolean--></defaultPassword>
    </WPA>
```

```
</WirelessSecurity>
<DHCPEnabled opt="true,false"><!--opt, xs: boolean--></DHCPEnabled>
<ipVersion opt="v4,v6"><!--opt, xs:string--></ipVersion>
<HostIpAddress><!--opt-->
  <ipAddress><!--dep, xs:string--></ipAddress>
  <ipv6Address><!--dep, xs:string--></ipv6Address>
</HostIpAddress>
<IPMask><!--opt-->
  <subnetMask><!--dep, xs:string, subnet mask for IPv4 address--></subnetMask>
  <bitMask><!--dep, xs:integer, bitmask IPv6 address--></bitMask>
</IPMask>
<AddressPool><!--opt-->
  <startIPv4Address><!--dep, xs:string--></startIPv4Address>
  <endIPv4Address><!--dep, xs:string--></endIPv4Address>
  <startIPv6Address><!--dep, xs:string--></startIPv6Address>
  <endIPv6Address><!--dep, xs:string--></endIPv6Address>
</AddressPool>
<DNSAddressList size="2"><!--opt-->
  <DNSAddress><!--opt-->
    <id><!--opt,xs:string,start from 1--></id>
    <ipAddress><!--dep, xs: string--></ipAddress>
    <ipv6Address><!--dep, xs: string--></ipv6Address>
  </DNSAddress>
</DNSAddressList>
<GatewayAddress>
  <ipAddress><!--dep, xs:string--></ipAddress>
  <ipv6Address><!--dep, xs:string--></ipv6Address>
</GatewayAddress>
<wifiApModeType opt="true,false,auto">
  <!--opt, xs: string, current wireless access point (AP) mode, if this node is returned, it indicates that the device
  supports auto mode-->
</wifiApModeType>
</WirelessServer>
```

Remarks

For Client supports auto AP, the node <<**wifiApModeType**> is valid; for Client does not supports auto AP, the node <**wifiApEnabled**> is valid; the values of these two nodes will effect each other, and when <**wifiApModeType**> equals to "auto", the value of <**wifiApEnabled**> is "true".

XML_Cap_WirelessServerStatus

WirelessServerStatus capability message in XML format

```
<WirelessServerStatus version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <wifiApEnabled>
    <!--dep, xs:boolean-->
  </wifiApEnabled>
  <linkDevices min="0" max="20">
    <!--dep, xs:integer, maximum number of linked devices-->
  </linkDevices>
```



```
<DeviceInfoList size="20">
  <!--opt-->
  <DeviceInfo>
    <!--opt-->
    <IpAddress>
      <!--req-->
      <ipAddress>
        <!--dep, xs:string-->
      </ipAddress>
      <ipv6Address>
        <!--dep, xs:string-->
      </ipv6Address>
    </IpAddress>
    <hostName>
      <!--opt, xs:string-->
    </hostName>
    <macAddress>
      <!--opt, xs:string-->
    </macAddress>
  </DeviceInfo>
</DeviceInfoList>
</WirelessServerStatus>
```

XML_Cap_WirelessStatus

WirelessStatus capability message in XML format

```
<WirelessStatus version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled>
    <!--dep, xs:boolean-->
  </enabled>
  <connectionEnabled>
    <!--dep, xs:boolean-->
  </connectionEnabled>
  <IpAddress>
    <!--req-->
    <ipAddress>
      <!--dep, xs:string-->
    </ipAddress>
    <ipv6Address>
      <!--dep, xs:string-->
    </ipv6Address>
  </IpAddress>
  <SubnetMask>
    <!--req-->
    <ipAddress>
      <!--dep, xs:string-->
    </ipAddress>
    <ipv6Address>
      <!--dep, xs:string-->
    </ipv6Address>
  </SubnetMask>
</WirelessStatus>
```

```
</SubnetMask>
<RouterAddress>
  <!--req-->
  <ipAddress>
    <!--dep, xs:string-->
  </ipAddress>
  <ipv6Address>
    <!--dep, xs:string-->
  </ipv6Address>
</RouterAddress>
<DNSAddressList size="2">
  <!--opt-->
  <DNSAddress>
    <!--opt-->
    <id>
      <!--opt, xs:string, DNS address ID, which starts from 1-->
    </id>
    <ipAddress>
      <!--dep, xs:string-->
    </ipAddress>
    <ipv6Address>
      <!--dep, xs:string-->
    </ipv6Address>
  </DNSAddress>
</DNSAddressList>
</WirelessStatus>
```

XML_Cap_WPS

WPS capability message in XML format

```
<WPS version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enable><!--required, xs:boolean, whether to enable WPS--></enable>
</WPS>
```

XML_CaptureMode

CaptureMode message in XML format

```
<CaptureMode version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <mode>
    <!--req, xs: string, video input mode: 1-640×480@30fps, 2-4CIF@30fps, 3-720P@25fps, 4-720P@30fps,
    5-720P@60fps, 6-1280×960@15fps, 7-1280×960@25fps, 8-1280×960@30fps, 9-1280×1024@30fps,
    10-1600×900@15fps, 11-1600×1200@15fps, 12-1080P@15fps, 13-1080P@25fps, 14-1080P@30fps,
    15-1080P@50fps, 16-1080P@60fps, 17-2048×1536@15fps, 18-2048×1536@20fps, 19-2048×1536@24fps,
    20-2048×1536@25fps, 21-2048×1536@30fps, 22-2560×2048@25fps, 23-2560×2048@30fps, 24-2560×1920@7.5fps,
    25-3072×2048@30fps, 26-1944×1212@24fps, 27-1944×1212@25fps-->
```

```
</mode>
</CaptureMode>
```

XML_CertificateInfo

CertificateInfo message in XML format

```
<CertificateInfo version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs: string--></id>
  <version><!--opt, xs: string--></version>
  <IssuerDN><!--req, isapi:DN-->
    <countryName><!--req, xs: string--></countryName>
    <stateOrProvinceName><!--opt, xs: string--></stateOrProvinceName>
    <localityName><!--opt, xs: string--></localityName>
    <organizationName><!--opt, xs: string--></organizationName>
    <organizationUnitName><!--opt, xs: string--></organizationUnitName>
    <commonName><!--req, xs: string--></commonName>
    <email><!--opt, xs: string--></email>
  </IssuerDN>
  <SubjectDN><!--req, isapi:DN-->
    <countryName><!--req, xs: string--></countryName>
    <stateOrProvinceName><!--opt, xs: string--></stateOrProvinceName>
    <localityName><!--opt, xs: string--></localityName>
    <organizationName><!--opt, xs: string--></organizationName>
    <organizationUnitName><!--opt, xs: string--></organizationUnitName>
    <commonName><!--req, xs: string--></commonName>
    <email><!--opt, xs: string--></email>
  </SubjectDN>
  <signatureAlgorithm><!--req, xs: string, signature algorithm: RSA_3, RSA_F4--></signatureAlgorithm>
  <keyAlgorithm><!--opt, xs: string--></keyAlgorithm>
  <startDate><!--req, xs: time, in ISO8601 time format--></startDate>
  <endDate>!--req, xs: time, in ISO8601 time format--></endDate>
  <serialNumber><!--req, xs: string, uuid--></serialNumber>
</CertificateInfo>
```

XML_CertificateReq

CertificateReq message in XML format

```
<CertificateReq version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs: string--></id>
  <SubjectDN><!--req, isapi:DN-->
    <countryName><!--req, xs: string--></countryName>
    <stateOrProvinceName><!--opt, xs: string--></stateOrProvinceName>
    <localityName><!--opt, xs: string--></localityName>
    <organizationName><!--opt, xs: string--></organizationName>
    <organizationUnitName><!--opt, xs: string--></organizationUnitName>
    <commonName><!--req, xs: string --></commonName>
    <email><!--opt, xs: string--></email>
```

```
</SubjectDN>
<validity><!--opt, xs: integer, expiry date, which is between 1 and 5000--></validity>
<passwd><!--opt, xs: string, password to protect private key--></passwd>
<RSAKeyLength><!--opt, xs: integer--></RSAKeyLength>
</CertificateReq>
```

XML_CertificateReq_ImportCert

CertificateReq message in XML format

```
<CertificateReq version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <certificateMode><!--opt, xs:string, importing method: "signingRequest"-signature request (import self-signed
certificate), "privateKey"-private key--></certificateMode>
  <privateKeyMode><!--dep, xs:string, private key mode: "seperateKey", "PKCS#12"--></privateKeyMode>
  <seperateKeyPassword><!--dep, xs:string, password in seperateKey mode, it should be encrypted--></
seperateKeyPassword>
  <PKCSPassword><!--dep, xs:string, password in PKCS#12 mode, it should be encrypted--></PKCSPassword>
  <dataType><!--dep, xs:string, data type: "certificate", "privateKey"-private key--></dataType>
</CertificateReq>
```

XML_CertificateReqInfo

CertificateReqInfo message in XML format

```
<CertificateReqInfo version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs: string--></id>
  <SubjectDN><!--req, isapi:DN-->
    <countryName><!--req, xs: string--></countryName>
    <stateOrProvinceName><!--opt, xs: string--></stateOrProvinceName>
    <localityName><!--opt, xs: string--></localityName>
    <organizationName><!--opt, xs: string--></organizationName>
    <organizationUnitName><!--opt, xs: string--></organizationUnitName>
    <commonName><!--req, xs: string --></commonName>
    <email><!--opt, xs: string--></email>
  </SubjectDN>
  <version><!--opt, xs: string--></version>
  <validity><!--opt, xs:int, expiry date, which is between 1 and 5000--></validity>
  <keyAlgorithm><!--opt, xs: string--></keyAlgorithm>
  <passwd><!--opt, xs: string, password to protect private key--></passwd>
  <RSAKeyLength><!--opt, xs: integer--></RSAKeyLength>
</CertificateReqInfo>
```

XML_CertificateResult

CertificateResult message in XML format

```
<CertificateResult version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <formatType><!--req, xs:string, certificate format, e.g., "PEM"--></formatType>
  <detailInfo><!--req, xs:string, certificate details--></detailInfo>
</CertificateResult>
```

XML_Challenge

Challenge message in XML format.

```
<Challenge version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <key><!--req, xs:string--></key>
</Challenge>
```

XML_ChanCtrl

ChanCtrl message (for traffic) in XML format

```
<ChanCtrl version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <streamGetType><!--streaming mode: 0-live view mode (default), 1-data receiving mode--></streamGetType>
</ChanCtrl>
```

XML_Channel_VideoOverlay

Message about the text overlay parameters of a specific video digital input channel in XML format.

```
<VideoOverlay version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <normalizedScreenSize><!--required-->
    <normalizedScreenWidth><!--required, read-only, xs:integer, normalized width--></normalizedScreenWidth>
    <normalizedScreenHeight><!--required, read-only, xs:integer, normalized height--></normalizedScreenHeight>
  </normalizedScreenSize>
  <attribute><!--optional-->
    <transparent><!--required, xs:boolean, whether it is transparent--></transparent>
    <flashing><!--required, xs:boolean, whether it is flickering--></flashing>
  </attribute>
  <TextOverlayList><!--optional-->
    <TextOverlay><!--list-->
      <id><!--required, xs:string, text No.--></id>
      <enabled><!--required, xs:boolean, whether to enable--></enabled>
      <positionX><!--required, xs:integer, X-coordinate--></positionX>
      <positionY><!--required, xs:integer, Y-coordinate--></positionY>
      <displayText><!--required, xs:string, overlaid text--></displayText>
    </TextOverlay>
  </TextOverlayList>
  <DateTimeOverlay><!--optional-->
    <enabled><!--required, xs:boolean, whether to display the date--> </enabled>
    <positionX> <!--required, xs:integer, X-coordinate--> </positionX>
    <positionY> <!--required, xs:integer, Y-coordinate--> </positionY>
```

```
<dateStyle><!--optional, xs:string, date format: "YYYY-MM-DD", "MM-DD-YYYY", "DD-MM-YYYY", "CHR-YYYY-MM-DD", "CHR-MM-DD-YYYY", "CHR-DD-MM-YYYY", "CHR-YYYY/MM/DD", "CHR-MM/DD/YYYY", "CHR-DD/MM/YYYY"--> </dateStyle>
<timeStyle> <!--optional, xs:string, time format: "12hour", "24hour"--> </timeStyle>
<displayWeek> <!--optional, xs:boolean, whether to display the day of the week--> </displayWeek>
</DateTimeOverlay>
<channelNameOverlay><!--optional-->
  <enabled> <!--required, xs:boolean, whether to display the channel name--> </enabled>
  <positionX> <!--required, xs:integer, X-coordinate--> </positionX>
  <positionY> <!--required, xs:integer, Y-coordinate--> </positionY>
</channelNameOverlay>
<fontSize><!--optional, xs:string, font size: "adaptive", "16*16", "32*32", "48*48", "64*64"--></fontSize>
<frontColorMode><!--optional, string, font color mode: "auto", "customize"--></frontColorMode>
<frontColor><!--dependency, xs:hexBinary, font color--></frontColor>
<BatteryPowerOverlay><!--optional, battery power overlay-->
  <enabled><!--required, xs:boolean, whether to display the overlaid battery power--></enabled>
  <positionX> <!--required, xs:integer, X-coordinate--> </positionX>
  <positionY> <!--required, xs:integer, Y-coordinate--> </positionY>
</BatteryPowerOverlay>
<alignment><!--optional, xs:string, alignment mode: "customize"-custom, "alignRight"-align to right, "alignLeft"-align to left--></alignment>
<publicSecurity><!--required, xs:boolean, security surveillance--></publicSecurity>
</VideoOverlay>
```

XML_ChannelCap_VideoOverlay

Message about the text overlay configuration capability of a specific video digital input channel in XML format.

```
<VideoOverlay version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <normalizedScreenSize><!--required-->
    <normalizedScreenWidth><!--required, read-only, xs:integer, normalized width--></normalizedScreenWidth>
    <normalizedScreenHeight><!--required, read-only, xs:integer, normalized height--></normalizedScreenHeight>
  </normalizedScreenSize>
  <attribute><!--optional-->
    <transparent opt="true,false"><!--required, xs:boolean, whether it is transparent--></transparent>
    <flashing opt="true,false"><!--required, xs:boolean, whether it is flickering--></flashing>
  </attribute>
  <TextOverlayList size="8"><!--optional-->
    <TextOverlay><!--list-->
      <id><!--required, xs:string, text No.--></id>
      <enabled><!--required, xs:boolean, whether to enable--></enabled>
      <positionX><!--required, xs:integer, X-coordinate--></positionX>
      <positionY><!--required, xs:integer, Y-coordinate--></positionY>
      <displayText><!--required, xs:string, overlaid text--></displayText>
      <isPersistentText><!--optional, xs:boolean, whether the OSD overlaid text is the same as the originally configured one after the device reboots--></isPersistentText>
    </TextOverlay>
  </TextOverlayList>
  <DateTimeOverlay>
```

```
<enabled opt="true,false"><!--required, xs:boolean, whether to display the date--></enabled>
<positionX><!--required, xs:integer, X-coordinate--></positionX>
<positionY><!--required, xs:integer, Y-coordinate--></positionY>
<dateStyle opt="YYYY-MM-DD,MM-DD-YYYY,DD-MM-YYYY,CHR-YYYY-MM-DD,CHR-MM-DD-YYYY,CHR-DD-MM-
YYYY,CHR-YYYY/MM/DD,CHR-MM/DD/YYYY,CHR-DD/MM/YYYY"><!--optional, xs:string, date format--></dateStyle>
<timeStyle opt="12hour,24hour"><!--optional, xs:string, time format--></timeStyle>
<displayWeek opt="true,false"><!--optional, xs:boolean, whether to display the day of the week--></displayWeek>
</DateTimeOverlay>
<channelNameOverlay>
  <enabled><!--required, xs:boolean, whether to display the channel name--></enabled>
  <positionX><!--required, xs:integer, X-coordinate--></positionX>
  <positionY><!--required, xs:integer, Y-coordinate--></positionY>
</channelNameOverlay>
<fontSize opt="16*16,32*32,48*48,64*64,adaptive"><!--optional, xs:string, font size, unit: pixel--></fontSize>
<frontColorMode opt="auto,customize"><!--optional, string, font color mode--></frontColorMode>
<frontColor><!--dependency, xs:hexBinary, font color--></frontColor>
<BatteryPowerOverlay/><!--optional, battery power overlay-->
<DeviceStatusDisplay><!--optional, device status display-->
  <batteryCapacityDisplay><!--optional, xs:boolean, whether to display battery capacity--></batteryCapacityDisplay>
  <chargingStatusDisplay><!--optional, xs:boolean, whether to display charging status--></chargingStatusDisplay>
  <bluetoothStatusDisplay><!--optional, xs:boolean, whether to display bluetooth status--></bluetoothStatusDisplay>
  <dialStatusDisplay><!--optional, xs:boolean, whether to display dial-up status--></dialStatusDisplay>
</DeviceStatusDisplay>
<alignment opt="customize,alignRight,alignLeft,GB,allRight,allLeft"><!--optional, xs:string, alignment mode:
"customize"-custom, "alignRight"-align to right, "alignLeft"-align to left, "allRight"-align all to right (align all OSD texts
to the right, including characters, time, title, and so on), "allLeft"-align all to left (align all OSD texts to the left,
including characters, time, title, and so on)--></alignment>
<lowRightTextNumber min="1" max="6"/><!--optional, number of characters in the lower-left corner-->
<lowLeftTextNumber min="7" max="8"/><!--optional, number of characters in the lower-right corner-->
<boundary min="0" max="2"><!--optional, xs:integer, boundary--></boundary>
<TextOverlaySize min="0" max="0"><!--optional, xs:integer, number of overlaid texts--></TextOverlaySize>
<FloorNumOverlay><!--optional-->
  <enabled><!--required, xs:boolean, whether to enable displaying the overlaid floor No.--></enabled>
  <positionX><!--required, xs:integer, X-coordinate--></positionX>
  <positionY><!--required, xs:integer, Y-coordinate--></positionY>
</FloorNumOverlay>
<upDownboudary><!--optional, xs:integer, minimum size of the upper or lower boundary, unit: number of
characters, range: [0,2] (the value can be 0, 1, or 2)--></upDownboudary>
<leftRightboudary><!--optional, xs:integer, minimum size of the left or right boundary, unit: number of characters,
range: [0,2] (the value can be 0, 1, or 2)--></leftRightboudary>
<Angle><!--optional, device angle-->
  <TiltAngle><!--optional, tilting angle-->
    <enabled opt="true,false"><!--required, xs:boolean, whether to enable: true, false (default)--></enabled>
    <positionX min="0" max="1000"><!--required, xs:integer,X-coordinate--></positionX>
    <positionY min="0" max="1000"><!--required,xs:integer, Y-coordinate--></positionY>
  </TiltAngle>
</Angle>
</VideoOverlay>
```

XML_ChannelEventCap

Message about event capability of a single channel in XML format

```
<ChannelEventCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <eventType opt=""/>
  <!--required, xs:string, event types supported by the channel-->
  <shieldEventType opt="behavior,faceSnap,humanRecognition,faceCapture,targetCapture"/>
  <!--required, xs:string, event types that support area shield-->
  <channelID><!--optional, xs:integer, current device channel No.--></channelID>
  <id><!--required, xs:integer--></id>
</ChannelEventCap>
```

Remarks

The values of node **<eventType>** are shown in the table below.

eventType	Description
VMD (motionDetection)	Motion detection
Shelteralarm (tamperDetection)	Video tampering alarm
videoLoss	Video loss
ROI	Region of interest
facedetection (faceDetection)	Face detection
intelliTrace	Auto-track
fielddetection (fieldDetection)	Intrusion detection
defocus (defocusDetection)	Defocus detection
audioexception (audioDetection)	Audio exception detection
scenechangedetection (sceneChangeDetection)	Sudden scene change detection
linedetection (lineDetection)	Line-crossing detection
regionEntrance	Region entrance detection
regionExiting	Region exiting detection
loitering	Loitering detection
group	People gathering detection
rapidMove	Fast moving detection
parking	Parking detection
unattendedBaggage	Unattended baggage

eventType	Description
attendedBaggage	Object removal detection
peopleDetection	Human detection
storageDetection	HDD health detection
behavior	Behavior analysis
faceCapture(faceSnap)	Face capture
ANPR(vehicleDetection)	Vehicle detection
fireDetection	Fire detection
shipsDetection	Ship detection
PIR	PIR alarm
targetCapture (humanRecognition)	Target capture
alarmResult (faceContrast)	Face comparison
framesPeopleCounting	People counting in single frame
hide	Tampering detection
audioLoss	Audio loss
cameraAngleAnomaly	Image exception
violentMotion	Violent motion
trial	Tailing
humanEnter	Human entrance
operateOverTime	Operation timed out
stickUp	Sticking scrip
installScanner	Installing scanner
faceDetect	ATM face detection
temperature (temperatureDetection)	Temperature difference detection
PeopleCounting	People counting
personQueueDetection	People queuing-up detection
heatmap	Heat map
mixedTargetDetection	Multi-target-type detection
MTD_face	Supported face attributes of multi-target-type detection (MTD)

eventType	Description
MTD_human	Supported human attributes of multi-target-type detection (MTD)
MTD_vehicle	Supported motor vehicle attributes of multi-target-type detection (MTD)
MTD_nonMotor	Supported non-motor vehicle attributes of multi-target-type detection (MTD)
faceSnapModeling	Face capture modeling
HVTVehicleDetection	Mixed-traffic detection
PictureCaptureComparision	N:1 face comparison
IO	Sensor alarm
smokeDetection	Smoke detection
smokeAndFireDetection	Smoke and fire detection
diskfull	HDD is full
diskerror	HDD error
nicbroken	Network disconnected
ipconflict	IP address conflicted
illaccess	Illegal access
personDensityDetection	People density detection
vehicleControl	Upload alarm of vehicle in whitelist and blacklist
vehicleRcogResult	Vehicle secondary recognition
cardMatch	Authentication
overSpeed	Overspeed alarm
highTempAlarm	High temperature alarm
abnormalAcceleration	Abnormal accelerated speed alarm
failDown	People falling down
leavePosition	Absence detection
peopleNumChange	The number of people changed
retention	Overstay detection
running	Running

eventType	Description
thermometry	Temperature measurement
heatmapPDC	Heat map people counting
heatmapDuration	Heat map people staying duration
intersectionAnalysis	Intersection analysis
AID_abandonedObject	Thrown object in traffic events
AID_pedestrian	Pedestrian detection in traffic events
AID_congestion	Congestion detection in traffic events
AID_roadBlock	Roadblock detection in traffic events
AID_construction	Construction detection in traffic events
AID_trafficAccident	Traffic accident detection in traffic events
AID_fogDetection	Fog detection in traffic events
TFS_illegalParking	Illegal parking of enforcement events
TFS_wrongDirection	Wrong-way driving of enforcement events
TFS_crossLane	Driving on lane line of enforcement events
TFS_laneChange	Lane change of enforcement events
TFS_vehicleExist	Vehicle existing of enforcement events
TFS_turnRound	Turning round of enforcement events
TFS_parallelParking	Parallel parking of enforcement events
TPS	Traffic data collection configuration
luma	Brightness exception
chroma	Color cast detection
snow	Snow noise
streak	Stripe noise
freeze	Image freeze
sigLose	Signal loss
clarity	Clarity exception
jitter	Image flutter
block	Video tampering
flowers	Blurred screen detection

eventType	Description
noise	Image noise
ghost	Abnormal light spot detection
purple	Image purple edge detection
ICR	ICR exception detection
protectiveFilm	Protective film unremoved
certificateRevocation	Certificate is expired
dataPreAlarm	Traffic pre-alarm
vibrationDetection	Vibration detection
radarFieldDetection	Radar intrusion detection
radarLineDetection	Radar line crossing detection

The value in the bracket is the old version, and considering the compatibility, both new and old value will be returned.

XML_ChannelEventCapList

Message about event capability of all channels in XML format

```
<ChannelEventCapList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ChannelEventCap/>
</ChannelEventCapList>
```

See Also

XML_ChannelEventCap

XML_ChannelInfo

ChannelInfo message in XML format.

```
<ChannelInfo version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--required, xs:integer, channel number--></id>
  <EagleEye>
    <!--optional, the channel supports the functions of PanoVu series camera-->
    <StreamFusion>
      <!--optional, "panoramic + PTZ camera streaming channel, it is not returned when not support-->
      <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
    </StreamFusion>
    <PanoramicMetaData>
      <!--optional, panoramic camera streaming channel, it is not returned when not support-->
```

```
<InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
</PanoramicMetaData>
</EagleEye>
<FishEye>
  <!--optional, the channel supports the functions of fisheye camera-->
  <FishEyeStream>
    <!--optional, fisheye camera stream, it is used for software decoding of play library, it is not returned when not
support-->
    <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
    </FishEyeStream>
  </FishEye>
</Thermal>
  <!--optional, the channel supports the functions of thermal products-->
  <TemperatureMetaData>
    <!--optional, temperature measurement metadata, it is not returned when not support-->
    <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
  </TemperatureMetaData>
  <ThermometryDetection>
    <!--optional, temperature measurement detection, it is not returned when not support-->
    <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
  </ThermometryDetection>
  <ThermometryDiffDetection>
    <!--optional, temperature difference detection, it is not returned when not support-->
    <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
  </ThermometryDiffDetection>
  <FireDetection>
    <!--optional, fire detection, it is not returned when not support-->
    <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
  </FireDetection>
  <SmokeDetection>
    <!--optional, smoke detection, it is not returned when not support-->
    <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
  </SmokeDetection>
  <ShipsDetection>
    <!--optional, ship detection, it is not returned when not support-->
    <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
  </ShipsDetection>
  <IntelligentRuleDisplay>
    <!--optional, intelligent rule display parameters of thermal application, it is not returned when not support-->
    <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
  </IntelligentRuleDisplay>
  <IntelligentBehaviorRule>
    <!--optional, intelligent behavior rule configuration, it is not returned when not support-->
    <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
  </IntelligentBehaviorRule>
  <DPC>
    <!--optional, defective pixel correction, it is not returned when not support-->
    <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
  </DPC>
</Metadata>
  <!--optional, metadata extraction, it is not returned when not support-->
  <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
```

```
</Metadata>
<ThermometryShieldMask>
  <!--optional, temperature measurement shielded area, it is not returned when not support-->
  <InsertChanNo><!--optional,xs:integer, access channel number--></InsertChanNo>
</ThermometryShieldMask>
<FaceThermometry>
  <!--optional, face thermography, it is not returned when not support-->
  <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
</FaceThermometry>
<BlackBody>
  <!--optional, black body, it is not returned when not support-->
  <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
</BlackBody>
<BodyTemperatureCompensation>
  <!--optional, body temperature compensation, it is not returned when not support-->
  <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
</BodyTemperatureCompensation>
<LensCorrection>
  <!--optional, lens correction, when this node is not returned, it indicates not support-->
  <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
</LensCorrection>
<BurningPrevention>
  <!--optional, burning prevention, when this node is not returned, it indicates not support-->
  <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
</BurningPrevention>
<FaceSnapThermometry>
  <!--optional, upload the captured face picture with temperature information, it is not returned when this function
is not supported-->
  <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
</FaceSnapThermometry>
</Thermal>
<isSupportChangedUpload/><!--optional, xs: boolean, whether supports uploading status changes-->
<PanoramaCamera><!--optional, PTZ camera channel-->
  <PTZCtrl>
    <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
  </PTZCtrl>
</PanoramaCamera>
<GlobalCamera><!--optional, Panoramic camera channe-->
  <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
</GlobalCamera>
</ChannelInfo>
```

XML_ChannelInfoList

ChannelInfoList message in XML format

```
<ChannelInfoList>version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ChannelInfo/><!--see details in the message of XML_ChannelInfo-->
</ChannelInfoList>
```

See Also

XML_ChannelInfo

XML_channelNameOverlay

channelNameOverlay message in XML format

```
<channelNameOverlay version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled>
    <!--req, xs:boolean, whether to enable channel name overlay-->
  </enabled>
  <name>
    <!--req, xs:string, channel name-->
  </name>
  <positionX>
    <!--req, xs:integer, X-coordinate-->
  </positionX>
  <positionY>
    <!--req, xs:integer, Y-coordinate-->
  </positionY>
</channelNameOverlay>
```

XML_CheckInfo

CheckInfo message in XML format

```
<CheckInfo version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <randomString><!--req, xs: string, random string--></randomString>
  <verificationCode>
    <!--opt, xs: string, verification code to be sent, which will be encrypted by sha256, and then convert the
    hexadecimal code to a string, finally it will be encoded by Base64. The format is first 32-byte of randomString + actual
    verification code-->
  </verificationCode>
</CheckInfo>
```

XML_CMSearchDataPackage

CMSearchDataPackage message in XML format.

```
<CMSearchDataPackage version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <metald>
    <!--required, string type, metadata ID, its format is log.std-cgi.com+/major type+/minor type-->
  </metald>
  <timeSpanList>
    <timeSpan>
      <startTime><!--required, start time with ISO8601 time format, e.g., 2017-08-02T00:00:00Z--></startTime>
    </timeSpan>
  </timeSpanList>
</CMSearchDataPackage>
```

```
<endTime><!--required, end time with ISO8601 time format, e.g., 2017-08-02T00:00:00Z--></endTime>
</timeSpan>
</timeSpanList>
</CMSearchDataPackage>
```

XML_CMSearchDataPackageResult

CMSearchDataPackageResult message in XML format.

```
<CMSearchDataPackageResult version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <url>
    <!--required, string type, file URL, e.g., http://ip:port//ISAPI/ContentMgmt/logSearch/dataPackage/data?xx-->
  </url>
</CMSearchDataPackageResult>
```

XML_CMSearchProfile

CMSearchProfile message in XML format

```
<CMSearchProfile version="1.0" xmlns="ISAPIIlianxce.org:resourcedescription">
  <searchProfile><!--req, xs: string, video search conditions--></searchProfile>
  <textSearch>
    <!--req, xs: string, text type to be searched, such as recording information text and metadata text-->
  </textSearch>
  <maxSearchTimespans>
    <!--req, xs: integer, maximum time period of results returned in each search-->
  </maxSearchTimespans>
  <maxSearchTracks><!--req, xs: integer, maximum tracks can be searched--></maxSearchTracks>
  <maxSearchSources>
    <!--req, xs: integer, maximum recording channels can be searched-->
  </maxSearchSources>
  <maxSearchMetadatas>
    <!--req, xs: integer, maximum number of metadata can be searched-->
  </maxSearchMetadatas>
  <maxSearchMatchResults>
    <!--req, xs: integer, maximum search results can be returned-->
  </maxSearchMatchResults>
  <maxSearchTimeout><!--req, xs: integer, maximum search timeout--></maxSearchTimeout>
  <maxConcurrentSearches>
    <!--req, xs: integer, maximum concurrent search operations-->
  </maxConcurrentSearches>
</CMSearchProfile>
```

XML_VideoPic_CMSearchResult

XML message about search results of videos and pictures


```

<CMSearchResult version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <searchID><!--req, xs:string, search ID--></searchID>
  <responseStatus><!--req, xs:string--></responseStatus>
  <responseStatusStrg><!--req, xs:string--></responseStatusStrg>
  <numOfMatches><!--opt, xs:integer--></numOfMatches>
  <matchList>
    <searchMatchItem>
      <sourceID></sourceID>
      <trackID></trackID>
      <timeSpan>
        <startTime></startTime>
        <endTime></endTime>
      </timeSpan>
      <mediaSegmentDescriptor>
        <contentType><!--req, xs:string, file type, corresponding to search condition--></contentType>
        <codecType><!--req, xs:string, video encoding format--></codecType>
        <rateType><!--opt, xs:string, video bit rate,4Mbps--></rateType>
        <playbackURI><!--opt, xs:string, video file URL, "name=" and "size=" are file name and file size respectively, e.g.,
rtsp://10.65.130.168/Streaming/tracks/201/?
starttime=20190213T091134Z&amp;endtime=20190213T092116Z&amp;name=01010000093000100&amp;size=7323
35288--></playbackURI>
        <lockStatus><!--opt, xs:string, status of video file: lock-locked file, unlock-unlocked file--></lockStatus>
        <remainLockTime><!--dep, xs:integer, unit:s, remaining locking time, which is returned only when lockStatus is
set to lock--></remainLockTime>
        <name><!--opt, xs:string, file name--></name>
        <size><!--opt, xs:integer, file size--></size>
        <mediaID><!--opt, xs:integer, file ID--></mediaID>
        <remark><!--opt, xs:string, file tag--></remark>
        <GPSInfo><!--opt, xs:string, file GPS information--></GPSInfo>
        <recorderCode><!--opt, xs:string, body camera No.--></recorderCode>
        <uploadState><!--opt, xs:string, status: uploaded, nonUploaded, uploadFailed, uploading--></uploadState>
        <uploadTime><!--dep, xs:datetime, time of completing video uploading, e.g., "2004-05-03T17:30:08+08:00"--></
uploadTime>
        <policeCode><!--opt, xs:string, No. of the policeman who recorded the video--></policeCode>
        <policeName><!--opt, xs:string, name of the policeman who recorded the video--></policeName>
        <shootingTime><!--opt, xs:datetime, captured or recorded time, e.g., "2004-05-03T17:30:08+08:00"--></
shootingTime>
        <FileTime><!--opt, xs:integer, time duration of the video file, unit:min--></FileTime>
        <remarks><!--opt, xs:string, file remarks--></remarks>
        <cloudStorageType><!--opt, xs:string, storage server type: 0-cloud storage, 1-cvr, 2-pstor, which should be
consistent with cloudStorageType in /ISAPI/ContentMgmt/channels/<ID>/cloudStorage/<ID>--></cloudStorageType>
        <fileUrl><!--opt, xs:string, file URL--></fileUrl>
        <thumbnailUrl><!--opt, xs:string, thumbnail URL of picture or video--></thumbnailUrl>
        <encryption>
          <enabled><!--opt, xs:boolean, whether to encrypt the file--></enabled>
          <encryptionkey><!--opt, xs:string, encryption key, which is returned only when enabled is set to true--></
encryptionkey>
          <encryptionLevel><!--opt, xs:integer, encryption level: 1-AES128, 2-AES256, which is returned only when
enabled is set to true--></encryptionLevel>
        </encryption>
      </mediaSegmentDescriptor>
    </searchMatchItem>
  </matchList>
  <metadataMatches>

```

```
<metadataDescriptor><!--searching conditions:
"evidence,illegalParking,crosslane,vehicleexist,lanechange,wrongdirection,congestion,turnround,pedestrian,constructi
on,roadBlock,abandonedObject,trafficAccident,fogDetection"--></metadataDescriptor>
</metadataMatches>
</searchMatchItem>
</matchList>
</CMSearchResult>
```

XML_CMSRecordProfile

CMSRecordProfile message in XML format

```
<CMRecordProfile version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <trackType><!--opt, xs: string, video track type--></trackType>
</CMRecordProfile>
```

XML_Color

XML message about color adjustment parameters

```
<Color version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <brightnessLevel><!--opt, xs:integer, brightness--></brightnessLevel>
  <contrastLevel><!--opt,xs:integer, contrast--></contrastLevel>
  <saturationLevel><!--opt,xs:integer, saturation--></saturationLevel>
  <hueLevel><!--opt, xs:integer, hue--></hueLevel>
  <grayScale>
    <grayScaleMode><!--opt,xs:string, gray scale mode: "indoor,outdoor"--></grayScaleMode>
  </grayScale>
  <nightMode>
    <!--opt, xs:boolean, enable night mode, when its value is "true", the saturation can be adjusted, otherwise, the
saturation cannot be adjusted-->
  </nightMode>
</Color>
```

XML_corridor

corridor message in XML format

```
<corridor version="2.0" xmlns="http://www.std-cgi.org/ver20/XMLSchema">
  <enabled><!--req, xs: boolean, whether to enable rotate mode--></enabled>
  <mirror><!--opt, xs: string, mirror mode: "on,off"--></mirror>
</corridor>
```

XML_DateTimeOverlay

DateTimeOverlay message in XML format

```
<DateTimeOverlay version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled>
    <!--req, xs:boolean, whether to enable date and time overlay-->
  </enabled>
  <positionX>
    <!--req, xs:integer, X-coordinate-->
  </positionX>
  <positionY>
    <!--req, xs:integer, Y-coordinate-->
  </positionY>
  <dateStyle>
    <!--opt, xs:string, date format: "YYYY-MM-DD, MM-DD-YYYY, DD-MM-YYYY, CHR-YYYY-MM-DD, CHR-MM-DD-YYYY,
CHR-DD-MM-YYYY"-->
  </dateStyle>
  <timeStyle>
    <!--opt, xs:string, "12hour, 24hour"-->
  </timeStyle>
  <displayWeek>
    <!--opt, xs:boolean, whether to display week information-->
  </displayWeek>
  <displayMilliSecond>
    <!--opt, xs:boolean, whether to display millisecond-->
  </displayMilliSecond>
</DateTimeOverlay>
```

XML_DDNS

DDNS message in XML format

```
<DDNS version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id>
    <!--req, xs: string-->
  </id>
  <enabled>
    <!--req, xs:boolean, "true,false"-->
  </enabled>
  <ethernetIfId>
    <!--opt, xs:string, ID. This node is used to link the DDNS to a network interface which is used by the DDNS client to
register-->
  </ethernetIfId>
  <provider>
    <!--req, xs:string, "IPServer, DynDNS, PeanutHall, NO-IP, HiDDNS..."-->
  </provider>
  <serverAddress><!--DDNS address. This node is required when <provider> is "IPServer"-->
  <addressingFormatType>
```

```
<!--req, xs:string, method used to locate the NTP server: "ipaddress, hostname"-->
</addressingFormatType>
<hostname>
  <!--dep, xs:string-->
</hostname>
<ipAddress>
  <!--dep, xs:string-->
</ipAddress>
<ipv6Address>
  <!--dep, xs:string-->
</ipv6Address>
</serverAddress>
<portNo>
  <!--opt, xs:integer, port No-->
</portNo>
<deviceDomainName>
  <!--dep, xs:string, domain name of the device-->
</deviceDomainName>
<username>
  <!--dep, xs:string-->
</username>
<password>
  <!--wo, dep, xs:string-->
</password>
</DDNS>
```

XML_DDNSList

DDNSList message in XML format

```
<DDNSList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <DDNS/><!--req, see details in the message of XML_DDNS-->
</DDNSList>
```

See Also

XML_DDNS

XML_DefaultParam

DefaultParam message in XML format

```
<DefaultParam version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <DialNum><!--opt, xs:string--></DialNum>
  <Username><!--opt, xs:string--></Username>
  <Password><!--opt, xs:string--></Password>
  <APNname><!--opt, xs:string--></APNname>
  <VerifyProto><!--req, xs:string, "auto, CHAP, PAP"--></VerifyProto>
</DefaultParam>
```

XML_Defog

Defog message in XML format

```
<Defog version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled>
    <!--req, xs:boolean-->
  </enabled>
</Defog>
```

XML_De haze

Dehaze message in XML format

```
<Dehaze version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <DehazeMode/><!--opt, xs: string, "open,close,auto"-->
  <DehazeLevel> <!--opt, xs: integer--></DehazeLevel>
</Dehaze>
```

XML_DeviceCap

XML message about device capability

```
<DeviceCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <SysCap><!--optional-->
    <isSupportDst><!--optional, xs: boolean, whether it supports daylight saving time--></isSupportDst>
    <NetworkCap/><!--optional, xs: boolean, network capability-->
    <IOCap/><!--optional, IO capability-->
    <SerialCap/><!--optional, serial port capability-->
    <VideoCap/><!--optional, video capability, see details in the message of XML_VideoCap-->
    <AudioCap/><!--optional, audio capability-->
    <isSupportHolidy><!--optional, xs:boolean--></isSupportHolidy>
    <RebootConfigurationCap>
      <Genetec><!--optional, xs:boolean--></Genetec>
      <ONVIF><!--optional, xs:boolean--></ONVIF>
      <RTSP><!--optional, xs:boolean--></RTSP>
      <HTTP><!--optional, xs:boolean--></HTTP>
      <SADP>
        <ISDiscoveryMode><!--optional, xs:boolean--></ISDiscoveryMode>
        <PcapMode><!--optional, xs:boolean--></PcapMode>
      </SADP>
      <IPCAddStatus><!--optional, xs:boolean--></IPCAddStatus>
    </RebootConfigurationCap>
    <isSupportExternalDevice><!--optional, xs:boolean--></isSupportExternalDevice>
    <isSupportChangedUpload>
      <!--optional, xs: boolean, whether it supports uploading status changes-->
    </isSupportChangedUpload>
    <isSupportGettingWorkingStatus>
```

```
<!--optional, xs:boolean, whether it supports getting device status-->
</isSupportGettingWorkingStatus>
<isSupportGettingChannelInfoByCondition>
  <!--optional, xs:boolean-->
</isSupportGettingChannelInfoByCondition>
<isSupportDiagnosedDataParameter>
  <!--optional, xs:boolean-->
</isSupportDiagnosedDataParameter>
<isSupportSimpleDevStatus>
  <!--optional, xs: boolean, whether it supports getting device working status-->
</isSupportSimpleDevStatus>
<isSupportFlexible>
  <!--optional, xs: boolean, whether it supports getting channel status by condition-->
</isSupportFlexible>
<isSupportPTZChannels>
  <!--optional, xs:boolean, whether it supports returning PTZ channel (which is different from the video channel)-->
</isSupportPTZChannels>
<isSupportSubscribeEvent>
  <!--optional, xs:boolean, whether it supports alarm or event subscription: "true,false"-->
</isSupportSubscribeEvent>
<isSupportDiagnosedData>
  <!--optional, xs:boolean, "true,false", whether it supports diagnosis data-->
</isSupportDiagnosedData>
<isSupportTimeCap>
  <!--optional, xs:boolean, whether it supports time capability-->
</isSupportTimeCap>
<isSupportThermalStreamData>
  <!--optional, xs:boolean, whether it supports uploading thermal stream data in real-time. If it is supported, the
returned value is "true"; otherwise, this node will not be returned-->
</isSupportThermalStreamData>
<isSupportPostUpdateFirmware>
  <!--optional,xs:boolean,"true,false", whether it supports upgrading the firmware-->
</isSupportPostUpdateFirmware>
<isSupportPostConfigData>
  <!--optional, xs:boolean,"true,false", whether it supports importing or exporting the configuration file-->
</isSupportPostConfigData>
<isSupportUserLock>
  <!--optional, xs:boolean,"true,false", whether it supports locking user-->
</isSupportUserLock>
<isSupportModuleLock><!--optional, xs:boolean, whether it supports locking the module: "true,false"--></
isSupportModuleLock>
<isSupportSoundCfg><!--optional, xs:boolean--></isSupportSoundCfg>
<isSupportMetadata>
  <!--optional, xs:boolean, if it is supported, return "true", otherwise, this node will not be returned-->
</isSupportMetadata>
<isSupportShutdown><!--optional, xs:boolean, whether it supports shutdown configuration--></
isSupportShutdown>
  <supportSmartOverlapChannles opt="1"/><!--optional, xs:boolean, whether it supports stream configuration of
smart events. If this function is supported, this node and the corresponding channel ID will be returned; otherwise,
this node will not be returned-->
</SysCap>
<voicetalkNums><!--optional, xs:integer, the number of two-way audio channels--></voicetalkNums>
```

```
<isSupportSnapshot><!--optional, xs:boolean, whether it supports capture: "true, false"--></isSupportSnapshot>
<SecurityCap/><!--optional, security capability-->
<EventCap/><!--optional, event capability-->
<ITCCap><!--opt--></ITCCap>
<ImageCap/><!--optional, image capability-->
<RacmCap/><!--optional, storage capability-->
<PTZCtrlCap>
  <isSupportPatrols><!--optional, xs:boolean--></isSupportPatrols>
</PTZCtrlCap>
<SmartCap/><!--optional, intelligent capability-->
<isSupportEhome><!--optional, xs:boolean--></isSupportEhome>
<isSupportStreamingEncrypt><!--optional, xs:boolean--></isSupportStreamingEncrypt>
<TestCap>
  <isSupportEmailTest><!--optional, xs:boolean--></isSupportEmailTest>
</TestCap>
<ThermalCap/><!--optional, temperature measurement capability-->
<WLAAlarmCap/><!--optional, wireless alarm capability-->
<SecurityCPCapabilities/><!--optional, security control panel capability-->
<isSupportGIS>
  <!--optional, xs:boolean, whether it supports GIS capability-->
</isSupportGIS>
<isSupportCompass>
  <!--optional, xs:boolean-->
</isSupportCompass>
<isSupportRoadInfoOverlays>
  <!--optional, xs:boolean-->
</isSupportRoadInfoOverlays>
<isSupportFaceCaptureStatistics>
  <!--optional, xs:boolean-->
</isSupportFaceCaptureStatistics>
<isSupportExternalDevice>
  <!--optional, xs:boolean-->
</isSupportExternalDevice>
<isSupportElectronicsEnlarge>
  <!--optional, xs:boolean, whether it supports digital zoom-->
</isSupportElectronicsEnlarge>
<isSupportRemoveStorage>
  <!--optional, xs:boolean-->
</isSupportRemoveStorage>
<isSupportCloud>
  <!--optional, xs:boolean-->
</isSupportCloud>
<isSupportRecordHost>
  <!--optional, xs:boolean-->
</isSupportRecordHost>
<isSupportEagleEye>
  <!--optional, xs:boolean, whether it supports PanoVu series camera-->
</isSupportEagleEye>
<isSupportPanorama>
  <!--optional, xs:boolean, whether it supports panorama-->
</isSupportPanorama>
<isSupportFirmwareVersionInfo>
```

```
<!--optional, xs:boolean, whether it supports displaying firmware version information-->
</isSupportFirmwareVersionInfo>
<isSupportExternalWirelessServer>
  <!--optional, xs: boolean-->
</isSupportExternalWirelessServer>
<isSupportSetupCalibration>
  <!--optional, xs:boolean, whether it supports setting calibration-->
</isSupportSetupCalibration>
<isSupportGetmutexFuncErrMsg>
  <!--optional, xs:boolean, whether it supports getting mutex information-->
</isSupportGetmutexFuncErrMsg>
<isSupportTokenAuthenticate><!--optional, xs:boolean--></isSupportTokenAuthenticate>
<isSupportStreamDualVCA><!--optional, xs:boolean--></isSupportStreamDualVCA>
<isSupportlaserSpotManual>
  <!--optional, boolean, whether it supports laser spot configuration-->
</isSupportlaserSpotManual>
<isSupportRTMP><!--optional, xs:boolean--></isSupportRTMP>
<isSupportTraffic><!--optional, xs:boolean--></isSupportTraffic>
<isSupportLaserSpotAdjustment>
  <!--optional, boolean, whether it supports adjusting laser spot size-->
</isSupportLaserSpotAdjustment>
<VideoIntercomCap/><!--optional, video intercom capability-->
<isSupportSafetyCabin>
  <!--optional, xs:boolean-->
</isSupportSafetyCabin>
<isSupportPEA>
  <!--optional, xs:boolean, whether it supports one-touch security control panel capability-->
</isSupportPEA>
<isSupportCurrentLock>
  <!--optional, xs:boolean, whether it supports locking current configuration-->
</isSupportCurrentLock>
<isSupportGuardAgainstTheft>
  <!--optional, xs:boolean, whether it supports device anti-theft configuration-->
</isSupportGuardAgainstTheft>
<isSupportPicInfoOverlap>
  <!--optional, xs:boolean, whether it supports picture information overlay-->
</isSupportPicInfoOverlap>
<isSupportPlay>
  <!--optional, xs: boolean, whether it supports live view: "true,false"-->
</isSupportPlay>
<isSupportPlayback>
  <!--optional, xs: boolean, whether it supports playback: "true,false"-->
</isSupportPlayback>
<UHFRFIDReader>
  <!--optional, supported capability of UHF RFID card reader-->
  <isSupportBasicInformation>
    <!--optional, xs:boolean, whether it supports basic parameters of UHF RFID card reader-->
  </isSupportBasicInformation>
  <isSupportHardDiskStorageTest>
    <!--optional, xs:boolean, whether it supports hard disk storage test of UHF RFID card reader-->
  </isSupportHardDiskStorageTest>
</UHFRFIDReader>
```



```
<isSupportIntelligentStructureAnalysis>
  <!--optional, xs:boolean, whether it supports structured VCA-->
</isSupportIntelligentStructureAnalysis>
<isSupportIntelligentAnalysisEngines>
  <!--optional, xs:boolean, whether it supports VCA engine configuration-->
</isSupportIntelligentAnalysisEngines>
<PreviewDisplayNum>
  <!--optional, xs:integer, the number of live view windows, which is the number of simultaneous live view windows
controlled by the device. Limited by the performance of DeepinMind series network video recorder, currently only live
view of a network camera is supported, and playback is not supported-->
</PreviewDisplayNum>
<isSupportBoard opt="true,false">
  <!--optional, xs:boolean, whether it supports protocol related to sub-board-->
</isSupportBoard>
<ResourceSwitch>
  <workMode opt="4KPreview,educationRecord">
    <!--req, xs:string, device working mode : "4KPreview"-4K live view mode, "educationRecord"-education recording
mode-->
  </workMode>
</ResourceSwitch>
<isSupportCustomStream><!--optional, xs:boolean--></isSupportCustomStream>
<isSupportTriggerCapCheck>
  <!--optional, xs:boolean, whether it supports verifying capability of alarm linkage actions-->
</isSupportTriggerCapCheck>
<isSupportActiveMulticast>
  <!--optional, xs: boolean, whether it supports active multicast-->
</isSupportActiveMulticast>
<isSupportChannelEventCap>
  <!--optional, xs:boolean, whether it supports getting event capability by channel-->
</isSupportChannelEventCap>
<isSupportPictureServer>
  <!-- opt, xs:boolean, whether it supports picture storage server-->
</isSupportPictureServer>
<isSupportVideoCompositeAlarm>
  <!--optional, xs:boolean, whether it supports video double check alarm-->
</isSupportVideoCompositeAlarm>
<isSupportSensorCalibrating>
  <!--optional, xs:boolean, whether it supports double sensor calibration-->
</isSupportSensorCalibrating>
<isSupportChannelEventListCap>
  <!--optional, xs:boolean, whether it supports getting event capability of all channels-->
</isSupportChannelEventListCap>
<VCAResourceChannelsCap>
  <!--optional, whether it supports independently switching to another VCA resource by channel-->
  <ChannelsList>
    <channelsID>
      <!--req, xs:integer, channel No. supported by the device-->
    </channelsID>
  </ChannelsList>
</VCAResourceChannelsCap>
<SensorCap/><!--optional, intelligent cabinet capability-->
<isSupportSecurityCP/>
```

```
<!--optional, xs:boolean, whether it supports the applications of security control panel: "true, false"-->
</isSupportSecurityCP>
<isSupportClientProxyWEB>
  <!--optional, xs:boolean, whether it supports the function that the client proxy passes through the remote web
configuration: "true"-->
</isSupportClientProxyWEB>
<WEBLocation>
  <!--optional, string type, web page location: "local"-local device, "remote"-remote location. If this node is not
returned, the web page will be in the local device by default-->
</WEBLocation>
<isSupportTime/>
  <!--optional, xs:boolean, "true, false", whether it supports time configuration-->
</isSupportTime>
<isSupportTimeZone/>
  <!--optional, xs:boolean, "true, false", whether it supports daylight saving time (DST) configuration-->
</isSupportTimeZone>
<isSupportMixedTargetDetection>
  <!--optional, xs:boolean, "true, false", whether it supports multi-target-type detection-->
</isSupportMixedTargetDetection>
<isSupportFaceContrastMode>
  <!--optional, xs:boolean, whether it supports face picture comparison mode-->
</isSupportFaceContrastMode>
<isSupportPictureCaptureComparision>
  <!--optional, xs:boolean, whether it supports face picture N:1 comparison between face pictures captured by the
camera and imported face pictures-->
</isSupportPictureCaptureComparision>
<isSupportGPSCalibratation>
  <!--optional, xs:boolean, whether it supports GPS calibration capability-->
</isSupportGPSCalibratation>
<isSupportChannelFullEventListCap>
  <!--optional, xs:boolean, whether it supports getting event list capability of all channels-->
</isSupportChannelFullEventListCap>
<isSupportAUXInfoCap>
  <!--optional, xs:boolean, whether it supports getting property capability of all channels-->
</isSupportAUXInfoCap>
<isSupportCalibrationFile>
  <!--optional, xs:boolean, whether it supports importing calibration file-->
</isSupportCalibrationFile>
<isSupportDisplayTrajectory>
  <!--optional, xs:boolean, whether it supports displaying trajectory-->
</isSupportDisplayTrajectory>
<maximumSuperPositionTime opt="5,10,20,30">
  <!--dep,xs:integer, the maximum time of trajectory displaying, unit: second, it is valid only when displaying
trajectory is supported-->
</maximumSuperPositionTime>
<isSupportUnitConfig>
  <!--optional, xs:boolean, whether it supports unit configuration-->
</isSupportUnitConfig>
<isSupportAutoMaintenance>
  <!--optional, xs:boolean, whether it supports automatic maintenance. When this node exists and values "true", it
indicates support-->
</isSupportAutoMaintenance>
```

```

<isSupportGetLinkSocketIP>
  <!--optional, xs: boolean, "true,false", whether it supports getting the SocketIP of current connection-->
</isSupportGetLinkSocketIP>
<isSupportIntelligentSearch>
  <!--optional, xs:boolean, whether it supports intelligent search-->
</isSupportIntelligentSearch>
<IOTCap><!--optional, xs:boolean, IoT device access capability-->
  <supportChannelNum>
    <!--req, xs:integer, number of supported channels of IoT device-->
  </supportChannelNum>
  <startChannelNo>
    <!--optional, xs:integer, initial channel ID, if this node is not inputted, it indicates that the initial channel ID is 1-->
  </startChannelNo>
  <isSupportlinkageChannelsSearch>
    <!--optional, boolean, returns "true" if support, returns "false" if not support-->
  </isSupportlinkageChannelsSearch>
</IOTCap>
<isSupportEncryption>
  <!--optional, xs: boolean, stream encryption capability-->
</isSupportEncryption>
<AIDEventSupport opt="abandonedObject, pedestrian, congestion, roadBlock, construction, trafficAccident,
fogDetection, wrongDirection, illegalParking, SSharpDriving, lowSpeed, dragRacing">
  <!--optional, xs:string, supported traffic incident type: "abandonedObject"-objects dropped down, "pedestrian"-
pedestrian, "congestion"-congestion, "roadBlock"-roadblock, "construction"-construction, "trafficAccident"-traffic
accident, "fogDetection"-fog, "wrongDirection"-wrong-way driving, "illegalParking"-illegal parking, "SSharpDriving"-
slalom driving, "lowSpeed"-driving in low speed, "dragRacing"-street racing-->
</AIDEventSupport>
<TFSEventSupport
opt="illegalParking ,wrongDirection,crossLane,laneChange,vehicleExist,turnRound,parallelParking,notKeepDistance,not
SlowZebraCrossing,overtakeRightSide,lowSpeed,dragRacing,changeLaneContinuously,SSharpDriving,largeVehicleOccup
yLine,jamCrossLine">
  <!--optional, xs:string, supported enforcement event type: "illegalParking"-illegal parking, "wrongDirection"-wrong-
way driving, "crossLane"-driving on the lane line, "laneChange"-illegal lane change, "vehicleExist"-motor vehicle on
non-motor vehicle lane, "turnRound"-illegal U-turn, "parallelParking"-parallel parking, "notKeepDistance"-not keeping
vehicle distance, "notSlowZebraCrossing"-not slowing down at zebra corssing, "overtakeRightSide"-overtaking on the
right, "lowSpeed"-driving in low speed, "dragRacing"-street racing, "changeLaneContinuously"-continuous lane
change, "SSharpDriving"-slalom driving, "largeVehicleOccupyLine"-lane occupation by large-sized vehicle,
"jamCrossLine"-queue jumping-->
</TFSEventSupport>
<isVehicleStatisticsSupport>
  <!--optional, xs: boolean, whether it supports setting parameters for traffic data collection-->
</isVehicleStatisticsSupport>
<isSupportIntersectionAnalysis>
  <!--optional, xs: boolean, whether it supports intersection analysis-->
</isSupportIntersectionAnalysis>
<supportRemoteCtrl opt="up,down,left,right,enter,menu,num,power,esc,edit,F1,.prev,rec,play,stop,notSupport"/><!--
whether it supports remote control-->
<isSptDiagnosis>
  <!--optional, xs:boolean, whether it supports device diagnosis: "true", "false"-->
</isSptDiagnosis>
<isSptSerialLogCfg>
  <!--optional, xs:boolean, whether it supports configuring serial port log redirection: "true", "false"-->

```

```
</isSptSerialLogCfg>
<isSptFileExport>
  <!--optional, xs:boolean, whether it supports exporting files from the device: "true", "false"-->
</isSptFileExport>
<isSptCertificationStandard>
  <!--optional, xs:boolean, whether it supports configuring authentication standard for security control panel: "true",
"false"-->
</isSptCertificationStandard>
<isSptKeypadLock>
  <!--optional, xs:boolean, whether it supports locking keypad: "true", "false"-->
</isSptKeypadLock>
<MixedTargetDetection><!--optional, whether the device supports recognizing specific target among mixed targets-->
  <isSupportFaceRecognition><!--optional, xs:boolean, whether it supports face recognition--></
isSupportFaceRecognition>
  <isSupportHumanRecognition><!--optional, xs:boolean, whether it supports human body recognition--></
isSupportHumanRecognition>
  <isSupportVehicleRecognition><!--optional, xs:boolean, whether it supports vehicle recognition--></
isSupportVehicleRecognition>
</MixedTargetDetection>
<isSupportDiscoveryMode><!--optional, xs:boolean--></isSupportDiscoveryMode>
<streamEncryptionType>
  <!--dep, xs:string, stream encryption type: "RTP/TLS", "SRTP/UDP", "SRTP/MULTICAST". This node is valid when
<isSupportEncryption> is "true", and the device can support one or more stream encryption types-->
</streamEncryptionType>
<isSupportLms><!--optional, xs:boolean, whether it supports laser--></isSupportLms>
<isSupportLCDScreen><!--optional, xs:boolean, whether it supports LCD screen--></isSupportLCDScreen>
<isSupportBluetooth><!--optional, xs:boolean, whether it supports bluetooth--></isSupportBluetooth>
<isSupportAcsUpdate>
  <!--optional, whether it supports upgrading slave access control devices or peripheral modules: "true"-yes, this
node is not returned-no-->
</isSupportAcsUpdate>
<isSupportAccessControlCap>
  <!--optional, whether it supports access control capability: "true"-yes, this node is not returned-no-->
</isSupportAccessControlCap>
<isSupportIDCardInfoEvent><!--optional, whether it supports ID card swiping event: "true"-yes. This node will not be
returned if this function is not supported--></isSupportIDCardInfoEvent>
<OpenPlatformCap><!--optional, embedded open platform capability, refer to the message XML_OpenPlatformCap
for details-->
<isSupportInstallationAngleCalibration>
  <!--optional, xs:boolean, whether it supports installation angle calibration-->
</isSupportInstallationAngleCalibration>
<isSupportZeroBiasCalibration>
  <!--optional, xs:boolean, whether it supports zero bias calibration-->
</isSupportZeroBiasCalibration>
<isSupportDevStatus><!--optional, xs:boolean, whether device supports getting device status--></
isSupportDevStatus>
<isSupportRadar><!--optional, xs:boolean, whether it supports the security radar--></isSupportRadar>
<isSupportRadarChannels><!--optional, xs:boolean, whether it supports getting radar channels--></
isSupportRadarChannels>
  <radarIPDForm><!--optional, xs:string, radar form: "single"-single radar, "double_diagonal"-two radars forming an
180° diagonal, "double_vertical"-two radars forming a 90° vertical angle--></radarIPDForm>
  <isSupportRadarFieldDetection><!--optional, xs:boolean, whether it supports intrusion detection (radar)--></
```

```
isSupportRadarFieldDetection>
  <isSupportRadarLineDetection><!--optional, xs:boolean, whether it supports line crossing detection (radar)--></
isSupportRadarLineDetection>
  <mixedTargetDetectionWebNoDisplay><!--optional, xs:boolean, whether to enable not displaying multi-target-type
recognition--></mixedTargetDetectionWebNoDisplay>
  <SHMCap><!--opt-->
    <isSupportHighHDTemperature><!--optional, xs:boolean, whether it supports HDD high temperature detection--></
isSupportHighHDTemperature>
    <isSupportLowHDTemperature><!--optional, xs:boolean, whether it supports HDD low temperature detection--></
isSupportLowHDTemperature>
    <isSupportHDImpact><!--optional, xs:boolean, whether it supports HDD impact detection--></isSupportHDImpact>
    <isSupportHDBadBlock><!--optional, xs:boolean, whether it supports HDD bad sector detection--></
isSupportHDBadBlock>
    <isSupportSevereHDFailure><!--optional, xs:boolean, whether it supports HDD severe fault detection--></
isSupportSevereHDFailure>
  </SHMCap>
  <isSupportBVCorrect><!--optional, xs:boolean, whether it supports configuring camera correction parameters--></
isSupportBVCorrect>
  <guideEventSupport opt="linkageCapture">
    <!--optional,xs:string, events which support quick setup by instruction, "linkageCapture"-capture by linkage-->
  </guideEventSupport>
  <isSupportAutoSwitch><!--optional, xs:boolean, whether it supports auto switch--> true</isSupportAutoSwitch>
  <isSupportDataPrealarm><!--optional,xs:boolean, whether it supports traffic pre-alarm event--></
isSupportDataPrealarm>
  <supportGISEvent opt="AID,TPS,ANPR,mixedTargetDetection">
    <!--optional, xs:string, event types that support GIS information access: AID (corresponding SDK event:
COMM_ALARM_AID_V41), TPS (corresponding SDK event: COMM_ALARM_TPS_REAL_TIME), ANPR (corresponding
SDK event: COMM_ITS_PLATE_RESULT), mixedTargetDetection-mixed targets detection-->
  </supportGISEvent>
  <isSupportIntelligentMode><!--optional, xs:boolean, whether it supports intelligent scene switch (related URI:/ISAPI/
System/IntelligentSceneSwitch?format=json)--></isSupportIntelligentMode>
  <isSupportCertificateCaptureEvent><!--optional, xs:boolean, whether it supports certificate capture and comparison
events: true=yes. If this function is not supported, this node will not be returned--></
isSupportCertificateCaptureEvent>
  <isSupportAlgorithmsInfo><!--optional, xs:boolean, whether it supports getting the algorithm library version
information: true=yes. If this function is not supported, this node will not be returned--></isSupportAlgorithmsInfo>
  <isSupportVibrationDetection><!--optional, xs:boolean, whether it supports vibration detection--></
isSupportVibrationDetection>
  <isSupportFaceTemperatureMeasurementEvent><!--optional, xs:boolean, whether it supports uploading face
thermography events (eventType: "FaceTemperatureMeasurementEvent")--></
isSupportFaceTemperatureMeasurementEvent>
  <isSupportQRCodeEvent><!--optional, xs:boolean, whether it supports uploading QR code events (eventType:
"QRCodeEvent")--></isSupportQRCodeEvent>
  <isSupportPersonArmingTrack><!--optional, xs:boolean, whether device supports person arming (related URI: /ISAPI/
Intelligent/channels/<ID>/personArmingTrack/capabilities?format=json)--></isSupportPersonArmingTrack>
  <isSupportManualPersonArmingTrack><!--optional, xs:boolean, whether device supports manual person arming
(related URI: /ISAPI/Intelligent/channels/<ID>/manualPersonArmingTrack?format=json)--></
isSupportManualPersonArmingTrack>
  <isSupportGPSCalibrationMode><!--optional, xs:boolean, whether device supports GPS calibration (related URI: /
ISAPI/System/GPSCalibration/channels/<ID>/mode?format=json)--></isSupportGPSCalibrationMode>
  <isSupportGPSVerification><!--optional, xs:boolean, whether device supports GPS verification (related URI: /ISAPI/
System/GPSVerification/channels/<ID>/points?format=json)--></isSupportGPSVerification>
```

```
<isSupportHBDLib><!--optional, xs:boolean, whether device supports human body picture library (related URI: /ISAPI/Intelligent/HBDLib/capabilities?format=json)--></isSupportHBDLib>
<isSupportFireEscapeDetection><!--optional, xs:boolean, whether the device supports fire engine access detection (related URI: /ISAPI/Intelligent/channels/<ID>/fireEscapeDetection/capabilities?format=json)--></isSupportFireEscapeDetection>
<isSupportTakingElevatorDetection><!--optional, xs:boolean, whether the device supports elevator detection (related URI: /ISAPI/Intelligent/channels/<ID>/takingElevatorDetection/capabilities?format=json)--></isSupportTakingElevatorDetection>
</DeviceCap>
```

XML_DeviceInfo

XML message about device information

```
<DeviceInfo version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <deviceName><!--required, xs:string--></deviceName>
  <deviceId><!--required, read-only, xs:string, uuid--></deviceId>
  <deviceDescription>
    <!--optional, xs:string, description about the device defined in RFC1213. For network camera, this node is set to "IPCamera"; for network speed dome, this node is set to "IPDome"; for DVR or DVS, this node is set to "DVR" or "DVS"-->
  </deviceDescription>
  <deviceLocation><!--optional, xs:string, actual location of the device--></deviceLocation>
  <deviceStatus><!--optional, read-only, xs:string, device status: "normal", "abnormal"-->
  <DetailAbnormalStatus>
    <!--dependent, error status details, it is valid only when deviceStatus is "abnormal"-->
    <hardDiskFull>
      <!--optional, read-only, xs: boolean, whether the error of "HDD full" occurred: "true"-yes,"false"-no-->
    </hardDiskFull>
    <hardDiskError>
      <!--optional, read-only, xs:boolean, whether the error of "HDD error" occurred: "true"-yes,"false"-no-->
    </hardDiskError>
    <ethernetBroken>
      <!--optional, read-only, xs: boolean, whether the error of "network disconnected" occurred: "true"-yes,"false"-no-->
    </ethernetBroken>
    <ipaddrConflict>
      <!--optional, read-only, xs: boolean, whether the error of "IP address conflicted" occurred: "true"-yes,"false"-no-->
    </ipaddrConflict>
    <illegalAccess>
      <!--optional, read-only, xs: boolean, whether the error of "illegal login" occurred: "true"-yes,"false"-no-->
    </illegalAccess>
    <recordError>
      <!--optional, read-only, xs: boolean, whether the error of "recording exception" occurred: "true"-yes,"false"-no-->
    </recordError>
    <raidLogicDiskError>
      <!--optional, read-only, xs: boolean, whether the error of "RAID exception" occurred: "true"-yes,"false"-no-->
    </raidLogicDiskError>
    <spareWorkDeviceError>
      <!--optional, read-only, xs: boolean, whether the error of "working device exception" occurred: "true"-yes,"false"-no-->
    </spareWorkDeviceError>
  </DetailAbnormalStatus>
</DeviceInfo>
```

```
</spareWorkDeviceError>
</DetailAbnormalStatus>
<systemContact><!--optional, xs:string, contact information of the device--></systemContact>
<model><!--required, read-only, xs:string--></model>
<serialNumber><!--required, read-only, xs:string--></serialNumber>
<macAddress><!--required, read-only, xs:string--></macAddress>
<firmwareVersion><!--required, read-only, xs:string--></firmwareVersion>
<firmwareReleasedDate><!--optional, read-only, xs:string--></firmwareReleasedDate>
<bootVersion><!--optional, read-only, xs:string--></bootVersion>
<bootReleasedDate><!--optional, read-only, xs:string--></bootReleasedDate>
<hardwareVersion><!--optional, read-only, xs:string--></hardwareVersion>
<encoderVersion><!--optional, read-only, xs:string--></encoderVersion>
<encoderReleasedDate><!--optional, read-only, xs:string--></encoderReleasedDate>
<decoderVersion><!--optional, read-only, xs:string--></decoderVersion>
<decoderReleasedDate><!--optional, read-only, xs:string--></decoderReleasedDate>
<softwareVersion><!--optional, read-only, xs:string, software version--></softwareVersion>
<capacity><!--optional, read-only, xs:integer, unit: MB, device capacity--></capacity>
<usedCapacity><!--optional, read-only, xs:integer, unit: MB, capacity usage--></usedCapacity>
<deviceType>
  <!--required, read-only, xs:string, device type: "IPCamera", "IPDome", "DVR", "HybirdNVR", "NVR", "DVS", "IPZoom",
  "CVR", "Radar", "PerimeterRadar"-perimeter radar, "ACS", "PHA"-Axiom hybrid security control panel-->
  <deviceType>
    <telecontrolID><!--optional, xs:integer, keyfob control ID, the value is between 1 and 255--></telecontrolID>
    <supportBeep><!--optional, xs:boolean--></supportBeep>
    <supportVideoLoss><!--optional, xs:boolean, whether it supports video loss detection--></supportVideoLoss>
    <firmwareVersionInfo><!--optional, read-only, xs:string, firmware version information--></firmwareVersionInfo>
    <actualFloorNum>
      <!--required, xs: integer, actual number of floors, which is between 1 and 128-->
    </actualFloorNum>
    <subChannelEnabled><!--optional, xs:boolean, whether to support sub-stream live view: "true"-yes, "false"-no--></
subChannelEnabled>
    <thrChannelEnabled><!--optional, xs:boolean, whether to support third stream live view: "true"-yes, "false"-no--></
thrChannelEnabled>
    <radarVersion><!--optional, xs:string, radar version--></radarVersion>
    <cameraModuleVersion><!--read-only, xs:string, camera module version--></cameraModuleVersion>
    <mainversion><!--optional, xs:integer, main version No. which is between 1 and 255--></mainversion>
    <subversion><!--optional, xs:integer, sub version No. which is between 1 and 255--></subversion>
    <upgradeversion><!--optional, xs:integer, upgraded version No. which is between 1 and 255--></upgradeversion>
    <customizeversion><!--optional, xs:integer, customized version No. which is between 1 and 255--></
customizeversion>
    <companyName><!--optional, xs:string, the manufacturing company's abbreviation--></companyName>
    <copyright><!--optional, xs:string, copyright information--></copyright>
    <systemName><!--optional, xs:string, storage system name: "storageManagement"-storage management system,
"distributedStorageManagement"-distributed storage management system--></systemName>
    <systemStatus><!--optional, xs:string, system status: "configured"-configured, "unConfigured"-not configured--></
systemStatus>
    <isLeaderDevice><!--optional, xs:boolean, whether it is the corresponding device of the resource IP address--></
isLeaderDevice>
    <clusterVersion><!--dependent, xs:string, system cluster version. This node is valid when <isLeaderDevice> is true--
></clusterVersion>
    <manufacturer><!--optional, xs:string, manufacturer information: "hikvision"-Hikvision devices; for neutral devices,
this node should be empty--></manufacturer>
```

```
<customizedInfo><!--optional, xs:string, order No. of the customization project. For baseline devices, this node is
empty; for custom devices, the order No. of the customization project will be returned by this node--></
customizedInfo>
<localZoneNum><!--optional, xs:integer, number of local zones--></localZoneNum>
<alarmOutNum><!--optional, xs:integer, number of alarm outputs--></alarmOutNum>
<distanceResolution><!--optional, xs:float, resolution of distance, unit: meter--></distanceResolution>
<angleResolution><!--optional, xs:float, resolution of angle, unit: degree--></angleResolution>
<speedResolution><!--optional, xs:float, resolution of speed, unit: m/s--></speedResolution>
<detectDistance><!--optional, xs:float, detection distance, unit: meter--></detectDistance>
<languageType opt="chinese,english,spanish,portuguese,italian,french,russian,german,polish"><!--optional,
xs:string--></languageType>
<relayNum><!--optional, xs:integer, number of local relays--></relayNum>
<electroLockNum><!--optional, xs:integer, number of local electronic locks--></electroLockNum>
<RS485Num><!--optional, xs:integer, number of local RS-485--></RS485Num>
<powerOnMode><!--optional, xs:string, device startup mode: "button"-press button to power on (default), "adapter"-
connect adapter to power on--></powerOnMode>
<DockStation>
<!--optional, dock station configuration-->
<Platform>
<!--optional, platform configuration-->
<type><!--required, xs:string, platform type: none, 9533, 8618, ISAPI--></type>
<ip><!--optional, xs:string, IP address --></ip>
<port><!--optional, xs:integer, communication port--></port>
<userName><!--required, xs:string, user name, which is used for the dock station to log in to platform--></
userName>
<password><!--required, xs:string, password, which is used for the dock station to log in to platform, it should be
encrypted--></password>
</Platform>
<centralStorageBackupEnabled><!--optional, xs:boolean, whether to enable central storage backup--></
centralStorageBackupEnabled>
</DockStation>
</DeviceInfo>
```

XML_DeviceLanguage

DeviceLanguage message in XML format

```
<DeviceLanguage version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
<language>
<!--required, xs:string, language supported by the device: "SimChinese"-simplified Chinese, "TraChinese"-traditional
Chinese, "English", "Russian", "Bulgarian", "Hungarian", "Greek", "German", "Italian", "Czech", "Slovakia", "French",
"Polish", "Dutch", "Portuguese", "Spanish", "Romanian", "Turkish", "Japanese", "Danish", "Swedish", "Norwegian",
"Finnish", "Korean", "Thai", "Estonia", "Vietnamese", "Hebrew", "Latvian", "Arabic", "Sovenian"-Slovenian, "Croatian",
"Lithuanian", "Serbian", "BrazilianPortuguese"-Brazilian Portuguese, "Indonesian", "Ukrainian"-->
</language>
</DeviceLanguage>
```


XML_DeviceStatus

DeviceStatus message in XML format

```
<DeviceStatus version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <currentDeviceTime><!--opt, xs: datetime--></currentDeviceTime>
  <deviceUpTime><!--opt, xs: integer, seconds--></deviceUpTime>
  <TemperatureList><!--opt-->
    <Temperature>
      <tempSensorDescription><!--req, xs: string--></tempSensorDescription>
      <temperature><!--req, xs: float--></temperature>
    </Temperature>
  </TemperatureList>
  <FanList><!--opt-->
    <Fan>
      <fanDescription><!--req, xs: string--></fanDescription>
      <speed><!--req, xs: integer--></speed>
    </Fan>
  </FanList>
  <PressureList><!--opt-->
    <Pressure>
      <pressureSensorDescription><!--req, xs: string--></pressureSensorDescription>
      <pressure><!--req, xs: integer--></pressure>
    </Pressure>
  </PressureList>
  <TamperList><!--opt-->
    <Tamper><tamperSensorDescription><!--req, xs: string--></tamperSensorDescription>
    <tamper><!--req, xs: boolean--> </tamper>
  </Tamper>
  </TamperList>
  <CPUList><!--opt-->
    <CPU>
      <cpuDescription><!--req, xs: string--></cpuDescription>
      <cpuUtilization><!--req, xs: integer, percentage, which is between 0 and 100--></cpuUtilization>
    </CPU>
  </CPUList>
  <MemoryList><!--opt-->
    <Memory>
      <memoryDescription><!--req, xs: string--></memoryDescription>
      <memoryUsage><!--req, xs: float, unit: MB--></memoryUsage>
      <memoryAvailable><!--req, xs: float, unit: MB--></memoryAvailable>
    </Memory>
  </MemoryList>
  <openFileHandles><!--opt, xs: integer--></openFileHandles>
  <CameraList><!--opt-->
    <Camera>
      <zoomReverseTimes><!--req, xs: integer--></zoomReverseTimes>
      <zoomTotalSteps><!--req, xs: integer--></zoomTotalSteps>
      <focusReverseTimes><!--req, xs: integer--></focusReverseTimes>
      <focusTotalSteps><!--req, xs: integer--></focusTotalSteps>
      <irisShiftTimes><!--req, xs: integer--></irisShiftTimes>
    </Camera>
  </CameraList>
</DeviceStatus>
```

```

<irisTotalSteps><!--req, xs: integer--></irisTotalSteps>
<icrShiftTimes><!--req, xs: integer--></icrShiftTimes>
<icrTotalSteps><!--req, xs: integer--></icrTotalSteps>
<lensIntirTimes><!--req, xs: integer--></lensIntirTimes>
<cameraRunTotalTime><!--req, xs: integer--></cameraRunTotalTime>
</Camera>
</CameraList>
<DomeInfoList><!--opt-->
<DomeInfo>
  <domeRunTotalTime><!--opt, xs: integer--></domeRunTotalTime>
  <runTimeUnderNegativetwenty><!--opt, xs: integer--></runTimeUnderNegativetwenty>
  <runTimeBetweenNttwentyPforty><!--opt, xs: integer--></runTimeBetweenNttwentyPforty>
  <runtimeOverPositiveforty><!--opt, xs: integer--></runtimeOverPositiveforty>
  <panTotalRounds><!--opt, xs: integer--></panTotalRounds>
  <tiltTotalRounds><!--opt, xs: integer--></tiltTotalRounds>
  <heatState><!--opt, xs: integer--></heatState>
  <fanState><!--opt, xs: integer--></fanState>
</DomeInfo>
</DomeInfoList>
<deviceStatus><!--req, xs: string, device working status: "normal, abnormal 故障"--></deviceStatus>
<dialSignalStrength>
  <!--optional, xs:integer, 4G signal strength, it is between 1 and 3-->
</dialSignalStrength>
<USBStatusList>
<USBStatus>
  <!--list-->
  <id>
    <!--required, xs:integer-->
  </id>
  <state>
    <!--optional, xs:string, USB status: "connected", "fullCapacity", "deviceException", "disconnected"-->
  </state>
</USBStatus>
</USBStatusList>
<WifiStatusList>
<WifiStatus>
  <!--list-->
  <id>
    <!--required, xs:integer-->
  </id>
  <state>
    <!--optional, xs:string, Wi-Fi status: "disabled"-Wi-Fi is disabled, "enable"-Wi-Fi is enabled, "disconnected"-Wi-Fi
is disconnected, "enableHotspot"-Wi-Fi hot spot is enabled-->
  </state>
</WifiStatus>
</WifiStatusList>
<AlertStreamServerList><!--opt, user of arming device-->
<AlertStreamServer><!--list-->
  <id><!--opt, xs:integer--></id>
  <protocolType><!--opt, xs:string, "SDK,ISAPI", arming protocol type--></protocolType>
  <ipAddress><!--dep, xs:string, IP address of platform, server, NVR/CVR, and so on--></ipAddress>
</AlertStreamServer>

```

```
</AlertStreamServerList>
</DeviceStatus>
```

XML_DiagnosedDataParameter

XML message about parameters of exporting diagnose information

```
<?xml version="1.0" encoding="utf-8"?>
<DiagnosedDataParameter version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled>
    <!--required, xs:boolean, whether to enable exporting diagnose information: true=yes, false=no-->
  </enabled>
  <HardwareInfo>
    <!--optional, xs:boolean, opt="true,false", whether to enable exporting hardware information, such as running
information of motor, fan, heater, compressor, supplement light, wiper, and so on-->
  </HardwareInfo>
  <SoftwareInfo>
    <!--optional, xs:boolean, opt="true,false", whether to enable exporting software information, such as software
status, running log, system information-->
  </SoftwareInfo>
  <logInfo>
    <!--optional, xs:boolean, opt="true,false", whether to enable exporting startup log--></logInfo>
</DiagnosedDataParameter>
```

XML_Dial

Dial message in XML format

```
<Dial version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--req, xs:boolean--></enabled>
  <DialMethod><!--req, xs:string, "auto, manual"--></DialMethod>
  <SwitchMethod>
    <!--req, xs:string, "auto,4GFirst,3GFirst, manualto2G, manualto3G, manualto4G"-->
  </SwitchMethod>
  <OfflineTime><!--opt, xs:integer--></OfflineTime>
  <UIMCardNum><!--opt, xs:string--></UIMCardNum>
  <DialNum><!--opt, xs:string--></DialNum>
  <Username><!--opt, xs:string--></Username>
  <Password><!--opt, xs:string--></Password>
  <APNname><!--opt, xs:string--></APNname>
  <SIMNum><!--opt, xs:string, SIM card No. (mobile phone number)--></SIMNum>
  <MTU><!--opt, xs: integer--></MTU>
  <VerifyProto><!--req, xs:string, "auto, CHAP, PAP"--></VerifyProto>
  <DefaultParam><!--opt, default parameters, see details in the message of XML_DefaultParam-->
  <netAPN><!--opt, xs:string, APN configuration of the private network--></netAPN>
  <Flow><!--opt, network traffic flow configuration-->
    <limitEnabled><!--opt, xs:boolean, whether to enable traffic flow limitation--></limitEnabled>
    <consumeFlow><!--opt, xs:float, unit: MB--></consumeFlow>
    <threshold><!--opt, xs:integer, unit: MB--></threshold>
```

```
</Flow>
<pinCode><!--opt, wo, xs:string--></pinCode>
<enabled4G><!-- opt, xs:boolean, whether to enable 4G--></enabled4G>
<enabledDNS><!-- opt, xs:boolean, whether to enable DNS manual configuration--></enabledDNS>
</Dial>
```

See Also

XML_DefaultParam

XML_Dialstatus

Dialstatus message in XML format

```
<Dialstatus version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <RealtimeMode>
    <!-- ro, xs:string, real time mode,
    "CDMA1x,EVDO,HYBRID,GSM,GPRS,EDGE,WCDMA,HSDPA,HSUPA,HSPA,TDSCDMA,TD-LTE,FDD-LTE,LTE,UNKNOWN"-->
  </RealtimeMode>
  <UIMInfo>
    <!-- ro, xs:string, UIM cord status: "UNKNOWN"-unknown, "VALID"-valid, "NOVALID"-invalid, "ROAM"-roam,
    "NOEXIST"-not exist, "WRONGPINCODE"-PIN code error, "NONNETWORKSERVICE"-connectionless network service,
    "ONLYSUPPORTEMERGENCYCALL"-the current network only supports emergency calls, "PINCODELOCKED"—PIN code
    locked-->
  </UIMInfo>
  <SignalStrength><!-- ro, xs: integer, signal strength--></SignalStrength>
  <Dialstat><!-- ro, xs:string, dial-up status: "disconnect"-disconnected, "dialing"-dialing, "success"-dialed, "shmError"-
  dialing exception, "certificationFailure"-authentication failed, "invalidDialingCharacters"-invalid dialing characters,
  "networkTimeout"-network timeout, "wirelessModemError"-wireless module error, "SIMError"-SIM card error,
  "networkError"-network error, "unknown"-unknown error--></Dialstat>
  <IpAddress> <!-- req -->
    <IpAddress><!-- dep, xs:string--></IpAddress>
    <ipv6Address><!-- dep, xs:string --></ipv6Address>
  </IpAddress>
  <SubnetMask><!-- req -->
    <IpAddress><!-- dep, xs:string--></IpAddress>
    <ipv6Address><!-- dep, xs:string--></ipv6Address>
  </SubnetMask>
  <Gateway><!-- req -->
    <IpAddress><!-- dep, xs:string--></IpAddress>
    <ipv6Address><!-- dep, xs:string--></ipv6Address>
  </Gateway>
  <DNSAddress> <!-- req -->
    <IpAddress><!-- dep, xs:string--></IpAddress>
    <ipv6Address><!-- dep, xs:string--></ipv6Address>
  </DNSAddress>
  <subDNSAddress><!-- opt, sub DNS address-->
    <IpAddress><!-- dep, xs:string --></IpAddress>
    <ipv6Address><!-- dep, xs:string--></ipv6Address>
  </subDNSAddress>
  <operator><!-- ro, opt,xs:string,operator--></operator>
```

```
<enableMessageConfig><!--opt, xs:boolean,whether to enable SMS service--></enableMessageConfig>
<registrationStatus><!--ro,opt, xs:boolean, registration status--></registrationStatus>
</DialStatus>
```

XML_Discovery

Discovery message in XML format

```
<Discovery version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <UPnP><!--req-->
    <enabled><!--req, xs:boolean, whether to enable UPnP: "true, false"--></enalbed>
  </UPnP>
  <ZeroConf><!--opt, channel-zero configuration-->
    <enabled><!--req, xs:boolean, "true, false"--></enabled>
  </ZeroConf>
</Discovery>
```

XML_diskQuota

diskQuota message in XML format

```
<diskQuota version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs: integer, channel ID--></id>
  <useVideoQuota><!--read-only, xs: integer, quota for video files, unit: MB--></useVideoQuota>
  <usePictureQuota><!--read-only, xs: integer, quota for picture files, unit: MB--></usePictureQuota>
  <totalDiskVolume><!--read-only, xs: integer, total disk capacity, unit: MB--></totalDiskVolume>
  <videoQuota><!--req, xs: integer, unit: MB--></videoQuota>
  <pictureQuota><!--opt, xs: integer, unit: MB--></pictureQuota>
  <type><!--opt, xs: string, "volume,ratio,absent-ratio,otherwise-volume"--></type>
  <videoQuotaRatio><!--dep, xs: integer, video quota ratio (%): [0,100]--></videoQuotaRatio>
  <pictureQuotaRatio><!--dep, xs: integer, picture quota ratio (%): [0,100]--></pictureQuotaRatio>
  <totalVideoVolume><!--read-only, xs: integer, unit: MB--></totalVideoVolume>
  <totalPictureVolume><!--ro, xs: integer, unit: MB--></totalPictureVolume>
  <freeVideoQuota><!--ro, xs: integer, free space for video files, unit: MB--></freeVideoQuota>
  <freePictureQuota><!--ro, xs: integer, free space for picture files, unit: MB--></freePictureQuota>
</diskQuota>
```

XML_DoubleLensParkingCap

DoubleLensParkingCap capability message in XML format

```
<DoubleLensParkingCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <cameraType opt = "masterCamera, slaveCamera">
    <!--req, xs:string, dual-lens camera type: "masterCamera"-fixed dual-lens camera, "slaveCamera"-PTZ dual-lens
camera-->
  </cameraType>
</DoubleLensParkingCap>
```

XML_DownloadAbility

DownloadAbility message in XML format

```
<DownloadAbility version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <isSupportDownloadbyTime>
    <!--opt, xs:boolean, whether it supports download by time-->
  </isSupportDownloadbyTime>
  <isSupportDownloadbyFileName>
    <!--opt, xs:boolean, whether it supports download by file name -->
  </isSupportDownloadbyFileName>
  <isSupportDownloadToUSB>
    <!--opt, xs:boolean, whether it supports exporting files to devices via USB-->
  </isSupportDownloadToUSB>
</DownloadAbility>
```

XML_downloadRequest

XML message about downloading request

```
<downloadRequest version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <playbackURI>
    <!--required, xs:string, playback URL, returned by the search service. It contains the information of file name and
size, e.g., <playbackURI>rtsp://IpAddress/Streaming/tracks/101?starttime=2016-07-18
00:00:00&endtime=2016-07-18 00:25:04Z&name=00000000721000000&size=1065437356</playbackURI>-->
  </playbackURI>
  <downloadCourseFile>
    <!--optional, xs:boolean, whether to download the course files-->
  </downloadCourseFile>
</downloadRequest>
```

Remarks

When download file by name, the **playbackURI** inputs the file start time, end time, file name, and size; when download file by time, the **playbackURI** inputs the start time and end time.

XML_DynamicCap

DynamicCap capability message in XML format.

```
<DynamicCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ResolutionAvailableDscriptorList><!--req-->
    <ResolutionAvailableDscriptor><!--req-->
      <videoResolutionWidth>
        <!--req, xs:integer-->
      </videoResolutionWidth>
      <videoResolutionHeight>
        <!--req, xs:integer-->
      </videoResolutionHeight>
    </ResolutionAvailableDscriptor>
  </ResolutionAvailableDscriptorList>
</DynamicCap>
```

```
</videoResolutionHeight>
<supportedFrameRate>
  <!--req, xs:string, frame rate supported by this resolution, and its value is multiplied by 100 to be returned, e.g., if
the returned values are "2000, 2200, 2500", the actual frame rates are "20, 22, 25"-->
</supportedFrameRate>
<supportedBitRate min="" max=""/>
<!--opt, xs:integer,in kbps-->
</ResolutionAvailableDescriptor>
</ResolutionAvailableDescriptorList>
<CodecParamDescriptorList>
  <CodecParamDescriptor>
    <videoCodecType><!-- req, xs: string, "MPEG4,MJPEG,3GP,H.264,HK.264"--></videoCodecType>
    <isSupportProfile><!--dep, xs: boolean--></isSupportProfile>
    <CBRCap><!--constant bit rate-->
      <isSupportSmooth><!--dep, xs: boolean--></isSupportSmooth>
    </CBRCap>
    <VBRcap><!--variable bit rate-->
      <isSupportSmooth><!--dep, xs: boolean--></isSupportSmooth>
    </VBRcap>
    <isSupportSVC><!--opt, xs: boolean--></isSupportSVC>
    <isSupportCABAC><!--opt, xs: boolean--></isSupportCABAC>
    <SmartCodecCap><!--opt-->
      <readOnlyParams opt="keyFrameInterval,Profile,SVC,fixedQuality">
        <!--opt, ro, xs: string, read-only options: I frame interval, encoding complexity, SVC, picture quality-->
      </readOnlyParams>
    </SmartCodecCap>
  </CodecParamDescriptor>
</CodecParamDescriptorList>
<BitrateType>
  <Constant><!--opt, constant bit rate-->
    <support opt="videoBitrate">
      <!--opt, xs: string, "averageVideoBitrate"-average bit rate, "videoBitrate"-upper-limit of bit rate-->
    </support>
    <hiddenAbility opt="averageVideoBitrate">
      <!--opt, xs:string, "averageVideoBitrate"-average bit rate, "videoBitrate"-upper-limit of bit rate-->
    </hiddenAbility>
  </Constant>
  <Variable><!--opt, variable bit rate-->
    <support opt="averageVideoBitrate">
      <!--opt, xs: string, "averageVideoBitrate"-average bit rate, "videoBitrate"-upper-limit of bit rate-->
    </support>
    <readOnlyAbility opt="videoBitrate">
      <!--opt, xs: string, "averageVideoBitrate"-average bit rate, "videoBitrate"-upper-limit of bit rate-->
    </readOnlyAbility>
  </Variable>
</BitrateType>
<vbrAverageDefault><!--dep, xs: integer, recommended average bit rate, unit: Kbps--></vbrAverageDefault>
<smart264EnabledPrompt opt="prompt1,prompt2,prompt3, prompt6, prompt25">
  <!--opt, wo, xs: string, audible prompt for Smart264 enabled-->
</smart264EnabledPrompt>
<smart265EnabledPrompt opt="prompt1,prompt2, prompt3, prompt25">
  <!--opt, wo, xs: string, audible prompt for Smart265 enabled-->
</smart265EnabledPrompt>
</SmartCodecCap>
</CodecParamDescriptor>
```

```
</CodecParamDescriptorList>
<AudioDescriptorList>
  <AudioDescriptor>
    <audioCompressionType>
      <!--req, xs: string, audio encoding type: "G.711alaw,G.711ulaw,G.726,G.729,G.729a,G.
729b,PCM,MP3,AC3,AAC,ADPCM,MP2L2"-->
    </audioCompressionType>
    <audioItemList>
      <audioItem>
        <audioSamplingRate default=""><!--opt, xs: string--></audioSamplingRate>
        <audioBitRate opt=""><!--dep, xs: integer--></audioBitRate>
        <noiseReduce opt="true,false"><!--req, xs: string--></noiseReduce>
      </audioItem>
    </audioItemList>
  </AudioDescriptor>
</AudioDescriptorList>
</DynamicCap>
```

XML_Ehome

Ehome message in XML format

```
<Ehome version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--opt, xs: boolean--></enabled>
  <addressingFormatType>
    <!--req, xs: string, address format type: "ipaddress"-IP address, "hostname"-host name-->
  </addressingFormatType>
  <hostName><!--dep, xs: string, host name, this node is valid only when <addressingFormatType> is "hostname"--></
hostName>
  <ipAddress><!--dep, xs: string, IP address, this node is valid only when <addressingFormatType> is "ipaddress"--></
ipAddress>
  <ipv6Address><!--dep, xs: string, IPv6 address, this node is valid only when <addressingFormatType> is "ipaddress"--
></ipv6Address>
  <portNo><!--opt, xs: integer, port No.--></portNo>
  <deviceId><!--req, xs: string, device ID--></deviceId>
  <registerStatus><!--ro, xs: boolean, registration status--></registerStatus>
  <key><!--opt, xs: string, encryption key of ISUP version 5.0, which is required for ISUP version 50--></key>
  <version><!--ro, xs: string, version No.--></version>
  <netWork><!--opt, xs:integer, 0-make no sense, 1-automatic, 2-wired network preferred, 3-wired network, 4-3G/4G/
GPRS--></netWork>
  <voiceDeviceType><!--opt, xs:string, two-way audio device type: "bluetooth"-bluetooth device, "client", "local"--></
voiceDeviceType>
  <protocolVersion><!--opt, xs:string, protocol version: "v2.0,v2.6,v4.0,v5.0". If this node is set to "v2.0", the device can
only use protocol v2.0 to register; if this node is set to "v2.6", "v4.0" or "v5.0", the device will firstly use this protocol
to register. If this node is not returned, the protocol version will be determined by <version--></protocolVersion>
</Ehome>
```


XML_EPTZ

EPTZ message in XML format

```
<EPTZ version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--req, xs:string, whether to enable e-PTZ: "true"-yes, "false"-no--></enabled>
  <streamType><!--opt, xs:integer, stream type: 1-main stream, 2-sub-stream, 3-third stream, 4-fourth stream--></streamType>
</EPTZ>
```

XML_EventCap

EventCap capability message in XML format

```
<EventCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <isSupportHDFull><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportHDFull>
  <isSupportHDError><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportHDError>
  <isSupportNicBroken><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportNicBroken>
  <isSupportIpConflict><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportIpConflict>
  <isSupportIllaAccess><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportIllaAccess>
  <isSupportViException><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportViException>
  <isSupportViMismatch><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportViMismatch>
  <isSupportRecordException><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportRecordException>
  <isSupportTriggerFocus><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportTriggerFocus>
  <isSupportMotionDetection><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportMotionDetection>
  <isSupportVideoLoss><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportVideoLoss>
  <isSupportTamperDetection><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportTamperDetection>
  <isSupportStudentsStoodUp><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportStudentsStoodUp>
  <isSupportFramesPeopleCounting><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportFramesPeopleCounting>
  <isSupportRaidException><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportRaidException>
  <isSupportSpareException><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportSpareException>
  <isSupportPoePowerException><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportPoePowerException>
  <isSupportRegionEntrance><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportRegionEntrance>
  <isSupportRegionExiting><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportRegionExiting>
  <isSupportLoitering><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportLoitering>
  <isSupportGroup><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportGroup>
  <isSupportRapidMove><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportRapidMove>
  <isSupportFireDetection><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportFireDetection>
  <isSupportIllegalParking><!--optional, xs:boolean, whether it supports illegal parking detection: "true"-support, "false"-not support--></isSupportIllegalParking>
  <isSupportUnattendedBaggage><!--optional, xs:boolean --></isSupportUnattendedBaggage>
```

```
<isSupportAttendedBaggage><!--optional, xs:boolean, "true"-support, "false"-not support--></
isSupportAttendedBaggage>
<isSupportHumanAttribute><!--optional, xs:boolean, "true"-support, "false"-not support--></
isSupportHumanAttribute>
<isSupportFaceContrast><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportFaceContrast>
<isSupportFaceLib><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportFaceLib>
<isSupportWhiteListFaceContrast><!--optional, xs:boolean, "true"-support, "false"-not support--></
isSupportWhiteListFaceContrast>
<isSupportBlackListFaceContrast><!--optional, xs:boolean, whether it supports blacklist face comparison: "true"-
support, "false"-not support--></isSupportBlackListFaceContrast>
<isSupportHumanRecognition><!--optional, xs:boolean, "true"-support, "false"-not support--></
isSupportHumanRecognition>
<isSupportFaceSnap><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportFaceSnap>
<isSupportPersonDensityDetection><!--optional, xs:boolean, "true"-support, "false"-not support--></
isSupportPersonDensityDetection>
<isSupportMixedTargetDetection><!--optional, xs:boolean, whether it supports multi-target-type detection alarm:
"true"-support, "false"-not support--></isSupportMixedTargetDetection>
<isSupportPedestrian><!--optional, xs:boolean, whether it supports pedestrian detection: "true"-support, "false"-not
support--></isSupportPedestrian>
<isSupportTrafficAccident><!--optional, xs:boolean, whether it supports traffic accident detection: "true"-support,
"false"-not support--></isSupportTrafficAccident>
<isSupportConstruction><!--optional, xs:boolean, whether it supports construction detection: "true"-support, "false"-
not support--></isSupportConstruction>
<isSupportRoadBlock><!--optional, xs:boolean, whether it supports roadblock detection: "true"-support, "false"-not
support--></isSupportRoadBlock>
<isSupportAbandonedObject><!--optional, xs:boolean, whether it supports thrown object detection: "true"-support,
"false"-not support--></isSupportAbandonedObject>
<isSupportParallelParking><!--optional, xs:boolean, whether it supports parallel parking detection: "true"-support,
"false"-not support--></isSupportParallelParking>
<isSupportParkingState><!--optional, xs:boolean, whether it supports parking space status detection: "true"-support,
"false"-not support, currently this node is not supported--></isSupportParkingState>
<isSupportCongestion><!--optional, xs:boolean, whether it supports congestion detection: "true"-support, "false"-
not support--></isSupportCongestion>
<isSupportVehicleStatistics><!--optional, xs:boolean, whether it supports data collection: "true"-support, "false"-not
support--></isSupportVehicleStatistics>
<isSupportWrongDirection><!--optional, xs:boolean, whether it supports wrong-way driving detection: "true"-
support, "false"-not support--></isSupportWrongDirection>
<isSupportTrunRound><!--optional, xs:boolean, whether it supports U-turning detection: "true"-support, "false"-not
support--></isSupportTrunRound>
<isSupportCrossLane><!--optional, xs:boolean, whether it supports driving on the lane line detection: "true"-support,
"false"-not support--></isSupportCrossLane>
<isSupportLaneChange><!--optional, xs:boolean, whether it supports illegal lane change detection: "true"-support,
"false"-not support--></isSupportLaneChange>
<isSupportVehicleExist><!--optional, xs:boolean, whether it supports motor vehicle on non-motor vehicle lane
detection: "true"-support, "false"-not support--></isSupportVehicleExist>
<isSupportFogDetection><!--optional, xs:boolean, whether it supports fog detection: "true"-support, "false"-not
support--></isSupportFogDetection>
<isSupportIntersectionAnalysis><!--optional, xs: boolean, whether it supports configuring intersection analysis alarm:
"true"-support, "false"-not support--></isSupportIntersectionAnalysis>
<isSupportVoltageInstable><!--optional, xs:boolean, whether it supports supply voltage exception alarm: "true"-
support, "false"-not support--></isSupportVoltageInstable>
<isSupportSafetyHelmetDetection><!--optional, xs:boolean, whether it supports hard hat detection: "true"-support,
```

```
"false"-not support--></isSupportSafetyHelmetDetection>
<isSupportCertificateRevocation><!--optional, xs:boolean, whether it supports certificate expiry alarm--></
isSupportCertificateRevocation>
<isSupportNoMaskDetection><!--optional, xs:boolean, whether device supports no wearing mask detection--></
isSupportNoMaskDetection>
<isSupportTMPA><!--optional, xs:boolean, whether device supports temperature measurement pre-alarm--></
isSupportTMPA>
<RuleScheduleCap><!--optional, capability of setting arming schedule by rule-->
<isSupportCityManagement>
  <!--optional, xs:boolean, whether the device supports setting arming schedule by rule for intelligent city
management; if supports, the value is true, otherwise, this node will not be returned-->
</isSupportCityManagement>
</RuleScheduleCap>
</EventCap>
```

XML_EventNotificationAlert_AlarmEventInfo

EventNotificationAlert message with alarm/event information in XML format.

```
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
<ipAddress><!--dep, xs:string, device IPv4 address--></ipAddress>
<ipv6Address><!--dep, xs:string, device IPv6 address--></ipv6Address>
<portNo><!--opt, xs:integer, device port number--></portNo>
<protocol><!--opt, xs:string, protocol type for uploading alarm/event information, "HTTP,HTTPS"--></protocol>
<macAddress><!--opt, xs:string, MAC address--></macAddress>
<channelID><!--dep, xs:string, device channel No., starts from 1--></channelID>
<dateTime><!--req, alarm/event triggered or occurred time, format: 2017-07-19T10:06:41+08:00--></dateTime>
<activePostCount><!--req, xs:integer, alarm/event frequency, starts from 1--></activePostCount>
<eventType><!--req, xs:string, alarm/event type, "peopleCounting, ANPR,..."--></eventType>
<eventState>
  <!--req, xs:string, durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is
detected,
  the alarm/event information will be uploaded continuously unit the status is set to "inactive"-->
</eventState>
<eventDescription><!--req, xs:string, alarm/event description--></eventDescription>
<...><!--opt, for different alarm/event types, the nodes are different, see the message examples in different
applications--></...>
</EventNotificationAlert>
```

XML_EventNotificationAlert_AlarmInputAlarmMsg

XML Message about Alarm Input Alarm Details

```
<?xml version="1.0" encoding="utf-8"?>
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
<ipAddress><!--dependent, xs:string, IPv4 address--></ipAddress>
<ipv6Address><!--dependent, xs:string, IPv6 address--></ipv6Address>
<portNo><!--optional, xs:integer, device's communication port No.--></portNo>
<protocolType><!--optional, xs:string, transport protocol type: HTTP, HTTPS, EHome--></protocolType>
```

```
<macAddress><!--optional, xs:string, MAC address--></macAddress>
<dynChannelID><!--optional, xs:string, digital channel No.--></dynChannelID>
<channelID><!--dependent, xs:string, channel No.--></channelID>
<dateTime>
  <!--required, xs:datetime, event occurred time, which is in IOS 8601 format, e.g., 2017-04-22T15:39:01+08:00-->
</dateTime>
<activePostCount><!--required, xs:integer, event occurred times--></activePostCount>
<eventType><!--required, xs:string, event types, here it should be set to "IO"--></eventType>
<eventState><!--required, xs:string, event status (for persistent event): active, inactive--></eventState>
<eventDescription><!--required, xs:string, event description--></eventDescription>
<inputIOPortID><!--optional, xs:integer, local alarm input ID--></inputIOPortID>
<dynInputIOPortID><!--optional, xs:integer, dynamic alarm input ID--></dynInputIOPortID>
<channelName><!--optional, xs:string, channel name--></channelName>
<deviceId><!--optional, xs:string, device ID--></deviceId>
<Extensions>
  <serialNumber><!--required, xs:string, serial No.--></serialNumber>
  <eventPush><!--required, xs:string, event or alarm notification details--></eventPush>
</Extensions>
</EventNotificationAlert>
```

XML_EventNotificationAlert_AudioExceptionAlarmMsg

XML Message about Audio Exception Alarm Details

```
<?xml version="1.0" encoding="utf-8"?>
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ipAddress><!--dependent, xs:string, IPv4 address--></ipAddress>
  <ipv6Address><!--dependent, xs:string, IPv6 address--></ipv6Address>
  <portNo><!--optional, xs:integer, device's communication port No.--></portNo>
  <protocolType><!--optional, xs:string, transport protocol type: HTTP, HTTPS, EHome--></protocolType>
  <macAddress><!--optional, xs:string, MAC address--></macAddress>
  <dynChannelID><!--optional, xs:string, digital channel No.--></dynChannelID>
  <channelID><!--dependent, xs:string, channel No.--></channelID>
  <dateTime>
    <!--required, xs:datetime, event occurred time, which is in IOS 8601 format, e.g., 2017-04-22T15:39:01+08:00-->
  </dateTime>
  <activePostCount><!--required, xs:integer, event occurred times--></activePostCount>
  <eventType><!--required, xs:string, event types, here it should be set to "audioexception"--></eventType>
  <eventState><!--required, xs:string, event status (for persistent event): active, inactive--></eventState>
  <eventDescription><!--required, xs:string, event description--></eventDescription>
  <channelName><!--optional, xs:string, channel name--></channelName>
  <deviceId><!--optional, xs:string, device ID--></deviceId>
  <Extensions>
    <serialNumber><!--required, xs:string, serial No.--></serialNumber>
    <eventPush><!--required, xs:string, event or alarm notification details--></eventPush>
  </Extensions>
</EventNotificationAlert>
```

XML_EventNotificationAlert_DefocusDetectionAlarmMsg

XML Message about Alarm Details of Defocus Detection

```
<?xml version="1.0" encoding="utf-8"?>
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ipAddress><!--dependent, xs:string, IPv4 address--></ipAddress>
  <ipv6Address><!--dependent, xs:string, IPv6 address--></ipv6Address>
  <portNo><!--optional, xs:integer, device's communication port No.--></portNo>
  <protocolType><!--optional, xs:string, transport protocol type: HTTP, HTTPS, EHome--></protocolType>
  <macAddress><!--optional, xs:string, MAC address--></macAddress>
  <dynChannelID><!--optional, xs:string, digital channel No.--></dynChannelID>
  <channelID><!--dependent, xs:string, channel No.--></channelID>
  <dateTime>
    <!--required, xs:datetime, event occurred time, which is in IOS 8601 format, e.g., 2017-04-22T15:39:01+08:00-->
  </dateTime>
  <activePostCount><!--required, xs:integer, event occurred times--></activePostCount>
  <eventType><!--required, xs:string, event types, here it should be set to "defocus"--></eventType>
  <eventState><!--required, xs:string, event status (for persistent event): active, inactive--></eventState>
  <eventDescription><!--required, xs:string, event description--></eventDescription>
  <channelName><!--optional, xs:string, channel name--></channelName>
  <deviceId><!--optional, xs:string, device ID--></deviceId>
  <Extensions>
    <serialNumber><!--required, xs:string, serial No.--></serialNumber>
    <eventPush><!--required, xs:string, event or event notification details--></eventPush>
  </Extensions>
</EventNotificationAlert>
```

XML_EventNotificationAlert_HDDErrorAlarmMsg

XML message about HDD error alarm details

```
<?xml version="1.0" encoding="utf-8"?>
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ipAddress><!--dependent, xs:string, IPv4 address--></ipAddress>
  <ipv6Address><!--dependent, xs:string, IPv6 address--></ipv6Address>
  <portNo><!--optional, xs:integer, device's communication port No.--></portNo>
  <protocolType><!--optional, xs:string, transport protocol type: HTTP, HTTPS, EHome--></protocolType>
  <macAddress><!--optional, xs:string, MAC address--></macAddress>
  <dynChannelID><!--optional, xs:string, digital channel No.--></dynChannelID>
  <channelID><!--dependent, xs:string, channel No.--></channelID>
  <dateTime>
    <!--required, xs:datetime, event occurred time, which is in IOS 8601 format, e.g., 2017-04-22T15:39:01+08:00-->
  </dateTime>
  <activePostCount><!--required, xs:integer, event occurred times--></activePostCount>
  <eventType><!--required, xs:string, event types, here it should be set to "diskerror"--></eventType>
  <eventState><!--required, xs:string, event status (for persistent event): active, inactive--></eventState>
  <eventDescription><!--required, xs:string, event description--></eventDescription>
```

```
<channelName><!--optional, xs:string, channel name--></channelName>
<deviceId><!--optional, xs:string, device ID--></deviceId>
<diskNo><!--optional, xs:integer, HDD No.--></diskNo>
<HDDList><!--dependent, xs:integer, HDD list, this node is valid when <eventType> is "diskfull, diskerror"-->
  <HDD>
    <id><!--required, xs:integer, index No., which starts from 1--></id>
    <diskNumber><!--required, xs:integer, HDD ID--></diskNumber>
  </HDD>
</HDDList>
</EventNotificationAlert>
```

XML_EventNotificationAlert_HeartbeatInfo

EventNotificationAlert message with heartbeat information (when there is no alarm is triggered) in XML format

```
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ipv6Address><!--dep, xs:string, device IPv6 address--></ipv6Address>
  <portNo><!--opt, xs:integer, device port number--></portNo>
  <protocol><!--opt, xs:string, protocol type for uploading alarm/event information, "HTTP,HTTPS"--></protocol>
  <macAddress><!--opt, xs:string, MAC address--></macAddress>
  <channelID><!--dep, xs:string, device channel No., starts from 1--></channelID>
  <dateTime><!--req, heartbeat uploaded time, format: 2017-07-19T10:06:41+08:00--></dateTime>
  <activePostCount><!--req, xs:integer, heartbeat frequency, starts from 1--></activePostCount>
  <eventType><!--req, xs:string, for heartbeat, it is "videoloss"--></eventType>
  <eventState>
    <!--req, xs:string, for heartbeat, it is "inactive"-->
  </eventState>
  <eventDescription><!--req, xs: string, description--></eventDescription>
</EventNotificationAlert>
```

Remarks

- For network camera or network speed dome with the version 5.5.0 and lower, the heartbeat frequency is 300 ms per heartbeat.
- For network camera or network speed dome with the version 5.5.0 and higher, the heartbeat frequency is 10 s per heartbeat. If no heartbeat received for continuous 30 s, it indicates that the heartbeat is timed out.

Example

Message Example of Heartbeat

```
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <ipAddress>10.17.133.46</ipAddress>
  <portNo>80</portNo>
  <protocol>HTTP</protocol>
  <macAddress>44:19:b6:6d:24:85</macAddress>
  <channelID>1</channelID>
  <dateTime>2017-05-04T11:20:02+08:00</dateTime>
  <activePostCount>0</activePostCount>
```

```
<eventType>videoloss</eventType>
<eventState>inactive</eventState>
<eventDescription>videoloss alarm</eventDescription>
</EventNotificationAlert>
```

XML_EventNotificationAlert_IllegalLoginAlarmMsg

XML message about illegal login alarm details

```
<?xml version="1.0" encoding="utf-8"?>
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ipAddress><!--dependent, xs:string, IPv4 address--></ipAddress>
  <ipv6Address><!--dependent, xs:string, IPv6 address--></ipv6Address>
  <portNo><!--optional, xs:integer, device's communication port No.--></portNo>
  <protocolType><!--optional, xs:string, transport protocol type: HTTP, HTTPS, EHome--></protocolType>
  <macAddress><!--optional, xs:string, MAC address--></macAddress>
  <dynChannelID><!--optional, xs:string, digital channel No.--></dynChannelID>
  <channelID><!--dependent, xs:string, channel No.--></channelID>
  <dateTime>
    <!--required, xs:datetime, event occurred time, which is in IOS 8601 format, e.g., 2017-04-22T15:39:01+08:00-->
  >
</dateTime>
  <activePostCount><!--required, xs:integer, event occurred times--></activePostCount>
  <eventType><!--required, xs:string, event types, here it should be set to "illaccess"--></eventType>
  <eventState><!--required, xs:string, event status (for persistent event): active, inactive--></eventState>
  <eventDescription><!--required, xs:string, event description--></eventDescription>
  <channelName><!--optional, xs:string, channel name--></channelName>
  <deviceId><!--optional, xs:string, device ID--></deviceId>
</EventNotificationAlert>
```

XML_EventNotificationAlert_MotionDetectionMsg

XML Message about Details of Motion Detection Event

```
<?xml version="1.0" encoding="utf-8"?>
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ipAddress><!--dependent, xs:string, IPv4 address--></ipAddress>
  <ipv6Address><!--dependent, xs:string, IPv6 address--></ipv6Address>
  <portNo><!--optional, xs:integer, device's communication port No.--></portNo>
  <protocolType><!--optional, xs:string, transport protocol type: HTTP, HTTPS, EHome--></protocolType>
  <macAddress><!--optional, xs:string, MAC address--></macAddress>
  <dynChannelID><!--optional, xs:string, digital channel No.--></dynChannelID>
  <channelID><!--dependent, xs:string, channel No.--></channelID>
  <dateTime>
    <!--required, xs:datetime, event occurred time, which is in IOS 8601 format, e.g., 2017-04-22T15:39:01+08:00-->
  </dateTime>
  <activePostCount><!--required, xs:integer, event occurred times--></activePostCount>
  <eventType><!--required, xs:string, event types, here it should be set to "VMD"--></eventType>
  <eventState><!--required, xs:string, event status (for persistent event): active, inactive--></eventState>
```

```
<eventDescription><!--required, xs:string, event description--></eventDescription>
<DetectionRegionList><!--optional, detection region parameter list (for expert mode only)-->
  <DetectionRegionEntry><!--list, parameters of a detection region (for expert mode only)-->
    <regionID><!--required, xs:string, detection region ID--></regionID>
    <sensitivityLevel><!--required, xs:integer, sensitivity level, value range: [0,100]--></sensitivityLevel>
    <RegionCoordinatesList><!--required, detection region coordinates-->
      <RegionCoordinates>
        <positionX><!--required, xs:integer, x-coordinate--></positionX>
        <positionY><!--required, xs:integer, y-coordinate--></positionY>
      </RegionCoordinates>
    </DetectionRegionEntry>
  </DetectionRegionList>
<channelName><!--optional, xs:string, channel name--></channelName>
<deviceId><!--optional, xs:string, device ID--></deviceId>
<Extensions>
  <serialNumber><!--required, xs:string, serial No.--></serialNumber>
  <eventPush><!--required, xs:string, event notification details--></eventPush>
</Extensions>
</EventNotificationAlert>
```

XML_EventNotificationAlert_PIRAlarmMsg

XML Message about PIR (Passive Infrared) Alarm Details

```
<?xml version="1.0" encoding="utf-8"?>
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ipAddress><!--dependent, xs:string, IPv4 address--></ipAddress>
  <ipv6Address><!--dependent, xs:string, IPv6 address--></ipv6Address>
  <portNo><!--optional, xs:integer, device's communication port No.--></portNo>
  <protocolType><!--optional, xs:string, transport protocol type: HTTP, HTTPS, EHome--></protocolType>
  <macAddress><!--optional, xs:string, MAC address--></macAddress>
  <dynChannelID><!--optional, xs:string, digital channel No.--></dynChannelID>
  <channelID><!--dependent, xs:string, channel No.--></channelID>
  <dateTime>
    <!--required, xs:datetime, event occurred time, which is in IOS 8601 format, e.g., 2017-04-22T15:39:01+08:00-->
  </dateTime>
  <activePostCount><!--required, xs:integer, event occurred times--></activePostCount>
  <eventType><!--required, xs:string, event types, here it should be set to "PIR"--></eventType>
  <eventState><!--required, xs:string, event status (for persistent event): active, inactive--></eventState>
  <eventDescription><!--required, xs:string, event description--></eventDescription>
  <channelName><!--optional, xs:string, channel name--></channelName>
  <deviceId><!--optional, xs:string, device ID--></deviceId>
</EventNotificationAlert>
```

XML_EventNotificationAlert_SceneChangeDetectionAlarmMsg

XML Message about Alarm Details of Scene Change Detection


```
<?xml version="1.0" encoding="utf-8"?>
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ipAddress><!--dependent, xs:string, IPv4 address--></ipAddress>
  <ipv6Address><!--dependent, xs:string, IPv6 address--></ipv6Address>
  <portNo><!--optional, xs:integer, device's communication port No.--></portNo>
  <protocolType><!--optional, xs:string, transport protocol type: HTTP, HTTPS, EHome--></protocolType>
  <macAddress><!--optional, xs:string, MAC address--></macAddress>
  <dynChannelID><!--optional, xs:string, digital channel No.--></dynChannelID>
  <channelID><!--dependent, xs:string, channel No.--></channelID>
  <dateTime>
    <!--required, xs:datetime, event occurred time, which is in IOS 8601 format, e.g., 2017-04-22T15:39:01+08:00-->
  </dateTime>
  <activePostCount><!--required, xs:integer, event occurred times--></activePostCount>
  <eventType><!--required, xs:string, event types, here it should be set to "scenechangedetection"--></eventType>
  <eventState><!--required, xs:string, event status (for persistent event): active, inactive--></eventState>
  <eventDescription><!--required, xs:string, event description--></eventDescription>
  <channelName><!--optional, xs:string, channel name--></channelName>
  <deviceId><!--optional, xs:string, deviceId--></deviceId>
  <Extensions>
    <serialNumber><!--required, xs:string, serial No.--></serialNumber>
    <eventPush><!--required, xs:string, event notification details--></eventPush>
  </Extensions>
</EventNotificationAlert>
```

XML_EventNotificationAlert_SubscriptionHeartbeat

Heartbeat information message returned when subscribing alarm/event, and it is in XML format.

```
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <ipAddress>10.17.133.46</ipAddress>
  <portNo>80</portNo>
  <protocol>HTTP</protocol>
  <macAddress>44:19:b6:6d:24:85</macAddress>
  <channelID>1</channelID>
  <dateTime>2017-05-04T11:20:02+08:00</dateTime>
  <activePostCount>0</activePostCount>
  <eventType>heartBeat</eventType>
  <eventState>active</eventState>
  <eventDescription>heartBeat</eventDescription>
</EventNotificationAlert>
```

See Also

XML_EventNotificationAlert_AlarmEventInfo

XML_EventNotificationAlert_VideoLossAlarmMsg

XML Message about Video Loss Alarm Details

```
<?xml version="1.0" encoding="utf-8"?>
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ipAddress><!--dependent, xs:string, IPv4 address--></ipAddress>
  <ipv6Address><!--dependent, xs:string, IPv6 address--></ipv6Address>
  <portNo><!--optional, xs:integer, device's communication port No.--></portNo>
  <protocolType><!--optional, xs:string, transport protocol type: HTTP, HTTPS, EHome--></protocolType>
  <macAddress><!--optional, xs:string, MAC address--></macAddress>
  <dynChannelID><!--optional, xs:string, digital channel No.--></dynChannelID>
  <channelID><!--dependent, xs:string, channel No.--></channelID>
  <dateTime>
    <!--required, xs:datetime, event occurred time, which is in IOS 8601 format, e.g., 2017-04-22T15:39:01+08:00-->
  </dateTime>
  <activePostCount><!--required, xs:integer, event occurred times--></activePostCount>
  <eventType><!--required, xs:string, event types, here it should be set to "videoloss"--></eventType>
  <eventState><!--required, xs:string, event status (for persistent event): active, inactive--></eventState>
  <eventDescription><!--required, xs:string, event description--></eventDescription>
  <channelName><!--optional, xs:string, channel name--></channelName>
  <deviceId><!--optional, xs:string, device ID--></deviceId>
  <Extensions>
    <serialNumber><!--required, xs:string, serial No.--></serialNumber>
    <eventPush><!--required, xs:string, event notification details--></eventPush>
  </Extensions>
</EventNotificationAlert>
```

XML_EventNotificationAlert_VideoTamperingAlarmMsg

XML Message about Video Tampering Alarm Details

```
<?xml version="1.0" encoding="utf-8"?>
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ipAddress><!--dependent, xs:string, IPv4 address--></ipAddress>
  <ipv6Address><!--dependent, xs:string, IPv6 address--></ipv6Address>
  <portNo><!--optional, xs:integer, device's communication port No.--></portNo>
  <protocolType><!--optional, xs:string, transport protocol type: HTTP, HTTPS, EHome--></protocolType>
  <macAddress><!--optional, xs:string, MAC address--></macAddress>
  <dynChannelID><!--optional, xs:string, digital channel No.--></dynChannelID>
  <channelID><!--dependent, xs:string, channel No.--></channelID>
  <dateTime>
    <!--required, xs:datetime, event occurred time, which is in IOS 8601 format, e.g., 2017-04-22T15:39:01+08:00-->
  </dateTime>
  <activePostCount><!--required, xs:integer, event occurred times--></activePostCount>
  <eventType><!--required, xs:string, event types, here it should be set to "shelteralarm"--></eventType>
  <eventState><!--required, xs:string, event status (for persistent event): active, inactive--></eventState>
  <eventDescription><!--required, xs:string, event description--></eventDescription>
  <channelName><!--optional, xs:string, channel name--></channelName>
  <deviceId><!--optional, xs:string, device ID--></deviceId>
  <Extensions>
    <serialNumber><!--required, xs:string, serial No.--></serialNumber>
    <eventPush><!--required, xs:string, event notification details--></eventPush>
  </Extensions>
</EventNotificationAlert>
```

```
</Extensions>
</EventNotificationAlert>
```

XML_EventTrigger

Linkage parameter message in XML format

```
<EventTrigger version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--required, xs:string, ID--></id>
  <eventType>
    <!--required, xs:string, see details in the "Remarks" below-->
  </eventType>
  <eventDescription><!--optional, xs:string--></eventDescription>
  <inputIOPortID><!--dependent, xs:string, alarm input ID--></inputIOPortID>
  <dynInputIOPortID><!--dependent, xs:string, dynamic alarm input ID--></dynInputPortID>
  <videoInputChannelID>
    <!--dependent, xs:string, video input channel ID, it is valid when <eventType> is "VMD, videoloss, tamperdetection,
regionEntrance, regionExiting, loitering, group, rapidMove, parking, unattendedBaggage, attendedBaggage"-->
  </videoInputChannelID>
  <dynVideoInputChannelID><!--dependent, xs:string, dynamic video input channel ID--></dynVideoInputChannelID>
  <intervalBetweenEvents><!--optional, xs:integer, event time interval, unit: second--></intervalBetweenEvents>
  <WLSensorID><!--dependent, xs:string, ID--></WLSensorID>
  <EventTriggerNotificationList/><!--optional, alarm/event linkage actions, see details in the message of
XML_EventTriggerNotificationList-->
</EventTrigger>
```

Remarks

The node **<eventType>** can be the following values: IO, VMD, videoloss, raidfailure, recordingfailure, badvideo, POS, analytics, fanfailure, overheat, tamperdetection, diskfull, diskerror, nicbroken, ipconflict, illaccess, videomismatch, resolutionmismatch, radifailure, PIR, WLSensor, spareException, poePowerException, heatmap, counting, linedetection, fielddetection, regionEntrance, regionExiting, loitering, group,rapidMove, parking, unattendedBaggage, attendedBaggage, HUMANATTRIBUTE, blacklist, whitelist, peopleDetection, allVehicleList, otherVehicleList, vehicledetection, storageDetection, shipsDetection, humanAttribute, faceContrast, blacklistFaceContrast, whiteListFaceContrast, faceSnap, faceLib, personDensityDetection, personQueueDetecton, mixedTargetDetection, HVTVehicleDetection, illegalParking, pedestrian, trafficAccident, construction, roadblock, abandonedObject, parallelParking, parkingState, congestion, intersectionAnalysis, heatMap, thermometry, shipsFlowDetection, dredgerDetection, reverseEntrance, luma, highHDTemperature, lowHDTemperature, hdImpact, hdBadBlock, SevereHDFailure, safetyHelmetDetection, vibrationDetection, HBDLib,TMPA,faceThermometry,noMaskDetection

See Also

XML_EventTriggerNotificationList

XML_EventTriggerCapType

XML message about capability of alarm linkage action types

```
<EventTriggerCapType version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <isSupportCenter><!--optional, xs:boolean--></isSupportCenter>
  <isSupportRecord><!--optional, xs:boolean--></isSupportRecord>
  <isSupportMonitorAlarm><!--optional, xs:boolean--></isSupportMonitorAlarm>
  <isSupportBeep><!--optional, xs: boolean, whether it supports audible warning--></isSupportBeep>
  <isSupportIO><!--optional, xs:boolean--></isSupportIO>
  <isSupportFTP><!--optional, xs:boolean--></isSupportFTP>
  <isSupportEmail><!--optional, xs:boolean--></isSupEmail>
  <isSupportLightAudioAlarm><!--optional, xs:boolean--></isSupportLightAudioAlarm>
  <isSupportFocus><!--optional, xs:boolean--></isSupportFocus>
  <isSupportPTZ><!--optional, xs:boolean--></isSupportPTZ>
  <maxPresetActionNum>
    <!--dependent, xs:integer, it is valid only when <isSupportPTZ> is "true"-->
  </maxPresetActionNum>
  <maxPatrolActionNum>
    <!--dependent, xs:integer, it is valid only when <isSupportPTZ> is "true"-->
  </maxPatrolActionNum>
  <maxPatternActionNum>
    <!--dependent, xs:integer, it is valid only when <isSupportPTZ> is "true"-->
  </maxPatternActionNum>
  <isSupportTrack><!--optional, xs:boolean, whether it supports PTZ linked tracking--></isSupportTrack>
  <isSupportWhiteLight>
    <!--optional, xs: boolean, whether it supports supplement light alarm linkage-->
  </isSupportWhiteLight>
  <isSupportCloud><!--optional, xs:boolean, whether it supports upload to the cloud--></isSupportCloud>
  <targetNotificationInterval max="1000" min="0" default="30"><!--xs:integer, range: [0, 1000], the default value is 30,
unit: seconds, this node is valid for <MotionDetectionTriggerCap> and <TamperDetectionTriggerCap> and this node is
valid when <isSupportPTZ> is "true"--></targetNotificationInterval>
  <direction opt="both,forward,reverse"><!--xs:string, triggering direction, this node is valid for the node
<BlackListTriggerCap>, <WhiteListTriggerCap>, and <VehicleDetectionTriggerCap>--></direction>
  <presetDurationTime min="" max=""><!--dependent, xs:integer--></presetDurationTime>
  <isSupportSMS><!--optional, xs:boolean, whether to support SMS (Short Message Service)--></isSupportSMS>
  <maxCellphoneNum><!--dependent, xs:integer, the maximum number of cellphones, which is node is valid only
when <isSupportSMS> is "true"--></maxCellphoneNum>
  <isSupportOSD><!--optional, xs:boolean--></isSupportOSD>
  <isSupportAudio><!--optional, xs:boolean, whether it supports setting audio alarm independently. If this node is set
to "true", audio alarm and buzzer alarm can be linked separately, and the lineage method is audio--></isSupportAudio>
  <AudioAction><!--dependent, this node is valid when <isSupportBeep> is "true" or <isSupportAudio> is "true"-->
    <audioTypeList>
      <audioType><!--list-->
        <audioID><!--required, xs:integer, alarm sound type--></audioID>
        <audioDescription><!--required, xs:string, alarm sound description, it should correspond to the alarm sound type--
--></audioDescription>
      </audioType>
    </audioTypeList>
    <alarmTimes opt="0,1,2,3,4,5,6,7,8,9,255"><!--required, xs:integer, alarm times, it is between 0 and 9, 255-
continuous alarm, unit: time--></alarmTimes>
```

```
</AudioAction>
<isSupportSMS><!--optional, xs:boolean --></isSupportSMS>
<maxCellphoneNum><!--dependent, if <isSupportSMS> is true, xs:integer--></maxCellphoneNum>
<isNotSupportCenterModify><!--optional, xs:boolean, whether editing configuration parameters of the surveillance
center is not supported: "true"-yes (configuration parameters of the surveillance center cannot be edited), "false" or
this node is not returned-no (configuration parameters of the surveillance center can be edited)--></
isNotSupportCenterModify>
<isSupportMessageConfig>
  <!--optional, xs:boolean, whether it supports SMS configuration, if supports, set cellphoneNumber to null-->
</isSupportMessageConfig>
<isSupportAnalogOutput><!--optional, xs:boolean, whether it supports IO output of linkage analog channel--></
isSupportAnalogOutput>
  <isSupportIOOutputUnify><!--optional, xs:boolean, whether it supports configuration of IO output--></
isSupportIOOutputUnify>
  <isSupportFaceContrast><!--optional, xs:boolean, whether it supports face picture comparison linkage--></
isSupportFaceContrast>
</EventTriggerCapType>
```

XML_EventTriggerNotification

Event linkage notification message in XML format

```
<EventTriggerNotification><!--opt-->
  <id><!--required, xs:string, device ID--></id>
  <notificationMethod>
    <!--required, xs:string, linkage actions, opt="email,IM,IO,syslog,HTTP,FTP,beep,ptz,record, monitorAlarm, center,
LightAudioAlarm,focus,trace,cloud,SMS,whiteLight,audio,whiteLight,faceContrast"-->
  </notificationMethod>
  <notificationRecurrence>
    <!--optional, xs:string, "beginning,beginningandend,recurring"-->
  </notificationRecurrence>
  <notificationInterval><!--dependent, xs:integer, unit: millisecond--></notificationInterval>
  <outputIOPortID><!--dependent, xs:string, video output No., it is required only when notificationMethod is "IO"--></
outputIOPortID>
  <dynOutputIOPortID><!--dependent, xs:string, dynamic video output No., it is required only when
notificationMethod is "IO"--></dynOutputIOPortID>
  <videoInputID><!--dependent, xs:string, video input No., it is required only when notificationMethod is "record"--></
videoInputID>
  <dynVideoInputID><!--dependent, xs:string, dynamic video input No., it is required only when notificationMethod is
"record"--></dynVideoInputID>
  <ptzAction><!--dependent, it is required only when notificationMethod is "ptz"-->
    <ptzChannelID><!--required, xs:string, PTZ channel ID--></ptzChannelID>
    <actionName><!--required, xs:string, PTZ control type: "preset", "pattern", "patrol"--></actionName>
    <actionNum><!--dependent, xs:integer></actionNum>
  </ptzAction>
  <WhiteLightAction><!--dependent, white light linkage parameters, this node is valid when notificationMethod is
"whiteLight"-->
    <whiteLightDurationTime><!--required, xs:integer, white light flashing duration, it is between 1 and 60, unit:
second--></whiteLightDurationTime>
  </WhiteLightAction>
```

```
<cellphoneNumber><!--dependent, xs:string, min="0" max="11",cellphone number--></cellphoneNumber-->
</EventTriggerNotification>
```

XML_EventTriggerNotificationList

EventTriggerNotificationList message in XML format

```
<EventTriggerNotificationList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <EventTriggerNotification/><!--opt, see details in the message of XML_EventTriggerNotification-->
</EventTriggerNotificationList>
```

See Also

XML_EventTriggerNotification

XML_EventTriggersCap

XML message about linkage capabilities of different alarm categories

```
<EventTriggersCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <DiskfullTriggerCap><!--optional, xs: EventTriggerCapType--></DiskfullTriggerCap>
  <DiskerrorTriggerCap><!--optional, xs: EventTriggerCapType--></DiskerrorTriggerCap>
  <NicbrokenTriggerCap><!--optional, xs: EventTriggerCapType--></NicbrokenTriggerCap>
  <IpconflictTriggerCap><!--optional, xs: EventTriggerCapType--></IpconflictTriggerCap>
  <IllaccesTriggerCap><!--optional, xs: EventTriggerCapType--></IllaccesTriggerCap>
  <BadvideoTriggerCap><!--optional, xs: EventTriggerCapType--></BadvideoTriggerCap>
  <VideomismatchTriggerCap><!--optional, xs: EventTriggerCapType--></VideomismatchTriggerCap>
  <IOTriggerCap><!--optional, xs: EventTriggerCapType--></IOTriggerCap>
  <LineDetectTriggerCap><!--optional, xs: EventTriggerCapType--></LineDetectTriggerCap>
  <RegionEntranceTriggerCap><!--optional, xs: EventTriggerCapType--></RegionEntranceTriggerCap>
  <RegionExitingTriggerCap><!--optional, xs: EventTriggerCapType--></RegionExitingTriggerCap>
  <LoiteringTriggerCap><!--optional, xs: EventTriggerCapType--></LoiteringTriggerCap>
  <GroupDetectionTriggerCap><!--optional, xs: EventTriggerCapType--></GroupDetectionTriggerCap>
  <RapidMoveTriggerCap><!--optional, xs: EventTriggerCapType--></RapidMoveTriggerCap>
  <ParkingTriggerCap><!--optional, xs: EventTriggerCapType--></ParkingTriggerCap>
  <UnattendedBaggageTriggerCap><!--optional, xs: EventTriggerCapType--></UnattendedBaggageTriggerCap>
  <AttendedBaggageTriggerCap><!--optional, xs: EventTriggerCapType--></AttendedBaggageTriggerCap>
  <FireDetectionTriggerCap><!--optional, xs: EventTriggerCapType--></FireDetectionTriggerCap>
  <FireDetectionCap><!--optional, xs: EventTriggerCapType--></FireDetectionCap>
  <StorageDetectionTriggerCap><!--optional, xs: EventTriggerCapType--></StorageDetectionTriggerCap>
  <ShipsDetectionTriggerCap><!--optional, xs: EventTriggerCapType--></ShipsDetectionTriggerCap>
  <ThermometryCap><!--optional, xs: EventTriggerCapType--></ThermometryCap>
  <VandalProofTriggerCap><!--optional, xs: EventTriggerCapType--></VandalProofTriggerCap>
  <BlackListTriggerCap><!--optional, xs: EventTriggerCapType, configuration capability of blacklist arming linkage--></
BlackListTriggerCap>
  <WhiteListTriggerCap><!--optional, xs: EventTriggerCapType, configuration capability of whitelist arming linkage--></
WhiteListTriggerCap>
  <AllVehicleListTriggerCap><!--optional,xs:EventTriggerCapType, configuration capability of other list arming linkage--
></AllVehicleListTriggerCap>
  <OtherVehicleListTriggerCap><!--optional,xs:EventTriggerCapType--></OtherVehicleListTriggerCap>
```

```
<PeopleDetectionTriggerCap><!--optional,xs:EventTriggerCapType--></PeopleDetectionTriggerCap>
<PIRAlarmCap><!--optional, xs: EventTriggerCapType--></PIRAlarmCap>
<TamperDetectionTriggerCap><!--optional, xs: EventTriggerCapType--></TamperDetectionTriggerCap>
<DefocusDetectionTriggerCap><!--optional, xs: EventTriggerCapType--></DefocusDetectionTriggerCap>
<FaceDetectionTriggerCap><!--optional, xs: EventTriggerCapType--></FaceDetectionTriggerCap>
<SceneChangeDetectionTriggerCap><!--optional, xs: EventTriggerCapType--></SceneChangeDetectionTriggerCap>
<VandalProofAlarmCap><!--optional, xs: EventTriggerCapType--></VandalProofAlarmCap>
<JudgmentTriggerCap><!--optional, xs: EventTriggerCapType--></JudgmentTriggerCap>
<FightingTriggerCap><!--optional, xs: EventTriggerCapType--></FightingTriggerCap>
<RisingTriggerCap><!--optional, xs: EventTriggerCapType--></RisingTriggerCap>
<DozingTriggerCap><!--optional, xs: EventTriggerCapType--></DozingTriggerCap>
<CountingTriggerCap><!--optional, xs: EventTriggerCapType--></CountingTriggerCap>
<VideoLossTriggerCap><!--optional, xs: EventTriggerCapType--></VideoLossTriggerCap>
<HideTriggerCap><!--optional, xs:EventTriggerCapType--></HideTriggerCap>
<AlarmInTriggerCap><!--optional, xs: EventTriggerCapType--></AlarmInTriggerCap>
<VehicleDetectionTriggerCap><!--optional, xs: EventTriggerCapType--></VehicleDetectionTriggerCap>
<AudioExceptionCap><!--optional, xs: EventTriggerCapType--></AudioExceptionCap>
<FiledDetectTriggerCap><!--optional, xs: EventTriggerCapType--></FiledDetectTriggerCap>
<MotionDetectionTriggerCap><!--optional, xs: EventTriggerCapType--></MotionDetectionTriggerCap>
<TemperatureCap><!--optional, xs: EventTriggerCapType--></TemperatureCap>
<IntelligentTriggerCap><!--optional, xs: EventTriggerCapType--></IntelligentTriggerCap>
<FaceContrastTriggerCap><!--optional, xs: EventTriggerCapType, face picture comparison alarm linkage--></
FaceContrastTriggerCap>
<PersonDensityDetectionTriggerCap><!--optional, xs: EventTriggerCapType--></PersonDensityDetectionTriggerCap>
<PersonQueueDetectionTriggerCap><!--optional, xs: EventTriggerCapType, queue management alarm linkage--></
PersonQueueDetectionTriggerCap>
<WhiteListFaceContrastTriggerCap><!--optional, xs: EventTriggerCapType--></WhiteListFaceContrastTriggerCap>
<HumanRecognitionTriggerCap><!--optional,xs: EventTriggerCapType--></HumanRecognitionTriggerCap>
<FaceSnapTriggerCap><!--optional, xs: EventTriggerCapType--></FaceSnapTriggerCap>
<isSupportWhiteLightAction>
  <!--dependent, xs: boolean, see details in EventTriggerCapType, it is valid when isSupportWhiteLight is "true"-->
</isSupportWhiteLightAction>
<isSupportAudioAction>
  <!--dependent, xs: boolean, see details in EventTriggerCapType, it is valid when isSupportBeep is "true"-->
</isSupportAudioAction>
<HFPDTriggerCap><!--optional, xs: EventTriggerCapType--></HFPDTriggerCap>
<MixedTargetDetectionCap><!--optional, xs: EventTriggerCapType--></MixedTargetDetectionCap>
<HVTVehicleDetectionTriggerCap><!--optional, xs: EventTriggerCapType--></HVTVehicleDetectionTriggerCap>
<VCATriggerCap><!--optional, xs: EventTriggerCapType--></VCATriggerCap>
<PIRCap><!--optional, xs: EventTriggerCapType--></PIRCap>
<IllegalParkingTriggerCap><!--optional, xs: EventTriggerCapType, whether it supports illegal parking detection--></
IllegalParkingTriggerCap>
<PedestrianTriggerCap><!--optional, xs: EventTriggerCapType, whether it supports pedestrian detection--></
PedestrianTriggerCap>
<TrafficAccidentTriggerCap><!--optional, xs: EventTriggerCapType, whether it supports traffic accident detection--></
TrafficAccidentTriggerCap>
<ConstructionTriggerCap><!--optional, xs: EventTriggerCapType, whether it supports construction detection--></
ConstructionTriggerCap>
<RoadBlockTriggerCap><!--optional, xs: EventTriggerCapType, whether it supports roadblock detection--></
RoadBlockTriggerCap>
<AbandonedObjectTriggerCap><!--optional, xs: EventTriggerCapType, whether it supports objects dropped down
detection--></AbandonedObjectTriggerCap>
```

```
<ParallelParkingTriggerCap><!--optional, xs: EventTriggerCapType, whether it supports parallel parking detection--></ParallelParkingTriggerCap>
<ParkingStateTriggerCap><!--optional, xs: EventTriggerCapType, whether it supports parking space status detection, currently this node is not supported--></ParkingStateTriggerCap>
<CongestionTriggerCap><!--optional, xs: EventTriggerCapType, whether it supports congestion detection--></CongestionTriggerCap>
<IntersectionAnalysisCap><!--optional, xs: EventTriggerCapType, whether it supports intersection analysis--></IntersectionAnalysisCap>
<ShipsFlowDetectionTriggerCap><!--optional,xs:EventTriggerCapType, ship flow detection--></ShipsFlowDetectionTriggerCap>
<dredgerDetectionTriggerCap><!--optional,xs:EventTriggerCapType, dredger detection--></dredgerDetectionTriggerCap>
<voltageInstableTriggerCap><!--optional,xs:EventTriggerCapType, supply voltage exception--></voltageInstableTriggerCap>
<HighHDDTemperatureTriggerCap><!--optional, xs:EventTriggerCapType, HDD high temperature detection--></HighHDDTemperatureTriggerCap>
<LowHDDTemperatureTriggerCap><!--optional, xs:EventTriggerCapType, HDD low temperature detection--></LowHDDTemperatureTriggerCap>
<HDImpactTriggerCap><!--optional, xs:EventTriggerCapType, HDD impact detection--></HDImpactTriggerCap>
<HDBadBlockTriggerCap><!--optional, xs:EventTriggerCapType, HDD bad sector detection--></HDBadBlockTriggerCap>
<SevereHDFailureTriggerCap><!--optional, xs:EventTriggerCapType, HDD severe fault detection--></SevereHDFailureTriggerCap>
<HUMANATTRIBUTECap><!--optional, xs:EventTriggerCapType--></HUMANATTRIBUTECap>
<HumanAttributeTriggerCap><!--optional, xs:EventTriggerCapType, human body attribute--></HumanAttributeTriggerCap>
<BlackListFaceContrastTriggerCap><!--optional, xs:EventTriggerCapType, alarm linkage capability of blacklist face comparison--></BlackListFaceContrastTriggerCap>
<FaceLibTriggerCap><!--optional, xs:EventTriggerCapType--></FaceLibTriggerCap>
<SafetyHelmetDetectionTriggerCap><!--optional, xs:EventTriggerCapType, alarm linkage capability of hard hat detection--></SafetyHelmetDetectionTriggerCap>
<VibrationDetectionTriggerCap><!--optional, xs:EventTriggerCapType, alarm linkage capability of vibration detection--></VibrationDetectionTriggerCap>
<RadarLineDetectionTriggerCap><!--optional, xs:EventTriggerCapType, alarm linkage capability of radar line crossing detection--></RadarLineDetectionTriggerCap>
<RadarFieldDetectionTriggerCap><!--optional, xs:EventTriggerCapType, alarm linkage capability of radar intrusion detection--></RadarFieldDetectionTriggerCap>
<HBDLibTriggerCap><!--optional, xs:EventTriggerCapType, alarm linkage capability of human body picture library--></HBDLibTriggerCap>
<FaceThermometryCap><!--optional, xs:EventTriggerCapType--></FaceThermometryCap>
<NoMaskDetectionTriggerCap><!--optional, xs:EventTriggerCapType, alarm linkage capability of no wearing mask detection--></NoMaskDetectionTriggerCap>
<TMPATriggerCap><!--optional, xs:EventTriggerCapType, alarm linkage capability of temperature measurement pre-alarm--></TMPATriggerCap>
<FireEscapeDetectionTriggerCap><!--optional, xs:EventTriggerCapType, alarm linkage capability of fire engine access detection--></FireEscapeDetectionTriggerCap>
<TakingElevatorDetectionTriggerCap><!--optional, xs:EventTriggerCapType, alarm linkage capability of elevator detection--></TakingElevatorDetectionTriggerCap>
<RuleTriggerCap><!--optional, linkage capability of rule triggered alarm -->
  <isSupportCityManagement>
    <!--optional, xs:boolean, whether the city management supports setting linkage actions by area; if supports, the value is true, otherwise, this node will not be returned-->
```



```
</isSupportCityManagement>
</RuleTriggerCap>
</EventTriggersCap>
```

See Also

XML_EventTriggerCapType

XML_ExportStatus

ExportStatus message in XML format

```
<ExportStatus version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <status><!--opt, xs:string, export status: "export"-exporting, "unexport"-not export--></status>
  <message><!--dep, xs:string, it is valid when status is "export"--></message>
</ExportStatus>
```

XML_Exposure

Exposure message in XML format

```
<Exposure version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ExposureType/><!--req, xs:string, "auto, IrisFirst, ShutterFirst, gainFirst, manual, plris, T5280-PQ1, T5289-PQ1,
T1140-PQ1, T2712-PQ1, HV1250P-MPIR, plris-General"-->
  <autoIrisLevel>
    <!--dep, xs:integer-->
  </autoIrisLevel>
  <OverexposeSuppress><!--opt-->
    <enabled>
      <!--req, xs:boolean-->
    </enabled>
    <Type>
      <!--dep, xs:string, this node depends on <enabled>, "AUTO,MANUAL"-->
    </Type>
    <DistanceLevel>
      <!--dep, xs: integer, this node depends on <Type>-->
    </DistanceLevel>
    <shortIRDistanceLevel>
      <!--dep, xs: integer, this node depends on <Type>-->
    </shortIRDistanceLevel>
    <longIRDistanceLevel>
      <!--dep, xs: integer, this node depends on <Type>-->
    </longIRDistanceLevel>
  </OverexposeSuppress>
  <plris><!--opt-->
    <plrisType/><!--dep, xs:string, this node depends on <ExposureType>, "AUTO,MANUAL"-->
    <IrisLevel>
      <!--dep, xs:integer, this node depends on <plrisType>-->
    </IrisLevel>
  </plris>
```

```
<PlrisGeneral><!--dep, it is available when <ExposureType> is set to "plris-General"-->
  <irisLevel>
    <!--opt, iris level of general lens, which is between 1 and 100, the default level is 50-->
  </irisLevel>
</PlrisGeneral>
<exposureLevel>
  <!--opt, xs:integer , this node depends on <ExposureType>-->
</exposureLevel>
<faceExposure><!--opt, face exposure-->
  <enabled opt="true,false">
    <!--opt, xs:boolean, whether to enable face exposure-->
  </enabled>
  <sensitivity min="0" max="100">
    <!--opt, xs:integer, sensitivity-->
  </sensitivity>
</faceExposure>
</Exposure>
```

XML_externSecurityCap

externSecurityCap message in XML format

```
<externSecurityCap>
  <RestAdminPassWord>
    <isSupportWithSecurityQuestion>
      <!--opt, xs: boolean, whether supports resetting password by answering security questions, "true,false" -->
    </isSupportWithSecurityQuestion>
    <isSupportWithGUIDFileData>
      <!-- opt, xs: boolean, whether supports resetting password by importing GUID file, "true,false" -->
    </isSupportWithGUIDFileData>
    <isSupportWithSecurityEmail>
      <!-- opt, xs: boolean, whether supports resetting password by setting recovery email, "true,false" -->
    </isSupportWithSecurityEmail>
  </RestAdminPassWord>
  <SecurityLimits><!--opt-->
    <LoginPasswordLenLimit min="1" max="16">
      <!--opt, minimum and maximum lengths of login password-->
    </LoginPasswordLenLimit>
    <SecurityAnswerLenLimit min="1" max="128">
      <!--opt, minimum and maximum answer length of security questions-->
    </SecurityAnswerLenLimit>
  </SecurityLimits>
</externSecurityCap>
```

XML_ExtraInfo

ExtraInfo message in XML format

```
<ExtraInfo version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <diskMode><!--req, xs:string, disk mode: "singleDisk"-single disk, "RAID". The device needs to reboot after changing
the disk mode--></diskMode>
</ExtraInfo>
```

XML_EZVIZ

EZVIZ message in XML format.

```
<EZVIZ version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--req, xs:boolean--></enabled>
  <registerStatus><!--ro, opt, xs:boolean--></registerStatus>
  <redirect><!--opt, xs:boolean, whether to allow the device to redirect the server address--></redirect>
  <serverAddress><!--opt-->
    <addressingFormatType>
      <!--req, xs:string, "ipaddress,hostname"-->
    </addressingFormatType>
    <hostName><!--dep, xs:string--></hostName>
    <ipAddress><!--dep, xs:string--></ipAddress>
    <ipv6Address><!--dep, xs:string--></ipv6Address>
  </serverAddress>
  <verificationCode>
    <!--opt, xs:string, verification code. Only the "admin" user can edit the verification code using the control. The
verification code will be displayed for other users. If this node is returned, setting verification code is supported-->
  </verificationCode>
  <offlineStatus>
    <!--ro,dep,xs:string, it is valid when registerStatus values "false", device offline status, opt="secretKeyInvalid"-invalid
verification code-->
  </offlineStatus>
  <operateCode><!--opt, xs:string, operation code for binding device--></operateCode>
  <netWork>
    <!--opt, xs:integer, 0-null ; 1-automatic ; 2-wired network priority ; 3-wired network(by default), 4-3G/4G/GPRS-->
  </netWork>
</EZVIZ>
```

XML_FirmwareCodeList

Message about firmware code list in XML format

```
<?xml version="1.0" encoding="utf-8"?>
<FirmwareCodeList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <FirmwareCode><!--list-->
    <index><!--required, xs:integer, index No., which starts from 1--></index>
    <code><!--required, xs:string, recognition code--></code>
    <version><!--required, xs:string, firmware version--></version>
    <channel><!--optional, xs:integer, IP channel No.--></channel>
    <devTypeName><!--optional, xs:string, device type of the corresponding device of the IP channel--></
devTypeName>
```

```
</FirmwareCode>
</FirmwareCodeList>
```

XML_FocusConfiguration

FocusConfiguration message in XML format

```
<FocusConfiguration version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <focusStyle/><!--req, xs:string, focus type, "AUTO, MANUAL, SEMIAUTOMATIC"-->
  <focusLimited/><!--opt, xs:integer, the minimum focus length, unit: cm-->
  <focusPosition/><!--dep, depends on "FocusStyle", xs:integer-->
  <focusSpeed><!--opt, xs:integer--></focusSpeed>
  <focusSensitivity>
    <!--opt, xs:integer, focus sensitivity, ranges from 0 to 2, it is valid when "focusStyle" is "MANUAL" or
"SEMIAUTOMATIC"-->
  </focusSensitivity>
  <temperatureChangeAdaptEnabled><!--opt, xs:boolean, "true,false"--></temperatureChangeAdaptEnabled>
  <relativeFocusPos>
    <!--opt, xs:integer, relative focus value, it is valid when "focusStyle" is "MANUAL" or "SEMIAUTOMATIC"-->
  </relativeFocusPos>
  <highTemperaturePriority><!--opt,xs:boolean, whether to enable high temperature priority mode--></
highTemperaturePriority>
</FocusConfiguration>
```

XML_FocusData

FocusData message in XML format

```
<FocusData version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <focus>
    <!--req, xs: integer, a vector, whose value is the percentage of maximum focus speed: negative value-focus near;
positive value-focus far-->
  </focus>
</FocusData>
```

XML_formatStatus

formatStatus message in XML format

```
<formatStatus version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <formatting><!--ro, req, xs: boolean--></formatting>
  <percent><!--ro, req, xs: integer, formatting progress percentage, which is between 0 and 100--></percent>
</formatStatus>
```

XML_FTPNotification

FTPNotification message in XML format

```
<FTPNotification version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id>
    <!--req, xs:string, FTP ID-->
  </id>
  <enabled>
    <!--req, xs:boolean, whether to enable the FTP server-->
  </enabled>
  <useSSL>
    <!--opt, xs:boolean-->
  </useSSL>
  <addressingFormatType opt="ipaddress,hostname">
    <!--req, xs:string, this node can only be set to "ipaddress"-->
  </addressingFormatType>
  <hostName>
    <!--dep, xs:string-->
  </hostName>
  <ipAddress>
    <!--dep, xs:string-->
  </ipAddress>
  <ipv6Address>
    <!--dep, xs:string-->
  </ipv6Address>
  <portNo>
    <!--opt, xs:integer, FTP port No.-->
  </portNo>
  <userName>
    <!--req, xs:string, user name-->
  </userName>
  <password>
    <!--wo, xs:string, password-->
  </password>
  <passiveModeEnabled>
    <!--opt, xs:boolean-->
  </passiveModeEnabled>
  <annoyftp>
    <!--opt, xs:boolean-->
  </annoyftp>
  <uploadPicture>
    <!--opt, xs:boolean-->
  </uploadPicture>
  <uploadVideoClip>
    <!--opt, xs:boolean-->
  </uploadVideoClip>
  <uploadPath>
    <!--req-->
  <pathDepth>
    <!--req, xs:integer, directory levels, up to 16 levels can be supported-->
  </pathDepth>
</FTPNotification>
```

```
</pathDepth>
<topDirNameRule>
  <!--dep, xs:string, parent directory name: "none", "devName"-device name, "devId"-device ID, "devIp"-device IP
address, "positionInfo"-camera 1, "time_month"-usage date (YYYY-MM), "time_date"-usage date (YYYY-MM-DD),
"illegalType"-violation type, "direction"-direction, "site"-place, "chanName"-channel name, "chanId"-channel No.,
"laneId"-lane No., "customize"-custom, "time", "buildUnitNo"-building No. and unit No. This node is set to NULL by
default-->
</topDirNameRule>
<topDirName/><!--dep, xs:string, custom string for parent directory, the maximum string length is 32 bytes. This
node is valid when <topDirNameRule> is "customize"-->
<subDirNameRule>
  <!--dep, xs:string, child directory name: "none", "devName"-device name, "devId"-device ID, "devIp"-device IP
address, "positionInfo"-camera 1, "time_month"-usage date (YYYY-MM), "time_date"-usage date (YYYY-MM-DD),
"illegalType"-violation type, "direction"-direction, "site"-place, "chanName"-channel name, "chanId"-channel No.,
"laneId"-lane No., "customize"-custom, "time", "buildUnitNo"-building No. and unit No. This node is set to NULL by
default-->
</subDirNameRule>
<subDirName/><!--dep, xs:string, custom string for child directory, the maximum string length is 32 bytes. This node
is valid when <subDirNameRule> is "customize"-->
<threeDirNameRule>
  <!--dep, xs:string, the third directory name: "none", "devName"-device name, "devId"-device ID, "devIp"-device IP
address, "positionInfo"-camera 1, "time_month"-usage date (YYYY-MM), "time_date"-usage date (YYYY-MM-DD),
"illegalType"-violation type, "direction"-direction, "site"-place, "chanName"-channel name, "chanId"-channel No.,
"laneId"-lane No., "customize"-custom, "time", "buildUnitNo"-building No. and unit No. This node is set to NULL by
default-->
</threeDirNameRule>
<threeDirName/><!--req, xs:string, custom string for the third directory, the maximum string length is 32 bytes. This
node is valid when <threeDirNameRule> is "customize"-->
<fourDirNameRule>
  <!--dep, xs:string, the fourth directory name: "none", "devName"-device name, "devId"-device ID, "devIp"-device
IP address, "positionInfo"-camera 1, "time_month"-usage date (YYYY-MM), "time_date"-usage date (YYYY-MM-DD),
"illegalType"-violation type, "direction"-direction, "site"-place, "chanName"-channel name, "chanId"-channel No.,
"laneId"-lane No., "customize"-custom, "time", "buildUnitNo"-building No. and unit No. This node is set to NULL by
default-->
</fourDirNameRule>
<fourDirName/><!--req, xs:string, custom string for the fourth directory, the maximum string length is 32 bytes. This
node is valid when <fourDirNameRule> is "customize"-->
</uploadPath>
<picArchivingInterval>
  <!--opt, xs:integer, the value is between 1 and 30, 0-close-->
</picArchivingInterval>
<picNameRuleType>
  <!--opt, xs:string, picture name rule type: "default, prefix"-->
</picNameRuleType>
<picNamePrefix>
  <!--dep, xs:string, prefix length of the picture name, which is between 0 and 32-->
</picNamePrefix>
<ftpPicNameRuleType>
  <!--req, xs:string, type of FTP picture name rule: "videoIntercom"-rule used by video intercom products, "ITC"-rule
used by traffic cameras-->
</ftpPicNameRuleType>
<FTPPicNameRule>
```

```
<!--dep, picture name rule of a specific FTP-->
<ItemList/><!--req, see details in the message of XML_ItemList-->
<delimiter>
  <!--req, xs:string, delimiter, which is a single character and the default value is "_"-->
</delimiter>
<customStr>
  <!--req, xs:string, custom string, its length is between 1 and 128-->
</customStr>
</FTPPicNameRule>
<upDataType>
  <!-- opt, xs:integer, picture uploading type: 0-all, 1-checkpoint, 2-violation. When only one FTP server is enabled,
this node can only be set to 0. When two FTP servers are both enabled, you should set 1 for one FTP server and set 2
for another FTP server, which means that two FTP servers cannot be set to the same type-->
</upDataType>
<uploadPlateEnable>
  <!--opt, xs:boolean, whether to enable uploading license plate thumbnail-->
</uploadPlateEnable>
<site>
  <!--req, xs:string, place, the maximum string length is 128 bytes-->
</site>
<roadNum>
  <!--req, xs:string, intersection No., the maximum string length is 32 bytes-->
</roadNum>
<instrumentNum>
  <!--req, xs:string, device No., the maximum string length is 32 bytes-->
</instrumentNum>
<direction>
  <!--req, xs:string, direction No., the maximum string length is 32 bytes-->
</direction>
<directionDesc>
  <!--req, xs:string, direction description, the maximum string length is 32 bytes-->
</directionDesc>
<monitoringInfo1>
  <!--req, xs:string, camera 1 information, the maximum string length is 44 bytes-->
</monitoringInfo1>
<uploadAttachedInfomation>
  <!--req, xs:boolean, whether to upload additional information-->
</uploadAttachedInfomation>
<brokenNetHttp><!--opt, xs:boolean, whether to enable ANR (automatic network replenishment)--></
brokenNetHttp>
</FTPNotification>
```

See Also

XML_ItemList

XML_FTPNotificationList

FTPNotificationList message in XML format

```
<FTPNotificationList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <FTPNotification/><!--opt, see details in the message of XML_FTPNotification-->
</FTPNotificationList>
```

See Also

XML_FTPNotification

XML_FTPTestDescription

FTPTestDescription message in XML format

```
<FTPTestDescription version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <useSSL><!--opt, xs: boolean></useSSL>
  <addressingFormatType>
    <!--req, xs: string, "ipaddress,hostname" -->
  </addressingFormatType>
  <hostName><!--dep, xs: string--></hostName>
  <ipAddress><!--dep, xs: string--></ipAddress>
  <ipv6Address><!--dep, xs: string--></ipv6Address>
  <portNo><!--opt, xs: integer--></portNo>
  <userName><!--req, xs: string--></userName>
  <password><!--wo, xs: string--></password>
  <passiveModeEnabled><!--opt, xs: boolean--></passiveModeEnabled>
  <annoyftp><!--opt, xs: boolean--></annoyftp>
  <uploadPath><!--req-->
    <pathDepth><!--req, xs: integer, the value is from 0 to 2--></pathDepth>
    <topDirNameRule>
      <!--dep, xs: string, "devName, devId, devIp, customize"-->
    </topDirNameRule>
    <topDirName/><!--dep, xs: string-->
    <subDirNameRule>
      <!--dep, xs: string, "chanName, chanId, customize"-->
    </subDirNameRule>
    <subDirName/><!--dep, xs: string-->
  </uploadPath>
</FTPTestDescription>
```

XML_FTPTestResult

FTPTestResult message in XML format

```
<FTPTestResult version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <errorDescription><!--req, xs: string--></errorDescription>
</FTPTestResult>
```


XML_Gain

Gain message in XML format

```
<Gain version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <GainLevel/><!--dep, xs:integer, this node is valid when <ExposureType> in the message XML_Exposure is
  "GainFisrt"-->
  <GainWindow><!--opt-->
    <RegionCoordinatesList><!--opt-->
      <RegionCoordinates><!--opt-->
        <positionX><!--req, xs: integer; x-coordinate--></positionX>
        <positionY><!--req, xs: integer; y-coordinate--></positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </GainWindow>
</Gain>
```

See Also

XML_Exposure

XML_GuardAgainstTheft

GuardAgainstTheft message in XML format

```
<GuardAgainstTheft version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--required, xs:boolean, whether to support enabling device anti-theft--></enabled>
  <SMSEnabled><!--required, xs:boolean, whether to support enabling SMS--></SMSEnabled>
  <phoneNum><!--dependency, xs:string, phone No.--></phoneNum>
  <longitudeLatitudeEnabled><!--dependency, xs:boolean, whether to support enabling longitude and latitude--></
longitudeLatitudeEnabled>
  <siteLocationEnabled><!--optional, xs:boolean, whether to support enabling site location--></siteLocationEnabled>
  <siteLocationInfo><!--dependency, xs:string, site location information--></siteLocationInfo>
  <timeEnabled><!--optional, xs:boolean, whether to support enabling time--></timeEnabled>
  <customInfo><!--optional, xs:string, custom information--></customInfo>
</GuardAgainstTheft>
```

XML_HardwareService

HardwareService message in XML format

```
<HardwareService version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <IrLightSwitch><!--opt-->
    <mode><!--req, xs:string, "open,close"--></mode>
  </IrLightSwitch>
  <ABF><!--opt-->
    <enabled><!--req, xs:boolean--></enabled>
  </ABF>
```

```
<LED><!--opt-->
  <enabled><!--req, xs:boolean--></enabled>
</LED>
<Defog>
  <!--opt -->
  <enabled>
    <!--req, xs:boolean-->
  </enabled>
</Defog>
<SupplementLight>
  <!--opt-->
  <enabled>
    <!--req, xs:boolean-->
  </enabled>
  <isSupportFireLaserLight>
    <!--opt, xs:boolean, "true, false"-->
  </isSupportFireLaserLight>
  <isSupportSupplementLightWord>
    <!--opt, xs:boolean, "true, false", whether to support displaying supplement light prompt-->
  </isSupportSupplementLightWord>
  <captureWithSupplimentLightEnabled>
    <!--opt, xs:boolean, "true,false", enable snapshot supplement light or not-->
  </captureWithSupplimentLightEnabled>
</SupplementLight>
<Deicing><!--opt-->
  <enabled><!--req, xs:boolean--></enabled>
</Deicing>
<ManualDeicing><!--opt-->
  <enabled><!--req, xs:boolean--></enabled>
</ManualDeicing>
<mutexAbility>
  <!--req, "laserLight,deicing", mutual exclusion ability, the laser light and deicing (including manual deicing and
automatic deicing) are mutual exclusive-->
</mutexAbility>
<VisibleMovementPower><!--opt-->
  <enabled><!--req, xs:boolean--></enabled>
</VisibleMovementPower>
<ThermalMovementPower><!--opt-->
  <enabled><!--req, xs:boolean--></enabled>
</ThermalMovementPower>
<PtzPower><!--opt-->
  <enabled><!--req, xs:boolean--></enabled>
</PtzPower>
<powerSavingControl><!--opt, xs:string, power saving strategy: "sleepMode"-sleeping mode, lowConsumptionMode-
低功耗模式 --></powerSavingControl>
<HighTemperatureProtection><!--opt-->
  <enabled><!--req, xs:boolean --></enabled>
  <temperatureType><!--dep, xs:string,unit:°C, "90,100,110" --></temperatureType>
</HighTemperatureProtection>
</HardwareService>
```

XML_hdd

XML Message about HDD Parameters

```
<hdd version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--ro, req, xs: string; ID--></id>
  <hddName><!--ro, req, xs: string--></hddName>
  <hddPath><!--ro, opt, xs: string--></hddPath>
  <hddType><!--ro, req, xs: string, "IDE, SATA, eSATA, NFS, iSCSI, Virtual Disk"--></hddType>
  <status>
    <!--ro, req, xs: string,
    "ok,unformatted,error,idle,mismatch,offline,smartFailed,repairing,formatting,notexist,unRecordHostFormatted",unReco
rdHostFormatted--unformatted in education sharing system-->
  </status>
  <capacity><!--ro, req, xs: float, unit: MB--></capacity>
  <freeSpace><!--ro, req, xs: float, unit: MB--></freeSpace>
  <property><!--req, xs: string, HDD properties, "RW,RO,Redund"--></property>
  <group><!--opt, xs: string; HDD group ID--></group>
  <reservedDayCfg><!--optional, xs:integer, video saving duration, it is between 0 and 1000, unit: day--></
reservedTimeCfg>
  <DataModeList><!--opt, ro, current HDD allocation mode-->
    <DataMode>
      <type><!--req, xs: string, storage application type: recordStorage-video storage, pictureCloudStorage-picture to be
saved in cloud storage, fileStorage-file-storage--></type>
      <occupancyRate><!--req, xs: integer, HDD usage, range: [0,100]--></occupancyRate>
    </DataMode>
  </DataModeList>
  <formatType>
    <!--ro, opt, xs: string, formatting type: FAT32 (default) and EXT4, this node is only available for SD card; if this node
does not exist, the default formatting type is FAT32-->
  </formatType>
  <encryptionStatus><!--ro, opt, xs:string, encryption status: "unencrypted", "encrypted", "verifyFailed"-verification
failed--></encryptionStatus>
</hdd>
```

XML_hddList

hddList message in XML format

```
<hddList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0" >
  <hdd xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0" >
    <id>
      <!--read-only, required, xs:string, HDD ID-->
    </id>
    <hddName>
      <!--read-only, required, xs:string-->
    </hddName>
    <hddPath>
      <!--read-only, optional, xs:string-->
```

```
</hddPath>
<hddType>
  <!--read-only, required, xs:string,"IDE,SATA,eSATA,NFS,iSCSI,Virtual Disk,NTFS",etc-->
</hddType>
<status>
  <!--read-only, required,
xs:string,"ok,unformatted,error,idle,mismatch,offline,smartFailed,reparing,formatting,notexist,unRecordHostFormatted
,maintaining,maintainingPercent]-->
</status>
<capacity>
  <!--read-only, required, xs:float,in MB-->
</capacity>
<freeSpace>
  <!--read-only, required, xs:float,in MB-->
</freeSpace>
<property>
  <!--required, xs:string "RW,read-only, Redund"-->
</property>
<group>
  <!--optional, xs:string; id-->
</group>
<DataModeList>
  <!--optional,ro-->
  <DataMode>
    <!--list-->
    <type>
      <!--required, xs:string,opt="recordStorage,pictureCloudStorage,fileStorage"-->
    </type>
    <occupancyRate>
      <!--required, xs:integer [0,100]-->
    </occupancyRate>
  </DataMode>
</DataModeList>
<formatType>
  <!--read-only, optional, xs:string; "FAT32,EXT4"-->
</formatType>
<ipAddress>
  <!--optional, xs:string-->
</ipAddress>
<ipv6Address>
  <!--optional, xs:string -->
</ipv6Address>
<formatType>
  <!--read-only, optional, xs:string; "FAT32,EXT4"-->
</formatType>
<encryptionStatus>
  <!--read-only, optional, xs:string, encryption status: "unencrypted"-unencrypted, "encrypted"-encrypted,
"verifyFailed"-verification failed-->
</encryptionStatus>
<logStorageEnable>
  <!--read-only, optional, xs:boolean, whether it is a log HDD-->
</logStorageEnable>
```

```
</hdd>
</hddList>
```

See Also

XML_hdd

XML_HDDSMARTTest

HDDSMARTTest message in XML format

```
<HDDSMARTTest version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <testType><!--opt, xs: string, checking type: short,expanded,conveyance--></testType>
</HDDSMARTTest>
```

XML_Hello

Hello message in XML format

```
<?xml version="1.0" encoding="UTF-8"?>
<Hello>
  <Types>hello</Types>
  <!--device type value-->
  <DeviceType>1234</DeviceType>
  <!--device type description-->
  <DeviceDescription>DS-2CD4024F</DeviceDescription>
  <!--device serial No.-->
  <DeviceSN>DS-2CD4024F20130925CCCH435107838</DeviceSN>
  <!--command port number of private protocol-->
  <CommandPort>8000</CommandPort>
  <!--http port number-->
  <HttpPort>80</HttpPort>
  <!--MAC address-->
  <MAC>44-19-b7-10-98-32</MAC>
  <!--IPv4 address-->
  <IPv4Address>172.9.3.106</IPv4Address>
  <!--IPv4 subnet mask-->
  <IPv4SubnetMask>255.255.255.0</IPv4SubnetMask>
  <!--IPv4 gateway-->
  <IPv4Gateway>172.9.3.1</IPv4Gateway>
  <!--IPv6 address-->
  <IPv6Address>2000:1:2:3:4619:b6ff:fe01:dd9d</IPv6Address>
  <!--IPv6 gateway-->
  <IPv6Gateway>2000:1:2:3:4619:b6ff:fe01:1</IPv6Gateway>
  <!--size of IPv6 subnet mask-->
  <IPv6MaskLen>64</IPv6MaskLen>
  <!--enable/disable DHCP, true: enable, false: disable-->
  <DHCP>false</DHCP>
  <!--total number of analog channels-->
  <AnalogChannelNum>32</AnalogChannelNum>
```

```
<!--total number of digital channels-->
<DigitalChannelNum>32</DigitalChannelNum>
<!--device software version information, not encrypt-->
<SoftwareVersion>V4.1.0 build130126</SoftwareVersion>
<!--DSP version information, not encrypt-->
<DSPVersion>V4.0 build130111</DSPVersion>
<!--device started time-->
<BootTime>2014-01-06 11:39:00</BootTime>
<!--true: OEM device, false: baseline device-->
<OEMCode>true</OEMCode>
<!--OEM manufacturer information, optional-->
<OEMInfo>Axis</OEMInfo>
<!--device software version information, it is encrypted in AES128/AES256 CBC mode with fixed key-->
<SoftwareVersionEncrypt>V4.1.0 build130126</SoftwareVersionEncrypt>
<!--DSP version, it is encrypted in AES128/AES256 CBC mode with fixed key-->
<DSPVersionEncrypt>V4.0 build130111</DSPVersionEncrypt>
<!--OEM manufacturer information, it is encrypted in AES128/AES256 CBC mode with fixed key, optional-->
<OEMInfoEncrypt>Axis</OEMInfoEncrypt>
<!--true: encrypted device, false: normal device>
<Encrypt>true</Encrypt>
<!--security code generated by specific algorithm (reserved), which is to prevent piracy-->
<SafeCode>123456</SafeCode>
<!--support upgrading reset password or not, true: yes, false: no>
<ResetAbility>true</ResetAbility>
<!--number of HDDs>
<DiskNumber>1</DiskNumber>
<!--true: activated, false: inactivated>
<Activated>true</Activated>
<!--support resetting password or not, true: yes, false: no>
<PasswordResetAbility>true</PasswordResetAbility>
<!--support synchronizing password of network camera or not, true: yes, false: no>
<SyncIPCPassWord>true</SyncIPCPassWord>
<!--support password reset mode 2 or not, true: yes, false: no>
<PasswordResetModeSecond>true</PasswordResetModeSecond>
<!--OEMCode details-->
<DetailOEMCode>10101</DetailOEMCode>
<!--true: EZVIZ device, false: baseline device-->
<EZVIZCode>true</EZVIZCode>
<!--support locking device or not, true: yes, false: no-->
<DeviceLock>true</DeviceLock>
</Hello>
```

XML_HttpHostNotification

XML message about parameters of a HTTP listening server

```
<HttpHostNotification version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--required, xs:string, ID--></id>
  <url><!--required, xs:string, the absolute path, e.g., http://<ipAddress>:<portNo>/<uri>--></url>
  <protocolType><!--required, xs:string, "HTTP,HTTPS,EHome"--></protocolType>
  <parameterFormatType><!--required, xs:string, alarm/event information format, "XML,JSON"--></parameterFormatType>
```

```
parameterFormatType>
<addressingFormatType><!--required, xs:string, "ipaddress,hostname"--></addressingFormatType>
<hostName><!--dependent, xs:string--></hostName>
<ipAddress><!--dependent, xs:string--></ipAddress>
<ipv6Address><!--dependent, xs:string--></ipv6Address>
<portNo><!--optional, xs:integer--></portNo>
<userName><!--dependent, xs:string--></userName>
<password><!--dependent, xs:string--></password>
<httpAuthenticationMethod><!--required, xs:string, "MD5digest,none"--></httpAuthenticationMethod>
<ANPR><!--optional-->
  <detectionUploadPicturesType>
    <!--optional, xs:string, types of alarm picture to be uploaded: "all, licensePlatePicture, detectionPicture"-->
  </detectionUploadPicturesType>
</ANPR>
<eventType optional="AID,TFS,TPS"><!--required, xs:string--></eventType>
<uploadImagesDataType>
  <!--optional, xs:string, "URL", "binary" (default), for cloud storage, only "URL" is supported-->
</uploadImagesDataType>
<eventMode><!--optional, xs:string, "all,list"--></eventMode>
<EventList><!--dependent, it is valid only when eventMode is "list"-->
  <Event><!--required-->
    <type><!--required, xs:string--></type>
  </Event>
</EventList>
<channels><!--optional, xs:string, "1,2,3,4..."--></channels>
<SubscribeEvent/><!--optional, event subscription parameters, see details in the message of XML_SubscribeEvent-->
</HttpHostNotification>
```

XML_HttpHostNotificationCap

XML message about capability of HTTP listening server

```
<?xml version="1.0" encoding="utf-8"?>
<HttpHostNotificationCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <hostNumber>2</hostNumber>
  <urlLen max=""/>
  <protocolType opt="HTTP,HTTPS"/>
  <parameterFormatType opt="XML,querystring,JSON"/>
  <addressingFormatType opt="ipaddress,hostname"/>
  <ipAddress opt="ipv4,ipv6"/>
  <portNo min="" max=""/>
  <userNameLen min="" max=""/>
  <passwordLen min="" max=""/>
  <httpAuthenticationMethod opt="MD5digest,none"/>
  <Extensions>
    <intervalBetweenEvents min="" max=""/>
  </Extensions>
  <uploadImagesDataType opt="URL,binary"/>
  <ANPR><!--optional-->
    <detectionUploadPicturesType opt="all,licensePlatePicture,detectionPicture..."/><!--optional, xs:string, types of
alarm pictures to be uploaded-->
```

```

<alarmHttpPushProtocol opt="baseline,custom"/>
</ANPR>
<httpBroken opt="true,false" def="true" ><!--optional, xs:boolean, whether to enable global ANR: true, false--></
httpBroken>
<SubscribeEventCap>
  <heartbeat min="" max=""/><!--optional, heartbeat time interval, unit: second-->
  <channelMode opt="all,list"/><!--required, all-subscribe events of all channels, list-subscribe event by channel-->
  <eventMode opt="all,list"/><!--required, event subscription mode: all-subscribe all events of all channels, list-
subscribe events by type, channel, and target-->
  <!--if the values of the two nodes channelMode and eventMode are both "all", it indicates that the device does not
support subscribe events by type and channel-->
  <EventList><!--dependent, alarm uploading mode, this node is valid only when eventMode is "list"-->
    <Event><!--required-->
      <type><!--required, xs:string, event types--></type>
      <pictureURLType opt="binary,localURL,cloudStorageURL" def=""/>
      <!--optional, xs:string, transmission format of alarm picture: "binary"-picture binary data, "localURL"-picture URL
from local device, "cloudStorageURL"-picture URL from cloud storage-->
    </Event>
  </EventList>
  <pictureURLType opt="binary,localURL,cloudStorageURL" def=""/>
  <!--optional, xs:string, transmission format of all alarm pictures: "binary"-picture binary data (default for camera),
"localURL"-picture URL from local device (default for NVR/DVR), "cloudStorageURL"-picture URL from cloud storage;
this node is in highest priority-->
  <ChangedUploadSub>
    <interval/><!--optional, xs:integer, the life cycle of arming GUID, unit: second, the default life cycle is 5 minutes; if
the reconnection is not started during the life cycle, a new GUID will be generated-->
    <StatusSub>
      <all/><!--optional, xs:boolean, whether to subscribe all-->
      <channel/><!--optional, xs:boolean, subscribe channel status, this node is not required when the node all is "true"--
>
      <hd/><!--optional, xs:boolean, subscribe the HDD status, this node is not required when the node all is "true"-->
      <capability/><!--optional, xs:boolean, subscribe the capability changed status, this node is not required when the
node all is "true"-->
    </StatusSub>
  </ChangedUploadSub>
</SubscribeEventCap>
</HttpHostNotificationCap>

```

XML_HttpHostNotificationList

HttpHostNotificationList message in XML format

```

<HttpHostNotificationList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <HttpHostNotification>
    <id><!--req, xs:string, ID--></id>
    <url><!--req, xs:string--></url>
    <protocolType><!--req, xs:string, "HTTP,HTTPS"--></protocolType>
    <parameterFormatType><!--req, xs:string, alarm/event information format, "XML,JSON"--></parameterFormatType>
    <addressingFormatType><!--req, xs:string, "ipaddress,hostname"--></addressingFormatType>
    <hostName><!--dep, xs:string--></hostName>
    <ipAddress><!--dep, xs:string--></ipAddress>
  </HttpHostNotification>
</HttpHostNotificationList>

```



```
<ipv6Address><!--dep, xs:string--></ipv6Address>
<portNo><!--opt, xs:integer--></portNo>
<userName><!--dep, xs:string--></userName>
<password><!--dep, xs:string--></password>
<httpAuthenticationMethod><!--req, xs:string, "MD5digest,none"--></httpAuthenticationMethod>
<uploadImagesDataType>
  <!--opt, xs:string, "URL", "binary" (default), for cloud storage, only "URL" is supported-->
</uploadImagesDataType>
<eventMode><!--opt, xs:string, "all,list"--></eventMode>
<EventList><!--dep, it is valid only when eventMode is "list"-->
  <Event><!--req-->
    <type><!--req, xs:string--></type>
  </Event>
</EventList>
<channels><!--opt, xs:string, "1,2,3,4..."--></channels>
</HttpHostNotification>
</HttpHostNotificationList>
```

Example

HttpHostNotificationList Message Example

```
<HttpHostNotificationList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <HttpHostNotification>
    <id>1</id>
    <url></url>
    <protocolType>HTTP</protocolType>
    <parameterFormatType>XML</parameterFormatType>
    <addressingFormatType>ipaddress</addressingFormatType>
    <ipAddress>0.0.0.0</ipAddress>
    <portNo>80</portNo>
    <userName></userName>
    <httpAuthenticationMethod>none</httpAuthenticationMethod>
  </HttpHostNotification>
</HttpHostNotificationList>
```

XML_HttpHostTestResult

HttpHostTestResult message in XML format.

```
<HttpHostTestResult version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <errorDescription>
    <!--req, xs:string-->
  </errorDescription>
</HttpHostTestResult>
```

XML_IbeaconParam

IbeaconParam message in XML format

```
<lbeaconParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <UUID min="" max="">
    <!--req, xs:string, parking lot ID, which supports using digits, letters and hyphen as the tag-->
  </UUID>
  <major min="" max="">
    <!--req, xs:integer, number of floors in the parking lot, which cannot be 0x00 for all-->
  </major>
  <minor min="" max="">
    <!--req, xs:integer, number of parking spaces in each floor, which cannot be 0x00 for all-->
  </minor>
  <sendPower min="" max="">
    <!--req, xs:integer, transmitted power: 01-0 dbm, 02- -6 dbm, 03- -23 dbm-->
  </sendPower>
  <frequency min="" max="">
    <!--req, xs:integer, transmitted frequency, which is the broadcast time interval, unit: 625 μs. The broadcast time interval is between 32 (20 ms) to 8000 (5s), and the default value is 160 (100 ms)-->
  </frequency>
  <measurePower min="" max="">
    <!--req, xs:integer, rated power, which is the RSSI (Received Signal Strength Indication) in the position of one meter away, unit: dbm-->
  </measurePower>
</lbeaconParam>
```

XML_IEEE802_1x

IEEE802_1x message in XML format

```
<IEEE802_1x version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--req, xs: boolean--></enabled>
  <authenticationProtocolType>
    <!--req, xs: string, "EAP-TLS,EAP-TTLS,EAP-PEAP,EAP-LEAP,EAP-FAST,EAP-MD5"-->
  </authenticationProtocolType>
  <innerTTLSAuthenticationMethod>
    <!--dep, xs: string, "MS-CHAP,MS-CHAPv2,PAP,EAP-MD5", this node is required when <authenticationProtocolType> is "EAP-TLS"-->
  </innerTTLSAuthenticationMethod>
  <innerEAPProtocolType>
    <!--dep, xs: string, "EAP-POTP,MS-CHAPv2", this node is required when <authenticationProtocolType> is "EAP-PEAP" or "EAP-FAST"-->
  </innerEAPProtocolType>
  <validateServerEnabled><!--dep, xs: boolean--></validateServerEnabled>
  <userName><!--dep, xs: string--></userName>
  <password><!--dep, xs: string--></password>
  <anonymousID><!--opt, xs: string--></anonymousID>
  <autoPACProvisioningEnabled>
    <!--dep, xs: boolean, this node is required when <authenticationProtocolType> is "EAP-FAST"-->
  </autoPACProvisioningEnabled>
  <Extensions><!--opt-->
    <EAPOLVersion><!--opt, xs:string, "1, 2"--></EAPOLVersion>
```

```
</Extensions>
</IEEE802_1x>
```

Remarks

- For EAP-LEAP or EAP-MD5, the parameters **userName**, **password**, and **Extensions** can be configured.
- For EAP-TLS, the parameters **userName**, **password**, and **Extensions** can be configured, and the certificates with different types (i.e., CA, user certificate, key) can be uploaded.

XML_IllegalLoginLock

IllegalLoginLock message in XML format

```
<IllegalLoginLock version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled>true</enabled>
  <maxIllegalLoginTimes>
    <!--opt, xs: integer, maximum illegal login attempts, whose value is between 3 and 20, and the default value is 5-->
  </maxIllegalLoginTimes>
</IllegalLoginLock>
```

XML_ImageCap

ImageCap message in XML format

```
<ImageCap version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <isSupportRegionalExposure><!--opt, xs:boolean--></isSupportRegionalExposure>
  <isSupportRegionalFocus><!--opt, xs:boolean--></isSupportRegionalFocus>
</ImageCap>
```

XML_ImageChannel

ImageChannel message in XML format

```
<ImageChannel version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs: integer--></id>
  <enabled><!--req, xs: boolean--></enabled>
  <videoInputID><!--req, xs: integer--></videoInputID>
  <Defog><!--opt, electronic defogging parameters, see details in the message of XML_Defog-->
  <NoiseReduce2D><!--opt, 2D noise reduction parameters, see details in the message of XML_NoiseReduce2D-->
  <Focusconfiguration><!--opt-->
  <LensInitialization><!--opt-->
  <ImageFlip><!--opt, image automatic flipping parameters, see details in the message of XML_ImageFlip-->
  <ImageFreeze><!--opt-->
  <proportionalpan><!--opt-->
  <WDR><!--opt, WDR parameters, refer to the message XML_WDR for details-->
  <BLC><!--opt, BLC parameters, refer to the message XML_BLC for details-->
```

```
<NoiseReduce/><!--opt, 3D DNR parameters, see details in the message of XML_NoiseReduce-->
<ImageEnhancement/><!--opt, image enhancement parameters, see details in the message of
XML_ImageEnhancement-->
<DSS/><!--opt, low illumination electronic shutter parameters in exposure, see details in the message of XML_DSS-->
<WhiteBalance/><!--opt, WB parameters, see details in the message of XML_WhiteBalance-->
<Exposure/><!--opt, exposure parameters, see details in the message of XML_Exposure-->
<Sharpness/><!--opt, sharpness parameters, see details in the message of XML_Sharpness-->
<gammaCorrection/><!--opt, gamma correction parameters, see details in the message of XML_gammaCorrection-->
<powerLineFrequency/><!--opt, image standard parameters, refer to the message XML_powerLineFrequency for
details-->
<Color/><!--opt, image color parameters, see details in the message of XML_Color-->
<IrcutFilter/><!--opt, day/night auto switch parameters, see details in the message of XML_IrcutFilter-->
<ImageModeList/><!--opt, default image mode parameters, see details in the message of XML_ImageModeList-->
<BrightEnhance/><!--opt, brightness enhancement parameters, see details in the message of XML_BrightEnhance-->
<ISPMODE/><!--opt, day/night mode parameters, see details in the message of XML_ISPMODE-->
<Shutter/><!--opt, shutter parameters in exposure, see details in the message of XML_Shutter-->
<Gain/><!--opt, gain parameters, see details in the message of XML_Gain-->
<ImageIcrE/><!--opt, IR-cut filter parameters, see details in the message of XML_ImageIcrE-->
<ImageMultishut/><!--opt, multi-shutter parameters, see details in the message of XML_ImageMultishut-->
<PlateBright/><!--opt, license plate brightness compensation parameters, see details in the message of
XML_PlateBright-->
<JPEGParam/><!--opt, JPEG picture size parameters, see details in the message of XML_JPEGParam-->
<DarkEnhance/><!--opt, dark space enhancement parameters, see details in the message of XML_DarkEnhance-->
<Hdr/><!--opt, WDR (Wide Dynamic Range) parameters, see details in the message of XML_Hdr-->
<LSE/><!--opt, contrast enhancement parameters, see details in the message of XML_LSE-->
<MCE/><!--opt, memory color enhancement parameters, see details in the message of XML_MCE-->
<Svce/><!--opt, part contrast parameters, see details in the message of XML_Svce-->
<SectionCtrl/><!--opt, configuration parameters of picture exposure control by video segment, see details in the
message of XML_SectionCtrl-->
<AutoContrast/><!--opt, automatic contrast parameters, see details in the message of XML_AutoContrast-->
<GrayRange/><!--opt, grayscale range parameters, see details in the message of XML_GrayRange-->
<LSEDetail/><!--opt, contrast enhancement parameters, see details in the message of XML_LSE-->
<ITCImageSnap/><!--opt, captured picture parameters, see details in the message of XML_ITCImageSnap-->
<ImageRecord/><!--opt, image parameters in the recorded video, see details in the message of XML_ImageRecord-->
<Scene/><!--opt-->
<EPTZ/><!--opt-->
<EIS/><!--opt-->
<HLC/><!--opt-->
<ZoomLimit/><!--opt-->
<corridor/><!--opt, image rotate mode parameters, refer to the message XML_corridor for details-->
<Dehaze/><!--opt, defog mode parameters, refer to the message XML_DeHaze for details-->
<ImageMode/><!--opt, xs: string, image mode: "standard, indoor, outdoor, dimLight"-->
<enableImageLossDetection><!--opt, xs: boolean--></enableImageLossDetection>
<CaptureMode/><!--opt, video input mode parameters, refer to the message XML_CaptureMode for details-->
<IrLight/><!--opt-->
<LensDistortionCorrection/><!--opt-->
<ExposureSync/><!--opt, exposure synchronization settings-->
<BrightnessSuddenChangeSuppression/><!--opt-->
<TempRange/><!--opt, temperature range, refer to the message XML_tempRange for details-->
<SupplementLight/><!-- opt -->
</ImageChannel>
```

XML_ImageChannellist

ImageChannellist message in XML format

```
<ImageChannellist version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ImageChannel/><!--opt-->
</ImageChannellist>
```

See Also

XML_ImageChannel

XML_ImageFlip

ImageFlip message in XML format

```
<ImageFlip version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled/><!--req, xs:boolean-->
  <ImageFlipStyle/><!--opt, xs:string, "LEFTRIGHT, UPDOWN, CENTER, AUTO", it can be enabled only when the value is
"true"-->
  <flipAngle><!--opt, xs:string, "90, 180, 270"--></flipAngle>
</ImageFlip>
```

XML_ImageMode

ImageMode message in XML format

```
<ImageMode version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <type><!--req, xs:string, "standard, indoor, outdoor, dimLight"--></type>
  <recommendation><!--req, ro-->
    <brightnessLevel><!--opt, xs:integer, ranging from 0 to 100--></brightnessLevel>
    <contrastLevel><!--opt, xs:integer, ranging from 0 to 100--></contrastLevel>
    <sharpnessLevel><!--opt, xs:integer, ranging from 0 to 100--></sharpnessLevel>
    <saturationLevel><!--opt, xs:integer, ranging from 0 to 100--></saturationLevel>
    <hueLevel><!--opt, xs:integer, ranging from 0 to 100--></hueLevel>
    <deNoiseLevel><!--opt, xs:integer, ranging from 0 to 100--></deNoiseLevel>
  </recommendation>
</ImageMode>
```

XML_ImageModeList

ImageModeList message in XML format

```
<ImageModeList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ImageMode/><!--opt, see details in the message of XML_ImageMode-->
</ImageModeList>
```

See Also

XML_ImageMode

XML_InputProxyChannel

InputProxyChannel message in XML format

```
<InputProxyChannel version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:string, starts from 1--></id>
  <name><!--opt, xs:string--></name>
  <sourceInputPortDescriptor><!--req-->
    <adminProtocol><!--req, xs:string, "HIKVISION,SONY,ISAPI,ONVIF,..."--></adminProtocol>
    <addressingFormatType><!--req, xs:string, "ipaddress,hostname"--></addressingFormatType>
    <hostName><!--dep, xs:string, domain name--></hostName>
    <ipAddress><!--dep, xs:string, IP address--></ipAddress>
    <ipv6Address><!--dep, xs:string, IPv6 address--></ipv6Address>
    <managePortNo><!--req, xs:integer--></managePortNo>
    <srcInputPort><!--req, xs:string, channel No.--></srcInputPort>
    <userName><!--req, xs:string, user name, which should be encrypted--></userName>
    <password><!--req, wo, xs:string, password, which should be encrypted--></password>
    <streamType><!--opt, xs:string, opt="auto,tcp,udp"--></streamType>
    <deviceId><!--dep, xs:string--></deviceId>
    <deviceTypeName><!--ro, opt, xs:string, device type name--></deviceTypeName>
    <serialNumber><!--ro, opt, xs:string, device serial No.--></serialNumber>
    <firmwareVersion><!--ro, opt, xs:string, firmware version--></firmwareVersion>
    <firmwareCode><!--ro, opt, xs:string, firmware code--></firmwareCode>
  </sourceInputPortDescriptor>
  <enableAnr>
    <!--opt, xs:boolean, whether enables ANR funtion-->
  </enableAnr>
  <NVRInfo>
    <ipAddressNVR>
      <!--opt, xs:string, IP address of NVR-->
    </ipAddressNVR>
    <portNVR>
      <!--opt, xs:integer, port No. of NVR-->
    </portNVR>
    <ipcChannelNo>
      <!--opt, xs:integer, channel No. of the network camera in NVR-->
    </ipcChannelNo>
  </NVRInfo>
</InputProxyChannel>
```

XML_InputProxyChannelList

InputProxyChannelList message in XML format

```
<InputProxyChannelList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <InputProxyChannel/><!--opt, see details in
      XML_InputProxyChannel
  -->
</InputProxyChannelList>
```

XML_Cap_InputProxyChannelListCap

InputProxyChannelListCap capability message in XML format

```
<InputProxyChannelListCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <InputProxyChannel>
    <id min="" max=""/><!--req, xs:string, starts from 1-->
    <name min="" max=""/><!--opt, xs:string-->
    <sourceInputPortDescriptor><!--req-->
      <adminProtocol opt="HIKVISION,SONY,ISAPI,ONVIF,..."/><!--req, xs:string-->
      <addressingFormatType opt="ipaddress,hostname"/><!--req, xs:string-->
      <hostName min="" max=""/><!--dep, xs:string, domain name-->
      <ipAddress min="" max=""/><!--dep, xs:string, IP address-->
      <ipv6Address min="" max=""/><!--dep, xs:string, IPv6 address-->
      <managePortNo min="" max=""/><!--req, xs:integer, port number-->
      <srcInputPort min="" max=""/><!--req, xs:string, channel No.-->
      <userName min="" max=""/><!--req, xs:string-->
      <password min="" max=""/><!--req, wo, xs:string-->
      <streamType opt="auto,tcp,udp"/><!--opt, xs:string-->
      <deviceId min="" max=""/><!--dep, xs:string-->
    </sourceInputPortDescriptor>
  </InputProxyChannel>
</InputProxyChannelListCap>
```

XML_InputProxyChannelStatus

InputProxyChannelStatus message in XML format

```
<InputProxyChannelStatus version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:string--></id>
  <sourceInputPortDescriptor/><!--req-->
  <online><!--req, xs:boolean, whether the camera is online--></online>
  <streamingProxyChannelIdList><!--req-->
    <streamingProxyChannelId>
      <!--req, xs:string, stream channel No., e.g., 101-main stream of channel 1, 102-sub-stream of channel 1-->
    </streamingProxyChannelId>
  </streamingProxyChannelIdList>
  <chanDetectResult>
    <!--opt, xs:string, network camera status: "connect"-connected, "overSysBandwidth"-insufficient bandwidth,
    "domainError"-incorrect domain name, "ipcStreamFail"-getting stream failed, "connecting", "chacnNoError"-incorrect
    channel No., "cipAddrConflictWithDev": IP address is conflicted with device address, "ipAddrConflcWithIpc"-IP
    address conflicted, "errorUserNameOrPasswd"-incorrect user name or password, "netUnreachable"-invalid network
```

```
address, "unknownError"-unknown error, "notExist"-does not exist, "ipcStreamTypeNotSupport"-the stream
transmission mode is not supported, "ipcResolutionNotSupport"-the resolution of network camera is not supported-->
</chanDetectResult>
</InputProxyChannelStatus>
```

XML_InputProxyChannelStatusList

InputProxyChannelStatusList message in XML format

```
<InputProxyChannelStatusList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <InputProxyChannelStatus/><!--opt, see details in XML_InputProxyChannelStatus-->
</InputProxyChannelStatusList>
```

See Also

XML_InputProxyChannelStatus

XML_IntelliCap

XML message about intelligent capability

```
<IntelliCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <isFaceSupport><!--opt, xs:boolean, whether it supports face detection--></isFaceSupport>
  <isBehaviorSupport><!--opt, xs:boolean, whether it supports behavior analysis--></isBehaviorSupport>
  <isLineDetectionSupport><!--opt, xs:boolean, whether it supports line crossing detection--></
isLineDetectionSupport>
  <isFieldDetectionSupport><!--opt, xs:boolean, whether it supports intrusion detection--></isFieldDetectionSupport>
  <isRegionEntranceSupport><!--opt, xs:boolean, whether it supports region entrance detection--></
isRegionEntranceSupport>
  <isRegionExitingSupport><!--opt, xs:boolean, whether it supports region exiting detection--></
isRegionExitingSupport>
  <isLoiteringSupport><!--opt, xs:boolean, whether it supports loitering detection--></isLoiteringSupport>
  <isGroupSupport><!--opt, xs:boolean, whether it supports people gathering detection--></isGroupSupport>
  <isRapidMoveSupport><!--opt, xs:boolean, whether it supports fast moving detection--></isRapidMoveSupport>
  <isParkingSupport><!--opt, xs:boolean, whether it supports parking detection--></isParkingSupport>
  <isUnattendedBaggageSupport><!--opt, xs:boolean, whether it supports unattended baggage detection--></
isUnattendedBaggageSupport>
  <isAttendedBaggageSupport><!--opt, xs:boolean, whether it supports object removal detection--></
isAttendedBaggageSupport>
  <isTeacherSupport><!--opt, xs:boolean, whether it supports teacher's behavior detection--></isTeacherSupport>
  <isStudentSupport><!--opt, xs:boolean, whether it supports student's behavior detection--></isStudentSupport>
  <isCombinedSupport><!--opt, xs:boolean, whether it supports combined detection--></isCombinedSupport>
  <isTrafficSupport><!--opt, xs:boolean, whether it supports traffic detection--></isTrafficSupport>
  <RestoreLib>
    <libName opt="studentsStoodUp,peopleCounting,shipDetection"><!--opt,xs:string,--></libName>
  </RestoreLib>
  <RestoreDefParamForbid>
    <!--opt, xs:boolean "true, false", forbid restoring algorithm library to default. When the node does not exist, it
indicates restoring to default is not supported. When the node exists and its value is "ture", it indicates support, when
the value is "false", it indicates not support-->
```



```
</RestoreDefParamForbid>
<RestoreAlgLibParam>
<!--opt, the capabilities of restoring algorithm library parameters according to algorithm library name-->
  <libName opt="faceSnap,HMS,behaviorAnalysis"><!--opt, xs:string--></libName>
</RestoreAlgLibParam>
<isFaceCaptureStatisticsSupport><!--whether it supports face picture statistics--></isFaceCaptureStatisticsSupport>
<isSupportPersonQueueDetection><!--whether it supports queue management--></
isSupportPersonQueueDetection>
<isSupportIntersectionAnalysis>
  <!--opt, xs: boolean, whether it supports intersection analysis-->
</isSupportIntersectionAnalysis>
<mixedTargetDetectionWithoutAttribute><!--opt, xs: boolean--></mixedTargetDetectionWithoutAttribute>
<isSupportUploadFacePictureByForm><!--opt, xs:boolean, whether it supports uploading face pictures by form--></
isSupportUploadFacePictureByForm>
  <isSupportUploadFacePictureByUrl><!--opt, xs: boolean, whether it supports uploading face pictures by URL--></
isSupportUploadFacePictureByUrl>
  <isSupportUploadHumanPictureByForm><!--opt, xs:boolean, whether it supports uploading human pictures in form--
></isSupportUploadHumanPictureByForm>
  <isSupportFaceScore><!--opt,xs:boolean, whether it supports face grading configuration (camera)--></
isSupportFaceScore>
  <HumanRecognitionModeSearchCap>
    <searchTargetsNumMax><!--opt, xs:integer, maximum number of sample pictures that can be imported for
searching by picture--></searchTargetsNumMax>
    <HumanMode>
      <searchCond opt="age_group,gender,jacet_color,glass,bag,ride,unlimit"/><!--setting multiple search conditions is
supported-->
      <similarity min="0.0" max="100.0"/><!--opt, xs:float, similarity, range: [0.0,100.0]-->
    </HumanMode>
    <positive opt="true,false"/><!--opt, xs:boolean, whether it is false human body recognition alarm: "true"-yes,
"false"-no-->
    <eventType opt="unlimit,humanRecognition">
      <!--opt, xs:string, event type: "unlimit"-no limit, "humanRecognition"-human body detection alarm-->
    </eventType>
    <isSupportMultiChannelSearch>
      <!--opt, xs:boolean, whether it supports multi-channel search-->
    </isSupportMultiChannelSearch>
    <isSupportTotalSearchResult>
      <!--opt, xs:boolean, whether it supports limiting number of results that can be obtained after a single search-->
    </isSupportTotalSearchResult>
  </HumanRecognitionModeSearchCap>
  <VehicleRecognitionModeSearchCap>
    <searchTargetsNumMax><!--opt, xs:integer, maximum number of sample pictures that can be imported for
searching by picture--></searchTargetsNumMax>
    <eventType opt="unlimit,vehicleBlackList,vehicleWhiteList"/><!--xs:string, event type: "unlimit"-no limit,
"vehicleBlackList"-vehicle blacklist, "vehicleWhiteList"-vehicle whitelist-->
    <VehicleMode>
      <searchCond opt="licensePlate,vehicleLogo,vehicleSubLogoRecog,vehicleType,vehicleColor,unlimit"/>
      <similarity min="0.0" max="100.0"/><!--opt, xs:float, similarity, range: [0.0,100.0]-->
    </VehicleMode>
    <isSupportMultiChannelSearch>
      <!--opt, xs:boolean, whether it supports multi-channel search-->
    </isSupportMultiChannelSearch>
```

```
<isSupportTotalSearchResult>
  <!--opt, xs:boolean, whether it supports limiting number of results that can be obtained after a single search-->
</isSupportTotalSearchResult>
</VehicleRecognitionModeSearchCap>
<FaceContrastPersonInfoExtend><!--configuration capability of face comparison tag-->
  <personInfoCap>
    <maxPersonInfo min="0" max="4">
      <!--opt, xs:integer, maximum number of person tags-->
    </maxPersonInfo>
  </personInfoCap>
  <personInfoFDlibCap>
    <maxPersonInfo min="0" max="4">
      <!--opt, xs:integer, maximum number of person tags-->
    </maxPersonInfo>
  </personInfoFDlibCap>
</FaceContrastPersonInfoExtend>
<isSupportSafetyHelmetDetection>
  <!-- opt, xs:boolean, whether it supports hard hat detection-->
</isSupportSafetyHelmetDetection>
</IntelliCap>
```

XML_IOCap

IOCap capability message in XML format

```
<IOCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <IOInputPortNums>
    <!--opt, xs:integer-->
  </IOInputPortNums>
  <IOOutputPortNums>
    <!--opt, xs:integer-->
  </IOOutputPortNums>
  <isSupportStrobeLamp>
    <!--opt, xs:integer-->
  </isSupportStrobeLamp>
  <SoftIOInputPortNums>
    <!--opt, xs:integer-->
  </SoftIOInputPortNums>
  <isSupportIOOutputAdvanceParameter>
    <!--opt, xs:boolean, whether supports configuration of alarm input advanced parameters-->
  </isSupportIOOutputAdvanceParameter>
  <isSupportCombinationAlarm>
    <!--opt, xs:boolean, whether supports composite alarm-->
  </isSupportCombinationAlarm>
</IOCap>
```

XML_IOInputPort

IOInputPort message in XML format

```
<IOInputPort version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:integer--></id>
  <enabled><!--req, Boolean, "true,false"--></enabled>
  <IODescriptor><!--opt, I/O port description-->
    <proxyProtocol>
      <!--req, string, access protocol, "HIKVISION, AXIS, PANASONIC, BOSCH, PELCO, SONY..."-->
    </proxyProtocol>
    <userName><!--req, string, user name--></userName>
    <addressingFormatType>
      <!--req, string, address type: "ipaddress,hostname"-->
    </addressingFormatType>
    <hostName><!--dep, xs:string--></hostName>
    <ipAddress><!--dep, xs:string--></ipAddress>
    <ipv6Address><!--dep, xs:string--></ipv6Address>
    <managePortNo><!--req, integer, manage port--></managePortNo>
    <innerIOPortID><!--req, integer, I/O port--></innerIOPortID>
  </IODescriptor>
  <triggering><!-- req, xs:string, "high,low"--></triggering>
  <name><!--opt,xs:string--></name>
  <IOUseType>
    <!-- opt, xs:string, "disable,openDoor,doorStatus,custom" -->
  </IOUseType>
  <inputType>
    <!--opt, xs:string, opt="switch,semaphore"-->
  </inputType>
  <CombinationAlarm><!--opt, composite alarm list-->
    <channel><!--req, xs:integer, channel No.--></channel>
    <EventTypeList><!--req, event type list-->
      <eventType><!--req, xs:string, event type--></eventType>
    </EventTypeList>
  </CombinationAlarm>
</IOInputPort>
```

XML_IOInputPortList

IOInputPortList message in XML format

```
<IOInputPortList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <IOInputPort/><!--opt, alarm input, see details in XML_IOInputPort-->
</IOInputPortList>
```

See Also

XML_IOInputPort

XML_IOOutputPort

IOOutputPort message in XML format

```
<IOOutputPort version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!-- req, xs:integer, "2"--></id>
  <PowerOnState>
    <!--req, output port configuration parameters when the device is powered on-->
    <defaultState>
      <!--ro, req, xs:string, default output port signal when it is not triggered, "high,low"-->
    </defaultState>
    <outputState>
      <!--ro, req, output port signal when it is being triggered, xs:string, "high,low,pulse"-->
    </outputState>
    <pulseDuration>
      <!--dep, xs:integer, duration of a output port signal when it is being triggered, it is valid when outputState is
"pulse", unit: milliseconds -->
    </pulseDuration>
  </PowerOnState>
  <name><!--opt, xs:string--></name>
  <IOUseType><!--opt, xs:string, "disable,electricLock,custom"--></IOUseType>
  <normalStatus><!--opt, xs:string, normal status: open-remain open, close-remain closed--></normalStatus>
</IOOutputPort>
```

XML_IOOutputPortList

IOOutputPortList message in XML format

```
<IOOutputPortList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <IOOutputPort/><!--opt, see details in XML_IOOutputPort-->
</IOOutputPort>
```

See Also

XML_IOOutputPort

XML_IOPortData

IOPortData message in XML format

```
<IOPortData xmlns="http://www.isapi.org/ver20/XMLSchema">
  <outputState><!--req, xs:string, output level: "high, low"--></outputState>
</IOPortData>
```

XML_IOPortStatus

IOPortStatus message in XML format

```
<IOPortStatus version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <!--req-->
  <ioPortID><!--req, xs: integer, I/O No.: 1, 2--></ioPortID>
  <ioPortType><!--req, xs: string, I/O type: "input", "output"--></ioPortType>
  <ioState><!--req, xs: string, I/O status: "active", "inactive"--></ioState>
</IOPortStatus>
```

XML_IOPortStatusList

IOPortStatusList message in XML format

```
<IOPortStatusList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <IOPortStatus><!--req-->
    <ioPortID><!--req, xs: integer, I/O No.: 1, 2--></ioPortID>
    <ioPortType><!--req, xs: string, I/O type: input, output--></ioPortType>
    <ioState><!-- req, xs: string, I/O status: active, inactive--></ioState>
  </IOPortStatus>
</IOPortStatusList>
```

XML_IOProxyInputPort

IOProxyInputPort message in XML format

```
<IOProxyInputPort version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:string--></id>
  <enabled><!--req, xs:boolean--></enabled>
  <IODescriptor><!--req, xs:string, description about the IO port connected to the front-end device-->
    <proxyProtocol><!--req, xs:string, "HIKVISION, AXIS, PANASONIC, BOSCH, PELCO, SONY, ..."--></proxyProtocol>
    <userName><!--req, wo, xs:string --></userName>
    <password><!--req, wo, xs:string --></password>
    <addressingFormatType><!--dep, req, xs:enumeration, "ipaddress, hostname, ..."--></addressingFormatType>
    <hostName><!--dep, xs:string--></hostName>
    <ipAddress><!--dep, xs:string--></ipAddress>
    <ipv6Address><!--dep, xs:string--></ipv6Address>
    <managePortNo><!--req, xs:integer--></managePortNo>
    <innerIOPortID><!--req, xs:string, ID--></innerIOPortID>
  </IODescriptor>
  <triggering><!--req, xs:string, "high, low, rising, falling"--></triggering>
  <name><!--opt, xs:string--></name>
  <CombinationAlarm><!--opt, information list of the combined alarm-->
    <channel><!--req, xs:integer, channel No.--></channel>
    <EventTypeList><!--req, event type list-->
      <eventType>
        <!--list, xs:string, event type-->
```

```
</eventType>
</EventTypeList>
</CombinationAlarm>
</IOProxyInputPort>
```

XML_IOProxyInputPortList

IOProxyInputPortList message in XML format

```
<IOProxyInputPortList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <IOProxyInputPort/><!--opt, input configuration of one digital channel, see details in the message of
XML_IOProxyInputPort-->
</IOProxyInputPortList>
```

See Also

XML_IOProxyInputPort

XML_IOProxyOutputPort

IOProxyOutputPort message in XML format

```
<IOProxyOutputPort version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:string--></id>
  <IODescriptor/><!--req-->
  <PowerOnState><!--req-->
    <defaultState><!--req, xs:string, "high, low"--></defaultState>
    <outputState><!--req, xs:string, "high, low, pulse"--></outputState>
    <pulseDuration><!--dep, xs:integer, unit: milliseconds--></pulseDuration>
  </PowerOnState>
  <name><!--opt, xs:string--></name>
</IOProxyOutputPort>
```

XML_IOProxyOutputPortList

IOProxyOutputPortList message in XML format

```
<IOProxyOutputPortList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <IOProxyOutputPort/><!--opt, output configuration of one digital channel, see details in the message of
XML_IOProxyOutputPort-->
</IOProxyOutputPortList>
```

See Also

XML_IOProxyOutputPort

XML_IOTTriggersCap

IOTTriggersCap message in XML format

```
<?xml version="1.0" encoding="utf-8"?>
<IOTTriggersCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <AccessController><!--opt, security control panel-->
    <XXTriggerCap><!--opt,xs: EventTriggerCapType--></XXTriggerCap>
  </AccessController>
  <VideoIntercom><!--opt, video intercom-->
    <XXTriggerCap><!--opt,xs: EventTriggerCapType --></XXTriggerCap>
  </VideoIntercom>
  <GJD><!--opt, GJD security control panel-->
    <XXTriggerCap><!--opt,xs: EventTriggerCapType--></XXTriggerCap>
  </GJD>
  <Luminite><!--opt, Luminite security control panel-->
    <XXTriggerCap><!--opt,xs: EventTriggerCapType--></XXTriggerCap>
  </Luminite>
  <OPTEX><!--opt, OPTEX security control panel-->
    <XXTriggerCap><!--opt,xs: EventTriggerCapType--></XXTriggerCap>
  </OPTEX>
  <CameraDetector><!--opt, detector-->
    <XXTriggerCap><!--opt,xs: EventTriggerCapType--></XXTriggerCap>
  </CameraDetector>
</IOTTriggersCap>
```

Remarks

The XX in the node **<XXTriggerCap>** corresponds to detailed event type. E.g., if the event type is humanRecognition, then the returned node is <HumanRecognitionTriggerCap>.

See Also

XML_EventTriggerCapType

XML_IPAddress

IPAddress message in XML format

```
<IPAddress version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ipVersion><!--req, xs:string, "v4,v6,dual"--></ipVersion>
  <addressingType><!--req, xs:string, "static,dynamic,api"--></addressingType>
  <ipAddress><!--dep, xs:string, ipv4 address--></ipAddress>
  <subnetMask><!--dep, xs:string, subnet mask for IPv4 address--></subnetMask>
  <ipv6Address><!--dep, xs:string--></ipv6Address>
  <bitMask><!--dep, xs:integer, bitmask IPv6 address--></bitMask>
  <DefaultGateway><!--dep-->
    <ipAddress><!--dep, xs:string--></ipAddress>
    <ipv6Address><!--dep, xs:string--></ipv6Address>
  </DefaultGateway>
```

```
<PrimaryDNS><!--dep-->
  <ipAddress><!--dep, xs:string--></ipAddress>
  <ipv6Address><!--dep, xs:string--></ipv6Address>
</PrimaryDNS>
<SecondaryDNS><!--dep-->
  <ipAddress><!--dep, xs:string--></ipAddress>
  <ipv6Address><!--dep, xs:string--></ipv6Address>
</SecondaryDNS>
<Ipv6Mode><!--opt-->
  <ipV6AddressingType><!--dep, xs:string, "router,ra,manual,dhcp"--></ipV6AddressingType>
  <ipv6AddressList>
    <v6Address>
      <id><!--dep, xs:string; id--></id>
      <type><!--dep, xs:string, "router,ra,manual,dhcp"--></type>
      <address><!--dep, xs:string--></address>
      <bitMask><!--dep, xs:integer--></bitMask>
    </v6Address>
  </ipv6AddressList>
</Ipv6Mode>
</IPAddress>
```

XML_IPFilter

IPFilter message in XML format

```
<IPFilter version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--req, xs:boolean--></enabled>
  <permissionType><!--opt, xs:string, "deny,allow"--></permissionType>
  <IPFilterAddressList size = "32"/><!--opt, the character size indicates the max. number of supported IP address. See
XML_IPFilterAddressList for details-->
</IPFilter>
```

See Also

XML_IPFilterAddressList

XML_IPFilterAddress

IPFilterAddress message in XML format

```
<IPFilterAddress version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:string;id--></id>
  <permissionType><!--dep, xs:string, "deny,allow" --></permissionType>
  <addressFilterType><!--ro, xs:string, "mask, range"--></addressFilterType>
  <AddressRange><!--dep, it is valid when <addressFilterType> is "range"-->
    <startIPAddress><!--dep, xs:string--></startIPAddress>
    <endIPAddress><!--dep, xs:string--></endIPAddress>
    <startIPv6Address><!--dep, xs:string--></startIPv6Address>
    <endIPv6Address><!--dep, xs:string--></endIPv6Address>
  </AddressRange>
```



```
<AddressMask><!--dep, it is valid when <addressFilterType> is "mask"-->
  <ipAddress><!--dep, xs:string--></ipAddress>
  <ipv6Address><!--dep, xs:string--></ipv6Address>
  <bitMask><!--opt, xs:string--></bitMask>
</AddressMask>
</IPFilterAddress>
```

XML_IPFilterAddressList

IPFilterAddressList message in XML format

```
<IPFilterAddressList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <IPFilterAddress/><!--opt, see XML_IPFilterAddress for details-->
</IPFilterAddressList>
```

See Also

XML_IPFilterAddress

XML_IrcutFilter

IrcutFilter message in XML format

```
<IrcutFilter version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <IrcutFilterType>
    <!--opt, xs: string, day/night auto switch mode: "auto, day, night, schedule, eventTrigger, darkFighterX,
darkFighterXAuto, darkFighterXSchedule"-->
  </IrcutFilterType>
  <dayToNightFilterLevel>
    <!--opt, xs: string, level of switching day to night: "low, normal, high"-->
  </dayToNightFilterLevel>
  <dayToNightFilterTime>
    <!--opt, xs: integer, time interval of switching day to night-->
  </dayToNightFilterTime>
  <nightToDayFilterLevel>
    <!--opt, xs: string, level of switching night to day: "low, normal, high"-->
  </nightToDayFilterLevel>
  <nightToDayFilterTime>
    <!--opt, xs: integer, time interval of switching night to day-->
  </nightToDayFilterTime>
  <Schedule><!--dep-->
    <scheduleType><!--req, xs: string, schedule type: "day,night"></scheduleType>
    <TimeRange><!--req, time period on schedule-->
      <beginTime><!--req, xs: time, which is in ISO8601 time format--></beginTime>
      <endTime><!--req, xs:time, which is in ISO8601 time format--></endTime>
    </TimeRange>
  </Schedule>
  <EventTrigger><!--dep-->
    <eventType><!--req, xs: string, event type: "IO,VMD"></eventType>
    <IrcutFilterAction><!--req, xs:string, "day,night"></IrcutFilterAction >
```

```
</EventTrigger>
</IrcutFilter>
```

XML_IrisData

IrisData message in XML format

```
<IrisData version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <iris>
    <!--req, xs: integer, a vector, whose value is the percentage of the maximum iris adjusting speed: negative value-
close iris; positive value-open iris-->
  </iris>
</IrisData>
```

XML_ISPMode

ISPMode message in XML format

```
<ISPMode version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <mode><!--opt, xs:string, "auto,schedule"--></mode>
  <Schedule><!--dep-->
    <scheduleType><!--req, xs:string, "day,night"--></scheduleType>
    <TimeRange><!--req-->
      <beginTime><!--req, xs:time, ISO8601 time--></beginTime>
      <endTime><!--req, xs:time, ISO8601 time--></endTime>
    </TimeRange>
  </Schedule>
</ISPMode>
```

XML_ItemList

ItemList message in XML format

```
<ItemList>
  <Item>
    <itemID>
      <!--req, xs:string, item ID, which is between 1 and 15-->
    </itemID>
    <itemOrder>
      <!--req, xs:string, name element: "none", "devIp"-device IP address, "time", "buildUnitNo"-building No. and unit
No., "outDoorDevNo"-door station, "unlockType"-unlocking type, "devName"-device name, "deviceNo"-device No.,
"channelName"-channel name, "channelNo"-channel No., "plateNo"-license plate number, "plateColor"-license plate
color, "laneNo"-lane No., "carSpeed"-vehicle speed, "positionInfo1"-camera 1, "pictureNo"-picture No., "CarNo"-
vehicle No., "speedLimit"-speed limit, "illegalCode"-violation code, "siteNo"-intersection No., "directionNo"-direction
No., "carColor"-vehicle color, "platePosition"-license plate coordinates, "carType"-vehicle type, "illegalType"-violation
type, "custom"-->
    </itemOrder>
```

```
<itemCustomStr>
  <!--req, xs:string, element custom string, which is between 1 and 32, unit: bytes. This node is valid only when
<itemOrder> is "custom". Currently traffic cameras only support one custom name-->
</itemCustomStr>
</Item>
</ItemList>
```

XML_Language

Language message in XML format

```
<Language version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <type><!--req, xs: string, "GBK,EUC-KR", def="GBK"--></type>
</Language>
```

XML_LensDistortionCorrection

LensDistortionCorrection message in XML format

```
<PrivacyMaskRegion version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs: integer--></id>
  <enabled><!--req, xs: boolean--></enabled>
  <RegionCoordinatesList><!--req-->
    <RegionCoordinates><!--list-->
      <positionX><!--req, xs: integer; coordinate--></positionX>
      <positionY><!--req, xs: integer; coordinate--></positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
  <privacymaskName><!--opt, xs: string--></privacymaskName>
  <maskType>
    <!--opt, xs: string, "gray,red,yellow,blue,orange,green,transparent,half-transparent,mosaic"-->
  </maskType>
  <zoomdoorlimit><!--opt, xs: integer, the value is between 10 and 1000--></zoomdoorlimit>
</PrivacyMaskRegion>
```

XML_Link

Link message in XML format

```
<Link version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <MACAddress><!--req, xs:string--></MACAddress>
  <autoNegotiation><!--req, xs:boolean--></autoNegotiation>
  <speed><!--req, xs:integer, "10, 100, 1000"--></speed>
  <duplex><!--req, xs:string, "half, full"--></duplex>
  <MTU><!--req, xs:integer--></MTU>
</Link>
```

XML_ListMember

ListMember message in XML format

```
<ListMember version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!-- req, xs:string--></id>
  <phoneNumber><!-- req, xs:string, cellphone number, which is sensitive information that needs to be encrypted--></phoneNumber>
  <SupportEntryList> <!-- req -->
    <SupportEntry>
      <entry><!--req,xs:string,control types: "SMSAlarm"-message linkage, "SMSCtrl"-SMS control, "CallCtrl"-calling control--></entry>
      <enabled><!-- opt, xs:boolean,"true,false" --></enabled>
    </SupportEntry>
  </SupportEntryList>
  <SMSAlarmTypeList> <!-- dep -->
    <SMSAlarmType>
      <type><!-- req, xs:string, "diskfull, diskerror, nicbroken, ipconflict,illaccess, AlarmInErr, tamper, vmd, wireless, pir, callhelp, AudioDetection,scenechangeDetection, defocusDetection, facedetection, LineDetection, FieldDetection,regionEntrance, regionExiting, loitering, group, rapidMove, parking, unattendedBaggage,attendedBaggage,dataPrealarm,all", event type that triggers message alarms, "all"-determine the event type that triggers message alarms based on event linkage actions, related URI (/ISAPI/Event/triggers/<ID>) --></type>
      <enabled><!-- opt, xs:boolean,"true,false" --></enabled>
    </SMSAlarmType>
  </SMSAlarmTypeList>
  <SMSCtrlTypeList>
    <SMSCtrlType>
      <type><!-- req,xs:string, message operations: "messageReboot"-reboot message, "dial"-enable/disable dial-up, "getDialStatus"-get dial-up status, "SMSAlarm"- receive/ reject message linkage--></type>
      <enabled><!-- opt, xs:boolean,"true,false"--></enabled>
    </SMSCtrlType>
  </SMSCtrlTypeList>
  <name><!-- opt, xs:string,contact name, which is sensitive information that needs to be encrypted--></name>
</ListMember>
```

XML_localPermission

localPermission message in XML format

```
<localPermission version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <backup><!--opt, xs:boolean--></backup>
  <record><!--opt, xs:boolean--></record>
  <playBack><!--opt, xs:boolean--></playBack>
  <preview><!--opt, xs:boolean--></preview>
  <videoChannelPermissionList><!--opt-->
    <videoChannelPermission><!--opt-->
      <id><!--req, corresponds to the video input channel ID--></id>
      <playBack><!--opt, xs:boolean--></playBack>
```

```
<preview><!--opt, xs:boolean--></preview>
<record><!--opt, xs:boolean--></record>
<backup><!--opt, xs:boolean--></backup>
<playBackDoubleVerification>
  <!--opt, xs:boolean, whether supports secondary authentication for playback-->
</playBackDoubleVerification>
<backupDoubleVerification>
  <!--opt, xs:boolean, whether supports secondary authentication for backup-->
</backupDoubleVerification>
</videoChannelPermission>
</videoChannelPermissionList>
<ptzControl>
  <!--req, xs:boolean-->
</ptzControl>
<ptzChannelPermissionList><!--opt-->
  <ptzChannelPermission><!--req-->
    <id><!--req, corresponds to PTZ channel ID--></id>
    <ptzControl><!--opt, xs:boolean--></ptzControl>
  </ptzChannelPermission>
</ptzChannelPermissionList>
<logOrStateCheck><!--opt, xs:boolean--></logOrStateCheck>
<parameterConfig><!--opt, xs:boolean--></parameterConfig>
<restartOrShutdown><!--opt, xs:boolean--></restartOrShutdown>
<upgrade><!--opt, xs:boolean--></upgrade>
</localPermission>
```

XML_localPermissionCap

localPermissionCap capability message in XML format

```
<localPermissionCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <backup><!--opt, xs:boolean--></backup>
  <record><!--opt, xs:boolean--></record>
  <playBack><!--opt, xs:boolean--></playBack>
  <preview><!--opt, xs:boolean--></preview>
  <videoChannelPermissionList><!--opt-->
    <videoChannelPermission><!--opt-->
      <id><!--req, corresponds to the video input channel ID--></id>
      <playBack><!--opt, xs:boolean--></playBack>
      <preview><!--opt, xs:boolean--></preview>
      <record><!--opt, xs:boolean--></record>
      <backup><!--opt, xs:boolean--></backup>
      <playBackDoubleVerification>
        <!--opt, xs:boolean, whether supports secondary authentication for playback-->
      </playBackDoubleVerification>
      <backupDoubleVerification>
        <!--opt, xs:boolean, whether supports secondary authentication for backup-->
      </backupDoubleVerification>
    </videoChannelPermission>
  </videoChannelPermissionList>
  <ptzControl>
```

```
<!--req, xs:boolean-->
</ptzControl>
<ptzChannelPermissionList><!--opt-->
  <ptzChannelPermission><!--req-->
    <id><!--req, corresponds to PTZ channel ID--></id>
    <ptzControl><!--opt, xs:boolean--></ptzControl>
  </ptzChannelPermission>
</ptzChannelPermissionList>
<logOrStateCheck><!--opt, xs:boolean--></logOrStateCheck>
<parameterConfig><!--opt, xs:boolean--></parameterConfig>
<restartOrShutdown><!--opt, xs:boolean--></restartOrShutdown>
<upgrade><!--opt, xs:boolean--></upgrade>
</localPermissionCap>
```

XML_LockByName

XML message about parameters of locking and unlocking video by file name

```
<?xml version="1.0" encoding="utf-8"?>
<LockByName version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <TrackId><!--required, xs:string, channel ID, e.g., 101 indicates the main stream of channel No.1--></TrackId>
  <recordName><!--required, xs:string, file name, the name length is between 1 and 100--></recordName>
  <command><!--required, xs:string, operation command: "lock" and "unlock"--></command>
  <duration>
    <!--dependent, xs:integer, locking duration, unit: second; if the value is 0, it indicates that the video will be
    permanently locked, this node is valid only when command is set to "lock"-->
  </duration>
  <recorderCode><!--optional, xs:string, body camera No.--></recorderCode>
  <policeCode><!--optional, xs:string, No. of police that collects data, it should be encrypted--></policeCode>
</LockByName>
```

XML_LockCap

XML message about video locking and unlocking capability

```
<?xml version="1.0" encoding="utf-8"?>
<LockCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <LockByName><!--optional, lock or unlock video by file name-->
    <TrackId><!--required, xs:string--></TrackId>
    <recordName min="1" max="100"><!--required, xs:string, video file name--></recordName>
    <command opt="lock,unlock"><!--required, xs:string, operation command: "lock" and "unlock"--></command>
    <duration min="" max=""><!--dependent, xs:integer, locking duration, unit: second--></duration>
    <recorderCode min="1" max="100"><!--optional, xs:string, body camera No.--></recorderCode>
    <policeCode min="1" max="100">
      <!--optional, xs:string, No. of police that collects data, it should be encrypted-->
    </policeCode>
  </LockByName>
  <LockByTime><!--optional, lock or unlock video by time-->
    <TrackId><!--required, xs:string--></TrackId>
```

```
<recordType opt= "ALL, MANUAL, CMR, MOTION, ALARM, EDR, ALARMANDMOTION, COMMAND, SMART">
  <!--required, xs:string-->
</recordType>
<startDateTime><!--dependent, xs:time, ISO8601 time format, e.g., 2004-05-03T17:30:08+08:00--></startDateTime>
<endDateTime><!--dependent, xs:time, ISO8601 time format, e.g., 2004-05-03T17:30:08+08:00--></endDateTime>
<command opt="lock,unlock"><!--required, xs:string--></command>
<duration min="" max=""><!--dependent, xs:integer--></duration>
</LockByTime>
</LockCap>
```

XML_LockPTZ

LockPTZ message in XML format

```
<LockPTZ version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <lockTime>
    <!--req, xs:integer, PTZ is unlocked when this node is set to 0, unit: second-->
  </lockTime>
</LockPTZ>
```

XML_LogConfig

LogConfig message in XML format.

```
<LogConfig version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--required, boolean type, whether to enable log--></enabled>
  <level>
    <!--required, string type, log types: "none,debug,info,error,fault,all", multiple type can be selected, and each type
    should be separated by comma-->
  </level>
</LogConfig>
```

XML_LogServer

XML message about log server information

```
<LogServer version="2.0" xmlns="http://www.std-cgi.org/ver20/XMLSchema">
  <enabled><!--required, xs: boolean, opt="true,false"--></enabled>
  <addressingFormatType>
    <!--required, xs: string, "ipaddress,hostname"-->
  </addressingFormatType>
  <hostName><!--dependent, xs: string--></hostName>
  <ipAddress><!--dependent, xs: string--></ipAddress>
  <ipv6Address><!--dependent, xs: string--></ipv6Address>
  <portNo><!--optional, xs: integer--></portNo>
  <transmissionEncryption>
    <!--optional, xs:boolean, whether to enable transmission encryption: "true"-yes, "false"-no (default). If this field is
```

```
not supported, the default encryption method is TLS-->
</transmissionEncryption>
<checkCertificate>
  <!--optional, xs:boolean,whether to enable checking certificates-->
</checkCertificate>
<uploadInterval>
  <!--optional, xs:integer, log uploading interval, which is 1 hour by default, range: [1,24], unit: hour-->
</uploadInterval>
</LogServer>
```

XML_LogServerCap

XML message about log server capability

```
<LogServerCap version="2.0" xmlns="http://www.std-cgi.org/ver20/XMLSchema">
  <enabled opt="true,false"></enabled>
  <addressingFormatType opt="ipaddress,hostname">
    <!--required, xs:string,"ipaddress,hostname"-->
  </addressingFormatType>
  <hostName min="" max="">
    <!--dependent, xs:string-->
  </hostName>
  <ipAddress min="" max=""><!--dep, xs:string--></ipAddress>
  <ipv6Address min="" max=""><!--dep, xs:string--></ipv6Address>
  <portNo min="" max=""><!--opt, xs:integer--></portNo>
  <transmissionEncryption opt="true,false">
    <!--optional, xs:boolean, whether to enable transmission encryption: "true"-yes, "false"-no (default). If this field is
not supported, the default encryption method is TLS-->
  </transmissionEncryption>
  <checkCertificate opt="true,false">
    <!--optional, xs:boolean,whether to enable checking certificates-->
  </checkCertificate>
  <uploadInterval min="1" max="24" def="1">
    <!--optional, xs:integer, log uploading interval, which is 1 hour by default, range: [1,24], unit: hour-->
  </uploadInterval>
</LogServerCap>
```

XML_LogServerTestDescription

LogServerTestDescription message in XML format

```
<LogServerTestDescription>
  <addressingFormatType>
    <!--required, xs:string, "ipaddress,hostname"-->
  </addressingFormatType>
  <hostName><!--dependent, xs:string--></hostName>
  <ipAddress><!--dependent, xs:string--></ipAddress>
  <ipv6Address><!--dependent, xs:string--></ipv6Address>
  <portNo><!--optional, xs:integer--></portNo>
```



```
<transmissionEncryption><!--optional, xs:boolean, whether to enable transmission encryption, and it is disabled by default. If this field is not supported, the default encryption method used by the device is TLS--></transmissionEncryption>
<checkCertificate><!--optional, xs:boolean, whether to enable checking certificate--></checkCertificate>
<uploadInterval><!--optional, xs:integer, log upload interval, which is 1 hour by default, range: [1,24], unit: hour--></uploadInterval>
</LogServerTestDescription>
```

XML_Log_CMSearchDescription

XML message about log search condition

```
<CMSearchDescription version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <searchID>
    <!--required, xs:string, search ID, e.g., {812F04E0-4089-11A3-9A0C-0305E82C2906}-->
  </searchID>
  <trackIDList>
    <trackID>
      <!--required, xs:integer, ID, XX01-search for the main stream video of channel XX, XX03-search for the pictures of channel XX-->
    </trackID>
  </trackIDList>
  <metald>
    <!--required, xs:string, format: log.std-cgi.com/<majorType>/<minorType>; <majorType> represents the major log type and <minorType> is the minor log type, see details in Log Type for ISAPI-->
  </metald>
  <timeSpanList>
    <timeSpan>
      <startTime><!--required, xs:time, ISO8601 time, start time of search, e.g.: 2017-08-02T00:00:00Z--></startTime>
      <endTime><!--required, xs:time, ISO8601 time, end time of search, e.g.: 2017-08-02T23:59:59Z--></endTime>
    </timeSpan>
  </timeSpanList>
  <searchResultPostion><!--required, xs:integer, start position of search--></searchResultPostion>
  <maxResults><!--optional, xs:integer, maximum number of returned results--></maxResults>
</CMSearchDescription>
```

See Also

Log Types for ISAPI

XML_Log_CMSearchResult

XML message about log search results

```
<CMSearchResult version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <searchID><!--required, xs:string, search ID--></searchID>
  <responseStatus><!--required, xs:string, search status--></responseStatus>
  <responseStatusStrg><!--required, xs:string, search status string: Ok, NO MATCHES, More--></responseStatusStrg>
  <totalMatches><!--optional, xs:integer, total number of matched records--></totalMatches>
  <numOfMatches><!--optional, xs:integer, number of returned records--></numOfMatches>
```

```
<matchList>
  <searchMatchItem>
    <logDescriptor>
      <metaId><!--required, xs:string, format: log.std-cgi.com/<majorType>/<minorType>; <majorType> represents the
major log type and <minorType> is the minor log type, see details in Log Type for ISAPI--></metaId>
      <StartDateTime><!--required, xs:time, ISO8601 time--></StartDateTime>
      <localID><!--optional, xs:string, channel No.--></localID>
      <paraType><!--optional, xs:string--></paraType>
      <userName><!--optional, xs:string--></userName>
      <logInfo><!--dependent, log details-->
        <OpenDoorRecord>
          <type><!--required, xs: string, unlocking type: password, hijack (unlock under duress), card, resident, center--
--></type>
        </OpenDoorRecord>
        <VisAlarmRecord>
          <type>
            <!--required, xs: string, alarm category: zone (zone alarm), dismantle (tampering alarm), hijack (duress alarm),
passwordErr (wrong password alarm), doorNotOpen-(door closed alarm), doorNotClose (door open alarm), SOS,
callReq (call request alarm), smartLockHijackFingerPrint (fingerprint duress alarm), smartLockHijackPassword
(password duress alarm), smartLockBreaking (forced-open door alarm), smartLockBeLocked (door locked alarm),
smartLockLowBattery (low battery alarm)-->
          </type>
        </VisAlarmRecord>
      </logInfo>
      <ipAddress><!--optional, xs:string--></ipAddress>
      <object><!--optional, xs:string, "network", "keypad", "remoteCtrl"-keyfob, "card"--></object>
      <params><!--optional, xs:string, parameters, such as zone No. and so on--></params>
      <seq><!--optional, xs:string, serial No.--></seq>
      <additionInformation><!--optional, xs:string, additional information--></additionInformation>
    </logDescriptor>
  </searchMatchItem>
</matchList>
</CMSearchResult>
```

See Also

Log Types for ISAPI

XML_MACFilter

MACFilter message in XML format

```
<MACFilter version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--req, xs:boolean--></enabled>
  <permissionType><!--req, xs:string,"deny, allow"--></permissionType>
  <MACFilterAddressList>
    <MACFilterAddress>
      <id><!--req, xs:string, id--></id>
      <MACAddress><!--req, xs:string--></MACAddress>
    </MACFilterAddress>
  </MACFilterAddressList>
```

```
</MACFilterAddressList>
</MACFilter>
```

XML_mailing

mailing message in XML format.

```
<mailing version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:string--></id>
  <enabled><!--opt, xs:boolean--></enabled>
  <sender><!--req-->
    <name><!--req, xs:string--></name>
    <emailAddress><!--req, xs:string--></emailAddress>
    <smtp><!--req-->
      <enableAuthorization><!--req, xs:boolean--></enableAuthorization>
      <enableSSL><!--opt, xs:boolean--></enableSSL>
      <addressingFormatType>
        <!--req, xs:string,"ipaddress,hostname"-->
      </addressingFormatType>
      <hostName><!--dep, xs:string--></hostName>
      <ipAddress><!--dep, xs:string--></ipAddress>
      <ipv6Address><!--dep, xs:string--></ipv6Address>
      <portNo><!--opt, xs:integer--></portNo>
      <accountName><!--dep, xs:string--></accountName>
      <password><!--dep, xs:string--></password>
      <enableTLS><!--opt, xs:boolean--></enableTLS>
      <startTLS><!--dep, xs:boolean--></startTLS>
    </smtp>
  </sender>
  <receiverList><!--req-->
    <receiver><!--req-->
      <id><!--req, xs:string--></id>
      <name><!--req, xs:string--></name>
      <emailAddress><!--req, xs:string--></emailAddress>
    </receiver>
  </receiverList>
  <attachment><!--opt-->
    <snapshot><!--opt-->
      <enabled><!--req, xs:boolean--></enabled>
      <interval><!--req, xs:integer, unit: second--></interval>
    </snapshot>
  </attachment>
</mailing>
```

XML_mailingList

mailingList message in XML format

```
<mailingList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <mailing><!--opt, xs:string--></mailing>
</mailingList>
```

XML_mailingTestResult

mailingTestResult message in XML format

```
<mailingTestResult version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <errorDescription><!--req, xs:string--></errorDescription>
</mailingTestResult>
```

XML_mailingTestDescription

mailingTestDescription message in XML format.

```
<mailingTestDescription version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <sendName><!--opt, xs:string--></sendName>
  <sendEmailAddress><!--req, xs:string--></sendEmailAddress>
  <addressingFormatType>
    <!--req, xs:string, "ipaddress,hostname"-->
  </addressingFormatType>
  <hostName><!--dep, xs:string--></hostName>
  <ipAddress><!--dep, xs:string--></ipAddress>
  <ipv6Address><!--dep, xs:string--></ipv6Address>
  <portNo><!--req, xs:integer--></portNo>
  <enableSSL><!--opt, xs:boolean--></enableSSL>
  <enableAuthorization><!--req, xs:boolean--></enableAuthorization>
  <accountName><!--dep, xs:string--></accountName>
  <password><!--dep, xs:string--></password>
  <receiverList><!--req-->
    <receiver><!--req-->
      <id><!--req, xs:string--></id>
      <name><!--req, xs:string--></name>
      <emailAddress><!--req, xs:string--></emailAddress>
    </receiver>
  </receiverList>
</mailingTestDescription>
```

XML_MasterSlaveTrackingCap

MasterSlaveTrackingCap capability message in XML format

```
<MasterSlaveTrackingCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <isSupportSlaveCameraCfg>
    <!--optional, xs:boolean, whether to support PTZ camera configuration-->
  </isSupportSlaveCameraCfg>
```

```
<isSupportSlaveCameraStatus>
  <!--optional, xs:boolean, whether to support PTZ camera status configuration-->
</isSupportSlaveCameraStatus>
<isSupportSlaveCameraTrackingRatio>
  <!--optional, xs:boolean, whether to support PTZ camera zoom ratio configuration-->
</isSupportSlaveCameraTrackingRatio>
<isSupportSlaveCameraTracking>
  <!--optional, xs:boolean, whether to support PTZ camera tracking-->
</isSupportSlaveCameraTracking>
<isSupportLinkedTracking>
  <!--optional, xs:boolean, whether device supports advanced parameters configuration of linked tracking (related
URI: /ISAPI/MasterSlaveTracking/linkedTracking?format=json)-->
</isSupportLinkedTracking>
</MasterSlaveTrackingCap>
```

XML_MaxElevation

MaxElevation message in XML format

```
<MaxElevation version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <mElevation><!--req, xs:integer, the lower limit of max. tilt-angle--></mElevation>
</MaxElevation>
```

XML_messageConfig

messageConfig message in XML format

```
<messageConfig version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled/> <!-- req, xs:boolean, "true,false" -->
  <SMSWhiteList/>
</messageConfig>
```

XML_messageConfigCap

messageConfigCap message in XML format

```
<messageConfigCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <supportEntry opt="SMSAlarm,SMSCtrl,CallCtrl">
    <!--req,xs:string,control types: "SMSAlarm"-message linkage, "SMSCtrl"-SMS control, "CallCtrl"-calling control-->
  </supportEntry>
  <SMSAlarmType
opt="diskfull,diskerror,nicbroken,ipconflict,illaccess,AlarmInErr,tamper,vmd,wireless,pir,callhelp,AudioDetection,scene
changeDetection,defocusDetection,facedetection,LineDetection,FieldDetection,regionEntrance,regionExiting,loitering,
group,rapidMove,parking,unattendedBaggage,attendedBaggage,dataPrealarm,all"/>
    <!--req, event type that triggers message alarms, "all"-determine the event type that triggers message alarms based
on event linkage actions related URL: /ISAPI/Event/triggers/<ID>-->
  <SMSCtrlType opt="messageReboot,dial,getDialStatus,SMSAlarm"/>
```

```
<!-- req,xs:string, message operations: "messageReboot"-reboot message, "dial"-enable/disable dial-up,
"getDialStatus"-get dial-up status, "SMSAlarm"- receive/ reject message linkage-->
<name min="0" max="32"><!--opt, xs:string, contact name, which is sensitive information that needs to be
encrypted--></name>
<maxListMemberNum><!--opt, xs:integer,max. number of SMS contacts, which is 8 by default--></
maxListMemberNum>
</messageConfigCap>
```

XML_MotionDetection

XML message about motion detection parameters

```
<MotionDetection version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--required, xs:boolean, whether to enable motion detection--></enabled>
  <enableHighlight><!--optional, xs:boolean, whether to enable highlight--></enableHighlight>
  <samplingInterval><!--optional, xs:integer, number of frames--></samplingInterval>
  <startTriggerTime><!--optional, xs:integer, unit: millisecond--></startTriggerTime>
  <endTriggerTime><!--opt, xs:integer, unit: millisecond--></endTriggerTime>
  <regionType><!--required, read-only, xs:string, region type: "grid", "roi", "none", "region"--></regionType>
  <Grid><!--dependency-->
    <rowGranularity><!--required, read-only, xs:integer--></rowGranularity>
    <columnGranularity><!--required, read-only, xs:integer--></columnGranularity>
  </Grid>
  <ROI><!--dependency-->
    <normalizedScreenWidth><!--required, read-only, xs:integer--></normalizedScreenWidth>
    <normalizedScreenHeight><!--required, read-only, xs:integer--></normalizedScreenHeight>
  </ROI>
  <MotionDetectionLayout><!--required, rule region configuration of motion detection-->
    <sensitivityLevel><!--required, xs:integer, sensitivity, which is between 0 and 100--></sensitivityLevel>
    <layout>
      <gridMap>
        <!--dependent, xs:string, the image is divided in to N x N grids, each grid can be enabled (value: 1) or disabled
        (value 0) motion detection, each four horizontal grids consist of a hexadecimal number, e.g., the corresponding
        hexadecimal number of "0011" is 3-->
      </gridMap>
      <roiMap><!--dependent,xs:string--></roiMap>
      <RegionList size="4"><!--this node is available when the regionType is set to "region"-->
        <Region><!--list-->
          <id><!--required, xs:string--></id>
          <RegionCoordinatesList><!--opt-->
            <RegionCoordinates><!--list-->
              <positionX><!--required, xs:integer; X-coordinate--></positionX>
              <positionY><!--required, xs:integer; Y-coordinate--></positionY>
            </RegionCoordinates>
          </RegionCoordinatesList>
        </Region>
      </RegionList>
    </layout>
    <targetType opt="human,vehicle"><!--optional, xs:string, target type, "human", "vehicle"--></targetType>
  </MotionDetectionLayout>
```

```
<enableWithMoving><!--optional, xs:boolean--></enableWithMoving>
</MotionDetection>
```

XML_MotionDetectionGridLayout

XML message about grid layout parameters of motion detection

```
<?xml version="1.0" encoding="utf-8"?>
<MotionDetectionGridLayout version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <sensitivityLevel><!--required, xs:integer, sensitivity, which is between 0 and 100--></sensitivityLevel>
  <layout>
    <gridMap>
      <!--dependent, xs:string, the image is divided in to N × N grids, each grid can be enabled (value: 1) or disabled
      (value 0) motion detection, each four horizontal grids consist of a hexadecimal number, e.g., the corresponding
      hexadecimal number of "0011" is 3-->
    </gridMap>
    <roiMap><!--dep,xs:string--></roiMap>
    <RegionList size="4"><!--this node is available when the regionType is set to "region"-->
      <Region><!--list-->
        <id><!--required, xs:string--></id>
        <RegionCoordinatesList><!--opt-->
          <RegionCoordinates><!--list-->
            <positionX><!--required, xs:integer; X-coordinate--></positionX>
            <positionY><!--required, xs:integer; Y-coordinate--></positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </Region>
    </RegionList>
  </layout>
  <targetType opt="human,vehicle"><!--optional, xs:string, target type, "human", "vehicle"--></targetType>
</MotionDetectionGridLayout>
```

XML_MotionDetectionLayout

XML message about rule region parameters of motion detection

```
<?xml version="1.0" encoding="utf-8"?>
<MotionDetectionLayout version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <sensitivityLevel><!--required, xs:integer, sensitivity, which is between 0 and 100--></sensitivityLevel>
  <layout>
    <gridMap>
      <!--dependent, xs:string, the image is divided in to N × N grids, each grid can be enabled (value: 1) or disabled
      (value 0) motion detection, each four horizontal grids consist of a hexadecimal number, e.g., the corresponding
      hexadecimal number of "0011" is 3-->
    </gridMap>
    <roiMap><!--dependent,xs:string--></roiMap>
    <RegionList size="4"><!--this node is available when the regionType is set to "region"-->
      <Region><!--list-->
        <id><!--required, xs:string--></id>
```

```
<RegionCoordinatesList><!--opt-->
  <RegionCoordinates><!--list-->
    <positionX><!--required, xs:integer; X-coordinate--></positionX>
    <positionY><!--required, xs:integer; Y-coordinate--></positionY>
  </RegionCoordinates>
</RegionCoordinatesList>
</Region>
</RegionList>
</layout>
<targetType opt="human,vehicle"><!--optional, xs:string, target type, "human", "vehicle"--></targetType>
</MotionDetectionLayout>
```

XML_MotionDetectionScheduleList

XML message about arming schedules of motion detection

```
<?xml version="1.0" encoding="utf-8"?>
<MotionDetectionScheduleList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <Schedule/><!--see details in the message XML_Schedule-->
</MotionDetectionScheduleList>
```

See Also

XML_Schedule

XML_MountingScenario

MountingScenario message in XML format

```
<MountingScenario version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <mode>
    <!--req, xs:string,
"indoor,outdoor,day,night,morning,nightfall,mode1,mode2,mode3,mode4,highway,road2,faceSnap,backlight,frontlight
", "lowIllumination"-low illumination, "backlight"-back light, "frontlight"-front light, "faceSnap"-face picture capture-->
  </mode>
</MountingScenario>
```

XML_MountList

MountList message in XML format

```
<MountList version="1.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <Mount>
    <id><!--req, xs: integer, storage node ID--></id>
    <path><!--req, xs: string, saving path--></path>
    <dir><!--req, xs: string--></dir>
    <size><!--req, xs: string,, storage size--></size>
    <descr><!--req, xs: string, storage description--></descr>
```



```
</Mount>
</MountList>
```

XML_NetworkCap

NetworkCap capability message in XML format

```
<NetworkCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <isSupportWireless>
    <!--req, xs:boolean, whether it supports accessing via wireless network, "true"-yes, "false"-no-->
  </isSupportWireless>
  <isSupportPPPoE>
    <!--req, xs:boolean, whether it supports PPPoE, "true"-yes, "false"-no-->
  </isSupportPPPoE>
  <isSupportBond>
    <!--req, xs:boolean, whether it supports NIC bonding, "true"-yes, "false"-no-->
  </isSupportBond>
  <isSupport802_1x>
    <!--req, xs:boolean, whether it supports 802_1x-->
  </isSupport802_1x>
  <isSupportNtp>
    <!--opt, xs:boolean, whether it supports NTP-->
  </isSupportNtp>
  <isSupportFtp>
    <!--opt, xs:boolean, whether it supports FTP-->
  </isSupportFtp>
  <isSupportUpnp>
    <!--opt, xs:boolean, whether it supports UPnP-->
  </isSupportUpnp>
  <isSupportPNP>
    <!--opt, xs:boolean, whether it supports PnP-->
  </isSupportPNP>
  <isSupportDdns>
    <!--opt, xs:boolean, whether it supports DNS-->
  </isSupportDdns>
  <isSupportHttps>
    <!--opt, xs:boolean, whether it supports HTTPS-->
  </isSupportHttps>
  <SnmpCap><!--opt, whether it supports SNMP-->
    <isSupport><!--req, xs:boolean--></isSupport>
  </SnmpCap>
  <isSupportExtNetCfg>
    <!--opt, xs:boolean, whether it supports configuring extended network parameters-->
  </isSupportExtNetCfg>
  <isSupportIPFilter>
    <!--opt, xs:boolean, whether it supports filtering IP address-->
  </isSupportIPFilter>
  <isSupportSSH opt="true"><!--opt, xs:boolean, whether it supports SSH--></isSupportSSH>
  <isSupportEZVIZ>
    <!--opt, xs:boolean, whether it supports Hik-Connect-->
  </isSupportEZVIZ>
```

```
<isSupportEhome>
  <!--opt, xs:boolean, whether it supports EHome-->
</isSupportEhome>
<isSupportWirelessServer>
  <!--opt, xs:boolean, whether it supports Wi-Fi hotspot-->
</isSupportWirelessServer>
<isSupportWirelessDial>
  <!--opt, xs:boolean, whether it supports wireless dial configuration-->
</isSupportWirelessDial>
<WPS><!--opt, Wi-Fi Protected Setup configuration-->
  <NetworkInterfaceList size="2">
    <NetworkInterface>
      <id><!--req, xs:string, NIC ID--></id>
      <enabled><!--req, xs:boolean, whether the NIC is enabled--></enabled>
      <isSupportAutoConnect><!--opt, xs:boolean--></isSupportAutoConnect>
      <isSupportDevicePinCode>
        <!--opt, xs:boolean, whether it supports device PIN code-->
      </isSupportDevicePinCode>
      <isSupportDevicePinCodeUpdate>
        <!--opt, xs:boolean, whether it supports updating device PIN code-->
      </isSupportDevicePinCodeUpdate>
      <ApPinCode><!--opt-->
        <ssid min="" max="">
          <!--opt, xs:string, maximum and minimum SSID length that can be returned by device-->
        </ssid>
        <pinCode min="" max="">
          <!--opt, xs:string, maximum and minimum PIN code length that can be returned by device-->
        </pinCode>
      </ApPinCode>
    </NetworkInterface>
  </NetworkInterfaceList>
</WPS>
<isSupportMACFilter>
  <!--opt, xs:boolean, whether it supports filtering MAC address-->
</isSupportMACFilter>
<verificationCode max="">
  <!--opt, xs:string, the maximum length of verificationCode that can be returned by device-->
</verificationCode>
<WPSCap><!--opt-->
  <isSupport><!--req, xs: boolean--></isSupport>
  <isSupportAutoConnect><!--req, xs: boolean--></isSupportAutoConnect>
</WPSCap>
<NetWorkMode>
  <workMode><!--opt, xs:string, network mode: "close,wifi,wifiAp"--></workMode>
</NetWorkMode>
<VerificationCodeModification><!--opt, xs:string, whether the verification code can be edited by the admin user-->
  <verificationCodeType opt="normal,empty"></verificationCodeType>
  <isSupportDeclarationURL><!--opt, xs:boolean, whether it supports URL declared by the service--></
isSupportDeclarationURL>
  <isSupportPrivacyPolicyURL><!--opt, xs:boolean, whether it supports the privacy policy URL--></
isSupportPrivacyPolicyURL>
  <verificationCodeModify opt="true,false">
```

```
<!--opt, whether the verification code is edited: "true"-yes, "false"-no, no return-not support-->
</verificationCodeModify>
<Hyperlinks><!--opt-->
  <declarationURL><!--opt,xs:string--></declarationURL>
  <privacyPolicyURL><!--opt,xs:string--></privacyPolicyURL>
</Hyperlinks>
<isSupportVerificationCodeCheck>
  <!--opt, xs: boolean, whether it supports verifying and configuring the verification code, true=yes, if this node is not
returned or the value of the returned node is false, it indicates that not support-->
  </isSupportVerificationCodeCheck>
  <isSupportOldVerificationCode><!--opt, xs:boolean, whether it supports old EZVIZ password configuration. The old
password contains six uppercase letters--></isSupportOldVerificationCode>
</VerificationCodeModification>
<EZVIZSecretKey>
  <!--opt, whether it supports capability of editing verification code for Hik-Connect-->
  <offlineStatus>
    <!--ro,dep,xs:string; it is valid when registerStatus values "false", device offline status, opt="secretKeyInvalid"-
invalid verification code-->
    </offlineStatus>
    <secretKey min="0" max="64"><!--opt, xs:string, verification code for Hik-Connect--></secretKey>
  </EZVIZSecretKey>
  <isSupportplatformAccess><!--opt, xs:boolean, capability of accessing the platform, whether it supports filtering IP
addresses that access to the platform--></isSupportplatformAccess>
  <isSupportIntegrate><!--opt, xs:boolean--></isSupportIntegrate>
  <isSupportIntelligentBoost><!--opt, xs:boolean, whether it supports bandwidth adaption--></
isSupportIntelligentBoost>
  <isSupportWebSocket><!--opt, xs:boolean--></isSupportWebSocket>
  <isSupportWebSocketS><!--opt, xs:boolean--></isSupportWebSocketS>
  <isSupportHostName>
    <!--optional, xs:boolean, whether it supports host name, related URI: /ISAPI/System/Network/interfaces/<ID>/
hostName/capabilities?format=json-->
    </isSupportHostName>
  <isSupportResourceStatistics><!--opt, xs:boolean, whether supports network resource information--></
isSupportResourceStatistics>
  <isSupportBandwidthLimit><!--opt, xs:boolean--></isSupportBandwidthLimit>
  <isSupportPOEPortsDisableServer><!--opt, xs:boolean--></isSupportPOEPortsDisableServer>
  <isSupportPOEConfiguration><!--opt, xs:boolean--></isSupportPOEConfiguration>
  <Adaption>
    <streamType opt="0,1,2,3,4,5,7,8,9,10">
      <!--stream types that support network self-adaptive during live view: 0-main stream, 1-sub-stream, 2-third stream,
3-virtual stream, 4-stream 5, 5-stream 6, 7-stream 7, 8-stream 8, ..., and so on-->
    </streamType>
    <isSupportPlayback><!--opt, xs: boolean, whether it supports self-adaptive during playback--></isSupportPlayback>
  </Adaption>
  <isSupportWifiProbe><!--opt, xs:boolean, whether it supports Wi-Fi probe configuration--></isSupportWifiProbe>
  <isSupportRFIDData><!--opt, xs:boolean, whether it supports configuration of RFID data collection--></
isSupportRFIDData>
  <isSupportwifiProbeSSID><!--opt, xs:boolean, whether it supports SSID configuration of Wi-Fi probe--></
isSupportwifiProbeSSID>
  <isSupportPOEPortsDisableAdaptiveServer><!--opt, xs:boolean--></isSupportPOEPortsDisableAdaptiveServer>
  <isSupportGetLinkSocketIP><!--opt,xs:boolean, whether it supports get the current linked SocketIP--></
isSupportGetLinkSocketIP>
```

```
<isSupportTrafficMonitor><!--opt, xs:boolean, whether it supports traffic monitoring of 4G network card--> </isSupportTrafficMonitor>
<isSupportWiredandWirelessTrafficMonitor><!--opt, xs:boolean, whether it supports monitoring traffic of wired NIC and wireless NIC--> </isSupportWiredandWirelessTrafficMonitor>
</NetworkCap>
```

XML_NetworkInterface

NetworkInterface message in XML format

```
<NetworkInterface version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:string--></id>
  <IPAddress/><!--req-->
  <Wireless/><!--opt-->
  <Discovery/><!--opt-->
  <Link/><!--opt-->
  <defaultConnection><!--opt, xs:boolean, default network connection, this node is required when the device has multiple network interfaces--></defaultConnection>
  <ActiveMulticast>
    <enabled><!--req, xs:boolean--></enabled>
    <streamID opt="main"><!--req, xs:string--></streamID>
    <ipV4Address><!--opt, xs:string--></ipV4Address>
    <ipV6Address><!--opt, xs:string--></ipV6Address>
    <port min="" max=""><!--opt, xs:integer--></port>
  </ActiveMulticast>
  <macAddress min="" max=""><!--opt, xs:string--></macAddress>
  <EthernetPortList size="4"><!--opt, network interface information-->
    <EthernetPort><!--opt-->
      <id><!--req, xs: integer, min="1",max="4"--></id>
      <MACAddress><!--req, xs:string--></MACAddress>
      <status><!--opt, xs:string; opt="connected, disconnect"--></status>
      <speed><!--req, xs:integer, "10, 100, 1000,10000"--></speed>
    </EthernetPort>
  </EthernetPortList>
</NetworkInterface>
```

XML_NetworkInterfaceList

NetworkInterfaceList message in XML format

```
<NetworkInterfaceList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <NetworkInterface/><!--opt, see details in the message of XML_NetworkInterface-->
</NetworkInterfaceList>
```

See Also

XML_NetworkInterface

XML_NetWorkMode

NetWorkMode message in XML format

```
<NetWorkMode version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <workMode><!--opt, xs:string, working mode: "close", "wifi"-Wi-Fi configuration, "wifiAp"-Wi-Fi access point (Wi-Fi
server) configuration--></workMode>
</NetWorkModeParam>
```

XML_NoiseReduce

NoiseReduce message in XML format

```
<NoiseReduce version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <mode><!--req, xs: string, 3D DNR mode: "close, general, advanced"--></mode>
  <GeneralMode><!--dep, this node is valid only when <mode> is "general"-->
    <generalLevel><!--req, xs: integer--></generalLevel>
  </GeneralMode>
  <AdvancedMode><!--dep, this node is valid only when <mode> is "advanced"-->
    <FrameNoiseReduceLevel><!--req, xs: integer--></FrameNoiseReduceLevel>
    <InterFrameNoiseReduceLevel><!--req, xs: integer--></InterFrameNoiseReduceLevel>
  </AdvancedMode>
</NoiseReduce>
```

XML_NTPServer

NTPServer message in XML format

```
<NTPServer version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:string, NTP server ID--></id>
  <addressingFormatType>
    <!--req, xs:string, which field will be used to locate the NTP server: "ipaddress, hostname"-->
  </addressingFormatType>
  <hostName><!--dep, xs:string--></hostName>
  <ipAddress><!--dep, xs:string--></ipAddress>
  <ipv6Address><!--dep, xs:string--></ipv6Address>
  <portNo><!--opt, xs:integer--></portNo>
  <synchronizeInterval>
    <!--opt, xs:integer, NTP time synchronization interval, unit: minute-->
  </synchronizeInterval>
</NTPServer>
```

XML_NTPServerList

NTPServerList message in XML format

```
<NTPServerList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <NTPServer/><!--opt, see details in the message of XML_NTPServer-->
</NTPServerList>
```

See Also

XML_NTPServer

XML_NTPTestDescription

NTPTestDescription message in XML format

```
<NTPTestDescription version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <addressingFormatType>
    <!--req, xs:string, "ipaddress,hostname"-->
  </addressingFormatType>
  <hostName><!--dep, xs:string--></hostName>
  <ipAddress><!--dep, xs:string--></ipAddress>
  <ipv6Address><!--dep, xs:string--></ipv6Address>
  <portNo><!--req, xs:integer--></portNo>
</NTPTestDescription>
```

XML_NTPTestResult

NTPTestResult message in XML format

```
<NTPTestResult version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <errorDescription><!--req, xs:string--></errorDescription>
</NTPTestResult>
```

XML_OnlineUpgradeCap

OnlineUpgradeCap message in XML format

```
<OnlineUpgradeCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <firmwareNum max="" />
  <!--req, the number of online upgrade packages, including full package and incremental package. Full package is
  used for upgrading whole firmware, while incremental package is used for upgrade certain unit, such as openssl
  library-->
  <firmwareCode max="" />
  <!--req, the maximum length of firmware code-->
  <firmwareVersion max="" />
  <!--req, the maximum length of version-->
  <firmwareCodeNumOnce max="" />
  <!--req, the maximum number of firmware codes can be obtained each time-->
  <upgradePercent min="" max="" />
  <!--req-->
```

```
<Version>
  <!--req, upgrade package version information-->
  <newVersion max="" />
  <!--req-->
  <changeLog max="" />
  <!--req-->
</Version>
<DeviceParameter>
  <!--opt, online upgrade parameters-->
  <isSupportAutoDownloadPackage>
    <!--opt,xs:boolean,"true,false", whether supports automatic download of upgrade package-->
  </isSupportAutoDownloadPackage>
  <notSupportAutoUpgrade>
    <!--opt,xs:boolean,"true,false", whether not support automatic download of upgrade package and automatic
upgrade-->
  </notSupportAutoUpgrade>
  <isSupportTimingUpgrade>
    <!--opt,xs:boolean,"true,false", whether supports scheduled upgrade-->
  </isSupportTimingUpgrade>
</DeviceParameter>
<ManualDownloadPackage>
  <!--opt, manually download upgrade package-->
  <supportOperation opt="start,cancel,pause,resume" />
  <!--opt, supported operations: "start,cancel,pause,resume"-->
</ManualDownloadPackage>
<isSupportIgnoreCurrentVersion>
  <!--opt, xs:boolean, "true,false", whether supports ignoring current version-->
</isSupportIgnoreCurrentVersion>
</OnlineUpgradeCap>
```

XML_OnlineUpgradeStatus

OnlineUpgradeStatus message in XML format

```
<OnlineUpgradeStatus version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <status>
    <!--ro, req, xs:string,
"notUpgrade,upgrading,successful,languageMismatch,writeFlashError,packageTypeMismatch,packageVersionMismatch,
h,netUnreachable,unknownError"-->
  </status>
  <percent><!-- ro, req, xs:integer "0-100" --></percent>
</OnlineUpgradeStatus>
```

XML_OnlineUpgradeVersion

OnlineUpgradeVersion message in XML format

```
<OnlineUpgradeVersion version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <newVersionAvailable><!--ro,req,xs:boolean, whether there is new verion of upgrade package--></
```

```
newVersionAvailable>
  <newVersion><!--ro, dep,xs:string, new version No--></newVersion>
  <changeLog><!--ro, dep,xs:string, update content of new version--></changeLog>
</OnlineUpgradeVersion>
```

XML_OnlineUpgradeServer

OnlineUpgradeServer message in XML format

```
<OnlineUpgradeServer version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <connectStatus><!--ro, req xs:boolean, online upgrade server connection status--></connectStatus>
</OnlineUpgradeServer>
```

XML_Palettes

Palettes message in XML format

```
<Palettes version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <mode><!--opt, xs:string,
  "WhiteHot,BlackHot,Fusion1,Rainbow,Fusion2,Ironbow1,Ironbow2,Sepia,Color1,Color2,IceFire,Rain,RedHot,GreenHot,
  DeepBlue,Color3"--></mode>
  <ColorateTarget><!--dep, colorate target, it is valid when mode is "WhiteHot"-->
    <ColorateTargetModelList>
      <ColorateTargetMode>
        <id><!--req, xs:integer, serial number, starts from 1--></id>
        <mode>
          <!--req, xs:string, colorate target mode; colorateHotAreae-colorate area with temperature high than configured
          threshold, colorateIntervalArea-colorate area with temperature between the configured threshold, colorateColdArea-
          colorate area with temperature lower than the configured threshold-->
        </mode>
        <enabled><!--req, xs:bool, "true,false"--></enabled>
        <TemperatureLimit><!--req, temperature limit-->
          <minTemperature><!--dep, it is valid when mode is "colorateHotAreae"/"colorateIntervalArea", xs: float--></
          minTemperature>
          <maxTemperature><!--dep, it is valid when mode is "colorateColdArea"/"colorateIntervalArea", xs: float--></
          maxTemperature>
        </TemperatureLimit>
        <Color><!--req, area color-->
          <R><!--req, xs:integer--></R>
          <G><!--req, xs:integer--></G>
          <B><!--req, xs:integer--></B>
        </Color>
      </ColorateTargetMode>
    </ColorateTargetModelList>
  </ColorateTarget>
</Palettes>
```


XML_ParkAction

ParkAction message in XML format

```
<ParkAction version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--required, xs: boolean, whether to enable parking--></enabled>
  <Parktime><!--req, xs:integer, parking time, unit: second--></Parktime>
  <Action><!--parking actions-->
    <ActionType opt="autoscan,framescan,randomscan,panoramascan,patrol,pattern,preset,areaScan"><!--required,
xs:string, action type: "atuoscan,framescan,randomscan,panoramascan,patrol,pattern,preset,areaScan"--></
ActionType>
    <ActionNum><!--required, xs:integer, parking No. This node will be used when <ActionType> is "patrol", "pattern",
or "preset". For other parking action types, this node should be set to 0--></ActionNum>
  </Action>
</ParkAction>
```

XML_ParkingParam

XML message about rule parameters of parking detection

```
<ParkingParam><!--dep-->
  <durationTime>
    <!--req, xs:integer, duration time, from 5 seconds to 100 seconds, default value: 5s-->
  </durationTime>
</ParkingParam>
```

XML_pingTestDescription

pingTestDescription message in XML format

```
<pingTestDescription version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ipAddress><!--dep, xs:string--></ipAddress>
</pingTestDescription>
```

XML_pingTestResult

pingTestResult message in XML format

```
<pingTestResult version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <status><!--req, xs:string, "used, not used"--></status>
</pingTestResult>
```

XML_Probe

Probe message in XML format.

```
<?xml version="1.0" encoding="UTF-8"?>
<Probe>
  <!--the UUID will be returned in the device response message for matching, if not matched, it will not be handled.-->
  <Uuid>8d2091bc-1dd2-11b2-807b-8ce748cf9334</Uuid>
  <Types>inquiry</Types>
</Probe>
```

XML_ProxyDevicesInfoCondition

Message about search condition of device parameters (all digital channels) in XML format

```
<ProxyDevicesInfoCondition version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <searchResultPosition><!--required, xs:integer--></searchResultPosition>
  <maxResults><!--required, xs:integer--></maxResults>
</ProxyDevicesInfoCondition>
```

XML_ProxyDevicesInfoResult

Message about search result of device parameters (all digital channels) in XML format

```
<ProxyDevicesInfoResult version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <responseStatus><!--required, xs:string, "true,false", search status--></responseStatus>
  <responseStatusStrg><!--required, xs:string, "OK,NO MATCHES,MORE", search status description--></responseStatusStrg>
  <numOfMatches><!--required, xs:integer, number of returned records--></numOfMatches>
  <totalMatches><!--required, xs:integer, number of matched records--></totalMatches>
  <DeviceInfoList><!--optional, list of task information-->
    <!--refer to XML_DeviceInfo-->
    <DeviceInfo>
      <deviceName><!--required, xs:string--></deviceName>
      <deviceId><!--required, read-only, xs:string, uuid--></deviceId>
      <deviceDescription><!--optional, xs:string--></deviceDescription>
      <deviceLocation><!--optional, xs:string--></deviceLocation>
      <deviceStatus><!--optional, read-only, xs:string,opt="normal,abnormal"--></deviceStatus>
      <DetailAbnormalStatus><!--dependent, this field is valid only when deviceStatus is "abnormal"-->
        <hardDiskFull><!--optional, read-only, xs:boolean, "true,false", whether disk is full--></hardDiskFull>
        <hardDiskError><!--optional, read-only, xs:boolean, "true,false", whether disk has error--></hardDiskError>
        <ethernetBroken><!--optional, read-only, xs:boolean, "true,false", whether it is disconnected--></ethernetBroken>
        <ipaddrConflict><!--optional, read-only, xs:boolean, "true,false", whether IP address conflicts--></ipaddrConflict>
        <illegalAccess><!--optional, read-only, xs:boolean, "true,false", whether it is illegal access--></illegalAccess>
        <recordError><!--read-only, optional, xs:boolean, "true,false", whether video exception occurred--></recordError>
        <raidLogicDiskError><!--read-only, optional, xs:boolean, "true,false", whether raid logic disk exception occurred--></raidLogicDiskError>
      </DeviceInfo>
    </DeviceInfoList>
</ProxyDevicesInfoResult>
```

```

    <spareWorkDeviceError><!--read-only, optional, xs:boolean, "true,false", whether spare device exception
occurred--></spareWorkDeviceError>
  </DetailAbnormalStatus>
  <systemContact><!--optional, xs:string--></systemContact>
  <model><!--required, read-only, xs:string--></model>
  <serialNumber><!--required, read-only, xs:string--></serialNumber>
  <macAddress><!--required, read-only, xs:string--></macAddress>
  <firmwareVersion><!--required, read-only, xs:string--></firmwareVersion>
  <firmwareReleasedDate><!--optional, read-only, xs:string--></firmwareReleasedDate>
  <encoderVersion><!--optional, read-only, xs:string--></encoderVersion>
  <encoderReleasedDate><!--optional, read-only, xs:string--></encoderReleasedDate>
  <bootVersion><!--optional, read-only, xs:string--></bootVersion>
  <bootReleasedDate><!--optional, read-only, xs:string--></bootReleasedDate>
  <hardwareVersion><!--optional, read-only, xs:string--></hardwareVersion>
  <decoderVersion><!--optional, read-only, xs:string--></decoderVersion>
  <decoderReleasedDate><!--optional, read-only, xs:string--></decoderReleasedDate>
  <softwareVersion><!--optional, read-only, xs:string, software version--></softwareVersion>
  <capacity><!--optional, read-only, xs:integer, device capacity (MB)--></capacity>
  <usedCapacity><!--optional, read-only, xs:integer, used capacity (MB)--></usedCapacity>
  <deviceType><!--required, read-only, xs:string, "IPCamera,IPDome,DVR,HybirdNVR,NVR,DVS,IPZoom,CVR"--></
deviceType>
  <telecontrolID><!--optional, xs:integer, "1-255"--></telecontrolID>
  <supportBeep><!--optional, xs:boolean--></supportBeep>
  <supportVideoLoss><!--optional, xs:boolean--></supportVideoLoss>
  <firmwareVersionInfo><!--read-only, optional, xs:string--></firmwareVersionInfo>
  <actualFloorNum><!--optional, xs:integer, "1-128"--></actualFloorNum>
  <radarVersion><!--optional, xs:string--></radarVersion>
  <cameraModuleVersion><!--read-only, xs:string--></cameraModuleVersion>
  <mainversion><!--optional, xs:integer, "1-255"--></mainversion>
  <subversion><!--optional, xs:integer, "1-255"--></subversion>
  <upgradeversion><!--optional, xs:integer, "1-255"--></upgradeversion>
  <customizeversion><!--optional, xs:integer, "1-255"--></customizeversion>
  <companyName><!--optional, xs:string, manufacturer name--></companyName>
  <copyright><!--optional, xs:string, copyright information--></copyright>
  <systemName><!--optional, xs:string, storage system, "storageManagement,distributedStorageManagement",
"storageManagement"-storage management system, "distributedStorageManagement"-distributed storage
management system--></systemName>
  <systemStatus><!--optional, system status, xs:string, "configured,unConfigured", "configured"-system has been
configured, "unConfigured"-system has not been configured--></systemStatus>
  <isLeaderDevice><!--optional, xs:boolean, whether it is the corresponding device of the resource IP--></
isLeaderDevice>
  <clusterVersion><!--dependent, xs:string, system cluster version, this field is valid only when isLeaderDevice is
"true"--></clusterVersion>
  <manufacturer><!--optional, xs:string,manufacturer information used for differentiating OEM devices--></
manufacturer>
  <customizedInfo><!--optional, xs:string, custom project information--></customizedInfo>
  <channel><!--optional, xs:integer, when IPC device information is obtained via NVR, this node refers to the channel
of the NVR--></channel>
</DeviceInfo>
</DeviceInfoList>
</ProxyDevicesInfoResult>

```

See Also

XML_DeviceInfo

XML_port

port message in XML format

```
<port version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id/><!--req, xs: string, ID-->
  <enabled/><!--req, xs: boolean-->
  <internalPort/><!--req, xs: string, "http,admin,rtsp,https,WebSocket,WebSocket,SDK_OVER_TLS,SRTP..."-->
  <externalPort/><!--req, xs:integer-->
</port>
```

XML_portStatus

portStatus message in XML format

```
<portStatus version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id/><!--req, xs: string, ID-->
  <enabled/><!--req-->
  <internalPort/><!--req, xs: string, "http,admin,rtsp,https,WebSocket,WebSocket,SDK_OVER_TLS,SRTP..."-->
  <externalPort/><!--req, xs: integer-->
  <status/><!--req, xs: string, "inactive, active, conflict, ..."-->
</portStatus>
```

XML_ports

ports message in XML format

```
<ports version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled/><!--req-->
  <mapmode><!--req, xs: string, "auto,manual"--></mapmode>
  <natRouterLanAddr><!--opt-->
    <ipVersion><!--req, xs: string, "v4,v6,dual"--></ipVersion>
    <ipAddress><!--dep, xs: string--></ipAddress>
    <ipv6Address><!--dep, xs: string--></ipv6Address>
  </natRouterLanAddr>
  <portList><!--req-->
    <port/><!--see details in the message XML_port-->
  </portList>
  <natType><!--req, xs: string, "manual, auto"--></natType>
</ports>
```

See Also

XML_port

XML_PortMapParam

PortMapParam message in XML format

```
<PortMapParam version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <userip><!--IP address of user terminal, or IP address of PoEO or Eth0--></userip>
</PortMapParam>
```

XML_PortMapParamRet

PortMapParamRet message in XML format

```
<PortMapParamRet version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <mapResult><!--whether port mapping is required: 0-no, 1-yes, 2-unknown--></mapResult>
  <MapPortGop>
    <MapPort>
      <mapPortIdx><!--index No. of mapping ports--></mapPortIdx>
      <mapPortName><!--mapping port name--></mapPortName>
      <mapPortValue><!--mapping port number--></mapPortValue>
    </MapPort>
  </MapPortGop>
</PortMapParamRet>
```

XML_portsStatus

portsStatus message in XML format.

```
<portsStatus version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled/><!--req-->
  <natRouterLanAddr><!--req-->
    <ipVersion><!--req, xs: string, "v4,v6,dual"--></ipVersion>
    <ipAddress><!--dep, xs: string--></ipAddress>
    <ipv6Address><!--dep, xs: string--></ipv6Address>
  </natRouterLanAddr>
  <natRouterWanAddr><!--req-->
    <ipVersion><!--req, xs: string, "v4,v6,dual"--></ipVersion>
    <ipAddress><!--dep, xs: string--></ipAddress>
    <ipv6Address><!--dep, xs: string--></ipv6Address>
  </natRouterWanAddr>
  <portStatusList><!--req-->
    <portStatus/><!--req, see details in the message XML_portStatus-->
  </portStatusList>
</portsStatus>
```

See Also

XML_portStatus

XML_powerLineFrequency

powerLineFrequency message in XML format

```
<powerLineFrequency version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <powerLineFrequencyMode/><!--opt, xs: string, standard mode: "50hz, 60hz"-->
</powerLineFrequency>
```

XML_PreviewSwitch

PreviewSwitch message in XML format

```
<PreviewSwitch version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <DisplayWindowList size="64">
    <!--req, if the number of channels is larger or equal to 64, 64 channels should be applied by a group for one time;
otherwise, apply the maximum number of channels-->
    <DisplayWindow>
      <id><!--req, xs: string, actual window No., which equals to id × groupNo--></id>
      <displayChannelNo><!--req, xs: string, 0-not display, 1 to 64-displayed channel No., min="0" max="64"--></
displayChannelNo>
    </DisplayWindow>
  </DisplayWindowList>
  <previewFrameNo>
    <!--req, xs: string, number of live view windows: 1, 4, 6, 8, 9, 25, 32, 36, auto1-custom window division 1, auto2-
custome window division 2, auto3-custome window division 3, auto4-custome window division 4-->
  </previewFrameNo>
  <sound><!--req, xs: boolean, whether to turn on audio during live view: true=yes, false=no--></sound>
  <switchTime><!--req, xs: string, switching interval: 0, 5, 10, 20, 30, 60, 120, and 300, unit:s--></switchTime>
  <sameSource>
    <!--dep, xs: boolean, whether the output is homologous: true=yes, false=no, this node is valid only when
VideoOutType is "noSameSourceVGA1", "noSameSourceVGA2", "noSameSourceVGA3", and "noSameSourceVGA4"-->
  </sameSource>
</PreviewSwitch>
```

XML_PreviewSwitchVideoOutCap

PreviewSwitchVideoOutCap message in XML format

```
<PreviewSwitchVideoOutCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <DisplayWindowList size="64">
    <DisplayWindow>
      <id><!--req, xs: string, window No., which equals to id × groupNo--></id>
      <displayChannelNo min="0" max="64"><!--req, xs: string, 0-not display, 1 to 64-displayed channel No.--></
```

```
displayChannelNo>
</DisplayWindow>
</DisplayWindowList>
<previewFrameNo opt="1,4,6,8,9,25,32,36,auto1,auto2,auto3,auto4">
  <!--req, xs: string, number of live view windows, auto1-custom window division 1, auto2-custome window division
2, auto3-custome window division 3, auto4-custome window division 4-->
</previewFrameNo>
<sound><!--req, xs: boolean, whether to turn on audio during live view: true=yes, false=no--></sound>
<switchTime opt="0,5,10,20,30,60,120,300"><!--req, xs: string, switching interval, unit: s--></switchTime>
<sameSource>
  <!--dep, xs: boolean, whether the output is homologous: true=yes, false=no, this node is valid only when
VideoOutType is "noSameSourceVGA1", "noSameSourceVGA2", "noSameSourceVGA3", and "noSameSourceVGA4"-->
</sameSource>
<supportGetByPreviewNum>
  <!--opt, whether supports setting number of live view windows in the URL, true=yes-->
</supportGetByPreviewNum>
</PreviewSwitchVideoOutCap>
```

XML_PrivacyMask

PrivacyMask message in XML format

```
<PrivacyMask version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--req, xs: boolean--></enabled>
  <normalizedScreenSize><!--opt-->
    <normalizedScreenWidth><!--req, xs: integer--></normalizedScreenWidth>
    <normalizedScreenHeight><!--req, xs: integer--></normalizedScreenHeight>
  </normalizedScreenSize>
  <PrivacyMaskRegionList size=8/><!--opt-->
    <regionType><!--opt, xs: string, "quadrilateral"--></regionType>
  </PrivacyMask>
```

XML_PrivacyMaskCap

PrivacyMaskCap message in XML format

```
<PrivacyMaskCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <maskTypeDescriptor>
    <!--opt, xs: string, privacy mask type: "gray,red,yellow,blue,orange,green,transparent,half-transparent,mosaic"-->
  </maskTypeDescriptor>
  <minZoomdoorlimit><!--opt, xs: integer></minZoomdoorlimit>
  <maxZoomdoorlimit><!--opt, xs: integer></maxZoomdoorlimit>
  <videoPrivacyType opt="privacyMask,privacyCover">
    <!--opt, xs: string, "privacyMask"-video tampering, "privacyCover"-privacy mask-->
  </videoPrivacyType>
</PrivacyMaskCap>
```

XML_PrivacyMaskRegion

PrivacyMaskRegion message in XML format

```
<PrivacyMaskRegion version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs: integer--></id>
  <enabled><!--req, xs: boolean--></enabled>
  <RegionCoordinatesList><!--req-->
    <RegionCoordinates><!--req-->
      <positionX><!--req, xs: integer; coordinate--></positionX>
      <positionY><!--req, xs: integer; coordinate--></positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
  <privacymaskName><!--opt, xs: string--></privacymaskName>
  <maskType>
    <!--opt, xs:string "gray,red,yellow,blue,orange,green,transparent,half-transparent,mosaic,black"-->
  </maskType>
  <zoomdoorlimit><!--opt, xs: integer, the value is between 10 and 1000--></zoomdoorlimit>
</PrivacyMaskRegion>
```

XML_PrivacyMaskRegionList

PrivacyMaskRegionList message in XML format

```
<PrivacyMaskRegionList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <PrivacyMaskRegion/><!--opt, refer to the message XML_PrivacyMaskRegion for details-->
</PrivacyMaskRegionList>
```

See Also

XML_PrivacyMaskRegion

XML_PTZAux

PTZAux message in XML format

```
<?xml version="1.0" encoding="utf-8"?>
<PTZAux version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req,xs:string,id--></id>
  <type><!--req, ro, xs:string,"LIGHT,WIPER,FAN,HEATER",auxiliary type: light, wiper, fan, heater--></type>
  <status><!--req, xs:string,"on,off", auxiliary status: turned on, turned off--></status>
</PTZAux>
```

XML_PTZAuxList

PTZAuxList message in XML format


```
<?xml version="1.0" encoding="utf-8"?>
<PTZAuxList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <PTZAux><!--list-->
    <id><!--req, xs:string, id--></id>
    <type><!--req, ro, xs:string, "LIGHT,WIPER,FAN,HEATER", auxiliary type: light, wiper, fan, heater--></type>
    <status><!--req, xs:string, "on,off", auxiliary status: turned on, turned off--></status>
  </PTZAux>
</PTZAuxList>
```

XML_PTZChannel

PTZChannel message in XML format

```
<PTZChanel version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:integer--></id>
  <enabled><!--ro, req, xs:boolean--></enabled>
  <serialNumber><!--req, xs:integer--></serialNumber>
  <videoInputID><!--req, xs:integer--></videoInputID>
  <panMaxSpeed><!--ro, opt, xs:integer, degrees/sec--></panMaxSpeed>
  <tiltMaxSpeed><!--ro, opt, xs:integer, degrees/sec--></tiltMaxSpeed>
  <presetSpeed><!--opt, xs:integer, 1..8--></presetSpeed>
  <autoPatrolSpeed><!--opt, xs:integer, 0..100--></autoPatrolSpeed>
  <keyBoardControlSpeed><!--opt, xs:integer, 0..100--></keyBoardControlSpeed>
  <controlProtocol><!--opt, xs:string, "pelco-d, modbus-RTU, modbus-ASCII"--></controlProtocol>
  <controlAddress><!--opt-->
    <enabled><!--req, xs:boolean--></enabled>
    <Address><!--opt, xs:string, 1-255--></Address>
  </controlAddress>
  <defaultPresetID><!--opt, xs:string, id--></defaultPresetID>
  <PTZRs485Para><!--opt-->
    <baudRate><!--req, xs:integer--></baudRate>
    <dataBits><!--req, xs:integer--></dataBits>
    <parityType><!--req, xs:string, "none, even, odd, mark, space"--></parityType>
    <stopBits><!--req, xs:string, "1, 1.5, 2"--></stopBits>
    <flowCtrl><!--req, xs:string, "none, software, hardware"--></flowCtrl>
  </PTZRs485Para>
  <manualControlSpeed>
    <!--opt, xs:string, "pedestrian, nonMotorVehicle, motorVehicle, selfadaptive, compatible"-->
  </manualControlSpeed>
</PTZChannel>
```

XML_PTZChanelCap

XML message about PTZ channel capability

```
<PTZChanelCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <AbsolutePanTiltPositionSpace><!--optional-->
    <XRange/><!--required-->
```

```
<YRange/><!--required-->
</AbsolutePanTiltPositionSpace>
<AbsoluteZoomPositionSpace><!--optional-->
  <ZRange/><!--required-->
</AbsoluteZoomPositionSpace>
<RelativePanTiltSpace><!--optional-->
  <XRange/><!--required-->
  <YRange/><!--required-->
</RelativePanTiltSpace>
<RelativeZoomSpace><!--optional-->
  <ZRange/><!--required-->
</RelativeZoomSpace>
<ContinuousPanTiltSpace><!--optional-->
  <XRange/><!--required-->
  <YRange/><!--required-->
</ContinuousPanTiltSpace>
<ContinuousZoomSpace><!--optional-->
  <ZRange/><!--required-->
</ContinuousZoomSpace>
<MomentaryPanTiltSpace><!--optional-->
  <XRange/><!--required-->
  <YRange/><!--required-->
</MomentaryPanTiltSpace>
<MomentaryZoomSpace><!--optional-->
  <ZRange/><!--required-->
</MomentaryZoomSpace>
<homePostionSupport><!--required, xs:boolean--></homePostionSupport>
<maxPresetNum>
  <!--required, xs:integer, max. supported preset number-->
</maxPresetNum>
<maxPatrolNum>
  <!--required,xs:integer, max. supported patrol number-->
</maxPatrolNum>
<maxPatternNum>
  <!--required,xs:integer, max. supported pattern number-->
</maxPatternNum>
<maxLimitesNum>
  <!--required,xs:integer, max. supported limit number-->
</maxLimitesNum>
<maxTimeTaskNum>
  <!--required,xs:integer, max. supported timing task number-->
</maxTimeTaskNum>
<serialNumber min="1" max="4">
  <!--configuration capability of RS-485 serial port supported by current channel-->
</serialNumber>
<controlProtocol>
  <!--optional, xs:string,"pelco-d,modbus-RTU,modbus-ASCII", supported PTZ control protocol-->
</controlProtocol>
<controlAddress>
  <!--optional, xs:string, 0-255, address-->
</controlAddress>
<PTZRs485Para>
```

```
<!--optional, PTZ RS485 parameters capability-->
<baudRate>
  <!--required, xs:integer, baud rate-->
</baudRate>
<dataBits>
  <!--required, xs:integer, data bit-->
</dataBits>
<parityType>
  <!--required, xs:string, "none,even,odd,mark,space", verification type-->
</parityType>
<stopBits>
  <!--required, xs:string, "1,1.5,2", stop bit-->
</stopBits>
<flowCtrl>
  <!--required, xs:string, "none, software, hardware", stream control type-->
</flowCtrl>
</PTZRs485Para>
<PresetNameCap>
  <!--optional, preset name capability-->
  <presetNameSupport>
    <!--optional,xs:boolean, whether to support preset name?-->
  </presetNameSupport>
  <maxPresetNameLen>
    <!--dependent,xs:integer, preset name length-->
  </maxPresetNameLen>
  <specialNo/>
  <!--dependent, special preset-->
</PresetNameCap>
<isSupportPosition3D>
  <!--optional, xs:boolean, whether to support 3D position-->
</isSupportPosition3D>
<isSupportManualTrack>
  <!--optional,xs:boolean, whether to support manual tracking position(NET_DVR_PTZ_MANUALTRACE)?-->
</isSupportManualTrack>
<manualControlSpeed opt="compatible,pedestrian,nonMotorVehicle,motorVehicle,selfadaptive">
  <!--optional,xs:string, manual control speed: "compatible"-compatible mode, "pedestrian"-pedestrian,
"nonMotorVehicle"-non-motor vehicle, "motorVehicle"-motor vehicle, "selfadaptive"-self adaptive-->
</manualControlSpeed>
<isSupportPtzlimiteds>
  <!--optional,xs:boolean, whether to support PTZ limitation-->
</isSupportPtzlimiteds>
<ParkAction/><!--see details in the message XML_ParkAction-->
<TimeTaskList/><!--see details in the message XML_TimeTaskList-->
<Thermometry>
  <maxThermometryPresetNum>
    <!--optional, xs:integer-->
  </maxThermometryPresetNum>
</Thermometry>
<isSupportPtzEagleFocusing>
  <!--optional, xs:boolean-->
</isSupportPtzEagleFocusing>
<TrackingRatio/>
```

```
<!--optional, tracking zoom ratio-->
<coefficient min="1" max="10" default="5">
  <!--optional, xs:integer, zoom ratio-->
</coefficient>
</TrackingRatio>
<TrackInitPosition>
  <!--optional, tracking initial position-->
  <slaveCameraID>
    <!--optional, xs:integer, slave camera ID, 1..4 -->
  </slaveCameraID>
</TrackInitPosition>
<isSupportAbsoluteEx><!--optional,xs:boolean, whether to support extended capability for PTZ absolute position--></
isSupportAbsoluteEx>
<isSupportCruise><!--optional,xs:boolean, whether to support auto-switch mode--></isSupportCruise>
<isSupportAreaScan><!--optional,xs:boolean, whether to support area scan--></isSupportAreaScan>
<isSupportFaceSnap3D><!--optional, xs: boolean--></isSupportFaceSnap3D>
<ManualPTZLockCap><!--optional, PTZ manual locking capability-->
  <lockStatus opt="lock,unlock"><!--optional, xs:string, PTZ locking status, the default status is locked--></lockStatus>
  <unlockTime><!--optional, xs:integer, remaining unlocking time duration (the automatic locking time duration is
300 s in the unlocked status*--></unlockTime>
</ManualPTZLockCap>
<isSupportOnepushSynchronizeFOV>
  <!--optional, xs:boolean, whether supports one-touch synchronize FOV, return true for supports, and no return
indicates not support-->
</isSupportOnepushSynchronizeFOV>
<isSupportLensCorrection>
  <!--optional, xs:boolean, whether supports lens correction, return true for supports, and no return indicates not
support-->
</isSupportLensCorrection>
<isSupportPTZTrackStatus>
  <!--optional, xs:boolean, whether to support getting PTZ tracking linkage status-->
</isSupportPTZTrackStatus>
<ptzctrlSupportSerialNumber opt="1,3"><!--optional, xs:string, RS-485 serial port range supported in PTZ control
mode--></ptzctrlSupportSerialNumber>
<LFPositionCap><!--optional, capability of calibrating position for smart linkage-->
  <elevation><!--optional, xs:integer, range: [-900,2700]--></elevation>
  <azimuth><!--optional, xs:integer, range: [0,3600]--></azimuth>
  <absoluteZoom><!--optional, xs:integer, range: [1,1000]--></absoluteZoom>
</LFPositionCap>
<pqrsZoom min="" max=""/><!--optional, xs: integer, zooming coordinates of Sony zoom camera module-->
<mnstFocus min="" max=""/><!--optional, xs: integer, focus coordinates of Sony zoom camera module-->
<isSupportPTZSave>
  <!--optional, xs: boolean, whether supports saving the current PTZ position information-->
</isSupportPTZSave>
<isSupportPTZSaveGet>
  <!--optional, xs:boolean, whether to support saving the PTZ position information of the current channel by GET
method-->
</isSupportPTZSaveGet>
<isSupportAutoGotoCfg>
  <!--optional, xs: boolean, whether supports automatically restoring to saved PTZ position: "true"-yes, this node is
not returned-no-->
<isSupportAutoGotoCfg>
```

```
<lockTime><!--optional, xs:integer, PTZ locking time, unit: second--></lockTime>
<isSupportPTZCtrlStatus><!--optional, xs:boolean, whether it supports getting the absolute PTZ position of the
current channel: "true"-yes, "false"-no. If this node is not returned, it does not necessarily indicate that the device
does not support this function--></isSupportPTZCtrlStatus>
<isSupportInitializationCtrl><!--optional, xs:boolean, whether it supports electric PTZ self-checking and control--></
isSupportInitializationCtrl>
<isSupportComponentCtrl><!--optional, xs:boolean, whether it supports PTZ component control--></
isSupportComponentCtrl>
<DoubleMirror><!--optional-->
  <enabled opt="true,fasle" def="false"><!--required, xs:boolean--></enabled>
</DoubleMirror>
<MoveAutoTracking><!--optional, moving object tracking-->
  <enabled opt="true,false"><!--required, xs:boolean, whether to enable this function--></enabled>
</MoveAutoTracking>
<isSupportTrackingRatioCfg><!--optional, xs:boolean, whether it supports configuring tracking parameters--></
isSupportTrackingRatioCfg >
<isSupportGPSCtrl>
  <!--optional, xs:boolean, whether current channel supports controlling pan and tilt by GPS, if supports, this node is
returned and the value is true; if not, this node is not returned (related URI: /ISAPI/PTZCtrl/channels/<ID>/GPS?
format=json)-->
</isSupportGPSCtrl>
<isSupportGyroVerify>
  <!--optional, xs:boolean, whether device supports gyroscope attitude calibration (related URI: /ISAPI/PTZCtrl/
channels/<ID>/gyroVerify?format=json)-->
</isSupportGyroVerify>
</PTZChanelCap>
```

See Also

XML_ParkAction

XML_TimeTaskList

XML_PTZOSDDisplay

PTZOSDDisplay message in XML format

```
<PTZOSDDisplay version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <zoomlable><!--req, xs:string, "2sec, 5sec, 10sec, alwaysclose, alwaysopen"--></zoomlable>
  <azimuth><!--req, xs:string, "2sec, 5sec, 10sec, alwaysclose, alwaysopen"--></azimuth>
  <presetlable><!--req, xs:string, "2sec, 5sec, 10sec, alwaysclose, alwaysopen"--></presetlable>
  <actionStatusDisplayEnabled><!--opt, xs:boolean, display status or not--></actionStatusDisplayEnabled>
</PTZOSDDisplay>
```

XML_PTZPreset

XML message about preset parameters

```
<PTZPreset version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--required, xs:boolean, whether to enable preset configuration--></enabled>
```

```
<id><!--required, xs:string, preset No.--></id>
<presetName><!--required, xs:string, preset name--></presetName>
<AbsoluteHigh>
  <elevation><!--optional, xs:integer, tilting parameter, the value is between -900 and 2700--></elevation>
  <azimuth><!--optional, xs:integer, panning parameter, the value is between 0 and 3600--></azimuth>
  <absoluteZoom><!--optional, xs:integer, zooming parameter, the value is between 1 and 1000--></absoluteZoom>
</AbsoluteHigh>
</PTZPreset>
```

XML_PTZPresetList

PTZPresetList message in XML format.

```
<PTZPresetList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <PTZPreset/><!--opt, see details in
                        XML_PTZPreset
                        -->
</PTZPresetList>
```

XML_PTZStatus

PTZStatus message in XML format

```
<PTZStatus version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <AbsoluteHigh><!--high-accuracy positioning which is accurate to one decimal place-->
    <elevation>
      <!--opt, xs:integer, tilting parameter, the value is between -900 and 2700, which corresponds to the range from
-90.0 degree to 270.0 degree vertically-->
    </elevation>
    <azimuth>
      <!--opt, xs:integer, panning parameter, the value is between 0 and 3600, which corresponds to the range from 0.0
degree to 360.0 degree horizontally-->
    </azimuth>
    <absoluteZoom>
      <!--opt, xs:integer, zooming parameter, the value is between 0 and 1000, which corresponds to the focus range
from 0.0 to 100.0-->
    </absoluteZoom>
  </AbsoluteHigh>
</PTZStatus>
```

XML_PublicKey

PublicKey message in XML format.

```
<PublicKey version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <key><!--req, xs:string--></key>
</PublicKey>
```

XML_RacmCap

XML message about device storage capability

```
<?xml version="1.0" encoding="utf-8"?>
<RacmCap version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <isSupportZeroChan>
    <!--optional, xs:boolean, whether it supports channel-zero, "true"-yes, "false"-no-->
  </isSupportZeroChan>
  <inputProxyNums><!--optional, xs:integer, number of supported digital channels--></inputProxyNums>
  <eSATANums><!--optional, xs:integer, number of supported eSATAs--></eSATANums>
  <miniSASNums><!--optional, xs:integer, number of supported miniSATAs--></miniSASNums>
  <nasNums><!--optional, xs:integer, number of supported NASs--></nasNums>
  <ipSanNums><!--optional, xs:integer, number of supported IPSANs--></ipSanNums>
  <isSupportRaid>
    <!--optional, xs:boolean, whether it supports RAID, "true"-yes, "false"-no-->
  </isSupportRaid>
  <isSupportExtHdCfg>
    <!--optional, xs:boolean, whether it supports HDD advanced management, "true"-yes, "false"-no-->
  </isSupportExtHdCfg>
  <isSupportTransCode><!--optional, xs:boolean, whether it supports auto-switch, "true"-yes, "false"-no--></
isSupportTransCode>
  <isSupportIpclImport>
    <!--optional, xs:boolean, whether it supports importing configuration files to network camera, "true"-yes, "false"-
no-->
  </isSupportIpclImport>
  <NasMountType><!--optional-->
    <isNFSSupportAuthentication/><!--optional, xs:boolean, whether it supports NFS authentication, "true"-yes, "false"-
no-->
    <isCIFSSupportAuthentication/><!--optional, xs:boolean, whether it supports CIFS authentication, "true"-yes, "false"-
no-->
  </NasMountType>
  <isSupportIpcStreamType/><!--optional, xs:boolean-->
  <isSupportIOInputProxy/><!--optional, xs:boolean, whether it supports dynamic alarm input channel: "true"-yes,
"false"-no-->
  <isSupportIOOutputProxy/><!--optional, xs:boolean, whether it supports dynamic alarm output channel: "true"-yes,
"false"-no-->
  <isSupportPTZRs485Proxy/><!--optional, xs:boolean, whether it supports dynamic PTZ485 channel, "true"-yes,
"false"-no-->
  <isSupportSrcIDSearch/><!--optional, xs:boolean, whether it supports search by stream ID, "true"-yes, "false"-no-->
  <isSupportReversePlayback/><!--optional, xs:boolean, whether it supports reverse playback, "true"-yes, "false"-no-->
  <isSupportSMARTTest/><!--optional, xs:boolean, whether it supports HDD checking, "true"-yes, "false"-no-->
  <isSupportDownloadByTime/><!--optional, xs:boolean, whether it supports downloading by time, "true"-yes, "false"-
no-->
  <pictureSearchType
opt="AllEvent,CMR,MOTION,ALARM,EDR,ALARMANDMOTION,Command,pir,wlsensor,callhelp,facedetection,FieldDete
ction,scenechangedetection,LineDetection,regionEntrance,regionExiting,loitering,group,rapidMove,parking,unattende
dBaggage,attendedBaggage,vehicleDetection>manual>manualSnapshot,playSnapshot,allPic,evidence,illegalParking,cro
sslane,vehicleexist,lanchange,wrongdirection,congestion,turnround,parallelParking, pedestrian, construction,
roadblock, abandonedObject, trafficAccident, fogDetection, personQueueCounting,personQueueTime,
mixedTargetDetection, safetyHelmet, accessController, videoInterCom, GJD, Luminite, OPTEX, securityControlPanel,
```

```
playCellphone, vibrationDetection, ATMPanel, ATMSurround, ATMFace, ATMSafetyCabin"/>
  <!--optional, xs:string, picture search conditions-->
  <recordSearchType
opt="AllEvent,CMR,MOTION,ALARM,EDR,ALARMANDMOTION,Command,pir,wlsensor,callhelp,facedetection,FieldDete
ction,scenechangedetection,LineDetection,regionEntrance,regionExiting,loitering,group,rapidMove,parking,unattende
dBaggage,attendedBaggage,vehicleDetection>manual>manualSnapshot,playSnapshot,AudioDetection,playCellphone,ru
nning,failDown,violentMotion,retention,allPerimeterEvent, allBehaviorEvent, vibrationDetection, ATMPanel,
ATMSurround, ATMFace, ATMSafetyCabin"/>
  <!--optional, xs:string, video search conditions-->
  <isSupportActivatelpc/><!--optional, xs:boolean, whether it supports activating network camera, "true"-yes, "false"-
no-->
  <isSupportCheckIpcSecurity><!--optional-->
  <isSupportCheckPassword/><!--optional, xs:boolean, whether it supports password verification, "true"-yes, "false"-
no-->
  </isSupportCheckIpcSecurity>
  <isSupportMainAndSubRecord/>
  <!--optional, xs:boolean, whether it supports recording in main stream or sub-stream, "true"-yes, "false"-no-->
  <isSupportSyncIPCPassword>
  <!--optional, xs:boolean, whether it supports synchronizing network camera's password, "true"-yes, "false"-no-->
  </isSupportSyncIPCPassword>
  <isSupportTransferIPC>
  <!--optional, xs:boolean, whether it supports network camera passthrough function, "true"-yes, "false"-no-->
  </isSupportTransferIPC>
  <isSupportPOS><!--optional, xs:boolean, whether it supports POS, "true"-yes, "false"-no--></isSupportPOS>
  <isSupportPassBackBasicCfg>
  <!--optional, xs:boolean, whether it supports the ANR basic function of CVR, "true"-yes, "false"-no-->
  </isSupportPassBackBasicCfg>
  <PassBackTaskConfig><!--optional, task management function of ANR-->
  <isSupportAddTask><!--optional, xs:boolean, whether it supports adding task, "true"-yes, "false"-no--></
isSupportAddTask>
  <isSupportSearchTask><!--optional, xs:boolean, whether it supports task search, "true"-yes, "false"-no--></
isSupportSearchTask>
  <isSupportControlTask><!--optional, xs:boolean, whether it supports task control, "true"-yes, "false"-no--></
isSupportControlTask>
  <isSupportDeleteTask><!--optional, xs:boolean, whether it supports deleting task, "true"-yes, "false"-no--></
isSupportDeleteTask>
  </PassBackTaskConfig>
  <PassBackPlanConfig><!--optional, task plan of ANR-->
  <isSupportAddPlan><!--optional, xs:boolean, whether it supports adding plan, "true"-yes, "false"-no--></
isSupportAddPlan>
  <isSupportSearchPlan><!--optional, xs:boolean, whether it supports plan search, "true"-yes, "false"-no--></
isSupportSearchPlan>
  <isSupportDeletePlan><!--optional, xs:boolean, whether it supports deleting plan, "true"-yes, "false"-no--></
isSupportDeletePlan>
  </PassBackTaskConfig>
  <IRAIDCap/><!--optional, network RAID-->
  <isSupportStorageExtraInfo>
  <!--optional, xs:boolean, whether it supports configuring storage additional information, "true"-yes, "false"-no-->
  </isSupportStorageExtraInfo>
  <isSupportRecordStatus><!--optional, xs:boolean--></isSupportRecordStatus>
  <supportAIDTFSType opt="illegalParking,wrongdirection crosslane,vehicleexist,lanchange, turnround, evidence"/>
  <!--optional, xs:string, event picture search conditions supported by both AID and TFS-->
```



```
<isSupportRacmChannelsCap><!--optional, xs:boolean--></isSupportRacmChannelsCap>
<LockCap/><!--optional, lock or unlock video-->
<isSupportForamtAll><!--optional, xs:boolean, "true,false"--></isSupportForamtAll>
<isSupportExtendCabinetCfg>
  <!--optional, xs:boolean, whether it supports enclosure configuration-->
</isSupportExtendCabinetCfg>
<diskGroupNums><!--optional, xs:integer, number of supported HDD groups--></diskGroupNums>
<isSupportCountingSearchByUTC><!--optional, xs:boolean, whether it supports searching people counting results by
UTC time--></isSupportCountingSearchByUTC>
<isSupportPlaybackReverseByUTC><!--optional, xs:boolean, whether it supports reverse playback by UTC time--></
isSupportPlaybackReverseByUTC>
<isSupportWebPrivatePlaybackByUTC>
  <!--optional, xs: boolean, whether it supports playback based on Web private protocol-->
</isSupportWebPrivatePlaybackByUTC>
<isSupportFindCommonFileByUTC>
  <!--optional, xs: boolean, whether it supports extending the time zone for searching files-->
</isSupportFindCommonFileByUTC>
<isSupportFindEventFileByUTC><!--optional, xs: boolean, whether it supports extending the time zone for searching
files by event-->
</isSupportFindEventFileByUTC>
<isSupportSmartSearchRecordByUTC>
  <!--optional, xs: boolean, whether it supports extending the time zone for VCA search-->
</isSupportSmartSearchRecordByUTC>
<isSupportMRDSearchByTimeZone>
  <!--optional, xs: boolean, whether it supports extending the time zone for searching files by calendar-->
</isSupportMRDSearchByTimeZone>
<isSupportSearchRecordLabelByUTC>
  <!--optional, xs: boolean, whether it supports extending the time zone for searching video tags-->
</isSupportSearchRecordLabelByUTC>
<isSupportSearchPictureByUTC>
  <!--optional, xs: boolean, whether it supports extending the time zone for searching pictures-->
</isSupportSearchPictureByUTC>
<isSupportSmartSearchPictureByUTC>
  <!--optional, xs: boolean, whether it supports extending the time zone for searching pictures with smart
information-->
</isSupportSmartSearchPictureByUTC>
<isSupportFindLogByUTC>
  <!--optional, xs: boolean, whether it supports extending the time zone for searching log-->
</isSupportFindLogByUTC>
<isSupportUploadRecordByUTC>
  <!--optional, xs: boolean, whether it supports extending the time zone for uploading files to cloud storage-->
</isSupportUploadRecordByUTC>
<isSupportPlaybackByUTC>
  <!--optional, xs: boolean, whether it supports extending the time zone for playback by time and locating by time-->
</isSupportPlaybackByUTC>
<SecurityLog>
  <isSupportSecurityLog><!--optional, boolean, whether it supports security log--></isSupportSecurityLog>
  <isSupportLogServer><!--optional, boolean, whether it supports log server configuration--></isSupportLogServer>
  <isSupportLogServerTest><!--optional, xs: boolean, whether it supports log server test--></isSupportLogServerTest>
  <SecurityLogTypeList><!--required, xs: list, supported log type list-->
    <SecurityLogType>
      <primaryType><!--required, xs: string, major type--></primaryType>
```

```
<secondaryType optional=""><!--required, xs: string, minor type--></secondaryType>
</SecurityLogType>
</SecurityLogTypeList>
</SecurityLog>
<iSptInputProxyChanCap>
  <!--optional, xs:boolean, whether it supports getting the capability of the digital channel-->
</iSptInputProxyChanCap>
<isSupportLogDataPackage>
  <!--optional, xs:boolean, whether it supports exporting logs: "true, false"-->
</isSupportLogDataPackage>
<logSearchTimeSpanNums>
  <!--optional, xs:integer, supported number of time periods for log search-->
</logSearchTimeSpanNums>
<isSupportManualRecord>
  <!--optional, xs:boolean, whether the device supports manual recording-->
</isSupportManualRecord>
<isSupportRemark><!--optional, xs:boolean, whether it supports file remarks--></isSupportRemark>
<FileUpload><!--optional, upload file-->
  <enabled><!--required, xs:boolean, whether it supports enabling file uploading--></enabled>
</FileUpload>
<PoliceInfoUpload><!--optional, upload police information-->
  <enabled><!--required, xs:boolean, whether it supports enabling police information uploading--></enabled>
</PoliceInfoUpload>
<PoliceInfo>
  <policeID><!--optional, xs:boolean, whether it supports police ID--></policeID>
  <policeCode><!--optional, xs:boolean, whether it supports police No.--></policeCode>
  <policeName><!--optional, xs:boolean, whether it supports police name--></policeName>
  <password><!--optional, xs:boolean, whether it supports police password--></password>
  <deviceID><!--optional, xs:boolean, whether it supports device ID--></deviceID>
</PoliceInfo>
<behaviorEventPicSearch
opt="allBehaviorEvent,running,group,violentMotion,failDown,playCellphone,peopleNumChange,leavePosition,retentio
n,sleepOnDuty">
  <!--optional, xs:string, behavior analysis events supported by picture search, which is used for web display,
"allBehaviorEvent"-all events in behaviorEventPicSearch-->
</behaviorEventPicSearch>
<perimeterEventPicSearch
opt="allPerimeterEvent,linedetection,fieldddetection,regionEntrance,regionExiting,loitering">
  <!--optional, xs:string, perimeter protection events that support picture search, which is used for web display,
"allPerimeterEvent"-all events in perimeterEventPicSearch-->
</perimeterEventPicSearch>
<isSupportAssignChannelID><!--optional, xs:boolean, whether it supports specify channel No. when adding IPC--></
isSupportAssignChannelID>
<isSupportAssignStreamID><!--optional, xs:boolean, whether it supports specify stream ID of the channel when
adding IPC--></isSupportAssignStreamID>
<isSupportTimeSearch><!--optional, xs:boolean, whether it supports searching for recording start and end time by
channel--></isSupportTimeSearch>
<CloudStorageServerCap><!--optional-->
  <isSupportCloudStorageParameter><!--optional, xs:boolean, whether it supports configuring cloud storage
parameters--></isSupportCloudStorageParameter>
  <isSupportCloudStoragePool><!--optional, xs:boolean, whether it supports configuring cloud storage pool
parameters--></isSupportCloudStoragePool>
```

```
</CloudStorageServerCap>
<CMSearchCount/><!--optional, xs:boolean, whether it supports searching for file quantity of dock station-->
</RacmCap>
```

XML_RegionalFocus

RegionalFocus message in XML format

```
<RegionalFocus version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <StartPoint><!--0-1000-->
    <positionX><!--req, xs:integer--></positionX>
    <positionY><!--req, xs:integer--></positionY>
  </StartPoint>
  <EndPoint><!--0-1000-->
    <positionX><!--req, xs:integer--></positionX>
    <positionY><!--req, xs:integer--></positionY>
  </EndPoint>
</RegionalFocus>
```

XML_RegionClip

RegionClip message in XML format

```
<RegionClip version="2.0" xmlns="http://www.std-cgi.org/ver20/XMLSchema">
  <id><!--req, xs: string--></id>
  <enabled><!--req, xs: boolean--></enabled>
  <normalizedScreenSize><!--req, read-only-->
    <normalizedScreenWidth><!--req, read-only, xs: integer--></normalizedScreenWidth>
    <normalizedScreenHeight><!--req, read-only, xs: integer--></normalizedScreenHeight>
  </normalizedScreenSize>
  <regionType><!--req, ro, xs:string, opt="rectangle,convexPolygon,concavePolygon"--></regionType>
  <videoResolutionWidth><!--opt="704"--></videoResolutionWidth>
  <videoResolutionHeight><!--opt="576"--></videoResolutionHeight>
  <ClipRegionList>
    <ClipRegion>
      <RegionCoordinatesList>
        <RegionCoordinates><!--req, size=4-->
          <positionX><!--req, xs: integer; x-coordinate--></positionX>
          <positionY><!--req, xs: integer; y-coordinate--></positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </ClipRegion>
  </ClipRegionList>
</RegionClip>
```

XML_remotePermission

remotePermission message in XML format

```
<remotePermission version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <record><!--opt, xs: boolean--></record>
  <boolean><!--opt, xs: boolean--></boolean>
  <preview><!--opt, xs: boolean--></preview>
  <videoChannelPermissionList><!--opt-->
    <videoChannelPermission><!--opt-->
      <id><!--req, corresponds to the video input channel ID--></id>
      <preview><!--opt, xs: boolean--></preview>
      <palyBack><!--opt, xs: boolean--></palyBack>
      <record><!--opt, xs:boolean--></record>
      <playBackDoubleVerification>
        <!--opt, xs:boolean, whether supports secondary authentication for playback and download-->
      </playBackDoubleVerification>
    </videoChannelPermission>
  </videoChannelPermissionList>
  <ptzControl><!--opt, xs: boolean--></ptzControl>
  <ptzChannelPermissionList><!--opt-->
    <ptzChannelPermission><!--opt-->
      <id><!--req, corresponds to PTZ channel ID--></id>
      <ptzControl><!--opt, xs: boolean--></ptzControl>
    </ptzChannelPermission>
  </ptzChannelPermissionList>
  <logOrStateCheck><!--opt, xs: boolean, permission to view log and status--></logOrStateCheck>
  <parameterConfig><!--opt, xs: boolean, parameter configuration permission--></parameterConfig>
  <restartOrShutdown><!--opt, xs: boolean, permission to reboot and shutdown--></restartOrShutdown>
  <upgrade><!--opt, xs: boolean, permission to upgrade--></upgrade>
  <voiceTalk><!--opt, xs: boolean--></voiceTalk>
  <transParentChannel><!--opt, xs: boolean--></transParentChannel>
  <contorlLocalOut><!--opt, xs: boolean--></contorlLocalOut>
  <alarmOutOrUpload><!--opt, xs: boolean, permission to upload and output alarm--></alarmOutOrUpload>
  <factoryReset><!--opt, xs:boolean, restore default parameters--></factoryReset>
  <arm><!--opt, xs:boolean, arm--></arm>
  <disarm><!--opt, xs:boolean, disarm--></disarm>
  <accessControl><!--opt, xs:boolean, access control--></accessControl>
  <subSysOrZoneArm><!--opt, xs:boolean, partition or zone arming permission--></subSysOrZoneArm>
  <subSysOrZoneDisarm><!--opt, xs:boolean, partition or zone disarming permission--></subSysOrZoneDisarm>
  <subSysOrZoneClearArm><!--opt, xs:boolean, permission to clear partition or zone alarms--></
subSysOrZoneClearArm>
  <zoneBypass><!--opt, xs:boolean, permission to bypass zone--></zoneBypass>
  <zoneBypassRecover><!--opt, xs:boolean, permission to recover bypassing zone--></zoneBypassRecover>
  <IRAID>
    <rapidConfiguration><!--opt, xs:boolean--></rapidConfiguration>
    <reset><!--opt, xs:boolean--></reset>
  </IRAID>
  <subSystemList><!--opt, partitions linked to the operator when adding an operator or setting operator parameters-->
    <subSystem><!--opt, xs:integer, partition linked to the operator. When adding an operator or setting operator
parameters, one or more partitions can be linked to an operator. An operator should be linked to at least one partition
```

```
and up to four partitions--></subSystem>
</subSystemList>
</remotePermission>
```

XML_remotePermissionCap

remotePermissionCap capability message in XML format

```
<remotePermissionCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <record><!--opt, xs: boolean--></record>
  <boolean><!--opt, xs: boolean--></boolean>
  <preview><!--opt, xs: boolean--></preview>
  <videoChannelPermissionList><!--opt-->
    <videoChannelPermission><!--opt-->
      <id><!--req, corresponds to the video input channel ID--></id>
      <preview><!--opt, xs: boolean--></preview>
      <palyBack><!--opt, xs: boolean--></palyBack>
      <record><!--opt, xs:boolean--></record>
      <playBackDoubleVerification>
        <!--opt, xs:boolean, whether supports secondary authentication for playback and download-->
      </playBackDoubleVerification>
    </videoChannelPermission>
  </videoChannelPermissionList>
  <ptzControl><!--opt, xs: boolean--></ptzControl>
  <ptzChannelPermissionList><!--opt-->
    <ptzChannelPermission><!--opt-->
      <id><!--req, corresponds to PTZ channel ID--></id>
      <ptzControl><!--opt, xs: boolean--></ptzControl>
    </ptzChannelPermission>
  </ptzChannelPermissionList>
  <logOrStateCheck><!--opt, xs: boolean, permission to view log and status--></logOrStateCheck>
  <parameterConfig><!--opt, xs: boolean, parameter configuration permission--></parameterConfig>
  <restartOrShutdown><!--opt, xs: boolean, permission to reboot and shutdown--></restartOrShutdown>
  <upgrade><!--opt, xs: boolean, permission to upgrade--></upgrade>
  <voiceTalk><!--opt, xs: boolean--></voiceTalk>
  <transParentChannel><!--opt, xs: boolean--></transParentChannel>
  <contorlLocalOut><!--opt, xs: boolean--></contorlLocalOut>
  <alarmOutOrUpload><!--opt, xs: boolean, permission to upload and output alarm--></alarmOutOrUpload>
  <factoryReset><!--opt, xs:boolean, restore default parameters--></factoryReset>
  <arm><!--opt, xs:boolean, arm--></arm>
  <disarm><!--opt, xs:boolean, disarm--></disarm>
  <accessControl><!--opt, xs:boolean, access control--></accessControl>
  <subSysOrZoneArm><!--opt, xs:boolean, partition or zone arming permission--></subSysOrZoneArm>
  <subSysOrZoneDisarm><!--opt, xs:boolean, partition or zone disarming permission--></subSysOrZoneDisarm>
  <subSysOrZoneClearArm><!--opt, xs:boolean, permission to clear partition or zone alarms--></
subSysOrZoneClearArm>
  <zoneBypass><!--opt, xs:boolean, permission to bypass zone--></zoneBypass>
  <zoneBypassRecover><!--opt, xs:boolean, permission to recover bypassing zone--></zoneBypassRecover>
  <IRAID>
    <rapidConfiguration><!--opt, xs:boolean--></rapidConfiguration>
    <reset><!--opt, xs:boolean--></reset>
```

```
</IRAID>
<subSystemNo min="" max=""><!--opt, xs:integer, range of partition No. When adding an operator or setting
operator parameters, one or more partitions can be linked to the operator--></subSystemNo>
<subSystem min="1" max="3"><!--opt, xs:integer, number of partitions that can be linked to the operator. The
attributes min="1" and max="3" are sample settings, which indicates that at least one partition and up to three
partitions can be linked. When adding an operator or setting operator parameters, one or more partitions can be
linked to an operator. Actually, an operator should be linked to at least one partition and up to four partitions--></
subSystemNo>
</remotePermissionCap>
```

XML_ResponseStatus

ResponseStatus message in XML format

```
<ResponseStatus version="2.0" xmlns="http://www.std-cgi.org/ver20/XMLSchema">
<requestURL>
<!--req, ro, xs:string, request URL-->
</requestURL>
<statusCode>
<!--req, ro, xs:integer, status code: 0,1-OK, 2-Device Busy, 3-Device Error, 4-Invalid Operation, 5-Invalid XML Format,
6-Invalid XML Content, 7-Reboot Required-->
</statusCode>
<statusString>
<!--req, ro, xs:string, status description: OK, Device Busy, Device Error, Invalid Operation, Invalid XML Format, Invalid
XML Content, Reboot-->
</statusString>
<subStatusCode>
<!--req, ro, xs:string, describe the error reason in detail-->
</subStatusCode>
</ResponseStatus>
```



Note

See **ISAPI Response Codes** for details about sub status codes and corresponding error codes.

XML_ResponseStatus_AuthenticationFailed

ResponseStatus message in XML format for failed authentication.

```
<ResponseStatus version="1.0" xmlns="http://www.std-cgi.org/ver20/XMLSchema">
<requestURL><!-- req, ro,xs:string --></requestURL>
<statusCode><!-- req, ro,xs:integer --></statusCode>
<statusString><!-- req, ro,xs:string --></statusString>
<subStatusCode><!-- req, ro,xs:string --></subStatusCode>
<lockStatus><!-- opt, ro,xs:string , "unlock,locked", locking status--></lockStatus>
<retryTimes><!-- opt, ro,xs:integer, remaining authentication attempts--></retryTimes>
<resLockTime><!-- opt, ro,xs:integer, remaining locking time, unit: second--></resLockTime>
</ResponseStatus>
```

XML_RTSPCertificate

XML message about parameters of RTSP authentication mode

```
<RTSPCertificate xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0" >
  <certificateType>
    <!--required, xs:string, authentication mode: digest (default), digest/basic-->
  </certificateType>
  <SecurityAlgorithm>
    <!--dependent, authentication algorithm, it is valid when certificateType is "digest" or "digest/basic"-->
    <algorithmType>
      <!--optional, xs:string, algorithm types: MD5, SHA256, MD5/SHA256-->
    </algorithmType>
  </SecurityAlgorithm>
</RTSPCertificate>
```

XML_RTSPCertificateCap

XML message about capability of RTSP authentication mode

```
<RTSPCertificateCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0" >
  <certificateType def="digest" opt="digest,digest/basic" >
    <!--required, xs:string-->
  </certificateType>
  <SecurityAlgorithm>
    <!--depend, authentication algorithm, it is valid when certificateType is "digest" or "digest/basic"-->
    <algorithmType opt="MD5,SHA256,MD5/SHA256" >
      <!--optional, xs:string, algorithm type-->
    </algorithmType>
  </SecurityAlgorithm>
</RTSPCertificateCap>
```

XML_Schedule

Schedule message in XML format

```
<Schedule version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--required, xs:string, ID--></id>
  <eventType>
    <!--optional, xs:string, alarm/event types, see details in the "Remarks" below-->
  </eventType>
  <inputIOPortID><!--read-only, dependent, xs:string, alarm input No.--></inputIOPortID>
  <outputIOPortID><!--read-only, dependent, xs:string, alarm output No.--></inputIOPortID>
  <videoInputChannelID><!--read-only, dependent, xs:string, video input channel ID--></videoInputChannelID>
  <TimeBlockList size="8"><!--required-->
    <TimeBlock><!--list-->
      <dayOfWeek>
        <!--optional, xs:integer, day of the week based on ISO8601, "1"=Monday, ...-->
```

```
</dayOfWeek>
<TimeRange><!--required-->
  <beginTime><!--required, xs:time, ISO8601 time--></beginTime>
  <endTime><!--required, xs:time, ISO8601 time--></endTime>
</TimeRange>
<CustomExtension>
  <vehicleDetectSceneID>
    <!--required, xs:interger-->
  </vehicleDetectSceneID>
</CustomExtension>
</TimeBlock>
</TimeBlockList>
<HolidayBlockList><!--optional-->
  <TimeBlock><!--list-->
    <TimeRange><!--required-->
      <beginTime><!--required, xs:time, ISO8601 time--></beginTime>
      <endTime><!--required, xs:time, ISO8601 time--></endTime>
    </TimeRange>
  </TimeBlock>
</HolidayBlockList>
</Schedule>
```

Remarks

The node **<eventType>** can be set to the following values: IO, VMD, videoloss, PIR, linedetection, fielddetection, audioexception, facedetection, regionEntrance, regionExiting, loitering, group, rapidMove, parking, unattendedBaggage, attendedBaggage, storageDetection, shipsDetection, HUMANATTRIBUTE, humanAttribute, faceContrast, faceSnap, faceLib, whiteListFaceContrast, personDensityDetection, personQueueDetection, mixedTargetDetection, fireDetection, illegalParking, pedestrian, trafficAccident, construction, roadblock, abandonedObject, parallelParking, parkingState, congestion, intersectionAnalysis, heatMap, reverseEntrance, vehicledetect, safetyHelmetDetection, vibrationDetection, TMPA, faceThermometry, HBDLib.

XML_SecurityCap

SecurityCap capability message in XML format

```
<SecurityCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <supportUserNums><!--optional, xs:integer, number of supported users--></supportUserNums>
  <userBondIpNums><!--optional, xs:integer, number of bound IP addresses supported by the user--></
userBondIpNums>
  <userBondMacNums><!--optional, xs:integer, number of bound MAC addresses supported by the user--></
userBondMacNums>
  <isSupCertificate><!--optional, xs:boolean, whether it supports certification: "true", "false"--></isSupCertificate>
  <issupIllegalLoginLock><!--optional, xs: boolean, whether it supports locking illegal login: "true,false"--
><issupIllegalLoginLock>
  <isSupportOnlineUser><!--optional, xs: boolean, "true,false"--><isSupportOnlineUser>
  <isSupportAnonymous><!--optional, xs: boolean, "true,false"--><isSupportAnonymous>
  <isSupportStreamEncryption><!--optional, xs:boolean, whether it supports stream encryption: "true", "false"--></
isSupportStreamEncryption>
```



```
<securityVersion opt="1,2"/><!--optional, xs:integer, encryption capability, each version contains encryption
algorithm and node range to be encrypted. "1"-encrypt by AES128, "2"-encrypt by AES256-->
<keyIterateNum>
  <!--dependent xs:integer, iteration times, this node depends on securityVersion, and the value is usually between
100 and 1000-->
</keyIterateNum>
<isSupportUserCheck>
  <!--dependent optional, xs:boolean, whether it supports verifying password when editing/adding/deleting user
parameters, this node depends on securityVersion-->
</isSupportUserCheck>
<isSupportGUIDFileDataExport><!--optional, xs:boolean, "true,false"--></isSupportGUIDFileDataExport>
<isSupportSecurityQuestionConfig><!--optional, xs:boolean, "true,false"--></isSupportSecurityQuestionConfig>
<supportSecurityNode opt="wirelessServer,wirelessDial"><!--optional, xs:string, "true,false"--></
supportSecurityNode>
<isSupportGetOnlineUserListSC><!--optional, xs:boolean, "true,false"--></isSupportGetOnlineUserListSC>
<SecurityLimits><!--optional-->
  <LoginPasswordLenLimit min="1" max="16"><!--opt--></LoginPasswordLenLimit>
  <SecurityAnswerLenLimit min="1" max="128"><!--opt--></SecurityAnswerLenLimit>
</SecurityLimits>
<RSAKeyLength opt="512,1024,2048" def=2048/><!--optional, RSA key length of the HTTPS certificate-->
<isSupportONVIFUserManagement><!--optional, xs:boolean, whether it supports user management by ONVIF
protocol: "true", "false"--><isSupportONVIFUserManagement>
<WebCertificateCap><!--opt-->
  <CertificateType opt="basic, digest, digest/basic"><!--required, xs:string--></CertificateType>
  <SecurityAlgorithm><!--dependent, this node is valid when certificateType is "digest" or "digest/basic"-->
    <algorithmType opt="MD5,SHA256,MD5/SHA256"><!--optional, xs:string, algorithm type: "MD5", "SHA256",
"MD5/SHA256"--></algorithmType>
  </SecurityAlgorithm>
</WebCertificateCap>
<isSupportConfigFileImport>
  <!--optional, xs: boolean, whether it supports importing configuration files securely: "true"-yes. If it is not
supported, this node will not be returned-->
</isSupportConfigFileImport>
<isSupportConfigFileExport>
  <!-- optional, xs:boolean, whether it supports exporting configuration files securely: "true"-yes. If it is not
supported, this node will not be returned-->
</isSupportConfigFileExport>
<cfgFileSecretKeyLenLimit min="0" max="16">
  <!--optional, length limit of the configuration files' encryption key-->
</cfgFileSecretKeyLenLimit>
<isSupportDeviceCertificatesManagement>
  <!--optional, xs: boolean, whether it supports device certificate management: true=yes, if not support, this node will
not be returned-->
</isSupportDeviceCertificatesManagement>
<supportIPCActivatePassword><!--optional, xs:boolean, whether it supports configuring password for activating the
network camera--></supportIPCActivatePassword>
<isIrreversible>
  <!--dependent optional, xs:boolean, whether it supports irreversible password storage. If this node does not exist,
irreversible password storage is not supported-->
</isIrreversible>
<salt><!--optional, xs:string, salt of user name--></salt>
<keypadPassword min="1" max=""/><!--optional, xs:string, keypad password length. If different types of users have
```

different keypad password length, this node only indicates the administrator's keypad password length, otherwise this node indicates that all types of users have the same keypad password length-->

```
<installerKeypadPassword min="1" max="" /><!--optional, xs:string, installer's keypad password length-->
<operatorKeypadPassword min="1" max="" /><!--optional, xs:string, operator's keypad password length-->
<userOperateType opt="1,2,3">
  <!--optional, xs:string, user operation type: "1"-network user, "2"-keypad user, "3"-network user and keypad user-->
</userOperateType>
<isSupportOnvifInfo><!--optional, xs:boolean, whether it supports the API of getting ONVIF protocol information. If
this node is returned and its value is "true", in indicates supporting this function; if this node is not returned, it
indicates that this function is not supported--></isSupportOnvifInfo>
<isSupportPictureURICertificate opt="true,false"><!--optional, xs:boolean, whether it supports configuring picture
URL authentication. If this node is not returned, it indicates that the default picture authentication method of the
device is digest--></isSupportPictureURICertificate>
<isSupportUnloggedUserPermissionConfig opt="true,false"><!--optional, xs:boolean, whether it supports configuring
permissions for users that have not logged in--></isSupportUnloggedUserPermissionConfig>
<isSupportUserNamePasswordCheckUpgrade><!--optional, xs:boolean, whether it supports upgrading user name
and password verification. If this node is not returned, it indicates that this function is not supported--></
isSupportUserNamePasswordCheckUpgrade>
<isSupportDeviceSelfSignCertExport><!--optional, xs:boolean--></isSupportDeviceSelfSignCertExport>
<isSupportSecurityEmail><!--optional, xs:boolean, whether it supports configuring security E-mail address. If this
node is not returned, it indicates that this function is not supported--></isSupportSecurityEmail>
<isSupportRTSPCertificate opt="true,false"><!--optional, xs:boolean--></isSupportRTSPCertificate>
<isSptUserEnabled><!--optional, xs:boolean, whether it supports configuration of enabling user: "true"-yes, "false"-
no--></isSptUserEnabled>
<isSptAdminCap><!--optional, xs:boolean, whether it supports getting administrator permission capability: "true"-
yes, "false"-no--></isSptAdminCap>
<DoubleVerificationCap><!--optional, double verification capability-->
  <isSupportUsersConfig>
    <!--optional, xs: boolean, "true,false", whether it supports configuring double verification user-->
  </isSupportUsersConfig>
  <isSupportUsersPermissionConfig>
    <!--optional, xs: boolean, "true,false", whether it supports configuring the permission of double verification user--
  >
</isSupportUsersPermissionConfig>
</DoubleVerificationCap>
<isSupportCertificateCustomID><!--optional, xs:Boolean, whether it supports certificate configuration with custom
ID--></isSupportCertificateCustomID>
<maxIllegalLoginTimes min="3" max="20"def="5">
  <!--dependentxs:integer,"maximum illegal login attempts" -->
</maxIllegalLoginTimes>
<SecurityAdvanced>
  <securityEnhanced> <!--optional,xs:boolean,security reinforcement--> </securityEnhanced>
  <noOperationEnabled><!--required, xs:Boolean, whether to enable the timeout of no operation--></
noOperationEnabled>
  <noOperationTime min="1" max="60"def="15"><!--required, xs:integer,timeout of no operation, unit: minute--></
noOperationTime>
  <isSupportDigestStatus><!--optional, Boolean—digest status></isSupportDigestStatus>
</SecurityAdvanced>
<LoginLinkNum><maxLinkNum min="1" max="128"def="50"><!--required, xs:integer,maximum number of logged in
accounts--></maxLinkNum><LoginLinkNum>
  <isSupportCCClientCertificate><!--optional,Boolean—whether to enable the CC client certificate></
isSupportCCClientCertificate>
```

```
<passwordValidity min="0" max="365"><!--optional, xs:string, password validity period, unit: day. If this node is
returned, it indicates supporting configuring password validity period. The password validity period can only be edited
by the administrator user--></passwordValidity>
</SecurityCap>
```

XML_SecurityQuestion

SecurityQuestion message in XML format

```
<SecurityQuestion version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <QuestionList>
    <Question><!--the number of security question must be 3-->
      <id><!--required, xs: integer, question ID, which corresponds to that of device--></id>
      <answer>
        <!--write-only, xs: string, answer of the security question, it will not be returned when getting question; when
setting question, this node will be encrypted (BASE64-->AES128CBE) for transmission-->
      </answer>
      <mark>
        <!--required, read-only, xs: boolean, whether the configured security question is marked-->
      </mark>
    </Question>
  </QuestionList>
  <password>
    <!--write-only, xs: string, password that encrypted by CBC, this node will not be returned when getting question; but
it is required when setting question-->
  </password>
</SecurityQuestion>
```

See Also

XML_SerialCap

Serial port capability message in XML format

```
<SerialCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <rs485PortNums min="1" max="5" def="1">
    <!--optional, xs: integer-->
  </rs485PortNums>
  <supportRS23Cconfig>
    <!--optional, xs: boolean-->
  </supportRS23Cconfig>
  <rs422PortNums opt="0">
    <!--optional, xs: integer-->
  </rs422PortNums>
  <rs232PortNums opt="1">
    <!--optional, xs: integer, range of RS-232 serial port No. supported by the device-->
  </rs232PortNums>
  <rs485WorkMode opt="Led, CaptureTrigger">
    <!--optional, xs: string, range of RS-485 serial port No. supported by the device-->
  </rs485WorkMode>
</SerialCap>
```

```
</rs485WorkMode>
<isSupportAuthenticationService>
  <!--optional, xs:boolean, whether the device supports serial port authentication-->
</isSupportAuthenticationService>
</SerialCap>
```

XML_SerialCommand

SerialCommand message in XML format

```
<SerialCommand version="1.0" xmlns="urn:psalliance-org">
  <chainNo>
    <!--opt, xs:string-->
  </chainNo>
  <command>
    <!--req, xs:hexBinary-->
  </command>
</SerialCommand>
```

XML_SerialPorList

SerialPorList message in XML format

```
<SerialPortList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <SerialPort/><!--opt, see details in the message of XML_SerialPort-->
</SerialPortList>
```

See Also

XML_SerialPort

XML_SerialPort

SerialPort message in XML format

```
<SerialPort version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id>
    <!--req, xs:string, serial port ID-->
  </id>
  <enabled>
    <!--req, xs:boolean, whether to enable the serial port: "true, false"-->
  </enabled>
  <serialPortType>
    <!--req, xs:string, serial port type: "RS485", "RS422", "RS232"-->
  </serialPortType>
  <duplexMode>
    <!--req, xs:string, duplex mode of the serial port: "half", "full"-->
  </duplexMode>
```

```
<direction>
  <!--req, xs:string, "monodirectional, bdirectional"-->
</direction>
<baudRate>
  <!--req, xs:integer-->
</baudRate>
<dataBits>
  <!--req, xs:integer-->
</dataBits>
<parityType>
  <!--req, xs:string, parity type: "none,even,odd,mark,space"-->
</parityType>
<stopBits>
  <!--req, xs:string, stop bit: "1,1.5,2"-->
</stopBits>
<workMode>
  <!--dep, xs:string, working mode: "console","transparent","audiomixer","stairsControl"-elevator
control,"cardReader"-card reader,"disabled","custom". This node is required only when <serialPortType> is set to
"RS232"-->
</workMode>
<flowCtrl>
  <!--req, xs:string, "none, software, hardware"-->
</flowCtrl>
<rs485WorkMode>
  <!--opt, xs:string, working mode of RS-485 serial port, which is used for LED display or triggering transmission of
captured pictures: "Led, CaptureTrigger". This node is valid only when <serialPortType> is "RS485"-->
</rs485WorkMode>
<audiomixerProtocolType>
  <!--opt, xs:string, "HIKVISION". This node is required only when <workMode> is "audiomixer"-->
</audiomixerProtocolType>
<deviceName><!--deq, xs:string, device name, the maximum length is 32 bytes, this node is valid only when
<serialPortType> is "RS485"--></deviceName>
<deviceType><!--deq, xs:integer, device type, this node is valid only when <serialPortType> is "RS485"--></
deviceType>
<deviceProtocol><!--deq, xs:integer, device protocol, this node is valid only when <serialPortType> is "RS485"--></
deviceProtocol>
<mode><!--deq, xs:string, working mode: "readerMode,clientMode,externMode,accessControlHost,disabled", this
node is valid only when <serialPortType> is "RS485"--></mode>
<outputDataType><!--dep, xs:string, output data type: "cardNo,employeeNo", this node is valid when <mode>is
"accessControlHost"--></outputDataType>
</SerialPort>
```

XML_SIPInfo

SIPInfo message in XML format

```
<SIPInfo version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <videoID>
    <!--req, xs:string-->
  </videoID>
  <AlarmInList>
```

```
<AlarmIn>
  <id>
    <!--req, xs:string-->
  </id>
  <alarmInID>
    <!--req, xs:string-->
  </alarmInID>
</AlarmIn>
</AlarmInList>
</SIPInfo>
```

XML_SIPInfoList

SIPInfoList message in XML format

```
<SIPInfoList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <SIPInfo>
    <videoID><!--req, xs: string--><videoID>
    <AlarmInList>
      <AlarmIn>
        <id><!--req, xs: string--></id>
        <alarmInID><!--req, xs: string--><alarmInID>
      </AlarmIn>
    </AlarmInList>
    <VideoInputList><!--opt-->
      <VideoInput>
        <id><!--req, xs: string--></id>
        <videoInputID><!--req, xs: string--></videoInputID>
      </VideoInput>
    </VideoInputList>
    <AudioOutputList><!--opt,ID list of audio outputs-->
      <AudioOutput>
        <id><!--req, xs: string--></id>
        <audioOutputID><!--req, xs: string--></audioOutputID>
      </AudioOutput>
    </AudioOutputList>
    <platformNo><!-- opt, xs:integer--></platformNo>
  </SIPInfo>
</SIPInfoList>
```

XML_SIPServer

SIPServer message in XML format

```
<SIPServer version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id>
    <!--req, xs:interger, the maximum ID depends on the number of channels-->
  </id>
  <localPort>
```

```
<!--req, xs:integer, local port No., which is between 1 and 65535-->
</localPort>
<streamID>
  <!--req, xs:integer, stream type: 1-main stream, 2-sub-stream, 3-third stream-->
</streamID>
<Standard>
  <!--opt-->
  <registerStatus>
    <!--ro, req, xs:boolean, registration status: "false"-unregistered, "true"-registered-->
  </registerStatus>
  <enabled>
    <!--req, xs:string, "true"-log in, "false"-log out-->
  </enabled>
  <registrar>
    <!--req, xs:string-->
  </registrar>
  <registrarPort>
    <!--req, xs:integer-->
  </registrarPort>
  <proxy>
    <!--req, xs:string-->
  </proxy>
  <proxyPort>
    <!--req, xs:integer-->
  </proxyPort>
  <displayName>
    <!--req, xs:string-->
  </displayName>
  <username>
    <!--req, xs:string-->
  </userName>
  <authID>
    <!--req, xs:string-->
  </authID>
  <password>
    <!--wo, req, xs:string-->
  </password>
  <expires>
    <!--req, xs:integer-->
  </expires>
  <voiceTalk>
    <!--optional, xs:boolean, whether to enable SIP two-way audio-->
  </voiceTalk>
</Standard>
<GB28181>
  <!--opt-->
  <registerStatus>
    <!--req, xs:boolean, "true,false"-->
  </registerStatus>
  <enabled>
    <!--req, xs:boolean, "true,false"-->
  </enabled>
```

```
<registrar>
  <!--req, xs:string-->
</registrar>
<registrarPort>
  <!--req, xs:integer-->
</registrarPort>
<serverId>
  <!--req, xs:string-->
</serverId>
<serverDomain>
  <!--req, xs:string-->
</serverDomain>
<username>
  <!--req, xs:string-->
</userName>
<authID>
  <!--req, xs:string-->
</authID>
<password>
  <!--wo, req, xs:string-->
</password>
<expires>
  <!--req, xs:integer-->
</expires>
<liveTime>
  <!--req, xs:integer-->
</liveTime>
<heartbeatTime>
  <!--req, xs:integer-->
</heartbeatTime>
<heartbeatCount>
  <!--req, xs:integer-->
</heartbeatCount>
</GB28181>
</SIPServer>
```

XML_SIPServerList

SIPServerList message in XML format

```
<SIPServerList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <SIPServer/><!--opt, see details in the message of XML_SIPServer-->
</SIPServerList>
```

See Also

XML_SIPServer

XML_Sharpness

Sharpness message in XML format

```
<Sharpness version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <SharpnessMode><!--opt, xs:string, "manual", "auto"--></SharpnessMode>
  <SharpnessLevel><!--req, xs:integer--></SharpnessLevel>
</Sharpness>
```

XML_Shutter

Shutter message in XML format

```
<Shutter version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ShutterLevel>
    <!--dep, xs: string, shutter level: "1/1, 1/2, 1/3, 1/6, 1/12, 1/25, 1/50, 1/75, 1/100, 1/120, 1/125, 1/150, 1/175,
    1/215, 1/225, 1/300, 1/400, 1/425, 1/600, 1/1000, 1/1250, 1/1750, 1/2500, 1/3500, 1/6000, 1/10000", this node is
    valid when <ExposureType> in the message XML_Exposure is "ShutterFisrt"-->
    <ShutterLevel>
  </Shutter>
```

See Also

XML_Exposure

XML_SlaveCamera

SlaveCamera message in XML format

```
<SlaveCameraInfo version="2.0" xmlns="http://www.std-cgi.org/ver20/XMLSchema">
  <id>
    <!--req, xs:string-->
  </id>
  <serverAddress>
    <addressingFormatType>
      <!--req, xs:string, "ipaddress,hostname"-->
    </addressingFormatType>
    <hostName>
      <!--dep, xs:string-->
    </hostName>
    <ipAddress>
      <!--dep, xs:string-->
    </ipAddress>
    <ipv6Address>
      <!--dep, xs:string-->
    </ipv6Address>
  </serverAddress>
  <userName>
```

```
<!--req, xs:string-->
</userName>
<passWord>
  <!--wo, xs:string-->
</passWord>
<portNo>
  <!--req, xs:integer-->
</portNo>
<loginStatus>
  <!--req, xs:string, "login,logout"-->
</loginStatus>
</SlaveCameraInfo>
```

XML_SlaveCameraStatus

SlaveCameraStatus message in XML format

```
<SlaveCameraStatus version="2.0" xmlns="http://www.std-cgi.org/ver20/XMLSchema">
  <SlaveCameraLinkStatusList>
    <!--req-->
    <SlaveCameraLinkStatus>
      <id>
        <!--req, xs:string-->
      </id>
      <linkStatus/>
      <!--ro, xs:string, connection status: "online,offline"-->
    </SlaveCameraLinkStatus>
  </SlaveCameraLinkStatusList>
</SlaveCameraStatus>
```

XML_SmartCap

SmartCap capability message in XML format

```
<SmartCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <isSupportROI><!--opt, xs:boolean, whether to support ROI (Region of Interest)--></isSupportROI>
  <isSupportFaceDetect><!--opt, xs:boolean, whether to support face detection--></isSupportFaceDetect>
  <isSupportIntelliTrace><!--opt, xs:boolean--></isSupportIntelliTrace>
  <isSupportFieldDetection><!--opt, xs:boolean, whether to support region detection--></isSupportFieldDetection>
  <isSupportDefocusDetection><!--opt, xs:boolean--></isSupportDefocusDetection>
  <isSupportAudioDetection><!--opt, xs:boolean--></isSupportAudioDetection>
  <isSupportSceneChangeDetection><!--opt, xs:boolean, whether to support scene change detection--></
isSupportSceneChangeDetection>
  <isSupportLineDetection><!--opt, xs:boolean--></isSupportLineDetection>
  <isSupportRegionEntrance><!--opt, xs:boolean--></isSupportRegionEntrance>
  <isSupportRegionExiting><!--opt, xs:boolean--></isSupportRegionExiting>
  <isSupportLoitering><!--opt, xs:boolean--></isSupportLoitering>
  <isSupportGroup><!--opt, xs:boolean--></isSupportGroup>
  <isSupportRapidMove><!--opt, xs:boolean--></isSupportRapidMove>
```

```
<isSupportParking><!--opt, xs:boolean--></isSupportParking>
<isSupportUnattendedBaggage><!--opt, xs:boolean--></isSupportUnattendedBaggage>
<isSupportAttendedBaggage><!--opt, xs:boolean--></isSupportAttendedBaggage>
<isSupportPeopleDetection><!--opt, xs:boolean--></isSupportPeopleDetection>
<isSupportStorageDetection><!--opt, xs:boolean--></isSupportStorageDetection>
<isSupportShipsDetection><!--opt, xs:boolean--></isSupportShipsDetection>
<isSupportSmartCalibration><!--opt, xs:boolean--></isSupportSmartCalibration>
<isSupportShield><!--opt, xs:boolean, whether to support shielded area--></isSupportShield>
<isSupportAlgVersion><!--opt, xs:boolean, whether to support algorithm library version--></isSupportAlgVersion>
<isSupportVideoOverlap><!--opt, xs:boolean, whether to support text overlay--></isSupportVideoOverlap>
<isSupportParkingState><!--opt, xs:boolean, whether to support parking space status detection--></
isSupportParkingState>
<isSupportChannelResource><!--opt, xs:boolean--></ isSupportChannelResource>
<isSupportAnalysisUnitSwitch opt="true,false"><!--opt, xs:boolean, whether to support analysis unit switch--></
isSupportAnalysisUnitSwitch>
<isSupportHFPD><!--opt, xs:boolean, whether to support frequently appeared person detection--></isSupportHFPD>
<isSupportImageROI<!--opt, xs:boolean, whether to support smartJpeg (image ROI (Region of Interest))--></
isSupportImageROI>
</SmartCap>
```

XML_SmartSearchDescription

SmartSearchDescription message in XML format

```
<SmartSearchDescription version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <searchID>
    <!--req, xs:string, search ID, which is used to confirm the upper-level platform or system. If the platform or system
are same during two times of search, the search history will be recorded in the memory for next fast search-->
  </searchID>
  <searchResultPosition><!--req, xs: integer--></searchResultPosition>
  <maxResults><!--req, xs: integer--></maxResults>
  <trackID><!--req, xs: integer, channel ID, e.g., 101-main stream if first channel--></trackID>
  <startTime><!--req, xs: datetime, start time of validity period, e.g., 2013-06-10T 12:00:00Z--></startTime>
  <endTime><!--req, xs: datetime, end time of validity period, e.g., 2013-06-10T 13:00:00Z--></endTime>
  <type>
    <!--req, xs: string; VCA event type: "motionDetection, traversePlaneDetection,FieldDetection, faceDetection"-->
  </type>
  <MotionDetection><!--dep, it is valid when type is "motionDetection"-->
    <Grid>
      <rowGranularity></rowGranularity>
      <columnGranularity></columnGranularity>
    </Grid>
    <MotionDetectionLayout>
      <layout>
        <gridMap></gridMap>
      </layout>
    </MotionDetectionLayout>
    <sensitivity><!--opt, xs: integer, value range: [1,100]--></sensitivity>
  </MotionDetection>
  <TraversePlane><!--dep, it is valid when type is "traversePlaneDetection"-->
    <TraversePlaneParam>
```

```
<LineList>
  <Line>
    <StartPoint>
      <x><!--req, xs: float--></x>
      <y><!--req, xs: float--></y>
    </StartPoint>
    <EndPoint>
      <x><!--req, xs: float--></x>
      <y><!--req, xs: float--></y>
    </EndPoint>
  </Line>
</LineList>
<crossDirection><!--opt, xs: string, crossing direction: "bothDirection, leftToRight, rightToLeft"--><crossDirection>
<sensitivity><!--opt, xs: integer, value range: [1,100]--></sensitivity>
<planeHeight><!--opt, xs: integer--></planeHeight>
<detectionTarget><!--req, xs: string, detection target type: "all, human, vehicle"--><detectionTarget>
</TraversePlaneParam>
<dwPreTime><!--opt, xs: integer, pre-recording time--></dwPreTime>
<dwDelayTime><!--opt, xs: integer, post-recording time--></dwDelayTime>
<PTZPos>
  <P><!--opt, xs: integer--></P>
  <T><!--opt, xs: integer--></T>
  <Z><!--opt, xs: integer--></Z>
</PTZPos>
<advancedType><!--opt, xs: string, target sub type: "humanFeature, objectFeature"--></advancedType>
<advance><!--dep, it is valid when type is "motionDetection"-->
  <HumanFeature>
    <age><!--opt, xs: integer, value range: [0,100]--></age>
    <sex><!--opt, xs: string, gender: "male,female"--></sex>
    <eyeGlass><!--opt, xs: boolean--></eyeGlass>
  </HumanFeature><!--dep, it is valid when advancedType is "humanFeature"-->
  <ObjectFeature>
    <colorRatel><!--opt, xs: integer--></colorRatel>
    <R><!--opt, xs: integer--></R>
    <G><!--opt, xs: integer--></G>
    <B><!--opt, xs: integer--></B>
  </ObjectFeature><!--dep, it is valid when advancedType is "objectFeature"-->
</advance>
</TraversePlane>
<FieldDetection><!--dep, it is valid when type is "FieldDetection"-->
  <normalizedScreenSize>
    <normalizedScreenWidth><!--opt, xs: integer--></normalizedScreenWidth>
    <normalizedScreenHeight><!--opt, xs: integer--></normalizedScreenHeight>
  </normalizedScreenSize>
  <param>
    <region>
      <pointList>
        <point>
          <x><!--opt, xs: integer--></x>
          <y><!--opt, xs: integer--></y>
        </point>
      </pointList>
```

```
</region>
</param>
<duration><!--opt, xs: integer, value range: [1,100]--></duration>
<sensitivity><!--opt, xs: integer, value range: [1,100]--></sensitivity>
<rate><!--opt, xs: integer, value range: [1,100]--></rate>
<detectionTarget><!--req, xs: string, detection target type: "all, human, vehicle"--><detectionTarget>
<dwPreTime><!--opt, xs: integer, pre-recording time--></dwPreTime>
<dwDelayTime><!--opt, xs: integer, post-recording time--></dwDelayTime>
<PTZPos>
  <P><!--opt, xs: integer--></P>
  <T><!--opt, xs: integer--></T>
  <Z><!--opt, xs: integer--></Z>
</PTZPos>
<advancedType><!--opt, xs: string, target sub type: "humanFeature, objectFeature"--></advancedType>
<advance><!--dep, it is valid when type is "motionDetection"-->
<HumanFeature>
  <age><!--opt, xs: integer, value range: [0,100]--></age>
  <sex><!--opt, xs: string, gender: "male,female"--></sex>
  <eyeGlass><!--opt, xs: boolean--></eyeGlass>
</HumanFeature><!--dep, it is valid when advancedType is "humanFeature"-->
<ObjectFeature>
  <colorRatel><!--opt, xs: integer--></colorRatel>
  <R><!--opt, xs: integer--></R>
  <G><!--opt, xs: integer--></G>
  <B><!--opt, xs: integer--></B>
</ObjectFeature><!--dep, it is valid when advancedType is "objectFeature"-->
</advance>
</FieldDetection>
<FaceDetection><!--dep, it is valid when type is "faceDetection"-->
<param>
  <region>
    <pointList>
      <point>
        <x/>
        <y/>
      </point>
    </pointList>
  </region>
</param>
<duration><!--opt, xs: integer, value range: [1,100]--></duration>
<sensitivity><!--opt, xs: integer, value range: [1,100]--></sensitivity>
<rate><!--opt, xs: integer, value range: [1,100]--></rate>
<detectionTarget><!--req, xs: string, detection target type: "all, human, vehicle"--><detectionTarget>
<dwPreTime><!--opt, xs: integer, pre-recording time--></dwPreTime>
<dwDelayTime><!--opt, xs: integer, post-recording time--></dwDelayTime>
<PTZPos>
  <P><!--opt, xs: integer--></P>
  <T><!--opt, xs: integer--></T>
  <Z><!--opt, xs: integer--></Z>
</PTZPos>
<advancedType><!--opt, xs: string, target sub type: "humanFeature, objectFeature"--></advancedType>
<advance><!--dep, it is valid when type is "motionDetection"-->
```

```
<HumanFeature>
  <age><!--opt, xs: integer, value range: [0,100]--></age>
  <sex><!--opt, xs: string, gender: "male,female"--></sex>
  <eyeGlass><!--opt, xs: boolean--></eyeGlass>
</HumanFeature><!--dep, it is valid when advancedType is "humanFeature"-->
<ObjectFeature>
  <colorRatel><!--opt, xs: integer--></colorRatel>
  <R><!--opt, xs: integer--></R>
  <G><!--opt, xs: integer--></G>
  <B><!--opt, xs: integer--></B>
</ObjectFeature><!--dep, it is valid when advancedType is "objectFeature"-->
</advance>
</FaceDetection>
</SmartSearchDescription>
```

XML_SmartSearchResult

SmartSearchResult message in XML format

```
<SmartSearchResult version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <searchID>
    <!--req, xs:string, search ID, which is used to confirm the upper-level platform or system. If the platform or system
are same during two times of search, the search history will be recorded in the memory for next fast search-->
  </searchID>
  <responseStatus><!--req, xs: string, "true,false"--></responseStatus>
  <responseStatusStrg><!--req, xs: string, "OK,NO MATCHES,MORE"--></responseStatusStrg>
  <numOfMatches><!--req, xs: integer--></numOfMatches>
  <totalMatches><!--req, xs: integer--></totalMatches>
  <matchList>
    <searchMatchItem><!--list-->
      <id><!--req, xs: integer--></id>
      <timeSpan>
        <startTime><!--req, xs: string, start time of validity period, e.g., 2013-06-10T 12:00:00Z--></startTime>
        <endTime><!--req,xs:string, end time of validity period, e.g., 2013-06-10T 12:00:00Z--></endTime>
      </timeSpan>
    </searchMatchItem>
  </matchList>
</SmartSearchResult>
```

XML_SMARTTestConfig

SMARTTestConfig message in XML format

```
<SMARTTestConfig version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enable opt="true,false"><!--req, xs:string, whether to continue using the HDD when the HDD SMART detection
failed--></enable>
</SMARTTestConfig>
```

XML_SMARTTestStatus

SMARTTestStatus message in XML format

```
<SMARTTestStatus version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs: string--></id>
  <temprature><!--req, xs: integer, unit: °C--></temprature>
  <powerOnDay><!--req, xs: integer, unit: day--></powerOnDay>
  <selfEvaluatingStatus><!--req, xs: string, "ok, error"--></selfEvaluatingStatus>
  <allEvaluatingStatus><!--req, xs: string, "unctional,badsectors,fault"--></allEvaluatingStatus>
  <selfTestPercent><!--req, xs: integer--></selfTestPercent>
  <selfTestStatus>
    <!--req, xs: string,
    "ok,aborted,interrupted,failed,unkown,electronic_element_error,servo_error,read_failed,progress,not_tested,not_rec
    ognized"-->
  </selfTestStatus>
  <testType><!--req, xs: string, "short,expanded,conveyance"--></testType>
  <TestResultList><!--req, list, the maximum size is 30 bytes-->
    <TestResult><!--list-->
      <attributeID><!--req, xs: string--></attributeID>
      <attributeName><!--opt, xs: string--></attributeName>
      <status><!--req, xs: string, "ok,illegal"--></status>
      <flags><!--req, xs: integer--></flags>
      <thresholds><!--req, xs: integer--></thresholds>
      <value><!--req, xs: string--></value>
      <worst><!--req, xs: integer--></worst>
      <rawValue><!--req, xs: integer--></rawValue>
    </TestResult>
  </TestResultList>
</SMARTTestStatus>
```

XML_SMSWhiteList

SMSWhiteList message in XML format

```
<SMSWhiteList version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema" >
  <ListMember/>
</SMSWhiteList>
```

XML_SNMP

XML message about SNMP configuration

```
<SNMP version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <SNMPv1c/><!--dependent -->
  <SNMPv2c/><!--dependent-->
  <SNMPAdvanced/><!--dependent, advanced parameters-->
```

```
<listenPort> <!--optional, xs:integer, SNMP port--></listenPort>
</SNMP>
```

XML_SNMPPAdvanced

XML message about SNMP advanced parameter configuration

```
<SNMPPAdvanced xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0" >
  <localEngineID>
    <!--required, xs:hexBinary, local device engine-->
  </localEngineID>
  <authenticationNotificationEnabled>
    <!--optional, xs:boolean, notification of failure to enable authentication-->
  </authenticationNotificationEnabled>
  <SNMPUserList/>
    <!--optional -->
  <SNMPNotificationFilterList/>
    <!--optional, filtering table based on OIDS-->
  <notificationEnabled>
    <!--optional, xs:boolean -->
  </notificationEnabled>
  <SNMPNotificationReceiverList/>
    <!--optional-->
  <enabled>
    <!--required, xs:boolean -->
  </enabled>
  <SNMPTrapReceiverList/>
    <!--optional, all trap address information of SNMP in version 3-->
</SNMPPAdvanced>
```

XML_SNMPTrapReceiver

XML message about parameters of single trap address of SNMP

```
<SNMPTrapReceiver xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0" >
  <id>
    <!--required, xs:string, ID -->
  </id>
  <ReceiverAddress>
    <!--required -->
  <addressingFormatType>
    <!--required, xs:string, "ipaddress,hostname"-->
  </addressingFormatType>
  <hostName>
    <!--dependent, xs:string -->
  </hostName>
  <ipAddress>
    <!--dependent, xs:string -->
  </ipAddress>
```



```
<ipv6Address>
  <!--dependent, xs:string -->
</ipv6Address>
<portNo>
  <!--optional, xs:integer -->
</portNo>
</ReceiverAddress>
<notificationType>
  <!--required, xs:string, "trap,inform" -->
</notificationType>
<communityString>
  <!--optional, xs:string -->
</communityString>
</SNMPTrapReceiver>
```

XML_SNMPTrapReceiverList

XML message about trap address parameters of SNMP

```
<SNMPTrapReceiverList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <SNMPTrapReceiver/> <!--optional-->
</SNMPTrapReceiverList>
```

XML_SoftwareService

SoftwareService message in XML format

```
<SoftwareService version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ThirdStream><!--opt, third stream configuration-->
    <enabled><!--req, xs: boolean, "true,false", whether to enable third stream configuration--></enabled>
  </ThirdStream>
  <PanoramaDisplay><!--opt, display the image range on panorama view-->
    <enabled>
      <!--req, xs: boolean, "true,false", whether to enable image range display on panorama view-->
    </enabled>
  </PanoramaDisplay>
  <MotionDetect>
    <enabled><!--whether to enable motion detetion--></enabled>
  </MotionDetect>
</SoftwareService>
```

XML_SocketIP

SocketIP message in XML format

```
<SocketIP version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ipAddress>
```

```
<!--opt, xs: string-->
</ipAddress>
<ipv6Address>
<!--opt, xs: string-->
</ipv6Address>
</SocketIP>
```

XML_sourceCapability

sourceCapability message in XML format

```
<sourceCapability version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <videoInputNums><!--req, xs:integer--></videoInputNums>
  <audioInputNums><!--opt, xs:integer--></audioInputNums>
</sourceCapability>
```

XML_sourceDescriptor

sourceDescriptor message in XML format

```
<sourceDescriptor version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <adminProtocol><!--req, xs:string, "HIKVISION,SONY,ISAPI,ONVIF,..."--></adminProtocol>
  <addressingFormatType><!--req,xs:string, opt="ipaddress,hostname"--></addressingFormatType>
  <hostName><!--dep,xs:string, domain name--></hostName>
  <ipAddress><!--dep,xs:string, IP address--></ipAddress>
  <ipv6Address><!--dep,xs:string, IPv6 address--></ipv6Address>
  <managePortNo><!--req,xs:integer, port number--></managePortNo>
  <userName><!--req,xs:string--></userName>
  <password><!--req,wo,xs:string--></password>
</sourceDescriptor>
```

XML_SSH

SSH message in XML format

```
<SSH version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--req, xs:boolean--></enabled>
</SSH>
```

XML_staticRouteList

staticRouteList message in XML format

```
<staticRouteList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <staticRoute><!--opt-->
    <id>
```

```
<!--req, xs: integer, this node must be configured when deleting-->
</id>
<enabled><!--req, xs: boolean--></enabled>
<dstIpAddr><!--req, xs: string--></dstIpAddr>
<dstNetmask><!--req, xs: string--></dstNetmask>
<dstGateway><!--req, xs: string--></dstGateway>
<ifName><!--req, xs: string, 0: G1, 1: G2--></ifName>
</staticRoute>
</staticRouteList>
```

XML_storageExtension

XML message about storage strategy.

```
<storageExtension xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0" >
  <LoopEnable,opt="true,false">
    <!--optional, xs:boolean, whether to enable overwriting-->
  </LoopEnable>
  <enableDormant>
    <!--optional, xs:boolean-->
  </enableDormant>
  <packDuration>
    <!--optional, xs:integer, unit: minute-->
  </packDuration>
  <logStorageMode>
    <!--optional, xs:string, log storage mode: "system"-system default mode, "custom"-custom mode-->
  </logStorageMode>
  <logStorageId>
    <!--dependent, xs:integer, disk ID (saving logs in a disk) or array number (saving logs in disk arrays), which are valid
only when logStorageMode is "custom"; -->
  </logStorageId>
  <logStorageCycle>
    <!--dependent, xs:integer, log storage period, which is valid only when logStorageMode is "custom"-->
  </logStorageCycle>
</storageExtension>
```

XML_StreamingChannel

StreamingChannel message in XML format

```
<StreamingChannel version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req,xs:string;id--></id>
  <channelName><!--req,xs:string--></channelName>
  <enabled><!--req,xs:boolean--></enabled>
  <Transport><!--req-->
    <maxPacketSize><!--opt, xs: integer--></maxPacketSize>
    <audioPacketLength><!--opt, xs: integer--></audioPacketLength>
    <audioInboundPacketLength><!--opt, xs: integer--></audioInboundPacketLength>
    <audioInboundPortNo><!--opt, xs: integer--></audioInboundPortNo>
```

```

<videoSourcePortNo><!--opt, xs: integer--></videoSourcePortNo>
<audioSourcePortNo><!--opt, xs: integer--></audioSourcePortNo>
<ControlProtocolList><!--req, protocol types for streaming-->
  <ControlProtocol><!--req-->
    <streamingTransport>
      <!--req,xs:string,"HTTP,RTSP,SHTTP,SRTP"-->
    </streamingTransport>
  </ControlProtocol>
</ControlProtocolList>
<Unicast><!--opt-->
  <enabled><!--req, xs: boolean--></enabled>
  <interfaceID><!--opt, xs: string--></interfaceID>
  <rtpTransportType>
    <!--opt, xs: string, "RTP/UDP, RTP/TCP"-->
  </rtpTransportType>
</Unicast>
<Multicast><!--opt-->
  <enabled><!--req, xs: boolean--></enabled>
  <userTriggerThreshold><!--opt, xs: integer--></userTriggerThreshold>
  <destIPAddress><!--dep, xs: string--></destIPAddress>
  <videoDestPortNo><!--opt, xs: integer--></videoDestPortNo>
  <audioDestPortNo><!--opt, xs: integer--></audioDestPortNo>
  <destIPv6Address><!--dep, xs: string--></destIPv6Address>
  <ttl><!--opt, xs:integer--></ttl>
  <activeMulticastEnabled>
    <!--opt, xs: boolean, "true,false", whether to enable active multicast, which is mutual exclusion with passive
multicast-->
  </activeMulticastEnabled>
  <packagingFormat><!--opt, xs: string, container format--></packagingFormat>
  <FecInfo><!--opt-->
    <fecRatio>
      <!--req, read-only, xs: integer, extra bandwidth occupation ratio of forward error correction (FEC) data, the value
is between 0 and 100, the default value is 0-->
    </fecRatio>
    <fecDestPortNo>
      <!--opt, xs: integer, Port No. of FEC multicast, the default port No. is specified by device-->
    </fecDestPortNo>
  </FecInfo>
</Multicast>
<Security><!--opt-->
  <enabled><!--req, xs: boolean--></enabled>
  <certificateType><!--req, xs: string, opt="digest,digest/baisc" def="digest"--></certificateType>
  <SecurityAlgorithm><!--dependency, this node is valid when certificateType is "digest" or "digest/basic"-->
    <algorithmType><!--optional, xs:string, algorithm type: "MD5", "SHA256", "MD5/SHA256"--></algorithmType>
  </SecurityAlgorithm>
</Security>
<SRTPMulticast><!--opt-->
  <SRTPVideoDestPortNo><!--opt, xs:integer--></SRTPVideoDestPortNo>
  <SRTPAudioDestPortNo><!--opt, xs:integer--></SRTPAudioDestPortNo>
</SRTPMulticast>
</Transport>
<Video><!--opt-->

```

```
<enabled><!--req, xs:boolean--></enabled>
<videoInputChannelID>
  <!--req, xs:string;id-->
</videoInputChannelID>
<videoCodecType>
  <!--req, xs:string, "MPEG4,MJPEG,3GP,H.264,HK.264,MPNG,SVAC,H.265"-->
</videoCodecType>
<videoResolutionWidth>
  <!--req, xs:integer-->
</videoResolutionWidth>
<videoResolutionHeight>
  <!--req, xs:integer-->
</videoResolutionHeight>
<videoQualityControlType>
  <!--opt, xs:string, "CBR,VBR"-->
</videoQualityControlType>
<constantBitRate>
  <!--dep, xs:integer, constant bit rate, unit: kbps-->
</constantBitRate>
<vbrUpperCap>
  <!--dep, xs:integer, upper limit of variable bit rate, unit: kbps-->
</vbrUpperCap>
<vbrLowerCap>
  <!--dep, xs:integer, lower limit of variable bit rate, unit: kbps-->
</vbrLowerCap>
<maxFrameRate>
  <!--req, xs:integer, maximum frame rate, the value is multiplied by 100 to be returned-->
</maxFrameRate>
<keyFrameInterval><!--opt, xs:integer, milliseconds--></keyFrameInterval>
<rotationDegree><!--opt, xs:integer, degrees, 0..360--></rotationDegree>
<mirrorEnabled><!--opt, xs:boolean--></mirrorEnabled>
<snapshotImageType><!--opt, xs:string, "JPEG,GIF,PNG"--></snapshotImageType>
<Mpeg4Profile> <!--dep, xs:string, "SP,ASP"--></Mpeg4Profile>
<H264Profile><!--dep, xs:string, "Baseline,Main,High, Extended"--></H264Profile>
<SVACProfile><!--dep, xs:string, "Baseline,Main,High,Extended"--></SVACProfile>
<GovLength> <!--opt, xs:integer--></GovLength>
<SVC>
  <enabled><!--req, xs:boolean--></enabled>
  <SVCMODE><!--dep, xs:string, "manual,auto"--></SVCMODE>
</SVC>
<smoothing><!--opt, xs:integer--></smoothing>
<SmartCodec><!--dep, -->
  <enabled><!--req, xs:boolean--></enabled>
</SmartCodec>
<vbrAverageCap><!--dep, xs:integer, in kbps--></vbrAverageCap>
<IntelligentInfoDisplayMethod>
  <!--opt, intelligent information displaying method, xs:string, "player,non-player"-->
</IntelligentInfoDisplayMethod>
</Video>
<Audio><!--opt-->
  <enabled><!--req, xs:boolean--></enabled>
  <audioInputChannelID><!--req, xs:string;id--></audioInputChannelID>
```

```
<audioCompressionType>
  <!--req, xs:string, "G.711alaw,G.711ulaw,G.726,G.729,G.729a,G.729b,PCM,MP3,AC3,AAC,ADPCM, MP2L2"-->
</audioCompressionType>
<audioInboundCompressionType>
  <!--opt, xs:string, "G.711alaw,G.711ulaw,G.726,G.729,G.729a,G.729b,PCM,MP3,AC3,AAC,ADPCM,MP2L2"-->
</audioInboundCompressionType>
<audioBitRate><!--opt, xs:integer, in kbps--></audioBitRate>
<audioSamplingRate><!--opt, xs:float, in kHz--></audioSamplingRate>
<audioResolution><!--opt, xs:integer, in bits--></audioResolution>
<VoiceChanger><!--opt, xs:integer, -12..0..12-->
  <enabled><!--req, xs:boolean--></enabled>
  <level><!--req, xs:integer, "-12..12"--></level>
</VoiceChanger>
</Audio>
<enableCABAC><!--opt, xs: boolean--></enableCABAC>
<subStreamRecStatus><!--opt, xs: boolean--></subStreamRecStatus>
<customStreamEnable><!--opt, xs: boolean, whether the stream is custom stream: "true"-yes, this node is not
returned-no--></customStreamEnable>
</StreamingChannel>
```

XML_StreamingChannelList

StreamingChannelList message in XML format

```
<StreamingChannelList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <StreamingChannel/><!--opt, see details in XML_StreamingChannel-->
</StreamingChannelList>
```

See Also

XML_StreamingChannel

XML_StreamingStatus

StreamingStatus message in XML format

```
<StreamingStatus version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <totalStreamingSessions><!--req, xs: integer, number of streaming sessions--></totalStreamingSessions>
  <StreamingSessionStatusList/><!--dep, this node is valid only when the number of sessions is larger than 0, refer to
XML_StreamingSessionStatusList for details-->
</StreamingStatus>
```

See Also

XML_StreamingSessionStatusList

XML_StreamingSessionStatusList

StreamingSessionStatusList message in XML format

```
<StreamingSessionStatusList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <StreamingSessionStatus>
    <clientAddress><!--req-->
      <ipAddress><!--dep, xs: string--></ipAddress>
      <ipv6Address><!--dep, xs: string--></ipv6Address>
    </clientAddress>
  </StreamingSessionStatus>
</StreamingSessionStatusList>
```

XML_SubscribeEvent

SubscribeEvent message in XML format

```
<SubscribeEvent version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema" >
  <heartbeat>
    <!--optional, xs:integer, heartbeat interval, unit: second, the default value is 30s-->
  </heartbeat>
  <eventMode>
    <!--required, xs:string, "all"-upload all alarms/events, "list"-upload specified alarm/event-->
  </eventMode>
  <EventList>
    <Event><!--uploading mode of specified alarm/event, this node exists only when eventMode is "list"-->
      <type>
        <!--required, xs:string, alarm/event types, which are obtained from the capability, refer to Alarm/Event Types for
Subscription for its values-->
      </type>
      <minorAlarm>
        <!--opt, xs:string, minor alarm type: "0x400,0x401,0x402,0x403", see details in Access Control Event Type. This
node is required when type is "AccessControllerEvent"-->
      </minorAlarm>
      <minorException>
        <!--opt, xs:string, minor exception type: "0x400,0x401,0x402,0x403", see details in Access Control Event Type.
This node is required when type is "AccessControllerEvent"-->
      </minorException>
      <minorOperation>
        <!--opt, xs:string, minor operation type: "0x400,0x401,0x402,0x403", see details in Access Control Event Type.
This node is required when type is "AccessControllerEvent"-->
      </minorOperation>
      <minorEvent>
        <!--opt, xs:string, minor event type: "0x01,0x02,0x03,0x04", see details in Access Control Event Type. This node is
required when type is "AccessControllerEvent"-->
      </minorEvent>
      <pictureURLType>
        <!--opt, xs:string, alarm picture format: "binary"-binary, "localURL"-device local URL, "cloudStorageURL"-cloud
storage URL-->
      </pictureURLType>
    </Event>
  </EventList>
</SubscribeEvent>
```

```
</pictureURLType>
</Event>
</EventList>
<channels>
  <!--optional, xs:string, event linked channel information, and multiple channels can be linked, each channel is
separated by comma, e.g., "1,2,3,4..."-->
</channels>
<channels>
  <!--optional, xs:string, specify channels (each channel is separated by comma, e.g., "1,2,3,4...") to be armed, this
node does not exist if you want to arm all channels, and if this node exists, the sub node <channels> in the node
<Event> is invalid-->
</channels>
<identityKey max="64"/>
<!--opt, xs: string, interaction command of subscription, supports subscribing comparison results of face picture
library (importing with this command), the maximum length is 64-->
</SubscribeEvent>
```

XML_SubscribeEventCap

SubscribeEventCap capability message in XML format

```
<SubscribeEventCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <heartbeat min="" max="" />
  <!--optional, heartbeat time interval, unit: second-->
  <format opt="xml,json"/><!--req, supported message format-->
  <channelMode opt="all,list" />
  <!--required, channel subscription mode: "all"-subscribe events/alarms of all channels, "list"-subscribe events/
alarms of specific channels-->
  <eventMode opt="all,list" />
  <!--required, event subscription mode: "all"-subscribe all event types (must be supported), "list"-subscribe specific
event types, if "list" is returned, "all" will also be returned-->
  <!--if both the channelMode and eventMode returns "all", it indicates that the device does not support subscribing
event/alarm by event type or channel-->
  <EventList><!--required, dependent, upload mode of specified alarms/events, it is valid only when eventMode is
"list"-->
  <Event><!--required-->
    <type><!--required, xs:string, refer to Supported Alarm/Event Types for details--></type>
    <minorAlarm opt="0x400,0x401,0x402,0x403">
      <!--opt, xs:string, minor alarm type, see details in Access Control Event Type. This node is required when type is
"AccessControllerEvent"-->
    </minorAlarm>
    <minorException opt="0x400,0x401,0x402,0x403">
      <!--opt, xs:string, minor exception type, see details in Access Control Event Type. This node is required when type
is "AccessControllerEvent"-->
    </minorException>
    <minorOperation opt="0x400,0x401,0x402,0x403">
      <!--opt, xs:string, minor operation type, see details in Access Control Event Type. This node is required when type
is "AccessControllerEvent"-->
    </minorOperation>
    <minorEvent opt="0x01,0x02,0x03,0x04">
      <!--opt, xs:string, minor event type, see details in Access Control Event Type. This node is required when type is
```



```
"AccessControllerEvent"-->
  </minorEvent>
  <pictureURLType opt="binary,localURL,cloudStorageURL" def=""/>
  <!--opt, xs:string, alarm picture format: "binary"-binary, "localURL"-device local URL, "cloudStorageURL"-cloud
storage URL, and the def is followed by the default format-->
</Event>
</EventList>
<pictureURLType opt="binary,localURL,cloudStorageURL" def=""/>
  <!--opt, xs:string, alarm picture format: "binary"-binary picture, "localURL"-device local URL, "cloudStorageURL"-
cloud storage URL. This node is the method of uploading all pictures related to the event. If this node is applied,
<pictureURLTypeEventall> is
"true"-->
      <hd/><!--opt, xs:boolean, whether to subscribe disk status. This node is not required when <all> is "true"-->
      <capability/><!--opt, xs:boolean, whether to subscribe capability change status. This node is not required when
<all> is "true"-->
    </StatusSub>
  </ChangedUploadSub>
  <identityKey max="64"/>
  <!--opt, xs: string, interaction command of subscription, supports subscribing comparison results of face picture
library (importing with this command), the maximum length is 64-->
</SubscribeEventCap>
```

XML_SubscribeEventResponse

SubscribeEventResponse message in XML format

```
<SubscribeEventResponse>
  <id><!--req, xs:integer, subscription ID--></id>
  <FailedEventList>
    <!--opt, list of subscription failed events. When subscription failed, it should be returned, and the upper layer can
check whether all event/alarm subscriptions are succeeded via the existence of node FailedEventList-->
    <Event>
      <type>
        <!--req, xs:string, refer to Supported Alarm/Event Types for details-->
      </type>
      <minorAlarm>
        <!--opt, xs:string, minor alarm type: "0x400,0x401,0x402,0x403", see details in Access Control Event Type. This
node is required when type is "AccessControllerEvent"-->
      </minorAlarm>
      <minorException>
        <!--opt, xs:string, minor exception type: "0x400,0x401,0x402,0x403", see details in Access Control Event Type.
This node is required when type is "AccessControllerEvent"-->
      </minorException>
```

```
<minorOperation>
  <!--opt, xs:string, minor operation type: "0x400,0x401,0x402,0x403", see details in Access Control Event Type.
This node is required when type is "AccessControllerEvent"-->
</minorOperation>
<minorEvent>
  <!--opt, xs:string, minor event type: "0x01,0x02,0x03,0x04", see details in Access Control Event Type. This node is
required when type is "AccessControllerEvent"-->
</minorEvent>
<pictureURLType>
  <!--opt,xs:string, opt="binary,localURL,cloudStorageURL", alarm picture transmission mode: "binary"-binary,
"localURL"-device local URL, "cloudStorageURL"-cloud storage URL-->
</pictureURLType>
<channels>
  <!--opt, xs:string, "1,2,3,4...", event related channel ID, supports multiple channels, and the channel ID is
separated by commas-->
</channels>
<subStatusCode>
  <!--req, string, subscription failure error code-->
</subStatusCode>
</Event>
</FailedEventList>
</SubscribeEventResponse>
```

XML_SupplementLight

SupplementLight message in XML format

```
<SupplementLight><!--opt-->
<mode><!--opt, xs: string, adjustment mode of supplement light, opt="schedule,off,on,auto"--></mode>
<Schedule>
  <TimeRange><!--req-->
    <beginTime><!--req, xs: time, ISO8601 time--></beginTime>
    <endTime><!-- req, xs: time, ISO8601 time--></endTime>
  </TimeRange>
</Schedule>
<brightnessLimit>
  <!--dep, xs: integer, brightness of supplement light, which is between 0 and 100; the brightness can be adjust when
&ltb>mode> is set to "on"-->
</brightnessLimit>
<supplementLightMode>
  <!--opt, xs: string, illumination mode: "mixed"-hybrid, "whitelight"-white light, "close"-disabled-->
</supplementLightMode>
<irLightBrightness>
  <!--dep, xs: integer, brightness of IR supplement light, which is between 0 and 100; this node is valid only when
&ltb>supplementLightMode> is set to "mixed" and &ltb>mixedLightBrightnessRegulatMode> is set to "manual"-->
</irLightBrightness>
<mixedLightBrightnessRegulatMode>
  <!--dep, xs: string, brightness adjustment mode of hybrid supplement light, opt="manual,auto"; this node is valid
only when &ltb>supplementLightMode> is set to "mixed"-->
</mixedLightBrightnessRegulatMode>
<highIrLightBrightness>
```

```
<!--dep, xs: integer, brightness of far IR light, which is between 0 and 100; this node cannot be configured together with node <irLightBrightness-->
</highIrLightBrightness>
<highWhiteLightBrightness>
<!--dep, xs: integer, brightness of far white light, which is between 0 and 100; this node cannot be configured together with node <brightnessLimit-->
</highWhiteLightBrightness>
<lowIrLightBrightness>
<!--dep, xs: integer, brightness of near IR light, which is between 0 and 100; this node cannot be configured together with node <irLightBrightness-->
</lowIrLightBrightness>
<lowWhiteLightBrightness>
<!--dep, xs: integer, brightness of near white light, which is between 0 and 100; this node cannot be configured together with node <brightnessLimit-->
</lowWhiteLightBrightness>
<whiteLightBrightness><!--dep, xs: integer, white light brightness, which is between 0 and 100--></whiteLightBrightness>
<irLightbrightnessLimit><!--dep, xs:integer, IR light brightness limit, range: [0,100]; this node is valid when the value of mode "auto"--></irLightbrightnessLimit>
<whiteLightbrightnessLimit><!--dep, xs:integer, white light brightness limit, range: [0,100]; this node is valid when the value of mode "auto"--></whiteLightbrightnessLimit>
<unscheduledBrightness><!--optional, xs:integer, brightness in the unscheduled time, by default it is 0 (the supplement light is disabled) and the value is between 0 and 100--></unscheduledBrightness>
</SupplementLight>
```

XML_SyncSignalOutputList

SyncSignalOutputList message in XML format

```
<SyncSignalOutputList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <SyncSignalOutput>
    <IOWorkMode>
      <!--req, xs:string, IO output mode: "flashLight"-strobe light control mode, "polarizer"-polarizer control mode, "continuousLight"-solid light control mode, "flashLightSwitch"-strobe light switching mode (used to switch between the strobe supplement light mode and IR mode). When this node is set to "polarizer", only <outputStatus>, <detectBrightnessEnable>, <brightnessThreshold>, <flashEnabled>, <startHour>, <startMinute>, <endHour>, and <endMinute> are valid, and the <outputStatus> can only be set to "high" or "low". When this node is set to "continuousLight", only <detectBrightnessEnable>, <brightnessThreshold>, <flashEnabled>, <startHour>, <startMinute>, <endHour>, and <endMinute> are valid-->
    </IOWorkMode>
    <id>
      <!--req, xs:integer, number of IO outputs, which is between 1 and 8-->
    </id>
    <defaultStatus>
      <!--req, xs:string, IO default status: "high, low"-->
    </defaultStatus>
    <outputStatus>
      <!--req,xs:string, IO effective status: "high, low, pulse"-->
    </outputStatus>
    <aheadTime>
      <!--req, xs:integer, IO pre-output time, unit: microsecond-->
```

```
</aheadTime>
<timeDelay>
  <!--req, xs:integer, IO effective duration, unit: microsecond-->
</timeDelay>
<freqMultiplyulti>
  <!--req, xs:integer, frequency multiplication, which is between 1 and 15-->
</freqMultiplyulti>
<dutyRate>
  <!--req, xs:integer, duty ratio, which is between 0 and 40, unit: %-->
</dutyRate>
<postFlashEnable>
  <!--req, xs:boolean, checkpoint output: 0-enable, 1-disable-->
</postFlashEnable>
<illegalFlashEnable>
  <!--req, xs:boolean, violation output: 0/"true"-enable, 1/"false"-disable-->
</illegalFlashEnable>
<videoFlashEnable>
  <!--req, xs:boolean, video output: 0/"true"-enable, 1/"false"-disable-->
</videoFlashEnable>
<detectBrightnessEnable>
  <!--req, xs:boolean, whether to enable flash light for automatic brightness detection-->
</detectBrightnessEnable>
<brightnessThreshold>
  <!--dep, xs:integer, brightness threshold of the enabled flash light, which is between 0 and 100-->
</brightnessThreshold>
<flashEnabled>
  <!--req, xs:boolean, whether to enable flash light-->
</flashEnabled>
<startHour>
  <!--dep, xs:integer, start time in hour-->
</startHour>
<startMinute>
  <!--dep, xs:integer, start time in minute-->
</startMinute>
<endHour>
  <!--dep, xs:integer, end time in hour-->
</endHour>
<endMinute>
  <!--dep, xs:integer, end time in minute-->
</endMinute>
<plateBrightness>
  <!--req, xs:boolean, whether to enable flash light by license plate brightness: 0-disable, 1-enable. Only one node
among <detectBrightness>, <flashEnabled>, <manualBrightnessEnable> and <plateBrightness> can be enabled-->
</plateBrightness>
<incrBrightEnable>
  <!--opt, xs:boolean, whether to enable brightness enhancement mode (for solid light mode)-->
</incrBrightEnable>
<incrBrightTime>
  <!--req, xs:integer, brightness enhancement duration. This node is valid only when <incrBrightEnable> is "true"-->
</incrBrightTime>
<incrBrightPercent>
  <!--req, xs:integer, percentage of brightness enhancement, which is between 0 and 100. This node is valid only
```

```
when <incrBrightEnable> is "true"-->
  </incrBrightPercent>
  <brightness>
    <!--dep, xs:integer, solid light brightness (for solid light mode), which is between 0 and 100-->
  </brightness>
  <delayCaptureTime>
    <!--dep, xs:integer, delayed capture time, which is between 1 and 1000, unit: millisecond. This node is valid only
when <incrBrightEnable> is "true"-->
  </delayCaptureTime>
  <manualBrightnessEnable>
    <!--req, xs:boolean, whether to enable adjusting brightness manually-->
  </manualBrightnessEnable>
  <manualBrightness>
    <!--dep, xs:integer, brightness adjusted manually, which is between 0 and 100-->
  </manualBrightness>
</SyncSignalOutput>
</SyncSignalOutputList>
```

XML_TargetEnhancement

TargetEnhancement message in XML format

```
<TargetEnhancement version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--req,xs:boolean--></enabled>
</TargetEnhancement>
```

XML_TargetEnhancementCap

TargetEnhancementCap message in XML format

```
<TargetEnhancementCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled opt="true,false"><!--req, xs:boolean--></enabled>
</TargetEnhancementCap>
```

XML_Telnetd

Telnetd message in XML format

```
<Telnetd version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled>
    <!--req, xs:boolean, whether to enable telnet: "true, false"-->
  </enabled>
  <radarMsgEnabled>
    <!--req, xs:boolean, whether to enable uploading radar debugging information: "true, false"-->
  </radarMsgEnabled>
  <illegalLoginLock>
    <!--req, xs:boolean, "true, false"-->
  </illegalLoginLock>
```

```
</illegalLoginLock>
</Telnetd>
```

XML_tempRange

tempRange message in XML format

```
<?xml version="1.0" encoding="utf-8"?>
<TempRange version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <mode>
    <!--req, xs:string, temperature range mode: "automatic"-automatic mode, "manual"-manual mode-->
  </mode>
  <temperatureUpperLimit>
    <!--dep, xs:float, maximum temperature, corrects to one decimal, the minimum value is "-20", the maximum value is "550", unit: Celsius degree; its value should be larger than temperatureLowerLimit; it is valid when the value of mode is "manual"-->
  </temperatureUpperLimit>
  <temperatureLowerLimit>
    <!--dep, xs:float, minimum temperature, corrects to one decimal, the minimum value is "-20", the maximum value is "550", unit: Celsius degree; its value should be smaller than temperatureUpperLimit; it is valid when the value of mode is "manual"-->
  </temperatureLowerLimit>
</TempRange>
```

XML_TextOverlay

TextOverlay message in XML format

```
<TextOverlay version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:string, ID--></id>
  <enabled><!--req, xs:boolean--></enabled>
  <positionX><!--req, xs:float--></positionX>
  <positionY><!--req, xs:float--></positionY>
  <displayText><!--req, xs:string--></displayText>
</TextOverlay>
```

XML_TextOverlayList

TextOverlayList message in XML format

```
<TextOverlayList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <TextOverlay/><!--opt, see details in the message of XML_TextOverlay-->
</TextOverlayList>
```

See Also

XML_TextOverlay

XML_ThermalCap

ThermalCap message in XML format.

```
<ThermalCap version="2.0" xmlns="http://www.std-cgi.org/ver20/XMLSchema">
  <isSupportFireDetection><!--opt, xs:boolean, whether supports fire detection--></isSupportFireDetection>
  <isSupportThermometry><!--opt, xs:boolean, whether supports thermometry--></isSupportThermometry>
  <isSupportRealtimeThermometry><!--opt, xs:boolean, whether supports uploading real-time thermometry data--></
isSupportRealtimeThermometry>
  <isFireFocusZoomSupport><!--opt, xs:boolean, whether supports visible light lens zooming--></
isFireFocusZoomSupport>
  <isSupportManualRanging>
    <!--opt, xs:boolean, this node will be returned if one or more channels of device supports this function, see detailed
channel capability in the response information"-->
  </isSupportManualRanging>
  <isSupportPower><!--opt, xs:boolean, whether supports power on/off capability--></isSupportPower>
  <isSupportRealtimeTempHumi><!--opt, xs:boolean, whether supports real-time detection of temperature and
humidity--></isSupportRealtimeTempHumi>
  <ManualThermCap>
    <manualThermRuleNum>
      <!--opt, xs:integer, the max. number of supported rules for manual thermometry. If this node is not returned, it
indicates manual thermometry is not supported-->
    </manualThermRuleNum>
  </ManualThermCap>
  <isSupportManualThermBasic>
    <!--opt, xs:boolean, whether supports basic configuration of manual thermometry-->
  </isSupportManualThermBasic>
  <isSupportFireShieldMask><!--opt, xs:boolean--></isSupportFireShieldMask>
  <isSupportSmokeShieldMask><!--opt, xs:boolean--></isSupportSmokeShieldMask>
  <isSupportThermometryMode>
    <!--opt, xs:boolean, whether supports the configuration of temperature measurement mode-->
  </isSupportThermometryMode>
  <isSupportThermalPip>
    <!--opt, xs:boolean, whether supports the PIP configuration-->
  </isSupportThermalPip>
  <isSupportThermalIntelRuleDisplay><!--opt, xs:boolean, whether supports VCA rule configuration--></
isSupportThermalIntelRuleDisplay>
  <AlgVersionInfo><!--opt, whether supports getting the version information of thermal algorithms library-->
    <thermometryAlgName min = "1" max = "128">
      <!--ro, xs:string, version information of temperature measurement algorithms library-->
    </thermometryAlgName>
    <shipsAlgName min = "1" max = "128"><!--ro, xs:string, version name of ship detection algorithms library--></
shipsAlgName>
  </AlgVersionInfo>
  <isSupportFaceThermometry><!--opt, xs:boolean, whether supports body thermometry configuration--></
isSupportFaceThermometry>
  <isSupportThermalBlackBody><!--opt, xs:boolean, whether supports black body configuration--></
isSupportThermalBlackBody>
  <isSupportThermalStreamParam><!--opt, xs:boolean, whether supports stream configuration--></
isSupportThermalStreamParam>
  <isSupportBodyTemperatureCompensation>
```

```
<!--opt, xs:boolean, whether supports temperature compensation configuration-->
</isSupportBodyTemperatureCompensation>
<isSupportTemperatureCorrection><!--opt, xs:boolean, whether device supports thermometry correction--></
isSupportTemperatureCorrection>
<isSupportClickToThermometry><!--opt, xs:boolean, whether device supports clicking for thermometry--></
isSupportClickToThermometry>
<isSupportThermometryHistorySearch><!--opt, xs:boolean--></isSupportThermometryHistorySearch>
<isSupportBurningPrevention><!--opt, xs:boolean, whether device supports burning prevention--></
isSupportBurningPrevention>
<isSupportTemperatureCollection><!--opt, xs:boolean, whether device supports temperature ANR--></
isSupportTemperatureCollection>
<isSupportJpegPicWithAppendData>
<!--opt, xs:boolean, whether device supports getting JPEG picture with pixel-to-pixel thermometry data. If supports,
it is returned and values true, if not support, it is not returned-->
</isSupportJpegPicWithAppendData>
<isSupportRealTimethermometryForHTTP>
<!--opt, xs:boolean, whether device supports real-time thermometry. If supports, it is returned and its value is true,
if not support, it is not returned-->
</isSupportRealTimethermometryForHTTP>
<isSupportShipsDetectionWithScene>
<!--opt, xs:boolean, whether device supports ship detection by scene, this node and isSupportShipsDetection in
XML_SmartCap are mutually exclusive-->
</isSupportShipsDetectionWithScene>
<isSupportthermometryOffLineCapture>
<!--opt, xs:boolean, whether device supports offline capture. If supports, this node returned and its value is true; if
not, it is not returned-->
</isSupportthermometryOffLineCapture>
</ThermalCap>
```

Remarks

When getting thermal product capabilities, **isSupportShipsDetectionWithScene** has a higher priority than **isSupportShipsDetection**. That is, firstly check if the node **isSupportShipsDetectionWithScene** exists and its value is "true", that indicates ship detection according to scene is supported, otherwise, check the if the node **isSupportShipsDetection** exists.

XML_Time

Time message in XML format

```
<Time version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
<timeMode><!--req, xs:string, "manual, NTP, local, satellite, timecorrect"--></timeMode>
<localTime>
<!--req, xs:datetime, ISO 8601 time format, device time set manually, e.g.: 2018-02-01T19:54:04. This node is
required when <timemode> is "manual" or "local"-->
</localTime>
<timeZone>
<!-- req, xs:string, POSIX time zone string, device time synchronized with NTP, this node is required when
<timemode> is "manual", "local" or "NTP". The value of this node should be the same as that configured by the
request URI: /ISAPI/System/time/timeZone-->
</timeZone>
```



```
<satelliteInterval><!--dep, xs:integer, unit: minute--></satelliteInterval>
</Time>
```

XML_TimeTaskList

TimeTaskList message in XML format

```
<TimeTaskList size="">
  <enabled><!--req, xs: boolean--></enabled>
  <Parktime min="" max=""><!--req, xs:integer, seconds--></Parktime>
  <TimeTaskBlock>
    <dayOfWeek><!--req, xs: integer, day of the week based on ISO8601, "1"-Monday, ...--></dayOfWeek>
    <TimeTaskRange>
      <TaskID min="" max=""><!--req, xs: string; ID--></TaskID>
      <beginTime><!--req, xs: time, ISO8601 time--></beginTime>
      <endTime><!--req, xs: time, ISO8601 time--></endTime>
      <Task>
        <TaskType
          opt="disable,autoscan,framescan,randomscan,panoramascan,patrol,pattern,preset,tiltscan,periodreboot,periodadjust,
auxoutput">
          <!--req, xs: strings-->
        </TaskType>
        <patrolTaskNum min="" max=""><!--dep, xs: integer, from 0 to 8--></patrolTaskNum>
        <patternTaskNum min="" max=""><!--dep, xs: integer, from 0 to 8--></patternTaskNum>
        <presetTaskNum min="" max=""><!--dep, xs: integer, from 0 to 8--></presetTaskNum>
        <auxoutputTaskNum min="" max=""><!--dep, xs: integer, from 0 to 8--></
auxoutputTaskNum>
      </Task>
    </TimeTaskRange>
  </TimeTaskBlock>
  <isSupportTimeTaskCopy><!--req, xs: boolean--></isSupportTimeTaskCopy>
</TimeTaskList>
```

XML_trackDailyParam

trackDailyParam message in XML format

```
<trackDailyParam version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <year><!--req, xs: integer, year--></year>
  <monthOfYear><!--req, xs: integer, month of the year--></monthOfYear>
</trackDailyParam>
```

XML_trackDailyDistribution

trackDailyDistribution message in XML format

```
<trackDailyDistribution version="1.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <dayList>
    <day>
      <id><!--req, xs: integer, ID--></id>
      <dayOfMonth><!--req, xs: integer, day of the month, starts from 1st--></dayOfMonth>
      <record><!--req, xs: boolean, true-with video, false-no video--></record>
      <recordType>
        <!--req, xs: string, record type: time-continuous recording; event-record based on event-->
      </recordType>
    </day>
  </dayList>
</trackDailyDistribution>
```

XML_TrackList

TrackList message in XML format

```
<TrackList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <Track/><!--opt, recording schedule list, refer to the message XML_Track for details-->
</TrackList>
```

See Also

XML_Track

XML_Track

Track message in XML format

```
<Track version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs: integer--></id>
  <Channel><!--req, xs: integer--></Channel>
  <Enable><!--req, xs: boolean--></Enable>
  <Description><!--req, xs: string--></Description>
  <TrackGUID><!--req, xs: string--></TrackGUID>
  <Size><!--opt, xs: integer--></Size>
  <Duration min="" max=""><!--opt, xs: string--></Duration>
  <DefaultRecordingMode><!--req, xs: string--></DefaultRecordingMode>
  <LoopEnable><!--opt, xs: string--></LoopEnable>
  <SrcDescriptor>
    <SrcGUID><!--req, xs: string--></SrcGUID>
    <SrcChannel><!--req, xs: integer--></SrcChannel>
    <StreamHint><!--req, xs: string--></StreamHint>
    <SrcDriver><!--req, xs: string--></SrcDriver>
    <SrcType><!--req, xs: string--></SrcType>
    <SrcUrl><!--req, xs: string--></SrcUrl>
    <SrcUrlMethods><!--req, xs: string--></SrcUrlMethods>
    <SrcLogin><!--req, xs: string--></SrcLogin>
  </SrcDescriptor>
```

```
<TrackSchedule>
  <ScheduleBlockList>
    <ScheduleBlock>
      <ScheduleBlockGUID><!--req, xs: string--></ScheduleBlockGUID>
      <ScheduleBlockType><!--req, xs: string--></ScheduleBlockType>
      <ScheduleAction>
        <id><!--req, xs: integer--></id>
        <ScheduleActionStartTime>
          <DayOfWeek><!--req, xs: string--></DayOfWeek>
          <TimeOfDay><!--req, xs: string--></TimeOfDay>
        </ScheduleActionStartTime>
        <ScheduleActionEndTime>
          <DayOfWeek><!--req, xs: string--></DayOfWeek>
          <TimeOfDay><!--req, xs: string--></TimeOfDay>
        </ScheduleActionEndTime>
        <ScheduleDSTEnable><!--req, xs: boolean--></ScheduleDSTEnable>
        <Description><!--req, xs: string--></Description>
        <Actions>
          <Record><!--req, xs: boolean--></Record>
          <Log><!--req, xs: boolean--></Log>
          <SaveImg><!--req, xs: boolean--></SaveImg>
          <ActionRecordingMode><!--req, xs: string--></ActionRecordingMode>
        </Actions>
      </ScheduleAction>
    </ScheduleBlock>
  </ScheduleBlockList>
</TrackSchedule>
<CustomExtensionList>
  <CustomExtension>
    <CustomExtensionName><!--req, xs: string--></CustomExtensionName>
    <enableSchedule><!--req, xs: boolean--></enableSchedule>
    <SaveAudio><!--req, xs: boolean--></SaveAudio>
    <PreRecordTimeSeconds><!--req, xs: integer--></PreRecordTimeSeconds>
    <PostRecordTimeSeconds><!--req, xs: integer--></PostRecordTimeSeconds>
    <HolidaySchedule>
      <ScheduleBlock>
        <ScheduleBlockGUID><!--req, xs: string--></ScheduleBlockGUID>
        <ScheduleBlockType><!--req, xs: string--></ScheduleBlockType>
      </ScheduleBlock>
    </HolidaySchedule>
  </CustomExtension>
</CustomExtensionList>
<IntelligentRecord><!--opt, xs:boolean, whether to enable VCA recording function: 0-no, 1-yes--></IntelligentRecord>
<delayTime><!--opt, xs:integer, capture delay time, unit: second--></delayTime>
<durationEnabled><!--opt, xs:boolean, whether to enable video expiry time. If this function is not supported, this
node will not be returned. If this function is supported, the video expiry date will be set by the node <Duration>--></
durationEnabled>
</Track>
```

XML_Tracking

Tracking message in XML format

```
<Tracking version="2.0" xmlns="http://www.std-cgi.org/ver20/XMLSchema">
  <enabled>
    <!--req, xs:boolean-->
  </enabled>
  <mode>
    <!--dep, xs:string, "auto,manual"-->
  </mode>
  <trackingTime>
    <!--dep, xs:integer-->
  </trackingTime>
  <normalizedScreenSize>
    <!--req, ro-->
    <normalizedScreenWidth>
      <!--req, ro, xs:integer-->
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, ro, xs:integer-->
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <ManualRegionCoordinatesList min="" max="">
    <!--dep-->
    <RegionCoordinates>
      <!--opt-->
      <positionX>
        <!--req, xs:integer, X-coordinate-->
      </positionX>
      <positionY>
        <!--req, xs:integer, Y-coordinate-->
      </positionY>
    </RegionCoordinates>
  </ManualRegionCoordinatesList>
  <Schedule><!--opt, schedule, this node is valid only when enabled is "true". By default the schedule is from 00:00 to 24:00 if no schedule is configured-->
    <TimeRange>
      <beginTime>
        <!--UTC time, e.g.: "2018-03-13T19:42:27+08:00"--->
      </beginTime>
      <endTime>
        <!--UTC time, e.g.: "2018-03-13T19:42:27+08:00"--->
      </endTime>
    </TimeRange>
  </Schedule>
</Tracking>
```

XML_TwoWayAudioChannel

TwoWayAudioChannel message in XML format

```
<TwoWayAudioChannel version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id>
    <!--req, xs:string, two-way audio channel ID-->
  </id>
  <enabled>
    <!--req, xs:boolean, whether to enable two-way audio: "true"-yes, "false"-no-->
  </enabled>
  <audioCompressionType>
    <!--req, xs:string, audio output encoding type: "G.711alaw,G.711ulaw,G.726,G.729,G.729a,G.729b,PCM,MP3,AC3,AAC,ADPCM,MP2L2"-->
  </audioCompressionType>
  <audioInboundCompressionType>
    <!--opt, xs:string, audio input encoding type: "G.711alaw,G.711ulaw,G.726,G.729,G.729a,G.729b,PCM,MP3,AC3,AAC,ADPCM"-->
  </audioInboundCompressionType>
  <speakerVolume>
    <!--opt, xs:integer, input volume-->
  </speakerVolume>
  <microphoneVolume>
    <!--opt, xs:integer, output volume-->
  </microphoneVolume>
  <noisereduce>
    <!--opt, xs:boolean, whether to enable noise reduction: "true, false"-->
  </noisereduce>
  <audioBitRate>
    <!--opt, xs:integer, audio frame rate, unit: kbs-->
  </audioBitRate>
  <audioInputType>
    <!--opt, xs:string, audio input type: "MicIn, LineIn"-->
  </audioInputType>
  <associateVideoInputs><!--opt-->
    <enabled>
      <!--req, xs:boolean-->
    </enabled>
    <videoInputChannelList>
      <!--req-->
      <videoInputChannelID>
        <!--opt, xs:string, ID-->
      </videoInputChannelID>
    </videoInputChannelList>
  </associateVideoInputs>
  <lineOutForbidden>
    <!--read-only, xs:boolean, whether the audio output is not supported, if this node is not returned or the value is "false", it represents that audio output is supported; if the value is "true", it represents that audio output is not supported-->
  </lineOutForbidden>
  <micInForbidden>
```

```
<!--read-only, xs: boolean, whether the audio input is not supported, if this node is not returned or the value is
"false", it represents that audio input is supported; if the value is "true", it represents that audio input is not
supported-->
</micInForbidden>
</TwoWayAudioChannel>
```

XML_TwoWayAudioChannelCap

TwoWayAudioChannelCap capability message in XML format

```
<TwoWayAudioChannelCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id opt="1">
    <!--req, xs:string, two-way audio channel ID-->
  </id>
  <enabled opt="true,false">
    <!--req, xs:boolean, whether to enable two-way audio: "true"-yes, "false"-no-->
  </enabled>
  <audioCompressionType opt="G.711alaw,G.711ulaw,G.726,G.729,G.729a,G.
729b,PCM,MP3,AC3,AAC,ADPCM,MP2L2">
    <!--req, xs:string, audio output encoding type-->
  </audioCompressionType>
  <audioInboundCompressionType opt=" G.711alaw,G.711ulaw,G.726,G.729,G.729a,G.
729b,PCM,MP3,AC3,AAC,ADPCM">
    <!--opt, xs:string, audio input encoding type-->
  </audioInboundCompressionType>
  <speakerVolume opt="0-100">
    <!--opt, xs:integer, input volume, which is between 0 and 100-->
  </speakerVolume>
  <microphoneVolume opt="0-100">
    <!--opt, xs:integer, output volume, which is between 0 and 100-->
  </microphoneVolume>
  <noisereduce opt="true,false">
    <!--opt, xs:boolean, whether to enable noise reduction: "true, false"-->
  </noisereduce>
  <audioBitRate opt="32,64,128">
    <!--opt, xs:integer, audio frame rate, unit: kbs-->
  </audioBitRate>
  <audioInputType opt="MicIn,LineIn">
    <!--opt, xs:string, audio input type-->
  </audioInputType>
  <associateVideoInputs>
    <!--opt-->
    <enabled opt="true,false">
      <!--req, xs:boolean-->
    </enabled>
    <videoInputChannelList>
      <!--req-->
      <videoInputChannelID opt="1">
        <!--opt, xs:string, ID-->
      </videoInputChannelID>
    </videoInputChannelList>
```

```
</associateVideoInputs>
</TwoWayAudioChannelCap>
```

XML_TwoWayAudioChannelList

TwoWayAudioChannelList message in XML format

```
<TwoWayAudioChannelList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <TwoWayAudioChannel/><!--opt, see details in the message of XML_TwoWayAudioChannel-->
</TwoWayAudioChannelList>
```

See Also

XML_TwoWayAudioChannel

XML_TwoWayAudioSession

TwoWayAudioSession message in XML format

```
<TwoWayAudioSession version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <sessionId><!--req, xs: string, communication session ID--></sessionId>
</TwoWayAudioSession>
```

XML_upgradeStatus

upgradeStatus message in XML format

```
<upgradeStatus version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <upgrading><!--ro, req, xs:boolean--></upgrading>
  <percent><!--ro, req, xs:integer, upgrading progress, the value is between 0 and 100--></percent>
</upgradeStatus>
```

XML_UPnP

UPnP message in XML format

```
<UPnP version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--req-->
  <ports><!--opt-->
</UPnP>
```

XML_User

User message in XML format

```
<User version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:integer, the value is between 1 and 16, the ID of the administrator account is 1 and the administrator
account cannot be deleted--></id>
  <enabled><!--opt, xs:boolean, whether to enable the user: "true"-yes, "false"-no--></enabled>
  <userName><!--req, xs:string, the sensitive information should be encrypted--></userName>
  <password><!--wo, req, xs:string, the sensitive information should be encrypted--></password>
  <keypadPassword><!--wo, opt, xs:string, keypad password, the sensitive information should be encrypted--></
keypadPassword>
  <loginPassword><!--wo, dep, xs:string, this node depends on security in the URL and it is required when security
exists. The sensitive information should be encrypted, and the password is the administrator password--></
loginPassword>
  <userOperateType><!--opt, xs:integer, user operation type: "1"-network user, "2"-keypad user, "3"-network user and
keypad user. When this node is set to 1 or NULL, password is valid, and password is required when the method is
POST, otherwise it is optional. When this node is set to 2, keypadPassword is valid, and keypadPassword is required
when the method is POST, otherwise it is optional. When this node is set to 3, both password and keypadPassword
are valid, and they are required when the method is POST, otherwise they are optional--></userOperateType>
  <bondIpAddressList><!--opt-->
    <bondIpAddress><!--opt-->
      <id><!--req, xs:integer--></id>
      <ipAddress><!--dep, xs:string--></ipAddress>
      <ipv6Address><!--dep, xs:string--></ipv6Address>
    </bondIpAddress>
  </bondIpAddressList>
  <bondMacAddressList><!--opt-->
    <bondMacAddress><!--opt-->
      <id><!--req, xs:integer--></id>
      <macAddress><!--opt, xs:string--></macAddress>
    </bondMacAddress>
  </bondMacAddressList>
  <userLevel><!--opt, xs:string, "Administrator,Operator,Viewer,installer,manufacturer"--></userLevel>
  <attribute><!--opt-->
    <inherent><!--xs:boolean--></inherent>
  </attribute>
</User>
```

XML_userCheck

userCheck message in XML format

```
<userCheck version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <statusValue><!--req, xs:integer, status code: "200"-succeeded, "401"-failed--></statusValue>
  <statusString><!--opt, xs:string, status: "OK", "Unauthorized"--></statusString>
  <isDefaultPassword><!--opt, xs:boolean--></isDefaultPassword>
  <isRiskPassword><!--opt, xs:boolean--></isRiskPassword>
  <isActivated><!--opt, xs:boolean--></isActivated>
  <lockStatus><!--opt, xs:string, locking status: "unlock", "locked"--></lockStatus>
  <unlockTime><!--opt, xs:integer, unlocking remaining time, unit: second--></unlockTime>
  <retryLoginTime><!--opt, xs:integer, remaining login attempts--></retryLoginTime>
</userCheck>
```


XML_UserList

UserList message in XML format

```
<UserList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <User/><!--opt, see details in the message of XML_User-->
</UserList>
```

See Also

XML_User

XML_UserPermission

UserPermission message in XML format

```
<UserPermission version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:string--></id>
  <userID><!--req, xs:string, user ID--></userID>
  <userType>
    <!--req, xs:string, user type: "admin"-administrator, which has all permissions and can review and edit user's
    permission, "operator"-operator, which has default permissions, "viewer"-viewer, which has default permissions,
    "installer", "manufacturer"-->
  </userType>
  <localPermission/>
    <!--opt, local permission, see details in the message of XML_localPermission -->
  <remotePermission/>
    <!--opt, remote permission, see details in the message of XML_remotePermission -->
</UserPermission>
```

See Also

XML_localPermission

XML_remotePermission

XML_UserPermissionCap

UserPermissionCap capability message in XML format

```
<UserPermissionCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <userType><!--req, xs:string, "admin"-administrator, "operator", "viewer", "installer", "manufacturer"--></userType>
  <localPermissionCap><!--opt, see details in the message of XML_localPermissionCap--></localPermissionCap>
  <remotePermissionCap><!--opt, see details in the message of XML_remotePermissionCap--></
remotePermissionCap>
</UserPermissionCap>
```

See Also

XML_localPermissionCap

XML_remotePermissionCap

XML_UserPermissionList

UserPermissionList message in XML format

```
<UserPermissionList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <UserPermission/>
  <!--opt, user permission, see details in the message of XML_UserPermission-->
</UserPermissionList>
```

See Also

XML_UserPermission

XML_VideoPic_CMSearchDescription

XML message about search conditions of video and picture

```
<CMSearchDescription version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <searchID>
    <!--required, xs:string, search ID, e.g., {812F04E0-4089-11A3-9A0C-0305E82C2906}-->
  </searchID>
  <trackIDList>
    <trackID>
      <!--required, xs:integer, ID, XX01-search for the main stream video of channel XX, XX03-search for the pictures of
channel XX-->
    </trackID>
  </trackIDList>
  <timeSpanList>
    <timeSpan>
      <startTime><!--required, xs:time, ISO8601 time, start time of search, e.g.: 2017-08-02T00:00:00Z--></startTime>
      <endTime><!--required, xs:time, ISO8601 time, end time of search, e.g.: 2017-08-02T23:59:59Z--></endTime>
    </timeSpan>
  </timeSpanList>
  <searchTypeList>
    <searchKeyword><!--required, xs:string, keyword list for search, the maximum length is 32 and the minimum length
is 1--></searchKeyword>
  </searchTypeList>
  <contentTypeList>
    <contentType><!--required, xs:string, file type to be searched for: "video", "audio", "metadata"-picture, "text",
"mixed", "other"--></contentType>
  </contentTypeList>
  <uploadState><!--optional, xs:string, uploading status: all, uploaded, nonUploaded, uploadFailed--></uploadState>
  <lockState><!--optional, xs:string, locking status: lock, unlock, all--></lockState>
  <searchResultPostion><!--required, xs:integer, start position of search--></searchResultPostion>
  <maxResults><!--optional, xs:integer, maximum number of returned results--></maxResults>
  <metadataList>
    <metadataDescriptor>
      <!--required, xs:string, description of file to be searched for, format: "recordType.meta.hikvision.com/
```

<eventType>", <eventType> is the event type to which related video should be searched for; for details, see the table below in remarks. For example, to search for the video/picture of motion detection, the value of **contentType** is "video"/"metadata" and the value of **metadataDescriptor** is "recordType.meta.hikvision.com/MOTION"-->

<!--to search for all videos, the value of **contentType** is "video" and value of **metadataDescriptor** is "recordType.meta.hikvision.com"; to search for all pictures, the value of **contentType** is "metadata" and value of **metadataDescriptor** is "recordType.meta.hikvision.com/allPic"-->

<!--the supported video/picture type of device is returned in the **recordSearchType/pictureSearchType** of XML_RacmCap (related URI: */ISAPI/ContentMgmt/capabilities*)-->

```

</metadataDescriptor>
<SearchProperty>
  <plateSearchMask>
    <!--optional, xs:string, ranges from 1 to 31-->
  </plateSearchMask>
  <stateOrProvince>
    <!--optional, xs:integer-->
  </stateOrProvince>
  <country>
    <!--optional, xs:string, country or region ID: 0-the algorithm library does not support the country or region ID,1-
(CZ-Czech Republic),2-(FRA-France),3-(DE-Germany),4-(E-Spain),5-(IT-Italy),6-(NL-Netherlands),7-(PL-Poland),8-(SVK-
Slovakia), 9-(BY-Belorussia), 10-(MDA-Moldova),11-(RU-Russia),12-(UA-Ukraine), 0xff-(All)-->
  </country>
  <subType><!--optional, xs:string, sub type: all,motorVehicle,nonMotorVehicle,pedestrian--></subType>
  <capTypeDescriptor><!--optional, xs:string, capability type descriptor: anyType, event, evidence--></
capTypeDescriptor>
</SearchProperty>
</metadataList>
</CMSearchDescription>

```

Remarks

eventType	Description
AllEvent	All events
safetyHelmet	Hard hat detection
smokeDetection	Smoke detection
leavePosition	Absence detection
peopleNumChange	The number of people changed
linedetection	Line crossing detection
fieldddetection	Intrusion detection
regionEntrance	Region entrance detection
regionExiting	Region exiting detection
loitering	Loitering detection
group	People gathering detection
rapidMove	Fast moving detection

eventType	Description
unattendedBaggage	Unattended baggage detection
attendedBaggage	Object removal detection
CMR	Scheduled recording or capture
ALARM	Alarm
EDR	Motion detection or alarm
ALARMANDMOTION	Motion detection and alarm
Command	Command is triggered
pir	PIR
wlsensor	Wireless alarm
callhelp	Calling alarm
facedetection	Face detection
parking	Parking
vehicleDetection	Vehicle detection
manual	Manual
manualSnapShot	Manual capture
playSnapShot	Capture of playback
dredgerDetection	Dredger detection
accessController	Access controller event
securityControlPanel	Security control panel event
violentMotion	Violent motion alarm
advReachHeight	Climbing detection
toiletTarry	In-Toilet overtime
audioAbnormal	Sudden change of sound intensity
standUp	Standing up
getUp	Getting up
vehicleMonitor	Vehicle arming
playCellphone	Playing cellphone
retention	Overstay detection
failDown	People falling down alarm

eventType	Description
sleepOnDuty	Sleeping on duty alarm
allPerimeterEvent	All perimeter events
allBehaviorEvent	All behavior analysis events
running	People running
spacingChange	Distance changing detection
faceSnapModeling	Face capture modeling
reverseEntrance	Reverse entering
studentsStoodUp	Student standing up detection
ATMPanel	ATM panel mode
ATMSurround	ATM environment mode
ATMFace	ATM human face mode
ATMSafetyCabin	ATM safety cabin mode

XML_WDR

WDR message in XML format

```
<WDR version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <mode><!--req, xs: string, WDR mode: "open,close,auto"--></mode>
  <WDRLevel><!--opt, xs: integer--></WDRLevel>
  <WDRContrastLevel><!--opt, xs: integer--></WDRContrastLevel>
  <WDRLevel1><!--opt, xs: integer--></WDRLevel1>
</WDR>
```

XML_WhiteBalance

WhiteBalance message in XML format

```
<WhiteBalance version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <WhiteBalanceStyle/><!--req, xs:string, "auto, manual, indoor, outdoor, autotrace, onece, sodiumlight,
  mercurylight,auto0, atuo1, fluorescent, natural, warm, incandescent"-->
  <WhiteBalanceRed/><!--dep,xs:integer, this node depends on <WhiteBalanceStyle-->
  <WhiteBalanceBlue/><!--dep,xs:integer, this node depends on <WhiteBalanceStyle-->
</WhiteBalance>
```

Remarks

Setting **WhiteBalanceRed** and **WhiteBalanceBlue** is available only when **WhiteBalanceStyle** is set to "manual".

XML_WirelessServer

WirelessServer message in XML format

```
<WirelessServer version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <wifiApEnabled><!--opt,xs:boolean,"true,false"--></wifiApEnabled>
  <broadcastEnabled><!--opt,xs:boolean,"true,false"--></broadcastEnabled>
  <wlanShareEnabled><!--opt,xs:boolean,"true,false"--></wlanShareEnabled>
  <ssid><!--opt, xs:string--></ssid>
  <WirelessSecurity><!--req-->
    <securityMode>
      <!--opt, xs: string, security mode: "disable,WEP,WPA-personal,WPA2-personal,WPA-RADIUS,WPA-enterprise,WPA2-
enterprise"-->
    </securityMode>
    <WEP><!--dep, depends on <securityMode>-->
      <authenticationType>
        <!--req, xs: string, authentication type: "open,sharedkey,auto"-->
      </authenticationType>
      <defaultTransmitKeyIndex>
        <!--req, xs: integer-->
      </defaultTransmitKeyIndex>
      <wepKeyLength><!--opt, xs: integer, the key length should be 64 and 128--></wepKeyLength>
      <EncryptionKeyList>
        <encryptionKey>
          <!--req, xs: hexBinary, WEP encryption key in hexadecimal format-->
        </encryptionKey>
      </EncryptionKeyList>
    </WEP>
    <WPA><!--dep, depends on <securityMode>-->
      <algorithmType>
        <!--req, xs: string, algorithm type: "TKIP,AES,TKIP/AES"-->
      </algorithmType>
      <sharedKey><!--opt, xs: string, shared key used in WPA--></sharedKey>
      <wpaKeyLength><!--opt, xs: integer, the key length is between 8 and 63--></wpaKeyLength>
      <defaultPassword><!--opt, xs: boolean--></defaultPassword>
    </WPA>
  </WirelessSecurity>
  <DHCPEnabled><!--opt,xs:boolean,"true,false"--></DHCPEnabled>
  <ipVersion><!--opt, xs:string, opt="v4,v6"--></ipVersion>
  <HostIpAddress><!--opt-->
    <ipAddress><!--dep, xs:string--></ipAddress>
    <ipv6Address><!--dep, xs:string--></ipv6Address>
  </HostIpAddress>
  <IPMask><!--opt-->
    <subnetMask><!--dep, xs:string, subnet mask for IPv4 address--></subnetMask>
  </IPMask>
</WirelessServer>
```

```
<bitMask><!--dep, xs:integer, bitmask IPv6 address--></bitMask>
</IPMask>
<AddressPool><!--opt-->
  <startIPv4Address><!--dep, xs:string--></startIPv4Address>
  <endIPv4Address><!--dep, xs:string--></endIPv4Address>
  <startIPv6Address><!--dep, xs:string--></startIPv6Address>
  <endIPv6Address><!--dep, xs:string--></endIPv6Address>
</AddressPool>
<DNSAddressList><!--opt-->
  <DNSAddress><!--opt>
    <id><!--opt,xs:string,start from 1--></id>
    <ipAddress><!--dep, xs:string--></ipAddress>
    <ipv6Address><!--dep, xs:string--></ipv6Address>
  </DNSAddress>
</DNSAddressList>
<GatewayAddress>
  <ipAddress><!--dep, xs:string--></ipAddress>
  <ipv6Address><!--dep, xs:string--></ipv6Address>
</GatewayAddress>
<wifiApModeType>
  <!--opt, xs: string, current wireless Access Point (AP) mode: "true,false,auto", if this node is returned, it indicates
that the device supports auto mode-->
  </wifiApModeType>
</WirelessServer>
```

Remarks

For Client supports auto AP, the node <<**wifiApModeType**> is valid; for Client does not supports auto AP, the node <**wifiApEnabled**> is valid; the values of these two nodes will effect each other, and when <**wifiApModeType**> equals to "auto", the value of <**wifiApEnabled**> is "true".

XML_WirelessServerStatus

WirelessServerStatus message in XML format

```
<WirelessServerStatus version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <wifiApEnabled>
    <!--dep, xs:boolean-->
  </wifiApEnabled>
  <linkDevices>
    <!--dep, xs:integer, maximum number of linked devices-->
  </linkDevices>
  <DeviceInfoList>
    <!--opt-->
    <DeviceInfo>
      <!--opt-->
      <IpAddress>
        <!--req-->
        <IpAddress>
          <!--dep, xs:string-->
        </IpAddress>
      </DeviceInfo>
    </DeviceInfoList>
  </WirelessServerStatus>
```

```
<ipv6Address>
  <!--dep, xs:string-->
</ipv6Address>
</IpAddress>
<hostName>
  <!--opt, xs:string-->
</hostName>
<macAddress>
  <!--opt, xs:string-->
</macAddress>
</DeviceInfo>
</DeviceInfoList>
</WirelessServerStatus>
```

XML_WirelessStatus

WirelessStatus message in XML format

```
<WirelessStatus version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled>
    <!--dep, xs:boolean-->
  </enabled>
  <connectionEnabled>
    <!--dep, xs:boolean-->
  </connectionEnabled>
  <IpAddress>
    <!--req-->
    <ipAddress>
      <!--dep, xs:string-->
    </ipAddress>
    <ipv6Address>
      <!--dep, xs:string-->
    </ipv6Address>
  </IpAddress>
  <SubnetMask>
    <!--req-->
    <ipAddress>
      <!--dep, xs:string-->
    </ipAddress>
    <ipv6Address>
      <!--dep, xs:string-->
    </ipv6Address>
  </SubnetMask>
  <RouterAddress>
    <!--req-->
    <ipAddress>
      <!--dep, xs:string-->
    </ipAddress>
    <ipv6Address>
      <!--dep, xs:string-->
    </ipv6Address>
  </RouterAddress>
</WirelessStatus>
```



```
</RouterAddress>
<DNSAddressList>
  <!--opt-->
  <DNSAddress>
    <!--opt-->
    <id>
      <!--opt, xs:string, DNS address ID, which starts from 1-->
    </id>
    <ipAddress>
      <!--dep, xs:string-->
    </ipAddress>
    <ipv6Address>
      <!--dep, xs:string-->
    </ipv6Address>
  </DNSAddress>
</DNSAddressList>
</WirelessStatus>
```

XML_Wireless

Wireless message in XML format

```
<Wireless version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--req, xs:boolean--></enabled>
  <wirelessNetworkMode>
    <!--opt, xs:string, "infrastructure,adhoc"-->
  </wirelessNetworkMode>
  <channel><!--opt, xs:string, "1 to 14,auto"--></channel>
  <ssid><!--opt, xs:string--></ssid>
  <wmmEnabled><!--opt, xs:boolean--></wmmEnabled>
  <WirelessSecurity><!--opt-->
    <securityMode>
      <!--opt, xs:string, "disable,WEP,WPA-personal,WPA2-personal,WPA-RADIUS, WPA-enterprise,WPA2-enterprise, WPA/
WPA2"-->
    </securityMode>
    <WEP><!--dep, depends on <securityMode-->
      <authenticationType>
        <!--req, xs:string, "open,sharedkey,auto"-->
      </authenticationType>
      <defaultTransmitKeyIndex><!--req, xs:integer--></defaultTransmitKeyIndex>
      <wepKeyLength><!--opt, xs:integer "64,128"--></wepKeyLength>
      <EncryptionKeyList>
        <encryptionKey>
          <!--req, xs:hexBinary, WEP encryption key in hexadecimal format-->
        </encryptionKey>
      </EncryptionKeyList>
    </WEP>
    <WPA><!--dep, depends on <securityMode-->
      <algorithmType><!--req, xs:string, "TKIP,AES,TKIP/AES"--></algorithmType>
      <sharedKey><!--req, xs:string, pre-shared key used in WPA--></sharedKey>
      <wpaKeyLength><!--req, xs: integer, "8-63"--></wpaKeyLength>
```

```
</WPA>
</WirelessSecurity>
<workScene><!--opt, xs:string, "computerRoom,monitorTerminal"--></workScene>
<protocol><!--req,xs:string, "802.11ac"--></protocol>
<protocolRealTime><!--req,xs:string, real-time mode--></protocolRealTime>
<hideSsid><!--opt, xs:boolean--></hideSsid>
<ChannelConfig><!--opt-->
  <width><!--opt, xs:string, "auto,20,40,80", bandwidth, unit: MHz--></width>
  <autoWidth>
    <!--dep,ro, xs:string, it is valid only when width is "auto"-->
  </autoWidth>
  <channel><!--opt, dep, xs:string, channel frequency, unit: MHz--></channel>
  <autoChannel><!--dep,ro,xs:string, it is valid only when channel is "auto"--></autoChannel>
  <transmitPower> <!--opt, xs:integer, "9,12,15,18,21,24,27", power, unit: dBm--><transmitPower>
  <transmitPowerRealTime> <!--opt, xs:integer, real-time power, unit: dBm--><transmitPowerRealTime>
  <countryID><!--opt,xs:integer, country code--></countryID>
</ChannelConfig>
</Wireless>
```

XML_WPS

WPS message in XML format

```
<WPS version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enable><!--req, xs:boolean--></enable>
</WPS>
```

XML_WpsApPincode

WpsApPincode message in XML format

```
<WpsApPincode version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ssid><!--req, xs:string--></ssid>
  <pinCode><!--req, xs:string--></pinCode>
</WpsApPincode>
```

XML_VCResource

Message about intelligent resources switch in XML format

```
<VCResource version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <type>
    <!--required, xs:string,
    "basicBehavior,fullBehavior,facesnapBehavior,facesnap,TFS,smartVehicleDetection,smartHVTDDetection,smart,judicial,s
    mart264AndRoadDetection,smart264AndFaceDetection,smart264AndHeatMap,smartIntelligentMonitor,smartTrafficD
    ataCollection,roadDetection,humanRecognition,perimeterCapture,vehicleDetection,HVTDDetection,mixedTargetDetecti
    on,trackingCaptureMode,nonTrackingCaptureMode,close,faceHumanModelingContrast,cityManagement,teacherBeha
```

```
    vior,12MPLiveView,personQueueDetection,verticalPeopleCounting,safetyHelmet,faceCounting,personArming"-->
  </type>
</VCAResource>
```

XML_VideoCap

VideoCap message in XML format

```
<VideoCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <videoInputPortNums><!--opt, xs: integer--><videoInputPortNums>
  <videoOutputPortNums><!--opt, xs: integer--><videoOutputPortNums>
  <isSupportHeatmap><!--opt, xs: boolean, whether to support heat map function--></isSupportHeatmap>
  <isSupportCounting><!--opt, xs: boolean--></isSupportCounting>
  <countingType><!--dep, xs: string, "human,object"--></countingType>
  <isSupportPreviewSwitch><!--opt, xs: boolean, whether to support live view switch--></isSupportPreviewSwitch>
  <isSupportRecodStatus>
    <!--opt, xs: boolean, whether supports getting recording status-->
  </isSupportRecodStatus>
  <isSupportPrivacyMask>
    <!--opt, xs: boolean, whether supports priavte mask function-->
  </isSupportPrivacyMask>
  <isSupportBinocularPreviewSwitch>
    <!--opt, xs: boolean, whether supports auto-switch during the live view of dual lens camera-->
  </isSupportBinocularPreviewSwitch>
  <isSupportCalibCheck>
    <!--opt, xs: boolean, whether supports calibration verification-->
  </isSupportCalibCheck>
  <isSupportPIP><!--opt, xs: boolean, "true, false"--></isSupportPIP>
  <channelFlexible opt="name,enable,online,linknum">
    <!--capability of getting channel status by condition-->
  </channelFlexible >
  <isSupportFocusVideoMode>
    <!--opt, xs: boolean, video focus mode for installation and debug-->
  </isSupportFocusVideoMode>
  <isSupportExternalChannel>
    <!--opt, xs: boolean, whether supports extending analog channel-->
  </isSupportExternalChannel>
  <isSupportMultiChannelCounting>
    <!--opt, xs: boolean, whether supports people counting of multiple channels-->
  </isSupportMultiChannelCounting>
  <isSupportCountingCollection>
    <!--opt, xs:boolean, whether supports people counting data replenishment-->
  </isSupportCountingCollection>
  <isSupportHeatmapCollection>
    <!--opt, xs:boolean, whether supports heat map data replenishment-->
  </isSupportHeatmapCollection>
  <OSDLanguage opt="GBK,EUC-KR" def="GBK"/>
  <isSupportInitLens><!--req, xs:boolean, whether to support initializing lens--><isSupportInitLens>
  <isSupportOneFocus><!--req, xs:boolean, whether to support one-touch focusing--><isSupportOneFoucs>
  <notSupportFocus><!--req, xs:boolean, the focus capability is not supported--><notSupportFoucs>
  <notSupportIris><!--req, xs:boolean, the iris capability is not supported--><notSupportIris>
```

```
<isSupportCapturePicOverlays><!--opt, xs:boolean, whether to support text overlay on the captured picture--></isSupportCapturePicOverlays>
<isSupportMergePicOverlays><!--opt, xs:boolean, whether to support text overlay on the composite picture--></isSupportMergePicOverlays>
</VideoCap>
```

XML_VideoInput

VideoInput message in XML format

```
<VideoInput version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <VideoInputChannelList/><!--op, see details in the message of XML_VideoInputChannelList-->
</VideoInput>
```

See Also

XML_VideoInputChannelList

XML_VideoInputChannel

VideoInputChannel message in XML format

```
<VideoInputChannel version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id>
    <!--req, xs:string-->
  </id>
  <inputPort>
    <!--req, xs:string-->
  </inputPort>
  <videoInputEnabled>
    <!--opt, xs:boolean-->
  </videoInputEnabled>
  <name>
    <!--opt, xs:string-->
  </name>
  <videoFormat>
    <!--opt, xs:string, "PAL, NTSC"-->
  </videoFormat>
  <portType>
    <!--opt, xs:string, "SDI, OPT, VGA, HDMI, YPbPr"-->
  </portType>
  <resDesc>
    <!--opt, xs:string-->
  </resDesc>
</VideoInputChannel>
```

XML_VideoInputChannelList

VideoInputChannelList message in XML format

```
<VideoInputChannelList version="2.0"
xmlns="http://www.isapi.org/ver20/XMLSchema">
  <VideoInputChannel/><!--opt, see details in the message of XML_VideoInputChannel-->
</VideoInputChannelList>
```

See Also

XML_VideoInputChannel

XML_VideoOutput

VideoOutput message in XML format

```
<VideoOutput version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <VideoOutputChannelList/><!--opt, refer to the message XML_VideoOutputChannelList for details-->
</VideoOutput>
```

See Also

XML_VideoOutputChannelList

XML_VideoOutputChannel

VideoOutputChannel message in XML format

```
<VideoOutputChannel version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs: string; ID--></id>
  <type><!--req, xs: string, video output type: "VGA,CVBS,HDMI,Spot,SDI"--></type>
  <menu><!--dep, read-only-->
    <mirrorMenu><!--req, xs: boolean--></mirrorMenu>
  </menu>
  <mode><!--opt,xs:string, video output mode: "close,clip,scale,open,SDI_1080P25,
...,HDMI_1080P,HDMI_720P,HDMI_2160P"--></mode>
  <resolution><!--opt, xs: string; video solution: "1920*1080/60HZ,1280*720/50HZ,..."--></resolution>
</VideoOutputChannel>
```

XML_VideoOutputChannelList

VideoOutputChannelList message in XML format

```
<VideoOutputChannelList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <VideoOutputChannel/><!--opt, refer to the message XML_VideoOutputChannel for details-->
</VideoOutputChannelList>
```

See Also

XML_VideoOutputChannel

XML_VideoOverlay

VideoOverlay message in XML format

```
<VideoOverlay version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <normalizedScreenSize><!--req, reference value of the region coordinates, which is 704*576 for PAL and 704*480 for NTSC-->
    <normalizedScreenWidth><!--ro, req, xs: integer--></normalizedScreenWidth>
    <normalizedScreenHeight><!--ro, req, xs: integer--></normalizedScreenHeight>
  </normalizedScreenSize>
  <attribute><!--opt, OSD properties-->
    <transparent><!--req, xs: boolean--></transparent>
    <flashing><!--req, xs: boolean--></flashing>
  </attribute>
  <TextOverlayList>
    <TextOverlay><!--text overlay parameter-->
      <id><!-- req, xs: string; text ID--></id>
      <enabled><!--req, xs: boolean, whether to enable text overlay--></enabled>
      <positionX><!--req, xs: float, x-coordinate--></positionX>
      <positionY><!-- req, xs: float, y-coordinate--></positionY>
      <displayText><!-- req, xs: string, displayed text contents--></displayText>
    </TextOverlay>
  </TextOverlayList>
  <DateTimeOverlay>
    <enabled><!--req, xs: boolean, whether to enable date and time overlay--></enabled>
    <positionX><!--req, xs: integer; x-coordinate--></positionX>
    <positionY><!--req, xs: integer; y-coordinate--></positionY>
    <dateStyle>
      <!--opt, xs: string, date display format: YYYY-MM-DD, MM-DD-YYYY, DD-MM-YYYY, CHR-YYYY-MM-DD, CHR-MM-DD-YYYY, CHR-DD-MM-YYYY, CHR-YYYY/MM/DD, CHR-MM/DD/YYYY, CHR-DD/MM/YYYY-->
    </dateStyle>
    <timeStyle><!--opt, xs: string, time format: 12hour, 24hour--></timeStyle>
    <displayWeek><!--opt, xs: boolean, whether to display day of the week--></displayWeek>
  </DateTimeOverlay>
  <channelNameOverlay>
    <enabled><!--req, xs: boolean, whether to enable channel name overlay--></enabled>
    <positionX><!--req, xs: integer; x-coordinate--></positionX>
    <positionY><!--req, xs: integer; y-coordinate--></positionY>
  </channelNameOverlay>
  <fontSize opt="adaptive,16*16,32*32,48*48,64*64,80*80,96*96,112*112,128*128"><!--opt, xs: string, font size of video OSD: "adaptive,16*16,32*32,48*48,64*64,80*80,96*96,112*112,128*128", unit: pixel. Currently only 16*16 and 32*32 are supported--></fontSize>
  <frontColorMode opt="auto,customize"><!--opt, string, front color mode: "auto,customize"--></frontColorMode>
  <frontColor><!--dep, xs: hexBinary, front color--></frontColor>
  <alignment><!--opt, xs:string, "customize,alignRight,alignLeft"--></alignment>
  <BatteryPowerOverlay/><!--opt-->
  <DeviceStatusDisplay>
```

```
<!--opt, display control of the device status on the video of portable speed dome-->
<batteryCapacityDisplay>
  <!--opt, xs:boolean, whether to display battery capacity-->
</batteryCapacityDisplay>
<chargingStatusDisplay>
  <!--opt, xs:boolean, whether to display charging status-->
</chargingStatusDisplay>
<bluetoothStatusDisplay>
  <!--opt, xs:boolean, whether to display bluetooth status-->
</bluetoothStatusDisplay>
<dialStatusDisplay>
  <!--opt, xs:boolean, whether to display dial-up status-->
</dialStatusDisplay>
</DeviceStatusDisplay>
<boundary><!--opt, xs:integer--></boundary>
<alignment><!--opt, xs:string, alignment mode: "customize, alignRight, alignLeft"--></alignment>
<publicSecurity><!--req, xs:boolean--></publicSecurity>
</VideoOverlay>
```

XML_VideoSourceActivation

XML Message about Parameters of Activating Network Device

```
<?xml version="1.0" encoding="utf-8"?>
<VideoSourceActivation version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <raterat><!--required, xs:string--></raterat>
  <passwordType>
    <!--required, xs:string, password types: "default"-NVR password, "selfDefine", "IPCActivatePasswd"-activation password-->
  </passwordType>
  <password><!--depend, xs:string--></password>
  <ActivationDevice>
    <mode><!--required, xs:string, "channel, descriptor"--></mode>
    <channel><!--depend-->
      <dynVideoInputChannelID><!--required, xs:integer--></dynVideoInputChannelID>
    </channel>
    <descriptor><!--depend-->
      <macAddress><!--required, xs:string--></macAddress>
      <addressingFormatType><!--required, xs:string, "ipaddress, hostname"--></addressingFormatType>
      <raterat><!--depend, xs:string--></raterat>
      <ipAddress><!--depend, xs:string--></ipAddress>
      <ipv6Address><!--depend, xs:string--></ipv6Address>
      <portNo><!--required, xs:integer--></portNo>
      <proxyProtocol><!--required, xs:string, "HIKVISION, SONY, ISAPI, ONVIF, ..."--></proxyProtocol>
    </descriptor>
  </ActivationDevice>
</VideoSourceActivation>
```

XML_VideoSourceActivationCapability

XML Message about Capability of Activating Network Device

```
<?xml version="1.0" encoding="utf-8"?>
<VideoSourceActivationCapability version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ActivateByChannel>
    <proxyProtocol opt="HIKVISION"/>
  </ActivateByChannel>
  <ActivateByAddress>
    <proxyProtocol opt="HIKVISION"/>
  </ActivateByAddress>
</VideoSourceActivationCapability>
```

XML_VideoSourceList

XML Message about Searched Network Devices

```
<VideoSourceList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <VideoSourceDescriptor><!--list-->
    <id><!--req, xs: string;id--></id>
    <proxyProtocol><!--req, xs: string, "HIKVISION, SONY, ISAPI, ONVIF"--></proxyProtocol>
    <addressingFormatType><!--req, xs: string, address type: "ipaddress, hostname"--></addressingFormatType>
    <hostName><!--dep, xs: string--></hostName>
    <ipAddress><!--dep, xs: string--></ipAddress>
    <subnetMask><!--opt, xs: string, subnet mask for IPv4 address--></subnetMask>
    <ipv6Address><!--dep, xs: string--></ipv6Address>
    <bitMask><!--opt, xs: integer, bitmask IPv6 address--></bitMask>
    <serialNumber><!--opt, xs: string--></serialNumber>
    <macAddress><!--opt, xs: string--></macAddress>
    <firmwareVersion><!--opt, req, xs: string--></firmwareVersion>
    <managePortNo><!--opt, xs: integer--></managePortNo>
    <userName><!--opt, xs: string--></userName>
    <password><!--opt, xs: string--></password>
    <srcInputPortNums><!--req, xs: integer--></srcInputPortNums>
    <deviceId><!--dep, xs: string--></deviceId>
    <deviceModel><!--opt, xs: string, device model--></deviceModel>
  </VideoSourceDescriptor>
</VideoSourceList>
```

XML_WebCertificate

WebCertificate message in XML format

```
<WebCertificate version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <CertificateType>
    <!--required, xs:string, authentication type: "basic", "digest", "digest/baisc"-->
  </CertificateType>
```



```
<SecurityAlgorithm><!--dependent, this node is valid when certificateType is "digest" or "digest/basic"-->
  <algorithmType>
    <!--optional, xs:string, algorithm type: "MD5", "SHA256", "MD5/SHA256"-->
    </algorithmType>
  </SecurityAlgorithm>
</WebCertificate>
```

XML_WirelessDialInterface

WirelessDialInterface message in XML format

```
<WirelessDialInterface version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id> <!--req, xs:string, only one network card is supported--> </id>
  <Dial/> <!--opt, refer to the URI: /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/dial -->
  <Schedule> <!-- opt, refer to the URI: /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/schedule -->
  <Dialstatus/> <!-- opt, refer to the URI: /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/dialstatus -->
  <messageConfig/> <!-- opt, refer to the URI: /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/messageConfig -->
  <messageList/> <!-- opt -->
</WirelessDialInterface>
```

XML_WirelessDialInterfaceList

WirelessDialInterfaceList message in XML format

```
<WirelessDialInterfaceList version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <WirelessDialInterface/>
</WirelessDialInterfaceList>
```

XML_ZoomFocus

ZoomFocus message in XML format

```
<ZoomFocus version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <pqrsZoom/><!--opt, xs: integer, zoom coordinates of Sony zoom camera module-->
  <mnstFocus/><!--opt, xs: integer, focus coordinates of Sony zoom camera module-->
</ZoomFocus>
```

A.2 Log Types for ISAPI

There are four major log types, i.e., alarm log, exception log, operation log, and event log. And each major type contains multiple minor types, see details in the following contents.

Alarm Logs

Log Type	Description
shortCircuit	Short Circuit Alarm
brokenCircuit	Open Circuit Alarm
alarmReset	Alarm Reset
alarmNormal	Return to Normal
passwordError	Incorrect Password (3 Tmes in a Row)
idCardIllegally	Invalid Card ID
keyPADRemove	Keypad Tampered
keyPADRemoveRestore	Keypad Restored
devRemove	Device Tampered
devRemoveRestore	Device Restored
belowAlarmLimit1	Sensor Value is Lower than Alarm Limit Value 1
belowAlarmLimit2	Sensor Value is Lower than Alarm Limit Value 2
belowAlarmLimit3	Sensor Value is Lower than Alarm Limit Value 3
belowAlarmLimit4	Sensor Value is Lower than Alarm Limit Value 4
aboveAlarmLimit1	Sensor Value is Higher than Alarm Limit Value 1
aboveAlarmLimit2	Sensor Value is Higher than Alarm Limit Value 2
aboveAlarmLimit3	Sensor Value is Higher than Alarm Limit Value 3
aboveAlarmLimit4	Sensor Value is Higher than Alarm Limit Value 4
UrgencyBtnON	Panic Button Triggered
UrgencyBtnOFF	Panic Button Restored
virtualDefenceBandit	Virtual Zone Burglary Alarm
virtualDefenceFire	Virtual Zone Fire Alarm
virtualDefenceUrgent	Virtual Zone Panic Alarm
motDetStart	Motion Detection Alarm Started
motDetStop	Motion Detection Alarm Stopped
hideAlarmStart	Device Blocked
hideAlarmStop	Device Blocking Alarm Restored

Log Type	Description
UPSAAlarm	UPS Alarm
electricityMeterAlarm	Coulombmeter Alarm
switchPowerAlarm	Switch Power Supply Alarm
GasDetectSys	Gas Detection Alarm
transformerTempAlarm	Transformer Temperature Alarm
tempHumiAlarm	Temperature and Humidity Sensor Alarm
UPSAAlarmRestore	UPS Alarm Restored
electricityMeterAlarmRestore	Coulombmeter Alarm Restored
switchPowerAlarmRestore	Switch Power Supply Alarm Restored
GasDetectSysRestore	Gas Detection Alarm Restored
transformerTempAlarmRestore	Transformer Temperature Alarm Restored
tempHumiAlarmRestore	Temperature-Humidity Sensor Alarm Restored
waterLevelSensorAlarm	Temperature-Humidity Sensor Alarm Restored
waterLevelSensorAlarmRestore	Flood Sensor Restored
dustNoiseAlarm	Dust and Noise Sensor Alarm
dustNoiseAlarmRestore	Dust and Noise Sensor Alarm Restored
environmentalLogger	Environmental Data Collector Alarm
environmentalLoggerAlarm	Environmental Data Collector Restored
triggerTemper	Detector Tampered
triggerTemperRestore	Detector Restored
emergencyCallHelp	Panic Alarm
emergencyCallHelpRestore	Panic Alarm Restored
consult	Consultation Alarm
consultRestore	Consultation Alarm Restored
deviceMoveAlarm	Device Motion Alarm
deviceMoveAlarmRestore	Device Motion Alarm Restored
earlyWarningAlarm	Early Warning Zone Alarm
earlyWarningAlarmRestore	Early Warning Zone Restored
warningAlarm	Warning Zone Alarm

Log Type	Description
warningAlarmRestore	Warning Zone Restored
wirelessOutputModTamperEvident	Wireless Output Expander Tampered
wirelessOutputModTamperEvidentReset	Wireless Output Expander tamper Restored
wirelessRepeaterTamperEvident	Wireless Repeater Tampered
wirelessRepeaterTamperEvidentReset	Wireless Repeater tamper Restored
wirelessSirenTamperEvident	Wireless Siren Tampered
wirelessSirenTamperEvidentReset	Wireless Siren Tamper Restored
wirelessKeypadTamperEvident	Wireless Keypad Tampered
wirelessKeypadTamperEvidentReset	Wireless Keypad Tamper Restored
wirelessCardReaderTamperEvident	Wireless Card Reader Tampered
wirelessCardReaderTamperEvidentReset	Wireless Card Reader Tamper Restored
softZoneMedicalAlarm	Virtual Zone Medical Alarm
accessControllerEvent	Access Controller Event
videoIntercomEvent	Video Intercom Event
GJDEvent	GJD Security Control Panel Event
LuminateEvent	LUMINITE Security Control Panel Event
OPTEXEvent	OPTEX Security Control Panel Event
cameraDetectorEvent	Detector Event
securityControlPanelEvent	Security Control Panel Event
RS-485AlarmInputModuleEvident	RS-485 Zone Module Tampered
RS-485AlarmInputModuleTamperReset	RS-485 Zone Module Tampering Reset
RS-485WirelessReceiverTamperEvident	RS-485 Wireless Receiver Module Tampered
RS-485WirelessReceiverTamperEvidentReset	RS-485 Wireless Receiver Module Tampering Reset
dredgerDetectionAlarm	Dredger Detection Alarm
crossLineAlarm	Line Crossing Alarm
crossLineAlarmRestore	Line Crossing Alarm Restored
HFPDAlarmStart	High Frequently Appeared Person Alarm Started
HFPDAlarmStop	High Frequently Appeared Person Alarm Stopped

Log Type	Description
LFPDAlarmStart	Low Frequency Person Alarm Started
LFPDAlarmStop	Low Frequency Person Alarm Stopped
safetyHelmetAlarmStart	Hard Hat Detection Alarm Started
safetyHelmetAlarmStop	Hard Hat Detection Alarm Stopped
dataPrealarm	Traffic Pre-alarm
playCellphoneStart	Playing Cellphone Alarm Started
playCellphoneStop	Playing Cellphone Alarm Stopped
sleepOnDutyStart	Sleeping on Duty Alarm Started
sleepOnDutyStop	Sleeping on Duty Alarm Stopped
vibrationDetectionStart	Vibration Detection Alarm Started
vibrationDetectionStop	Vibration Detection Alarm Stopped
fireEscapeDetectionStart	Fire Engine Access Detection Started
fireEscapeDetectionStop	Fire Engine Access Detection Ended
takingElevatorDetectionStart	Elevator Detection Started
takingElevatorDetectionStop	Elevator Detection Ended
unregisteredStreetVendorStart	Unlicensed Business Vendor Detection Started
unregisteredStreetVendorStop	Unlicensed Business Vendor Detection Ended
stallOutsideShopStart	Business Outside Store Detection Started
stallOutsideShopStop	Business Outside Store Detection Ended
stallOccupyingRoadStart	Business on Sidewalk Detection Started
stallOccupyingRoadStop	Business on Sidewalk Detection Ended
illegalHeapStart	Pile Goods or Materials in Chaos Detection Started
illegalHeapStop	Pile Goods or Materials in Chaos Detection Ended
illParkofNonMotorVehicleStart	Non-Motor Vehicles Parking in Chaos Detection Started
illParkofNonMotorVehicleStop	Non-Motor Vehicles Parking in Chaos Detection Ended
illegalOutdoorAdvertisementStart	Unauthorized Outdoor Advertisement Detection Started

Log Type	Description
illegalOutdoorAdvertisementStop	Unauthorized Outdoor Advertisement Detection Ended
packGarbageStart	Packaged Garbage Detection Started
packGarbageStop	Packaged Garbage Detection Ended
stallUnderUmbrellaStart	Running Business with Patio Umbrella Detection Started
stallUnderUmbrellaStop	Running Business with Patio Umbrella Detection Ended
dustbinOverflowStart	Overflowing Dustbin Detection Started
dustbinOverflowStop	Overflowing Dustbin Detection Ended
exposeGarbageStart	Exposed Garbage Detection Started
exposeGarbageStop	Exposed Garbage Detection Ended
hangClothingAlongStreetStart	Hanging Clothes On Street Detection Started
hangClothingAlongStreetStop	Hanging Clothes On Street Detection Ended
ATMPanelStart	ATM Panel Alarm Started
ATMPanelStop	ATM Panel Alarm Ended
ATMSurroundStart	ATM Surround Alarm Started
ATMSurroundStop	ATM Surround Alarm Ended
ATMFaceStart	ATM Face Alarm Started
ATMFaceStop	ATM Face Alarm Ended
ATMSafetyCabinStart	ATM Safety Cabin Alarm Started
ATMSafetyCabinStop	ATM Safety Cabin Alarm Ended

Exception Logs

Log Type	Description
powerOn	Power on
powerOff	Power off
WDTRreset	WDT Reset
lowBatteryVoltage	Low Battery Voltage
ACLoss	AC Power Disconnected

Log Type	Description
ACRestore	AC Power Restored
RTCException	RTC Real-time Clock Exception
netFailure	Network Disconnected
netRestore	Network Connected
telLineBroken	Telephone Line Disconnected
telLineRestore	Telephone Line Connected
expanderBusLoss	Bus Expander Disconnected
expanderBusRestore	Bus Expander Connected
keypadBusLoss	Keypad Expander Disconnected
keypadBusRestore	Keypad Expander Connected
sensorFailure	Analog Sensor Fault
sensorRestore	Analog Sensor Restored
RS485DisConnect	RS-485 Channel Disconnected
RS485Connect	RS-486 Channel Connected
batteryVoltageRestore	Battery Voltage Restored
wiredNetAbnormal	Wired Network Exception
wiredNetRestore	Wired Network Restored
GPRSAbnormal	GPRS Exception
GPRSRestore	GPRS Restored
3GAbnormal	3G Network Exception
3GRestore	3G Network Restored
SIMCardAbnormal	SIM Card Exception
SIMCardRestore	SIM Card Restored
VILost	Video Loss
illegalAccess	Illegal Login
HDFull	HDD Full
HDError	HDD Error
DCDLost	MODEM Disconnected
IPConflict	IP Address Conflicted

Log Type	Description
netbroken	Network Disconnected
recError	Recording Error
VIError	Video Input Exception(Only for Analog Channel)
formatHDDError	Remote HDD Formatting Failed
USBError	USB Communication Error
USBRestore	USB Communication Error Restored
printError	Printer Error
printRestore	Printer Error Restored
subsystemCommunicationError	Sub-board Communication Error
IPCIpconflict	Network Camera IP Address Conflicted
VIMisMatch	Video Standard Mismatches
MCURestart	MCU Restarted
GprsModuleFault	GPRS Module Fault
telephoneFault	Telephone Module Fault
wifiAbnormal	Wi-Fi Exception
wifiRestore	Wi-Fi Restored
RFAbnormal	RF Exception
RFRestore	RF Restored
detectorOnline	Detector Connected
detectorOffline	Detector Disconnected
detectorBatteryNormal	Detector Battery Restored
detectorBatteryLow	Detector Battery Low
dataTrafficOverflow	Cellular Network Data Exceeded
radarSignalFault	Radar Transmitter Fault
radarSignalFaultRestore	Radar Transmitter Restored
wirelessOutputModOffline	Wireless Output Expander Disconnected
wirelessOutputModOnline	Wireless Output Expander Connected
wirelessRepeaterOffline	Wireless Repeater Disconnected
wirelessRepeaterOnline	Wireless Repeater Connected

Log Type	Description
triggerOffline	Trigger Disconnected
triggerOnline	Trigger Connected
wirelessSirenOffline	Wireless Siren Disconnected
wirelessSirenOnline	Wireless Siren Connected
sirenLowPower	Siren Battery Low
sirenPowerRecovery	Siren Battery Restored
ipcDisconnect	Network Camera Disconnected
ipcConnectRecovery	Network Camera Connected
sendMailFailed	Sending Email Failed
eventUploadException	Uploading Event Failed or Uploaded Event Lost
keyfobLowPower	Low Keyfob Battery
keyfobPowerRecovery	Normal Keyfob Battery
detectorOvertime	Detector Heartbeat Timed Out
detectorOvertimeRecovery	Detector Heartbeat Timeout Restored
wSirenOvertime	Wireless Siren Heartbeat Timed Out
wSirenOvertimeRecovery	Wireless Siren Heartbeat Timeout Restored
wOutputOvertime	Wireless Output Module Heartbeat Timed Out
wOutputOvertimeRecovery	Wireless Output Module Heartbeat Timeout Restored
wRepeaterOvertime	Wireless Repeater Heartbeat Timed Out
wRepeaterOvertimeRecovery	Wireless Repeater Heartbeat Timeout Restored
rfJamming	RF Wireless Communication Blocked
rfJammingRecovery	RF Wireless Communication Blocking Restored
batteryMiss	Storage Battery Loss
batteryMissRecovery	Storage Battery Restored
ARCUUploadFailed	Uploading to ARC Failed
ARCUUploadSucceeded	Uploaded to ARC
ARCUUploadRecovery	Uploading to ARC Restored
wirelessKeypadOffline	Wireless Keypad Disconnected

Log Type	Description
wirelessKeypadOnline	Wireless Keypad Connected
wirelessCardReaderOffline	Wireless Card Reader Disconnected
wirelessCardReaderOnline	Wireless Card Reader Connected
keypadLowPower	Low Keypad Battery
keypadLowPowerRecovery	Low Keypad Battery Recovered
cardReaderLowPower	Low Card Reader Battery
cardReaderLowPowerRecovery	Low Card Reader Battery Recovered
wKeypadOvertime	Wireless Keypad Heartbeat Timed Out
wKeypadOvertimeRecovery	Wireless Keypad Heartbeat Timeout Recovered
wCardReaderOvertime	Wireless Card Reader Heartbeat Timed Out
wCardReaderOvertimeRecovery	Wireless Card Reader Heartbeat Timeout Recovered
ATSFailed	ATS Failed
ATSRecovery	ATS Recovered
LANPathFailed	Wired or Wireless Connection Failed
LANPathRecovery	Wired or Wireless Connection Recovered
mobileNetPathFailed	Mobile Network Connection Failed
mobileNetPathRecovery	Mobile Network Connection Recovered
RS-485AlarmInputModuleDisconnected	RS-485 Zone Module Offline
RS-485AlarmInputModuleConnected	RS-485 Zone Module Online
RS-485WirelessReceiverDisconnected	RS-485 Wireless Receiver Module Offline
RS-485WirelessReceiverConnected	RS-485 Wireless Receiver Module Online
keypadDisconnected	Keypad Offline
keypadConnected	Keypad Online
overvoltage	High Supply Voltage
undervoltage	Low Supply Voltage
highHDDTemperature	HDD High Temperature
lowHDDTemperature	HDD Low Temperature
hdImpact	HDD Impact
hdBadBlock	HDD Bad Sector

Log Type	Description
severeHDFailure	HDD Severe Fault
safetyHelmetException	Hard Hat Detection Exception

Operation Logs

Log Type	Description
guard	Normal Arming
unguard	Normal Disarming
bypass	Bypass
duressAccess	Duress
localReboot	Local Reboot
remoteReboot	Remote Reboot
localUpgrade	Local Upgrade
remoteUpgrade	Remote Upgrade
recoveryDefaultParam	Restore Default Settings
outputAlarm	Remote Alarm Output Control
accessOpen	Access Control: Open
accessClose	Access Control : Closed
sirenOpen	Siren: On
sirenClose	Siren: Off
modZoneConfig	Zone Settings
modAlarmoutConfig	Alarm Output Settings
modAnalogConfig	Sensor Settings
RS485Config	RS-485 Channel Settings
phoneConfig	Dialing Settings
addAdmin	Added Administrator
modAdminParam	Edited Administrator
delAdmin	Deleted Administrator
addNetUser	Added DVR/NVR Operator
modNetUserParam	Edited DVR/NVR Operator

Log Type	Description
delNetUser	Deleted DVR/NVR Operator
addOperator	Added Camera Operator
modOperatorPw	Edited Camera Operator Password
delOperator	Deleted Camera Operator Password
addKeyPadUser	Added Keypad/Card Reader User
delKeyPadUser	Deleted Keyboard/Card Reader User
remoteUserLogin	Remote Login
remoteUserLogout	Remote Logout
remoteGuard	Remote Arming
remoteUnguard	Remote Disarming
modHostConfig	Edited Control Panel Settings
restoreBypass	Bypass Restored
alarmOutOpen	Turned on Output
alarmOutClose	Turned off Output
modSubsystemParam	Edited Subsystem Parameters
groupBypass	Group Bypass
groupBypassRestore	Group Bypass Restored
modGprsParam	Edited GPRS Parameters
modNetReportParam	Edited Network Report Settings
modReportMode	Edited Uploading Mode
modGatewayParam	Edited Access Control Settings
remoteStartRec	Remote: Started Recording
remoteStopRec	Remote: Stopped Recording
transChanStart	Transparent Transmission Started
transChanStop	Transparent Transmission Stopped
startVoiceTalk	Two-way Audio Started
stopVoiceTalk	Two-way Audio Terminated
remotePlayByFile	Remote: Playback or Downloaded by File
remotePlayByTime	Remote: Playback by Time

Log Type	Description
remotePTZCtrl	Remote: PTZ Control
remoteLockFile	Remote: Locked File
remoteUnlockFile	Remote: Unlocked File
remoteFormatHd	Remote: Formatted HDD
remoteDownloadCfgFile	Remote: Exported Configuration Files
remoteUploadCfgFile	Remote: Imported Configuration Files
remoteDownloadRecFile	Remote: Exported File
stayArm	Stay Arming
quickArm	Instant Arming
keyswitchArm	Key Zone Arming
keyswitchDisarm	Key Zone Disarming
clearAlarm	Alarm Cleared
modFaultConfig	Edited System Fault Settings
modAlarmOutConfig	Edited Event Alarm Output Settings
searchExternalModule	Searched for External Module
registerExternalModule	Re-registered External Module
closeKeypadAlarm	Disabled Keypad Beep
mod3GConfig	Edited Mobile Parameters
modPrintConfig	Edited Printer Parameters
SDCardFormat	Formatted SD Card
upgradeSubsystem	Upgraded Sub-board
planArmConfig	Arming/Disarming Schedule Configuration
phoneArm	SMS Arming
phoneStayArm	SMS Stay Arming
phoneQuickArm	SMS Instant Arming
phoneDisarm	SMS Disarming
phoneClearAlarm	SMS Alarm Cleared
whiteConfig	Whitelist Settings
timeTriggerConfig	Enabled/Disabled Trigger Configuration by Schedule

Log Type	Description
pictureConfig	Capture Settings
tamperConfig	Zone Tamper-Proof Settings
remoteKeypadUpgrade	Remote: Upgraded Keypad
singlePartitionArmORDisarm	Single-Zone Arming/Disarming
cardConfiguration	Card Settings
cardArmORDisarm	Arming/Disarming by Card
extendNetCenterConfig	Extension Network Center Settings
netCardConfig	NIC Settings
DDNSConfig	DDNS Settings
RS485BusConfig	RS-485 Bus Settings
RS485BusReRegistration	RS-485 Bus Re-registration
remoteOpenElectricLock	Remote: Unlocked
remoteCloseElectricLock	Remote: Locked
localOpenElectricLock	Local: Unlocked
localCloseElectricLock	Local: Locked
openAlarmLamp	Remote: Turned On Alarm Lamp
closeAlarmLamp	Remote: Turned Off Strobe
temporaryPassword	Operation Record of Temporary Password
oneKeyAwayArm	One-Push Away Arming
oneKeyStayArm	One-Push Stay Arming
singleZoneArm	Single-Zone Arming
singleZoneDisarm	Single-Zone Disarming
HiDDNSConfig	HiDDNS Settings
remoteKeypadUpdata	Remote: Upgraded Keypad
zoneAddDetector	Added Detector
zoneDelDetector	Deleted Detector
queryDetectorSignal	Checked Detector Signal Strength on Security Control Panel

Log Type	Description
queryDetectorBattery	Checked Detector Remaining Battery on Security Control Panel
setDetectorGuard	Detector Arming
setDetectorUnguard	Detector Disarming
setWifiParm	Wi-Fi Settings
voiceOpen	Audio On
voiceClose	Mute
functionKeyEnable	Enabled Function Key
functionKeyDisable	Disabled Panel Function Button
readCard	Swiped Patrol Card
localDeviceActive	Activated Device Remotely
localFactoryDefault	Restored Factory Settings Locally
remoteFactoryDefault	Restored Factory Settings Remotely
addWirelessOutputMod	Added Wireless Output Module
delWirelessOutputMod	Deleted Wireless Output Module
addWirelessRepeater	Added Wireless Repeater
delWirelessRepeater	Deleted Wireless Repeater
telListConfig	Mobile Phone Number Settings
searchRFSignal	Checked RF Signal
addWirelessSiren	Added Wireless Siren
delWirelessSiren	Deleted Wireless Siren
flowConfig	Cellular Data Limit Settings
addRemoter	Added Keyfob
delRemoter	Deleted Keyfob
addCard	Added Card
delCard	Deleted Card
remoteAddIpc	Added Network Camera
remoteDelIpc	Deleted Network Camera
remoteSetIpc	Edited Network Camera

Log Type	Description
localAddressFilterConfig/ remoteAddressFilterConfig	Local/Remote Address Filter Configuration
enterProgramMode	Programming Mode Enabled for Keypad
existProgramMode	Programming Mode Disabled for Keypad
localIOTCfgFileInput	Local operation: import IoT configuration file
localIOTCfgFileOutput	Local operation: export IoT configuration file
remoteIOTCfgFileInput	Remote operation: import IoT configuration file
remoteIOTCfgFileOutput	Remote operation: export IoT configuration file
localIOTAdd	Local operation: add IoT channel
remoteIOTAdd	Remote operation: add IoT channel
localIOTDelete	Local operation: delete IoT channel
remoteIOTDelete	Remote operation: delete IoT channel
localIOTSet	Local operation: configure IoT channel
remoteIOTSet	Remote operation: configure IoT channel
armWithFault	Armed with Fault
entryDelay	Entering and Exiting Delay
modArmConfig	Edit Arming Parameters
modCertificateStandard	Edit Authentication Standard
entryPaceTest	Pacing Mode Entered
exitPaceTest	Pacing Mode Exited
addNetOperator	Add Operator
modNetOperator	Edit Operator Information
delNetOperator	Delete Operator
addNetInstaller	Add Installer
modNetInstaller	Edit Installer Information
delNetInstaller	Delete Installer
addManufacturer	Add Manufacturer
modManufacturer	Edit Manufacturer Information
delManufacturer	Delete Manufacturer

Log Type	Description
upgradeSucceeded	Upgraded
upgradeFailed	Upgrading Failed
zoneDisabled	Zone Shielded
localCfgSecurity	Security Parameter Configured Locally
remoteCfgSecurity	Security Parameter Configured Remotely
remoteGetParaSecurity	Security Parameters Obtained Remotely
delRS-485InputModule	RS-485 Zone Module Deleted
delRS-485OutputModule	RS-485 Output Module Deleted
delRS-485WirelessReceiver	RS-485 Wireless Receiver Module Deleted
enrollRS-485InputModule	RS-485 Zone Module Registered
enrollRS-485OutputModule	RS-485 Output Module Registered
delRS-485OutputModule	RS-485 Output Module Deleted
enrollRS-485WirelessReceiver	RS-485 Wireless Receiver Module Registered
enrollKeypad	Keypad Registered
delKeypad	Keypad Deleted
scheduledAngleCalibration	Scheduled Angle Calibration
addZone	Added Zone
modZone	Edited Zone
delZone	Deleted Zone
addAlarmLine	Added Trigger Line
modAlarmLine	Edited Trigger Line
delAlarmLine	Deleted Trigger Line
remoteHFPDconfig/localHFPDconfig	Remote/Local Configuration of Frequently Appeared Person Detection
remoteLFPDconfig	Remote Configuration of Low Frequency Person Detection
modifyUserPassword	Password Changed
logOut	Logged Out
indicatorStatusSettings	Indicator Switch Settings

Log Type	Description
wiredNetworkSettings	Wired Network Settings
notificationSettings	Message Notification Settings
alarmCenterSettings	Alarm Receiving Center Settings
videoRecordStrategySettings	Video Recording Strategy Settings
cameraLinkageSettings	Camera Linkage Settings
armingDisarmingScheduleSettings	Arming and Disarming Schedule Settings
alarmSpeedSettings	Speed Threshold Settings for Triggering Alarms
videoTrackingSwitchSettings	Video Tracking Switching Settings
frequencySettings	Frequency Band Settings
masterSlaveTrackingSettings	Master-Slave Tracking Settings
parkingPointSettings	Parking Point Settings
administratorEdited	Administrator Parameters Edited
securityConfigured	Security Settings
relayParametersEdited	Relay Parameters Edited
radarSensitivitySettings	Radar Sensitivity Settings
detectAngandDistanceSettings	Detector Angle and Range Settings
masterSlaveRadarSettings	Master Radar and Slave Radar Settings
remoteCheckTime	Remote: Manual Time Synchronization
remoteParamSimpleDefault	Remote: Partly Restore to Default Settings
remoteParamFactoryDefault	Remote: Restore All to Default Settings
localAutoSwitchConfig	Configure Auto Power On or Off Locally
remoteAutoSwitchConfig	Configure Auto Power On or Off Remotely
remoteCfgWirelessDialParam	Configure wireless dial-up parameters remotely
localCfgWirelessDialParam	Configure wireless dial-up parameters locally
remoteCfgWirelessSmsParam	Configure wireless message parameters remotely
localCfgWirelessSmsParam	Configure wireless message parameters locally
remoteCfgWirelessSmsSelfHelp	Configure SMS self-service parameters remotely
localCfgWirelessSmsSelfHelp	Configure SMS self-service parameters locally
remoteCfgWirelessNetFlowParam	Configure wireless traffic parameters remotely

Log Type	Description
localCfgWirelessNetFlowParam	Configure wireless traffic parameters locally
scaleCfg	Scale Settings
radarTrailCfg	Radar Pattern Settings
MapImportCfg	Map Importing Settings
radarCalibrationCfg	Radar Calibration Settings
LocalEzvizOperation	Local EZVIZ Operations
RemoteEzvizOperation	Remote EZVIZ Operations

Event Logs

Log Type	Description
SDKSchool	SDK Synchronization
selfTimeSchool	Time Synchronization by Schedule
insertSubsystem	Plugged in Sub-board
pullOutSubsystem	Pulled out Sub-board
autoArm	Auto Arming
autoDisarm	Auto Disarming
triggerOn	Activated Trigger by Schedule
triggerOff	Deactivated Trigger by Schedule
autoArmFailed	Auto Arming Failed
autoDisarmFailed	Auto Disarming Failed
triggerOnFailed	Activating Trigger Failed
triggerOffFailed	Deactivating Trigger Failed
mandatoryAlarm	Forced Arming
keyPADlocked	Keypad Locked
keyPADunlocked	Keypad Unlocked
insetUSB	Plugged in USB Flash Drive
pulloutUSB	Removed USB Flash Drive
lateRemind	Late to Disarm
keypadUnlocked	Unlocked Keypad

Log Type	Description
timeSynchronization	Time Synchronization
armFailed	Arming Failed
ARCStart	ARC Connected
locked	Locked

Additional Logs

Log Type	Description
doubleVerificationPass	Double Verification Completed
hdFormatStart	Formatting HDD Started
hdFormatStop	Formatting HDD Stopped
wirelessRunningStatus	Wireless Network Running Status

A.3 Supported Alarm/Event Types

This part lists the alarm or event types that can be received or subscribed via Intelligent Security API and the corresponding values to be set for getting the details.

Alarm/Event Type	Value of <type>	Alarm/Event Details
Intrusion Detection	fielddetection	Refer to <i>Intelligent Security API (Behavior Analysis) _Developer Guide</i> for details.
Line Crossing Detection	linedetection	
Region Entrance Detection	regionEntrance	
Region Exiting Detection	regionExiting	
Loitering Detection	loitering	
People Gathering Detection	group	
Fast Moving Detection	rapidMove	
Parking Detection	parking	
Unattended Baggage Detection	unattendedBaggage	
Object Removal Detection	attendedBaggage	

Alarm/Event Type	Value of <type>	Alarm/Event Details
Absence Alarm	leavePosition	
People Number Changed Alarm	peopleNumChange	
Fast Moving Alarm	running	
Violent Motion Alarm	violentMotion	
People Falling Down Alarm	failDown	
Overstay Alarm	retention	
People Counting Alarm	PeopleCounting	Refer to <i>Intelligent Security API (People Counting) _ Developer Guide</i> for details
Temperature Measurement Alarm	TMA	Refer to <i>Intelligent Security API (Thermal) _ Developer Guide</i> for details
Temperature Measurement Pre-Alarm	TMPA	
Temperature Difference Alarm	TDA	
Fire Source Detection	fireDetection	
Smoke Detection	smokeDetection	
Fire and Smoke Detection	smokeAndFireDetection	
Ship Detection	shipsDetection	
Face Detection Alarm	facedetection	Refer to <i>Intelligent Security API (Facial) _ Developer Guide</i> for details.
Face Capture Alarm	faceCapture	
Face Picture Comparison Result	alarmResult	
Stranger Detection Alarm	whiteFaceContrast	
High Frequently Person Detection	HFPD	
Low Frequency Person Detection	LFPD	

Intelligent Security API (General) Developer Guide

Alarm/Event Type	Value of <type>	Alarm/Event Details
Waiting Time Detection Alarm	personQueueTime	Refer to <i>Intelligent Security API (Queue Management) _Developer Guide</i> for details
People Queuing-Up Alarm	personQueueCounting	
Multi-Target-Type Detection Alarm	mixedTargetDetection	Refer to <i>Intelligent Security API (Multi-Target-Type Detection) _Developer Guide</i> for details
Traffic Incident Alarm	AID	Refer to <i>Intelligent Security API (Traffic Enforcement) _Developer Guide</i> for details
Traffic Violation Enforcement Alarm	TFS	
Traffic Data Statistics Alarm	TPS	
ANPR Alarm	ANPR	Refer to <i>Intelligent Security API (ANPR) _Developer Guide</i> for details
Blacklist and Whitelist ANPR Alarm	vehicleMatchResult	
Vehicle Feature Recognition Alarm	vehicleRcogResult	
Radar Measurement Data	radarMeasurement	
Heat Map Alarm	heatmap	Refer to <i>Intelligent Security API (Heat Map) _Developer Guide</i> for details
Face and ID Card Recognition Alarm	cardMatch	Refer to <i>Intelligent Security API (Access Control on Person) _Developer Guide</i> for details
Hot Spare Exception Alarm	hotSpare	The alarm or event details will be provided in the further versions.
Overspeed Alarm	overSpeed	The alarm or event details will be provided in the further versions.
Temperature Too High Alarm	highTempAlarm	The alarm or event details will be provided in the further versions.
Acceleration Exception Alarm	abnormalAcceleration	The alarm or event details will be provided in the further versions.
Collision Alarm	collision	The alarm or event details will be provided in the further versions.

Alarm/Event Type	Value of <type>	Alarm/Event Details
Rollover Detection	rollover	The alarm or event details will be provided in the further versions.
Abnormal Driving Behavior Detection	abnormalDriving	The alarm or event details will be provided in the further versions.
GPS Information Uploaded	GPSUpload	The alarm or event details will be provided in the further versions.
Video Standard Mismatched	PALMismatch	The alarm or event details will be provided in the further versions.
Alarm Input Alarm	IO	
Motion Detection Alarm	VMD	<i>XML_EventNotificationAlert_MotionDetectionMsg</i>
Video Loss Alarm	videoloss	<i>XML_EventNotificationAlert_VideoLossAlarmMsg</i>
Video Tampering Alarm	shelteralarm	<i>XML_EventNotificationAlert_VideoTamperingAlarmMsg</i>
Defocus Alarm	defocus	<i>XML_EventNotificationAlert_DefocusDetectionAlarmMsg</i>
Audio Exception Alarm	audioexception	<i>XML_EventNotificationAlert_AudioExceptionAlarmMsg</i>
Scene Changed Alarm	scenechangedetection	<i>XML_EventNotificationAlert_SceneChangeDetectionAlarmMsg</i>
PIR Alarm	PIR	<i>XML_EventNotificationAlert_PIRAlarmMsg</i>
HDD Full	diskfull	The alarm or event details will be provided in the further versions.
HDD Error	diskerror	<i>XML_EventNotificationAlert_HDDErrorAlarmMsg</i>
Network Disconnected	nicbroken	The alarm or event details will be provided in the further versions.
IP Address Conflicted	ipconflict	The alarm or event details will be provided in the further versions.
Illegal Login Alarm	illaccess	<i>XML_EventNotificationAlert_IllegalLoginAlarmMsg</i>
PoE Power Exception Alarm	poe	The alarm or event details will be provided in the further versions.

Alarm/Event Type	Value of <type>	Alarm/Event Details
Video Exception Alarm	videoException	The alarm or event details will be provided in the further versions.
Recording Exception Alarm	recordException	The alarm or event details will be provided in the further versions.
HDD Unformatted Alarm	diskUnformat	The alarm or event details will be provided in the further versions.
Supply Voltage Exception Alarm	voltageinstable	<i>JSON_EventNotificationAlert_voltageinstable</i>
HDD High Temperature	highHDTemperature	<i>JSON_EventNotificationAlert_HDDHighTemperatureEventMsg</i>
HDD Low Temperature	lowHDTemperature	<i>JSON_EventNotificationAlert_HDDLowTemperatureEventMsg</i>
HDD Impact	hdImpact	<i>JSON_EventNotificationAlert_HDDImpactEventMsg</i>
HDD Bad Sector	hdBadBlock	<i>JSON_EventNotificationAlert_HDDBadSectorEventMsg</i>
HDD Severe Fault	severeHDFailure	<i>JSON_EventNotificationAlert_HDDSevereFaultEventMsg</i>
Certificate Expiry Alarm	certificateRevocation	<i>JSON_EventNotificationAlert_CertificateExpiryAlarmMsg</i>
Network Traffic Pre-alarm	dataPrealarm	<i>JSON_EventNotificationAlert_NetworkTrafficPrealarmMsg</i>
Auto Person Arming	personArmingTrack	Refer to <i>Intelligent Security API (Person Arming) _ Developer Guide</i> for details
Manual Person Arming	manualPersonArmingTrack	

A.4 ISAPI Response Codes

The response codes returned during ISAPI integration is based on the status codes of HTTP protocol. 7 kinds of status codes are predefined, including 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid Message Format), 6 (Invalid Message Content), and 7 (Reboot Required). Each kind of status code contains multiple sub status codes, and the response codes are in a one-to-one correspondence with the sub status codes.

StatusCode=1

SubStatusCode	Error Code	Description
ok	0x1	Operation completed.
riskPassword	0x10000002	Risky password.
armProcess	0x10000005	Arming process.

StatusCode=2

Sub Status Code	Error Code	Description
noMemory	0x20000001	Insufficient memory.
serviceUnavailable	0x20000002	The service is not available.
upgrading	0x20000003	Upgrading.
deviceBusy	0x20000004	The device is busy or no response.
reConnectIpc	0x20000005	The video server is reconnected.
transferUpgradePackageFailed	0x20000006	Transmitting device upgrade data failed.
startUpgradeFailed	0x20000007	Starting upgrading device failed.
getUpgradeProcessfailed.	0x20000008	Getting upgrade status failed.
certificateExist	0x2000000B	The Authentication certificate already exists.

StatusCode=3

Sub Status Code	Error Code	Description
deviceError	0x30000001	Hardware error.
badFlash	0x30000002	Flash operation error.
28181Uninitialized	0x30000003	The 28181 configuration is not initialized.
socketConnectError	0x30000005	Connecting to socket failed.
receiveError	0x30000007	Receive response message failed.

Sub Status Code	Error Code	Description
deletePictureError	0x3000000A	Deleting picture failed.
pictureSizeExceedLimit	0x3000000C	Too large picture size.
clearCacheError	0x3000000D	Clearing cache failed.
updateDatabasError	0x3000000F	Updating database failed.
searchDatabaseError	0x30000010	Searching in the database failed.
writeDatabaseError	0x30000011	Writing to database failed.
deleteDatabaseError	0x30000012	Deleting database element failed.
searchDatabaseElementError	0x30000013	Getting number of database elements failed.
cloudAutoUpgradeException	0x30000016	Downloading upgrade packet from cloud and upgrading failed.
HBPEXception	0x30001000	HBP exception.
UDEPEXception	0x30001001	UDEP exception
elasticSearchException	0x30001002	Elastic exception.
kafkaException	0x30001003	Kafka exception.
HBaseException	0x30001004	Hbase exception.
sparkException	0x30001005	Spark exception.
yarnException	0x30001006	Yarn exception.
cacheException	0x30001007	Cache exception.
trafficException	0x30001008	Monitoring point big data server exception.
faceException	0x30001009	Human face big data server exception.
SSDFileSystemsIsError	0x30001013	SSD file system error (Error occurs when it is non-Ext4 file system)
insufficientSSDCapacityForFPD	0x30001014	Insufficient SSD space for person frequency detection
wifiException	0x3000100A	Wi-Fi big data server exception

Sub Status Code	Error Code	Description
structException	0x3000100D	Video parameters structure server exception.
calibrationTimeout	0x30002051	Calibration timed out.
captureTimeout	0x30006000	Data collection timed out.
lowScore	0x30006001	Low quality of collected data.
uploadingFailed	0x30007004	Uploading failed.

StatusCode=4

Sub Status Code	Error Code	Description
notSupport	0x40000001	Not supported.
lowPrivilege	0x40000002	No permission.
badAuthorization	0x40000003	Authentication failed.
methodNotAllowed	0x40000004	Invalid HTTP method.
notSetHdiskRedund	0x40000005	Setting spare HDD failed.
invalidOperation	0x40000006	Invalid operation.
hdFormatFail	0x40002049	Formatting HDD failed.
notActivated	0x40000007	Inactivated.
hasActivated	0x40000008	Activated.
certificateAlreadyExist	0x40000009	The certificate already exists.
operateFailed	0x4000000F	Operation failed.
USBNotExist	0x40000010	USB device is not connected.
upgradePackageMorethan2GB	0x40001000	Up to 2GB upgrade package is allowed to be uploaded.
IDNotExist	0x40001001	The ID does not exist.
synchronizationError	0x40001003	Synchronization failed.
synchronizing	0x40001004	Synchronizing.
importError	0x40001005	Importing failed.
importing	0x40001006	Importing.
fileAlreadyExists	0x40001007	The file already exists.
invalidID	0x40001008	Invalid ID.

Sub Status Code	Error Code	Description
backupnodeNotAllowe Log	0x40001009	Accessing to backup node is not allowed.
exportingError	0x4000100A	Exporting failed.
exporting	0x4000100B	Exporting.
exportEnded	0x4000100C	Exporting stopped.
exported	0x4000100D	Exported.
IPOccupied	0x4000100E	The IP address is already occupied.
IDAlreadyExists	0x4000100F	The ID already exists.
exportItemsExceedLimi t	0x40001010	No more items can be exported.
noFiles	0x40001011	The file does not exist.
beingExportedByAnoth erUser	0x40001012	Being exported by others.
needReAuthentication	0x40001013	Authentication is needed after upgrade.
unitAddNotOnline	0x40001015	The added data analysis server is offline.
unitControl	0x40001016	The data analysis server is already added.
analysis unitFull	0x40001017	No more data analysis server can be added.
unitIDError	0x40001018	The data analysis server ID does not exist.
unitExit	0x40001019	The data analysis server already exists in the list.
unitSearch	0x4000101A	Searching data analysis server in the list failed.
unitNotOnline	0x4000101B	The data analysis server is offline.
unitInfoError	0x4000101C	Getting data analysis server information failed.
unitGetNodeInfoError	0x4000101D	Getting node information failed.
unitGetNetworkInfoErr or	0x4000101E	Getting the network information of data analysis server failed
unitSetNetworkInfoErr or	0x4000101F	Setting the network information of data analysis server failed
setSmartNodeInfoError	0x40001020	Setting node information failed.
setUnitNetworkInfoErr or	0x40001021	Setting data analysis server network information failed.

Sub Status Code	Error Code	Description
unitRestartCloseError	0x40001022	Rebooting or shutting down data analysis server failed.
virtualIPnotAllowed	0x40001023	Adding virtual IP address is not allowed.
unitInstalled	0x40001024	The data analysis server is already installed.
badSubnetMask	0x40001025	Invalid subnet mask.
uintVersionMismatched	0x40001026	Data analysis server version mismatches.
deviceMOdelMismatched	0x40001027	Adding failed. Device model mismatches.
unitAddNotSelf	0x40001028	Adding peripherals is not allowed.
noValidUnit	0x40001029	No valid data analysis server.
unitNameDuplicate	0x4000102A	Duplicated data analysis server name.
deleteUnitFirst	0x4000102B	Delete the added data analysis server of the node first.
getLocalInfoFailed	0x4000102C	Getting the server information failed.
getClientAddedNodeFailed	0x4000102D	Getting the added node information of data analysis server failed.
taskExit	0x4000102E	The task already exists.
taskInitError	0x4000102F	Initializing task failed.
taskSubmitError	0x40001030	Submitting task failed.
taskDelError	0x40001031	Deleting task failed.
taskPauseError	0x40001032	Pausing task failed.
taskContinueError	0x40001033	Starting task failed.
taskSeverNoCfg	0x40001035	Full-text search server is not configured.
taskPicSeverNoCfg	0x40001036	The picture server is not configured.
taskStreamError	0x40001037	Streaming information exception.
taskRecSDK	0x40001038	History recording is not supported.
taskCasaError	0x4000103A	Cascading is not supported.
taskVCARuleError	0x4000103B	Invalid VCA rule.
taskNoRun	0x4000103C	The task is not executed.

Sub Status Code	Error Code	Description
unitLinksNoStorageNode	0x4000103D	No node is linked with the data analysis server. Configure the node first.
searchFailed	0x4000103E	Searching video files failed.
searchNull	0x4000103F	No video clip.
userScheOffline	0x40001040	The task scheduler service is offline.
updateTypeUnmatched	0x40001041	The upgrade package type mismatches.
userExist	0x40001043	The user already exists.
userCannotDelAdmin	0x40001044	The administrator cannot be deleted.
userInexistence	0x40001045	The user name does not exist.
userCannotCreatAdmin	0x40001046	The administrator cannot be created.
monitorCamExceed	0x40001048	Up to 3000 cameras can be added.
monitorCunitOverLimit	0x40001049	Adding failed. Up to 5 lower-levels are supported by the control center.
monitorReginOverLimit	0x4000104A	Adding failed. Up to 5 lower-levels are supported by the area.
monitorArming	0x4000104B	The camera is already armed. Disarm the camera and try again.
monitorSyncCfgNotSet	0x4000104C	The system parameters are not configured.
monitorFdSyncing	0x4000104E	Synchronizing. Try again after completing the synchronization.
monitorParseFailed	0x4000104F	Parsing camera information failed.
monitorCreatRootFailed	0x40001050	Creating resource node failed.
deleteArmingInfo	0x40001051	The camera is already . Disarm the camera and try again.
cannotModify	0x40001052	Editing is not allowed. Select again.
cannotDel	0x40001053	Deletion is not allowed. Select again.
deviceExist	0x40001054	The device already exists.
IPErrorConnectFailed	0x40001056	Connection failed. Check the network port.

Sub Status Code	Error Code	Description
cannotAdd	0x40001057	Only the capture cameras can be added.
serverExist	0x40001058	The server already exists.
fullTextParamError	0x40001059	Incorrect full-text search parameters.
storParamError	0x4000105A	Incorrect storage server parameters.
picServerFull	0x4000105B	The storage space of picture storage server is full.
NTPUnconnect	0x4000105C	Connecting to NTP server failed. Check the parameters.
storSerConnectFailed	0x4000105D	Connecting to storage server failed. Check the network port.
storSerLoginFailed	0x4000105E	Logging in to storage server failed. Check the user name and password.
searchSerConnectFailed	0x4000105F	Connecting to full-text search server failed. Check the network port.
searchSerLoginFailed	0x40001060	Logging in to full-text search server failed. Check the user name and password.
kafkaConnectFailed	0x40001061	Connecting to Kafka failed. Check the network port.
mgmtConnectFailed	0x40001062	Connecting to system failed. Check the network port.
mgmtLoginFailed	0x40001063	Logging in to system failed. Check the user name and password.
TDAConnectFailed	0x40001064	Connecting to traffic data access server failed. Checking the server status.
86sdkConnectFailed	0x40001065	Connecting to listening port of iVMS-8600 System failed. Check the parameters.
nameExist	0x40001066	Duplicated server name.
batchProcessFailed	0x40001067	Processing in batch failed.
IDNotExist	0x40001068	The server ID does not exist.
serviceNumberReachedLimit	0x40001069	No more service can be added.
invalidServiceType.	0x4000106A	Invalid service type.

Sub Status Code	Error Code	Description
clusterGetInfo	0x4000106B	Getting cluster group information failed.
clusterDelNode	0x4000106C	Deletion node failed.
clusterAddNode	0x4000106D	Adding node failed.
clusterInstalling	0x4000106E	Creating cluster...Do not operate.
clusterUninstall	0x4000106F	Reseting cluster...Do not operate.
clusterInstall	0x40001070	Creating cluster failed.
clusterIpError	0x40001071	Invalid IP address of task scheduler server.
clusterNotSameSeg	0x40001072	The master node and slave node must be in the same network segment.
clusterVirIpError	0x40001073	Automatically getting virtual IP address failed. Enter manually.
clusterNodeUnadd	0x40001074	The specified master(slave) node is not added.
clusterNodeOffline	0x40001075	The task scheduler server is offline.
nodeNotCurrentIP	0x40001076	The analysis node of the current IP address is required when adding master and slave nodes.
addNodeNetFailed	0x40001077	Adding node failed. The network disconnected.
needTwoMgmtNode	0x40001078	Two management nodes are required when adding master and slave nodes.
ipConflict	0x40001079	The virtual IP address and data analysis server's IP address conflicted.
ipUsed	0x4000107A	The virtual IP address has been occupied.
cloudAlalyseOnline	0x4000107B	The cloud analytic server is online.
virIP&mainIPnotSame NetSegment	0x4000107C	The virtual IP address is not in the same network segment with the IP address of master/slave node.
getNodeDispatchInfoFa iled	0x4000107D	Getting node scheduler information failed.
unableModifyManage mentNetworkIP	0x4000107E	Editing management network interface failed. The analysis board is in the cluster.
notSpecifyVirtualIP	0x4000107F	Virtual IP address should be specified for master and slave cluster.
armingFull	0x40001080	No more device can be armed.

Sub Status Code	Error Code	Description
armingNoFind	0x40001081	The arming information does not exist.
disArming	0x40001082	Disarming failed.
getArmingError	0x40001084	Getting arming information failed.
refreshArmingError	0x40001085	Refreshing arming information failed.
ArmingPlateSame	0x40001086	The license plate number is repeatedly armed.
ArmingParseXLSError	0x40001087	Parsing arming information file failed.
ArmingTimeError	0x40001088	Invalid arming time period.
ArmingSearchTimeError	0x40001089	Invalid search time period.
armingRelationshipReachesLimit	0x4000108A	No more relation can be created.
duplicateAarmingName	0x4000108B	The relation name already exists.
noMoreArmingListAdded	0x4000108C	No more blacklist library can be armed.
noMoreCamerasAdded	0x4000108D	No more camera can be armed.
noMoreArmingListAddedWithCamera	0x4000108E	No more library can be linked to the camera.
noMoreArmingPeriodAdded	0x4000108F	No more time period can be added to the arming schedule.
armingPeriodsOverlapped	0x40001090	The time periods in the arming schedule are overlapped.
noArmingAlarmInfo	0x40001091	The alarm information does not exist.
armingAlarmUnRead	0x40001092	Getting number of unread alarms failed.
getArmingAlarmError	0x40001093	Getting alarm information failed.
searchByPictureTimedOut	0x40001094	Searching picture by picture timeout. Search again.
comparisonTimeRangeError	0x40001095	Comparison time period error.
selectMonitorNumberUpperLimit	0x40001096	No more monitoring point ID can be filtered.

Sub Status Code	Error Code	Description
noMoreComparisonTasksAdded	0x40001097	No more comparison task can be executed at the same time.
GetComparisonResultFailed	0x40001098	Getting comparison result failed.
comparisonTypeError	0x40001099	Comparison type error.
comparisonUnfinished	0x4000109A	The comparison is not completed.
facePictureModelInvalid	0x4000109B	Invalid face model.
duplicateLibraryName.	0x4000109C	The library name already exists.
noRecord	0x4000109D	No record found.
countingRecordsFailed.	0x4000109E	Calculate the number of records failed.
getHumanFaceFrameFailed	0x4000109F	Getting face thumbnail from the picture failed.
modelingFailed.	0x400010A0	Modeling face according to picture URL failed.
1V1FacePictureComparisonFailed	0x400010A1	Comparison 1 VS 1 face picture failed.
libraryArmed	0x400010A2	The blacklist library is armed.
licenseExceedLimit	0x400010A3	Dongle limited.
licenseExpired	0x400010A4	Dongle expired.
licenseDisabled	0x400010A5	Unavailable dongle.
licenseNotExist	0x400010A6	The dongle does not exist.
SessionExpired	0x400010A7	Session expired .
beyondConcurrentLimit	0x400010A8	Out of concurrent limit.
stopSync	0x400010A9	Synchronization stopped.
getProgressFailed	0x400010AA	Getting progress failed.
uploadExtraCaps	0x400010AB	No more files can be uploaded.
timeRangeError	0x400010AC	Time period error.
dataPortNotConnected	0x400010AD	The data port is not connected.
addClusterNodeFailed	0x400010AE	Adding to the cluster failed. The device is already added to other cluster.

Sub Status Code	Error Code	Description
taskNotExist	0x400010AF	The task does not exist.
taskQueryFailed	0x400010B0	Searching task failed.
modifyTimeRuleFailed	0x400010B2	The task already exists. Editing time rule is not allowed.
modifySmartRuleFailed	0x400010B3	The task already exists. Editing VAC rule is not allowed.
queryHistoryVideoFailed	0x400010B4	Searching history video failed.
addDeviceFailed	0x400010B5	Adding device failed.
addVideoFailed	0x400010B6	Adding video files failed.
deleteAllVideoFailed	0x400010B7	Deleting all video files failed.
createVideoIndexFailed	0x400010B8	Indexing video files failed.
videoCheckTypeFailed	0x400010B9	Verifying video files types failed.
configStructuredAddressFailed	0x400010BA	Configuring IP address of structured server failed.
configPictureServerAddressFailed	0x400010BB	Configuring IP address of picture stored server failed.
storageServiceIPNotExist	0x400010BD	The storage server IP address does not exist.
syncBackupDatabaseFailed	0x400010BE	Synchronizing slave database failed. Try again.
syncBackupNTPTimeFailed	0x400010BF	Synchronizing NTP time of slave server failed.
clusterNotSelectLoopbackAddress	0x400010C0	Loopback address is not supported by the master or slave cluster.
addFaceRecordFailed	0x400010C1	Adding face record failed.
deleteFaceRecordFailed	0x400010C2	Deleting face record failed.
modifyFaceRecordFailed	0x400010C3	Editing face record failed.
queryFaceRecordFailed	0x400010C4	Searching face record failed.
faceDetectFailed	0x400010C5	Detecting face failed.

Sub Status Code	Error Code	Description
libraryNotExist	0x400010C6	The library does not exist.
blackListQueryExporting	0x400010C7	Exporting matched blacklists.
blackListQueryExported	0x400010C8	The matched blacklists are exported.
blackListQueryStopExporting	0x400010C9	Exporting matched blacklists is stopped.
blackListAlarmQueryExporting	0x400010CA	Exporting matched blacklist alarms.
blackListAlarmQueryExported	0x400010CB	The matched blacklists alarms are exported.
blackListAlarmQueryStopExporting	0x400010CC	Exporting matched blacklist alarms is stopped.
getBigDataCloudAnalysisFailed	0x400010CD	Getting big data cloud analytic information failed.
setBigDataCloudAnalysisFailed	0x400010CE	Configuring big data cloud analytic failed.
submitMapSearchFailed	0x400010CF	Submitting search by picture task failed.
controlRelationshipNotExist	0x400010D0	The relation does not exist.
getHistoryAlarmInfoFailed	0x400010D1	Getting history alarm information failed.
getFlowReportFailed	0x400010D2	Getting people counting report failed.
addGuardFailed	0x400010D3	Adding arming configuration failed.
deleteGuardFailed	0x400010D4	Deleting arming configuration failed.
modifyGuardFailed	0x400010D5	Editing arming configuration failed.
queryGuardFailed	0x400010D6	Searching arming configurations failed.
uploadUserSuperCaps	0x400010D7	No more user information can be uploaded.
bigDataServerConnectFailed	0x400010D8	Connecting to big data server failed.
microVideoCloudRequestInfoBuildFailed	0x400010D9	Adding response information of micro video cloud failed.

Sub Status Code	Error Code	Description
microVideoCloudResponseInfoBuildFailed	0x400010DA	Parsing response information of micro video cloud failed.
transcodingServerRequestInfoBuildFailed	0x400010DB	Adding response information of transcoding server failed.
transcodingServerResponseInfoParseFailed	0x400010DC	Parsing response information of transcoding server failed.
transcodingServerOffline	0x400010DD	Transcoding server is offline.
microVideoCloudOffline	0x400010DE	Micro video cloud is offline.
UPSServerOffline	0x400010DF	UPS monitor server is offline.
statisticReportRequestInfoBuildFailed	0x400010E0	Adding response information of statistics report failed.
statisticReportResponseInfoParseFailed	0x400010E1	Parsing response information of statistics report failed.
DisplayConfigInfoBuildFailed	0x400010E2	Adding display configuration information failed.
DisplayConfigInfoParseFailed	0x400010E3	Parsing display configuration information failed.
DisplayConfigInfoSaveFailed	0x400010E4	Saving display configuration information failed.
notSupportDisplayConfigType	0x400010E5	The display configuration type is not supported.
passError	0x400010E7	Incorrect password.
upgradePackageLarge	0x400010EB	Too large upgrade package.
sessionUserReachesLimit	0x400010EC	No more user can log in via session.
ISO8601TimeFormatError	0x400010ED	Invalid ISO8601 time format.
clusterDissolutionFailed	0x400010EE	Deleting cluster failed.
getServiceNodeInfoFailed	0x400010EF	Getting service node information failed.

Sub Status Code	Error Code	Description
getUPSInfoFailed	0x400010F0	Getting UPS configuration information failed.
getDataStatisticsReportFailed	0x400010F1	Getting data statistic report failed.
getDisplayConfigInfoFailed	0x400010F2	Getting display configuration failed.
namingAnalysisBoardNotAllowed	0x400010F3	Renaming analysis board is not allowed.
onlyDrawRegionsOfConvexPolygon	0x400010F4	Only drawing convex polygon area is supported.
bigDataServerResponseInfoParseFailed	0x400010F5	Parsing response message of big data service failed.
bigDataServerReturnFailed	0x400010F6	No response is returned by big data service.
microVideoReturnFailed	0x400010F7	No response is returned by micro video cloud service.
transcodingServerReturnFailed	0x400010F8	No response is returned by transcoding service.
UPSServerReturnFailed	0x400010F9	No response is returned by UPS monitoring service.
forwardingServerReturnFailed	0x400010FA	No response is returned by forwarding service.
storageServerReturnFailed	0x400010FB	No response is returned by storage service.
cloudAnalysisServerReturnFailed	0x400010FC	No response is returned by cloud analytic service.
modelEmpty	0x400010FD	No model is obtained.
mainAndBackupNodeCannotModifyManagementNetworkInterfaceIP	0x400010FE	Editing the management interface IP address of master node and backup node is not allowed.
IDTooLong	0x400010FF	The ID is too long.
pictureCheckFailed	0x40001100	Detecting picture failed.
pictureModelingFailed	0x40001101	Modeling picture failed.

Sub Status Code	Error Code	Description
setCloudAnalysisDefaultProvinceFailed	0x40001102	Setting default province of cloud analytic service failed.
InspectionAreasNumberExceedLimit	0x40001103	No more detection regions can be added.
picturePixelsTooLarge	0x40001105	The picture resolution is too high.
picturePixelsTooSmall	0x40001106	The picture resolution is too low.
storageServiceIPEmpty	0x40001107	The storage server IP address is required.
bigDataServerRequestInfoBuildFail	0x40001108	Creating request message of big data service failed.
analysisTimedOut	0x40001109	Analysis time out.
high-performanceModeDisabled.	0x4000110A	Please enable high-performance mode.
configuringUPSMonitoringServerTimedOut	0x4000110B	Configuring the UPS monitoring server time out. Check IP address.
cloudAnalysisRequestInformationBuildFailed	0x4000110C	Creating request message of cloud analytic service failed.
cloudAnalysisResponseInformationParseFailed	0x4000110D	Parsing response message of cloud analytic service failed.
allCloudAnalysisInterfaceFailed	0x4000110E	Calling API for cloud analytic service failed.
cloudAnalysisModelCompareFailed	0x4000110F	Model comparison of cloud analytic service failed.
cloudAnalysisFacePictureQualityRatingFailed	0x40001110	Getting face quality grading of cloud analytic service failed.
cloudAnalysisExtractFeaturePointsFailed	0x40001111	Extracting feature of cloud analytic service failed.
cloudAnalysisExtractPropertyFailed	0x40001112	Extracting property of cloud analytic service failed.
getAddedNodeInformationFailed	0x40001113	Getting the added nodes information of data analysis server failed.
noMoreAnalysisUnitsAdded	0x40001114	No more data analysis servers can be added.

Sub Status Code	Error Code	Description
detectionAreaInvalid	0x40001115	Invalid detection region.
shieldAreaInvalid	0x40001116	Invalid shield region.
noMoreShieldAreasAdded	0x40001117	No more shield region can be drawn.
onlyAreaOfRectangleShapeAllowed	0x40001118	Only drawing rectangle is allowed in detection area.
numberReachedLimit	0x40001119	Number reached the limit.
wait1~3MinutesGetIPAfterSetupDHCP	0x4000111A	Wait 1 to 3 minutes to get IP address after configuring DHCP.
plannedTimeMustbeHalfAnHour	0x4000111B	Schedule must be half an hour.
oneDeviceCannotBuildCluster	0x4000111C	Creating master and backup cluster requires at least two devices.
updatePackageFileNotUploaded	0x4000111E	Upgrade package is not uploaded.
highPerformanceTasksNotSupportDrawingDetectionRegions	0x4000111F	Drawing detection area is not allowed under high-performance mode.
controlCenterIDDoesNotExist	0x40001120	The control center ID does not exist.
regionIDDoesNotExist	0x40001121	The area ID does not exist.
licensePlateFormatError	0x40001122	Invalid license plate format.
managementNodesNotSupportThisOperation	0x40001123	The operation is not supported.
searchByPictureResourceNotConfiged	0x40001124	The conditions for searching picture by picture are not configured.
videoFileEncapsulationFormatNotSupported	0x40001125	The video container format is not supported.
videoPackageFailure	0x40001126	Converting video container format failed.
videoCodingFormatNotSupported	0x40001127	Video coding format is not supported.

Sub Status Code	Error Code	Description
monitorOfDeviceArmingDeleteArmingInfo	0x40001129	The camera is armed. Disarm it and try again.
getVideoSourceTypeFailed	0x4000112A	Getting video source type failed.
smartRulesBuildFailed	0x4000112B	Creating VAC rule failed.
smartRulesParseFailed	0x4000112C	Parsing VAC rule failed.
timeRulesBuildFailed	0x4000112D	Creating time rule failed.
timeRulesParseFailed	0x4000112E	Parsing time rule failed.
monitorInfoInvalid	0x4000112F	Invalid camera information.
addingFailedVersionMismatches	0x40001130	Adding failed. The device version mismatches.
theInformationReturnedAfterCloudAnalysisIsEmpty	0x40001131	No response is returned by the cloud analytic service.
selectingIpAddressOfHostAndSpareNodeFailedCheckTheStatus	0x40001132	Setting IP address for master node and backup node failed. Check the node status.
theSearchIdDoesNotExist	0x40001133	The search ID does not exist.
theSynchronizationIdDoesNotExist	0x40001134	The synchronization ID does not exist.
theUserIdDoesNotExist	0x40001136	The user ID does not exist.
theIndexCodeDoesNotExist	0x40001138	The index code does not exist.
theControlCenterIdDoesNotExist	0x40001139	The control center ID does not exist.
theAreaIdDoesNotExist	0x4000113A	The area ID does not exist.
theArmingLinkageIdDoesNotExist	0x4000113C	The arming relationship ID does not exist.
theListLibraryIdDoesNotExist	0x4000113D	The list library ID does not exist.
invalidCityCode	0x4000113E	Invalid city code.

Sub Status Code	Error Code	Description
synchronizingThePasswordOfSpareServerFailed	0x4000113F	Synchronizing backup system password failed.
editingStreamingTypeIsNotSupported	0x40001140	Editing streaming type is not supported.
switchingScheduledTaskToTemporaryTaskIsNotSupported	0x40001141	Switching scheduled task to temporary task is not supported.
switchingTemporaryTaskToScheduledTaskIsNotSupported	0x40001142	Switching temporary task to scheduled task is not supported.
theTaskIsNotDispatchedOrItIsUpdating	0x40001143	The task is not dispatched or is updating.
thisTaskDoesNotExist	0x40001144	This task does not exist in the cloud analytic service.
duplicatedSchedule	0x40001145	Schedule period cannot be overlapped.
continuousScheduleWithSameAlgorithmTypeShouldBeMerged	0x40001146	The continuous schedule periods with same algorithm type should be merged.
invalidStreamingTimeRange	0x40001147	Invalid streaming time period.
invalidListLibraryType	0x40001148	Invalid list library type.
theNumberOfMatchedResultsShouldBeLargerThan0	0x40001149	The number of search results should be larger than 0.
invalidValueRangeOfSimilarity	0x4000114A	Invalid similarity range.
invalidSortingType	0x4000114B	Invalid sorting type.
noMoreListLibraryCanBeLinkedToTheDevice	0x4000114C	No more lists can be added to one device.
InvalidRecipientAddressFormat	0x4000114D	Invalid address format of result receiver.

Sub Status Code	Error Code	Description
creatingClusterFailedTheDongleIsNotPluggedIn	0x4000114E	Insert the dongle before creating cluster.
theURLIsTooLong	0x4000114F	No schedule configured for the task.
noScheduleIsConfiguredForTheTask	0x40001150	No schedule configured for the task.
theDongleIsExpired	0x40001151	Dongle has expired.
dongleException	0x40001152	Dongle exception.
invalidKey	0x40001153	Invalid authorization service key.
decryptionFailed	0x40001154	Decrypting authorization service failed.
encryptionFailed	0x40001155	Encrypting authorization service failed.
AuthorizeServiceResponseError	0x40001156	Authorization service response exception.
incorrectParameter	0x40001157	Authorization service parameters error.
operationFailed	0x40001158	Operating authorization service error.
noAnalysisResourceOrNoDataInTheListLibrary	0x40001159	No cloud analytic resources or no data in the list library.
calculationException	0x4000115A	Calculation exception.
allocatingList	0x4000115B	Allocating list.
thisOperationIsNotSupportedByTheCloudAnalytics	0x4000115C	This operation is not supported by the cloud analytic service.
theCloudAnalyticsIsInterrupted	0x4000115D	The operation of cloud analytic service is interrupted.
theServiceIsNotReady	0x4000115E	The service is not ready.
searchingForExternalApiFailed	0x4000115F	Searching external interfaces failed.
noOnlineNode	0x40001160	No node is online.
noNodeAllocated	0x40001161	No allocated node.
noMatchedList	0x40001162	No matched list.
allocatingFailedTooManyFacePictureLists	0x40001163	Allocation failed. Too many lists of big data service.

Sub Status Code	Error Code	Description
searchIsNotCompleted SearchAgain	0x40001164	Current searching is not completed. Search again.
allocatingListIsNotCompleted	0x40001165	Allocating list is not completed.
searchingForCloudAnalyticsResultsFailed	0x40001166	Searching cloud analytic service overtime.
noDataOfTheCurrentLibraryFound	0x40001167	No data in the current library. Make sure there is data in the Hbase.
noFacePictureLibraryIsArmed	0x40001168	No face picture library is armed for big data service.
noAvailableDataSlicingVersionInformationArmedFirstAndSliceTheData	0x40001169	Invalid standard version information.
duplicatedOperationDataSlicingIsExecuting	0x4000116A	Slicing failed. Duplicated operation.
slicingDataFailedNoArmedFacePictureLibrary	0x4000116B	Slicing failed. No arming information in the face big data.
GenerateBenchmarkFileFailedSlicingAgain	0x4000116C	Generating sliced file failed. Slice again.
NonprimaryNodesProhibitedFromSlicingData	0x4000116D	Slicing is not allowed by the backup node.
NoReadyNodeToClusterServers	0x4000116E	Creating the cluster failed. No ready node.
NodeManagementServicesOffline	0x4000116F	The node management server is offline.
theCamera(s)OfTheControlCenterAreAlreadyArmed.DisarmThemFirst	0x40001170	Some cameras in control center are already armed. Disarm them and try again.
theCamera(s)OfTheAreaAreAlreadyArmed.DisarmThemFirst	0x40001171	Some cameras in this area are already armed. Disarm them and try again.
configuringHigh-frequencyPeopleDetectionFailed	0x40001172	Configuring high frequency people detection failed.

Sub Status Code	Error Code	Description
searchingForHigh-frequencyPeopleDetectionLogsFailed.	0x40001173	Searching detection event logs of high-frequency people detection failed.
gettingDetailsOfSearchHigh-frequencyPeopleDetectionLogsFailed.	0x40001174	Getting the search result details of high frequency alarms failed.
theArmedCamerasAlreadyExistInTheControlCenter	0x40001175	Some cameras in control center are already armed.
disarmingFailedTheCamerasNotArmed	0x40001177	Disarming failed. The camera is not armed.
noDataReturned	0x40001178	No response is returned by the big data service.
preallocFailure	0x40001179	Pre-allocating algorithm resource failed.
overDogLimit	0x4000117A	Configuration failed. No more resources can be pre-allocated.
analysisServicesDoNotSupport	0x4000117B	Not supported.
commandAndDispatchServiceError	0x4000117C	Scheduling service of cloud analytic service error.
engineModuleError	0x4000117D	Engine module of cloud analytic service error.
streamingServiceError	0x4000117E	Streaming component of cloud analytic service error.
faceAnalysisModuleError	0x4000117F	Face analysis module of cloud analytic service error.
vehicleAnalysisModuleError	0x40001180	Vehicle pictures analytic module of cloud analytic service error.
videoStructuralAnalysisModuleError	0x40001181	Video structuring module of cloud analytic service error.
postprocessingModuleError	0x40001182	Post-processing module of cloud analytic service error.
frequentlyAppearedPersonAlarmsAlreadyConfiguredForListLibrary	0x40001183	High frequency alarm is already armed for blacklist library.

Sub Status Code	Error Code	Description
creatingListLibraryFailed	0x40001184	Creating list library failed.
invalidIdentityKeyOfListLibrary	0x40001185	Invalid identity key of list library.
noMoreDevicesCanBeArmed	0x40001186	No more camera can be added.
settingAlgorithmTypeForDeviceFailed	0x40001187	Allocating task resource failed.
gettingHighFrequencyPersonDetectionAlarmInformationFailed	0x40001188	Setting high frequency alarm failed.
invalidSearchConfition	0x40001189	Invalid result.
theTaskIsNotCompleted	0x4000118B	The task is not completed.
resourceOverRemainLimit	0x4000118C	No more resource can be pre-allocated.
frequentlyAppearedPersonAlarmsAlreadyConfiguredForTheCameraDisarmFirstAndTryAgain	0x4000118D	The high frequency alarm of this camera is configured. Delete the arming information and try again.
switchtimedifflesslimit	0x4000123b	Time difference between power on and off should be less than 10 minutes.
noMorePeopleNumChangeRulesAdded	0x4000128A	Maximum number of people number changing rules reached.
noMoreViolentMotionRulesAdded	0x4000128D	Maximum number of violent motion rules reached.
noMoreLeavePositionRulesAdded	0x4000128E	Maximum number of leaving position rules reached.
SMRDiskNotSupportRaid	0x40001291	SMR disk does not support RAID.
OnlySupportHikAndCustomProtocol	0x400012A3	IPv6 camera can only be added via Device Network SDK or custom protocols.

Sub Status Code	Error Code	Description
vehicleEnginesNoResource	0x400012A6	Insufficient vehicle engine resources.
noMoreRunningRulesAdded	0x400012A9	Maximum number of running rules reached.
noMoreGroupRulesAdded	0x400012AA	Maximum number of people gathering rules reached.
noMoreFailDownRulesAdded	0x400012AB	Maximum number of people falling down rules reached.
noMorePlayCellphoneRulesAdded	0x400012AC	Maximum number of playing cellphone rules reached.
ruleEventTypeDuplicate	0x400012C8	Event type duplicated.
noMoreRetentionRulesAdded	0x400015AD	Maximum number of people retention rules reached.
noMoreSleepOnDutyRulesAdded	0x400015AE	Maximum number of sleeping on duty rules reached.
polygonNotAllowedCrossing	0x400015C2	Polygons are not allowed to cross.
alreadyRunning	0x40002026	The application program is running.
notRunning	0x40002027	The application program is stopped.
packNotFound	0x40002028	The software packet does not exist.
alreadyExist	0x40002029	The application program already exists.
noMemory	0x4000202A	Insufficient memory.
invalidLicense	0x4000202B	Invalid License.
noClientCertificate	0x40002036	The client certificate is not installed.
noCACertificate	0x40002037	The CA certificate is not installed.
authenticationFailed	0x40002038	Authenticating certificate failed. Check the certificate.
clientCertificateExpired	0x40002039	The client certificate is expired.
HDMIResolutionIllegal	0x40002040	The HDMI video resolution cannot be larger than that of main and sub stream.

Sub Status Code	Error Code	Description
clientCertificateRevocation	0x4000203A	The client certificate is revoked.
CACertificateExpired	0x4000203B	The CA certificate is expired.
CACertificateRevocation	0x4000203C	The CA certificate is revoked.
connectFail	0x4000203D	Connection failed.
loginNumExceedLimit	0x4000203F	No more user can log in.
formattingFailed	0x40002056	Formatting HDD failed.
encryptedFormattingFailed	0x40002057	Formatting encrypted HDD failed.
wrongPassword	0x40002058	Verifying password of SD card failed. Incorrect password.
audiolsPlayingPleaseWait	0x40002067	Audio is playing. Please wait.
twoWayAudioInProgressPleaseWait	0x40002068	Two-way audio in progress. Please wait.
calibrationPointNumFull	0x40002069	The maximum number of calibration points reached.
completeTheLevelCalibrationFirst	0x4000206A	The level calibration is not set.
completeTheRadarCameraCalibrationFirst	0x4000206B	The radar-camera calibration is not set.
pointsOnStraightLine	0x4000209C	Calibrating failed. The calibration points cannot be one the same line.
TValueLessThanOrEqualZero	0x4000209D	Calibration failed. The T value of the calibration points should be larger than 0.
HBDLibNumOverLimit	0x40002092	The number of human body picture libraries reaches the upper limit
theShieldRegionError	0x40002093	Saving failed. The shielded area should be the ground area where the shielded object is located.
theDetectionAreaError	0x40002094	Saving failed. The detection area should only cover the ground area.

Sub Status Code	Error Code	Description
invalidLaneLine	0x40002096	Saving failed. Invalid lane line.
noDetectionArea	0x400050df	No detection area
armingFailed	0x40008000	Arming failed.
disarmingFailed	0x40008001	Disarming failed.
clearAlarmFailed	0x40008002	Clearing alarm failed.
bypassFailed	0x40008003	Bypass failed.
bypassRecoverFailed	0x40008004	Bypass recovery failed.
outputsOpenFailed	0x40008005	Opening relay failed.
outputsCloseFailed	0x40008006	Closing relay failed.
registerTimeOut	0x40008007	Registering timed out.
registerFailed	0x40008008	Registering failed.
addedByOtherHost	0x40008009	The peripheral is already added by other security control panel.
alreadyAdded	0x4000800A	The peripheral is already added.
armedStatus	0x4000800B	The partition is armed.
bypassStatus	0x4000800C	Bypassed.
zoneNotSupport	0x4000800D	This operation is not supported by the zone.
zoneFault	0x4000800E	The zone is in fault status.
pwdConflict	0x4000800F	Password conflicted.
audioTestEntryFailed	0x40008010	Enabling audio test mode failed.
audioTestRecoveryFailed	0x40008011	Disabling audio test mode failed.
addCardMode	0x40008012	Adding card mode.
searchMode	0x40008013	Search mode.
addRemoterMode	0x40008014	Adding keyfob mode.
registerMode	0x40008015	Registration mode.
exDevNotExist	0x40008016	The peripheral does not exist.
theNumberOfExDevLimited	0x40008017	No peripheral can be added.

Sub Status Code	Error Code	Description
sirenConfigFailed	0x40008018	Setting siren failed.
chanCannotRepeatedBinded	0x40008019	This channel is already linked by the zone.
masterSlavesEnable	0x4000802c	The master-slave relationship has taken effect, the slave radar does not support this operation.
forceTrackNotEnabled	0x4000802d	Mandatory tracking is disabled.
isNotSupportZoneConfigByLocalArea	0x4000802e	This area does not support the zone type.
alarmLineCross	0x4000802f	Trigger lines are overlapped.
zoneDrawingOutOfRange	0x40008030	The drawn zone is out of detection range.
alarmLineDrawingOutOfRange	0x40008031	The drawn alarm trigger line is out of detection range.
hasTargetInWarningArea	0x40008032	The warning zone already contains targets. Whether to enable mandatory arming?
radarModuleConnectFail	0x40008033	Radar module communication failed.
importCfgFilePasswordErr	0x40008034	Incorrect password for importing configuration files.
inProgramMode	0x4000801B	The keypad is in programming mode.
inPaceTest	0x4000801C	In pacing mode.
arming	0x4000801D	Arming.
subscribeTypeErr	0x4000a016	This metadata type is not supported to be subscribed.
startAppFail	/	Starting running application program failed.
yuvconflict	/	The raw video stream conflicted.
overMaxAppNum	/	No more application program can be uploaded.
noFlash	/	Insufficient flash.
noFlash	/	The platform mismatches.

StatusCode=5

Sub Status Code	Error Code	Description
badXmlFormat	0x50000001	Invalid XML format.

StatusCode=6

Sub Status Code	Error Code	Description
badParameters	0x60000001	Invalid parameter.
badHostAddress	0x60000002	Invalid host IP address.
badXmlContent	0x60000003	Invalid XML content.
badIPv4Address	0x60000004	Invalid IPv4 address.
badIPv6Address	0x60000005	Invalid IPv6 address.
conflictIPv4Address	0x60000006	IPv4 address conflicted.
conflictIPv6Address	0x60000007	IPv6 address conflicted.
badDomainName	0x60000008	Invalid domain name.
connectSreverFail	0x60000009	Connecting to server failed.
conflictDomainName	0x6000000A	Domain name conflicted.
badPort	0x6000000B	Port number conflicted.
portError	0x6000000C	Port error.
exportErrorData	0x6000000D	Importing data failed.
badNetMask	0x6000000E	Invalid sub-net mask.
badVersion	0x6000000F	Version mismatches.
badDevType	0x600000010	Device type mismatches.
badLanguage	0x600000011	Language mismatches.
incorrentUserNameOrPasswor d	0x600000012	Incorrect user name or password.
invalidStoragePoolOfCloudServ er	0x600000013	Invalid storage pool. The storage pool is not configured or incorrect ID.
noFreeSpaceOfStoragePool	0x600000014	Storage pool is full.
riskPassword	0x600000015	Risky password.

Sub Status Code	Error Code	Description
UnSupportCapture	0x600000016	Capturing in 4096*2160 or 3072*2048 resolution is not supported when H.264+ is enabled.
userPwdLenUnder8	0x600000023	At least two kinds of characters, including digits, letters, and symbols, should be contained in the password.
userPwdNameSame	0x600000025	Duplicated password.
userPwdNameMirror	0x600000026	The password cannot be the reverse order of user name.
beyondARGSRangeLimit	0x600000027	The parameter value is out of limit.
DetectionLineOutOfDetectionRegion	0x600000085	The rule line is out of region.
DetectionRegionError	0x600000086	Rule region error. Make sure the rule region is convex polygon.
DetectionRegionOutOfCountingRegion	0x600000087	The rule region must be marked as red frame.
PedalAreaError	0x600000088	The pedal area must be in the rule region.
DetectionAreaABError	0x600000089	The detection region A and B must be in the a rule frame.
ABRegionCannotIntersect	0x60000008a	Region A and B cannot be overlapped.
customHBPIDError	0x60000008b	Incorrect ID of custom human body picture library
customHBPIDRepeat	0x60000008c	Duplicated ID of custom human body picture library
dataVersionsInHBDLibMismatches	0x60000008d	Database versions mismatches of human body picture library
invalidHBPID	0x60000008e	Invalid human body picture PID

Sub Status Code	Error Code	Description
invalidHBDID	0x6000008f	Invalid ID of human body picture library
humanLibraryError	0x60000090	Error of human body picture library
humanLibraryNumError	0x60000091	No more human body picture library can be added
humanImagesNumError	0x60000092	No more human body picture can be added
noHumanInThePicture	0x60000093	Modeling failed, no human body in the picture
analysisEnginesNoResourceError	0x60001000	No analysis engine.
analysisEnginesUsageExcced	0x60001001	The engine usage is overloaded.
PicAnalysisNoResourceError	0x60001002	No analysis engine provided for picture secondary recognition.
analysisEnginesLoadingError	0x60001003	Initializing analysis engine.
analysisEnginesAbnormaError	0x60001004	Analysis engine exception.
analysisEnginesFacelibImporting	0x60001005	Importing pictures to face picture library. Failed to edit analysis engine parameters.
analysisEnginesAssociatedChannel	0x60001006	The analysis engine is linked to channel.
smdEncodingNoResource	0x60001007	Insufficient motion detection encoding resources.
smdDecodingNoResource	0x60001008	Insufficient motion detection decoding resources.
diskError	0x60001009	HDD error.
diskFull	0x6000100a	HDD full.
facelibDataProcessing	0x6000100b	Handling face picture library data.
capturePackageFailed	0x6000100c	Capturing packet failed.
capturePackageProcessing	0x6000100d	Capturing packet.

Sub Status Code	Error Code	Description
noSupportWithPlaybackAbstract	0x6000100e	This function is not supported. Playback by video synopsis is enabled.
insufficientNetworkBandwidth	0x6000100f	Insufficient network bandwidth.
tapeLibNeedStopArchive	0x60001010	Stop the filing operation of tape library first.
identityKeyError	0x60001011	Incorrect interaction command.
identityKeyMissing	0x60001012	The interaction command is lost.
noSupportWithPersonDensityDetect	0x60001013	This function is not supported. The people density detection is enabled.
ipcResolutionOverflow	0x60001014	The configured resolution of network camera is invalid.
ipcBitrateOverflow	0x60001015	The configured bit rate of network camera is invalid.
tooGreatTimeDifference	0x60001016	Too large time difference between device and server.
noSupportWithPlayback	0x60001017	This function is not supported. Playback is enabled.
channelNoSupportWithSMD	0x60001018	This function is not supported. Motion detection is enabled.
channelNoSupportWithFD	0x60001019	This function is not supported. Face capture is enabled.
illegalPhoneNumber	0x6000101a	Invalid phone number.
illegalCertificateNumber	0x6000101b	Invalid certificate No.
linkedCameraOutLimit	0x6000101c	Connecting camera timed out.
achieveMaxChannelLimit	0x6000101e	No more channels are allowed.
humanMisInfoFilterEnabledChannelNumError	0x6000101f	No more channels are allowed to enable preventing false alarm.

Sub Status Code	Error Code	Description
humanEnginesNoResource	0x60001020	Insufficient human body analysis engine resources.
taskNumberOverflow	0x60001021	No more tasks can be added.
collisionTimeOverflow	0x60001022	No more comparison duration can be configured.
invalidTaskID	0x60001023	Invalid task ID.
eventNotSupport	0x60001024	Event subscription is not supported.
invalidEZVIZSecretKey	0x60001034	Invalid verification code for Hik-Connect.
needDoubleVerification	0x60001042	Double verification required
noDoubleVerificationUser	0x60001043	No double verification user
timeSpanNumOverLimit	0x60001044	Max. number of time buckets reached
channelNumOverLimit	0x60001045	Max. number of channels reached
noSearchIDResource	0x60001046	Insufficient searchID resources
noSupportDeleteStrangerLib	0x60001051	Deleting stranger library is not supported
noSupportCreateStrangerLib	0x60001052	Creating stranger library is not supported
behaviorAnalysisRuleInfoError	0x60001053	Behavior analysis rule parameters error.
safetyHelmetParamError	0x60001054	Hard hat parameters error.
OneChannelOnlyCanBindOneEngine	0x60001077	No more engines can be bound.
TransitionUseEmmc	0x60002000	Starting device failed. The EMMC is overused.
AdaptiveStreamNotEnabled	0x60002001	The stream self-adaptive function is not enabled.
AdaptiveStreamAndVariableBitrateEnabled	0x60002002	Stream self-adaptive and variable bitrate function cannot be enabled at the same time.

Sub Status Code	Error Code	Description
noSafetyHelmetRegion	0x60002023	The hard hat detection area is not configured (if users save their settings without configuring the arming area, they should be prompted to configure one).
unclosedSafetyHelmet	0x60002024	The hard hat detection is enabled (If users save their settings after deleting the arming area, they should be prompted to disable hard hat detection first and then delete the arming area).
width/ heightRatioOfPictureError	0x6000202C	The width/height ratio of the uploaded picture should be in the range from 1:2 to 2:1.
PTZNotInitialized	0x6000202E	PTZ is not initialized.
PTZSelfChecking	0x6000202F	PTZ is self-checking.
PTZLocked	0x60002030	PTZ is locked.
deployExceedMax	0x60006018	The arming connections exceed the maximum number.
detectorTypeMismatch	0x60008000	The detector type mismatched.
nameExist	0x60008001	The name already exists.
uploadImageSizeError	0x60008016	The size of the uploaded picture is larger than 5 MB.
laneAndRegionOverlap	/	The lanes are overlapped.
unitConfigurationNotInEffect	/	Invalid unit parameter.
ruleAndShieldingMaskConflict	/	The line-rule region overlaps with the shielded area.
wholeRuleInShieldingMask	/	There are complete temperature measurement rules in the shielded area.

Sub Status Code	Error Code	Description
LogDiskNotSetReadOnlyInGroupMode	0x60001100	The log HDD in the HDD group cannot be set to read-only.
LogDiskNotSetRedundancyInGroupMode	0x60001101	The log HDD in the HDD group cannot be set to redundancy.

StatusCode=7

SubStatusCode	Error Code	Description
rebootRequired	0x70000001	Reboot to take effect.

